

STATISTICAL METHODS

(Volume I)

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Tata McGraw-Hill Publishing Company Limited

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San Juan Santiago Singapore Sydney Tokyo Toronto



Tata McGraw-Hill

Published by the Tata McGraw-Hill Publishing Company Limited,
7 West Patel Nagar, New Delhi 110 008.

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This edition can be exported from India only by the publishers,
Tata McGraw-Hill Publishing Company Limited.

ISBN 13: 978-0-07-026350-5

ISBN 10: 0-07-026350-7

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Published by the Tata McGraw-Hill Publishing Company Limited, 7 West Patel Nagar, New Delhi 110 008, typeset at Bukprint India, B-180A, Guru Nanak Pura, Laxmi Nagar-110 092 and printed at Adarsh Printers, C-50-51, Mohan Park, Naveen Shahdara, Delhi 110 032

Cover: Rashtriya Printers

RALXRRBFRCXRY

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PREFACE

Statistics: The whole concept of this mathematical science revolves around *data*. Thus, **Statistical Methods** deal with techniques for collecting, analysing and drawing conclusions from data. Today, the subject has applications, wide and varied, spanning from disciplines like natural and social sciences, humanities, government and business to fields like economics and criminology. For ease of understanding of the subject, '*Statistical Methods*' has been published in two volumes—Volume I and II, providing requisite and thorough coverage.

PURPOSE

This first volume will serve the purpose of a text to the students of UG and PG courses such as B.Sc. (Statistics), B.Sc. (Maths), M.Sc. (Statistics), B.A. (Economics), B.B.A., B.Com, and M.B.A., wherever statistics features as a fundamental paper. It presents the mathematics underlying elementary statistical methods in as simple a manner as possible. Detailed solution of numerous questions (both theoretical and numerical) from C.A. (Intermediate), I.C.W.A. (Intermediate and Final), and M.Com., B.Sc. (Economics, Honours), B.Com (Honours and Pass), Diplomas in Social Welfare and Management examinations of various Universities have been incorporated to make the book more versatile and hence, will be of considerable value to these students as well.

FEATURES

The structuring of this book facilitates students to understand statistical methods as efficiently as possible. It succeeds at simplifying the subject and makes it easy even to an average student. A brief discussion of important theoretical aspects has been provided in a simple language. Special notes and techniques of working out numerical problems, with elaborate steps enhance the understanding of concepts. Chapter-end exercises with answers have been provided to enable students confirm their mastery over the subject.

In order to make the book more user-friendly, the following important features have been incorporated.

- ☐ *Around 1000 Solved and Unsolved Problems.*
- ☐ *All Problems graded in order of difficulty.*
- ☐ *A complete list of Statistical Formulae and Results in the Appendix.*
- ☐ *Special Supplement including Solved Problems and Additional Exercises with Answers.*
- ☐ *Abundant Exercises, Solved Examples, Theoretical Questions, Short Answer Questions and Relevant Tables and Figures.*

Excellent Pedagogy includes Solved Examination Questions of Calcutta University (C.U.), Burdwan University (B.U.), Institute of Chartered Accountants of India (C.A.), Institute of Cost & Works Accountants of India (I.C.W.A.), Diploma in Social Welfare (D.S.W.) of Calcutta University, Diploma in Management (D.M.) of Calcutta University and West Bengal Higher Secondary Council (W.B.H.S.).



ORGANISATION

The coherent arrangement of material in this book makes it a good reference source for statistical methods. The first volume is divided into 10 chapters. **Chapter 1** deals with Collection of Data, Classification and Tabulation. Charts and Diagrams and Useful Mathematical Devices are enumerated in **Chapters 2** and **3**. Graphical ways of summarising a group of data are dealt with in **Chapter 4** on Frequency Distributions. **Chapter 5** on Measures of Central Tendency covers ways of describing the central position of a frequency distribution for a group of data. Measures of Dispersion are elucidated in **Chapter 6**. **Chapter 7** focuses on Moments, Skewness and Kurtosis whereas **Chapter 8** explains Curve Fitting and Method of Least Squares. **Chapters 9** and **10** explain concepts of Correlation, Regression and Interpolation. Subsequently, the second volume spans over 9 Chapters covering topics like probability, distributions, sampling, tests of significance, variance, vital statistics and quality control. Readers are recommended to also study Volume II for sound understanding of the subject.

ACKNOWLEDGEMENTS

Thanks are due to The Institute of Cost and Works Accountants of India and the Institute of Chartered Accountants of India for the kind permission accorded to make use of their questions. Also, a note of acknowledgement to all those who have been directly or indirectly associated with the book.

Every effort has been made to make this text as error-free as possible. However, feedback from readers is always welcome.

1 COLLECTION OF DATA CLASSIFICATION AND TABULATION

1.1 MEANING OF 'STATISTICS'

Statistics is a familiar word now-a-days. It has two different meanings. In the plural sense, the word implies a set of numerical figures, usually obtained by measurement or counting. These are also collectively known as *data*. In the singular sense, 'Statistics' refers to the subject of scientific activity which deals with the theories and methods of collection, analysis and interpretation of such data.

Example 1.1 Distinguish between 'Statistics as data' and 'Statistics as tool'.
Illustrate your answer with examples. [C.U., M.Com. '66]

Or,

What are the two senses in which the term *Statistics* is generally used?
[B.U., B.A.(Econ) '73]

Solution The word *Statistics* is used in two different senses. Firstly, it refers to a collection of numerical figures or data obtained from the fields of human activity. For example, we frequently speak of Statistics of Price, Statistics of Food, Statistics of Educational Institutions, and so on, denoting thereby a set of facts and figures on the concerned topic. Secondly, the word is used to imply the subject of scientific activity dealing with the theories and methods of collection, analysis and interpretation of such data. It may be mentioned that unlike Physics, Chemistry, Biology, Geology etc., Statistics is not a separate branch of science. It is a scientific method or tool, like Mathematics, and is applied for handling numerical data in all branches of science, commerce, trade or humanities, where measurement or counting is possible. Heaps of apparently inconsistent data, resulting from a complex combination of many known or unknown factors, are analysed by applying statistical tools and techniques in order to find out the underlying pattern of variation. For example, the monthly sales figures of a company over several years, when plotted on a graph paper may show haphazard fluctuations; but when the time series data (see Chapter 14) are analysed, they may reveal important seasonal and cyclical movements superimposed on trend, which may be used with advantage for forecasting future sales.



Example 1.2 What are the characteristics and limitations of *Statistics*?
[C.A., Nov. '70, '72, '76]

Solution *Characteristics* 1. In Statistics, all available information are expressed in quantitative terms. In the study of 'variables' numerical figures arise as a direct consequence



of measurement, e.g. age in years—28, 34, 27, etc. If the character studied is an ‘attribute’, the number of cases in each category gives rise to figures, e.g. defective articles 23, non-defectives 389. Sometimes in the study of attributes like intelligence, scores or marks are assigned to certain suitably framed questions, and intelligence is then indicated by total scores, say 53, 48, etc. Even ‘dispersion’, ‘skewness’, ‘probability’, business activity, etc. are expressed numerically.

2. Statistics must be aggregates of facts. A single fact, say, 672 lakh metric tons production of coal in India during the year 1965, is of no interest to a statistician. But if production figures for several consecutive years are available, they may be statistically analysed to draw important conclusions.

3. Statistics must be related to some field of inquiry. In other words, statistical data will be collected with a definite object in view. Stray collection of figures will be of no use.

4. Statistics should be capable of being related to each other, so that some cause-and-effect relationship can be established. In regression analysis, these are useful for ‘prediction’.

5. Statistics are affected by a multiplicity of causes. In all fields of inquiry, the observed data are the result of a large number of factors, each of which contributes to the final figure. It is for the statistician to split up the effects due to component factors and study them.

6. Exactness cannot be guaranteed. No estimates or forecasts or statistical decisions can be taken to be exact. They are all subject to certain amount of inaccuracy whose limits may be stated in terms of probability. Almost in all cases, estimates and conclusions are arrived at on the basis of samples, and so it cannot be expected that they will lead to exact results.

Limitations 1. Statistics is applicable only to quantitative data. This cannot be used to study such events which cannot be expressed numerically.

2. Statistics can be used to analyse only collective matters, and not individual events.

3. Statistical decisions are applicable only on the average and in the long run. They may not hold in a particular case.

4. Statistical methods, if not applied in the proper perspective of the collected data, may lead to false conclusions. These should therefore be handled with utmost care and by experts only.

5. Statistical data must be uniform, in the sense that they should be subject to a stable causal system. There should not be much change in the group of factors responsible for variation in the data.



Example 1.3 *Illustrate with suitable examples the use of statistical methods in commerce, business and industry.*

[C.A., May '69 & '73; Nov. '70 & '77; I.C.W.A., June '73; C.U., M.Com., '74]

Solution With the gradual industrialisation and consequent expansion of trade, businessmen can no longer rely on the old system of hit-or-miss methods, or leave their future on chances. They have now to proceed on scientific principles, prepare themselves for competitive markets and plan their business accordingly. The traders and manufacturers have therefore to depend on a variety of factors, e.g. past sales records, present labour conditions and prices of raw materials used, prices of similar products already available in the market, etc. All these factors are statistically taken account of before fixing the price of a new commodity, so that it may find a suitable place in the market.

The following are some of the statistical techniques applied in Commerce, Business and Industry:

1. The various statistical measures of ‘average’, ‘dispersion’, ‘skewness’, ‘correlation’, etc. are generally useful for bringing out the main characteristics of available records. Their calculation enables comparison between different time periods or geographical areas.



2. 'Time series' analysis helps in isolating the components of variation noticed in the chronological records of sales, profits, prices, production, etc. These components—'trend', 'seasonal' and 'cyclical'—are used for future planning and forecasts.

3. 'Index numbers' of various types are constructed by the use of statistical methods. Index number of price, Index number of production, Index number of cost of living, Index number of business activity, etc. are so important in business and commerce that they have come to be known as 'business barometers'.

4. 'Regression' analysis enable establishment of relationship between various economic factors, e.g. input and output, production and sales, demand and per capita income. 'Regression equation' of a dependent variable upon one or several independent variables also helps in assessing the impact of each of the several factors, and may be used for forecasting.

5. Various sampling methods, e.g. 'random sampling', 'stratified sampling' etc. are available for conducting business and market surveys with the least amount of time, labour and money. These may also be used in test audit for checking the accuracy of records.

6. 'Statistical Quality Control' is now an indispensable tool for controlling the quality of manufactured products and in the maintenance of quality during production. 'Sampling inspection' is used for decision to accept or reject lots of manufactured product.

7. 'Vital statistics' and 'demography' are useful for the calculation of mortality rates in a community classified by age, sex, profession etc. These rates, or probabilities of death, are used in insurance business for fixing the rates of premium payable at different ages.

8. In the recruitment of right type of personnel for a job, the aptitude and efficiency of candidates are statistically determined by using 'test scores'.



1.2 VARIABLE AND ATTRIBUTE

The character of statistical information collected from a group of individuals or objects, is of two types—quantitative and qualitative. Information about the ages of a group of men is quantitative, because age is expressed in numbers, say 29 years, 43.5 years, etc. Religion of a group of men is qualitative, because religion cannot be stated in numerical terms; e.g. either Hindu or Buddhist or Christian etc. Age is a quantitative character and religion is a qualitative character. The quantitative character is technically called *variable* and the qualitative character is called *attribute*. A variable takes different 'values' and these values can be measured numerically in suitable units. An attribute cannot be measured, but can only be classified under different heads or 'categories'.

Example 1.4 Explain with illustration the distinction between (i) an attribute and a variable, and (ii) a continuous variable and a discrete variable.

[C.U., B.A.(Econ) '72; M.Com. '69, '74; I.C.W.A., June '74]

Solution Variable is a quantitative character of an individual or item and its values can always be measured. A variable whose values depend on chance and cannot be predicted is called a *random variable* or *variate*. Statistical theories deal with random variables only, and hence very often the term 'variable' is used in practice. Variates are of two types—(i) Continuous, and (ii) Discrete or Discontinuous.

A variate is said to be *continuous*, when it can take any value within a specified interval. For example, height of an individual may have any value between, say, 58 inches and 72 inches. It may be 65.2 inches, 65.23 inches, or even 65.22709 inches, provided we could measure the height so precisely. Hence, a continuous variate can take an infinite number of values within a given interval, however small it may be. Height, weight, temperature, density etc. are examples of a continuous variate.



A *discrete* variate, however can take only some isolated values. The number of children per family, the number of workers in a factory, the price of a cinema ticket sold during a week, etc. are examples of this type. The number of children per family can take only the values 0, 1, 2, 3, ..., i.e. whole numbers; it cannot take any value, e.g. a fractional value. Similarly, the price of a cinema ticket has only some specified values.



Example 1.5 Classify each of the following characters as either an attribute, or a discrete variable, or a continuous variable: —Family size, Family income (per month), Mother tongue of a student, size of agricultural holding of a family, Division obtained (by a student) at the Higher Secondary Examination. [B.U., B.A.(Econ) '73]

Solution (i) Family size—Discrete variable. Because it can take only some isolated values. The family size (i.e. number of members) may be either 1, or 2, or 3, or 4, etc. It cannot take any value. For example, we cannot speak of 2.73 members in a family.

(ii) Family income (per month)—Discrete variable. It can take values like Rs 874.27 P., or Rs. 874.28 P., etc. but not any value. For example, family income cannot take a value between the above two, because income in fractions of a Paisa is impossible.

(iii) Mother tongue—Attribute. It cannot be measured, but can only be classified as Bengali, Hindi, Tamil, etc.

(iv) Size of agricultural holding—Continuous variable. It can take any value within a given range, say 47.325697... acres. The sizes of two holdings may differ by any amount (theoretically), say, even by fractions of a square inch.

(v) Division obtained by a student—Attribute. Although it is expressed in numbers, e.g. Division 1, or 2, or 3, Division is not actually measured. It is only a means of classification into one of the categories.



1.3 PRIMARY DATA AND SECONDARY DATA

Statistical data may be of two types, viz. 'primary' and 'secondary'. *Primary data* are those which are collected for a specific purpose directly from the field of enquiry, and hence are original in nature. Such data are published by authorities who themselves are responsible for their collection. *Secondary data* are such numerical information which have previously been collected by some agency for one purpose and are merely compiled from that source for use in a different connection. In fact, data collected by someone when used by another, or collected for one purpose when used for another, will be called Secondary data. The same data are primary in the hands the collecting authority, but are secondary in the hands of another. For example, the census figures published by the Registrar General of India will be primary data, while the same data contained in any other publication will be called secondary data.

Sometimes, the sources of data are also classified as primary or secondary, depending on the type of data available therefrom. The authority who collects the data from the field of enquiry is known as *Primary source*, while some agency publishing the data originally collected by another is known as *Secondary source*.

The nature, scope and objects of a statistical investigation should be taken into account for deciding whether the data are to be collected originally or whether the available secondary data are to be utilised. It is generally preferable to make use of primary data from several standpoints—(i) Such data usually show detailed information



and a description regarding the definition of the terms used. (ii) Very often, a note on the method of collection and any approximations used are also available, so that while using these data it can be decided in advance how much reliance can be placed on these figures. (iii) Secondary data usually contain errors due to transcription, rounding etc. and hence are hardly reliable. In spite of all the advantages of primary data, secondary data are used when either due to limitations of time and money at the disposal of the investigator the data cannot be collected directly, or it becomes necessary to compare the data collected over a period of time, or utmost accuracy is not essential.

Publications containing Primary Data:

- (i) "Reserve Bank of India Bulletin" issued *monthly* by the Reserve Bank of India, Bombay.
- (ii) "Annual Report of the Chief Inspector of Mines in India", issued *annually* by the office of the Chief Inspector of Mines, Dhanbad.
- (iii) "Indian Textile Bulletin", issued *monthly* by the Textile Commissioner, Bombay.
- (iv) "Monthly Coal Bulletin", issued *monthly* by the Office of the Chief Inspector of Mines, Dhanbad.

Publications containing Secondary Data:

- (i) "Statistical Abstract of the Indian Union", issued *annually* by Central Statistical Organisation (C.S.O.), New Delhi.
- (ii) "Monthly Abstract of Statistics", issued by C.S.O.

Example 1.6 Describe the various methods of collecting primary data and comment on their relative advantages.

[I.C.W.A., Jan. '72; C.U., M.Com. '68 & '71; C.A., May '66 & '68, Nov. '69; D.S.W., Nov. '73]

Solution The following methods are generally used for collection of primary data:

- (a) Direct personal observation,
- (b) Indirect oral investigation,
- (c) Questionnaires sent by mail,
- (d) Schedules sent through investigators.

In '*direct personal observation*', the investigator collects the requisite information personally through observation or by measurement. For example, in order to study the conditions of students residing in Calcutta hostels, the investigator meets the students in their hostels and collects necessary data after a personal study. The method is time-consuming and costly, but yields very accurate results. It is therefore suitable for such studies when the field of enquiry is small.

In '*indirect oral investigation*', data are collected through indirect sources. Persons who are likely to have information about the problem, are interrogated and on the basis of their answers, factual data have to be compiled. Most of the Commissions of Enquiry or Committees appointed by Government collect primary data by this method. The accuracy of the method depends largely upon the type of persons interviewed and hence these persons have to be selected very carefully.

In the '*mailed questionnaire method*', the most important instrument is the questionnaire. This contains a set of questions, relevant to the subject of enquiry, answers where to are expected to yield the requisite information. Printed questionnaires are sent by mail to a selected list of persons, with the request to return them duly filled in. Supplementary instructions regarding the definitions of terms used and methods of filling up the forms, should also accompany the



questionnaire. The method is cheap and expeditious, and a large area can be covered within a limited cost. Two principal disadvantages of the method are—the low degree of reliability of the collected data, and a large number of non-responses.

'Schedules sent through investigators' is the most widely used method of collection of primary data. Here paid investigators (also called 'enumerators') are employed for data collection. The investigators carry with them printed 'schedules' specially designed for the purpose, interview the people concerned, and fill up the schedules themselves on the spot, based on answers received from the informants. The method is very popular and yields satisfactory results. Much of the accuracy of the collected data however depends on the ability and tactfulness of investigators, who are given special training as to how they should elicit the correct information through friendly discussions. This method is adopted during the decennial census of population in this country.



Example 1.7 Define "Secondary Data." State the chief sources, and point out the dangers involved in their use and what precautions are necessary before using them.

[C.A., Nov. '67 & '71, May '73; I.C.W.A., Dec. '73]

Solution Secondary data are such statistical information which have already been collected by someone for his own purpose and are now available for use by others for their purposes. Indeed, numerical data which are not gathered directly from the field of enquiry, but are merely compiled from other sources, are referred to as secondary data. The chief sources of such data are:

- (i) Official publications of State and Central Governments, Foreign Governments, and international bodies, such as U.N.O., I.L.O. etc.
- (ii) Publications and reports of trade associations, Chambers of Commerce, co-operative societies etc.
- (iii) Journals and magazines published by private agencies, e.g. Commerce, Capital, etc.
- (iv) Reports of Committees and Commissions of Enquiry.
- (v) Unpublished reports prepared by research scholars, labour and trade unions, etc.

One must be very careful in using secondary data, because of their many limitations. For example, the definition and coverage of the terms, and the units of measurement may not quite agree with those required by the user. The basis of classification of such data may also be different. Sometimes the available secondary data may be mere estimates and not facts. The method employed for the collection of data may not be satisfactory. Secondary data in most cases contain errors due to transcription, rounding etc.

Detailed scrutiny of secondary data is therefore extremely essential before putting them into use. The user must satisfy himself with (i) the scope and object of enquiry for which the data were originally collected, (ii) method of collection, (iii) time and area covered by the data, (iv) precise definitions of the terms used, (v) extent of accuracy of the data, and (vi) the integrity and experience of the authority who collected these data.



1.4 POPULATION (OR UNIVERSE) AND SAMPLE

The totality of statistical information on a particular character, from all members covered by an enquiry, is called *population* or *universe*; e.g. population of income, population of registered factories in West Bengal, etc. The *sample* is a selected part of the population and is used to throw light on the population characteristics. In a socio-economic investigation covering the inhabitants of Calcutta city, if the monthly income



figures of each and every Calcuttan could be obtained, the whole collection of figures of income would constitute the 'population of income.' This, however, is impracticable, and only a few thousand inhabitants would ordinarily be interviewed and their income figures obtained. These few thousand figures of income would comprise a 'sample of income', based on which the average income is estimated.

1.5 COMPLETE ENUMERATION (OR CENSUS) AND SAMPLE SURVEY

Example 1.8 *Distinguish between Census method and Sample Survey method.*
[C.A., Nov. '75 & '76; May '70; D.S.W., Nov. '69]

Solution There are two ways of collection of statistical data—(i) *Census* or *Complete enumeration*, and (ii) *Sample survey*. In census, information is collected from each and every member of the population, whereas sample survey refers to the study of population on the basis of information obtained from a sample. Census of population (human) in India and other countries of the world is an example of complete enumeration. The 'test audit' covering only a small number of entries, which is applied to verify the accuracy of all the entries in books of accounts, is an example of sample survey method. The same method is employed in estimating the crop yield of a country, or in counting the number of trees in a forest.

The census method, since the entire population is investigated, would require a large number of enumerators and other staff involving considerable time, money and labour. It is therefore generally applied for intensive investigations or when the number of members in the population is not large. Although the method is expected to yield accurate information, in practice the results are influenced by large non-sampling errors, whose magnitudes cannot be ascertained.

The sample survey method involving only a small part of the population, has in comparison with the census method, greater flexibility as regards intensity of data collection and the degree of accuracy of the final results, depending on the amount of time and money available for the purpose. Here sampling errors no doubt influence the final conclusions, but non-sampling errors can be kept to a minimum because of the possibility of deployment of highly trained personnel. On the other hand, the limits of sampling error can be stated in terms of probability. Sample survey method is suitable for all situations, but in some cases census method cannot be applied.



Example 1.9 *Discuss the advantages of Sampling methods over the Census method of collecting statistical information.*

[I.C.W.A., Jan. '65; C.U., B.A.(Econ) '69, '73;
B.U., B.A.(Econ) '67; C.A., Nov. '74 & '76]

Solution Sample survey method has definite advantages over census method of collecting statistical information from several standpoints:

1. Sample survey takes *less time, labour and money* than is necessary in complete enumeration, because only a part of the whole population is investigated. In most cases, time and money available for the completion of an investigation are limited, and as such it becomes necessary to employ sample survey method.

2. There is *greater scope* in sample survey than in census. In the former, only a small group of skilled investigators is employed for collection of information. Although the cost per investigator is larger in sample survey than in census, because of the specialized training to be imparted to these personnel, the amount of information collected by each is much larger. As



such, a larger geographical area can be covered and more intensive data collection can be made in sample survey with the same amount of money spent in census.

3. Sample survey yields *more precise results* than census, because of the deployment of specially trained investigators, and the possibility of better control and supervision over them. Census necessitates a large army of investigators all of whom cannot be expected to have the same standards of skill, intelligence and efficiency. Sample survey no doubt introduces sampling error, but non-sampling errors are practically absent here. On the other hand, although in the census method, there is no question of sampling error (because the whole population is investigated), yet large magnitudes of non-sampling errors affect the accuracy of results more severely than the sampling errors introduced in a sample survey.

4. An extra advantage of sample survey method is that the *magnitude of error is known*. Sample survey depends on the laws of probability, and as such the magnitude of sampling errors can be theoretically calculated. Since non-sampling errors do not follow any definite probability law, no such estimate of error is possible in census method.

5. In some cases, *complete enumeration is not feasible* and sample survey method is the only way. For example, a rice merchant can not afford to examine every single grain of rice he purchases. He has to depend only on a sample, based on which he forms an idea about the quality of rice in the whole consignment.

Because of these advantages, random sampling and various other sampling techniques are getting very popular now-a-days. A great limitation of sample survey method is that it yields results for the concerned phenomena as a whole or its broad subdivisions, but not for small sectors. Therefore, if time and cost are not important factors or the population is not very large, census method will be more appropriate than sample survey method.



1.6 STATISTICAL ENQUIRY

Example 1.10 Describe the different stages of a Statistical enquiry.

[C.U., M.Com. '70; C.A., May '77]

Solution 'Statistical enquiry' refers to some investigation wherein relevant information is collected, analysed and interpreted by the application of statistical methods. Broadly, the different stages of a statistical enquiry are as follows :

1. *Planning the Enquiry*—In any scientific investigation, planning is essential. Without proper planning the collected data may not be found suitable for the purpose and considerable time and money will be wasted. The following points should be considered at the planning stage:—
 (i) *Object and scope of the enquiry*—The first and most essential thing is to prepare a clear-cut statement of the purpose of the enquiry. This will determine its scope. (ii) *Source of data, whether primary or secondary*—Usually primary data are collected during statistical enquiries, and as such the method of collection must be decided in advance. This depends on the time and money available for the investigation. (iii) *Type of enquiry, whether census or sample survey*—Because of the many advantages of sample survey over census, most statistical enquiries use the former. The type of sampling to use, e.g. random, stratified, or multi-stage etc. and the selection of the actual sample, has to be finalised. (iv) *Definition of statistical units*—It is extremely essential that the units be accurately and clearly defined, and the same definitions be strictly adhered to during the course of the whole investigation. (v) *Degree of accuracy*—The standard of accuracy of the results and of the final conclusions derived therefrom must be determined in advance on the basis of available resources. Absolute accuracy is neither possible nor very necessary. The enquiry should be so planned as to obtain the maximum accuracy within the limited cost.

2. *Collection of Data*—Numerical data form the raw material for any statistical enquiry, and hence the collection of accurate data is the most important part in the whole investigation. The



method of collection depends to a large extent on the nature, object and scope of the enquiry on one hand and the availability of time and money on the other. For the collection of primary data, one of the following methods may be used—(i) Direct personal observation, (ii) Indirect oral investigation, (iii) Sending postal questionnaires, and (iv) Sending schedules through investigators. Of these, the first gives fairly accurate results; but it is very costly and has a limited scope. The third method has a very large coverage, but suffers from the disadvantage of a high proportion of non-response or incomplete returns. Collection of data through enumerators equipped with schedules is the most widely used method. This involves organisational problems associated with the selection of suitable enumerators and training of these personnel for tactful collection of data.

3. *Editing the Returns and Presentation of Data*—Soon after the collection of data begins, arrangements should be made to receive and scrutinize the completed forms in batches. These returns usually contain many inconsistencies, errors and omissions. The defective returns should be immediately referred back for correction. When all returns have been received and checked, the information is classified and tabulated. Usually many tables of various types have to be prepared and in large-scale enquiries, it is not possible to complete the work manually. Specially developed machines, e.g. electric tabulators, may then be employed for the purpose, where the work of sorting and tabulation is simultaneously done and at a great speed. However, when the enquiry does not involve a large number of returns, manual sorting and tabulation is resorted to. Besides tabular presentation, numerous charts and diagrams are also prepared.

4. *Analysis of Data and Interpretation of Results*—Statistical analysis is a method of abstracting significant facts from the large mass of numerical data collected during the enquiry. The process includes determination of various statistical measures, the estimation of statistical constants, and subsequent tests of significance. The techniques for analysis require specialised knowledge in statistical theories. The results of analysis are then interpreted and conclusions are drawn.

5. *Preparation of Report*—After the work of the enquiry has been completed, it is necessary that a detailed report is published. This report should give a complete description of all the stages of the enquiry, definitions of the terms used, coverage, degree of reliability of the data, and final results of the enquiry. Charts and tables are also included in the report. The report is helpful in planning any future enquiries on a related problem.

Example 1.11 *What do you mean by a questionnaire? State the essential points to be observed in drafting a good questionnaire.*

Describe carefully the merits and defects of the mailed questionnaire method of investigation and the precautions necessary in using this method of investigation.

[I.C.W.A., July '72, Dec. '73; C.A., Nov. '73 & May '71, '75 '76]

Solution The 'questionnaire' is a proforma containing a sequence of questions relevant to a statistical enquiry. It is used for collection of primary data from individual persons through their responses to the set of questions. The questionnaire is used in two ways. Very often, questionnaires are sent by post to individual informants who are required to write the answers against each question and return the completed proforma again by post. The method is very popular and is usually referred to as the 'mailed questionnaire method' of investigation. Sometimes, however, the enumerators (i.e. people employed for collection of data) themselves carry the questionnaires and note down the answers while interviewing the informants. This method is employed during the census of population.

The drafting of a good questionnaire requires utmost skill, and the success of any investigation depend to a large extent on tactful drafting of the questionnaire. The following points should be observed in drafting a questionnaire:

1. The questionnaire should be as short as possible. Many questions may arise during an investigation. But if all are included, the questionnaire will become unduly lengthy with the



consequence that the respondents (i.e. persons who are required to answer them) will feel bored and reluctant to answer all the questions.

2. The individual questions should be, simple, unambiguous and precise. Lengthy questions cause irritation, resulting in careless and inaccurate replies. Complicated questions should be split up into several smaller parts which can be easily answered by the respondents. Explanations and definitions of some of the terms used in the questionnaire must therefore accompany each proforma.

3. If possible, questions should be so set as to elicit only two possible definite answers—‘yes’ or ‘no’.

4. Questions affecting the pride and sentiments of informants must not be asked. Questions on personal matters like income and property should be avoided as much as possible.

5. Questions should be capable of yielding objective answers, avoiding matters of opinion.

6. The units in which the information is to be collected should be clearly and precisely mentioned in the questionnaire.

7. The arrangement of questions in the proforma should be such as to have an easy and systematic flow of answers in turn. Questions should not skip back and forth from one topic to another.

8. After the questionnaire has been devised, it is desirable to try it on a few individuals. The procedure, which is known as *pilot survey*, is useful in detecting the shortcomings of the questionnaire, so that necessary modifications may be made before it is used in the actual enquiry.

Merits—(i) Mailed questionnaire method of investigation is the least expensive. Since the enumerators do not have to travel to the informants’ places, the requisite information can be collected only at a fraction of the cost needed in the other methods. (ii) For the same reason, a vast area can be covered by the enquiry. (iii) The information can also be gathered at a comparatively short period of time; because the questionnaires can be sent out simultaneously to many informants. (iv) The method is useful for collecting information on such matters which the informants may not like to disclose in the presence of enumerators.

Defects—(i) Mailed questionnaire method is not very satisfactory due to the low proportion of response. In many cases, the informants do not care to take the trouble and the completed proformae are not returned. (ii) The quality of the collected information is also very poor. Many respondents do not carefully fill in the questionnaires, and haphazard answers are given. Incomplete returns are also not rare. This necessitates large follow-up correspondence. (iii) The respondents may not fully understand the meaning of some of the questions (inspite of the accompanying instructions) resulting in the low accuracy of the collected information.

Precautions—People have their own whims and feelings of pride and prejudice. Unless compelled to submit returns under an Act of law, they are usually reluctant to disclose any information or take unnecessary trouble. Much persuasion is therefore necessary to elicit information in such form as to be really useful for valid statistical analysis. The following precautions should be taken: (i) The purpose of the enquiry should be clearly explained and the informants requested for co-operation. They should be assured of complete secrecy of their identity and that the outcome of the investigation will in no way harm them. (ii) Utmost care should be taken in drafting the questionnaire. The questions should be simple and unambiguous, so that respondents do not find it difficult to answer them. Questions on personal matters should be avoided as much as possible. (iii) It should be made clear that no postage need be paid by the respondents and that the investigator is fully responsible for the cost of postage. All that is necessary is to fill up the proforma and post it as early as possible. (iv) Reminders in the form of courteous letters should be sent to persons who fail to return the completed questionnaires. Sometimes telegraphic or express reminders are also helpful. (v) As soon as each return is received, it should be scrutinized for any inconsistencies or anomalies. If necessary, some of the returns should be referred back immediately for correction.





1.7 USEFUL TERMS

(1) Law of Statistical Regularity

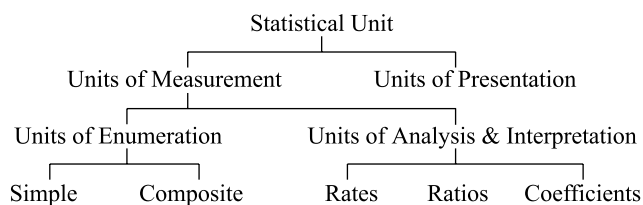
The Law of Statistical Regularity states that a moderately large number of items selected at random from a larger group (technically called 'population') are almost sure, on the whole, to possess the characteristics of the larger group. In other words, if a random sample (Chapter 13) of large size is taken, the sample will to a fairly accurate degree exhibit the characteristics of the entire population. Two points should be kept in mind—(i) The sample must be drawn 'at random'. This means that the items constituting the sample are obtained by giving equal chance of selection to all the items in the population. (ii) The number of items included in the sample should be sufficiently large. The larger the size of the sample, the more representative will the sample be of the population.

Note that the Law of Statistical Regularity does not guarantee that the large random sample must reflect all the characteristics of the population. A part can never represent the whole in toto. The law only states about the tendency of the sample to contain the characteristics of the population. If 10,000 persons are selected at random from a certain city and their income figures are obtained, these data are likely to form a miniature model of the income structure for the city. We are almost sure that the average of these 10,000 figures will be close to the average income of all the inhabitants, i.e. the true per capita income for the whole city. Again, if data from another random sample of 20,000 people were available, the average calculated from this larger sample is expected to be closer to the true average, thus reducing the error with increase in the size of sample. It is because of the operation of this law that sufficiently accurate estimates about the population are obtained by sample survey, replacing the costly and laborious method of complete enumeration. Insurance companies insure against death, fire, etc., banks and commercial firms can undertake risks in business, and gamblers bet money on the basis of this law.

(2) Statistical Unit

Example 1.12 Define a Statistical Unit and explain what should be the essential requisites of a good statistical unit. [C.A., May '73, '74 & '75]

Solution *Statistical Unit* is that thing in terms of which requisite information is collected and expressed during a statistical inquiry. For effective comparison and valid conclusions, it is necessary that the statistical unit is suitably chosen. A mistake in the choice of the appropriate unit is more harmful than mistakes in the collection of data.



Examples: 'Simple' units—yard, ton, etc.; 'Composite' units—manday, kilowatt-hour, passenger-mile, etc.; Units of Analysis and Interpretation—maund per acre (yield), per capita per annum (income), number per thousand (birth rate).



The requisites of a good statistical unit are:

1. It should be clear and unambiguous. For instance, during the census of population in India, 'household' is often employed as a unit. Whether all persons living in the same house, or those having a common kitchen, or only blood relations living in the same house would comprise the household, has to be clearly defined before the collection of data begins.
2. It should be rigid, easy to ascertain and homogeneous. In the collection of data for estimation of crop-yield, all enumerators should be told specifically whether per acre yield figures are to be given in maunds or tons; and if in tons, whether metric ton or imperial ton.
3. It should be suitable for the inquiry. In a survey of industries, volume of production may be stated either in tons, or in number, or even in rupees. If the survey covers different types of industries, naturally 'rupees' should be the appropriate unit.
4. It should be stable. If production of electric fans in India over several years is compared, the 'number' of fans produced would be a better unit than their values in 'rupees'. The former unit is more stable than the latter, which depends on the price per fan varying from year to year.



(3) Statistical Error

Example 1.13 Define Statistical Error. How is it different from a mistake? Write an essay on the different types of statistical errors and how they are measured.

[C.A., Nov. '71]

Solution In statistical terminology, the word 'error' is used in a special sense. Error shows the extent to which the observed value of a quantity exceeds (or falls short of) the true value.

$$\text{Error} = \text{Observed value} - \text{True value.}$$

It may be positive, negative or zero. Since statistical data are obtained either by measurement or by observation, it is never possible to attain absolute accuracy either in the collected data or in the final results. Inaccuracies are bound to occur. For example, data on income, for obvious reasons, are hardly accurate. Measurements on any continuous variable can never be exact, e.g. heights measured upto the nearest tenths of an inch. Personal prejudices of investigators, or faulty methods of collection of data also affect the accuracy. Again, it is not possible to assure absolute accuracy while making estimates about the population based on the analysis of samples. Such inaccuracies arising in the course of statistical investigations are called '*statistical errors*'. Statistical errors should not, however, be confused with 'mistakes'. Mistakes are caused by carelessness, e.g. following a wrong method of calculation, computational mistakes, or clerical lapses. Statistical errors are inherent in any statistical enquiry, and enter the process unconsciously. They can never be eliminated.

Statistical errors may be classified into

- (a) Biased error, (b) Unbiased error.

Biased errors are those which arise due to personal bias or prejudices of investigators and informants, and defects in the measuring instrument. The total effect of biased errors goes on increasing, as the number of observations increases. Hence these are also known as 'cumulative errors'. For example, while estimating crop-yield by inspection of the field, an investigator may have the habit of over-estimating the actual yield. Similarly, the usual habit of women understating their age results in a lower average age. Biased errors thus affect the accuracy of results in one direction only. They are the most undesirable and should be avoided as much as possible.

Unbiased errors are those which enter into a statistical enquiry due to chance causes. Since they are sometimes in one direction and sometimes in the other, unbiased errors are compensating. With increase in the number of observations, the total effect of unbiased errors therefore



diminishes. Unbiased errors follow the law of statistical regularity, and hence the magnitude of such errors can be predicted based on probability considerations. For example, while measuring heights of persons to the nearest inch, some recorded values are smaller, while some others are larger, than the true heights. The total of the observations do not therefore, differ much from the actual total. The larger the number of observations, the smaller will be the difference between the two. Thus unbiased errors can be minimised by taking a large number of observations.

(4) Statistical Fallacies

All statistical data and conclusions derived from their analysis must be interpreted in their proper perspective and by experts only. Mistakes in interpretation often give rise to wrong and absolutely baseless propositions being imputed into the results. Such erroneous conclusions reached on a misinterpretation of data are known as *Statistical Fallacies*. The chief causes of statistical fallacies are:

(a) The characteristics of a group may be wrongly assumed to hold in respect of each individual member. Example : (i) “The per capita national income in India increased from Rs 300 in 1950 to Rs 900 in 1970, and so the prosperity of Indians has increased three times” (C.A., May '72). (ii) Conversation between patient and doctor: “Patient — ‘What are the chances of getting better? Doctor — ‘One hundred per cent. Medical records show that 9 out of every 10 die of the disease you have. Yours is the tenth case I have treated; the others all died. Statistics are statistics; so you are bound to get well’.”

(b) An unrepresentative sample may be analysed and the results may be generalised to cover the whole population. Example: “80% of the people who die of cancer are found to be smokers in a hospital, and so it may be concluded that smoking causes cancer” (C.A., May '66). In order to justify the statement, it is necessary to examine the percentage of smokers amongst patients dying of other diseases and from all causes. Other factors like sex, age-group, etc. of the patients should also be considered. It may turn out that smokers among patients dying from another cause is greater than 80%, or that cancer patients resort to smoking habits, being aware of the fatality of the disease.

(c) The characteristics may not be properly defined as to make the data comparable. Example: (i) “In a war, 40% of the people of a State had to go to the field where 10% of them died; 12% of the people who remained at home died during the period. As the death-rate among those who did not go to the field was higher than those who went to the field, the conclusion was that joining the war was safer than remaining at home” (C.U., M.Com. '71). Here the composition of people in the two categories—those who went to the field, and those who remained at home—has been completely ignored. The risks of death at various ages, and for males and females are different. Healthy adult people have gone to the field, and the old and infirm have been left at home, whose risk of death from natural causes is high. (ii) “The average cost of production was Rs 1.50 in 1960 and Rs 1.75 in 1961, and so the factory has become inefficient” (C.A., May '70). The cost may have increased because of increase in prices of raw materials, but still the overall profits may have gone up.

(d) A cause-and-effect relationship may be wrongly assumed, which may not hold for all values of the variables. Example: “Since we had more than double the normal rainfall during the current year, we can expect a record yield in rice” (C.A., May '72). Obviously, rainfall increases rice yield to a limited extent; excessive rains may spoil the standing crop in the field.

(e) Sometimes, data transformed into percentages may show a misleading result. Example: “The production of a factory increased by 10% from 1972 to 1973, and by 15% during the next year; so production increased by 25% over the two-year period 1972–74.” Here the productions at the beginning of the two periods were not the same, and hence the percentages cannot be added.





1.8 CLASSIFICATION

Example 1.14 Define 'classification' and explain the various ways of classification adopted in Statistics. [C.A., Nov. '68]

Solution Classification is the process of arranging the collected statistical information under different categories or classes according to some common characteristic possessed by the individual members. Statistical data collected during the course of an investigation are so varied that it is not possible to appreciate, even after a careful study, the true significance of the figures, unless they are arranged properly. To make the data really useful, they must be *classified* or grouped into homogeneous categories, so that the like will go with the like and the unlike with the unlike. Classification prepares the ground for enabling comparison and analysis by instituting a logical and orderly arrangement of data. For example, during the population census, apart from the number of members in each family, various other information, e.g. age, sex, occupation etc. of all people in the country are collected. The total population is then classified according to sex into males and females; according to age groups 0–10 years, 10–20 years, etc.; according to livelihood into agricultural classes, production other than cultivation, commerce, transport, etc. If such classifications are not made, it will not be possible to analyse the data and reveal their true significance from the heaps of material collected during the population census.



Types of Classification

Broadly, there are four types of classification:

1. *On qualitative basis*—Classification of the total population of a country on the basis of sex, religion, occupation, etc. belongs to this type. This is also known as *classification by attributes*.
2. *On quantitative basis*—Classification of the total population according to age, or of industries according to the number of persons employed, etc. are included in this type. Here, the basis of classification is some variable, and hence this is also known as *classification by variables*.
3. *On time basis*—Some statistical data are arranged in order of their time of occurrence. Production of a factory may be shown by weeks, months, quarters or years. Statistical data classified according to time are known as *time series* (Chapter 14).
4. *On geographical basis*—The total population of a country may be classified by States or districts, or exports of a particular commodity from India may be classified by the country to which exported. The basis of classification in such cases is by geographical regions.

1.9 TABULATION

Tabulation may be defined as the logical and systematic organisation of statistical data in rows and columns, designed to simplify the presentation and facilitate comparisons. The advantages of tabulation are—(i) Tabulation enables the significance of data readily understood, and leaves a lasting impression than textual presentation. (ii) It facilitates quick comparison of statistical data shown between rows and columns. (iii) Errors and omissions can be readily detected when data are tabulated. (iv) Repetition of explanatory terms and phrases can be avoided, and the concise tabular form clearly reveals the characteristics of data.



Example 1.15 Define a Statistical Table and state the essentials of a good table.

[C.A., Nov. '68 & '69, May '74 & '75; ICWA, June '73 & '74]

Solution *Statistical Table* is a systematic arrangement of quantitative data under appropriate heads in rows and columns. After the data have been collected, they should be tabulated, i.e. put in the form of a table, so that the whole information can be had at a glance. There is no ideal method of tabulation. Skill in tabulation is generally attained after years of experience. Nevertheless, the following general rules should be followed while tabulating statistical information:

1. A table must contain a title. This title should be clear and convey in as few words as possible, the contents of the table. Columns should similarly have definite and, at the same time, sufficiently comprehensive headings.
2. When several tables are used, they should be numbered, e.g. Table 1, Table 2, etc., to facilitate future reference. If there are many columns, say more than four, they should also be serially numbered.
3. Units of measurement must be clearly shown in the column headings, e.g. price (Rs), height (inches), weight (lbs.), etc.
4. The table should be well-balanced in length and breadth. When the table is unusually long, a separate coarse grouping should be tried. If, however, this causes any serious loss of information, some breaks or extra spaces should be left at equal distances apart.
5. The arrangement of items in the table should have a logical sequence. Time series data, if any, must be arranged chronologically.
6. Columns of figures which are directly comparable should be kept as close as possible. If percentages or averages appear in a table, they should be placed near the figures to which they relate. Large numbers should be rounded and mentioned in thousands, lakhs or millions.
7. Light and heavy rulings may be used to distinguish between sub-columns and main columns. Double rulings may also be used instead of heavy ones.
8. Totals of columns may be shown at the bottom of the table. In cases where row totals are useful, they should also be shown.
9. If the tabulated data have been compiled from another source, this source should be mentioned in the foot-note.
10. Items which are in any way different from the rest, e.g. estimated figures, revised figures, etc. should be marked with asterisk or numbers and explanatory notes given in the foot-note.



Types of Tabulation

In very general terms, there are two types of tabulation—(i) *Simple tabulation*, and (ii) *Complex tabulation*. A simple table contains data in respect of one characteristic only, information relating to other characteristics being left out. A complex table contains figures relating to several characteristics. Tabulation may also be *Single*, *Double*, *Treble* or *Manifold*. A single tabulation is one that answers one or more groups of independent questions. Double, Treble or Manifold tabulation shows the subdivision of a total according to two, three or many characteristics respectively and answers as many mutually dependent questions.

Example 1.16 What are the different parts of a table?

[C.A., May '65; I.C.W.A., July '71 & Jan. '74]

Solution The different parts of a table are:

- (i) *Title*—This is a brief description of the contents and is shown at the top of the table.
- (ii) *Stub*—The extreme left part of the table, where descriptions of rows are shown, is called Stub.



(iii) *Caption and Box-head*—The upper part of the table which shows the description of columns and sub-columns is called Caption. The whole of the upper part, including caption, units of measurement and column-numbers, if any, is called Box-head.

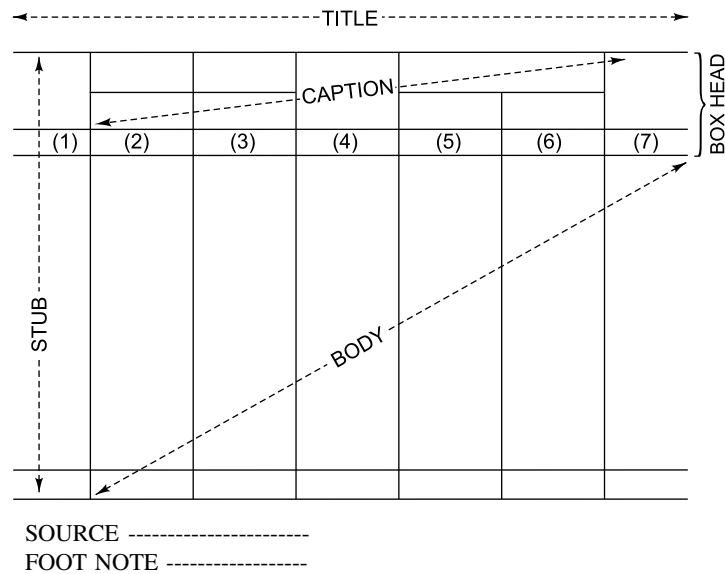


Fig. 1.1 Different Parts of Table

(iv) *Body*—It is that part of the table which shows the figures.

(v) *Footnote*—This is the part below the Body, where the source of data and any explanations are shown.



Example 1.17 Prepare a blank table showing the average wages of males and females classified into two age-groups of eighteen years and over, and under eighteen years in five industries at two different dates. [C.U., B.A.(Econ) '73; M.Com. '63]

Solution

Table 1.1 Average (Monthly) Wages (Rs) in Five Industries

Industry	As on... (date)				As on... (date)			
	Under 18 years		18 years & over		Under 18 years		18years & over	
	Male	Female	Male	Female	Male	Female	Male	Female
1.								
2.								
3.								
4.								
5.								
Total								





Example 1.18 Explain how you would tabulate statistics of death from principal diseases by sexes, in two different States in India for a period of five years.

[I.C.W.A., July '70]

Solution

Table 1.2 Statistics of Death from Principal Diseases in Bihar and U.P., by Sex (1961-65)

Disease/Sex	Bihar					U.P.				
	1961	1962	1963	1964	1965	1961	1962	1963	1964	1965
1. Fever: { Male { Female Total										
2. Cholera: { Male { Female Total										
3. Dysentery: { Male { Female Total										
4. Small-pox: { Male { Female Total										
5. All Cases: { Male { Female Grand Total										

Example 1.19 Prepare a blank table showing the particulars relating to students of the Bombay University classified according to their age, sex, faculty and three important religions.

[C.A., Nov. '69]



Solution

Table 1.3 Number of Students in Bombay University
(by Age, Sex, Faculty and Religion)

Faculty/ Age (years)	Hindu			Christian			Parsee			Total		
	Male	Female	Total	Male	Female	Total	Male	Female	Total	Male	Female	Total
1. Arts:												
{ Upto 15												
{ 16 – 20												
{ Above 20												
Total												
2. Science:												
{ Upto 15												
{ 16 – 20												
{ Above 20												
Total												
3. Medicine:												
{ Upto 15												
{ 16 – 20												
{ Above 20												
Total												
4. Agriculture:												
{ Upto 15												
{ 16 – 20												
{ Above 20												
Total												
All Faculties:												
{ Upto 15												
{ 16 – 20												
{ Above 20												
Grand Total												



Example 1.20 Represent the information contained in the following passage in a suitable tabular form:

‘The cropped area of vegetables (excluding potatoes) grown for human consumption in the United Kingdom rose in 1955–56 and was the highest since 1950–51. The cropped area increased to 509,000 acres, some 11,000 acres more than in 1954–55. The area of root vegetables increased by 8,100 acres to 62,400 acres, carrots alone increasing by 5,700 acres to 33,200 acres. The area of cabbage rose slightly, thus halting the steady decline since 1947–48; the cropped area was 75,700 acres with 74,800 acres in 1954–55. The cropped area of cauliflower and broccoli was 33,400 acres, 2,400 acres less than in 1954–55. Peas harvested dry decreased by about 9,800 acres to 121,800 acres,



but a larger area of beans, mainly broad beans and green peas, was grown. The area of broad beans increased by 2,600 acres to 7,300 acres and the area of green peas for canning and quick freezing rose by 7,000 acres to 50,400 acres'. [I.C.W.A., June '74]

Solution

Table 1.4 Cropped Area under Vegetables
(excl. Potatoes) in U.K., 1954-56
(Hundred acres)

Type	1954-55	1955-56	Increase (+) or Decrease (-)
1. Root vegetables:	543	624	+ 81
{ Carrots	{ 275	{ 332	{ + 57
{ Others	{ 268	{ 292	{ + 24
2. Cabbage	748	757	+ 9
3. Cauliflower & Broccoli	358	334	- 24
4. Peas (harvested dry)	1,316	1,218	- 98
5. Beans:	2,015	2,157	+ 142
{ Broad beans	{ 47	{ 73	{ + 26
{ Green Peas	{ 434	{ 504	{ + 70
{ Miscellaneous	{ 1,534	{ 1,580	{ + 46
Total	4,980	5,090*	+ 110

*Highest since 1950-51



Example 1.21 The total number of accidents in Southern Railway in 1960 was 3500, and it decreased by 300 in 1961 and by 700 in 1962. The total number of accidents in Metre Gauge section showed a progressive increase from 1960 to 1962. It was 245 in 1960; 346 in 1961; and 428 in 1962. In the metre gauge section, "Not compensated" cases were 49 in 1960, 77 in 1961, and 108 in 1962. "Compensated" cases in the broad gauge section were 2867, 2587 and 2152 in these three years respectively.

From the above report, you are required to prepare a neat table as per the rules of tabulation. [C.A., Nov. '71]

Solution

Table 1.5 Number of Accidents in Southern
Railway During 1960-62

Section	1960	1961	1962
Metre Gauge:	245	346	428
{ Not compensated	49	77	108
{ Compensated	196	269	320
Broad Gauge:	3,255	2,854	2,372
{ Not compensated	388	267	220
{ Compensated	2,867	2,587	2,152
Total:	3,500	3,200	2,800
{ Not compensated	437	344	328
{ Compensated	3,063	2,856	2,472





Example 1.22 In the White Paper on National Income and Expenditure in U.K. (1938–45) it was stated that the estimated number of individuals assessed to income tax in 1938 was 1745 thousand with annual gross income between £250 and £500, while in 1944 the corresponding number was 5200 thousand. In the higher range £500–£1000 the number rose from 500 thousand to 1400 thousand, between the same years, while in the £1000–£2000 group the increase was from 195 thousand to 520 thousand. The distribution of net income after deduction of tax gave for the first group 1820 thousand and 5050 thousand, for the second group 450 thousand and 830 thousand, and for the last group 155 thousand and 186 thousand respectively.

Calculate appropriate statistical measures to bring out the relative difference in the numbers in each range as well as between the years. Calculate further percentages to bring out the effect of taxation in the three groups. Tabulate your results together with the original data. [B.U., B.A.(Econ) '66; C.U., B.A. (Econ) '71]

Solution

Table 1.6 Number of Individuals Assessed to Income Tax (Estimated)–Years 1938 & 1944

Year	Income Range (£)			Total
	250–500	500–1000	1000–2000	
I. Number ('000) with { 1938 Gross Income { 1944 Increase (%)	1745 5200 198	500 1400 180	195 520 167	2440 7120 192
II. Number ('000) with Net Income { 1938 after taxation { 1944 Increase (%)	1820 5050 177	450 830 84	155 186 20	2425 6066 150
III. Increase (+) or Decrease (–) after taxation (%) { 1938 { 1944	(+) 4 (–) 3	(–) 10 (–) 41	(–) 21 (–) 64	(–) 1 (–) 15

Source: White Paper on National Income and Expenditure, U.K. (1938–45)



1.10 MECHANICAL TABULATION

In large-scale surveys, the manual process of classification and tabulation of various types of data would involve huge time and expenditure, and therefore mechanical aids to tabulation are resorted to by what are known as 'Punched Card Systems.' In this process, the information from the interviewers' schedules are transferred on to specially designed *Punched Cards*, and these are run through *Sorting* and *Tabulating Machines*, which give out the desired tables in printed form according to the desired classification.

The commontype 'Punched Cards' are thin cards approximately of size 3 inches by 7½ inches, having a number of columns, usually 80, with the column numbers



printed thereon. Under each column there are printed figures 0, 1, 2, ... 9, one below the other at equidistant intervals. Some cards have provision for two more positions in each column at the top. When the completed schedules are available, the first task is to transfer the information on to the cards by means of rectangular holes punched at appropriate places thereon. For this purpose, *Code Numbers* are assigned to the various characters, like age, sex, income, profession, etc., and their various categories. One column or a group of columns is allotted to each character, and the different categories or classes of a character are assigned different positions in that column represented by the printed numbers 0 to 9 according to a predetermined arrangement. For example, we may allot column number 1 to the character age (code : 1 for '0 to 20 years', 2 for '20 to 30 years', 3 for '30 to 40 years', and so on; column number 2 to sex (coded : 1 for male, and 2 for female), etc.

A separate card is used for each member, and holes are cut thereon at appropriate positions by means of the *Punching Machine*, which looks like a typewriter keyboard and is operated manually by means of keys. The accuracy of the transcribed information is then verified by *Punched Card Verifier*, also operated manually, on which the punching operations are repeated. If a card is punched at a wrong position or not punched, the machine locks. The verified cards are then passed through a *Sorting Machine*, which automatically sorts out the cards into different groups and sub-groups, as desired, at speeds upto 40,000 cards per hour. The sorted cards are now put through an electric *Tabulating Machine*, which 'senses' the cards one by one by means of electrical contacts through the holes, and transforms the information into printed figures and words, as needed. The same cards may be used over and over again through the Sorting and Tabulating machines for obtaining different types of tables. The tabulating machine does not only tabulate the data at high speed, but is also capable of printing totals and sub-totals, and may be so set as to reproduce any part of the data contained in the cards at speeds ranging from 5,000 to 12,000 cards per hour.



EXERCISES



1. Examine whether the following variables are discrete or continuous—Age of a person, Size of land holding, Size of family, Temperature, Monthly sales in a shop. [C.U., M.Com. '69]
2. Explain the terms 'classification' and 'tabulation' and point out their importance in a statistical investigation. What precautions would you take in tabulating statistical data? [I.C.W.A., Jan. '70]
3. State the various reasons why errors are made in the interpretation of statistical data, and indicate how to guard against such errors. [C.A., May '70 & '72]
4. What do you mean by an error in statistics? Classify the different types of errors in statistics and discuss their effect on statistical inference. [C.A., May '76]
5. Draw up a table to show the number of wholly unemployed, temporarily stopped and the total unemployed, each class being divided into males and females for the following industries: Fishing, Coal mining, Iron-ore mining, Engineering and Ship building.



6. Draw up a blank table in which could be shown the number of persons employed in six industries, on two different dates, distinguishing males from females and among the latter, singles, married and widows.
[I.C.W.A., Jan. '73]
7. Prepare a blank table showing the distribution of population according to sex and four religions in five age groups in seven different cities.
[C.U., B.A. (Econ) '72]
8. Draw up a blank table showing the Exports and Imports during the years 1960, 1961, 1962, 1963 and 1964 relating to the ports Mumbai, Kolkata, Madras and other ports. The table should provide for the values and the balance of trade and the totals for each year.
[C.A., Nov. '68]
9. You are given data on exports (both quantity and value) of Indian jute to U.K., U.S.A., U.S.S.R., Japan and Canada for 5 consecutive years. Suggest a suitable tabular representation by drawing a blank table. [B.U., B.A. (Econ) '72]
10. Draft a blank table to show the distribution of personnel in a manufacturing concern according to (a) Sex : male and female; (b) Three grades of salary: below Rs 300, Rs 300–Rs 500, Rs 500 and over; (c) Two periods : 1940 and 1950; (d) Three age groups : below 25, 25 and under 40, 40 and over.
[C.A., May '63]
11. Tabulate the following information which appeared as part of a newspaper article on dishwashing machines—“An idea of the slow growth of the industry during a period of soaring sale in other appliances in competitive price ranges is given in a recent study by the trade journal *Electrical Merchandising*: In 1947, the survey showed, 120,000 motor-driven dishwashers were sold at an average price of \$ 250. In 1948, sales rose to 225,000 at an average price of \$ 275. In 1949, with the average price the same as the year before, the sales dropped to 160,000. In 1950, they jumped to 230,000 at an average price of \$ 290 and in 1951 they continued to climb to 260,000 units even though the average price went up to \$ 300. In 1952, sales dropped to 175,000 units and in 1953 they rose to only 180,000 despite the fact that average price remained at \$ 300.”
12. Present the following information in a tabular form and suggest a suitable title:
‘The production of 10.95 lakh tons of rice in Maharashtra in 1962–1963 was the lowest in the period since 1955–56. In 1963–64, however, it has shown a spectacular recovery and reached the level of 15.14 lakh tons. During 1963–64, wheat and bajri output decreased. The production of bajri which was 5.50 lakh tons in 1962–63 declined to 4.51 lakh tons in 1963–64. The production of wheat also decreased from 4.63 lakh tons in 1962–63 to 3.43 lakh tons in 1963–64. The area under pulses has shown a decreasing trend and the production was less by 22,000 tons in 1963–64 than the production of 8.89 lakh tons in 1962–63.’
[I.C.W.A., Jan. '70]
13. An investigation conducted by the Education Department in a public library revealed the following facts. You are required to tabulate the information as neatly and clearly as you can:



“In 1950 the total number of the readers was 46,000 and they borrowed some 16,000 volumes. In 1960 the number of books borrowed increased by 4,000 and the borrowers by 50%. The classification was on the basis of three sections: Literature, Fiction and Illustrated News. There were 10,000 and 30,000 readers in the sections Literature and Fiction respectively in the year 1950. In the same year 2,000 and 10,000 books were lent in the sections Illustrated News and Fiction respectively. Marked changes were seen in 1960. There were 7,000 and 42,000 readers in the Literature and Fiction sections respectively. So also 4,000 and 13,000 books were lent in the sections Illustrated News and Fiction respectively.”

[C.A., Nov. '66]

14. Tabulate the following information which has been taken from *The Times Review of Industry*, 1958:

‘In 1958 France exported 1,66,000 cars as against 1,07,000 in 1957 of which 64,000 went to Europe as compared with 46,000 in the same period in 1957. Some 34,000 went to oversea French territories as compared with 26,000 and 68,000 elsewhere as against 35,000. The increase to countries other than Europe is 40 per cent. The proportion of French production going to foreign markets was 27 per cent, last year. Renault’s sales outside France amounted to 37 per cent of total production, while their total foreign sales of cars was 92,000.’

[C.U., B.A. (Econ) '69]

15. The total population of Hong Kong aged 15 and over rose from 1853 thousand in 1961 to 2528 thousand in 1971. The male population rose from 945 thousand to 1280 thousand and the female population from 908 thousand to 1248 thousand. The distribution of population (above 15 years) by conjugal status was as follows in 1961: 504 thousand persons were never married, 1202 thousand were currently married, 135 thousand widowed and only 12 thousand divorced or separated. The corresponding figures (in thousand) were 914, 1481, 125 and 8 in 1971. Among males aged 15 or more in 1961, 334 thousand were never married, 588 thousand currently married, 17 thousand widowed and 6 thousand divorced or separated; the corresponding figures for 1971 were 546, 710, 20 and 4 (in thousand).

Present the data in a tabular form with appropriate headings and sub-headings.

[C.U., B.A. (Econ) '78]

16. The following is an excerpt from the White Paper on the general budget for 1957-58, appearing in the publication “*Budget for 1957-58*” of the Ministry of Finance, Govt. of India:

“...The value of imports increased from Rs 418 crores in April-November, 1955, to Rs 535 crores in April-November, 1956. Of this increase of Rs 117 crores, Rs 96 crores was accounted for by the increase in the imports of machinery, iron and steel and other metals. Imports of machinery increased from Rs 73.5 crores in April-November, 1955 to Rs 105.8 crores in April-November, 1956; of iron and steel from Rs 34.3 crores to Rs 88 crores and of other metals from Rs 16.4 crores to Rs 26.2 crores.

The small decline in exports (from Rs 388.4 crores in April-November, 1955, to 378 crores in April-November, 1956) is mostly accounted for by

Represent the information contained in the above excerpt in a suitable tabular form.

[B.U., B.A.(Econ) '68]

- 'Lamps are rejected at several manufacturing stages for different faults. 12,000 glass tubes are supplied to make 40-watt, 60-watt and 100-watt lamps in the ratio 1 : 2 : 3. At the stage I, 10 per cent of the 40-watt, 4 per cent of the 60-watt and 5 per cent of the 100-watt bulbs are broken. At the stage II, about 1 per cent of the remainder of the lamps have broken filaments. At the stage III, 100 100-watt lamps have badly soldered caps, and half as many have crooked caps; twice as many 40-watt and 60-watt lamps have these faults. At the stage IV, about 3 per cent are rejected for bad type-marking and 1 in every 100 are broken in the packing which follows.'

- [I.C.W.A., July '71; C.U., B.A.(Econ) '67]

ANSWERS

- 10.

Distribution of Personnel in a Manufacturing Concern

[illegible]



11.

Sale of Motor-driven Dishwashers (1947-1953)

<i>Year</i>	<i>Number of Units sold ('000)</i>	<i>Average Price (\$)</i>
1947	120	250
1948	225	275
1949	160	275
1950	230	290
1951	260	300
1952	175	300
1953	180	300

Source: *Electrical Merchandising*

13.

Number of Readers in and Books Borrowed Library
(Years from a Public 1950 & 1960)

<i>Sections</i>	<i>Readers ('000)</i>		<i>Books Borrowed ('000)</i>	
	1950	1960	1950	1960
1. Literature	10	7	4	3
2. Fiction	30	42	10	13
3. Illustrated News	6	20	2	4
Total	46	69	16	20

14.

Export of Motor Cars from France, 1957-1958
(Number '000)

<i>Exported to</i>	<i>1957</i>	<i>1958</i>
1. Europe	46	64
2. Oversea French Territories	26	34
3. Other Countries	35	68
Total	107	166

15.

Production and Export of Motor Cars (France, 1957)

<i>Company</i>	<i>Production ('000)</i>	<i>Export ('000)</i>	<i>Per cent Exported</i>
(i) Renault	250	92	37
(ii) Other Companies	150	15	10
Total	400	107	27



16.

Values of Exports and Imports, Govt. of India
(Rs crores)

<i>Items</i>	<i>April–November 1955</i>	<i>April–November 1956</i>
I. Imports:		
Machinery	73.5	105.8
Iron & Steel	34.3	88.0
Other metals	16.4	26.2
Miscellaneous	293.8	315.0
Total	418.0	535.0
II. Exports:		
Oil	26.2	13.4
Cotton	24.6	10.9
Manganese ore	10.6	6.8
Cotton textiles	42.2	40.1
Jute manufactures	83.3	79.5
Miscellaneous	201.5	227.3
Total	388.4	378.0

Source: “Budget for 1957-58”, Ministry of Finance, Govt. of India.

17.

Wastage of Electric Lamps during Manufacture

<i>Power of Lamp (watt)</i>	<i>Total No. of Glass Tubes Supplied</i>	<i>Number Rejected at Stages</i>					<i>Rejections %</i>
		<i>I Broken bulbs</i>	<i>II Broken filament</i>	<i>III Bad cap</i>	<i>IV Bad type marking</i>	<i>Total</i>	
40	2,000	200	18	300	59	577	29
60	4,000	160	38	300	140	638	16
100	6,000	300	57	150	220	727	12
Total	12,000	600	113	750	419	1,942	16

40-watt lamps have the greatest wastage, viz. 29%.

2 CHARTS AND DIAGRAMS

2.1 OBJECTS OF DIAGRAMMATIC REPRESENTATION

Charts and diagrams are effective devices for vivid presentation of statistical data. The main object of diagrammatic representation is to emphasise the relative position of different subdivisions and not simply to record details.

Advantages—Diagrams are appealing to the eyes as well as to intellect, and are therefore helpful in assimilating the data readily and quickly. Moreover, a chart can clarify a complex problem and reveal hidden facts, which are not apparent from the tabular form. It is sometimes necessary in finding the trend in time series and also in finding relations between several sets of observations. In some cases, the graph may be used as a means of checking mistakes.

Disadvantages—Charts do not show details, which is possible in the tabular presentation. Graphical presentation can reveal only the approximate position. Also graphs and charts require much time to construct, whereas the desired information can be quickly conveyed by arranging the data in the form of a table.

2.2 TYPES OF CHARTS AND DIAGRAMS

The common types of charts and diagrams are:

1. Line diagram, or Graph
2. Bar diagram
3. Pie diagram
4. Pictogram
5. Histogram, Frequency polygon, and Ogive.

[**Note:** The words 'chart' and 'diagram' are used almost in the same sense. We also say *Line chart*, *Bar chart*, *Pie chart*, etc.]

(1) Line Diagram (or Graph)

This is the most common method of representing statistical data, especially used in business and commerce, where data are shown in accordance with the time of occurrence. The line diagram shows by means of a curve or a straight line, the



relationship between two variables. Two straight lines, one horizontal and the other vertical (known as *X-axis* and *Y-axis* respectively), are drawn on a graph paper, which intersect at a point, called *origin*. The given data are represented as points on the graph paper. The position of any point is determined by its distances from the axes on the basis of the given data. The consecutive points thus obtained are joined by pieces of straight lines, giving the line diagram.

Two types of line diagrams are used—(i) *natural scale*, and (ii) *ratio scale*. Equal distances represent equal *amounts* of change in the natural scale, but equal *ratios* of change in the ratio scale (see Fig. 2.3).

Hints for Drawing Line Diagrams

1. Graph papers usually available in the market are not equal in length and breadth. The line diagram should be so planned that the horizontal axis is longer than the vertical axis. In Fig. 2.1 note that the diagram is longer horizontally.
2. The scales should be indicated below the horizontal axis and on the left of the vertical axis. It is not necessary to write such expressions as “1 small division = 5 units.”
3. The variables represented by the two axes must be clearly shown. Note that in Fig. 2.1 the words ‘Year and Month’ appear below the horizontal axis, and the words ‘Cheque Clearance’ appear on the top of the vertical axis. The variable represented by the vertical axis is usually shown on the left of the axis (see Fig. 2.2).
4. The units of numbers indicated by the scales must be shown. Note the words ‘Rs Crores’ below ‘Cheque Clearance.’
5. The horizontal scale may not start from zero; but the zero point of the vertical scale must be shown. In cases where the minimum observation shown by the vertical scale is well above the zero point, a break in the vertical axis must be shown (see Figs 2.1 and 2.8).
6. A brief but clear title must accompany the diagram.
7. If several line diagrams are shown on the same graph paper, they should preferably be drawn by distinctly different lines, e.g. thick, thin, solid, and broken lines (see Figs 2.2 and 2.8).

Example 2.1 Draw a line diagram of the following data, relating to cheque clearance:

Cheque Clearance (Crores of Rs)

Month Year	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
1958	832	765	873	792	791	663	834	754	806	799	773	887
1959	894	828	946	923	849	—	—	—	—	—	—	—

[C.U., M.Com. '61]

**Solution**

Cheque
Clearance
(Rs Crores)

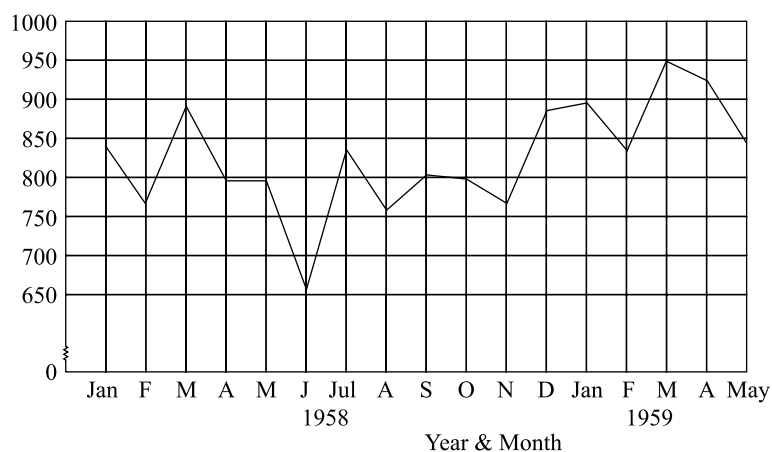


Fig. 2.1 Line Diagram showing Cheque Clearance



Example 2.2 Draw line diagrams of the following statistical information with regard to exports and imports, and comment on their relationship:

Year	Value of Exports at 1927-28 Prices (Rs Crores)	Value of Imports at 1927-28 Prices (Rs Crores)
1937-38	301	243
1938-39	295	226
1939-40	309	230
1940-41	260	184
1941-42	276	168
1942-43	184	85
1943-44	158	89
1944-45	156	160
1945-46	182	177

[C.U., B.A. (Econ) '64, '70]

Solution (See Fig. 2.2, next page).

Comments: (i) Values of exports generally exceed values of imports. (ii) Exports and imports closely follow the same pattern of change. (iii) While exports and imports showed a decreasing tendency since 1937-38 till 1943-44 an increasing tendency was noticed during the years that followed. This indicates the presence of cyclical fluctuations in both the series. Values of exports and imports are expected to rise further for some years after 1945-46.

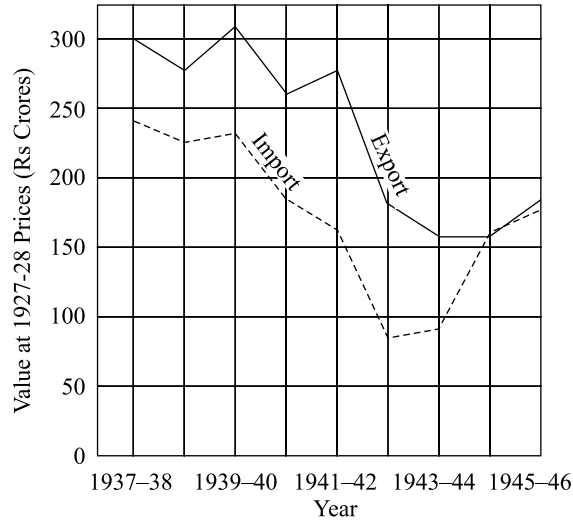


Fig. 2.2 Line Diagrams showing Exports & Imports (1937-1946)



Example 2.3 Explain clearly the distinction between the natural scale and the logarithmic scale used in graphical presentation of data.

[C.U., B.A.(Econ) '62; I.C.W.A., Jan. '71]

Solution In the usual type of graph papers, all rulings are shown equally apart, both horizontally and vertically. These are known as *natural scale* or *arithmetic scale* graph papers. There is a special type of graph paper (Fig. 2.5) in which the distances of rulings from the initial line are proportional to the logarithms of numbers, and hence the distances between consecutive rulings are not equal. Such a scale is known as *logarithmic scale* or *ratio scale*.

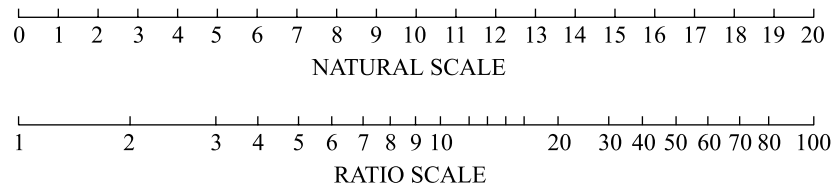


Fig. 2.3 Comparison of Natural Scale and Ratio Scale

The natural scale is used for showing absolute amount of change. In Fig. 2.1, an amount of 50 (Rs crores) increase either from 650 to 700, or from 900 to 950 is represented by the same distance in the vertical direction. However, in terms of percentage changes, these do not have the same significance. In the former, the increase is $\frac{50}{650} \times 100 = 7.7\%$, whereas in the latter case it is $\frac{50}{900} \times 100 = 5.6\%$. Such discrepancies in percentage changes can be readily detected when the data are shown on the ratio scale. Equal distances represent equal absolute changes in the natural scale, but equal percentage changes in the ratio scale. A series of observations having a common difference, i.e. in arithmetic progression, shows a straight line on the natural



scale; but a straight line is obtained on the ratio scale only when the consecutive observations have a common ratio, i.e. are in geometric progression (Fig. 2.4).

In the natural scale graph, the base line of the vertical scale must show zero. But since the ratio scale shows proportionate changes, there is no zero point. The base line must show either 1, or 10, or 100, or 1000, etc.

Some graph papers have a ratio scale in the vertical direction, but natural scale in the horizontal direction. These are known as *semi-logarithmic* (or briefly *semi-log*) graph papers (Fig. 2.5). There are some others which show ratio scales in both directions and are known as *double-logarithmic* (or briefly *double-log*) graph papers.



Example 2.4 Explain what is a semi-logarithmic graph. What purpose is served by such graphs and what are its uses?

[I.C.W.A., Jan. '69; C.U., B.A.(Econ) '71]

Solution *Semi-logarithmic Graph or Ratio Chart* is a line diagram drawn on a special type of graph paper which shows the natural scale in the horizontal direction and the logarithmic or ratio scale in the vertical direction (Fig. 2.5). It shows the logarithmic scale in one direction only, and hence the name 'semi-logarithmic' ('semi' denotes 'half', e.g. semi-circle). In the semi-logarithmic graph paper, the vertical rulings are equispaced, but the horizontal rulings are not, their distances from the base line being proportional to the logarithms of the numbers represented. If a semi-logarithmic graph paper is not available, the ratio chart may be drawn on a natural scale graph paper by plotting the logarithms of values of the dependent variable y against the corresponding values of the independent variable x (Fig. 2.6).

Uses of Semi-logarithmic graph—(1) The semi-log graph is used where ratios of change are more important than absolute amounts of change. If the relationship between the variables follows the compound interest law, i.e. there is a constant ratio between the successive values of y corresponding to equidistant values of x , the equation will be of the form

$$y = ab^x$$

and the semi-log graph will give a perfect straight line (Fig. 2.4). In the analysis of time series, if therefore a given series of observations is suspected to show a constant rate of growth or decline, the simplest way to test this is to draw the semi-log graph and see whether the points lie on or nearabout a straight line.

(2) Semi-logarithmic graphs are useful in comparing fluctuations within the same series, checking established ratios, and reading percentage changes without numerical calculations. In the statistical analysis of business and economic activities, cases very often arise when we are more interested in the relative amounts of growth or decline than the absolute amounts of change. It may also become necessary to compare the rates of change in two or more series (Fig. 2.7). In such cases it is very easy to compare the rates from the slopes of semi-logarithmic charts. Two parallel lines on a semi-logarithmic graph paper will mean that the series represented by the lines are both changing at the same constant rate. A steeper slope of the semi-logarithmic chart at a point indicates a higher rate of growth than at another (Fig. 2.7).

(3) Semi-logarithmic graphs are also useful in representing a series with widely differing values (Example 2.6) or in comparing the rates of change of several series of observations in different units (Fig. 2.7).



Example 2.5 From the following table, draw a ratio chart on a graph paper:

Year	1937	1938	1939	1940	1941	1942	1943	1944	1945
Units produced	2	4	8	16	32	64	128	256	512

[C.U., M.Com. '56]



Solution Since the number of units produced each year is exactly double the number in the previous year, there is a common ratio between the successive observations. The ratio chart (Fig. 2.4) is therefore, a straight line.

[Note that the base line does not represent zero as is necessary in the natural scale. Also, equal distances in the vertical direction represent equal ratios of change, viz. 2 to 4, 4 to 8, etc.]

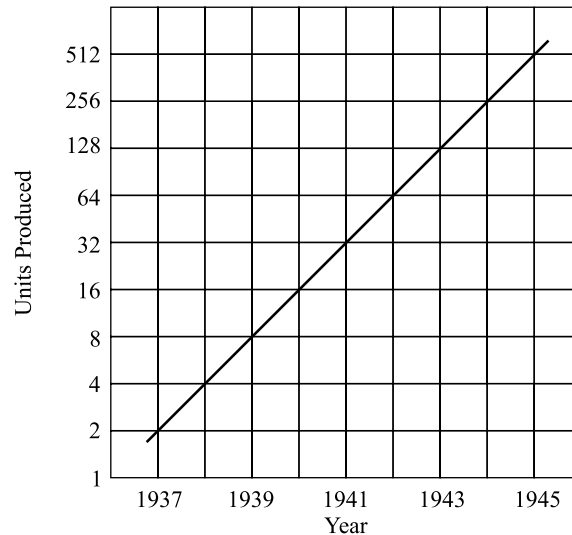


Fig. 2.4 Ratio Chart



Example 2.6 The following table gives the total units produced at the beginning of the different years. Represent the data graphically and estimate the mid-year value for 1949 and 1953.

Year	1947	1948	1949	1950	1951	1952	1953	1954	1955
Units Produced	20	62	147	300	536	811	1104	1425	1755

[I.C.W.A., Jan. '65]

Solution

Table 2.1 Calculation of Logarithms for Ratio Chart

Year	Units Produced <i>y</i>	Log <i>y</i> (from log table)	log <i>y</i> (rounded) to 2 decimals
1947	20	1.3010	1.30
1948	62	1.7924	1.79
1949	147	2.1673	2.17
1950	300	2.4771	2.48
1951	536	2.7292	2.73
1952	811	2.9090	2.91
1953	1104	3.0430	3.04
1954	1425	3.1538	3.15
1955	1755	3.2442	3.24



It is found that the number of units produced differ widely—from only 20 in 1947 to 1755 in the year 1955. The ratio chart is the most appropriate graphical representation of such data.

(Drawn on Semi-Logarithmic Graph Paper)

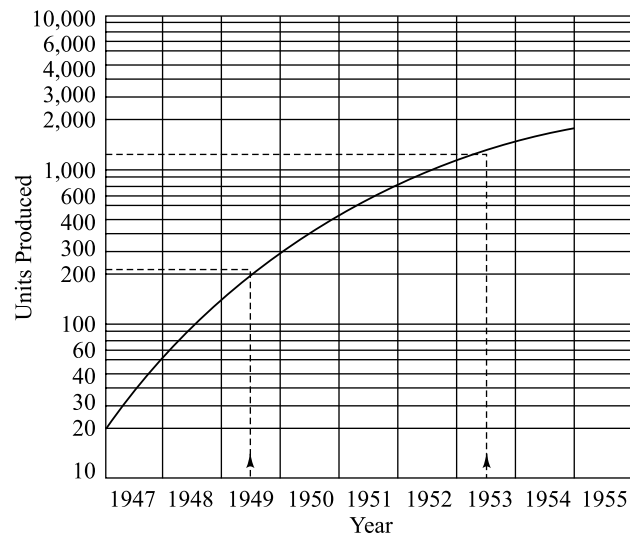


Fig. 2.5 Semi-Logarithmic Graph (or Ratio Chart)

(Drawn on Ordinary Graph Paper)

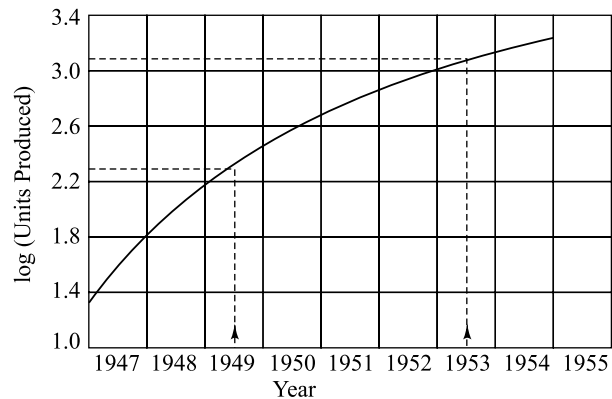


Fig. 2.6 Semi-Logarithmic Graph (or Ratio Chart)

If a semi-log graph paper is available, the ratio chart will be as shown in Fig. 2.5. However, the ratio chart may also be drawn on a natural scale graph paper by showing the logarithms of values along the vertical scale, as in Fig. 2.6. The values of $\log y$, shown in the last column of Table 2.1, are plotted on an ordinary graph paper against the corresponding years, giving the ratio chart (Fig. 2.6).

In order to estimate the mid-year values for 1949 and 1953, two vertical lines are drawn starting from the mid-points of intervals representing such years on the horizontal axis, until they meet the ratio chart. From these points of intersection, two horizontal lines are now drawn to meet the vertical scales on the left. If the ratio chart is drawn on a semi-log graph paper (Fig. 2.5), the values can be read directly from the scale, and found to be approximately 210 and 1260. However, if the ratio chart is drawn on an ordinary graph paper (Fig. 2.6) the values



of $\log y$ are read approximately to be 2.32 and 3.10. The antilogarithms of these values, viz. 209 and 1259, give the required estimates.



Example 2.7 The profits of a particular firm and of the whole industry are given in following table:

Year	1950	1951	1952	1953	1954	1955	1956
Firm (in Rs 10,000)	1.50	2.00	2.18	2.69	3.50	6.00	14.00
Industry (in Rs 10,00,000)	3.2	4.0	5.4	6.8	8.0	11.0	19.0

Compare trends of profit by semi-logarithmic graphs and comment on the performance of the firm in relation to that of the industry. [I.C.W.A., Jan. '69]

Solution We draw the semi-logarithmic graphs (Fig. 2.7) on an ordinary (arithmetic scale) graph paper. For this purpose, the logarithms of profits should be plotted against the corresponding years. These logarithms (using a four-figure log-table) are shown below. However, since it is difficult to use the four-decimal figures on a graph, they have been rounded to 2 decimals. The two semilogarithmic graphs will be found at Fig. 2.7.

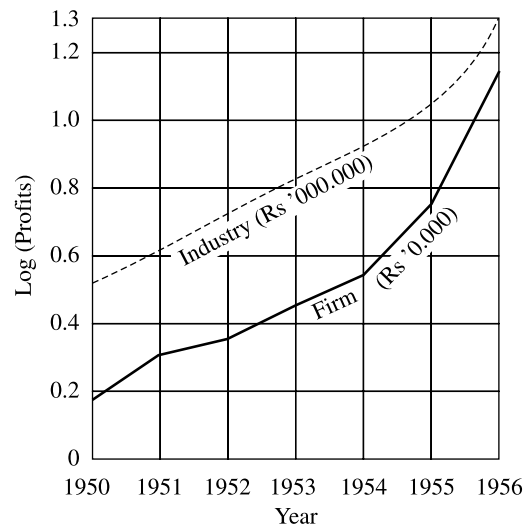


Fig. 2.7 Semi-Logarithmic Graphs

Table 2.2 Calculation of Logarithms

Year	Profits of Firm (Rs 10,000) y	Profits of Industry (Rs 10,00,00) Y	$\log y$	$\log Y$
1950	1.50	3.2	0.1761 = 0.18	0.5051 = 0.51
1951	2.00	4.0	.3010 = .30	.6021 = .60
1952	2.18	5.4	.3385 = .34	.7324 = .73
1953	2.69	6.8	.4298 = .43	.8325 = .83
1954	3.50	8.0	.5441 = .54	.9031 = .90
1955	6.00	11.0	.7782 = .78	1.0414 = 1.04
1956	14.00	19.0	1.1461 = 1.15	1.2788 = 1.28

Comments—The slope of the semi logarithmic graph for the firm is found to be smaller than that for the industry at the beginning, indicating a smaller rate of growth in profits for the firm. However, from 1953 the graph for the firm rose more rapidly, and this shows a favourable trend of profits for the firm in relation to that for the industry as a whole.

Example 2.8 Represent graphically from the following data the growth of the population of a State in India to show both relative growth and growth by absolute amount:

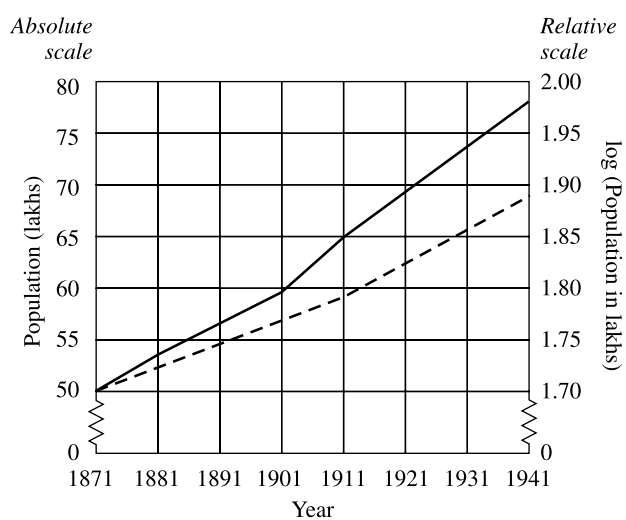
Census year	1871	1881	1891	1901	1911	1921	1931	1941
Population (in lakhs)	50.0	52.4	55.6	59.5	65.0	68.7	73.1	77.5

[B.U., B.A. (Econ) '69]

Solution In Fig. 2.8, two line charts have been drawn, one showing the absolute growth (corresponding to the natural scale on the left) and another showing the relative growth (corresponding to the logarithmic scale on the right).

Table 2.3 Calculation of Logarithms

Year	Absolute Value	Population (in lakh) Logarithm	
1871	50.0	1.6990 = 1.70	(approx.)
1881	52.4	1.7193 = 1.72	"
1891	55.6	1.7451 = 1.75	"
1901	59.5	1.7745 = 1.77	"
1911	65.0	1.8129 = 1.81	"
1921	68.7	1.8370 = 1.84	"
1931	73.1	1.8639 = 1.86	"
1941	77.5	1.8893 = 1.89	"



Solid Line—Absolute growth; Broken Line—Relative growth

Fig. 2.8 Graphs showing Population Growth (Absolute & Relative) of an Indian State



(2) Bar Diagram

Bar diagram consists of a group of equispaced rectangular bars, one for each category (or class) of given statistical data. The bars, starting from a common base line, must be of equal width and their lengths represent the values of statistical data. They are shaded or coloured suitably.

There are two types of bar diagrams—*vertical bar diagram* and *horizontal bar diagram*. Vertical bars (Fig. 2.10) are used to represent time series data or data classified by the values of a variable. Horizontal bars (Fig. 2.14) are used to depict data classified by attributes only.

For each of these types, we have again *grouped bar diagram*, *sub-divided* (or *component*) *bar diagram*, *paired bar diagram*, etc. Grouped bar diagrams (Fig. 2.12) are used to show the comparison of two or more sets of related statistical data. Sub-divided or component bar diagrams (Fig. 2.11) are useful for comparing the sizes of the different component parts among themselves, and also the relation between each part and the whole. The paired bar diagram consists of several pairs of horizontal bars—one extending towards the left and the other towards the right—starting from two vertical lines at the middle of the diagram. This may be used to represent the age and sex composition of the population of any geographical region.

(3) Pie Diagram

Pie diagram is a circle whose area is divided proportionately among the different components by straight lines drawn from the centre to the circumference of the circle (Fig. 2.15). When statistical data are given for a number of categories, and we are interested in the comparison of the various categories or between a part and the whole, such a diagram is very helpful in effectively displaying the data.

For drawing a pie diagram, it is necessary to express the value of each category as a percentage of the total. Since the full angle 360° around the centre of the circle represents the whole, i.e. 100%, the percentage figure of each component is multiplied by 36 degrees (Table 2.4) to find the angle of the corresponding sector at the centre of the circle. The diagram can be drawn with the help of a compass and a protractor.

(4) Pictogram

Pictogram consists of rows of picture symbols of equal size. Each symbol represents a definite numerical value. If a fraction of this value occurs, then the proportionate part of the picture from the left is shown. Pictograms are used for representing time series data, one row of pictures for each time period. It may also be used for displaying statistical data classified by attributes.

(5) Histogram, Frequency Polygon, and Ogive

These diagrams are used to depict statistical data given in the form of frequency distributions, and will be discussed later (Section 4.7).

Example 2.9 *The following table shows the numbers of agricultural and non-agricultural workers for the year 1860–1950. Graph the data using (a) line graphs, (b) bar charts, and (c) component bar chart. Add a small note on the diagrams.*



Year	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950
Agricultural Workers (in millions)	6.2	6.9	8.6	9.9	10.9	11.6	11.4	10.5	8.8	6.8
Non-Agricultural Workers (in millions)	4.3	6.1	8.8	13.4	18.2	25.4	31.0	38.4	42.9	52.2

[C.U., M.Com. '67]

Solution In the component bar chart, the length of each bar must represent the total number of workers (agricultural and non-agricultural combined), in which separate lengths have to be shown for each component, viz. agricultural and non-agricultural. These totals are calculated below:

Year	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950
Agricultural	6.2	6.9	8.6	9.9	10.9	11.6	11.4	10.5	8.8	6.8
Non-agricultural	4.3	6.1	8.8	13.4	18.2	25.4	31.0	38.4	42.9	52.2
Total	10.5	13.0	17.4	23.3	29.1	37.0	42.4	48.9	51.7	59.0

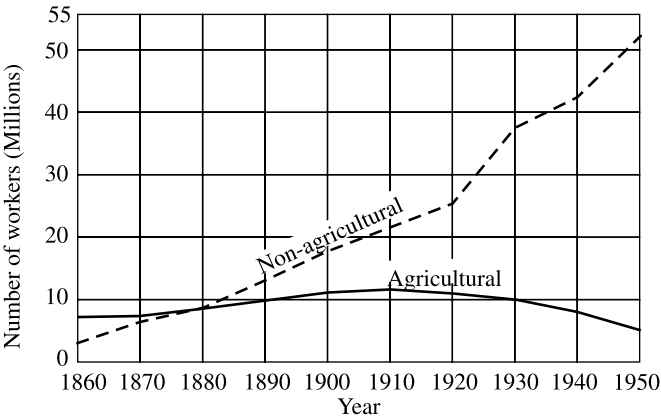


Fig. 2.9 Line Graphs showing Agricultural and Non-agricultural Workers (1860-1950)

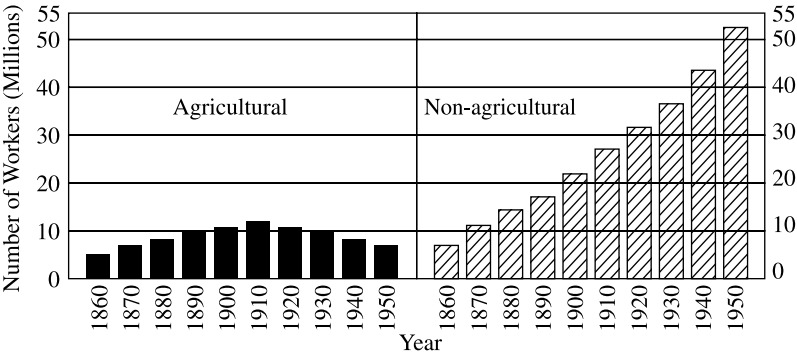


Fig. 2.10 Bar Charts showing Agricultural and Non-agricultural Workers (1860-1950)



It is observed from these diagrams that the number of agricultural workers systematically increased since the year 1860 to about double its figure in the year 1910, and then again it came down gradually. The number of non-agricultural workers, however, increased consistently, nearly 12 times in 90 years since 1860. The trend appears to be approximately a straight line since 1880, indicating a constant annual increase in the number of non-agricultural workers. A comparison of the diagrams for Agricultural and Non-agricultural shows that people are gradually leaving agriculture and engaging themselves more and more to other means of livelihood.

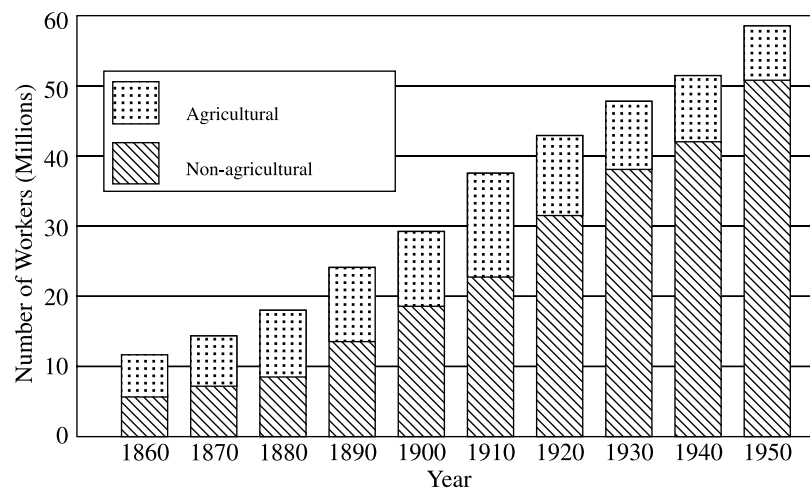


Fig. 2.11 Component Bar Chart showing Agricultural and Non-agricultural Workers (1860-1950)



Example 2.10 Represent the following data by a suitable diagram showing the difference between proceeds and costs:

Proceeds and Costs of a firm (in thousands of rupees)

Year	Total Proceeds	Total Costs
1950	22.0	19.5
1951	27.3	21.7
1952	28.2	30.0
1953	30.3	25.6
1954	32.7	26.1
1955	33.3	34.2

[I.C.W.A., Jan. '66]

Solution We draw a grouped bar diagram for the data.

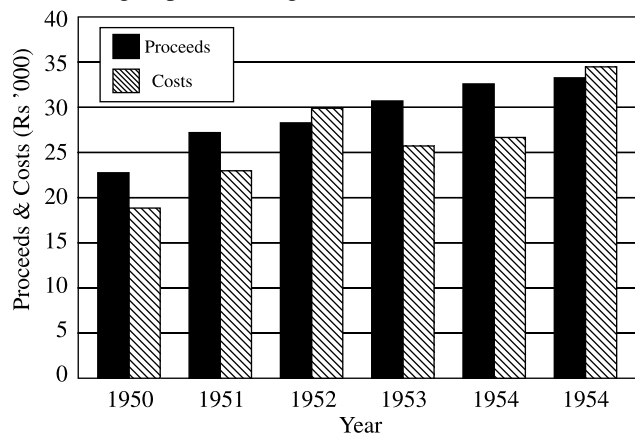


Fig. 2.12 Grouped Bar Diagram

Example 2.11 The following figures show the actual outlay for public sector in India's first two Five-year Plans under different heads of development expenditure:

Head of Development	Percentage of Plan I	Total Outlay Plan II
Agriculture & Community Development ...	14.8	11.7
Irrigation & Power ...	29.7	18.9
Industry & Mining ...	5.0	24.1
Transport & Communications ...	26.4	27.0
Social Services ...	21.0	16.5
Miscellaneous ...	3.1	1.8
Total	100.0	100.0

Represent the data by means of a sub-divided bar chart.

Solution

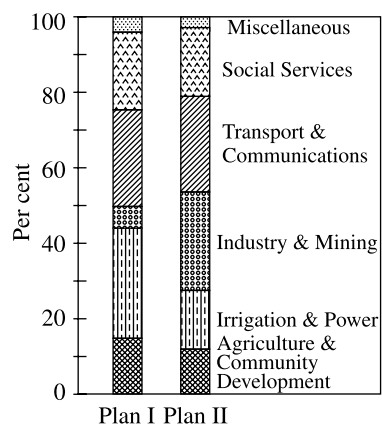


Fig. 2.13 Sub-Divided Bar Chart showing Actual Outlay in India's Five Year Plans I & II



Example 2.12 The following table gives the average approximate yield of rice in lbs. per acre in various countries of the world in 1938–39:

Country	India	Siam	U.S.A.	Italy	Egypt	Japan
Yield in lbs. per acre	728	943	1,469	2,903	2,153	2,276

Indicate this by a suitable diagram which will highlight the relative backwardness of India in this regard. [I.C.W.A., Jan. '64]



Example 2.13 Construct a pie chart for the following data:

*Principal Exporting Countries of Cotton
(1,000 bales)—1955–56*

U.S.A.	India	Egypt	Brazil	Argentina
6,367	2,999	1,688	650	202

[C.U., M.Com. '74]

Solution (Example 2.12)

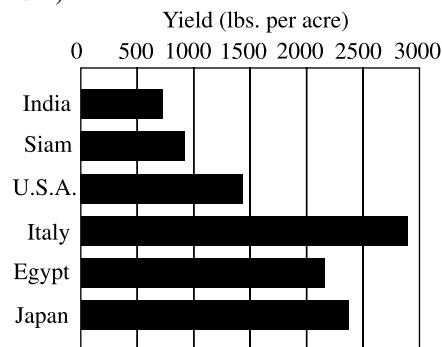


Fig. 2.14 Bar Diagram showing Average Yield of Rice in Various Countries (1938–39)

Solution (Example 2.13)

Table 2.4 Calculations for Pie Chart

Country	Exports ('000 bales)	Per cent of Total	Angle (degrees) at the Centre of Pie Chart col. (3) $\times 3.6$
(1)	(2)	(3)	(4)
U.S.A.	6,367	$(6367 \div 11906) \times 100 = 54$	194.4
India	2,999	$(2999 \div 11906) \times 100 = 25$	90.0
Egypt	1,688	$(1688 \div 11906) \times 100 = 14$	50.4
Brazil	650	$(650 \div 11906) \times 100 = 5$	18.0
Argentina	202	$(202 \div 11906) \times 100 = 2$	7.2
Total	11,906	100	360.0

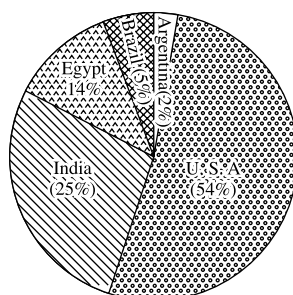


Fig. 2.15 Pie Chart showing Principal Exporting Countries of Cotton (1955-56)

EXERCISES

- Briefly describe one diagrammatic method for representing (i) a time series, (ii) distribution of a total among its components, (iii) a frequency distribution of individuals classified according to an attribute. [I.C.W.A., Dec. '73]
- Indicate the diagrams you would consider most appropriate to represent the following statistical data, stating briefly the reasons for your choice: (i) Marks scored by two selected candidates in each of six subjects tested at an examination; (ii) Total value of Indian exports and imports during the years 1956-1972; (iii) Middleclass cost of living index numbers in Bombay and Calcutta during the years 1961-1972; (iv) Distribution of civil conditions of persons enumerated in 1971-census. [B.U., B.A. (Econ) '73]
- Represent the following statistical information graphically:

Year	1924	1925	1926	1927	1928	1929	1930
Monthly Average Production	609	522	205	608	551	632	516

[C.U., B.Com. (Hons) '65]

- Plot the following data relating to population of India so as to indicate the proportionate increase in population from one period to another:

Year	1872	1881	1891	1901	1911	1921	1931	1941
Population (in millions)	210	250	290	295	315	320	350	390

[C.U., B.A. (Econ) '62]

- Represent the information contained in the following table in a component part chart:

Commodity pattern of India's Exports (Percentage)

	1956-'57	1957-'58	1958-'59
Capital Goods	0.29	0.31	0.30
Intermediate Goods	45.82	46.87	44.19
Consumer Goods	50.50	47.32	48.19
Unclassified	3.39	5.50	7.32
Total	100.00	100.00	100.00

[C.U., B.Com. (Hons.) '67]



6. The following table shows the values of a variable y corresponding to some given equidistant values of the independent variable x :

x	...	7	8	9	10	11	12
y	...	132	214	330	486	688	942

Draw a semi-logarithmic chart and find by graphical interpolation the value of y , when $x = 10.5$ [I.C.W.A., Jan. '71]

7. The following table shows the number of bushels of wheat and corn produced in a farm during the years 1950 to 1960.

Express the yearly number of bushels of wheat and corn as percentages of total annual production. Graph the percentages by component bar charts.

Year	1950	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960
No. of Bushels of Wheat	200	185	225	250	240	195	210	225	250	230	235
No. of Bushels of Corn	75	90	100	85	80	100	110	105	95	110	100

[Dip. Soc. Welfare, May '68]

8. Of the life insurance policy dividends paid in the United States 21% were taken in cash, 31% were used to pay premiums, 18% were used to purchase additional paid-up life insurance, 30% were left with life insurance companies to earn interest. Construct a pie diagram showing these different uses of policy dividends. [C.U., M.Com. '62]
9. A summary of the estimated receipts and expenditures of Government of India for a particular year, is given below:

Receipts	Millions of Rupees	Expenditure	Millions of Rupees
Direct taxes on income	2076.0	Interest on public debt	1143.3
Customs duties	1680.0	Education and health	272.6
Other taxes	2955.6	National defence	2774.5
Revenue from public undertakings	1660.7	Transfers to States	3755.3
Other receipts	1189.1	Other current expenditures	1754.4
		Capital expenditures	5702.2
		Other loans and advances	759.8
Total	9561.4	Total	16162.1

Draw suitable diagrams to show the relative importance of different heads of income and expenditure. [C.U., B.A.(Econ) '73]

**ANSWERS**

1. (i) Line diagram, or Vertical bar diagram; (ii) Component bar diagram, or Pie diagram; (iii) Horizontal bar diagram.
2. (i) Grouped horizontal bar diagram; (ii) Grouped vertical bar diagram; (iii) Two line diagrams drawn on the same graph paper; (iv) Pie diagram.
3. Line diagram.
4. Semi-logarithmic chart.
5. Component bar diagram.
6. Plot $\log y$ against x , and join the points by free-hand curve.
7. Component bar diagram using percentages.
8. Pie Diagram.
9. Two component bar diagrams, one for receipts and another for expenditure.

3 USEFUL MATHEMATICAL DEVICES

3.1 ROUNDING OF NUMBERS

It is known that the number 54.38 rounded to the nearest unit is 54, because the digit 3 next to the unit's place is less than 5; when rounded to one decimal, it is 54.4, because the digit 8 next to the first decimal place is more than 5. Difficulties arise when the next digit to be dropped is exactly 5. The rules are:

(a) When this 5 is followed by digits all of which are not zeros, then the preceding digit is increased by one. 72.5002 rounded to the nearest unit becomes 73 and 12.7537 rounded to one decimal is 12.8.

(b) When this 5 is the last digit or is followed by zeros only, it has become conventional to round the preceding digit to an *even* number. 73.5 rounded to the nearest unit is 74; 12.85 rounded to one decimal is 12.8; 41500 rounded to the nearest thousand is 42 thousand.

Example 3.1

<i>Quantity</i>	<i>Rounded to</i>	<i>Rounded Number</i>
(a) 3.245002	<i>nearest hundredth</i>	3.25
(b) 3.25501	" "	3.26
(c) 3.245	" "	3.24
(d) 3.255	" "	3.26
(e) 274.95	<i>nearest tenth</i>	275.0
(f) 5.99735	<i>nearest hundredth</i>	6.00
(g) 289,523	<i>nearest thousand</i>	290 thousand
(h) 8,500,000	<i>nearest million</i>	8 million
(i) 33.5	<i>nearest unit</i>	34

3.2 ABSOLUTE, RELATIVE AND PERCENTAGE ERRORS

Most of us are accustomed to think of 5.2 as meaning the same thing as 5.20 and 5.200000...to an unlimited number of decimal places. But 5.2 cm actually means a measurement between 5.15 and 5.25 cm. Similarly, 5.20 would mean a number between 5.195 and 5.205, i.e.,



5.2 means a number between $5.2 \pm .05$

5.20 means a number between $5.20 \pm .005$

Unless otherwise stated, *an observation of a continuous variate should be interpreted as extending half a unit of the last place below to half a unit above the recorded value.*

Absolute error of a measurement is its difference from the true value. *Relative error* is the ratio of absolute error to the true value. It is usually expressed as a percentage, and then called *percentage error*.

Absolute Error = Difference between measurement and true value.

$$\text{Relative Error} = \frac{\text{Absolute Error}}{\text{True value}}$$

$$\text{Percentage Error} = \text{Relative Error} \times 100$$

The maximum *Absolute error* in the sum or difference of two measurements is equal to the sum of the separate absolute errors. The maximum *relative error* in the product or quotient of two measurements is approximately equal to the sum of the separate relative errors. Thus, if the measurements x_1 and x_2 have absolute errors e_1 and e_2 respectively, then the maximum absolute error either in $(x_1 + x_2)$ or in $(x_1 - x_2)$ is $(e_1 + e_2)$. Again, the maximum relative error either in $x_1 x_2$ or in x_1/x_2 or in x_2/x_1 is

$$\left(\frac{e_1}{x_1} + \frac{e_2}{x_2} \right).$$

Example 3.2 The true value of π upto 5 decimals is 3.14159. Find the absolute error, relative error, and percentage error, if π is approximated by $22/7$.

Solution $22/7 = 3.14286$ (upto 5 decimals)

Hence, Absolute error = $3.14286 - 3.14159 = .00127$

$$\text{Relative error} = \left(\frac{.00127}{3.14159} \right) = .0004$$

$$\text{Percentage error} = .0004 \times 100 = .04$$



Example 3.3 The sides of a rectangular field are measured to be 127 ± 0.2 ft. and 237 ± 0.4 ft. Find the possible error in the sum of the two sides and also the limits of the true value of the sum of the two sides. [I.C.W.A., Jan. '65]

Solution The absolute errors are $e_1 = 0.2$ and $e_2 = 0.4$. Hence the possible error, i.e. the maximum absolute error, in the sum is given by

$$e_1 + e_2 = 0.2 + 0.4 = 0.6 \text{ ft.}$$

The limits of the true value of the sum are, therefore,

$$(127 + 237) - 0.6 \quad \text{and} \quad (127 + 237) + 0.6$$

i.e. 363.4 ft. and 364.6 ft.

Ans. 0.6 ft; 363.4 to 364.6 ft.



3.3 SIGNIFICANT FIGURES

The significant figures of a quantity are those starting from the first non-zero digit on the left to the end of the last digit accurately specified on the right.

**Example 3.4**

Quantity	Number of Significant Figures
(a) 58	two
(b) 604	three
(c) 1008	four
(d) 95080	five
(e) 7.0	two
(f) 0.80	two
(g) 18.00	four
(h) 0.003	one
(i) 0.0030	two
(j) 3.20	three

3.4 SOME SHORT PROCESSES OF CALCULATION

(1) *Multiplication by 5, 25, or 125*—Place 0, 00 or 000 after the multiplicand and divide by 2, 4, or 8.

Example 3.5 (a) $849 \times 5 = 8490 \div 2 = 4245$
 (b) $436 \times 25 = 43600 \div 4 = 10900$
 (c) $6874 \times 125 = 6874000 \div 8 = 859250$

Multiplication by 25 becomes easier in the following way: For 17×25 let us think of (17×25 paise), i.e. 17 twentyfive-paise coins which make Rs 4.25 P. = 425 P. Hence, $17 \times 25 = 425$. Similarly, 434×25 reminds us of 434 twentyfive-paise coins = Rs 108.50 P. leading to the result $434 \times 25 = 10850$.

(2) *Division by 5, 25, or 125*—Multiply by 2, 4, or 8 and shift the decimal point one, two, or three places to the left.

Example 3.6 (a) $849 \div 5 = (849 \times 2)$ and shift the decimal point one place = 169.8
 (b) $436 \div 25 = (436 \times 4)$ and shift the decimal point two places = 17.44
 (c) $6845 \div 125 = (6845 \times 8)$ and shift the decimal point three places = 54.760.

(3) *Multiplication (division) by 5, .5, .05, .005 etc.*—Divide (multiply) the given number by 2 and adjust the decimal point.

Example 3.7 (a) To find 8.49×5 , divide 849 by 2 mentally, getting 424.5 (forget the decimal point completely for the present). Our answer will be one amongst 4245, 424.5, 42.45, 4.245, .4245, etc., i.e. the non-zero digits from the left must be 4245; and it is now a matter of fixing the decimal point suitably. Can we not guess the result? Yes, we can; the result is nearly $8 \times 5 = 40$. The correct answer must be 42.45.

(b) $.068 \times .05 \rightarrow 68 \div 2 = 34 \rightarrow .0034$ (because $.07 \times .05 = .0035$)
 (c) $6.7 \times .005 \rightarrow 67 \div 2 \rightarrow 33.5 \rightarrow .0335$ (because $7 \times .005 = .035$)
 (d) $84.6 \times 50 \rightarrow 846 \div 2 = 423 \rightarrow 4230$ (since $80 \times 50 = 4000$)



$$(e) 8.49 \div 5 \rightarrow 849 \times 2 = 1698 \rightarrow 1.698 \quad (\text{since } 8 \div 5 = 1 \dots)$$

$$(f) 84.6 \div 50 = \frac{84.6 \times 2}{50 \times 2} = \frac{169.2}{100} = 1.692$$

$$(g) .846 \div .05 = \frac{.846 \times 2}{.05 \times 2} = \frac{1.692}{.1} = \frac{16.92}{1} = 16.92$$

(4) *Multiplication (division) by 25, 2.5, .25, .025 etc.*—Divide (multiply) by 4 and adjust the decimal point.

Example 3.8 (a) $48 \times 2.5 \rightarrow 48 \div 4 = 12 \rightarrow 120$ (since $48 \times 2 = 96$)

(b) $14.2 \times .25 \rightarrow 142 \div 4 \rightarrow 355 \rightarrow 3.55$ (since $14 \times .2 = 2.8$)

(c) $483.2 \times .025 \rightarrow 4832 \div 4 = 1208 \rightarrow 12.08$ (since $500 \times .025 = 12.5$)

$$(d) 48 \div 2.5 = \frac{48 \times 4}{2.5 \times 4} = \frac{192}{10} = 19.2$$

$$(e) 483.2 \div 250 = \frac{483.2 \times 4}{250 \times 4} = \frac{1932.8}{1000} = 1.9328$$

$$(f) 1.42 \div .025 = \frac{1.42 \times 4}{.025 \times 4} = \frac{5.68}{.1} = \frac{56.8}{1} = 56.8$$

(5) *Division by a number ending in 5*—If the divisor is a multiple of 5, multiply both the numerator and denominator by 2; in case the denominator is a multiple of 25, use 4 instead of 2.

Example 3.9 (a) $\frac{44}{3.5} = \frac{44 \times 2}{3.5 \times 2} = \frac{88}{7} = 12.57$

(b) $\frac{3.8}{75} = \frac{3.8 \times 4}{.75 \times 4} = \frac{15.2}{3} = 5.07$

(c) $\frac{382}{150} = \frac{382 \times 2}{150 \times 2} = \frac{764}{300} = 2.55$

(d) $\frac{246}{27.5} = \frac{246 \times 4}{27.5 \times 4} = \frac{984}{110} = \frac{98.4}{11} = 8.95$

(6) *Multiplication by 9 (or 11)*—Place a zero to the right of the multiplicand and subtract (add) the multiplicand.

Example 3.10 (a) $234 \times 9 \rightarrow$

$$\begin{array}{r} 2340 \\ -234 \\ \hline 2106 \end{array}$$

(b) $234 \times 11 \rightarrow$

$$\begin{array}{r} 2340 \\ +234 \\ \hline 2574 \end{array}$$

For multiplication by 99, place two zeros to the right and then subtract the multiplicand.

7. *Subtracting decimal fraction from whole number*—The integral part of the result will be one less than the difference between the whole number and the integral part of



the number to be subtracted. For the decimal part of the result, subtract from 9 each of the digits following the decimal point, except the last digit, which is subtracted from 10.

Example 3.11 $8 - 5.498063 = 2.501937$

Because, $8 - 5 - 1 = 2$; $9 - 4 = 5$; $9 - 9 = 0$; $9 - 8 = 1$; $9 - 0 = 9$; $9 - 6 = 3$; and finally $10 - 3 = 7$.

(8) *Fixing the position of decimal point in division*—It is a common practice that in divisions involving decimal fractions, the decimals of the numerator and the denominator are shifted to the right by the same number of places, so that both of them become whole numbers. If now the numerator is less than the denominator, many students find it difficult to decide how many zeros will follow the decimal point in the quotient before the first significant figure appears. The following simple rule may be found useful:

In the denominator take as many digits from the left, such that the number so formed just exceeds the numerator. The number of digits in the denominator in excess of this will give the exact number of zeros after the decimal point preceding the first significant figure of the quotient.

Example 3.12 (a) Find the number of zeros after the decimal point in the

$$\text{quotient } \frac{243}{51807}.$$

$$(b) \quad 23 \div 3798 = .00$$

$$(c) \quad 19 \div 1265 = .0$$

$$(d) \quad 173 \div 5297 = .0$$

$$(e) \quad 3.84 \div 423 = 384 \div 42300 = .00$$

$$(f) \quad 0.663 \div 0.83 = 663 \div 830 = .$$

$$(g) \quad .07 \div 664 = 7 \div 66400 = .000$$

$$(h) \quad .0002 \div .039 = 2 \div 390 = .00$$

$$(i) \quad 0.24 \div 72.113 = 240 \div 72113 = .00$$

Solution (a) The first digit from the left in the denominator 51807, viz. 5, is less than the numerator 243. Next consider the number 51 formed by the first two digits. This is also less than the numerator 243. Again consider the first 3 digits of denominator, giving 518. This is greater than 243 and hence here we stop. Since there are 2 more digits 0 and 7 remaining in the denominator, there will be exactly 2 zeros in the result after the decimal point.

$$\frac{243}{51807} = .0046 \dots$$

For obtaining the significant figures in the result, ignore all decimals both in the numerator and in the denominator; put sufficient number of zeros to the right of the numerator and then divide.

$$\begin{array}{r} 51907 \overline{) 243000} \quad (46\dots \\ \underline{207228} \\ 357720 \\ \underline{310842} \\ 46878 \end{array}$$





3.5 ROOTS AND RECIPROCAL EXPRESSED AS POWER

Example 3.13

(a) $\sqrt{x} = x^{\frac{1}{2}}$	(g) $\frac{1}{x} = x^{-1}$
(b) $\sqrt[3]{x} = x^{\frac{1}{3}}$	(h) $\frac{1}{x^2} = x^{-2}$
(c) $\sqrt[n]{x} = x^{\frac{1}{n}}$	(i) $\frac{1}{x^n} = x^{-n}$
(d) $x\sqrt{x} = x \cdot x^{\frac{1}{2}} = x^{\frac{3}{2}}$	(j) $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$
(e) $\sqrt[3]{x^2} = (x^2)^{\frac{1}{3}} = x^{\frac{2}{3}}$	(k) $\frac{1}{x\sqrt{x}} = x^{-\frac{3}{2}}$
(f) $\sqrt[n]{x^m} = (x^m)^{\frac{1}{n}} = x^{\frac{m}{n}}$	(l) $\frac{1}{\sqrt[3]{x^2}} = x^{-\frac{2}{3}}$

3.6 LOGARITHM

Every positive number can be expressed as a power of 10. This power is called *logarithm to the base 10* or simply *logarithm* of the number, briefly written as *log*. For example, since $100 = 10^2$, $\log 100 = 2$; $.001 = 10^{-3}$, therefore $\log .001 = -3$. The number 69 when expressed as a power of 10 will be

$$69 = 10^{1.8388}$$

This means that $\log 69 = 1.8388$

The result 1.8388 is not exact, but correct only upto 4 decimal places. Logarithms of numbers correct to certain decimals are obtained from log-tables. For example, a four-figure log table is used to obtain logarithms correct to 4 decimals only.

(i) How to find the logarithm?

The logarithm of any number consists of an integral part, called *characteristic*, and a decimal part, called *mantissa*.

The characteristic of any number greater than 1 is positive, and is equal to the number of digits before the decimal point minus one. The characteristic of a number less than 1 is negative, and is equal to the number of consecutive zeros immediately following the decimal point plus one. A negative characteristic, say -3 , is written as $\bar{3}$ (read *bar 3*).

Example 3.14

Number	Characteristic
(a) 176	2
(b) 69.58	1
(c) 3.47064	0
(d) 0.754	$\bar{1}$ (i.e. -1)
(e) .075403	$\bar{2}$
(f) .0000327	$\bar{5}$
(g) 1000	3



The mantissa of any number is obtained from log-tables. In a four-figure log-table, series of four-figure numbers are shown in rows against 10, 11, 12, ...99 at the extreme left of the table. These four-figure numbers are also arranged in columns with headings 0, 1, 2, ...9. At the extreme right of the table, the mean-differences are shown with column headings 1, 2, ...9. Such a log-table can be used to find the logarithm of a number not exceeding 4 significant figures.

The mantissa is independent of the position of decimal point in the number whose logarithm is required. Thus, the mantissa of 695.8 is the same as that of 6958 or of 0.6958 or even of .0006958.

In order to obtain the mantissa of 6958, we look up the number 69 (by the first two digits) at the extreme left column of the table. Then moving horizontally towards the right we come under the column-heading 5 (the third digit) to the figure 8420, which is the mantissa of 695. Moving further to the right under the mean-difference column with heading 8 (the fourth digit), we get 5, which is added to 8420, giving 8425. Thus the mantissa of 6958 is .8425.

Example 3.15

<i>Number</i>	<i>Mantissa</i>
(a) 176	.2455
(b) 3470	.5403
(c) 1254	.0983
(d) 7506	.8754
(e) 32	.5051
(f) 3	.4771
(g) 300,000	.4771

An important point to note is that the characteristic may be positive or negative, but the *mantissa is always positive*.

Example 3.16

<i>Number</i>	<i>Logarithm</i>	
(a) 6958	3.8425	
(b) 695.8	2.8425	
(c) 69.58	1.8425	
(d) 6.958	0.8425	
(e) .6958	$\bar{1}.8425$	(i.e. $-1 + .8425 = -.1575$)
(f) .06958	$\bar{2}.8425$	(i.e. $-2 + .8425 = -1.1575$)
(g) .006958	$\bar{3}.8425$	
(h) 176	2.2455	
(i) 0.75	$\bar{1}.8751$	

**(ii) Rules for logarithms**

(1) The logarithm of the product of two or more numbers is the sum of their logarithms.

$$\log (a \times b) = (\log a) + (\log b) \quad (3.6.1)$$

Example 3.17 (a) $\log (47 \times 8) = (\log 47) + (\log 8)$
 $= 1.6721 + 0.9031$
 $= 2.5752.$

This may be verified directly. $\log .376 = 2.5752$

(b) $\log (439.4 \times .0347) = (\log 439.4) + (\log .0347)$
 $= 2.6429 + \bar{2}.5403 = 2.6429 - 2 + .5403 = 1.1832$

(c) $\log (x_1, x_2, x_3) = \log x_1 + \log x_2 + \log x_3$

(2) The logarithm of the ratio of two numbers is equal to the logarithm of numerator minus the logarithm of denominator.

$$\log \left(\frac{a}{b} \right) = (\log a) - (\log b) \quad (3.6.2)$$

Example 3.18 (a) $\log \left(\frac{758.3}{44.78} \right) = (\log 758.3) - (\log 44.78)$
 $= 2.8799 - 1.6511 = 1.2288$

(b) $\log \left(\frac{.035}{44.78} \right) = \log .035 - \log 44.78$
 $= \bar{2}.5441 - 1.6511 = -2 + .5441 - 1.6511$
 $= -3.1070$

(3) The logarithm of any number raised to a certain power is equal to the product of the power and the logarithm of the number.

$$\log (a^b) = b(\log a) \quad (3.6.3)$$

Example 3.19 (a) $\log (4.8)^3 = 3 \times (\log 4.8)$
 $= 3 \times 0.6812 = 2.0436$

(b) $\log (4.8)^{\frac{1}{3}} = \frac{1}{3} \times (\log 4.8) = \frac{1}{3} \times 0.6812 = 0.2271$

(c) $\log \left(1 + \frac{r}{100} \right)^n = n \log \left(1 + \frac{r}{100} \right)$

(d) $\log P \left(1 + \frac{r}{100} \right)^n = \log P + \log \left(1 + \frac{r}{100} \right)^n$
 $= \log P + n \log \left(1 + \frac{r}{100} \right)$

(e) $\log \left(x_1^{f_1} \cdot x_2^{f_2} \dots x_n^{f_n} \right)^{\frac{1}{N}} = \frac{1}{N} \log \left(x_1^{f_1} \cdot x_2^{f_2} \dots x_n^{f_n} \right)$
 $= \frac{1}{N} \left[\log x_1^{f_1} + \log x_2^{f_2} + \dots + \log x_n^{f_n} \right]$



$$= \frac{1}{N} [f_1(\log x_1) + f_2(\log x_2) + \dots + f_n(\log x_n)]$$

(iii) How to find the antilogarithm?

The antilogarithm of a number is that quantity whose logarithm is the given number. Since $\log 695.8 = 2.8425$, therefore

$$\text{antilog } 2.8425 = 695.8.$$

In order to find antilog 2.8425 by using the tables, we first consider only the decimal part, viz, .8425. The process of reading the antilog-table is exactly similar to that of the log-table. The first two digits (including zeros) immediately after the decimal point of the number are found from the extreme left column of the table, the third digit from the column-headings, and the fourth digit from the mean-difference column, thus obtaining $6950 + 8 = 6958$. The decimal point has to be inserted at a place depending on the value of the characteristic. Since the integral part of the number 2.8425 is 2, hence we must have three digits in the integral part of the antilog, giving 695.8.

Example 3.20

<i>Number</i>	<i>Antilogarithm</i>
(a) 2.034	108.1
(b) -0.8734 (= 1.1266)	0.1339
(c) -3.8734 (= $\bar{4} - .1266$)	.0001339
(d) .0357	1.086
(e) 3.2678	1852

(iv) Uses of logarithms

Logarithms are used to find the product, quotient, powers and roots of numbers.

Example 3.21 Find the values of

$$\begin{array}{ll}
 (a) 439.4 \times .0347 \times 6.8 & (b) \frac{0.6745 \times 8.351}{.00737 \times 4905} \\
 (c) 35(.09)^4 (.91)^3 & (d) \frac{3.845}{(2.6408)^3} \\
 (e) \sqrt[3]{4.8} & (f) \frac{129}{\sqrt{108 \times 176}} \\
 (g) \frac{114.72}{\sqrt{548.76 \times 171.20}}
 \end{array}$$

Solution (a) $\log (439.4 \times .0347 \times 6.8)$
 $= \log 439.4 + \log .0347 + \log 6.8$
 $= 2.6429 + \bar{2}.5403 + 0.8325$
 $= 2.6429 + (-2 + .5403) + 0.8325$
 $= 2.0157$

$\therefore 439.4 \times .0347 \times 6.8 = \text{antilog } 2.0157 = 103.7$



$$\begin{aligned}
 \text{(b) Let } x &= \frac{0.6745 \times 8.351}{.00737 \times 4905} \\
 \log x &= \log (0.6745 \times 8.351) - \log (.00737 \times 4905) \\
 &= (\log 0.6745 + \log 8.351) - (\log .00737 + \log 4905) \\
 &= (\bar{1}.8290 + 0.9218) - (\bar{3}.8675 + 3.6906) \\
 &= (-1 + .8290 + .9218) - (-3 + .8675 + 3.6906) \\
 &= -1 + 1.7508 - 1.5581 \\
 &= -1 + .1927 = \bar{1}.1927
 \end{aligned}$$

$$\therefore x = \text{antilog } \bar{1}.1927 = 0.1559$$

$$\begin{aligned}
 \text{(c) Let } x &= 35(.09)^4 (.91)^3 \\
 \log x &= \log 35 + \log (.09)^4 + \log (.91)^3 \\
 &= \log 35 + 4(\log .09) + 3(\log .91) \\
 &= 1.5441 + 4(\bar{2}.9542) + 3(1.9590) \\
 &= 1.5441 + 4(-2 + .9542) + 3(-1 + .9590) \\
 &= 8.2379 - 11 = .2379 - 3 = \bar{3}.2379 \\
 x &= \text{antilog } \bar{3}.2379 = .001730
 \end{aligned}$$

$$\begin{aligned}
 \text{(d) Let } x &= \frac{3.854}{(2.6408)^3} \\
 \log x &= \log 3.854 - \log (2.6408)^3 \\
 &= \log 3.854 - 3(\log 2.6408) \\
 &= 0.5860 - 3(0.4218); \text{ using } \log 2.641 \text{ for } \log 2.6408 \\
 &= 0.5860 - 1.2654 = (0.5860 - 0.2654) - 1 \\
 &= .3206 - 1 = \bar{1}.3206 \\
 \therefore x &= \text{antilog } \bar{1}.3206 = .2092
 \end{aligned}$$

$$\text{(e) } \sqrt[3]{4.8} = (4.8)^{\frac{1}{3}}, \text{ see Sec. 3.5}$$

$$\log \sqrt[3]{4.8} = \log (4.8)^{\frac{1}{3}} = \frac{1}{3} (\log 4.8) = \frac{1}{3} (.6812) = 0.2271$$

$$\therefore \sqrt[3]{4.8} = \text{antilog } .2271 = 1.687$$

$$\begin{aligned}
 \text{(f) Let } x &= \frac{129}{\sqrt{108 \times 176}} = \frac{129}{(108 \times 176)^{\frac{1}{2}}} \\
 \log x &= \log 129 - \frac{1}{2} \log (108 \times 176) \\
 &= \log 129 - \frac{1}{2} (\log 108 + \log 176) \\
 &= 2.1106 - \frac{1}{2} (2.0334 + 2.2455) \\
 &= 2.1106 - 2.1394 = \bar{1}.9712
 \end{aligned}$$

$$\therefore x = \text{antilog } \bar{1}.9712 = 0.9358$$

(g) For using the four-figure log table, we round off the numbers to 4 significant digits.

$$\text{Let } x = \frac{114.72}{\sqrt{548.76 \times 171.20}} = \frac{114.7}{\sqrt{548.8 \times 171.2}} \text{ (approx.)}$$



$$\log x = \log 114.7 - \frac{1}{2} (\log 548.8 + \log 171.2)$$

$$= 2.0596 - \frac{1}{2} (2.7394 + 2.2335)$$

$$= 2.0596 - 2.4864 = 1.5732$$

$$\therefore x = \text{antilog } \bar{1}.5732 = .3743$$



3.7 A.P. SERIES AND G.P. SERIES

A series of observations is said to be in Arithmetic Progression (A.P.), if the consecutive values have a common difference.

- Example 3.22** (i) 1, 2, 3, 4, ...
 (ii) 5, 8, 11, 14, ...
 (iii) 7, 5, 3, 1, -1, -3, -5, ...

The A.P. series is algebraically represented by

$$a, (a + d), (a + 2d), (a + 3d), \dots \quad (3.7.1)$$

where a = initial term, and d = common difference. The n th term (t_n) and the sum (S_n) upto n terms of the A.P. series are given by

$$t_n = a + (n - 1)d, S_n = [2a + (n - 1)d] \frac{n}{2} \quad (3.7.2)$$

A series of observations is said to be in Geometric Progression (G.P.), if the consecutive values have a common ratio.

- Example 3.23** (i) 5, 10, 20, 40, 80, 160, ...

$$(ii) 4, 1, \frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \dots$$

The G.P. series is algebraically represented by

$$a, ar, ar^2, ar^3 \dots \quad (3.7.3)$$

where a = initial term, and r = common ratio. The n th term (t_n') and the sum (S_n') upto n terms of the G.P. series are given by

$$t_n' = ar^{n-1} \quad S_n' = \frac{a(1 - r^n)}{(1 - r)}$$

Example 3.24 Two men X and Y started working for a certain company at similar jobs on Jan. 1, 1950. X asked for an initial salary of Rs 300 with an annual increment of Rs 30. Y asked for an initial salary of Rs 200 with a rise of Rs 15 every six months. What are their respective salaries on Dec. 31, 1959, assuming that the arrangements remained unaltered? Find the total amount paid to each of them as salary during the period. [C.U., B.A. (Econ) '68]

Solution The initial monthly salary of X is Rs 300 and increases every year by Rs 30. Hence his salary figures in successive years will be in A.P. with a common difference of Rs 30. Similarly, the salary figures of Y in successive half-years will be in A.P. with a common difference of Rs 15. For X, Dec. 31, 1959 falls in the 10th year of service, and for Y it falls in the 20th half-year. Thus, for the two A.P. series, we have



Men	Initial Term (a)	Common Difference (d)	No. of Terms (n)
X	300	30	10
Y	200	15	20

Using the first formula of (3.7.2), the salaries as on Dec. 31, 1959 are

$$t_{10} = 300 + (10 - 1) \times 30 = 570 \text{ Rs (for X)}$$

$$t_{20} = 200 + (20 - 1) \times 15 = 485 \text{ Rs (for Y)}$$

For determining the total amount paid to each of them, we apply the second formula of (3.7.2). Taking one month's salary from each year, the total amount paid to X in each of the 12 months during the 10 year period is

$$S_{10} = \{2 \times 300 + (10 - 1) \times 30\} \frac{10}{2} = 4,350 \text{ Rs}$$

Therefore, the total amount paid to X throughout the period is

$$= 4,350 \times 12 = 52,200 \text{ Rs}$$

Similarly, taking one month's salary from each half-year, the total amount paid to Y in each of the 6 months during the 20 half-years is

$$S_{20} = \{2 \times 200 + (20 - 1) \times 15\} \frac{20}{2} = 6,850 \text{ Rs}$$

Therefore, the total amount paid to Y throughout the period is

$$= 6,850 \times 6 = 41,100 \text{ Rs}$$

Ans. (a) Rs 570, Rs 485; (b) Rs 52,200, Rs 41,100.



3.8 SUM AND SUM OF THE SQUARES OF NUMBERS

$$\text{Sum of the first } n \text{ numbers} = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Sum of the squares of the first n numbers

$$= 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

3.9 SIMPLE INTEREST LAW AND COMPOUND INTEREST LAW

If P denotes the principal invested, i the interest rate (i.e. interest on Re 1 for 1 period), n the number of periods, and A denotes the amount (i.e. principal + interest) after n periods, then

$$A = P(1 + ni) \quad \text{at Simple interest} \quad (3.9.1)$$

$$A = P(1 + i)^n \quad \text{at Compound interest} \quad (3.9.2)$$

If the rates of interest vary in different periods, say i_1 for n_1 periods, i_2 for the next n_2 periods, i_3 for the next n_3 periods then

$$A = P(1 + n_1 i_1 + n_2 i_2 + n_3 i_3) \quad \text{at Simple interest} \quad (3.9.3)$$

$$A = P(1 + i_1)^{n_1} (1 + i_2)^{n_2} (1 + i_3)^{n_3} \quad \text{at Compound interest} \quad (3.9.4)$$



3.10 PERMUTATION AND COMBINATION

The *permutation* of n things taken r at a time (denoted by ${}^n P_r$) is the total number of possible orders of a arrangement by considering r things at a time out of n things.

$${}^n P_r = n(n-1)(n-2)(n-3) \dots (n-r+1)$$

When $r = n$, we have ${}^n P_n = n(n-1)(n-2) \dots 3.2.1 =$ Product of all numbers from 1 to n . This is written as $n!$ (read 'factorial n ')

$$n! = 1 \times 2 \times 3 \times 4 \times \dots \times n$$

It may be verified that

$$1! = 1; 2! = 2; 3! = 6; 4! = 24; 5! = 120;$$

Customarily, we write $0! = 1$.

Example 3.25 (i) ${}^5 P_2 = 5 \times 4 = 20$; (ii) ${}^7 P_3 = 7 \times 6 \times 5 = 210$.

The *combination* of n things taken r at a time (denoted by ${}^n C_r$) is the total number of possible groups that can be formed by considering r things at a time out of n , without any regard to the order of arrangement.

$${}^n C_r = \frac{n(n-1)(n-2)(n-3) \dots (n-r+1)}{r!} = \frac{n!}{r!(n-r)!}$$

Putting $r = 0$ and n successively, we have

$${}^n C_0 = \frac{n!}{0!n!} = 1; \quad {}^n C_n = \frac{n!}{0!n!} = 1$$

A useful relation is ${}^n C_r = {}^n C_{n-r}$

Example 3.26 (i) ${}^6 C_2 = \frac{6 \times 5}{1 \times 2} = 15$

$$(ii) \quad {}^8 C_3 = \frac{8 \times 7 \times 6}{1 \times 2 \times 3} = 56$$

$$(iii) \quad {}^8 C_6 = {}^8 C_2 = \frac{8 \times 7}{1 \times 2} = 28$$

$$(iv) \quad {}^{20} C_{16} = {}^{20} C_4 = \frac{20 \times 19 \times 18 \times 17}{1 \times 2 \times 3 \times 4} = 4845$$

3.11 BINOMIAL SERIES AND BINOMIAL COEFFICIENTS

We know that

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

In general, when n is a positive integer,

$$(a+b)^n = ({}^n C_0)a^n + ({}^n C_1)a^{n-1}b + ({}^n C_2)a^{n-2}b^2 + \dots + ({}^n C_n)b^n$$

$$(a-b)^n = ({}^n C_0)a^n - ({}^n C_1)a^{n-1}b + ({}^n C_2)a^{n-2}b^2 - \dots + (-)^n ({}^n C_n)b^n \quad (3.11.1)$$

The series expansion on the right is known as *binomial series* and the coefficients

$${}^n C_0, {}^n C_1, {}^n C_2, \dots, {}^n C_n$$

are called *binomial coefficients*.

**Table 3.1** Binomial Coefficients

n	<i>Binomial Coefficients</i>										
1	1,	1									
2	1,	2,	1								
3	1,	3,	3,	1							
4	1,	4,	6,	4,	1						
5	1,	5,	10,	10,	5,	1					
6	1,	6,	15,	20,	15,	6,	1				
7	1,	7,	21,	35,	35,	21,	7,	1			
8	1,	8,	28,	56,	70,	56,	28,	8,	1		
9	1,	9,	36,	84,	126,	126,	84,	36,	9,	1	
10	1,	10,	45,	120,	210,	252,	210,	120,	45,	10,	1

Note that the binomial coefficients are symmetrical, i.e. the coefficients equidistant from either end are equal. Using the binomial coefficients in Table 3.1, we can easily write down the binomial series for any power upto 10.

Example 3.27 (i) $(1+x)^7 = 1 + 7x + 21x^2 + 35x^3 + 35x^4 + 21x^5 + 7x^6 + x^7$
(ii) $(x_i - c)^4 = x_i^4 - 4x_i^3c + 6x_i^2c^2 - 4x_ic^3 + c^4$
(iii) $(E-1)^5 = E^5 - 5E^4 + 10E^3 - 10E^2 + 5E - 1$

3.12 INEQUALITIES

The following symbols have meanings shown against each:

$<$ less than
 $\leq$ less than, or equal to
 $>$ greater than
 $\geq$ greater than, or equal to

(i) If $a > b$, then $a + c > b + c$; $a - c > b - c$

but $-a < -b$; $\frac{1}{a} < \frac{1}{b}$

(ii) Cauchy-Schwarz inequality:

$$(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2$$

The two sides are equal only when $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n}$

3.13 CONCEPT OF 'FUNCTION'

If two variables x and y are so related that the value of y depends on the value of x , then y is said to be a *function of x* . Sometimes, x is called the *independent variable* (to which we can assign values at pleasure), and y is called the *dependent variable* (whose values depend on those assigned to the independent variable).



- Example 3.28**
- (a) $y = 7 + 4x$
 - (b) $y = 9x^2 - 5x + 6$
 - (c) $y = x^3 - x - 5$
 - (d) $y = 5 \times 2x$
 - (e) $y = 4(\log x)^2 + (\log x) + 4$

In each of the above cases, y will be called a *function of x* , the functional form being given by the expression on the right. In general, any mathematical expression involving x is a function of x , and this is often denoted by writing $f(x)$, $g(x)$ etc. For example, if

$$f(x) = x^3 - x - 5,$$

then the value of the function when $x = 4$ is

$$f(4) = 4^3 - 4 - 5 = 55.$$

3.14 POLYNOMIAL

Polynomial in x is a special function in which the individual terms contain only positive integral powers of x , e.g. x , x^2 , x^3 , etc. The *degree* of the polynomial is the highest power of x contained in the polynomial.

- Example 3.29**
- (a) $4x + 7$ is a *1st degree polynomial or linear function*.
 - (b) $9x^2 - 5x + 6$ is a *2nd degree polynomial or parabolic function*.
 - (c) $x^3 + 5$ is a *3rd degree polynomial or cubic function*.
 - (d) $2x^4 + 7x^3 - 4x^2 + 5$ is a *4th degree polynomial*.
 - (e) $x(x-1)(x-2) + 8x(x-1) + 4x + 3$ is a *cubic function*.
 - (f) $a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$ is an *n -th degree polynomial, provided a_n is not zero*.

If $f(x)$ represents a polynomial in x of degree n , then $y = f(x)$ geometrically represents a curve, called the *n -th degree curve*.

- Example 3.30**
- (a) $y = 4x + 7$ represents a *1st degree curve or straight line*.
 - (b) $y = 9x^2 - 5x + 6$ represents a *2nd degree curve or parabola*.
 - (c) $y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ represents an *n -th degree curve, provided a_n is not zero*.

3.15 SIGMA (Σ) NOTATION

The Greek capital letter Σ , when placed before a mathematical expression, signifies

the *sum* of a series. For instance, $\sum_{r=1}^{20} (r^2 + 3)$ denotes the sum of a series of 20 terms

obtained by putting $r = 1, 2, 3, \dots, 20$ successively in the expression $(r^2 + 3)$, i.e.

$$\sum_{r=1}^{20} (r^2 + 3) = (1^2 + 3) + (2^2 + 3) + (3^2 + 3) \dots + (20^2 + 3).$$



The expression $(r^2 + 3)$, which follows Σ , may therefore be taken as a general term of the series.

Similarly, if $x_1, x_2, \dots, x_n$ represent a series of n values, their sum may be written as

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n \quad (3.15.1)$$

This is read as *summation of x sub i , for i equal to 1 to n* . Very often it happens that there is no necessity of writing the *limits of summation*, it being implied that the lower and the upper limits are 1 and n respectively. Thus,

$$\sum_1^n x_i, \quad \sum_i^n x_i, \quad \sum x_1, \quad \sum x$$

are all used to denote the sum (3.15.1)

Example 3.31 Write the expanded form:

$$\begin{array}{lll} (i) \sum_{i=2}^7 (x_i - 3)^2 & (ii) \sum_{i=5}^9 x_i^3 & (iii) \sum_{i=1}^{40} (x_i + 4) \\ (iv) \sum_{i=1}^n f_i x_i & (v) \sum_i a x_i & (vi) \sum_l^n \log x_i \\ (vii) \sum_{x=1}^n x & (viii) \sum_{x=1}^n x^2 & (ix) \sum_{x=1}^n (2x - 1) \\ (x) \sum_{x=0}^n {}^n C_x & (xi) \sum_{x=0}^n {}^n C_x p^x q^{n-x} & \end{array}$$

Solution (i) The limits of summation are from 2 to 7. Putting $i = 2, 3, 4, 5, 6, 7$ successively in the algebraic expression $(x_i - 3)^2$, we get $(x_2 - 3)^2, (x_3 - 3)^2, (x_4 - 3)^2, (x_5 - 3)^2, (x_6 - 3)^2, (x_7 - 3)^2$. The sum of these terms gives the expanded form.

$$(x_1 - 3)^2 = (x_2 - 3)^2 + (x_3 - 3)^2 + (x_4 - 3)^2 + (x_5 - 3)^2 + (x_6 - 3)^2 + (x_7 - 3)^2$$

$$(ii) x_5^3 + x_6^3 + x_7^3 + x_8^3 + x_9^3$$

$$(iii) (x_1 + 4) + (x_2 + 4) + (x_3 + 4) + \dots + (x_{50} + 4)$$

$$(iv) f_1 x_1 + f_2 x_2 + \dots + f_n x_n$$

$$(v) a x_1 + a x_2 + \dots + a x_n$$

$$(vi) \log x_1 + \log x_2 + \dots + \log x_n$$

$$(vii) 1 + 2 + 3 + \dots + n$$

$$(viii) 1^2 + 2^2 + 3^2 + \dots + n^2$$

$$(ix) 1 + 3 + 5 + \dots + (2n - 1)$$

$$(x) {}^n C_0 + {}^n C_1 + {}^n C_2 + \dots + {}^n C_n$$

$$(xi) {}^n C_0 q^n + {}^n C_1 p q^{n-1} + {}^n C_2 p^2 q^{n-2} + \dots + {}^n C_n p^n$$



Example 3.32 Write the contracted form with Σ notation:

$$\begin{array}{ll} (i) x_1 + x_2 + \dots + x_n & \\ (ii) (x_1 + y_1) + (x_2 + y_2) + \dots + (x_n + y_n) & \\ (iii) f_1 x_1^2 + f_2 x_2^2 + \dots + f_n x_n^2 & \\ (iv) (2x_1 + 7) + (2x_2 + 7) + \dots + (2x_9 + 7) & \\ (v) x_1(x_1 - 1) + x_2(x_2 - 1) + \dots + x_n(x_n - 1) & \\ (vi) f_1(\log x_1) + f_2(\log x_2) + \dots + f_n(\log x_n) & \end{array}$$



Solution

$$\begin{array}{lll}
 \text{(i)} \sum_{i=1}^n x_i & \text{(ii)} \sum_{i=1}^n (x_i + y_i) & \text{(iii)} \sum_{i=1}^n f_i x_i^2 \\
 \text{(iv)} \sum_{i=1}^n (2x_i + 7) & \text{(v)} \sum_{i=1}^n x_i (x_i - 1) & \text{(vi)} \sum_{i=1}^n f_i (\log x_i)
 \end{array}$$

If a series of values $x_1, x_2, \dots, x_n$ and the corresponding values of another series $f_1, f_2, \dots, f_n$ are shown vertically in the columns of a table, then

Σx denotes the total of x -column
 Σfx " " " " fx -column
 Σfx^2 " " " " fx^2 -column

Table 3.2 Meaning of Σx , Σfx , Σfx^2

	x	f	fx	fx^2
	x_1	f_1	$f_1 x_1$	$f_1 x_1^2$
	x_2	f_2	$f_2 x_2$	$f_2 x_2^2$
	...	...	...	...
	x_n	f_n	$f_n x_n$	$f_n x_n^2$
Total	Σx	Σf	Σfx	Σfx^2

Example 3.33 Given $x_1 = 6, \quad x_2 = -4, \quad x_3 = 5, \quad x_4 = 8,$
 $f_1 = 3, \quad f_2 = 1, \quad f_3 = 2, \quad f_4 = 4,$

find the values of

$$\begin{array}{lll}
 \text{(i)} \Sigma x & \text{(ii)} \Sigma fx & \text{(iii)} \Sigma x^2 \\
 \text{(iv)} \Sigma f(x + 4) & \text{(v)} \Sigma \left(\frac{f}{x} \right) & \text{(vi)} (\Sigma x)^2 \\
 \text{(vii)} \frac{\Sigma fx}{\Sigma f} & \text{(viii)} \Sigma fx^2 & \text{(ix)} \Sigma fx(x - 1)
 \end{array}$$

Solution

Table 3.3 Calculations for Σx , Σfx , etc.

	x	f	fx	x^2	$(x + 4)$	$f(x + 4)$	$\frac{f}{x}$	fx^2	$fx(x - 1)$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	6	3	18	36	10	30	0.5	108	90
	-4	1	-4	16	0	0	-.25	16	20
	5	2	10	25	9	18	.4	50	40
	8	4	32	64	12	48	.5	256	224
Total	15	10	56	141	31	96	1.15	430	374



Hints on calculation (Table 3.3):

$$\text{col. 3} = (\text{col. 1}) (\text{col. 2})$$

$$\text{col. 4} = (\text{col. 1})^2$$

$$\text{col. 5} = (\text{col. 1}) + 4$$

$$\text{col. 6} = (\text{col. 2}) (\text{col. 5})$$

$$\text{col. 7} = \frac{\text{col. 2}}{\text{col. 1}}$$

$$\text{col. 8} = (\text{col. 1}) (\text{col. 3}), \text{ or } (\text{col. 2}) (\text{col. 4})$$

$$\text{col. 9} = (\text{col. 3}) (\text{col. 1 minus 1})$$

$$(i) \quad \Sigma x = \text{Total of } x\text{-column} = 15$$

$$(ii) \quad \Sigma fx = \text{Total of } fx\text{-column} = 56$$

$$(iii) \quad \Sigma x^2 = \text{Total of } x^2\text{-column} = 141$$

$$(iv) \quad \Sigma f(x+4) = \text{Total of } f(x+4) \text{ column} = 96$$

$$(v) \quad \Sigma \left(\frac{f}{x} \right) = \text{Total of column with heading } \frac{f}{x} = 1.15$$

$$(vi) \quad (\Sigma x)^2 = (\text{Total of } x\text{-column})^2 = 15^2 = 225$$

$$(vii) \quad \frac{\Sigma fx}{\Sigma f} = \frac{\text{Total of } fx\text{-column}}{\text{Total of } f\text{-column}} = \frac{56}{10} = 5.6$$

$$(viii) \quad \Sigma fx^2 = \text{Total of } fx^2\text{-column} = 430$$

$$(ix) \quad \Sigma f(x-1) = \text{Total of column with heading } f(x-1) = 374$$



3 Rules for Σ notation

(Rule 1) If Σ appears before an expression containing two or more terms, then Σ may be placed before each term. (3.15.2)

Example 3.34

$$(i) \quad \Sigma (x_i + y_i) = \Sigma x_i + \Sigma y_i$$

$$(ii) \quad \Sigma (x_i - y_i) = \Sigma x_i - \Sigma y_i$$

$$(iii) \quad \Sigma (2f_i x_i - 3y_i + 8) = \Sigma 2f_i x_i - \Sigma 3y_i + \Sigma 8$$

$$(iv) \quad \Sigma (ax_i^2 + bx_i + c) = \Sigma ax_i^2 + \Sigma bx_i + \Sigma c$$

$$(v) \quad \Sigma f_i(x_i - 5) = \Sigma (f_i x_i - 5y_i) = \Sigma f_i x_i - \Sigma 5f_i$$

$$(vi) \quad \Sigma f_i(x_i - 5)^2 = \Sigma f_i x_i^2 - \Sigma 10f_i x_i + \Sigma 25f_i$$

$$\text{Solution } (i) \quad \Sigma (x_i + y_i) = (x_1 + y_1) + (x_2 + y_2) + \dots + (x_n + y_n)$$

$$= x_1 + y_1 + x_2 + y_2 + \dots + x_n + y_n$$

$$= (x_1 + x_2 + \dots + x_n) + (y_1 + y_2 + \dots + y_n)$$

$$= \Sigma x_i + \Sigma y_i$$

$$(ii) \quad \Sigma (x_i - y_i) = (x_1 - y_1) + (x_2 - y_2) + \dots + (x_n - y_n)$$

$$= x_1 - y_1 + x_2 - y_2 + \dots + x_n - y_n$$

$$= x_1 + x_2 + \dots + x_n - y_1 - y_2 - \dots - y_n$$

$$= (x_1 + x_2 + \dots + x_n) - (y_1 + y_2 + \dots + y_n)$$

$$= \Sigma x_i - \Sigma y_i$$





(Rule II) If Σ appears before a term containing a constant factor (i.e. which is independent of i), then the constant factor may come in front of Σ . (3.15.3)

Example 3.35 (i) $\Sigma cx_i = c \Sigma y_i$ Where c is a constant.

$$(ii) \Sigma 8(x_i + 3)^2 = 8 \Sigma (x_i + 3)^2.$$

$$\begin{aligned} \text{Solution (i)} \quad \Sigma cx_i &= cx_1 + cx_2 + \dots + cx_n \\ &= c(x_1 + x_2 + \dots + x_n) = c \Sigma x_i \end{aligned}$$



(Rule III) If Σ appears before a constant, then the constant is multiplied by the total number of terms included in the sum. (3.15.4)

$$\text{Example 3.36 (i)} \quad \sum_{i=1}^n c = cn.$$

$$(ii) \sum_8^8 13 \times 6 = 78$$

$$\begin{aligned} \text{Solution (i)} \quad \sum_{i=1}^n c &= c + c + \dots + c \text{ (one } c \text{ for each value of } i) \\ &= cn. \end{aligned}$$



Example 3.37 Show that

$$\Sigma f_i(x_i + 1)^4 = \Sigma f_i x_i^4 + 4 \Sigma f_i x_i^3 + 6 \Sigma f_i x_i^2 + 4 \Sigma f_i x_i + N,$$

where $N = \Sigma f_i$

$$\begin{aligned} \text{Solution} \quad \Sigma f_i(x_i + 1)^4 &= \Sigma f_i(x_i^4 + 4x_i^3 + 6x_i^2 + 4x_i + 1) \\ &= \Sigma f_i x_i^4 + 4 \Sigma f_i x_i^3 + 6 \Sigma f_i x_i^2 + 4 \Sigma f_i x_i + \Sigma f_i \\ &= \Sigma f_i x_i^4 + 4 \Sigma f_i x_i^3 + 6 \Sigma f_i x_i^2 + 4 \Sigma f_i x_i + \Sigma f_i && \text{(by Rule 1)} \\ &= \Sigma f_i x_i^4 + 4 \Sigma f_i x_i^3 + 6 \Sigma f_i x_i^2 + 4 \Sigma f_i x_i + N && \text{(by Rule II)} \end{aligned}$$



Example 3.38

$$\text{If } m_1 = \frac{1}{n} \sum_1^n x_i, m_2 = \frac{1}{n} \sum_2^{n+1} x_i \text{ and } m_3 = \frac{1}{n} \sum_3^{n+2} x_i, \text{ show that}$$

$$(i) m_2 = m_1 + \frac{x_{n+1} - x_1}{n}$$

$$(ii) m_3 = m_2 + \frac{x_{n+2} - x_2}{n} \quad \text{(C.U., M. Com. '63)}$$



Solution

$$\begin{aligned}
 \text{(i) Right hand side} &= m_1 + \frac{x_{n+1} - x_1}{n} \\
 &= \frac{1}{n} \sum_1^n x_i + \frac{1}{n} (x_{n+1} - x_1) \\
 &= \frac{1}{n} \left(\sum_1^n x_i + x_{n+1} - x_1 \right) \\
 &= \frac{1}{n} (x_1 + x_2 + \dots + x_n + x_{n+1} - x_1) \\
 &= \frac{1}{n} (x_2 + \dots + x_n + x_{n+1}) \\
 &= \frac{1}{n} \sum_2^{n+1} x_i = m_2 = \text{Left hand side.} \quad (\text{shown})
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) Right hand side} &= m_2 + \frac{x_{n+2} - x_2}{n} \\
 &= \frac{1}{n} \sum_2^{n+1} x_i + \frac{1}{n} (x_{n+2} - x_2) \\
 &= \frac{1}{n} \left(\sum_2^{n+1} x_i + x_{n+2} - x_2 \right) \\
 &= \frac{1}{n} (x_2 + x_3 + \dots + x_{n+1} + x_{n+2} - x_2) \\
 &= \frac{1}{n} (x_3 + \dots + x_{n+1} + x_{n+2}) \\
 &= \frac{1}{n} \sum_3^{n+2} x_i = m_3 = \text{Left hand side.} \quad (\text{shown})
 \end{aligned}$$



3.16 SIMPLE INTERPOLATION

Interpolation denotes the process of finding the intermediate value of a function from a set of given values of that function (Chapter 10). For example, suppose the following pairs of values of x and y are known:

x	y
4	17
7	32



What is the value of y when x has any specified value lying between the given values, say $x = 5.2$? This is a problem of interpolation. In *simple interpolation*, we assume that the values of y change uniformly with those of x . When x has increased by 3 (from 4 to 7) y has increased by 15 (from 17 to 32), i.e. y increases by 5, as x increases by unity. When $x = 5.2$, the increase in x is $5.2 - 4 = 1.2$, for which the increase in y should be $1.2 \times 5 = 6$. The required value of y is, therefore, $17 + 6 = 23$.

The following alternative method will be found very convenient. Arrange the problem in the tabular form shown below:

x		y
D_1	$\left[\begin{array}{c} 4 \\ 5.2 \rightarrow \\ 7 \end{array} \right]$	$\left[\begin{array}{c} 17 \\ \leftarrow Y \\ 32 \end{array} \right] D_2$
	$d_1 \left\{ \begin{array}{c} 4 \\ 5.2 \rightarrow \\ 7 \end{array} \right\}$	$d_2 \left\{ \begin{array}{c} 17 \\ \leftarrow Y \\ 32 \end{array} \right\}$

The partial differences in the values of x and y are

$$d_1 = 5.2 - 4, \quad d_2 = Y - 17$$

and the total differences are

$$D_1 = 7 - 4, \quad D_2 = 32 - 17$$

Since we assume a uniform rate of change in the given interval, the partial differences must be proportional to the corresponding total differences.

$$\frac{\text{Partial difference in } x}{\text{Total difference in } x} = \frac{\text{Partial difference in } y}{\text{Total difference in } y}$$

$$\text{i.e.} \quad \frac{d_1}{D_1} = \frac{d_2}{D_2} \quad (3.16.1)$$

Substituting the values,

$$\frac{5.2 - 4}{7 - 4} = \frac{Y - 17}{32 - 17}$$

$$\text{or,} \quad \frac{1.2}{3} = \frac{Y - 17}{15}$$

$$\text{or,} \quad Y - 17 = \frac{1.2}{3} \times 15 = 1.2 \times 5 = 6$$

$$\therefore Y = 17 + 6 = 23.$$

The relation (3.16.1) may also be used to find the unknown value of x corresponding to any specified y , intermediate between the given values.

Example 3.39 From the following table, find (i) the value of y , when $x = 19.3$, (ii) the value of x , when $y = 12.8$, using simple interpolation:

x	12	17	22	27	32
y	2.8	5.1	8.8	12.0	15.2

Solution (i) $x = 19.3$ lies between the tabulated values $x = 17$ and $x = 22$; hence, the corresponding value of y , say Y , lies between $y = 5.1$ and $y = 8.8$, as indicated by arrow marks. (ii) Similarly, $y = 12.8$ lies between the given values $y = 12.0$ and $y = 15.2$, and hence the corresponding value of x , say X , lies between $x = 27$ and $x = 32$, as indicated. Now using (3.16.1),



x	y
12	2.8
17	5.1
19.3 →	← Y
22	8.8
27	12.0
X →	← 12.8
32	15.2

$$(i) \frac{19.3 - 17}{22 - 17} = \frac{Y - 5.1}{8.8 - 5.1}$$

$$\text{or, } \frac{2.3}{5} = \frac{Y - 5.1}{3.7}$$

$$\therefore Y - 5.1 = \frac{2.3}{5} \times 3.7 = 1.702$$

$$\therefore Y = 5.1 + 1.702 = 6.802$$

$$(ii) \frac{X - 27}{32 - 27} = \frac{12.8 - 12.0}{15.2 - 12.0}$$

Solving this we get $X = 28.25$

Ans. (i) 6.802, (ii) 28.25.



4 FREQUENCY DISTRIBUTION

4.1 OBSERVATION, FREQUENCY

Suppose that the daily number of car accidents in a certain city recorded over a period of 30 days are as shown below:

Table 4.1 Daily Number of Car Accidents

4	6	5	4	4	6
3	5	4	5	3	5
4	4	5	6	4	3
5	5	3	3	5	4
7	6	5	5	4	5

The variable observed here is “the daily number of car accidents”, and the data obtained are known as *observations* or *observed values* or *measurements*. If n observations on a variable x are available, they are usually denoted by

$$x_1, x_2, \dots, x_n$$

x_1 denotes the first observation on x , x_2 denotes the second observation, etc. and finally x_n denotes the n th observation on x . In general, x_i denotes the i th observation on x ($i = 1, 2, \dots, n$). For example, in Table 4.1, $x_1 = 4$, $x_2 = 3$, $x_3 = 4$, ..., $x_{30} = 5$. The 30 observations are not all different; some of them are repeated. The distinct observations are known as *values of the variable*.

Frequency of a value of the variable is the number of times it occurs in a given series of observations. In Table 4.1, 3 occurs 5 times, 4 occurs 9 times, 5 occurs 11 times, 6 occurs 4 times, and 7 occurs once. Hence, the frequencies of the values 3, 4, 5, 6, 7 are respectively 5, 9, 11, 4, 1.

4.2 SIMPLE SERIES (OR UNGROUPED DATA) AND FREQUENCY DISTRIBUTION

‘Simple series’ or ‘ungrouped data’ are a series of observations recorded without any definite systematic arrangement. e.g. Table 4.1. If however, a large number of observations are available in the form of a simple series, the mind cannot properly



grasp their significance. Some method of condensation of data is therefore necessary. If the order in which the observations have been recorded is immaterial, one way of doing this is to write down the values of the variable in order of magnitude and show the corresponding frequencies side by side in a tabular form. Such a table is called *Frequency Table*, or more usually, *Frequency Distribution* of the variable.

'*Frequency Distribution*' is a statistical table which shows the values of the variable arranged in order of magnitude, either individually or in groups, and also the corresponding frequencies side by side. There are two types of frequency distributions:

- (a) Simple frequency distribution,
- (b) Grouped frequency distribution.

Simple frequency distribution shows the values of the variable individually (Table 4.2), whereas *grouped frequency distribution* (Table 4.3) shows the values of the variable in groups of intervals.

Table 4.2 Simple Frequency Distribution

<i>Daily Number of Car Accidents</i>	<i>Frequency (No. of days)</i>
3	5
4	9
5	11
6	4
7	1
Total	30

Consider the following data showing the ages (in years) of 200 unemployed persons who registered their names in an Employment Exchange on a certain day.

<i>Ages (in years) of 200 Unemployed Persons</i>									
33	26	33	30	24	25	28	22	17	30
26	17	29	22	18	40	20	33	21	25
25	20	21	29	27	26	19	37	32	21
20	40	20	20	27	19	20	20	29	20
21	49	34	26	29	31	25	33	23	16
21	20	23	29	23	17	23	27	43	22
24	26	29	29	20	17	32	23	24	25
36	19	28	26	31	52	25	57	16	26
27	19	24	19	39	23	24	25	17	19
31	18	26	23	19	30	20	19	50	40
22	27	17	22	43	33	22	17	21	25
24	46	27	20	26	20	21	24	22	21
25	22	24	32	24	20	22	15	34	22
22	20	37	19	24	21	28	27	19	31
22	24	17	29	24	17	23	21	21	31
31	18	22	19	33	25	21	25	23	21
24	18	15	18	24	19	22	22	16	21
17	18	25	31	26	29	24	31	23	18
31	24	23	21	24	18	21	19	30	24
24	22	30	19	25	21	21	25	45	24



The first step towards summarisation of data is to arrange the figures in the form of a simple frequency distribution as follows:

Age (yrs.)	Frequency	Age (yrs.)	Frequency	Age (yrs.)	Frequency
15	2	26	10	39	1
16	3	27	7	40	3
17	10	28	3	43	2
18	8	29	9	45	1
19	14	30	5	46	1
20	15	31	8	49	1
21	19	32	3	50	1
22	16	33	6	52	1
23	11	34	2	57	1
24	20	36	1		
25	14	37	2	Total	200

Although this table gives a somewhat clear picture, there are still too many figures to assimilate, and the age distribution of unemployed persons in general is not very prominent. The usual procedure in such cases is to group the observations into a number of intervals and show the frequency of occurrence of all values included therein, separately for each interval (Table 4.3). Such a frequency table is known as *Grouped Frequency Distribution* (Note that the frequency of each individual observation cannot be obtained from a grouped frequency distribution).

Table 4.3 Grouped Frequency Distribution

Age in Years	Frequency (No. of Persons)
15–19	37
20–24	81
25–29	43
30–34	24
35–44	9
45–59	6
Total	200

An inspection of the frequency distribution gives a quick idea about the central value (average) in the series, and shows how the observations vary ('dispersion') around the average. The histogram and frequency polygon drawn from a frequency distribution also throw light on the shape ('skewness' and 'kurtosis') of the frequency curve. Again, all recorded data may be thought of as a sample coming from a certain 'population'. The frequency distribution may be used to approximate the 'probability distribution' (Chapter 12) in the population, which is fundamental in solving statistical problems. Thus, the important uses of a frequency distribution are: (1) Condensation and summarization of a large mass of data; (2) Visualizing the important characteristics of data, like average, dispersion, skewness and kurtosis; (3) Approximating the nature of probability distribution in the population.

- Class interval, or Class;*
- Class frequency, and Total frequency;*
- Class limits—Lower class limit, Upper class limit;*
- Class boundaries—Lower class boundary, Upper class boundary;*
- Class mark, or Mid-value, or Mid-point of class interval;*
- Width, or Size of class interval;*
- Frequency density.*

(1) Class interval (or Class)

When one end of a class is not specified, the class is called an *open-end class*. A frequency distribution may have either one or two open-end classes (see Tables 4.13 and 4.14). The necessity of open-end classes arises when there are relatively few observations which are far apart from the rest. In such a case, it is not considered worth-while to show several classes with zero frequencies (called *empty class*) before reaching a class with a very small frequency.

The number of observations falling within a class is called its *Class frequency*, or simply *Frequency*. The sum of all the class frequencies is called *Total frequency*. In Table 4.4, column (2), the class frequencies are 37, 81, etc. and the total frequency is 200. Total frequency shows the total number of observations considered in the frequency distribution.

[illegible]



(3) Class limits

All recorded data or observations are discrete in character. For a discrete variable, the values themselves are isolated, e.g. records regarding the number of workers employed in a factory will show data such as 226, 105, 873, 66, 484, etc. (no fractional numbers are possible). Although a continuous variable can theoretically take any value, all observations are rounded to a certain unit for convenience. For example, if it is decided to record the height of persons to the nearest inch, we get observations like 58, 67, 63, etc.; if the decision is to record them to the nearest tenths of an inch we may get 57.8, 67.5, 63.0, etc. In any case, all observations will be uniformly rounded, either all whole numbers, or all showing upto one place of decimal, and so on.

When a grouped frequency distribution is constructed (Section 4.4) from these collected data, the values of the variable are shown in several classes. For determining the class frequencies it is necessary that these classes are mutually exclusive, i.e. be such that any observation is contained in only one class; for example 54–56, 57–59, 60–62, etc. (not like 54–56, 56–58, 58–60, etc.), or 54.1–56.0, 56.1–58.0, 58.1–60.0, etc. (not like 54.0–56.0, 56.0–58.0, 58.0–60.0 etc.). Note that classes will be written to the same degree of precision as the original data.

In the construction of a grouped frequency distribution, the class intervals must therefore be defined by pairs of numbers such that the upper end of one class does not coincide with the lower end of the immediately-following class. *The two numbers used to specify the limits of a class interval for the purpose of tallying the original observations into the various classes, are called 'Class Limits'.* The smaller of the pair is known as *Lower Class Limit* and the larger as *Upper Class Limit* with reference to the particular class. See Table 4.4, cols. (3) and (4). Class limits are used solely for the construction of grouped frequency distribution from ungrouped data.

(4) Class boundaries

When measurements are taken on a continuous variable, all data are recorded nearest to a certain unit. Thus, if ages are recorded to the nearest whole number of years, any age between 14.5 years and 15.5 years is recorded as 15 years. Similarly, '19 years' denotes an age between 18.5 years and 19.5 years. Hence, the class interval 15–19 actually includes all ages between 14.5 and 19.5 years. *These most extreme values which would ever be included in a class interval are called 'Class Boundaries'.* Class boundaries are, in fact, the real limits of a class interval. The lower extreme point is called *Lower Class Boundary*, and the upper extreme point is called *Upper Class Boundary* with reference to any particular class. See Table 4.4, columns (5) and (6).

Class boundaries may be calculated from class limits by applying the following rule:

$$\text{Lower class boundary} = \text{Lower class limit} - \frac{1}{2}d$$

$$\text{Upper class boundary} = \text{Upper class limit} + \frac{1}{2}d$$

where d is the common difference between the upper class limit of any class interval and the lower class limit of the next class interval. In fact, if observations are recorded to the nearest unit, $d = 1$; if to the nearest tenth of a unit, $d = 0.1$, etc. In Table 4.4, $d = 1$. Excepting the first and the last, all other class boundaries lie midway between the upper limit of a class and the lower limit of the next higher class.



Note that the upper boundary of any class coincides with the lower boundary of the next class; but the upper limit of any class is different from the lower limit of the next class. This fact may be utilized in deciding whether the pairs of numbers defining the class intervals in a given frequency distribution are really class limits or class boundaries. Class limits are used only for the construction of a grouped frequency distribution, but in all statistical calculations and diagrams involving end-points of classes (e.g. median, mode, quartiles, and histogram, ogive etc.), class boundaries are used.

(5) Class mark (or Mid-value or Mid-point)

The value exactly at the middle of a class interval is called *Class mark or Mid-value*. It lies half-way between the class limits or between the class boundaries. See Table 4.4, column (7).

$$\begin{aligned}\text{Class mark} &= (\text{Lower class limit} + \text{Upper class limit}) \div 2 \\ &= (\text{Lower class boundary} + \text{Upper class boundary}) \div 2\end{aligned}$$

Class mark is used as a representative value of the class interval, for the calculation of mean, standard deviation, mean deviation, etc.

(6) Width (or Size) of class

Width or Size of a class is the difference between the lower and the upper class boundaries (*not* class limits). See Table 4.4, column (8).

$$\text{Width of class} = \text{Upper class boundary} - \text{Lower class boundary}.$$

In the construction of a frequency distribution, it is generally preferable to have classes of equal width. This simplifies the calculation of some statistical measures (e.g. mean, standard deviation) by short-cut methods. Unequal width of classes is resorted to when some of the observations are few and far away from the rest. The use of equal width in such cases may result in some '*empty classes*', i.e. classes with zero frequency.

If all the classes are of equal width, this common width must be equal to the difference between (i) the successive lower class limits/boundaries, (ii) the successive upper class limits/boundaries, and (iii) the successive class marks. Therefore, in order to see, whether all the classes have a common width, the following test may be applied: Calculate mentally the differences between the successive lower class limits/boundaries, and also the successive upper class limits/boundaries. If these differences are all equal, the classes have a common width, which is equal to this common difference. If not, all classes do not have the same width.

If no confusion arises, sometimes 'class interval' is itself used to denote its width, e.g. 'equal class intervals', or 'class interval 10 paise' (see Example 4 : 6).

(7) Frequency density

Frequency density of a class is its frequency per unit width. It shows the concentration of frequency in a class and is given by the formula

$$\text{Frequency density} = \frac{\text{Class frequency}}{\text{Width of the class}}$$

See Table 4.4, column (9). Frequency density is used in drawing histogram, when the classes are of unequal width.

Example 4.1 (a) Simple Frequency Distribution:

Table 4.5 and 4.6 illustrate two simple frequency distributions, the former relating to a discrete variable and the latter to a continuous variable.



Table 4.5 Showing Number of Telephone Calls Received in 245 One-minute Intervals

<i>No. of Calls</i>	<i>Frequency</i>
0	14
1	21
2	25
3	43
4	51
5	40
6	39
7	12
<i>Total</i>	245

(Source: I.C.W.A., June '74)

Table 4.6 Showing Heights of School Children

<i>Height (inch)</i>	<i>No. of Children</i>
60	2
61	0
62	15
63	29
64	25
65	12
66	10
67	4
68	3
<i>Total</i>	100

(Source: I.C.W.A., Jan. '73)

(b) Grouped Frequency Distribution:

(i) Tables 4.7 and 4.8 relate to discrete variables and the classes have been defined by using class limits.

Table 4.7 Showing Wages of 65 Employees

<i>Wages (Rs)</i>	<i>No. of Employees</i>
50.00 – 59.99	8
60.00 – 69.99	10
70.00 – 79.99	16
80.00 – 89.99	14
90.00 – 99.99	10
100.00 – 109.99	5
110.00 – 119.99	2
<i>Total</i>	65

(Source: B.U., B.A. (Econ) '72)

**Table 4.8** Showing Marks Obtained in Statistics

Mark	No. of Candidates
30–39	2
40–49	3
50–59	11
60–69	20
70–79	32
80–89	25
90–99	7
Total	100

(Source: C.U., M.Com. '72)

(ii) Tables 4.9 and 4.10 relate to discrete variables, but the classes are defined by using class boundaries. Here, the number of values (if any) which are exactly equal to the class boundaries, is supposed to be small compared to the total frequency as to influence the pattern of the distribution to any appreciable extent. Table 4.9 also illustrates a frequency distribution with unequal width for class intervals.

Table 4.9 Showing Annual Sales of 213 Firms

Annual Sales (Rs '000)	No. of Firms
0–20	20
20–50	50
50–100	69
100–250	30
250–500	25
500–1000	19
Total	213

(Source: C.U., M.Com. '72)

Table 4.10 Showing Distribution of Daily Wages

Wages (Rs)	No. of Workers
2–3	15
3–4	30
4–5	44
5–6	60
6–7	30
7–8	14
8–9	7

(Source: C.A., Nov. '72)

(iii) Tables 4.11 and 4.12 relate to continuous variables, the classes having been defined by using class limits in the former and class boundaries in the latter.

**Table 4.11** Showing Weights of 100 School-children

<i>Weight (lbs.)</i>	<i>Frequency</i>
65–69	8
70–74	15
75–79	18
80–84	25
85–89	14
90–94	9
95–99	6
100–104	5
<i>Total</i>	<i>100</i>

(Source: I.C.W.A., June '74)

Table 4.12 Showing Hospitalisation Time

<i>Treatment Duration (days)</i>	<i>Number of Patients</i>
0–10	460
10–20	730
20–30	260
30–40	150
40–50	90
<i>Total</i>	<i>1690</i>

(Source: C.U., M.Com. '71)

(iv) Tables 4.13 and 4.14 illustrate grouped frequency distributions with open-end classes—either end or both ends may be open.

Table 4.13 Showing Income of Employees in an Industry

<i>Income (Rs)</i>	<i>Frequency</i>
0–50	90
50–100	150
100–150	100
150–200	80
200–250	70
250–over	10
<i>Total</i>	<i>500</i>

(Source: I.C.W.A., June '73)

**Table 4.14** Showing Income Distribution

Monthly Income (Rs)	Frequency
<i>Upto 100</i>	9
<i>101 – 150</i>	51
<i>151 – 200</i>	120
<i>201 – 300</i>	240
<i>301 – 500</i>	136
<i>501 – 750</i>	33
<i>751 – 1000</i>	9
<i>Above 1000</i>	2
<i>Total</i>	600

(Source: I.C.W.A., Jan. '72)

(v) Tables 4.15, 4.16 and 4.17 show class intervals in which the upper ends have not been stated explicitly (they should not be confused with open-end classes). In Tables 4.15 and 4.16, the class intervals are respectively 10–19, 20–29, ..., 70–79 and 25–26, 26–27, ..., 35–36. In Table 4.17, the class intervals should be taken to imply '57.5 and above, but below 60.0', '60.0 and above, but below 62.5' and so on.

Table 4.15

Age (years)	Frequency
<i>10 – 19</i>	3
<i>20 –</i>	61
<i>30 –</i>	223
<i>40 –</i>	137
<i>50 –</i>	53
<i>60 –</i>	19
<i>70 –</i>	4
	500

(Source: I.C.W.A., July '67)

Table 4.16

Weekly Earnings (Rs)	No. of Persons
<i>25–</i>	25
<i>26–</i>	70
<i>27–</i>	210
<i>28–</i>	275
<i>29–</i>	430
<i>30–</i>	550
<i>31–</i>	340
<i>32–</i>	130
<i>33–</i>	90
<i>34–</i>	55
<i>35–36</i>	25
	2200

(Source: I.C.W.A., Dec. '73)

**Table 4.17**

Heights (inch)	Number
57.5 –	6
60.0 –	26
62.5 –	190
65.0 –	281
67.5 –	412
70.0 –	127
72.5 –	38
	1080

(Source: I.C.W.A., July '65)

(vi) Table 4.18 shows 4 different forms of class intervals, in each of which the class boundaries are same, viz. as in case B. Note that an observation 25, if any, is included in the 2nd class interval of C, but included in the 1st class interval of D. In cases A and B, the true significance of class intervals is generally understood from the text of the collected data. However, no difficulty arises in statistical calculations, since class boundaries only will be used for the purpose.

Table 4.18 Different Forms of Class Intervals

A	B		C					D			
20–	20–25	20	and	above	but	below	25	Above	20	but	upto 25
25–	25–30	25	"	"	"	"	30	"	25	"	" 30
30–	30–35	30	"	"	"	"	35	"	30	"	" 35
35–	35–40	35	"	"	"	"	40	"	35	"	" 40



Example 4.2 Find the class boundaries, if the class limits are $9-10\frac{3}{4}$; $11-12\frac{3}{4}$; $13-14\frac{3}{4}$; ... the measurements being made nearest to $\frac{1}{8}$ of an unit. Explain why you should consider the class boundaries as the real limits for data of continuous type.

[C.U., B.A. (Econ) '68]

Solution (There is a mistake in the question). The measurements have been made nearest to $\frac{1}{4}$ of a unit, and not $\frac{1}{8}$, as stated. Hence, $d = \frac{1}{4}$. This may also be seen from the following:

$$d = 11 - 10\frac{3}{4} = 13 - 12\frac{3}{4} = \dots = \frac{1}{4} \text{ in each case.}$$

So $d/2 = \frac{1}{8}$. Subtracting $d/2$ from each of the lower class limits and adding $d/2$ to each of the upper class limits, the class boundaries are $\left(9 - \frac{1}{8}\right)$ and $\left(10\frac{3}{4} + \frac{1}{8}\right)$; $\left(11 - \frac{1}{8}\right)$ and



$\left(12\frac{3}{4} + \frac{1}{8}\right); \left(13 - \frac{1}{8}\right)$ and $\left(14\frac{3}{4} + \frac{1}{8}\right)$ etc. The required class boundaries are, therefore.

$$8\frac{7}{8} - 10\frac{7}{8}; 10\frac{7}{8} - 12\frac{7}{8}; 12\frac{7}{8} - 14\frac{7}{8}; \dots$$

When observations are taken in respect of a continuous variable, all measurements are made nearest to a certain unit, e.g. heights measured to the nearest inch. Hence, any height between 61.5 and 62.5 inches is recorded as 62 (to the nearest inch); similarly any height between 63.5 and 64.5 is recorded as 64. If 62–64 be a class interval of the distribution of heights (inches), then the class interval actually includes all heights between 61.5 and 64.5 inches, i.e. the class boundaries. As such, the class boundaries should be considered as the real limits for the class interval.



Example 4.3 Find the class boundaries in each of the following cases, when the class intervals given are:

- (a) 20–25; 25–30; 30–35; ...
- (b) 20–24; 25–29; 30–34; ...
- (c) 20–24.9; 25–29.9; 30–34.9; ...
- (d) 20.00–24.99; 25.00–29.99; 30.00–34.99; ...
- (e) 20.01–25.00; 25.01–30.00; 30.01–35.00; ...
- (f) 20– ; 25– ; 30– ; ... ; 55–60.
- (g) 20– ; 25– ; 30– ; ... ; 55–59.
- (h) 20– ; 25– ; 30– ; ... ; 55–.
- (i) 20– ; 25– ; 30– ; ... ; 55–. (Measurements are given to the nearest unit).
- (j) 20– ; 25– ; 30– ; ... ; 55–. (Measurements are given to the nearest 1/8 of a unit).
- (k) Above 20 but not exceeding 25;
 " 25; " " " 30;
 " 30 " " " 35; etc.
- (l) 20 and above but below 25;
 25 " " " " 30;
 30 " " " " 35; etc.
- (m) 20–24; 25–29; 30–34; ... (Age last birthday)

Solution (a), (f), (h), (k), (l), (m):–

20–25; 25–30; 30–35; ...

(b), (g), (i):

19.5–24.5; 24.5–29.5; 29.5–34.5; ...

(c) 19.95–24.95; 24.95–29.95; 29.95–34.95; ...

(d) 19.995–24.995; 24.995–29.995; 29.995–34.995; ...

(e) 20.005–25.005; 25.005–30.005; 30.005–35.005; ...

(j) $19\frac{15}{16} - 24\frac{15}{16}; 24\frac{15}{16} - 29\frac{15}{16}; 29\frac{15}{16} - 34\frac{15}{16}; \dots$





4.4 CONSTRUCTION OF FREQUENCY DISTRIBUTION

Example 4.4 Describe the various steps in the construction of a frequency distribution from unclassified data. [C.U., M.Com. '69, '74; B.A.(Econ) '72, '74]

Solution The steps in the construction of a frequency distribution from ungrouped data are as follows:—

1. Find the largest and the smallest observations in the given data, and then calculate the *range*, i.e. the difference between them.
2. Divide the range into a suitable number of class intervals, by means of class limits. The number of class intervals should not ordinarily be less than 5 nor more than 15, depending on the number of observations available. The more the number of observations, the larger will be the number of class intervals. Class limits should be so chosen that most of the observations lie near the class marks. The class intervals should preferably be of the same width. In special circumstances, class intervals of unequal width may also be used.
3. The number of observations falling in each class interval, i.e. class frequency, is determined by *tally marks* (see Table 4.19).
4. A table is now prepared showing the class intervals in the first column and the corresponding class frequencies in the second column. This is the required frequency distribution.



Example 4.5 Discuss the problems in the construction of a frequency distribution from raw data, with particular reference to the choice of number of classes and class limits. [I.C.W.A., Jan '72; C.A., May '72; C.U., M.Com. '71; B.A. (Econ)'64]

Solution In constructing a grouped frequency distribution, the two main problems involved are: (1) choice of the number of classes, and (b) choice of the class limits. Although there are no hard and fast rules in this regard, it is generally agreed that the number of classes should neither be very large (because, in that case the frequency distribution will be very lengthy), nor very small (because, in that case the true pattern of the distribution of observations will be obscured). As a working rule, this number should lie between 5 and 15, depending on the number of observations available. When the number of observations is small, some authors suggest the use of Sturges' formula

$$n = 1 + 3.3 (\log N)$$

where n denotes the number of classes and N the total frequency. In practice, the number of classes is fixed from other considerations depending on the particular situation.

The second problem, viz. the choice of class limits is to some extent dependent on the number of classes. For this purpose, we first find the maximum and the minimum of the observations and calculate the 'range', i.e. the difference between these two. If we like to have classes of equal width, then the approximate width of the classes may be obtained on dividing the range by the number of such classes. The lower limit of the first class may not coincide with the minimum observation. In most cases we prefer to have classes of width 5 or its multiple, e.g. 15–19, 20–24 etc., or 60.01–80.00, 80.01–100.00, 100.01–120.00 etc., one of the class limits being preferably a multiple of 5. Again, sometimes it is found that some particular values of the variable occur more frequently than others. In such cases, the class limits should be so chosen that the highly frequent values lie near the class marks. This will minimise the errors of grouping (see Section 7.6) in the calculation of various statistical measures like standard deviation, moments, etc. when class intervals are represented by their class marks.





Example 4.6 You are given below the wages paid to some workers in a small factory. Form a frequency distribution with class interval 10 paise:

Wages in Rs								
1.10	1.13	1.44	1.44	1.27	1.17	1.98	1.36	1.30
1.27	1.24	1.73	1.51	1.12	1.42	1.03	1.58	1.46
1.40	1.21	1.62	1.31	1.55	1.33	1.04	1.48	1.20
1.60	1.70	1.09	1.49	1.86	1.95	1.51	1.82	1.42
1.29	1.54	1.38	1.87	1.41	1.77	1.15	1.57	1.07
1.65	1.36	1.67	1.41	1.55	1.22	1.69	1.67	1.34
1.45	1.39	1.25	1.26	1.75	1.57	1.53	1.37	1.59
1.19	1.52	1.56	1.32	1.81	1.40	1.47	1.38	1.62
1.76	1.28	1.92	1.46	1.46	1.35	1.16	1.42	1.78
1.68	1.47	1.37	1.35	1.47	1.43	1.66	1.56	1.48

[C.A., May '67]

Solution From the given observations, we find that

Maximum value = 1.98

Minimum value = 1.03

The width of the classes is to be 10 paise (Note that in the question 'class interval 10 paise' implies that the *width* of the classes is to be 10 paise). Let us take the class limits 1.01–1.10; 1.11–1.20; 1.21–1.30; ...; 1.91–2.00. These are shown in a *tally sheet* (Table 4.19) and the given observations are represented by *tally marks* one by one against the appropriate classes. The tally marks are shown in groups of five to facilitate counting, every fifth tally mark in a class interval being placed across the preceding four.

Table 4.19 Tally Sheet

Class Limits	Tally Marks	Frequency
1.01–1.10		5
1.11–1.20		7
1.21–1.30		10
1.31–1.40		15
1.41–1.50		18
1.51–1.60		14
1.61–1.70		9
1.71–1.80		5
1.81–1.90		4
1.91–2.00		3
Total		90



The required frequency distribution is shown below:

Table 4.20 Frequency Distribution of Wages

<i>Wages (Rs)</i>	<i>Frequency</i>
1.01 – 1.10	5
1.11 – 1.20	7
1.21 – 1.30	10
1.31 – 1.40	15
1.41 – 1.50	18
1.51 – 1.60	14
1.61 – 1.70	9
1.71 – 1.80	5
1.81 – 1.90	4
1.91 – 2.00	3
Total	90



Example 4.7 In a test run of 50 cars of the same model, the average mileage covered per gallon of fuel consumed by each car was as follows:

18.6	33.4	25.3	27.8	30.6	31.9	33.0	26.3	24.9	29.4
20.0	26.2	28.1	33.1	37.5	22.5	39.1	32.9	33.8	52.6
32.5	34.6	32.7	9.5	38.5	29.6	25.3	49.5	30.1	27.9
26.9	23.8	36.0	38.0	27.5	32.3	34.2	23.1	34.7	29.0
34.1	38.6	25.9	40.6	53.8	29.3	36.8	27.1	34.9	31.6

Arrange the data in a frequency table with six classes. Clearly indicate the class boundaries and the class marks. [B.U., B.A.(Econ) '73]

Solution [Note: (i) Since the given data are rounded to one decimal place, the *class limits* must also be shown in one decimal (see Table 4.21). Class limits in whole numbers like 9–16, 17–24, ..., or in two decimals like 9.01–16.50, 16.51–24.00, ... will not do here.

(ii) Unlike Example 4.6, the *width of classes* is not specified here. So the common width has to be determined from the data, keeping in view the total number of classes in the proposed frequency distribution.]

In the given data, *Maximum value* = 53.8 and *Minimum value* = 9.5, so that *Range* = 53.8 – 9.5 = 44.3. If we are to use 6 classes of equal width, the common width will be approximately $44.3 \div 6 = 7.38 = 7.5$ (suppose). Using the class limits 9.1–16.5, 16.6–24.0, ... 46.6–54.0, and by tally marking, the frequency table has been obtained as shown below:

Table 4.21 Frequency Table showing Average Mileage Covered per Gallon of Fuel by 50 Cars

<i>Class Limits</i>	<i>Frequency</i>	<i>Class Boundaries</i>	<i>Class Mark</i>
9.1–16.5	1	9.05–16.55	12.8
16.6–24.0	5	16.55–24.05	20.3
24.1–31.5	18	24.05–31.55	27.8
31.6–39.0	21	31.55–39.05	35.3
39.1–46.5	2	39.05–46.55	42.8
46.6–54.0	3	46.55–54.05	50.3
Total	50	–	–





4.5 CUMULATIVE FREQUENCY DISTRIBUTION

In statistical investigations, sometimes we are interested in the number of observations smaller than (or greater than) a given value. In such cases, or chief concern is the accumulated frequency upto (or above) some value of the variable. This accumulated frequency is known as 'cumulative frequency'.

Cumulative Frequency corresponding to a specified value of the variable may be defined as the number of observations smaller than (or greater than) that value.

The number of observations 'upto' a given value is called *less-than cumulative frequency*; and the number of observations 'greater than' a value is called the *more-than cumulative frequency*. 'Cumulative frequency' only refers to the less-than type. When a frequency distribution is given, the cumulative totals of frequencies give the cumulative frequencies. When a grouped frequency distribution relates to a variable of the continuous type, the cumulative frequencies calculated therefrom must be shown against the class boundary points (i.e. end-points of classes). Cumulative frequency expressed as a percentage of total frequency, is known as *Cumulative Percentage*.

A table showing the cumulative frequencies against values of the variable systematically arranged in increasing (or decreasing) order is known as Cumulative Frequency Distribution. If cumulative percentages are shown, instead of cumulative frequencies, the table is called *Cumulative Percentage Distribution*.

Example 4.8 What do you mean by a cumulative frequency distribution? Point out its special advantages and uses. [C.A., May '67; I.C.W.A., Jan. '71]

Solution *Cumulative Frequency Distribution* is a statistical table which shows the values of the variable and the corresponding cumulative frequencies. It can be derived from a grouped frequency distribution by writing down the consecutive class boundary points and noting the number of observations less than (or more than) each class boundary point. A cumulative frequency distribution is of advantage in such studies as the number or percentage of observations less than (or more than) a specified value of the variable.

Uses—Cumulative frequency distribution is used (i) to find the number of observations less than (or more than) any given value; (ii) to find the number of observations falling between any two specified values of the variable; (iii) to find median, quartiles, deciles and percentiles (Section 5.18) graphically; i.e. in general, to find the value of the variable below (or above) which a specified number or percentage of the total frequency lies.



Example 4.9 Construct (a) the cumulative frequency distribution, and (b) the grouped frequency distribution, from the following data:

Value	Frequency
Less than 10	4
" " 20	16
" " 30	40
" " 40	76
" " 50	96
" " 60	112
" " 70	120
" " 80	125



Solution The cumulative frequency distribution shows the values of the variable (class boundaries) and the corresponding cumulative frequencies (less-than type):

Table 4.22 Cumulative Frequency Distribution

<i>Value</i>	<i>Cumulative Frequency (less-than)</i>
10	4
20	16
30	40
40	76
50	96
60	112
70	120
80	125

The grouped frequency distribution shows the values of the variable in class intervals and the corresponding class frequencies:

Table 4.23 Grouped Frequency Distribution

<i>Value</i>	<i>Frequency</i>
0 – 10	4
10 – 20	12
20 – 30	24
30 – 40	36
40 – 50	20
50 – 60	16
60 – 70	8
70 – 80	5
Total	125

Note: (i) Unless otherwise stated, or implied from the given data, the variable should be treated as continuous.

(ii) For a continuous variable, expressions like “less than 10”, “upto 10”, and “less than or equal to 10” are considered equivalent. However for a discrete variable, whereas the last two convey the same meaning, the first expression, which excludes the value ‘exactly 10’ should be treated as different.

(iii) For the value “less than 40” the ‘frequency’ is 76. However, if the words “less than” are omitted, we must write ‘cumulative frequency’ in place of ‘frequency’, e.g. for the value 40 the cumulative frequency is 76 (Table 4.22).

(iv) In the absence of any information to the contrary, the lower boundary of the first class interval in Table 4.23 is assumed to be 0; this makes all the classes of equal width, as is usual in practice.





Example 4.10 From the following data, calculate

- (a) “less-than” and “more than” cumulative frequencies,
- (b) cumulative frequency distribution, and
- (c) cumulative percentage distribution:

Age (years)	Frequency
15–19	37
20–24	81
25–29	43
30–34	24
35–44	9
45–59	6
Total	200

Solution [Working note: The class boundaries are 14.5–19.5, 19.5–24.5, etc. (Table 4.4). The class boundary points are therefore, 14.5, 19.5, 24.5, 29.5, 34.5, 44.5, 59.5. There is no observation below 14.5, and hence its cumulative frequency is 0; the frequency below 19.5 is 37; the frequency below 24.5 is $(37 + 81)$, i.e. 118; the frequency below 29.5 is $(37 + 81 + 43)$, i.e. 161; and so on. The “less-than” cumulative frequencies corresponding to the above class boundary points are therefore 0, 37, $(37 + 81)$, $(37 + 81 + 43)$, $(37 + 81 + 43 + 24)$, $(37 + 81 + 43 + 24 + 9)$, $(37 + 81 + 43 + 24 + 9 + 6)$, i.e. 0, 37, 118, 161, 185, 194, 200.

The “more-than”, cumulative frequencies are calculated by cumulating the class frequencies from the higher end of the values. Corresponding to the class boundary points in reverse order 59.5, 44.5, 34.5, 29.5, 24.5, 19.5, 14.5 these are respectively 0, 6, $(6 + 9)$, $(6 + 9 + 24)$, $(6 + 9 + 24 + 43)$, $(6 + 9 + 24 + 43 + 81)$, $(6 + 9 + 24 + 43 + 81 + 37)$, i.e. 0, 6, 15, 39, 82, 163, 200.

It should be noted that the “less-than” cumulative frequency corresponding to the highest class boundary point and the “more-than” cumulative frequency corresponding to the lowest class boundary point must be equal to the total frequency N . Also, the sum of the “less-than” and the “more-than” cumulative frequencies corresponding to any value is equal to the total frequency]

Table 4.24 Cumulative Frequencies

Age (years)	Cumulative Frequency	
	“less than”	“more-than”
14.5	0	$200 = N$
19.5	37	163
24.5	118	82
29.5	161	39
34.5	185	15
44.5	194	6
39.5	$200 = N$	0

The cumulative frequency distribution shows the values of the variable (class boundary points) and the corresponding cumulative frequencies (less-than type) side by side:

**Table 4.25** Cumulative Frequency Distribution

<i>Age (years)</i>	<i>Cumulative Frequency ("less-than")</i>
14.5	0
19.5	37
24.5	118
29.5	161
34.5	185
44.5	194
59.5	200 = N

The cumulative percentage distribution shows the values of the variable (class boundary points) and the corresponding cumulative percentages.

Table 4.26 Cumulative Percentage Distribution

<i>Age (years)</i>	<i>Cumulative Percentage ("less-than")</i>
14.5	0
19.5	18.5
24.5	59.0
29.5	80.5
34.5	92.5
44.5	97.0
59.5	100.0



Example 4.11 Estimate the number of candidates securing not more than 48 marks and number of candidates whose marks lie between 48 and 62:

<i>Marks Obtained</i>	<i>Candidates</i>
Not more than 45	447
Not more than 50	484
Not more than 55	505
Not more than 60	511
Not more than 65	514

[I.C.W.A., July '66]

Solution [Working note: The number of candidates securing not more than 48 marks is actually the cumulative frequency corresponding to 48. This cumulative frequency can be found by simple interpolation from a cumulative frequency distribution. Again, if we find the number of candidates securing not more than 62 marks, i.e. cumulative frequency corresponding to 62, then the difference between the cumulative frequencies corresponding to 48 and 62 would give the number of candidates whose marks lie between 48 and 62].

**Table 4.27** Cumulative Frequency Distribution

Marks	Cumulative Frequency ("less-than")
45	447
48→	←y
50	484
55	505
60	511
62→	←z
65	514

Now, 48 lies between the marks 45 and 50 in the first column, and hence its cumulative frequency (y) must lie between the corresponding cumulative frequencies 447 and 484 in the second column, as shown by the arrow mark. Applying formula (3.16.1) for simple interpolation (p. 77)

$$d_1/D_1 = d_2/D_2$$

i.e. $\frac{48 - 45}{50 - 45} = \frac{y - 447}{484 - 447}$

or, $\frac{3}{5} = \frac{y - 447}{37}$

or, $y - 447 = \frac{3}{5} \times 37 = 22.2$

or, $y = 447 + 22.2 = 469.2 = 469$ (approx.)

That is, the number of candidates securing not more than 48 marks is 469.

Again, 62 lies between 60 and 65 in the first column, and its cumulative frequency (z) must lie between the corresponding cumulative frequencies 511 and 514 in the second column. Applying simple interpolation, as before,

$$\frac{62 - 60}{65 - 60} = \frac{z - 511}{514 - 511}$$

or, $\frac{2}{5} = \frac{z - 511}{3}$

or, $z - 511 = \frac{2}{5} \times 3 = 1.2$

$\therefore z = 511 + 1.2 = 512.2 = 512$ (approx).

Therefore, the number of candidates securing marks between 48 and 62 is

$$z - y = 512 - 469 = 43.$$

Ans. 469; 43.

**Example 4.12** Calculate

- the number of cases between 112 and 134,
- Number less than 112,
- Number greater than 134, from the following:

Class limit	90–100	100–110	110–120	120–130	130–140	140–150	150–160
Frequency	16	22	45	60	50	24	10

[C.A., May '69]



Solution The required numbers can be obtained by applying simple interpolation in the cumulative frequency distribution.

- (a) Number of cases between 112 and 134
 = Number of cases less than 134 – Number of cases less than 112.
 = Cumulative frequency corresponding to 134 – Cumulative frequency corresponding to 112.
- (b) Number less than 112 = Cumulative frequency (less-than) corresponding to 112.
- (c) Number greater than 134 = Cumulative frequency (more-than) corresponding to 134.
 = Total frequency – Cumulative frequency (less-than) corresponding to 134.

Table 4.28 Cumulative Frequency Distribution

<i>Class Boundary</i>	<i>Cumulative Frequency (less-than)</i>
90	0
100	16
110	38
112→	←y
120	83
130	143
134→	←z
140	193
150	217
160	227 = N

To find the cumulative frequency y, we have

$$\frac{112 - 110}{120 - 110} = \frac{y - 38}{83 - 38}$$

or,
$$\frac{2}{10} = \frac{y - 38}{45}$$

or,
$$y - 38 = \frac{2}{10} \times 45 = 9$$

∴
$$y = 38 + 9 = 47$$

Again, to find the cumulative frequency z,

$$\frac{134 - 130}{140 - 130} = \frac{z - 143}{193 - 143}$$

or,
$$\frac{4}{10} = \frac{z - 143}{50}$$

or,
$$z - 143 = \frac{4}{10} \times 50 = 20$$

∴
$$z = 143 + 20 = 163$$

Therefore, (a) Number of cases between 112 and 134 = 163 – 47 = 116,

(b) Number less than 112 = 47,

(c) Number greater than 134 = 227 – 163 = 64.

Ans. (a) 116; (b) 47; (c) 64.





4.6 RELATIVE FREQUENCY DISTRIBUTION

Relative frequency denotes the class frequency expressed as a fraction of the total frequency.

$$\text{Relative frequency} = \frac{\text{Class frequency}}{\text{Total frequency}}$$

The sum of all the relative frequencies is equal to unity [see Table 4.4, column (10)]. When relative frequencies are shown against the corresponding classes, the table is known as *Relative Frequency Distribution*.

Table 4.29 Relative Frequency Distribution

(Data: Table 4.3)	
Age (years)	Relative Frequency
15–19	0.185
20–24	.405
25–29	.215
30–34	.120
35–44	.045
45–59	.030
Total	1.000

4.7 DIAGRAMMATIC REPRESENTATION OF FREQUENCY DISTRIBUTIONS

The diagrams commonly used to depict statistical data, given in the form of a frequency distribution, are:

1. Histogram,
2. Frequency polygon,
3. Ogive (or Cumulative frequency polygon)

(1) Histogram

Histogram is the most common form of diagrammatic representation of a grouped frequency distribution. It consists of a set of adjoining rectangles drawn on a horizontal base line, with *areas proportional to the class frequencies*. The width of rectangles, one for each class, extends over the class boundaries (*not* class limits) shown on the horizontal scale. When all classes have equal width, the heights of rectangles will be proportional to the class frequencies and it is then customary to take the heights numerically equal to the class frequencies (Fig. 4.1). If, however, the classes are of unequal width the rectangles will also be unequal width, and therefore the heights must be proportional to the frequency densities (Fig. 4.5). Because, then

$$\begin{aligned} \text{Area of each rectangle} &= \text{width} \times \text{height} \\ &= (\text{width of class}) \times (\text{frequency density}) \end{aligned}$$



$$\begin{aligned}
 &= (\text{width of class}) \times \frac{\text{class frequency}}{\text{width of class}} \\
 &= \text{class frequency.}
 \end{aligned}$$

Although vertical bar chart and histogram may appear some-what alike, the main point of distinction between them is that the consecutive rectangles in a histogram have no space in between, but the bar diagram must have equal spaces left between the consecutive rectangular bars. Also, the rectangles in a bar diagram must be of equal width, but those in a histogram are proportional to the widths of classes, and hence may also be unequal. The histogram is, after all, an *area diagram*, which emphasises the widths of rectangles between the class boundaries. But in a bar diagram only the height is all-important, the spacing and the width of bars being arbitrary.

Uses—(1) The series of rectangles in a histogram give a visual representation of the relative sizes of the various groups, and the entire distribution of total frequency among the different classes becomes at once visible. (2) The surface of the tops of rectangles also gives an idea of the nature of frequency curve (Section 4.8) for the population, from which the given data may be assumed to be a sample only. (3) The histogram may be used to find the mode graphically (Fig. 5.4).

(2) Frequency Polygon

Frequency Polygon is the graphical representation alternative to histogram and may be looked upon as derived from histogram by joining the mid-points of the tops of consecutive rectangles. It is generally used in cases when all the classes have a common width. In actual construction, therefore, the frequency polygon is obtained by joining the successive points whose abscissae represent the mid-values and ordinates represent the corresponding class frequencies. The two end-points are joined to the base line at the mid-values of the empty classes at each end of the frequency distribution. Thus the frequency polygon has the same area as the histogram, provided the width of all classes is the same. The frequency polygon is particularly useful in representing simple frequency distributions of a discrete variable (see Fig. 4.6).

Use—The frequency polygon gives us an approximate idea of the shape of frequency curve.

(3) Ogive (or Cumulative Frequency Polygon)

Ogive is the graphical representation of a cumulative frequency distribution, and hence is also called *Cumulative Frequency Polygon*. When cumulative frequencies are plotted against the corresponding class boundaries and the successive points are joined by straight lines, the line diagram obtained is known as *Ogive* or *Cumulative Frequency Polygon*. The ogive is of “less-than” or “more-than” type according as the cumulative frequencies used are of “less-than” or “more-than” type. The “less-than ogive” starts from the lowest class boundary on the horizontal axis and gradually rising upward ends at the highest class boundary corresponding to the cumulative frequency N , i.e. the total frequency. It looks like an elongated letter S (Fig. 4.2). The “more-than ogive” has the appearance of an elongated S , turned upside down (Fig. 5.3). Unequal widths of classes in the frequency distribution do not cause any difficulty in the construction of an ogive.



Uses—Since the ogive is only a diagrammatic representation of the cumulative frequency distribution, it has the same uses as those of the latter, but by graphical means. (1) The ogive is used to find the median, quartiles, deciles and percentiles, or the value of the variable such that its cumulative frequency is a specified number. (2) It is also useful in finding the cumulative frequency corresponding to a given value of the variable; and (3) to find the number of observations which are expected to lie between two given values.

Example 4.13 Draw histogram, frequency polygon and ogives (both “less-than” and “more-than” types) for the following frequency distribution:

Wages (Rs)	50–59	60–69	70–79	80–89	90–99	100–109	110–119
No. of employees	8	10	16	14	10	5	2

Solution [Note: Here the class intervals are defined by class limits and so we have to find the class boundaries for drawing the histogram. All the classes have the same width and therefore when drawing the histogram, heights of the rectangles may be represented by class frequencies. Also for drawing the ogives, we have to calculate both ‘less-than’ and ‘more-than’ cumulative frequencies.]

Table 4.30 Calculations for Drawing Histogram

Class Limits	Class Boundaries	Frequency
50–59	49.5–59.5	8
60–69	59.5–69.5	10
70–79	69.5–79.5	16
80–89	79.5–89.5	14
90–99	89.5–99.5	10
100–109	99.5–109.5	5
110–119	109.5–119.5	2

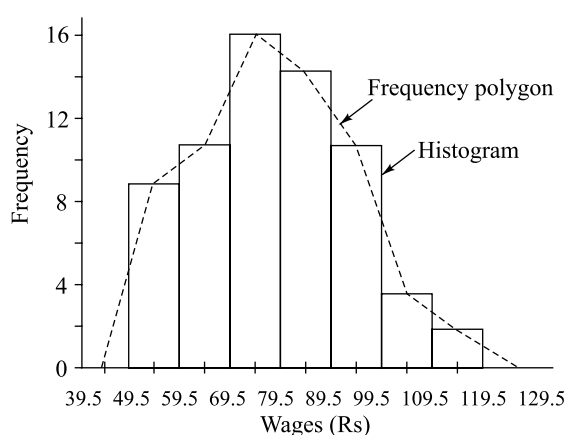
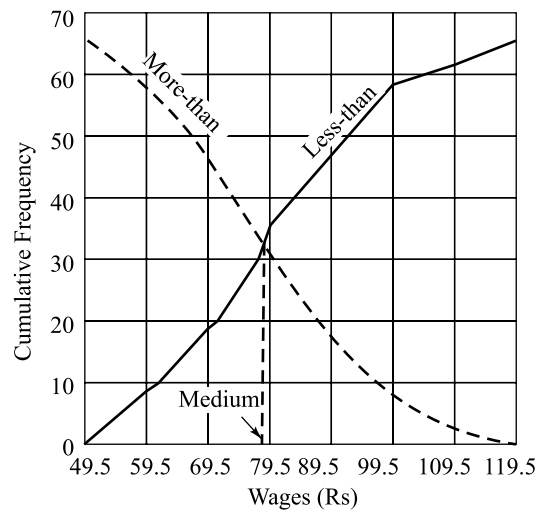


Fig. 4.1 Histogram and Frequency Polygon for Wage Distribution

**Table 4.31** Calculations for Drawing Ogives

Class Boundary	Cumulative Frequency	
	'less-than'	'more-than'
49.5	0	65 = N
59.5	8	57
69.5	18	47
79.5	34	31
89.5	48	17
99.5	58	7
109.5	63	2
119.5	65 = N	0

**Fig. 4.2** Ogives for Wage Distribution

Example 4.14 The following is an analysis of sales of 534 firms in an industry:

Values of Sales (£'000)	Number of Firms
0–500	3
500–1000	42
1000–1500	63
1500–2000	105
2000–2500	120
2500–3000	99
3000–3500	51
3500–4000	47
4000–4500	4



Draw the histogram and the less than cumulative frequency polygon of the above distribution. Find, graphically, the number of firms whose sales lie between £ 12,00,000 and £ 26,00,000. [C.U., B.A. (Econ.) '63]

Solution [Note: Here the class intervals have been defined by class boundaries, as evident from the fact that the upper end of any class coincides with the lower end of the next class. Therefore, it is not necessary to construct a table for calculation of class boundaries like Table 4.30].

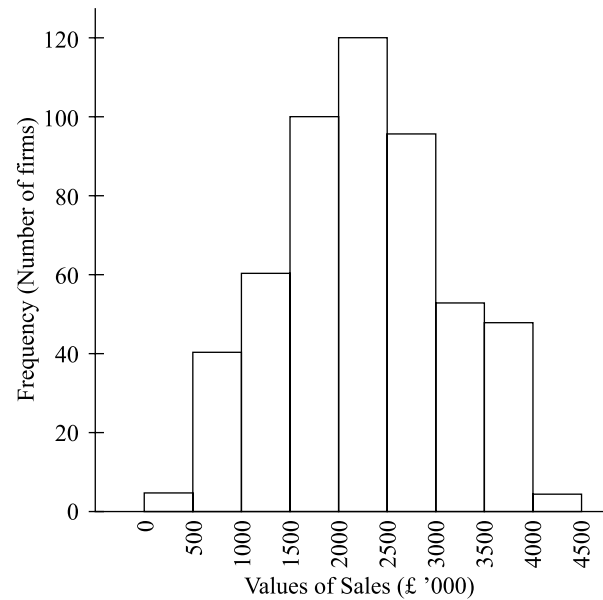


Fig. 4.3 Histogram

In order to draw the cumulative frequency polygon, we have to construct a cumulative frequency distribution.

Table 4.32 Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency (less-than)
0	0
500	3
1000	45
1500	108
2000	213
2500	333
3000	432
3500	483
4000	530
4500	534 = N



Taking the class boundaries along the horizontal axis and the less-than cumulative frequencies along the vertical axis, the data in Table 4.32 are plotted as points on a graph paper and the consecutive points are joined by straight lines. The line diagram obtained is the required less-than cumulative frequency polygon (Fig. 4.4).

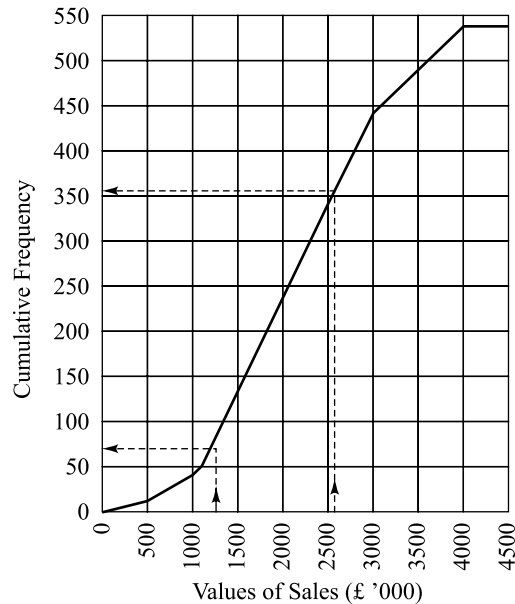


Fig. 4.4 Cumulative Frequency Polygon (less-than)

In order to obtain the number of firms whose sales lie between £12,00,000 and £26,00,000 we have to find graphically the cumulative frequencies corresponding to these values. For this purpose, two perpendiculars (broken lines in Fig. 4.4) are erected at the points 1200 and 2600 on the horizontal scale (since the units are £ '000). From the points of intersection of these perpendiculars with the cumulative frequency polygon, horizontal lines are then drawn meeting the vertical scale at two other points. Their values are read from the vertical scale, and found to be 70 and 353 approximately. This means that the cumulative frequencies corresponding to 1200 and 2600 units on the horizontal axis are respectively 70 and 353. The difference, viz. $353 - 70 = 283$, gives the frequency between 1200 and 2600 units. That is, the number of firms whose sales lie between £12,00,000 and £26,00,000 is 283.

Ans. 283



Example 4.15 Draw the histogram of the following frequency distribution and find the proportion of firms with annual sales greater than Rs 70,000:

Annual Sales (Rs '000) :	0-20	20-50	50-100	100-250	250-500	500-1000
No. of firms :	20	50	69	30	25	19

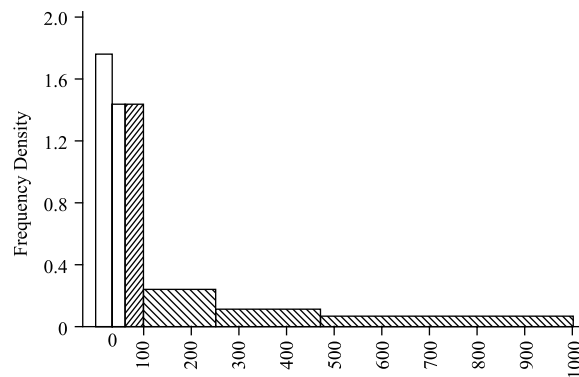
[C.U.,M.Com. '72]

Solution [Note: Here, class boundaries are given. The class intervals are also of unequal width, so that the heights of rectangles in the histogram must be proportional to *frequency densities*.]

**Table 4.33** Calculations for Drawing Histogram

Class Interval (Rs '000)	Frequency (<i>f</i>)	Width of Class (<i>w</i>)	Frequency Density ($f \div w$)
0–20	20	20	1.00
20–50	50	30	1.67
50–100	69	50	1.38
100–250	30	150	0.20
250–500	25	250	0.10
500–1000	19	500	0.04

The histogram is shown at Fig. 4.5.

**Fig. 4.5** Histogram for the Distribution of Sales

The proportion of firms with annual sales greater than Rs 70,000 is given by the proportion of area under the histogram which lies to the right of the vertical line at 70 units (Rs '000) on the horizontal axis. Assuming that frequency 69 in the class interval 50–100 is uniformly distributed in the whole interval, the proportional part of the area (i.e. frequency) between 70 and 100 units is

$$(100 \times 70) \times 1.38 = 41.4.$$

Therefore, the frequency above 70 units is

$$41.4 + 30 + 25 + 19 = 115.4.$$

Since the total frequency is 213,

$$\therefore \text{Required proportion of firms} = 115.4 \div 213 = 0.54$$



Example 4.16 In a sample survey of 60 workers' families living in a factory area, the following data were obtained, as regards the number of members in the families. Form a frequency distribution.

5	11	4	6	3	10	5	7	9	6
6	2	3	7	8	6	4	3	6	5
9	5	6	4	7	1	5	8	6	2
6	4	8	7	5	12	4	7	10	6
7	8	3	6	7	5	5	8	6	4
6	11	5	2	6	9	7	3	7	5

Draw a frequency polygon for the above distribution.

[I.C.W.A., July '69]



Solution

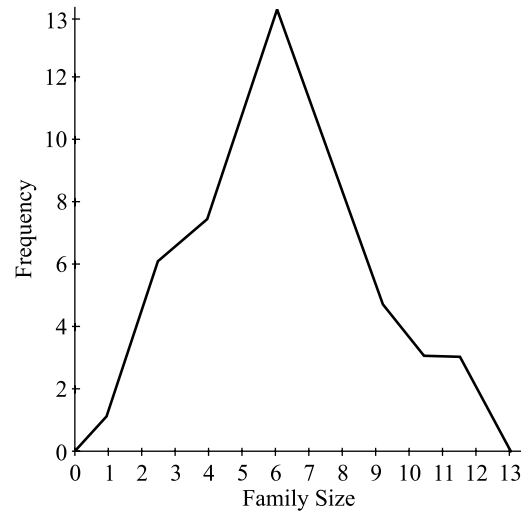


Fig. 4.6 Frequency Polygon for Distribution of Family Size

The frequency distribution, obtained by tally marking, is as follows:

Table 4.34 Frequency Distribution of Workers' Family Size

Family Size	1	2	3	4	5	6	7	8	9	10	11	12	Total
Frequency	1	3	5	6	10	13	9	5	3	2	2	1	60

The frequency polygon for the above distribution is shown at Fig. 4.6.



Example 4.17 Draw a suitable diagram to represent the frequency distribution of the daily number of car accidents shown in Table 4.2 (page 67). Also construct a cumulative frequency distribution and draw the cumulative frequency diagram.

Solution A simple frequency distribution relating to a discrete variable is represented by *Frequency Bar Diagram*. (In all other cases, a histogram is used.) This diagram consists essentially of a group of vertical thick lines (or bars) drawn at such points which represent values of the variable on the horizontal axis. The heights of bars are proportional to frequencies.

Table 4.35 Cumulative Frequency Distribution

No. of Car Accidents	Cumulative Frequency
3	5
4	14
5	25
6	29
7	30

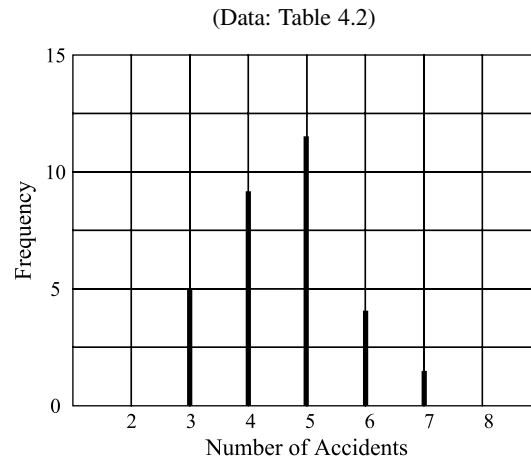


Fig. 4.7 Frequency Bar Diagram

It will be noticed that the cumulative frequency diagram (Fig. 4.8) for a discrete variable from a simple frequency distribution is a *Step Diagram*, the cumulative frequency remaining constant between two successive values of the variable.

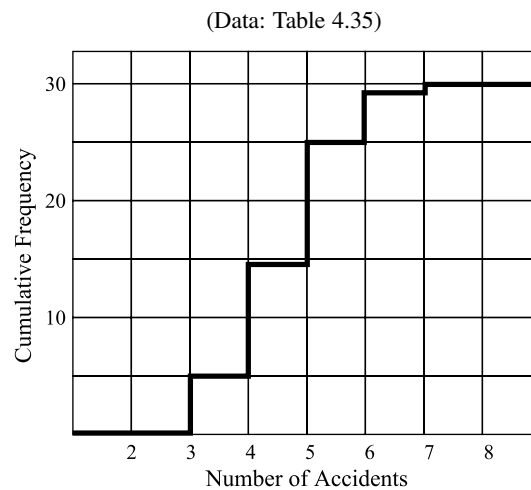


Fig. 4.8 Cumulative Frequency Diagram



4.8 FREQUENCY CURVE

The histogram gives us only an approximate idea of the nature of distribution with a limited number of observations. By drawing rectangles in a histogram, we assume that the observations in any class are uniformly distributed throughout the range of values within the class boundaries, although this may not be really true. The widths of classes could be made smaller; but, in that case some of the classes may not have any class frequency, and the true pattern of the distribution in the population will not



be known. If, however, the number of observations is very large, the position will improve and no class is expected to be empty, even when the widths of the classes are considerably small.

If the widths of classes be made smaller and smaller and at the same time the total frequency be also increased indefinitely, then the histogram and the frequency polygon will closely approach a smooth curve, known as *Frequency Curve*.

The frequency curve shows the probability distribution (Section 12.1) of the variable in the population and its area bounded by the ordinates at two specified points on the horizontal axis represents the probability that a value of the variable lies between those two limits. Like histogram, the frequency curve is therefore an area diagram.

Generally, there are 4 types of frequency curves:

(i) Symmetrical bell-shaped, (ii) Asymmetrical single-humped, (iii) J-shaped, and (iv) U-shaped. The histogram in Fig. 4.5 represents a J-shaped distribution.

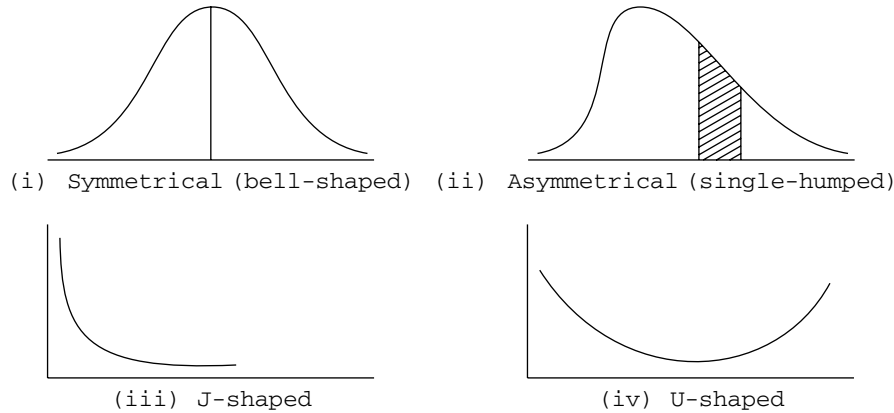


Fig. 4.9 Different Types of frequency Curves

For most of the distributions met with in practice, the frequency curve is bell-shaped, and in such cases three important characteristics are immediately apparent from the frequency curve:

1. The first characteristic is a measure of *central tendency* (p. 119). In particular, the 'mode' of the distribution is given by the abscissa of the highest point of the frequency curve.
2. The second characteristic is a measure of *dispersion* (p. 204). In particular, the 'range', i.e. the maximum possible discrepancy between any two values, is given by the distance between the two points at which the frequency curve meets the horizontal axis.
3. The third characteristic is the *shape* of the frequency curve, i.e. whether the curve is symmetrical or not; and if not, a measure of the degree of "skewness" (p. 271). A symmetrical curve indicates that mean, median and mode are equal. For asymmetrical curves this is not true, the mean being greater or less than the mode according as the longer tail of the curve lies to the right for to the left (Fig. 7.1).



A very important symmetrical (bell-shaped) frequency curve used in statistical theory is the so-called “Normal curve”; the corresponding probability distribution is called “Normal distribution” (Chapter 12). However, this curve does not adequately represent most of the data occurring in business and commerce.

EXERCISES

1. What is a frequency distribution? What considerations should be given in selecting the class-intervals while preparing a frequency distribution?

[I.C.W.A., June '75 (old)]

2. Define the terms: frequency, relative frequency, cumulative frequency (of less-than and more-than types), and frequency density.

[B.U., B.A.(Econ) '73]

3. Below is given the distribution of heights of a group of 60 students:

Height (in cm)	145.0–149.9	150.0–154.9	155.0–159.9	160.0–164.9
No. of students	2	5	9	15
	165.0–169.9	170.0–174.9	175.0–179.9	180.0–184.9
	16	7	5	1

Explain the terms ‘class limits’ and ‘class boundaries’ with reference to this distribution.

[I.C.W.A., Dec. '75]

4. Discuss in detail how you will construct a frequency table from raw data relating to a continuous variable.

[C.U., M.Com. '74; W.B.H.S. '78]

5. Form a frequency distribution with the following:

7, 4, 3, 5, 6; 3, 3, 2, 4, 3; 4, 3, 3, 4, 4;
3, 2, 2, 4, 3; 5, 4, 3, 4, 3; 4, 3, 1, 2, 3;

[C.U., B.Com. '71]

6. The following are the monthly salaries of 20 employees:

(Rs) 130, 62, 145, 118, 125; 76, 151, 142, 110, 98;
95, 116, 100, 103, 71; 85, 80, 122, 132, 95;

Form a frequency distribution with class intervals Rs 61–80, 81–100, 101–120, 121–140 and 141–160.

[C.U., B.Com. '74]

7. The data below give the marks secured by 70 candidates in a certain examination:

21	31	35	52	64	74	89	53	42	7
22	35	43	67	76	35	46	26	32	40
72	43	38	41	63	71	28	32	45	54
15	18	52	73	86	50	39	55	47	12
44	58	67	85	39	40	50	65	72	69
57	63	5	56	79	37	24	54	82	49
51	54	68	29	34	44	58	62	59	65

Construct a frequency distribution of the marks, taking classes of uniform width of 10 marks and 0 as the lower limit of the lower-most class.

[I.C.W.A., June '74]



8. Ages at death (years) of 50 persons of a town are given below:

36	48	50	45	49	31	50	48	43	42
37	32	40	39	41	47	45	39	43	47
38	39	37	40	32	52	56	31	54	36
51	46	41	55	58	31	42	53	32	44
53	36	60	59	41	53	58	36	38	60

Arrange the data in a frequency distribution in 10 class intervals, and obtain the percentage frequency in each class interval. [C.U., B.Com. '72]

9. Form an ordinary frequency table from the following cumulative frequency distribution of marks obtained by 22 students:

Marks	Number of students
Below 10	3
" 20	8
" 30	17
" 40	20
" 50	22

[I.C.W.A., June '77]

10. From the following data, calculate the "percentage" of workers getting wages:
(a) more than Rs 44, (b) between Rs 22 and Rs 58.

Wages (Rs) :	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	Total
No. of workers :	20	45	85	160	70	55	35	30	500

[C.A., May '76]

11. What do you mean by a histogram? Explain with the help of an example and illustration how it is constructed in the case of a continuous variable with unequal class intervals. [C.A., Nov. '73]
12. What is a Histogram? How is it constructed?
13. Draw the histogram of the following frequency distribution of heights of 100 college students:

Height (cm)	141-150	151-160	161-170	171-180	181-190	Total
Frequency	5	16	56	19	4	100

[W.B.H.S. '78]

14. Draw histogram and frequency polygon to present the following data:

Income (Rs)	100-149	150-199	200-249	250-299	300-349	350-399
No. of individuals	21	32	52	105	62	43
	400-449	450-499	Total			
	18	9	342			

[I.C.W.A., June '78]

15. Draw the histogram of the distribution given below and obtain the number of firms whose sales lie between Rs 12,00,000 and Rs 26,00,000:



<i>Values of Sales (Rs '000)</i>	<i>No. of Firms</i>
0 – 500	3
500 – 1000	42
1000 – 2500	288
2500 – 3500	150
3500 – 4500	51

Draw also the cumulative frequency polygon. [C.U., B.A. (Econ) '71]

16. What are cumulative frequencies? How do you represent them diagrammatically for discrete and continuous distributions? [I.C.W.A., June '74]

17. What is a cumulative frequency graph? Explain its uses. Draw a cumulative frequency graph and estimate the number of persons between the ages 30–32 in the following table:

<i>Age</i>	20–25	25–30	30–35	35–40	40–45	45–50	50–55	55–60
<i>No. of persons</i>	50	70	100	180	150	120	70	59

[C.U., M.Com. '68]

18. Describe the two types of 'ogive' and discuss their uses.

[I.C.W.A., Dec. '75 (old)]

19. Draw an ogive from the following data and find graphically the number of observations lying between 360 and 440:

<i>Value</i>	<i>Number of observations</i>
More than 200	400
More than 250	370
More than 300	315
More than 350	220
More than 400	115
More than 500	45
More than 600	15
More than 700	0

[I.C.W.A., Jan. '72]

20. Draw less-than Ogive on the basis of the data given below:

<i>Mid point</i>	18	25	32	39	46	53	60	
<i>Frequency</i>	10	15	32	42	26	12	9	$N = 146$

[C.A., May '74]



ANSWERS



3. *Class limits* : 145.0 and 149.9, 150.0 and 154.9, etc.
Class boundaries : 144.95 and 149.95, 149.95 and 154.95, etc.



5.	<i>Value:</i>	1	2	3	4	5	6	7	Total
	<i>Frequency:</i>	1	4	12	9	2	1	1	30

6.	<i>Salary (Rs):</i>	61–80	81–100	101–120	121–140	141–160	Total
	<i>Frequency:</i>	4	5	4	4	3	20

7.	<i>Marks:</i>	0–9	10–19	20–29	30–39	40–49	50–59	60–69
	<i>Frequency:</i>	2	3	6	11	12	15	10
		70–79	80–89	Total				
		7	4	70				

8.	<i>Age (years):</i>	31–33	34–36	37–39	40–42	43–45	46–48	49–51
	<i>Frequency:</i>	6	4	7	7	5	5	4
	<i>Percentage frequency:</i>	12	8	14	14	10	10	8
		52–54	55–57	58–60	Total			
		5	2	5	50			
		10	4	10	100			

9.	<i>Marks:</i>	0–9	10–19	20–29	30–39	40–49	Total
	<i>Frequency:</i>	3	5	9	3	2	22

10. (a) 32.4, (b) 68.4

15. 265 17. 40 19. 112

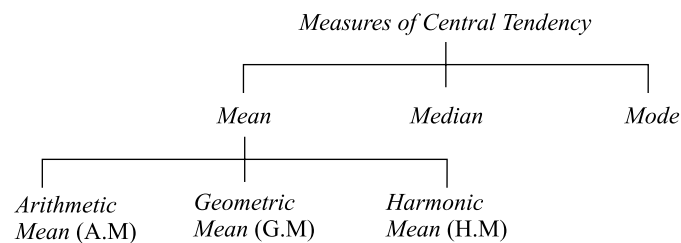
20. **Hint:** Class boundary points are 14.5, 21.5, 28.5; etc.

5 MEASURES OF CENTRAL TENDENCY

5.1 AVERAGES OR MEASURES OF CENTRAL TENDENCY

The word ‘average’ denotes a ‘representative’ or ‘typical value’ of a whole set of observations. It is a single figure which describes the entire series of observations with their varying sizes. Since a typical value usually occupies a central position, so that some observations are larger and some others are smaller than it, averages are also known as *measures of central tendency*.

There are 3 measures of central tendency—Mean, Median, and Mode. Again, Mean is of 3 types—Arithmetic Mean (A.M.), Geometric Mean (G.M.) and Harmonic Mean (H.M.). The words ‘mean’ and ‘average’ only refer to Arithmetic Mean.



Example 5.1 What are the desirable properties of a statistical average? Examine how far the arithmetic mean, the geometric mean, the median and the mode possess these properties.

[C.U., B.A.(Econ) '62; M.Com. '72; B.U., B.A.(Econ) '73; C.A., Nov. '67, May '74, May '76]

Solution According to G.U. Yule, the desirable properties of a satisfactory average are as follows:

- (i) It should be rigidly defined;
- (ii) It should be based on all observations;
- (iii) It should be readily comprehensible;
- (iv) It should be easily calculated;
- (v) It should be affected as little as possible by sampling fluctuations; and
- (vi) It should readily lend itself to algebraic treatment.



Arithmetic mean is the simplest and easiest to understand, and fulfils all the criteria of a satisfactory average. It is rigidly defined, based on all observations, and also easy to calculate. If the arithmetic mean and the number of observations in each of several groups are given, they can be algebraically combined to find the arithmetic mean of the composite group. Arithmetic mean is also the most stable of all the measures of central tendency, so far as sampling fluctuations are concerned. If many samples are drawn from the same statistical population, arithmetic mean will be found to fluctuate less than any other measure of central tendency.

Geometric mean is rigidly defined, based on all observations, and is suited for algebraic treatment. However, it is difficult to calculate and is not readily understood by common people.

Median is rigidly defined and based on all observations. But it is not so well-understood as arithmetic mean or mode. Median is easy to calculate. However, it shows greater fluctuation from sample to sample, and hence is less reliable than arithmetic mean. It can not be treated algebraically.

Although the technical term 'mode' may not be known to all, its concept is well-understood and easily comprehensible. A dealer of shoes will be more interested in the mode of different sizes of shoes, he sells, than arithmetic mean or median. Mode is that size of shoes which is sold most. It is rigidly defined, and based on all observations. However, it is very difficult to calculate mode accurately. Mode does not lend itself to algebraic treatment as arithmetic mean or geometric mean. It is less reliable and less stable as regards sampling fluctuations.



Example 5.2 Give a critical review of the different measures of central tendency, with examples.

[C.U., M.Com. '64, '71; B.Com(Hons) '66; B.A.(Econ) '67, '73, '75]

Solution There are three measures of central tendency: (1) Mean, (2) Median, (3) Mode. Again, Mean is of three types—(i) Arithmetic Mean, (ii) Geometric Mean, and (iii) Harmonic Mean.



(1) Mean

Arithmetic Mean (A.M.) of a set of observations is defined as their sum, divided by the number of observations.

Formula: (i) Simple A.M. $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

(ii) Weighted A.M.

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i, \quad \text{where } N = \sum_{i=1}^n f_i$$

Illustration 1. A.M. of 8, 1, 6 is

$$\bar{x} = \frac{1}{3} (8 + 1 + 6) = \frac{1}{3} \times 15 = 5$$

Illustration 2. A.M. of 8, 1, 6 with frequencies (or weights) 3, 2, 5 respectively is (here $N = 3 + 2 + 5 = 10$)

$$\bar{x} = \frac{1}{10} (3 \times 8 + 2 \times 1 + 5 \times 6) = 56/10 = 5.6$$



A.M. is simple to calculate and easy to understand. It has also certain algebraic properties, for which it is widely used. For instance, given the mean ($\bar{x}_i$), and the number of observations (n_i) in each of several groups, the mean ($\bar{x}$) of the composite group can be calculated by the formula

$$N\bar{x} = \sum n_i \bar{x}_i \quad \text{where } N = \sum n_i$$

Illustration 3. If 7, 4, 9 be the A.M. of three separate groups containing 6, 2, 5 observations respectively, then the A.M. of all the 13 observations is given by

$$13\bar{x} = 6 \times 7 + 2 \times 4 + 5 \times 9 = 95$$

$$\therefore \bar{x} = 95/13 = 7.31$$

An important property of A.M. is that the algebraic sum of deviations of observations from their A.M. is zero.

$$\begin{aligned} \sum (x_i - \bar{x}) &= 0, & \text{where } \bar{x} &= \text{simple A.M.} \\ \sum f_i (x_i - \bar{x}) &= 0, & \text{where } \bar{x} &= \text{weighted A.M.} \end{aligned}$$

Illustration 4. The algebraic sum of deviations of 8, 1, 6 from their A.M., viz 5 (see *Illustration 1*), is

$$(8 - 5) + (1 - 5) + (6 - 5) = 3 + (-4) + 1 = 0.$$

Illustration 5. When the observations 8, 1, 6 are weighted by 3, 2, 5 respectively, A.M. is 5.6 (see *Illustration 2*), and the algebraic sum of the deviations is

$$\begin{aligned} &3(8 - 5.6) + 2(1 - 5.6) + 5(6 - 5.6) \\ &= 3 \times 2.4 + 2 \times (-4.6) + 5 \times 0.4 \\ &= 7.2 - 9.2 + 2.0 = 0 \end{aligned}$$

However, A.M. suffers from the disadvantage that it is highly affected by the presence of extreme values, i.e. extremely large or small observations.

Illustration 6. A.M. of 2, 5, 8, 4, 6, 9 is $34/6 = 5.67$; but A.M. of 2, 5, 8, 4, 6, 9, 71 is $105/7 = 15$, which is nearly 3 times the previous A.M. 5.67, because of the presence of a very large observation 71.

Geometric Mean (G.M.) of a set of n observations is the n -th root of their product.

$$\begin{aligned} \text{Formula:} \quad & \text{(i) Simple G.M.} \quad g = \sqrt[n]{x_1 x_2 \dots x_n} \\ & \text{(ii) Weighted G.M.} \quad G = \sqrt[N]{x_1^{f_1} x_2^{f_2} \dots x_n^{f_n}} \end{aligned}$$

Illustration 7. (a) G.M. of 8, 1, 6 is

$$g = \sqrt[3]{8 \times 1 \times 6} = \sqrt[3]{48}$$

(b) G.M. of 8, 1, 6 with weights 3, 2, 5 respectively is

$$G = \sqrt[10]{8^3 \times 1^2 \times 6^5} = \sqrt[10]{3981312}$$

However, for numerical calculations, the formulae are written with logarithms:

$$\begin{aligned} \text{(i) } \log g &= \frac{1}{n} \sum_i \log x_i \\ \text{(ii) } \log G &= \frac{1}{N} \sum_i f_i (\log x_i) \end{aligned}$$



Illustration 8. (a) G.M. of 8, 1, 6 is calculated by

$$\begin{aligned}\log g &= \frac{1}{3} (\log 8 + \log 1 + \log 6) \\ &= \frac{1}{3} (0.9031 + 0 + 0.7782) = 0.5604.\end{aligned}$$

$\therefore g = \text{antilog } 0.5604 = 3.63.$

(b) G.M. of 8, 1, 6 weighted by 3, 2, 5 respectively is given by

$$\begin{aligned}\log G &= \frac{1}{10} (3 \times \log 8 + 2 \times \log 1 + 5 \times \log 6) \\ &= \frac{1}{10} (3 \times 0.9031 + 2 \times 0 + 5 \times 0.7782) \\ &= 0.6600\end{aligned}$$

$\therefore G = \text{antilog } 0.6600 = 4.57$

G.M. is more difficult to calculate than A.M. But, since G.M. is less affected by the presence of extreme values, it is sometimes used in the construction of index numbers.

Harmonic Mean (H.M.) is the reciprocal of the arithmetic mean of reciprocals of observations.

Formula: (i) Simple H.M.
$$h = \frac{n}{\sum_i \left(\frac{1}{x_i} \right)}$$

(ii) Weighted H.M.
$$H = \frac{N}{\sum_i \left(\frac{f_i}{x_i} \right)}$$

Illustration 9. H.M. of 8, 1, 6 is

$$h = \frac{3}{\frac{1}{8} + \frac{1}{1} + \frac{1}{6}} = \frac{3}{\frac{31}{24}} = \frac{72}{31} = 2.32$$

Illustration 10. H.M. of 8, 1, 6 weighted by 3, 2, 5 respectively is

$$H = \frac{10}{\frac{3}{8} + \frac{2}{1} + \frac{5}{6}} = \frac{10}{\frac{77}{24}} = \frac{240}{77} = 3.12$$

H.M. has a very limited use.

For any given set of observations, A.M. is never less than G.M., and G.M. is never less than H.M.

$$A.M. \geq G.M. \geq H.M.$$

Illustration 11. From *Illustrations 1, 8(a)* and *9* we find that for the 3 observations 8, 1, 6:

$$A.M. = 5; \quad G.M. = 3.63; \quad H.M. = 2.32;$$

which support the above theory.



Illustration 12. From *Illustrations 2, 8(b) and 10*, we find that for the observations 8, 1, 6, weighted by 3, 2, 5 respectively:

$$A.M. = 5.6; \quad G.M. = 4.57; \quad H.M. = 3.12$$

This verifies the theory.

(2) Median

Median of a set of observations is the value of the middle-most item when they are arranged in order of magnitude. It can be calculated from a grouped frequency distribution either (i) by using simple interpolation in a cumulative frequency distribution, or (ii) by using the formula:

$$\text{Median} = l_1 + \frac{(N/2) - F}{f_m} \times c$$

where

- l_1 = lower boundary of the median class;
- N = total frequency;
- F = cumulative frequency corresponding to l_1 ;
- f_m = frequency of the median class;
- c = width of the median class.

Median is, in a certain sense, the real measure of central tendency, as it gives the value of the most central observation. It is unaffected by extreme values, and can be easily calculated from frequency distributions with open-end classes.

Illustration 13. To find the median of 2, 5, 8, 4, 9, 6, 71, we arrange the observations in order of magnitude

$$2, 4, 5, (6), 8, 9, 71.$$

Here, the middle-most item is 6 : i.e. Median = 6. If the large observation 71 is replaced by a still large value, median remains unaltered.

(3) Mode

Mode of a set of observations is that value which occurs with the maximum frequency. It is the most typical or fashionable value and at times represents the true characteristic of the frequency distribution as a measure of central tendency. In the case of a simple frequency distribution, mode can be found by inspection only. However, in the case of a grouped frequency distribution, it is difficult to find the mode accurately. It is generally calculated by the formula:

$$\text{Mode} = l_1 + \frac{d_1}{d_1 + d_2} \times c$$

where

- l_1 = lower boundary of the modal class;
- d_1 = difference of frequencies in the modal class and the preceding class;
- d_2 = difference of frequencies in the modal class and the following class;
- c = common width of classes.

The formula is applicable only when all classes have the same width. Mode has certain peculiarities too. When all observations occur with equal frequency, mode does not exist. Again, there is more than one mode, if two or more values occur with the maximum frequency.



Illustration 14. (i) Mode of the observations 2, 5, 8, 4, 3, 4, 5, 2, 4 is 4, because 4 occurs the maximum number of times.

(ii) For the observations 5, 3, 6, 3, 5, 6, there is no mode.

(iii) For the observations 5, 3, 6, 3, 5, 10, 7, 2, there are two modes, viz. 3 and 5.

5.2 ARITHMETIC MEAN (A.M.)

Arithmetic Mean of a set of observations is defined to be their sum, divided by the number of observations.

Given n observations $x_1, x_2, \dots, x_n$, their A.M., denoted by the symbol $\bar{x}$ (read x -bar), is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \Sigma x \quad (5.2.1)$$

If $x_1, x_2, \dots, x_n$ have frequencies $f_1, f_2, \dots, f_n$ respectively, i.e. x_1 occurs f_1 times, x_2 occurs f_2 times and so on, then the sum of all the $f_1 + f_2 + \dots + f_n$ observations is

$$\underbrace{x_1 + x_1 + \dots + x_1}_{f_1 \text{ terms}} + \underbrace{x_2 + x_2 + \dots + x_2}_{f_2 \text{ terms}} + \dots + \underbrace{x_n + x_n + \dots + x_n}_{f_n \text{ terms}} \\ = f_1 x_1 + f_2 x_2 + \dots + f_n x_n$$

Hence, the arithmetic mean is

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} = \frac{\Sigma f x}{N} \quad (5.2.2)$$

where $N = \Sigma f$ is the total frequency. This is sometimes referred to as *Weighted arithmetic mean*, as distinct from *Simple arithmetic mean*.

Example 5.3 Distinguish between simple and weighted average, and state the circumstances under which the latter should be employed.

[I.C.W.A., June. '73; C.A., Nov. '66 & '73, May '70, '76]

Solution *Simple Average* (sometimes called *Unweighted Average*) of a set of observations is the average calculated by considering each different value as equally important. If however, in the context of the problem, all values are not considered as equally important, the *relative importance* of the different values is indicated by assigning certain numbers, called *weights*. Let $x_1, x_2, \dots, x_n$ denote the values and $w_1, w_2, \dots, w_n$ be the corresponding weights. Then

$$\text{Simple Average} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\Sigma x}{n}$$

$$\text{Weighted Average} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + \dots + w_n} = \frac{\Sigma w x}{\Sigma w}$$

When a frequency distribution is given, the frequencies of values are themselves treated as weights, and the formulae for averages (i.e. arithmetic means) are

$$\text{Simple A.M.} = \frac{\Sigma x}{n}; \text{ Weighted A.M.} = \frac{\Sigma f x}{N} \quad (5.2.3)$$

where $N = \Sigma f$. Thus for the simple average each different value is considered only once, while for weighted average each value is considered as many times as it occurs. Simple average is obtained on dividing the total of a set of observations by their number. Weighted average is obtained on multiplying the values of the variable by the corresponding weights, and then dividing the sum of products by the sum of weights.



Simple and weighted averages are equal only when all the weights are equal, i.e. $w_1 = w_2 = \dots = w_n$ or $f_1 = f_2 = \dots = f_n$. Even if the weight of one observation be different from the rest, the two averages will be unequal. Again, if each weight is multiplied (or divided) by a constant, the weighted average will remain the same.

Simple average is used when there is nothing to suspect that the values to be averaged are comparable. However, in cases where the values differ in accuracy in varying degrees or are conceptually such as would influence the final result differently, weighted average is used. If the weights are known, weighted average is more appropriate, and must be used.

[**Note:** (i) The word 'average' used in 'simple average' and 'weighted average' generally refers to A.M., G.M. or H.M., e.g. Simple A.M., Weighted G.M., etc. Expressions like 'simple median' and 'weighted mode' are not used.

(ii) Although the words 'frequency' and 'weight' are used almost synonymously, there is a slight difference between the two. Frequency of an observation, being the number of times it occurs, must always be a whole number. Weight of an observation is its '*relative importance*' as compared with the other values, and may therefore be expressed either in whole numbers or percentages or fractions. Weights and frequencies are proportional. Frequencies are generally used as weights.

(iii) Refer to *Illustrations 1* and *2* in Example 5.2.

$$\text{Simple A.M.} = \frac{1}{3}(8 + 1 + 6) = \frac{1}{3} \times 8 + \frac{1}{3} \times \frac{1}{3} \times 6$$

Simple A.M. of 3 observations is thus obtained by adding one-third part of each observation.

$$\text{Weighted A.M.} = \frac{1}{10}(3 \times 8 + 2 \times 1 + 5 \times 6) = \frac{2}{10} \times 8 + \frac{3}{10} \times 1 + \frac{5}{10} \times 6$$

This shows that in the weighted A.M., we do not take equal parts of the observations but $\frac{2}{10}$ part of 8, $\frac{3}{10}$ part of 1, $\frac{5}{10}$ part of 6. The individual parts must of course add to 1 in both

the cases, viz. $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$, and also $\frac{2}{10} + \frac{3}{10} + \frac{5}{10} = 1$.]



Example 5.4 Calculate the weighted mean price of a table from the following data, assuming that weights are proportional to the number of tables sold:

Price per Table (Rs)	36	40	44	48
Number of Tables Sold	14	11	9	6

[C.A., Nov. '62]

Solution

Table 5.1 Calculations for Weighted Mean

Price (Rs) per Table (x)	No. of Tables Sold (f)	fx
36	14	504
40	11	440
44	9	396
48	6	288
Total	40	1628



$$\text{Weighted Mean} = \frac{\sum fx}{N} = \frac{1628}{40} = 40.70 \text{ Rs}$$

$$[\text{Note: Simple Mean} = \frac{30 + 40 + 44 + 48}{4} = \frac{168}{4} = 42.00 \text{ Rs}] \quad \text{Ans. Rs 40.70 P}$$



Example 5.5 Calculate the simple and the weighted arithmetic average price per ton of coal purchased by an industry for the half year. Account for the difference between the two.

Month	Price per Ton	Tons Purchased
Jan.	Rs 42.50	25
Feb.	Rs 51.25	30
Mar.	Rs 50.00	40
Apr.	Rs 52.00	50
May	Rs 44.25	10
June	Rs 54.00	45

[C.A., Nov. '66]

Solution In order to find the weighted average price per ton, we have to use the 'tons purchased' as weights (Note the words 'average price'; so 'price' must be the variable x).

Table 5.2 Calculations for Simple and Weighted Arithmetic Average

Month	Price (Rs) per Ton (x)	Tons Purchased (f)	fx
Jan.	42.50	25	1,062.50
Feb.	51.25	30	1,537.50
Mar.	50.00	40	2,000.00
Apr.	52.00	50	2,600.00
May	44.25	10	442.50
June	54.00	45	2,430.00
Total	294.00	200 = N	10,072.50

$$\text{Simple arithmetic average} = \frac{\sum x}{n} = \frac{294.00}{6} = 49.00 \text{ Rs}$$

$$\text{Weighted arithmetic average} = \frac{\sum fx}{N} = \frac{10072.50}{200} = 50.36 \text{ Rs}$$

Simple average and weighted average are equal, only when all weights are equal. In the present case, the difference between simple and weighted arithmetic averages has arisen, because the weights are not all equal.

Ans. Rs 49.00; Rs 50.36 P





Example 5.6 The relative importance of the following eight groups of family expenditure was found to be: food 348, rent 88, clothing 97, fuel 65, household durable goods 71, miscellaneous goods 35, services 79, drink and tobacco 217. The corresponding increases in price for October 1950 based on 1947 gave the following eight percentages 25, 1, 22, 18, 14, 13, 11 and 4. Calculate the average percentage increase in price for the whole group.

[B.U., B.A.(Econ) '67]

Solution

[**Note:** Since the relative importance (i.e. weight) is given, we must calculate the weighted average].

Table 5.3 Calculations for Weighted Average

Group	Percentage Increase in Price (x)	Weight (f)	fx
1. Food	25	348	8,700
2. Rent	1	88	88
3. Clothing	22	97	2,134
4. Fuel	18	65	1,170
5. Household durable goods	14	71	994
6. Miscellaneous goods	13	35	455
7. Services	11	79	869
8. Drink & Tobacco	4	217	868
Total	—	1,000	15,278

$$\text{Average percentage increase} = \frac{\sum fx}{\sum f} = \frac{15,278}{1,000} = 15.278$$

Ans. 15.278



Example 5.7 The following table gives the marks of two candidates:

Subject	Weight	Marks of	
		Ist Candidate	2nd Candidate
A	1	70	80
B	2	65	64
C	3	58	56
D	4	63	60

Find the weighted average mark of each candidate. By what figure would the 2nd candidate have had to increase his marks in subject B, all other marks remaining same, in order that both the candidates have the same place?

[C.U., M.Com. '63]



Solution

Table 5.4 Calculations for Weighted Arithmetic Mean

Subject	Marks of		Weight (f)	fx	fy
	1st Candi. (x)	2nd Candi. (y)			
A	70	80	1	70	80
B	65	64	2	130	128
C	58	56	3	174	168
D	63	60	4	252	240
Total	—	—	10	626	616

Weighted averages (arithmetic mean) of marks are:

$$(a) \text{ 1st candidate} = \frac{\sum fx}{\sum f} = \frac{260}{10} = 62.6$$

$$(b) \text{ 2nd candidate} = \frac{\sum fy}{\sum f} = \frac{616}{10} = 61.6$$

(Second Part) Let the required increase in marks be K . Then the marks of 2nd candidate in subject B would be $(64 + K)$. Hence, in Table 5.4, the number 64 against subject B in the 3rd column would be $64 + K$; and consequently in the last column 128 should be replaced by $2 \times (64 + K) = 128 + 2K$. The total of the last column would then be $616 + 2K$. In order that both candidates have the same place, their total marks should be equal. Hence

$$616 + 2K = \text{Total marks of 1st candidate} = 626$$

or,

$$2K = 626 - 616 = 10$$

$\therefore$

$$K = 10/2 = 5.$$

Ans. (i) 62.6, 61.6, (ii) 5



5.3 IMPORTANT PROPERTIES OF A.M.

(a) The total of a set of observations is equal to the product of their number and the A.M.

$$(i) \sum x_i = n \bar{x}; \quad (ii) \sum f_i x_i = N \bar{x} \quad (5.3.1)$$

If we assume that observation x_i are all different, then the first relation in (5.3.1) implies that the simple sum is equal to the product of their number and the simple A.M. The second relation implies that the weighted sum is equal to the product of the sum of weights and the weighted A.M.

(b) The sum of the deviations of a set of observations from their A.M. is always zero.

$$(i) \sum (x_i - \bar{x}) = 0, \quad \text{where } \bar{x} = \sum x_i / n$$

$$(ii) \sum f_i (x_i - \bar{x}) = 0, \quad \text{where } \bar{x} = \sum f_i x_i / N \quad (5.3.2)$$

[Note: 'Deviation of A from B ' is defined to be the quantity $A - B$ (not $B - A$). Deviation may be positive, or negative, or zero, according as A is greater than, or less than, or equal to B . For example, deviation of 8 from 2 is $8 - 2 = 6$; deviation of 4 from 7 is $4 - 7 = -3$; deviation of x_i from A is $x_i - A$; deviation of x_i from $\bar{x}$ is $x_i - \bar{x}$.]



Relations (5.3.2) imply that the simple (weighted) sum of the deviations of observations from the simple (weighted) A.M. is always zero (Example 5:2, Illustrations 4 & 5).

(c) If two variables x and z are so related that $z = ax + b$ for each $x = x_i$, where a and b are constants, then

$$\bar{z} = a\bar{x} + b \quad (5.3.3)$$

In particular, if $y = \frac{x - c}{d}$, where c and d are constants, then

$$\bar{y} = c + d\bar{x} \quad (5.3.4)$$

Relation (5.3.4) implies that if each of the observations x_i is increased, decreased, multiplied or divided by a constant, then the mean $\bar{x}$ also will be similarly affected.

(d) If a group of n_1 observations has A.M. $\bar{x}_1$, and another group of n_2 observations has A.M. $\bar{x}_2$, then the A.M. ($\bar{x}$) of the composite group of $n_1 + n_2 (= N$, say) observations is given by

$$N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2 \quad (5.3.5)$$

This can be generalised to any number of groups:

$$N\bar{x} = \sum n_i \bar{x}_i \quad \text{where } N = \sum n_i \quad (5.3.6)$$

(e) The sum of the squares of deviations of a set of observations has the smallest value, when deviations are taken from their A.M.

(i) $\sum (x_i - A)^2$ is minimum, when $A =$ simple A.M.

(ii) $\sum f_i(x_i - A)^2$ is minimum, when $A =$ weighted A.M. (5.3.7)

Example 5.8 Show that if $\bar{x}$ is the arithmetic mean of the quantities $x_1, x_2, \dots, x_n$ then $\sum_{i=1}^n (x_i - \bar{x}) = 0$. [C.U., M.Com. '63, '65]

Or,

Prove that the sum of the deviations of $x_1, x_2, \dots, x_n$ from their mean $\bar{x}$ is equal to zero. [I.C.W.A., Dec. '76 (old)]

Solution (First method) Let $x_1, x_2, \dots, x_n$ be a series of numbers, and $\bar{x}$ their arithmetic mean, defined by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

The deviations of the numbers from their A.M. ($\bar{x}$) are

$$(x_1 - \bar{x}), (x_2 - \bar{x}), \dots, (x_n - \bar{x})$$

The algebraic sum of these deviations is

$$\begin{aligned} &= (x_1 - \bar{x}) + (x_2 - \bar{x}) + \dots + (x_n - \bar{x}) \\ &= x_1 + x_2 + \dots + x_n - \bar{x} - \bar{x} - \dots - \bar{x} \text{ (} n \text{ times)} \\ &= (x_1 + x_2 + \dots + x_n) - n\bar{x} \\ &= n\bar{x} - n\bar{x} \quad \text{by (5.3.1)} \\ &= 0. \end{aligned}$$

(Second method)

$$\text{By definition } \bar{x} = \frac{1}{n} \sum x_i$$

The deviation of x_i from the mean $\bar{x}$ is $(x_i - \bar{x})$, $i = 1, 2, \dots, n$.



Hence, the algebraic sum of all the deviations is

$$\begin{aligned}
 & \sum_{i=1}^n (x_i - \bar{x}) \\
 &= \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} \quad \text{by (3.15.2)} \\
 &= n\bar{x} - \sum_{i=1}^n \bar{x} \quad \text{by (5.3.1)} \\
 &= n\bar{x} - n\bar{x} \quad \text{by (3.15.4)} \\
 &= 0.
 \end{aligned}$$



Example 5.9 Show that if $\bar{x}$ be the arithmetic mean of the values x_i weighted by f_i ($i = 1, 2, \dots, n$), then

$$\sum_{i=1}^n f_i (x_i - \bar{x}) = 0.$$

Solution (First method) The arithmetic mean of the values x_i weighted by f_i ($i = 1, 2, \dots, n$) is, by definition,

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{N}$$

where $N = f_1 + f_2 + \dots + f_n$. Therefore,

$$\begin{aligned}
 \sum_{i=1}^n f_i (x_i - \bar{x}) &= f_1 (x_1 - \bar{x}) + f_2 (x_2 - \bar{x}) + \dots + f_n (x_n - \bar{x}) \\
 &= f_1 x_1 - f_1 \bar{x} + f_2 x_2 - f_2 \bar{x} + \dots + f_n x_n - f_n \bar{x} \\
 &= f_1 x_1 + f_2 x_2 + \dots + f_n x_n - f_1 \bar{x} - f_2 \bar{x} - \dots - f_n \bar{x} \\
 &= (f_1 x_1 + f_2 x_2 + \dots + f_n x_n) - \bar{x} (f_1 + f_2 + \dots + f_n) \\
 &= N\bar{x} - \bar{x}.N = N\bar{x} - N\bar{x} = 0.
 \end{aligned}$$

(Second method) $\bar{x} = \frac{1}{N} \sum f_i x_i$, where $N = \sum f_i$

$$\begin{aligned}
 \therefore \quad \sum f_i (x_i - \bar{x}) &= \sum (f_i x_i - f_i \bar{x}) \\
 &= \sum f_i x_i - \sum f_i \bar{x}, \quad \text{by (3.15.2)} \\
 &= \sum f_i x_i - \bar{x} \sum f_i, \quad \text{by (3.15.3)} \\
 &= \sum f_i x_i - \left(\frac{1}{N} \sum f_i x_i \right) \cdot N, \\
 &= \sum f_i x_i - \sum f_i x_i \\
 &= 0.
 \end{aligned}$$



Example 5.10 If $y_i = x_i - c$, ($i = 1, 2, \dots, n$) where c is a constant, prove that $\bar{x} = c + \bar{y}$.

Solution [Note that y_i is the deviation of x_i from c .]

Since $y_i = x_i - c$, therefore $x_i = c + y_i$

Multiplying both sides by f_i and then summing over all values of $i = 1, 2, \dots, n$ we have

$$\sum_{i=1}^n f_i x_i = \sum_{i=1}^n f_i (c + y_i)$$



$$= \Sigma(f_i c + f_i y_i) \quad \text{by (3.15.2)}$$

$$= \Sigma f_i c + \Sigma f_i y_i \quad \text{by (3.15.3)}$$

$$= c \Sigma f_i + \Sigma f_i y_i \quad \text{since } \Sigma f_i = N$$

$$= cN + \Sigma f_i y_i$$

Hence,

$$\bar{x} = \frac{1}{N} \Sigma f_i x_i = \frac{1}{N} (cN + \Sigma f_i y_i)$$

$$= c + \frac{1}{N} \Sigma f_i y_i = c + \bar{y}$$

The significance of this result is that if $y_1, y_2, \dots, y_n$ are the deviations of $x_1, x_2, \dots, x_n$ from an arbitrary constant c , then

$$\text{Mean of } x = c + \text{Mean of } y.$$



Example 5.11 If $y_i = \frac{x_i - c}{d}$ ($i = 1, 2, \dots, n$) where c and d are constants, prove that $\bar{x} = c + d\bar{y}$. [B.U., B.A. (Econ) '67; I.C.W.A., July '72]

Solution [Note: y_i may be called deviation of x_i from c in units of d . The constants c and d are known as *origin* and *scale* respectively of the values y_i]

Since $y_i = \frac{x_i - c}{d}$, we have $dy_i = x_i - c$

or,

$$x_i = c + dy_i$$

Multiplying both sides by f_i , we get $f_i x_i = f_i(c + dy_i)$

Now, summing over all values of $i = 1, 2, \dots, n$

$$\Sigma f_i x_i = \Sigma f_i (c + dy_i) = \Sigma (f_i c + d f_i y_i) \quad \text{by (3.15.2)}$$

$$= \Sigma f_i c + d \Sigma f_i y_i \quad \text{by (3.15.3)}$$

$$= c \Sigma f_i + d \Sigma f_i y_i \quad \text{writing } N = \Sigma f_i$$

Hence,
$$\bar{x} = \frac{1}{N} \Sigma f_i x_i = \frac{1}{N} (cN + d \Sigma f_i y_i)$$

$$= c + d \left(\frac{1}{N} \Sigma f_i y_i \right)$$

$$= c + d\bar{y}$$

This means that if $y_1, y_2, \dots, y_n$ be the deviations of $x_1, x_2, \dots, x_n$ from an arbitrary constant c , in units of another constant d , then

$$\text{Mean of } x = c + d \times (\text{Mean of } y)$$

It may be noted that the result of Example 5.10 may be obtained from this, on putting $d = 1$.



Example 5.12 Prove that $\sum_{i=1}^n (x_i - A)^2$ is the least when $A = \bar{x}$, where $x_1, x_2, \dots, x_n$ are the observations, A is any arbitrary constant and $\bar{x}$ the arithmetic mean. [C.U., B.A. (Econ) '65]

Or,

Prove that for a given set of observations the sum of the squares of deviations is the minimum, when deviations are taken from the arithmetic mean.

[I.C.W.A., Jan. '71]



Solution $\sum_{i=1}^n (x_i - A)^2 / n$ will be the least, when $\sum_{i=1}^n (x_i - A)^2$ is so. Now, we can write

$$\begin{aligned}
 x_i - A &= (x_i - \bar{x}) + (\bar{x} - A) \\
 \text{Therefore, } \Sigma(x_i - A)^2 &= \Sigma\{(x_i - \bar{x}) + (\bar{x} - A)\}^2 \\
 &= \Sigma(x_i - \bar{x})^2 + 2(x_i - \bar{x})(\bar{x} - A) + (\bar{x} - A)^2 \\
 &= \Sigma(x_i - \bar{x})^2 + \Sigma 2(x_i - \bar{x})(\bar{x} - A) + \Sigma(\bar{x} - A)^2, \text{ by (3.15.2)} \\
 &= \Sigma(x_i - \bar{x})^2 + 2(\bar{x} - A) \Sigma(x_i - \bar{x}) + n(\bar{x} - A)^2, \\
 &\hspace{15em} \text{by (3.15.3) \& (3.15.4)} \\
 &= \Sigma(x_i - \bar{x})^2 + 2(\bar{x} - A) \cdot 0 + n(\bar{x} - A)^2, \hspace{2em} \text{by (5.3.2)}
 \end{aligned}$$

i.e. $\Sigma(x_i - A)^2 = \Sigma(x_i - \bar{x})^2 + n(\bar{x} - A)^2$.

Both terms on the right are positive; because, the first is the sum of n squares $(x_i - \bar{x})^2$, and the second is the product of n (a positive integer) and a square. But we have only to choose the value of A which makes $\Sigma(x_i - A)^2$ the minimum possible, and this will be achieved when the second term on the right has the minimum possible value, viz. 0, i.e.

$$\begin{aligned}
 n(\bar{x} - A)^2 &= 0 \\
 \text{or, } (\bar{x} - A)^2 &= 0 \\
 \text{or, } (\bar{x} - A) &= 0 \\
 \therefore A &= \bar{x}.
 \end{aligned}$$



Example 5.13 Show that if

$S = f_k + (f_k + f_{k-1}) + (f_k + f_{k-1} + f_{k-2}) + \dots + (f_k + f_{k-1} + \dots + f_2)$ where $f_1, f_2, \dots, f_k$ are the class frequencies, then A.M. $= x_1 + \frac{hS}{N}$; x_1 = mid-point of the first class, h = class interval, and N = total frequency. [C.U., B.A. (Econ) '70]

Solution Let $x_1, x_2, \dots, x_k$ be the mid-points of the successive classes, and $f_1, f_2, \dots, f_k$ be the corresponding class frequencies. Since the classes have the same width h here 'class interval' has been used in the sense 'width of class', the successive mid-points $x_1, x_2, \dots, x_k$ must also have the same difference h , i.e. $x_2 = x_1 + h, x_3 = x_1 + 2h, \dots, x_k = x_1 + (k-1)h$.

Table 5.5 Calculations for Mean

Mid-point (x)	Class Frequency (f)	Cumulative Frequency "more-than"	$y = \frac{x - x_1}{h}$	fy
x_1	f_1	—	0	0
x_2	f_2	$F_2 = f_k + f_{k-1} + \dots + f_2$	1	f_2
x_3	f_3	$F_3 = f_k + f_{k-1} + \dots + f_3$	2	$2f_3$
...	...		...	...
x_{k-2}	f_{k-2}	$F_{k-2} = f_k + f_{k-1} + f_{k-2}$	$k-3$	$(k-3)f_{k-2}$
x_{k-1}	f_{k-1}	$F_{k-1} = f_k + f_{k-1}$	$k-2$	$(k-2)f_{k-1}$
x_k	f_k	$F_k = f_k$	$k-1$	$(k-1)f_k$
Total	N	S	—	Σfy

Let $F_1, F_2, \dots, F_1$ denote the "more-than" cumulative frequencies. It will be noticed from Table 5.5, that the given expression is

$$S = F_k + F_{k-1} + \dots + F_2$$



i.e. the sum of “more-than” cumulative frequencies excepting the first (viz. F_1). Also $f_2 = F_2 - F_3, f_3 = F_3 - F_4, \dots, f_{k-2} = F_{k-2} - F_{k-1},$
 $f_{k-1} = f_{k-1} - F_k$, and $f_k = F_k$... (i)

If we write $y_i = \frac{x_i - x_1}{h}$, ($i = 1, 2, \dots k$), then using (5.4.2, page 116),

$$\bar{x} = x_1 + h\bar{y} = x_1 + h \left(\frac{\sum fy}{N} \right) \quad \dots (ii)$$

But, $y_1 = 0, y_2 = 1, y_3 = 2, \dots, y_k = (k-1)$. Therefore, from the last column of Table 5.5, and using (i) we find

$$\begin{aligned} \sum fy &= 0 + f_2 + 2f_3 + \dots + (k-3)f_{k-2} + (k-2)f_{k-1} + (k-1)f_k \\ &= 0 + (F_2 - F_3) + 2(F_3 - F_4) + \dots + (k-3)(F_{k-2} - F_{k-1}) \\ &\quad + (k-2)(F_{k-1} - F_k) + (k-1)F_k \\ &= F_2 - F_3 + 2F_3 - 2F_4 + \dots + (k-3)F_{k-2} - (k-3)F_{k-1} \\ &\quad + (k-2)F_{k-1} - (k-2)F_k + (k-1)F_k \\ &= F_2 + F_3 + F_4 + \dots + F_{k-1} + F_k \\ &= S \end{aligned}$$

Hence, from (ii) we have $\bar{x} = x_1 + \frac{hS}{N}$

[**Note:** This relation gives us another method of calculating the A.M. from a grouped frequency distribution with a common width.]

$$\text{A.M.} = (\text{1st mid-point}) + \frac{(\text{Common width}) \times (\text{Total of more-than cumulative frequency, except the 1st})}{\text{Total frequency}}$$

Illustration: (Data of Example 5.15, page 117)

Mid-point	Class Frequency	Cumulative Frequency ("more-than")
55 = x_1	8	—
65	10	57
75	16	47
85	14	31
95	10	17
105	5	7
115	2	2
Total	65 = N	161 = S

Here, $x_1 = 55, h = 10, N = 65, S = 161$. Using the above relation

$$\bar{x} = 55 + 10 \times \frac{161}{65} = 79.77$$

This agrees with the result obtained in Example 5.15, using the direct method.



5.4 SIMPLIFIED CALCULATION FOR A.M.

Although A.M. can be calculated by applying (5.2.3) directly, the calculations can be considerably simplified based on the following theory (see Examples 5.10 and 5.11)–



- (I) If $y_1, y_2, \dots, y_n$ represent the deviations of $x_1, x_2, \dots, x_n$ from an arbitrary constant c , then

$$\text{Mean of } x = c + \text{Mean of } y$$

Stated in symbols,

$$\text{if } y = x - c, \text{ then } \bar{x} = c + \bar{y} \quad (5.4.1)$$

Thus, in order to find the A.M., we can always reduce the given observations x_i by subtracting a convenient number c and obtain deviations $y_i = x_i - c$. The mean of these deviation y_i , which can be calculated easily, is added to the constant c , giving the required mean of x (see Example 5.14).

- (II) If $y_1, y_2, \dots, y_n$ represent the deviations of $x_1, x_2, \dots, x_n$ from an arbitrary constant c , in units of another constant d , then

$$\text{Mean of } x = c + d (\text{Mean of } y)$$

Stated in symbols,

$$\text{if } y = \frac{x - c}{d}, \text{ then } \bar{x} = c + d\bar{y} \quad (5.4.2)$$

In some cases, the values of x_i are such that after subtracting a constant c , the deviations can further be reduced on division by a constant factor. If this is possible, the deviations $x_i - c$ are divided by the constant factor d , obtaining step-deviations $y_i = (x_i - c)/d$. These will be small numbers compared to the original values x_i . The A.M. of step-deviations is multiplied by the divisor d , and the product is added to the constant c , giving the required mean of x . This method is particularly useful when the consecutive values of x_i have a common difference (see Example 5.15). Then the step deviations can be reduced to 0, 1, 2, 3, ... and -1, -2, -3, ... whatever be the values of x_i .

For the calculation of mean from a grouped frequency distribution, see Example 5.16.

Example 5.14 The index numbers of 4 commodities were 92, 125, 180 and 80, and the weights 12, 7, 6 and 9 are assigned to these commodities. Find the combined arithmetic average index number. [C.A., May '65]

Solution We have to find the A.M. of 92, 125, 180, 80 weighted by 12, 7, 6, 9 respectively. By direct calculation

$$\begin{aligned} \bar{x} &= \frac{\sum fx}{N} = \frac{12 \times 92 + 7 \times 125 + 6 \times 180 + 9 \times 80}{12 + 7 + 6 + 9} \\ &= \frac{1104 + 875 + 1080 + 720}{34} = 3779/34 = 111.1. \end{aligned}$$

The calculations can be very much simplified, when we apply the method of deviations shown below:

Table 5.6 Calculations for Arithmetic Average

Commodity	Index Number x	Weight f	$y = x - 100$	fy
A	92	12	-8	-96
B	125	7	25	175
C	180	6	80	480
D	80	9	-20	180
Total	—	34	—	379



We know that, if $y = x - c$, then $\bar{x} = c + \bar{y}$. Here $c = 100$, and

$$\bar{y} = \Sigma fy/N = 379/34 = 11.1$$

Therefore, $\bar{x} = c + \bar{y} = 100 + 11.1 = 111.1$

Ans. 111.1



Example 5.15 Compute the mean weekly wages of the 65 employees working in a factory from the frequency table, using the coding method (i.e. transforming x to a new variate)–

x	55.00	65.00	75.00	85.00	95.00	105.00	115.00
f	8	10	16	14	10	5	2

[Dip. Soc. Welfare, May '68]

Solution [Note: When the successive values of x have a common difference, the calculation of mean will be very much simplified if we transform x to a new variable y , defined by

$$y = \frac{x - (\text{one of the given values of } x, \text{ preferably near the middle})}{\text{common difference}}$$

The values of y can be written down without calculation: Put 0 against the value of x which has been taken as the origin; $-1, -2, -3, \dots$ successively as we move away towards the smaller values of x ; and $1, 2, 3, \dots$ as we move away towards the higher values of x]

We introduce the new variable $y = \frac{x - 85.00}{10}$; so that

Table 5.7 Calculations for Mean

x	f	$y = \frac{x - 85.00}{10}$	fy
55.00	8	-3	-24
65.00	10	-2	-20
75.00	16	-1	-16
85.00	14	0	0
95.00	10	1	10
105.00	5	2	10
115.00	2	3	6
Total	65	–	– 34

$$\bar{x} = c + d\bar{y} = 85.00 + 10\bar{y}$$

$$\bar{y} = \Sigma fy/N = -34/65$$

$$\bar{x} = 85.00 + 10\left(\frac{-34}{65}\right)$$

$$= 85.00 - \frac{340}{65}$$

$$= 85.00 - 5.23 = 79.77$$

Ans. 79.77



Example 5.16 How do you calculate the mean of a grouped frequency distribution?



Solution In a grouped frequency distribution, let $x_1, x_2, \dots, x_n$ represent the mid-values of classes with class frequencies $f_1, f_2, \dots, f_n$ respectively. Let $y_1, y_2, \dots, y_n$ represent the deviations of these mid-values from an arbitrary origin c , in units of another constant d ; i.e. $y_i = (x_i - c)/d$. Then

$$\text{Mean} = c + d\bar{y} = c + d \left(\frac{\sum fy}{N} \right) \quad \text{where } N = \sum f.$$

Here, the mid-value is taken as representative of all observations falling within each class. The computation is done in the following tabular form:

Table 5.8 Calculations for Mean

Class Interval	Frequency f	Mid-value x	Deviation $= x - c$	$y = \frac{x - c}{d}$	fy
(1)	(2)	(3)	(4)	(5)	(6)
	f_1	x_1		y_1	
	f_2	x_2		y_2	
	$\vdots$	$\vdots$		$\vdots$	
	$\vdots$	$\vdots$		$\vdots$	
	f_n	x_n		y_n	
Total	$N = \sum f$	—	—	—	$\sum fy$

The steps in the calculation are as follows:

1. From the given data, the mid-values—col. (3)—are found for each class interval. This is done by averaging the two class limits or class boundaries, as given in the frequency distribution.
2. One of the mid-values is taken as the arbitrary origin c , preferably the one corresponding to the maximum frequency, if it lies near the middle of the distribution. The deviation of each mid-value from the arbitrary origin c is calculated, as shown in col. (4).
3. The deviations are now divided by their highest common factor d and the result, called 'step deviation', is shown in column (5). If the class intervals have a common width, d will be equal to this width. [See Table 5.9].
4. The step deviations are multiplied by the corresponding frequencies, and the total of these products is obtained; see col. (6).
5. This total is now multiplied by the divisor d , used in col. (5), and the product divided by the total frequency N .
6. The result is added to the arbitrary origin c , to give the arithmetic mean of the frequency distribution.

$$\text{Mean} = \text{Arbitrary origin} + \frac{\text{total of col. (6)}}{\text{total of col. (2)}} \times (\text{divisor})$$



Example 5.17 Given the following frequency distribution, calculate the mean:

Monthly Wages (Rs)	12.5–17.5	17.5–22.5	22.5–27.5	27.5–32.5	32.5–37.5
Number of Workers	2	22	10	14	3
	37.5–42.5	42.5–47.5	47.5–52.5	52.5–57.5	Total
	4	6	1	1	63

[I.C.W.A., Dec. '77]



Solution

Table 5.9 Calculations for Mean

Class Interval	Frequency (<i>f</i>)	Mid-value (<i>x</i>)	$y = \frac{x - 30.0}{5}$	<i>fy</i>
12.5–17.5	2	15.0	–3	–6
17.5–22.5	22	20.0	–2	–44
22.5–27.5	10	25.0	–1	–10
27.5–32.5	14	30.0	0	0
32.5–37.5	3	35.0	1	3
37.5–42.5	4	40.0	2	8
42.5–47.5	6	45.0	3	18
47.5–52.5	1	50.0	4	4
52.5–57.5	1	55.0	5	5
Total	63	–	–	–22

[**Note:** If the successive mid-values are found to have a common difference, it is unnecessary to calculate col. (4) of Table 5.8. The step deviation in col. (5) can be obtained easily by the following rule: Take *any one of the mid-values* (preferably one near the middle of the table) as origin *c*, and the *common differences of mid-values* as the value of *d*. Put 0 against the value of *x* which has been taken as origin *c*, put –1, –2, –3, ... successively as we move away towards the smaller values of *x*, and 1, 2, 3, ... towards the higher values).

We know that if $y = (x - c)/d$, then $\bar{x} = c + d\bar{y}$ (Example 5.11)

$$\text{Here } y = \frac{x - 30.0}{5} \text{ (i.e. } c = 30.0, d = 5\text{), and } \bar{y} = \frac{\sum fy}{N} = \frac{-22}{63}$$

$$\text{Therefore, } \bar{x} = c + d\bar{y} = 30.0 + 5 \times \left(\frac{-22}{63} \right) = 30.0 - \frac{110}{63}$$

$$= 30.0 - 1.75 = 28.25 \text{ Rs}$$

Ans. Rs 28.25



Example 5.18 Compute the mean from the following frequency distribution of earners by monthly income:

Income (Rs)	Below 200	200–399	400–599	600–799	800–999	1000–1199
No. of earners	25	72	47	22	13	7

[C.U., B.A.(Econ) '78]

Solution

Note: (i) Here 'class limits' have been used to define the class intervals, but in the Example 5.17 'class boundaries' were used.

(ii) It is generally assumed that the open-end class has the same width as the adjacent class.

**Table 5.10** Calculations for Mean

Class Interval	Frequency (<i>f</i>)	Mid-value (<i>x</i>)	$y = \frac{x - 499.5}{200}$	<i>fy</i>
0–199	25	99.5	–2	–50
200–399	72	299.5	–1	–72
400–599	47	499.5	0	0
600–799	22	699.5	1	22
800–999	13	899.5	2	26
1000–1199	7	1099.5	3	21
Total	186	–	–	–53

If $y = (x - c)/d$, then $\bar{x} = c + d\bar{y}$. Here $c = 499.5$, $d = 200$

$$\therefore \bar{x} = 499.5 + 200 \times \left(\frac{-53}{186} \right) = 499.5 - 56.99 = 442.51 \text{ Rs}$$



Example 5.19 Find the mean from the following distribution:

Age in years	15–19	20–24	25–29	30–34	35–44	45–59	Total
No. of persons	37	81	43	24	9	6	200

Solution

Table 5.11 Calculations for Arithmetic Mean

Class Interval	Frequency (<i>f</i>)	Mid-value <i>x</i>	Deviation (<i>x</i> – 27)	$y = \frac{x - 27}{5}$	<i>fy</i>
15–19	37	17	–10	–2	–74
20–24	81	22	–5	–1	–81
25–29	43	27	0	0	0
30–34	24	32	5	1	24
35–44	9	39.5	12.5	2.5	22.5
45–59	6	52	25	5	30
Total	200	–	–	–	–78.5

If $y = \frac{x - c}{d}$ then $\bar{x} = c + d\bar{y}$.

Here $c = 27$, $d = 5$, $\bar{y} = \frac{\sum fy}{N} = \frac{-78.5}{200}$

$$\therefore \bar{x} = 27 + 5 \left(\frac{-78.5}{200} \right) = 27 - 1.96 = 25.04 \text{ years.}$$

[**Note:** (i) Here the classes are of unequal width.

(ii) The deviations in Table 5.11 have no common factor; but since most of them are divisible by 5, we have taken $d = 5$ in order to reduce the calculations. This example shows that the values of y may not be whole numbers.

(iii) The A.M., or in fact any measure of central tendency, must be stated in the same unit in which the observations are given.]





Example 5.20 Find the missing frequencies in the following frequency distribution, when it is known that A.M. = 11.09

Class Limits	9.3–9.7	9.8–10.2	10.3–10.7	10.8–11.2	11.3–11.7
Frequency	2	5	f_3	f_4	14
	11.8–12.2	12.3–12.7	12.8–13.2	Total	
	6	3	1	60	

[C.U., M.Com. '74]

Solution We proceed to calculate the mean in the usual way.

Table 5.12 Calculations for Arithmetic Mean

Class Limits	Frequency (f)	Mid-value (x)	$y = \frac{x - 11.0}{0.5}$	fy
9.3–9.7	2	9.5	–3	–6
9.8–10.2	5	10.0	–2	–10
10.3–10.7	f_2	10.5	–1	– f_3
10.8–11.2	f_4	11.0	0	0
11.3–11.7	14	11.5	1	14
11.8–12.2	6	12.0	2	12
12.3–12.7	3	12.5	3	9
12.8–13.2	1	13.0	4	4
Total	60	–	–	$23 - f_3$

By the formula $\bar{x} = c + d\bar{y}$, and since $\bar{x} = 11.09$ is given, we have

$$11.09 = 11.0 + 0.5 \times \frac{23 - f_3}{60}$$

$$\text{or, } 11.09 - 11.0 = \frac{0.5}{60} \times (23 - f_3)$$

$$\text{i.e. } 0.09 = \frac{1}{120} (23 - f_3)$$

$$\text{or, } 10.8 = 23 - f_3$$

$$\therefore f_3 = 23 - 10.8 = 12.2$$

Since the frequency f_3 must be a whole number, $f_3 = 12$. Again, total frequency is

$$2 + 5 + f_3 + f_4 + 14 + 6 + 3 + 1 = 60;$$

$$\text{or, } 31 + f_3 + f_4 = 60$$

$$\text{or, } f_3 + f_4 = 60 - 31 = 29$$

$$\text{Since } f_3 = 12, \text{ we have } f_4 = 17.$$

$$\text{Ans. } f_3 = 12, f_4 = 17$$



Example 5.21 The following table gives the monthly income of 45 employees in a factory. The total monthly income of 5 employees in the class 'Rs 400 and above' is Rs 6000. Find the mean.



Income (Rs)	220–249	250–279	280–309	310–339	340–369	370–399
Frequency	6 400 & above 5	8 Total 45	12	5	5	4

[I.C.W.A., Dec. '75-old]

Solution [Note: Since the total income of the 5 employees in the open-end class 'Rs 400 and above' is given, it is unnecessary to make any assumption, as in Example 5.18]

We first calculate the mean income of the 40 employees, excluding those in the open-end class.

Putting $y = \frac{x - 294.5}{30}$, we find that $\Sigma fy = 7$ (The calculations are not shown here). Hence

$$\bar{x} = 294.5 + 30 \times \left(\frac{7}{40} \right) = 294.5 + 5.25 = \text{Rs } 299.75$$

The total income of the 40 employees is then Rs $299.75 \times 40 = \text{Rs } 11,990$. To this is added the total income Rs 6,000 (given) of the 5 employees in the open-end class, so that for all the 45 employees, Total income = Rs 11,990 + Rs 6,000 = Rs 17,990. Therefore,

$$\text{Mean income} = \frac{17,990}{45} = \text{Rs } 399.78$$



Example 5.22 Put the following information into a frequency distribution and obtain the arithmetic mean (assuming the range of salary is Rs 0–500):

“For a group of wage earners, 20%, 40%, 70% and 80% of the wage earners receive less than Rs 50, 120, 300 and 350 respectively; and 5% are receiving Rs 400 and over.”

[C.U., B.A. (Econ) '65]

Solution We first arrange the data in a cumulative percentage distribution (Table 5.13), from which the grouped frequency distribution (Table 5.14) has been obtained, by taking differences of successive cumulative percentages (see Example 4.9, page 95-96).

Table 5.13Cumulative Percentage
Distribution

Salary (Rs)	Cumulative Percentage
0	0
50	20
120	40
300	70
350	80
400	95
500	100

Table 5.14Frequency Distribution
of Wages

Salary (Rs)	Percentage of Wage Earners
0–50	20
50–120	20
120–300	30
300–350	10
350–400	15
400–500	5
Total	100

Now, we calculate the arithmetic mean from Table 5.14 using the percentage of wage earners as weights (Note that actual frequencies are not known; but this will not affect the calculation of mean).

**Table 5.15** Calculations for Arithmetic Mean

Salary (Rs)	Percentage of Wage Earners (f)	Mid-value (x)	Deviation $x - 210$	$y = \frac{x - 210}{5}$	fy
0–50	20	25	–185	–37	–740
50–120	20	85	–125	–25	–500
120–300	30	210	0	0	0
300–350	10	325	115	23	230
350–400	15	375	165	33	495
400–500	5	450	240	48	240
Total	100	–	–	–	–275

If $y = \frac{x - c}{d}$, then $\bar{x} = c + d\bar{y}$. Here $y = \frac{x - 210}{5}$, so that $c = 210$ and $d = 5$. Therefore,

$$\bar{x} = 210 + 5 \left(\frac{-275}{100} \right) = 210 - 13.75 = 196.25 \text{ Rs}$$

Ans. Rs 196.25



Example 5.23 (a) Show that the weighted arithmetic mean is unaffected, if all the weights are multiplied by some common factor. [C.U., B.A.(Econ) '77]

(b) Prove that simple A.M. and weighted A.M. are equal, when all the weights are equal.

Solution (a) Let the observations $x_1, x_2, \dots, x_n$ have weights $w_1, w_2, \dots, w_n$ respectively. Then

$$\text{Weighted A.M.} = \frac{\sum w_i x_i}{\sum w_i}$$

If each of the weights is multiplied by a constant k , then using the new weights $kw_1, kw_2, \dots, kw_n$ we have

$$\text{Weighted A.M.} = \frac{\sum (kw_i) x_i}{\sum (kw_i)} = \frac{k \sum w_i x_i}{k \sum w_i} = \frac{\sum w_i x_i}{\sum w_i}$$

which is the same as obtained earlier with the original weights. This shows that the weighted A.M. is unaffected, if all the weights are multiplied by a constant.

$$(b) \text{ Simple A.M.} = \frac{\sum x_i}{n} \quad \text{Weighted A.M.} = \frac{\sum w_i x_i}{\sum w_i}$$

If all the weights are equal, i.e. $w_1 = w_2 = \dots = w_n = w$ (suppose), then

$$\text{Weighted A.M.} = \frac{\sum wx_i}{\sum w} = \frac{w \sum x_i}{wn} = \frac{\sum x_i}{n} = \text{Simple A.M.}$$

This proves the statement.



Example 5.24 The table given below has been constructed from data obtained from a factory showing the distribution of the number of processed articles per day per person and the rate of payment:

**Table 5.16**

Daily No. of Articles Processed per Person	No. of Persons Processing	Rate of Payment per Article Processed (nP)
80–99	12	3.1
100–119	63	3.2
120–139	87	3.3
140–159	56	3.4
160–169	3	3.5

Calculate the rate of payment per person per article processed.

[I.C.W.A., July '64; C.U., B.A.(Econ.) '65, '69]

Solution Let us denote by

x : Rate of payment per article processed,

n : Number of persons processing

z : Daily number of articles processed per person.

Then, the number of articles processed at the rate x_i is

$$n_i z_i = f_i \text{ (suppose)}$$

(Here, z is the midvalue of the class interval). We have to find the weighted A.M. of x , the weights being f .

Since the successive values of x have a common difference, we apply the formula $\bar{x} = c + d\bar{y}$.

Table 5.17 Calculations for Weighted Arithmetic Mean

Rate of Payment per Article (nP)	No. of Persons Processing	Daily No. of Articles Processed	$nz = f$	$y = \frac{x - 3.3}{0.1}$	fy
x	n	z			
3.1	12	89.5	1,074.0	-2	-2,148.0
3.2	63	109.5	6,898.5	-1	-6,898.5
3.3	87	129.5	11,266.5	0	0
3.4	56	149.5	8,372.0	1	8,372.0
3.5	3	164.5	493.5	2	987.0
Total	—	—	28,104.5	—	312.5

$$\bar{x} = c + d\bar{y} = 3.3 + 0.1 (312.5/28104.5)$$

$$= 3.3 + 0.001$$

$$= 3.301 \text{ (approx.)}$$

Ans. 3.301 nP.



5.5 MEAN OF COMPOSITE GROUP

If two groups contain n_1 and n_2 observations with means $\bar{x}_1$ and $\bar{x}_2$ respectively, then the mean ($\bar{x}$) of the composite group of $n_1 + n_2$ observations is given by the relation

$$N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2 \quad (5.5.1)$$

where

$$N = n_1 + n_2.$$



It will be found convenient to arrange the data in the following tabular form:

Table 5.18 Data for Calculation of Composite Mean

Characteristics	Groups		Composite Group
	I	II	
No. of Observations	n_1	n_2	N
Mean	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}$

[**Note:** The product of the two quantities in Composite Group is equal to the sum of the products of quantities for all the Groups.]

In general, if there are several groups, the i -th group containing n_i observations with mean $\bar{x}_i$, then the mean ($\bar{x}$) of the composite group is given by the relation

$$N\bar{x} = \sum n_i \bar{x}_i \quad (5.5.2)$$

where $N = \sum n_i$. In this case also the data may be laid out in the form as shown in Table 5.18, all the groups being shown one after the other, followed by the composite group (see Table 5.21).

Example 5.25 There are two branches of an establishment employing 100 and 80 persons respectively. If the arithmetic means of the monthly salaries paid by the two branches are Rs 275 and Rs 225 respectively, find the arithmetic mean of the salaries of the employees of the establishment as a whole. [C.A., Nov. '63]

Solution

Table 5.19 Mean of Composite Group

Characteristics	Groups		Composite Group
	I	II	
No. of Observations	$n_1 = 100$	$n_2 = 80$	$N = 180$
Mean salary (Rs)	$\bar{x}_1 = 275$	$\bar{x}_2 = 225$	$\bar{x} = ?$

Applying (5.5.1), $N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$
 or, $180\bar{x} = 100 \times 275 + 80 \times 225 = 45,500$
 $\therefore \bar{x} = 45500/180 = 252.78 \text{ Rs}$



Example 5.26 The mean monthly salary paid to all employees in a certain company was rupees 500. The mean monthly salaries paid to male and female employees were 520 and 420 rupees respectively. Obtain the percentage of male to female employees in the company. [C.U., M.Com. '71]

Solution We apply the formula for mean of composite group:

$$N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$$

The given information is shown in the following table:

**Table 5.20** Mean of Composite Group

Characteristics	Groups		Composite Group (All employees)
	I (Males)	II (Females)	
No. of Employees	n_1	n_2	$N = n_1 + n_2$
Mean Salary (Rs)	$\bar{x}_1 = 520$	$\bar{x}_2 = 420$	$\bar{x} = 500$

Substituting the values in the formula,

$$(n_1 + n_2) \times 500 = n_1 \times 520 + n_2 \times 420$$

$$\text{or, } 500n_1 + 500n_2 = 520n_1 + 420n_2$$

$$\text{or, } 500n_1 - 520n_1 = 420n_2 - 500n_2$$

$$\text{or, } -20n_1 = -80n_2$$

$$\therefore \frac{n_1}{n_2} = \frac{-80}{-20} = \frac{4}{1}$$

$$\text{i.e. } n_1 : n_2 = 4 : 1$$

$$\text{Hence, percentage of male employees} = \frac{4}{4+1} \times 100 = 80$$

$$\text{Percentage of female employees} = 100 - 80 = 20$$

Ans. Males 80%, Females 20%.



Example 5.27 The mean age of a group of 100 children was 9.35 years. The mean age of 25 of them was 8.75 years and that of another 65 was 10.51 years. What was the mean age of the remainder? [C.U., M.Com. '65]

Solution We use the formula for mean of the composite group of 100, observations, the ages of 25 children with mean 8.75 years forming Group I, those of another 65 with mean 10.51 years forming Group II, and the remaining 10 (because in all there are 100) forming Group III.

Table 5.21 Mean of Composite Group

Characteristics	Groups			Composite Group
	I	II	III	
No. of Observations	$n_1 = 25$	$n_2 = 65$	$n_3 = 10$	$N = 100$
Mean (years)	$\bar{x}_1 = 8.75$	$\bar{x}_2 = 10.51$	$\bar{x}_3 = ?$	$\bar{x} = 9.35$

The formula is

$$N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3$$

$$\text{or, } 100 \times 9.35 = 25 \times 8.75 + 65 \times 10.51 + 10\bar{x}_3$$

$$\text{or, } 935 = 218.75 + 683.15 + 10\bar{x}_3$$

$$\text{or, } 10\bar{x}_3 = 935 - 218.75 - 683.15 = 33.10$$

$$\therefore \bar{x}_3 = 33.10/10 = 3.31 \text{ years.}$$

Ans. 3.31 years



5.6 GEOMETRIC MEAN (G.M.)

Geometric Mean of a group of n observations is the n -th root of their product. It is defined only when all observations have the same sign, and none of them is zero.



Given n observations $x_1, x_2, \dots, x_n$,

$$\text{G.M.} = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n} \quad (5.6.1)$$

This may also be written as $(x_1 x_2 \dots x_n)^{\frac{1}{n}}$, see Example 3.13 (c), p. 49.

If however, $x_1, x_2, \dots, x_n$ have frequencies $f_1, f_2, \dots, f_n$ respectively, the product of all the $N (= f_1 + f_2 + \dots + f_n)$ observations is

$$\underbrace{x_1 \cdot x_1 \dots x_1}_{f_1 \text{ terms}} \cdot \underbrace{x_2 \cdot x_2 \dots x_2}_{f_2 \text{ terms}} \dots \underbrace{x_n \cdot x_n \dots x_n}_{f_n \text{ terms}} = x_1^{f_1} x_2^{f_2} \dots x_n^{f_n}$$

so that

$$\text{G.M.} = \sqrt[N]{x_1^{f_1} x_2^{f_2} \dots x_n^{f_n}} \quad (5.6.2)$$

This may also be written in the form $(x_1^{f_1} x_2^{f_2} \dots x_n^{f_n})^{\frac{1}{N}}$ where $N = \sum f_i$ is the total frequency.

As in the case of A.M., we have *Simple geometric mean* and *Weighted geometric mean*, given by the formulae

$$\text{Simple G.M. (g)} = (x_1 x_2 \dots x_n)^{\frac{1}{n}}$$

$$\text{Weighted G.M. (G)} = (x_1^{f_1} x_2^{f_2} \dots x_n^{f_n})^{\frac{1}{N}} \quad (5.6.3)$$

They are equal, only when all weights are equal. For practical calculations, these formulae cannot be applied directly. Taking logarithms of both sides [see Examples 3.17(c), 3.19(e), 3.31(vi), 3.32(vi)] we have

$$\begin{aligned} \log g &= \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n) \\ &= \frac{1}{n} \sum \log x \\ \log G &= \frac{1}{N} [f_1(\log x_1) + f_2(\log x_2) + \dots + f_n(\log x_n)] \\ &= \frac{1}{N} \sum f_i (\log x_i) \end{aligned} \quad (5.6.4)$$

Uses—Geometric mean is useful in averaging ratios, rates and percentages. It is, therefore, considered as the most appropriate type in finding the average rate under compound interest, depreciation of machines and, sometimes, in the growth of living organism, when the compound interest law (p. 55) holds. G.M. is particularly useful in economics and business in the construction of index numbers. It makes the index numbers time reversible and gives equal weight to equal ratios of change. If a series of numbers are in geometric progression, either exactly or approximately, G.M. is the appropriate average to use.

5.7 PROPERTIES OF G.M.

(a) The product of a group of n observations is equal to the n -th power of their G.M.

$$x_1 x_2 \dots x_n = g^n \quad (5.7.1)$$



This follows from (5.6.3)

(b) The logarithm of G.M. of a set of observations is equal to the A.M. of their logarithms (see 5.6.4)

$$\log g = \frac{1}{N} \sum \log x_i; \quad \log G = \frac{1}{N} \sum f_i (\log x_i) \quad (5.7.2)$$

(c) If $G_1, G_2, \dots$ be the geometric means of several groups having $n_1, n_2, \dots$ observations respectively, then G.M. (G) of the composite group is given by their weighted geometric mean.

$$G = \sqrt[N]{G_1^{n_1} G_2^{n_2} \dots}$$

i.e.
$$\text{Log } G = \frac{1}{N} \sum n_i (\log G_i) \quad (5.7.3)$$

where $N = n_1 + n_2 + \dots$

Example 5.28 (a) If you have n distinct observations on a variable x given by $x_1, x_2, \dots, x_n$ with frequencies $f_1, f_2, \dots, f_n$ respectively, write the formulae for (i) the simple geometric mean, and (ii) the weighted geometric mean. Are these formulae directly applied for numerical computation? If not, show how this is done. State the usefulness of geometric mean in the construction of index numbers.

(b) Apply the geometric mean to find the general index from the following group indices by assigning the given weights:

Group	A	B	C	D	E	F
Group index	118	120	97	107	111	93
Weight	4	1	2	6	5	2

[I.C.W.A., July '67]

Solution (a) The formulae are

$$\text{Simple Geometric Mean } (g) = \sqrt[n]{x_1 x_2 \dots x_n}$$

$$\text{Weighted Geometric Mean } (G) = \sqrt[N]{x_1^{f_1} x_2^{f_2} \dots x_n^{f_n}}$$

where $N = f_1 + f_2 + \dots + f_n$. These formulae cannot be directly applied for numerical calculations. Taking logarithms, we have

$$\log g = \frac{1}{n} \sum \log x_i; \quad \log G = \frac{1}{N} \sum f_i (\log x_i)$$

That is, the logarithm of simple (weighted) G.M. is equal to simple (weighted) A.M. of the logarithms of observations. The antilog of A.M., so found, gives G.M.

G.M. is sometimes preferred to A.M. in the construction of index numbers, although it involves more laborious numerical calculations. This is so, because G.M. is less affected than A.M. by the presence of extremely large or small values. Since in the construction of index numbers, ratios of change are more important than absolute magnitudes of change, G.M. is likely to be more typical of the changing prices than A.M. or any other average.



(b)

Table 5.22 Calculations for Geometric Mean

Group	Group Index (x)	Weight (f)	log x	f(log x)
A	118	4	2.0719	8.2876
B	120	1	2.0792	2.0792
C	97	2	1.9868	3.9736
D	107	6	2.0294	12.1764
E	111	5	2.0453	10.2265
F	93	2	1.9685	3.9370
Total	—	20	—	40.6803

$$\log G = \frac{1}{N} \sum f_i (\log x_i) = \frac{40.6803}{20} = 2.0340$$

$$G = \text{antilog } 2.0340 = 108.1 \quad [\text{see Example 3.20(a)}]$$

Ans. 108.1



Example 5.29 The geometric mean of 10 observations on a certain variable was calculated to be 16.2. It was later discovered that one of the observations was wrongly recorded as 12.9, when in fact it was 21.9. Apply appropriate correction and calculate the correct geometric mean. [I.C.W.A. (Inter), June '74]

Solution Applying (5.7.1), the product of 10 observations (including the incorrect value 12.9) is $(16.2)^{10}$. When the incorrect 12.9 is replaced by the correct 21.9, the product would be

$$g^{10} = \frac{(16.2)^{10}}{12.9} \times 21.9$$

where g denotes the new geometric mean. Taking logarithms

$$\begin{aligned} 10 (\log g) &= 10 (\log 16.2) + \log 21.9 - \log 12.9 \\ &= 10 (1.2095) + 1.3404 - 1.1106 = 12.3248 \end{aligned}$$

$$\therefore \log g = 1.23248 = 1.2325 \text{ (approx.)}, \text{ so that } g = 17.08$$

Ans. 17.08



Example 5.30 The weighted geometric mean of the four numbers 8, 25, 17 and 30 is 15.3. If the weights of the first three numbers are 5, 3 and 4 respectively, find the weight of the fourth number. [I.C.W.A., Jan '71]

Solution We use the formula

$$\log G = \frac{1}{N} \sum f (\log x)$$

Denoting by w the weight of the fourth number 30, the calculations are shown below:

**Table 5.23** Calculations for Geometric Mean

x	f	$\log x$	$f(\log x)$
8	5	0.9031	4.5155
25	3	1.3979	4.1937
17	4	1.2304	4.9216
30	w	1.4771	$1.4771w$
Total	$12 + w$	—	$13.6308 + 1.4771w$

Since G.M. is given to be $G = 15.3$ and $\log 15.3 = 1.1847$, we have

$$1.1847 = \frac{13.6308 + 1.4771w}{12 + w}$$

By cross multiplication,

$$14.2164 + 1.1847w = 13.6308 + 1.4771w$$

$$\text{or, } 1.1847w - 1.4771w = 13.6308 - 14.2164$$

$$\text{or, } -0.2924w = -0.5856$$

$$\therefore w = 0.5856 \div 0.2924 = 2.$$

Ans. 2



Example 5.31 The production of portland cement increased from 99 million barrels in 1951 to 176 million in 1958. Assuming that the production increased at a constant annual rate, find the average annual rate of increase.

Solution Since the annual rate of increase was a constant, here the Compound Interest Law (3.9.2) holds: $A = P(1+i)^n$, where i denotes the rate per unit.

$$A = 176 \text{ million} = 176 \times 10^6$$

$$P = 99 \text{ million} = 99 \times 10^6$$

$$n = 7 \text{ (from 1951 to 1958).}$$

Substituting these values

$$176 \times 10^6 = 99 \times 10^6 (1+i)^7$$

$$\text{or, } 176 = 99 (1+i)^7$$

Taking logarithms of both sides,

$$\log 176 = \log 99 + 7 \log (1+i) \quad [\text{see Example 3.19(d)}]$$

$$\text{or, } 2.2455 = 1.9956 + 7 \log (1+i), \quad [\text{see Example 3.16(h)}]$$

$$\text{or, } 7 \log (1+i) = 2.2455 - 1.9956 = 0.2499$$

$$\therefore \log (1+i) = 0.2499 \div 7 = 0.0357$$

$$\text{or, } (1+i) = \text{antilog } 0.0357 = 1.086 \quad [\text{see Example 3.20(d)}]$$

$$\text{or, } i = 1.086 - 1 = 0.086 = 8.6\%$$

Ans. 8.6%.



Example 5.32 Compared to the previous year, the overhead expenses went up by 32% in 1961; it increased by 40% in the next year and by 50% in the following year. Calculate the average rate of increase in the overhead expenses over the three years. Explain clearly the reason for the choice of the average. [C.A., Nov. '67]

Solution Let i be the average rate of increase per year over the 3 year period. Here, since the increase in any year is based on the expenses of the previous year and *not* on any fixed expense, hence the Compound Interest Law (3.9.4, p 55) will hold.



$$\begin{aligned}
 P(1+i)^3 &= P \left(1 + \frac{32}{100}\right) \left(1 + \frac{40}{100}\right) \left(1 + \frac{50}{100}\right) \\
 &= P \times 1.32 \times 1.40 \times 1.50
 \end{aligned}$$

where P denotes the overhead expenses in 1960.

$$\text{or, } (1+i)^3 = 1.32 \times 1.40 \times 1.50$$

Taking logarithms of both sides

$$\begin{aligned}
 3 \log (1+i) &= \log 1.32 + \log 1.40 + \log 1.50 \\
 &= 0.1206 + 0.1461 + 0.1761 \\
 &= 0.4428
 \end{aligned}$$

$$\text{or, } \log (1+i) = 0.4428 \div 3 = 0.1476$$

$$\therefore 1+i = \text{antilog } 0.1476 = 1.405$$

$$\begin{aligned}
 \text{or, } i &= 1.405 - 1 = 0.405, \\
 &= 40.5\%.
 \end{aligned}$$

Ans. 40.5%



Example 5.33 A sum of money was invested for 4 years at 2% per annum and then the accumulated sum was invested for 3 years at 3% per annum. What is the average compound interest rate? [B.U., B.A.(Econ) '70]

Solution Let P be the sum invested and A the amount after the whole period of 7 years. Using (3.9.4)

$$\begin{aligned}
 A &= P(1+i_1)^{n_1} (1+i_2)^{n_2} = P \left(1 + \frac{2}{100}\right)^4 \left(1 + \frac{3}{100}\right)^3 \\
 &= P(1.02)^4 (1.03)^3.
 \end{aligned}$$

Had the compound interest rate remained a constant i throughout the 7 year period yielding the same amount as before, this rate i would be considered as the average compound interest rate.

$$\begin{aligned}
 A &= P(1+i)^n = P(1+i)^7 \\
 \text{Therefore, } P(1+i)^7 &= P(1.02)^4 (1.03)^3
 \end{aligned}$$

Cancelling P from both sides and then taking logarithms

$$\begin{aligned}
 7 \log (1+i) &= 4 (\log 1.02) + 3 (\log 1.03) \\
 &= 4(.0086) + 3(.0128) = .0728
 \end{aligned}$$

$$\text{or, } \log (1+i) = .0728 \div 7 = .0104$$

$$\text{or, } 1+i = \text{antilog } .0104 = 1.024$$

$$i = 1.024 - 1 = .024 = 2.4\%.$$

Ans. 2.4% per annum.



Example 5.34 A machine depreciates 25% of its value during the first two years, 10% for the next 3 years and 2% in value for the next five years, depreciation being calculated on the diminishing value. If the value of the machine is Rs 5000, find the depreciated value after 10 years. [C.U., B.A.(Econ) '64]

Solution In the calculation of depreciated values, the Compound Interest Law (3.9.4) should be applied, with i negative.

$$\begin{aligned}
 A &= P(1-i_1)^{n_1} (1-i_2)^{n_2} (1-i_3)^{n_3} \\
 &= 5000 \left(1 - \frac{25}{100}\right)^2 \left(1 - \frac{10}{100}\right)^3 \left(1 - \frac{2}{100}\right)^5 \\
 &= 5000 \times (0.75)^2 \times (0.90)^3 \times (0.98)^5
 \end{aligned}$$



$$\begin{aligned}
 \therefore \log A &= \log 5000 + 2(\log 0.75) + 3(\log 0.90) + 5(\log 0.98) \\
 &= 3.6990 + 2 \times \bar{1}.8751 + 3 \times \bar{1}.9542 + 5 \times \bar{1}.9912 \quad [\text{see Example 3.16(i)}] \\
 &= 3.6990 + 2(-1 + .8751) + 3(-1 + .9542) + 5(-1 + .9912) \\
 &= 3.6990 + (-2 + 1.7502) + (-3 + 2.8626) + (-5 + 4.9560) \\
 &= 13.2678 - 10 = 3.2678 \\
 \therefore A &= \text{antilog } 3.2678 = 1852 \text{ Rs} \quad [\text{see Example 3.20 (e)}] \\
 &\quad \text{Ans. Rs 1852}
 \end{aligned}$$



Example 5.35 A firm buys Rs 10,000 worth of plant on January 1, 1939, and arranges for depreciation according to the following schedule:

Category	Cost (Rs)	Depreciations rate
General Plant	4,000	5% on reducing balance
Electrical Plant	1,000	7.5% " " "
Process Plant	3,000	10% " " "
Lorries	2,000	20% " " "

Find the average depreciation rates for 1939 and 1940. Give reasons for the difference between the two averages. [C.U., B.A.(Econ) '66 ; I.C.W.A., June '73]

Solution Let P be the cost and i the rate of depreciation; then the depreciated value after 1 year would be $P(1 - i)$, and after 2 years would be $P(1 - i)^2$.

Table 5.24 Calculations for Depreciated Values

Category	Cost (Rs) (P)	Depreciation rate (i)	$(1-i)$	$(1-i)^2$	Depreciated 1-1-1940	Values on 1-1-1941
					$p(1-i)$	$p(1-i)^2$
General Plant	4,000	.05	.95	.9025	3,800	3,610
Electrical Plant	1,000	.075	.925	.856	925	856
Process Plant	3,000	.10	.90	.81	2,700	2,430
Lorries	2,000	.20	.80	.64	1,600	1,280
Total	10,000	—	—	—	9,025	8,176

Amount of depreciation during 1939 is $10,000 - 9,025 = 975$ Rs

$$\therefore \text{Average rate of depreciation} = \frac{975}{10,000} = .0975 = 9.75\%$$

Amount of depreciation during 1940 is $9025 - 8176 = 849$ Rs

$$\therefore \text{Average rate of depreciation} = \frac{849}{9025} = .0941 = 9.41\%$$

Average depreciation rates during the years 1939 and 1940 are different, because the rates for various categories are not equal. The two average rates will be the same, only when the rates of depreciation for different categories are equal.

Ans. (i) 9.75%, (ii) 9.41%





Example 5.36 Three groups of observations contain 8, 7 and 5 observations. Their geometric means are 8.52, 10.12 and 7.75 respectively. Find the geometric mean of the 20 observations in the single group formed by pooling the three groups. [I.C.W.A., Dec. '76]

Solution We apply the formula (5.7.3) for the G.M. of composite group.

$$\begin{array}{llll} \text{Here} & n_1 = 8 & n_2 = 7 & n_3 = 5 & N = 20 \\ & G_1 = 8.52 & G_2 = 10.12 & G_3 = 7.75 & G = ? \end{array}$$

Substituting the values in the formula,

$$\begin{aligned} \log G &= \frac{1}{20} [8(\log 8.52) + 7(\log 10.12) + 5(\log 7.75)] \\ &= \frac{1}{20} [8 \times 0.9304 + 7 \times 1.0052 + 5 \times 0.8893] \\ &= 18.9261 \div 20 = 0.9463 \end{aligned}$$

$$\text{Therefore, } G = \text{antilog } 0.9463 = 8.837$$



5.8 HARMONIC MEAN (H.M.)

Harmonic Mean of a set of observations is the reciprocal of the arithmetic mean of their reciprocals. Like G.M., H.M. is defined only when no observation is zero.

[**Note:** 'Reciprocal of k ' denotes $1/k$]

$$\text{Simple H.M.} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum \left(\frac{1}{x_i} \right)} \quad (5.8.1)$$

$$\text{Weighted H.M.} = \frac{N}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_n}{x_n}} = \frac{N}{\sum \left(\frac{f_i}{x_i} \right)} \quad (5.8.2)$$

They are equal only when all weights are equal (like A.M. and G.M.)

Uses—The use of H.M. is very limited. H.M. gives the largest weight to the smallest item and the smallest weight to the largest item. Hence, when there are a few extremely large or small values, H.M. is preferable to A.M. as an average. H.M. is also useful in averages involving time, rate and price.

Example 5.37 Point out the mistake or ambiguity in the following statement: 'A person goes from X to Y on cycle at 20 m.p.h. and returns at 24 m.p.h. His average speed was 22 m.p.h.' [C.A., May '70]

Solution Average speed will not be the A.M., viz $(20 + 24)/2 = 22$ m.p.h. Since the distance travelled in each case was the same, the appropriate average here will be the H.M.

$$\text{Average speed} = \frac{2}{\frac{1}{20} + \frac{1}{24}} = \frac{240}{11} = 21 \frac{9}{11} \text{ m.p.h.} = 21.82 \text{ m.p.h.}$$

The A.M. will be used only if the duration (i.e. time) of journey at 20 m.p.h. and also at 24 m.p.h. were the same. Ans. 21.82 m.p.h.





Example 5.38 If the interest paid on each of three different sums of money yielding 5 per cent, 6 per cent and 8 per cent simple interest per annum respectively is the same, what is the average yield per cent on the total sum invested? [I.C.W.A., Jan. '73]

Solution H.M. is the appropriate average, here.

$$\begin{aligned}\text{Average} = \text{H.M. of } 5, 6 \text{ and } 8 &= \frac{3}{\frac{1}{5} + \frac{1}{6} + \frac{1}{8}} = \frac{360}{59} \\ &= 6.1 \text{ per cent}\end{aligned}$$



Example 5.39 A motor car covered a distance of 50 miles four times. The first time at 50 m.p.h., the second at 20 m.p.h., the third at 40 m.p.h. and the fourth at 25 m.p.h. Calculate the average speed and explain the choice of the average.

[C.A., Nov. '67]

Solution The average speed will be given by the harmonic mean of the four speeds 50, 20, 40, 25, each weighted by the distance covered, viz. 50. However, since the weights are equal, simple H.M. and weighted H.M. will be the same; i.e. simple H.M. of the four speeds will give the required average speed.

$$\text{Average speed} = \frac{4}{\frac{1}{50} + \frac{1}{20} + \frac{1}{40} + \frac{1}{25}} = \frac{800}{27} = 30 \text{ m.p.h.}$$

[Important Note: Which average to use—A.M. or H.M.?

In averaging rates and ratios involving speed, time and distance; or price, quantity and amount of money spent, sometimes it becomes difficult to decide whether A.M. or H.M. is the appropriate average. In such cases, the following rule will be found helpful in coming to the right decision:

If the given ratios are stated as x units per y , then for finding the average ratio.

- use (i) H.M. when x 's are given
(ii) A.M. when y 's are given

For example, in order to find the average speed shown in *miles per hour* (m.p.h.) use H.M. when *miles* (i.e. distances covered) are given; but use A.M. when *hours* (i.e. time of journey) are given. The given miles or hours, as the case may be, should be used as weights when finding the average.]



Example 5.40 (a) A man travelled 12 miles at 4 m.p.h. and again 10 miles at 5 m.p.h. What was the average speed?

(b) A man travelled 12 hours at 4 m.p.h. and again 10 hours at 5 m.p.h. What was the average speed?

Solution (a) Here, the speeds are shown in *miles per hour* (m.p.h.); we are also given the *miles* travelled. Hence, H.M. of the speeds 4 and 5, weighted by miles 12 and 10, will be the appropriate average.

x	f	(f/x)
4	12	3
5	10	2
Total	22	5



$$\begin{aligned}\text{Average speed} &= \frac{N}{\sum \left(\frac{f}{x} \right)} \\ &= \frac{22}{5} = 4.40 \text{ m.p.h.}\end{aligned}$$

(b) Here, the speeds are shown in m.p.h., as before. But we are given the *hours* travelled. Hence, A.M. of the speeds 4 and 5, weighted by hours 12 and 10, will be the appropriate average.

x	f	fx
4	12	48
5	10	50
Total	22	98

$$\begin{aligned}\text{Average speed} &= \frac{\sum fx}{N} \\ &= \frac{98}{22} = 4.45 \text{ m.p.h.}\end{aligned}$$

Ans. (a) 4.40 m.p.h.; (b) 4.45 m.p.h.



Example 5.41 A person bought 6 rupees worth of orange from five markets at 15 p., 20 p., 25 p., 30 p. and 50 p. per orange respectively. What is the average price of an orange?

What would be the average price, if he had purchased 20 oranges from each market?

[B.U., B.A.(Econ) '73]

Solution [Note: Here, the variable is *price per orange* (in paise per number). In the first case, each value of the variable has weight 600 p, and hence H.M. is the appropriate average. In the second case, each value has weight 20 oranges, and so A.M. is to be used. Again, weights being equal for all values, simple and weighted averages are the same.]

Case I. Average price = H.M. of 15, 20, 25, 30 and 50

$$= \frac{5}{\frac{1}{15} + \frac{1}{20} + \frac{1}{25} + \frac{1}{30} + \frac{1}{50}} = \frac{1500}{63} = 24 \text{ p.}$$

Case II. Average price = A.M. of 15, 20, 25, 30 and 50

$$= (15 + 20 + 25 + 30 + 50) \div 5 = 28 \text{ p.}$$



5.9 ADVANTAGES AND DISADVANTAGES OF A.M., G.M., H.M.

(1) Arithmetic Mean

Advantages—(1) A.M. is readily understood, and hence needs no explanation, when used.



(2) The computation of A.M. is easy and does not involve any laborious numerical calculations. Even if all observations are not known individually, A.M. can be found, provided their sum and the number of observations are known.

(3) It can be calculated without arranging the data in order of magnitude (as required for median), or in the form of a frequency distribution (as required for mode).

(4) A.M. can be treated algebraically. Given the A.M. and the number of observations in each of several groups, A.M. of the composite group can be easily determined by using an algebraic formula.

(5) It is a very stable and reliable average as regards sampling fluctuations. If many samples are drawn from the same population and each time several measures of central tendency calculated, it will be found that A.M. fluctuates less from sample to sample than any other measures.

Disadvantages—(1) A.M. cannot be obtained by inspection, as in the case of median or mode.

(2) It cannot be calculated unless the exact magnitude of all observations and their number is known accurately. If some of the extreme values are missing, the accuracy of A.M. is greatly affected. A.M. cannot therefore be calculated from a grouped frequency distribution with open-end classes, unless some assumptions are made regarding the sizes of these classes.

(3) The greatest disadvantage of A.M. is that it is highly affected by the presence of even a few extremely large or small observations.

(4) A.M. may not be an actual value of the variable. (For example, A.M. of the number of children per family may come to 2.54. However, although there may be 2 children or 3 children in a family, 2.54 children is meaningless.)

(2) Geometric Mean

Advantages—(1) Given the product and the number of observations, G.M. can be calculated, even if the magnitude of each observation is not known individually.

(2) G.M. is particularly suitable for averaging rates, ratios and percentages. It is therefore used in the construction of index numbers.

(3) Unlike A.M., G.M. is not affected very much by the presence of extremely large or small observations.

(4) It can be treated algebraically. G.M. of the composite group can be determined, if the G.M. and the number of observations in each of the constituent groups are known.

Disadvantages—(1) The significance of G.M. is not easy to understand.

(2) It cannot be calculated, if any of the observations is zero. Also, if all the observations do not have the same sign, G.M. will be meaningless.

(3) It is difficult to calculate. The calculation of G.M. needs knowledge of logarithms.

(4) Like A.M., G.M. may not be an actual value of the variable.

(3) Harmonic Mean

Advantages—(1) H.M. is advantageous when it is desired to give greater weight to smaller observations and less weight to the larger.

(2) It is used in averages involving time, rate and price.



Disadvantages—(1) H.M has an abstract character, and its meaning is not generally understood by common people. As such, it has only a limited use.

(2) Like G.M., it cannot be calculated, if any of the observations is zero.

(3) It may not be an actual value of the variable.

5.10 RELATIONS BETWEEN A.M., G.M., H.M.

(1) For any given set of observations, A.M. is greater than or equal to G.M., and G.M. is greater than or equal to H.M.

$$A.M. \geq G.M. \geq H.M. \quad (5.10.1)$$

They are equal, only when all observations are equal.

(2) For two observations only,

$$\frac{A.M.}{G.M.} = \frac{G.M.}{H.M.} \quad (5.10.2)$$

This means that G.M. not only lies between A.M. and H.M., but $(G.M.)^2 = A.M. \times H.M.$, provided there are only two observations.

Example 5.42 Prove that

$$A.M. \geq G.M. \geq H.M.$$

where A.M., G.M., and H.M. represent arithmetic, geometric and harmonic means respectively. [C.U., B.Com. (Hons) '64, '68]

Or,

Prove that the arithmetic mean is greater than or equal to the geometric mean, and state the case when the two averages are equal. [C.U., B.A.(Econ) '66, '68]

Solution Let $x_1, x_2, \dots, x_n$ be a set of n observations (all positive). Their A.M., G.M., and H.M. (denoted by A , G and H respectively) are

$$A = \frac{x_1 + x_2 + \dots + x_n}{n}; \quad G = (x_1 x_2 \dots x_n)^{\frac{1}{n}}$$

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

Considering only two observations x_1 and x_2 , we see that

$$(\sqrt{x_1} - \sqrt{x_2})^2 \geq 0,$$

because the left side is a square quantity.

$$\text{or,} \quad x_1 + x_2 - 2\sqrt{x_1 x_2} \geq 0$$

$$\text{or,} \quad x_1 + x_2 \geq 2\sqrt{x_1 x_2}$$

$$\text{or,} \quad \frac{x_1 + x_2}{2} \geq \sqrt{x_1 x_2} \quad \dots (i)$$

$$\text{i.e.} \quad A.M. \geq G.M., \quad \text{when } n = 2. \quad \dots (ii)$$

Similarly, considering only the observations x_3 and x_4 , we have

$$\frac{x_3 + x_4}{2} \geq \sqrt{x_3 x_4} \quad \dots (iii)$$



If we now consider the two quantities $\frac{x_1 + x_2}{2}$ and $\frac{x_3 + x_4}{2}$, we must have, by (ii),

$$\frac{\frac{x_1 + x_2}{2} + \frac{x_3 + x_4}{2}}{2} \geq \sqrt{\frac{x_1 + x_2}{2} \cdot \frac{x_3 + x_4}{2}}$$

or,

$$\frac{x_1 + x_2 + x_3 + x_4}{4} \geq \sqrt{\frac{x_1 + x_2}{2} \cdot \frac{x_3 + x_4}{2}} \quad \dots (iv)$$

But,

$$\frac{x_1 + x_2}{2} \cdot \frac{x_3 + x_4}{2} \geq \sqrt{x_1 x_2} \cdot \sqrt{x_3 x_4} \quad \dots (v)$$

because each of the two terms on the left is greater than or equal to the corresponding term on the right, by (i) and (iii).

Substituting from (v) in (iv),

$$\frac{x_1 + x_2 + x_3 + x_4}{4} \geq \sqrt{\sqrt{x_1 x_2} \cdot \sqrt{x_3 x_4}}$$

or,

$$\frac{x_1 + x_2 + x_3 + x_4}{4} \geq \sqrt[4]{x_1 x_2 x_3 x_4} \quad \dots (vi)$$

i.e., $A.M. \geq G.M.$, when $n = 4$.

Proceeding this way, it can be shown that $A.M. \geq G.M.$, whenever $n = 2$ or 4 or 8 or 16 or 32 etc. i.e., of the form $n = 2^m$, where m is a positive integer. But we have to prove the result for any value of n . For this purpose, let us suppose that the given value of n lies between two such values 2^{m-1} and 2^m , i.e. $2^{m-1} < n < 2^m$.

We now consider 2^m ($= N$, say) values, consisting of the n given observations $x_1, x_2, \dots, x_n$, and $(N - n)$ further values each equal to A , i.e. $(x_1 + x_2 + \dots + x_n)/n$.

$$\underbrace{x_1, x_2, \dots, x_n}_{n \text{ terms}} \quad \underbrace{A, A, \dots, A}_{(N-n) \text{ terms}}$$

A.M. of these N values is

$$\frac{x_1 + x_2 + \dots + x_n + A + A \dots + A}{N} = \frac{nA + (N - n)A}{N} = A$$

Also, G.M. of these N values is

$$(x_1 x_2 \dots x_n \cdot A \cdot A \dots A)^{\frac{1}{N}} = (G^n \cdot A^{N-n})^{\frac{1}{N}}$$

Since $A.M. \geq G.M.$ for $N = 2^m$ values, therefore, in the present case

$$A \geq (G^n \cdot A^{N-n})^{\frac{1}{N}}$$

Raising both sides to the power N , we have

$$A^N \geq G^n \cdot A^{N-n}$$

Simplifying, we get

$$A^n \geq G^n$$

or,

$$A \geq G$$

This completes the proof that $A.M. \geq G.M.$ for any number of observations.

Using this result, we shall now prove that $G \geq H$. For this purpose, let us consider the n

values, $\frac{1}{x_1}, \frac{1}{x_2}, \dots, \frac{1}{x_n}$

The A.M. of these values is

$$\frac{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}{n} = \frac{1}{H}$$



and their G.M. is

$$\left(\frac{1}{x_1} \cdot \frac{1}{x_2} \cdots \frac{1}{x_n} \right)^{\frac{1}{n}} = \left(\frac{1}{x_1 x_2 \cdots x_n} \right)^{\frac{1}{n}} = \left(\frac{1}{G^n} \right)^{\frac{1}{n}} = \frac{1}{G}$$

Since, we have proved that $A.M. \geq G.M.$, in the present case

$$\frac{1}{H} \geq \frac{1}{G}$$

i.e.

$$G \geq H$$

[Section 3.12(i)]

Combining the two results $A \geq G$ and $G \geq H$, we have $A \geq G \geq H$;

i.e. in general,

$$A.M. \geq G.M. \geq H.M.$$

We shall now prove that $A.M. = G.M. = H.M.$, only when all the observations have the same value, i.e. $x_1 = k, x_2 = k, \dots, x_n = k$. In such a situation

$$A.M. = \frac{k + k + \cdots + k}{n} = \frac{nk}{n} = k$$

$$G.M. = (k.k.\cdots k)^{\frac{1}{n}} = (k^n)^{\frac{1}{n}} = k$$

$$H.M. = \frac{n}{\frac{1}{k} + \frac{1}{k} + \cdots + \frac{1}{k}} = \frac{n}{(n/k)} = k$$

and hence $A.M. = G.M. = H.M.$, when all the observations are equal.



Example 5.43 If x_1 and x_2 are two positive values of a variate, prove that their geometric mean is equal to the geometric mean of their arithmetic and harmonic means. [I.C.W.A., Jan. '66]

Solution Let A , G and H represent respectively $A.M.$, $G.M.$ and $H.M.$ of the two values x_1 and x_2 . Then, by definition,

$$A = \frac{x_1 + x_2}{2}; G = \sqrt{x_1 x_2}$$

$$H = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}} = \frac{2}{\frac{x_1 + x_2}{x_1 x_2}} = \frac{2x_1 x_2}{x_1 + x_2}$$

$\therefore$

$$A \times H = \frac{x_1 + x_2}{2} \times \frac{2x_1 x_2}{x_1 + x_2} = x_1 x_2 = (\sqrt{x_1 x_2})^2 = G^2$$

Thus,

$$A \times H = G^2$$

or,

$$G^2 = A \times H$$

or,

$$G = \sqrt{A \times H}$$

= Geometric mean of A and H .

(proved)



Example 5.44 The arithmetic mean of two observations is 25 and their geometric mean is 15. Find (i) their harmonic mean and (ii) the two observations.

[I.C.W.A., Jan. '72]



Solution For two observations $A.M. \times H.M. = (G.M.)^2$

Substituting the values of $A.M.$ and $G.M.$,

$$25 \times H.M. = (15)^2,$$

so that

$$H.M. = (15)^2/25 = 225/25 = 9.$$

Again, if x and y be the two observations

$$A.M. = \frac{x+y}{2} = 25 \text{ and } G.M. = \sqrt{xy} = 15$$

i.e. $x + y = 50$ and $xy = 225$. Substituting the value of y from the first equation into the second, we have

$$\begin{array}{ll} x(50-x) = 225 & \text{or, } 50x - x^2 = 225 \\ \text{or, } x^2 - 50x + 225 = 0; & \text{or, } (x-45)(x-5) = 0 \\ \therefore x = 45, 5. & \text{Hence, } y = 5, 45. \end{array}$$

In any case, the two observations are 45 and 5.

[**Note:** Alternatively, first calculate the two observations by the above method and then calculate the H.M. directly]. Ans. (i) 9, (ii) 45, 5.

5.11 MEDIAN

Median of a set of observations is the middle-most value when the observations are arranged in order of magnitude. The number of observations smaller than Median is the same as the number greater than it. Thus, Median divides the observations into two equal parts. Median is, in a certain sense, the real measure of central tendency, as it gives the value of the most central observation. It is unaffected by the presence of extremely large or small observations and can be calculated from frequency distributions with open-end classes. Median finds the largest application in psychological and achievement tests, e.g. to find the boy of average intelligence, the candidates may be ranked in order of intelligence and the median employed.

An important property of Median is that for any given set of observations the sum of absolute deviations from median is the least.

5.12 CALCULATION OF MEDIAN

The median is calculated as follows:

(a) *From simple series*—The given data are arranged in order of magnitude. If the number of observations be odd, the value of the middle-most item is the median. However, if the number be even, the arithmetic mean of the two middle-most items is taken as median (Example 5.45).

(b) *From simple frequency distribution*—The cumulative frequency (“less than” type) corresponding to each distinct value of the variable is calculated. If the total frequency be N , the value of the variable corresponding to cumulative frequency $(N+1)/2$ gives the median.

(c) *From grouped frequency distribution*—Median from a grouped frequency distribution is that value which corresponds to cumulative frequency $N/2$ [Sometimes $(N+1)/2$ is used as in (b) above; but this procedure is not correct]. Median from a grouped frequency distribution can be calculated by any of the following methods:



(i) *By the application of formula for median:* The cumulative frequencies are calculated. The class in which cumulative frequency $N/2$ lies, is called the *median class*. Now we apply the formula:

$$\text{Median} = l_1 + \frac{N/2 - F}{f_m} \times c \quad (5.12.1)$$

where l_1 = lower boundary of median class;
 N = total frequency;
 F = cumulative frequency below l_1 ;
 f_m = frequency of median class;
 c = width of median class.

(ii) *By the application of simple interpolation in a cumulative frequency distribution:* If F_1 and F_2 be the cumulative frequencies shown in the table which are just smaller than and just larger than $N/2$, and they correspond to the class boundaries l_1 and l_2 respectively, then

$$\frac{\text{Median} - l_1}{l_2 - l_1} = \frac{(N/2) - F_1}{F_2 - F_1} \quad (5.12.2)$$

[**Note:** l_1 and l_2 are the lower and the upper boundaries of the median class, and $N/2$ lies between F_1 and F_2].

(iii) *Graphical method*—An approximate value of median can be obtained graphically from ogive, or cumulative frequency polygon (page 104). Draw a horizontal line from the point $N/2$ on the vertical scale showing the cumulative frequencies, until it meets the ogive (either less-than or more-than type). From the point of intersection, a perpendicular is now drawn on the horizontal axis. The position of the foot of the perpendicular is read from the horizontal scale showing values of the variable, and this gives the median. If both ogives (less-than and more-than) are available on the same graph paper, the position of the foot of the perpendicular drawn from the point of intersection of the two ogives, gives the median (Figs 5.2, 5.3 and 4.2 at pages 148, 149, 90).

5.13 ADVANTAGES AND DISADVANTAGES OF MEDIAN

Advantages—(1) Median is not difficult to understand, although it is not so popular as the arithmetic mean.

(2) It is easy to calculate. Even when all the observations are not known, median can be calculated, provided the general location of all observations and values near the middle are available. Median can also be calculated without difficulty from grouped frequency distributions with classes of unequal width or with open-end classes (Example 5.50).

(3) Median is applicable to qualitative data in psychological and social studies, where numerical measurements may not be available, but it is possible to rank the objects in some order.

Disadvantages—(1) For the calculation of median, the data must be arranged.

(2) Unlike A.M. or G.M., it cannot be treated algebraically. Given the medians of several groups of observations, median of the composite group can not be determined.

(3) If it is desired to give greater importance to large or small values, median is unsuitable.



(4) The calculation of median from a grouped frequency distribution is based on simple interpolation, which assumes that the observations in the median class are uniformly distributed. But in reality, this may not be true.

(5) Median is affected more by sampling fluctuations than the arithmetic mean, and is therefore, less reliable.

Example 5.45 Find the median of (i) 32, 22, 29, 17, 40, 26, 21; (ii) 3.1, 2.6, 5.0, 4.7, 2.4, 3.9, 5.1, 3.6.

Solution (i) The data when arranged are: 17, 21, 22, (26), 29, 32, 40. Since the middlemost item is 26, we have *Median* = 26.

(ii) The data are arranged: 2.4, 2.6, 3.1, (3.6), (3.9), 4.7, 5.0, 5.1. There are two middlemost items, viz. 3.6 and 3.9. So,

$$\text{Median} = \frac{3.6 + 3.9}{2} = 3.75$$



Example 5.46 Find the median from the following simple frequency distribution:

<i>x</i>	0	1	2	3	4	5	6	Total
<i>f</i>	7	44	35	16	9	4	1	116

Solution

Table 5.25 Calculation for Median

<i>x</i>	<i>f</i>	Cumulative Frequency
0	7	7
1	44	51
2	←35	←86
3	16	102
4	9	111
5	4	115
6	1	116 = <i>N</i>

Here, $(N + 1)/2 = 117/2 = 58.5$. From the last column of Table 5.25 it is found that $(N + 1)/2 = 58.5$ is greater than cumulative frequency 51, but is smaller than the next cumulative frequency 86, corresponding to $x = 2$. Hence, median is 2. (Also see Example 5.57).

[**Note:** Median is the value of x corresponding to cumulative frequency 58.5. This implies that when all the 116 observations are arranged in order of magnitude, median will be just midway between the 58th and 59th observations. Last column of Table 5.25 shows that the smallest 7 observations are each 0; 8th, 9th, ... 51st observations are each 1; 52nd, 53rd, 86th observations are each 2; and so on. Thus, 58th and 59th observations are each 2, so that 58.5th observation, viz. median, is $(2 + 2)/2 = 2$].

Ans. 2.



Example 5.47 Show that the expression for median is

$$\text{Median} = l_i + \frac{\left(\frac{N}{2} - \Sigma f_i\right)}{f_{med}} \times C$$



where l_1 is the lower limit of the class containing the median, Σf_1 is the summation of the frequencies in all classes below l_1 , f_{med} is the frequency of the class in which the median falls and c is the class interval, and N is the total number of observations.

[C.U., M.Com. '65]

Solution If l_1 and l_2 represent the lower and the upper boundaries (note that l_1 is the *lower boundary*, and NOT the “lower limit”, as stated in the question) of the class containing median, then cumulative frequency below l_1 , viz. Σf_1 , is smaller than $N/2$; and cumulative frequency below l_2 is greater than $N/2$. Also $l_2 - l_1 = c$. Therefore, the frequency between l_1 and the median is $\left(\frac{N}{2} - \Sigma f_1\right)$. Since the frequency in the median class of width c is f_{med} , the width per

unit frequency is $\frac{c}{f_{med}}$, and the difference of median beyond l_1 for a further increase of

$\left(\frac{N}{2} - \Sigma f_1\right)$ in frequency is proportionately $\frac{c}{f_{med}} \times \left(\frac{N}{2} - \Sigma f_1\right)$

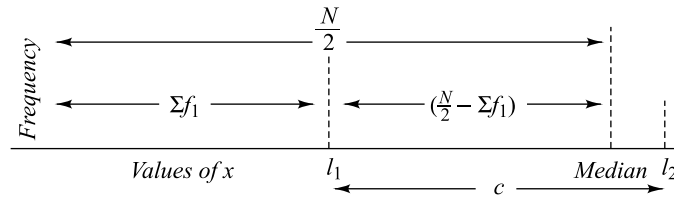


Fig. 5.1 Diagram Explaining Formula for Median

Hence,

$$\begin{aligned} \text{Median} &= l_1 + \frac{c}{f_{med}} \times \left(\frac{N}{2} - \Sigma f_1\right) \\ &= l_1 + \frac{\left(\frac{N}{2} - \Sigma f_1\right)}{f_{med}} \times c \end{aligned}$$



Example 5.48 Find the median and the median class of the data given below:

Class Boundaries	15–25	25–35	35–45	45–55	55–65	65–75
Frequency	4	11	19	14	0	2

[I.C.W.A., Jan. '65]

Solution (First method) For a grouped frequency distribution

Median = Value of the variable corresponding to cumulative frequency $N/2$.

This can be found from a cumulative frequency distribution by using simple interpolation (page 63) as shown below:

**Table 5.26** Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency (less-than)
15	0
25	4
35	15
Median →	← $N/2 = 25$
45	34
55	48
65	48
75	50 = N

Since $N/2 = 25$ lies between the cumulative frequencies 15 and 34, the corresponding value of the variable, viz. median, must lie in the interval between 35 and 45. The *median class* is, therefore, (35 – 45). Now applying simple interpolation

$$\frac{\text{Median} - 35}{45 - 35} = \frac{25 - 15}{34 - 15}$$

or,
$$\frac{\text{Median} - 35}{10} = \frac{10}{19}$$

or,
$$\text{Median} - 35 = \frac{10}{19} \times 10 = \frac{100}{19} = 5.26$$

or,
$$\text{Median} = 35 + 5.26 = 40.26$$

(Second method)

We first calculate the cumulative frequencies against each class interval as shown below:

Table 5.27 Calculation of Cumulative Frequency

Class Boundaries	Frequency	Cumulative Frequency
15–25	4	4
25–35	11	15 = F
(35–45)	19 = f_m	34
45–55	14	48
55–65	0	48
65–75	2	50 = N

From the last col. of the table it will be seen that $N/2 = 25$ is more than cumulative frequency 15, but is less than the next cumulative frequency 34; hence the *median class* is 35–45. Now, we can apply formula (5.12.1), page 141; here $l_1 = 35$, $N = 50$, $F = 15$, $f_m = 19$, $c = 45 - 35 = 10$.

$$\therefore \text{Median} = 35 + \frac{25 - 15}{19} \times 10 = 35 + 5.26 = 40.26$$

Ans. (i) 40.26; (ii) 35–45



Example 5.49 The following is the table which gives you the distribution of marks secured by some students in an examination:



Marks	0–20	21–30	31–40	41–50	51–60	61–70	71–80
No. of Students	42	38	120	84	48	36	31

Find: (i) Median marks. (ii) The percentage of failure if the minimum for a pass is 35 marks. [C.A., Nov. '69]

Solution [Note: Here class limits are given, and therefore we have to find class boundaries for the cumulative frequency distribution.]

Table 5.28 Cumulative Frequency Distribution

Marks (class boundaries)	Cumulative Frequency
20.5	42
30.5	80
34.5→	←F
Median→	←N/2 = 199.5
40.5	200
50.5	284
60.5	332
70.5	368
80.5	399 = N

(i) Median is the value of the variable corresponding to cumulative frequency $N/2$, i.e. 199.5. Using simple interpolation in the above cumulative frequency distribution,

$$\frac{\text{Median} - 30.5}{40.5 - 30.5} = \frac{199.5 - 80}{200 - 80}$$

Solving this, we get

$$\text{Median} = 30.5 + \frac{119.5 - 80}{120} = 40.46.$$

(ii) The minimum for a pass is 35. Assuming that 'marks' is a continuous variable and that marks are rounded to the nearest whole number, the mark 35 actually represents the interval (34.5–35.5). The minimum for a pass on the continuous scale is therefore 34.5. The number (F) of students obtaining less than 34.5 is the cumulative frequency corresponding to 34.5 marks. Using simple interpolation in Table 5.28 again, we have

$$\frac{34.5 - 30.5}{40.5 - 30.5} = \frac{F - 80}{200 - 80}$$

$$\text{Solving, } F = 80 + \frac{4}{10} \times 120 = 128$$

$$\text{The percentage of failures is therefore } \frac{128}{399} \times 100 = 32.08$$

Ans. (i) 40.46, (ii) 32.08



Example 5.50 You are given below a certain statistical distribution:

Value Less than	100	100–200	200–300	300–400	400 & above	Total
Frequency	40	89	148	64	39	380

Calculate the most suitable average giving reasons for your choice,

[C.A., Nov. '73]



Solution The most suitable average from a frequency distribution with open-end classes is Median. Here, the total frequency $N = 380$; so $N/2 = 190$.

Table 5.29 Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency (less-than)
100	40
200	129
Median →	← $N/2 = 190$
300	277
400	341
...	$380 = N$

Using simple interpolation (page 63, formula 3.16.1),

$$\frac{\text{Median} - 200}{300 - 200} = \frac{190 - 129}{277 - 129}$$

Solving this, we get

$$\text{Median} = 200 + \frac{61}{148} \times 100 = 241.2$$



Example 5.51 The table below gives the diastolic blood pressure of 250 men. The readings were made to the nearest millimetre and the central value of each group is given:

Blood Pressure (mm):	60	65	70	75	80	85	90	95
Number of Men:	4	5	31	39	114	30	25	2

Calculate from the data the mean and the median. [I.C.W.A., July '70]

Solution [Note: For the calculation of mean, we have to use the given midvalues. However, for the calculation of median, the class boundaries are required. Since the given midvalues are found to have a common difference, the classes must be of equal width and the class boundaries will lie exactly midway between the successive midvalues.]

Table 5.30 Calculations for Mean and Median

Mid-values x	Class Boundaries	Frequency f	$y = \frac{x - 75}{5}$	fy	Cumulative Frequency
60	57.5–62.5	4	–3	–12	4
65	62.5–67.5	5	–2	–10	9
70	67.5–72.5	31	–1	–31	40
75	72.5–77.5	39	0	0	79
80	(77.5–82.5)	114	1	114	193
85	82.5–87.5	30	2	60	223
90	87.5–92.5	25	3	75	248
95	92.5–97.5	2	4	8	250 = N
Total	—	250 = N	—	204	—



Using (5.4.2)

$$\text{Mean} = 75 + 5 \left(\frac{204}{250} \right) = 79.08 \text{ mm.}$$

Again, it is found that $N/2 = 125$ exceeds the cumulative frequency 79, but is less than the next higher cumulative frequency 193. Hence the median class is (77.5–82.5). Now using formula (5.12.1)

$$\text{Median} = 77.5 + \frac{125 - 79}{114} (82.5 - 77.5) = 79.52 \text{ mm.}$$

Ans. Mean = 79.08 mm., Median = 79.52 mm.



Exmaple 5.52 Draw one of the ogives for the following data and find the median wage:

Weekly Wages in Rs	0–20	20–40	40–60	60–80	80–100
Number of Workers	40	51	64	38	7

[C.A.Nov.'63]

Solution Let us draw the ogive of 'less-than' type. For this purpose, we have to construct a cumulative frequency distribution (Note that in the data class boundaries are given) as shown below.

Class Boundary	0	20	40	60	80	100
Cumulative Frequecny	0	40	91	155	193	200 = N

The cumulative frequencies are plotted on a graph paper against class boundaries, and the ogive (Fig. 5.2) is drawn. From the point $N/2 = 200/2 = 100$ on the vertical scale, a horizontal line is drawn meeting the ogive. From the point of intersection, a perpendicular (dotted line in the figure) is now drawn on the horizontal axis. The point showing the foot of the perpendicular is now read from the scale, and is found to be approximately 43. Thus, median = 43.

Ans. 43.

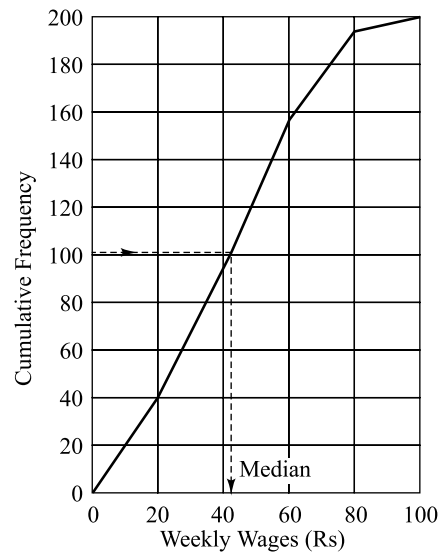


Fig. 5.2 Ogive (Less-than) for Wage Distribution





Example 5.53 Draw an ogive of 'more than' type on the data given below. Deduce from it the median of the distribution.

Wt. in gms.	410–419	420–429	430–439	440–449	450–459	460–469	470–479
Frequency	14	20	42	54	45	18	7

[I.C.W.A., July '68]

Solution

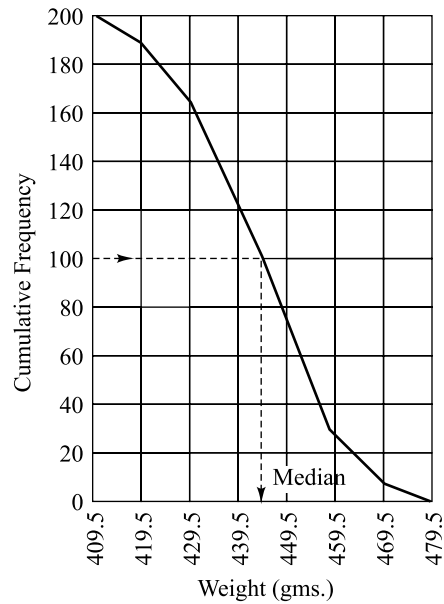


Fig. 5.3 Ogive (More-than) for Weight Distribution

We construct a cumulative frequency distribution of 'more-than' type (Note that here class limits are given) as shown below:

Class Boundary	409.5	419.5	429.5	439.5	449.5	459.5	469.5	479.5
Cumulative Frequency	200	186	166	124	70	25	7	0

The data are plotted on a graph paper and the ogive is drawn.

As in the previous example, a horizontal line is drawn from the point $N/2 = 100$ on the vertical scale, and proceeding the same way, median is read from the horizontal scale, and found to be 444 gms.

Ans. 444 gms.

[**Note:** The median must be stated in the same units as the original data].



Exmample 5.54 Find the median by simple interpolation or graphically from the following distribution:



X :	1	2	3	4-6	7-9	10-12	13-20	21-28	29-36
$Freq.$:	10	5	3	9	6	2	1	10	15

Should interpolation and graphic method give the same value of median? If so, why? [C.U., B.A. (Econ) '68]

Solution We have to construct a cumulative frequency distribution from the given data:

<i>Class</i>	Median ↓								
<i>Boundary:</i>	1.5	2.5	3.5	6.5	9.5	12.5	20.5	28.5	36.5
<i>Cumulative</i>									
<i>Frequency:</i>	10	15	18	27	↑ 33	35	36	46	61 = N
	$N/2 = 30.5$								

Now, $N/2 = 30.5$ lies between the consecutive cumulative frequencies 27 and 33 in the cumulative frequency distribution. Therefore, Median (i.e. the value of the variable corresponding to cumulative frequency 30.5) must lie between the class boundaries 6.5 and 9.5. Applying simple interpolation, we have

$$\frac{\text{Median} - 6.5}{9.5 - 6.5} = \frac{30.5 - 27}{33 - 27}$$

$$\text{or, } \frac{\text{Median} - 6.5}{3} = \frac{3.5}{6}$$

$$\text{or, } \text{Median} - 6.5 = \frac{3.5}{6} \times 3 = 1.75$$

$$\therefore \text{Median} = 6.5 + 1.75 = 8.25.$$

[**Note:** If we apply formula (5.12.2), $l_1 = 6.5$, $l_2 = 9.5$, $F_1 = 27$, $F_2 = 33$].

Simple interpolation and graphic method should both give the same value of median; because they are based on the same principles, viz. uniform distribution of frequencies in the median class. Simple interpolation involves arithmetic calculation and the graphic method uses the ogive. In practical calculations, however, the graphic method gives only an approximate result.



Example 5.55 You are given the following incomplete frequency distribution. It is known that the total frequency is 1,000 and that the median is 413.11. Estimate by calculation the missing frequencies.

<i>Values</i>	300-325	325-350	350-375	375-400	400-425	425-450
<i>Frequency</i>	5	17	80	?	326	?
					450-475	475-500
					88	9

[C.A., May '77]

Solution Let a and b denote the missing frequencies of classes 375-400 and 425-450 respectively. Then the total frequency is

$$5 + 17 + 80 + a + 326 + b + 88 + 9 = 1000 \text{ (given)}$$

$$\text{or, } 525 + a + b = 1000$$



$$\therefore a + b = 475 \quad \dots \quad (i)$$

Again, since the given value of median, viz. 413.11, lies in the class interval 400–425, this must be the median class. So, in the formula for median (5.12.1) at p. 141 we have $l_1 = 400$, $f_m = 326$, $F = 5 + 17 + 80 + a = 102 + a$, $c = 25$ and $N = 1000$.

$$\text{Median} = l_1 + \frac{(N/2) - F}{f_m} \times c$$

$$\text{or,} \quad 413.11 = 400 + \frac{500 - (102 + a)}{326} \times 25$$

$$\text{or,} \quad 413.11 = 400 + \frac{398 - a}{326} \times 25$$

$$\text{or,} \quad (413.11 - 400) \times \frac{326}{25} = (398 - a)$$

$$\text{or,} \quad 171 = 398 - a,$$

$$\therefore a = 227$$

... (ii)

Using (i) and (ii), we get $b = 248$.

Ans. 227, 248



Example 5.56 Show that $\sum_{i=1}^n (x_i - A)^2$ is least if $A = \bar{x}$, but $\sum_{i=1}^n |x_i - A|$ is least

if $A = \text{median}$.

[C.U., B.Com. (Hons.)' 63; B.A. (Econ)' 71]

Solution (First Part) See Example 5.12, page 113

(Second Part) Let us arrange the given observations $x_1, x_2, \dots, x_n$ in increasing order of magnitude and let the ordered series be denoted by

$$X_1, X_2, \dots, X_n \quad \dots \quad (1)$$

X_1 is the smallest and X_n the largest among the given observation.

Since $X_1, X_2, \dots, X_n$ is only a different arrangement of the same observation $x_1, x_2, \dots, x_n$ we have

$$\sum_{i=1}^n |x_i - A| = \sum_{i=1}^n |X_i - A|$$



The number of observations may be either odd or even, and we shall prove the result separately for each case:

Case I. When n is odd, say $n = 2m + 1$.

$$\begin{aligned} \sum_{i=1}^n |x_i - A| &= \sum_{i=1}^{2m+1} |x_i - A| \\ &= |X_1 - A| + |X_2 - A| + \dots + |X_m - A| \\ &\quad + |X_{m+1} - A| + |X_{m+2} - A| + \dots + |X_{2m} - A| \\ &\quad + |X_{2m+1} - A| \quad \dots \quad (2) \end{aligned}$$

There are $2m + 1$ terms on the right, and we group them in pairs, the terms equidistant from either end being paired together. Now the sum of the first and the last terms, viz.

$$|X_1 - A| + |X_{2m+1} - A|$$



has the minimum value, only when A lies between X_1 and X_{2m+1} , i.e. $X_1 \leq A \leq X_{2m+1}$.
The sum of the 2nd terms from either end, viz.

$$|X_2 - A| + |X_{2m} - A|$$

has the minimum value, only when A lies between X_2 and X_{2m} , i.e. $X_2 \leq A \leq X_{2m}$.
Continuing this way, the sum of the last pair, viz.

$$|X_m - A| + |X_{m+2} - A|$$

has the minimum value, only when A lies between X_m and X_{m+2} , i.e. $X_m \leq A \leq X_{m+2}$.
There is still another term, viz.

$$|X_{m+1} - A|$$

which has the minimum value zero, only when $A = X_{m+1}$.

It will now be noticed that (i) X_{m+1} is the median, because it occupies the middle-most position of the ordered series $X_1, X_2, \dots, X_m, X_{m+1}, X_{m+2}, \dots, X_{2m+1}$; and (ii) if $A = X_{m+1}$, it satisfies all the conditions for minimising the sums of the terms in pairs, and also the remaining term. Thus for odd n ,

$$\sum_{i=1}^n |x_i - A| \text{ is minimum, when } A = \text{median.}$$

Case II. When n is even, say $n = 2m$

$$\begin{aligned} \sum_{i=1}^n |x_i - A| &= \sum_{i=1}^{2m} |X_i - A| \\ &= |X_1 - A| + |X_2 - A| + \dots + |X_m - A| + \\ &\quad |X_{m+1} - A| + |X_{m+2} - A| + \dots + |X_{2m} - A| \quad \dots (3) \end{aligned}$$

As in Case I, we group the terms in pairs, there being exactly m pairs now. Proceeding the same way as before, we find that the sum on the right of (3) is minimum, only when A lies between the two middle-most values X_m and X_{m+1} of the ordered

series $X_1, X_2, \dots, X_m, X_{m+1}, \dots, X_{2m}$. In particular, if we take $A = \text{median} = \frac{1}{2}(X_m + X_{m+1})$,

we see that it lies between X_m and X_{m+1} , and as such minimises (3). Thus, for even n ,

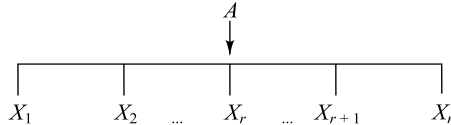
$$\sum_{i=1}^n |x_i - A| \text{ is minimum, when } A = \text{median.}$$

Combining the results of Case I and Case II, we conclude that in general

$$\sum_{i=1}^n |x_i - A| \text{ is the least, when } A = \text{median.}$$

(Alternative Proof for the Second Part)

Let $\Delta = \sum_{i=1}^n |x_i - A| = \text{Sum of the distances from } A,$
 $r = \text{Number of observations smaller than } A.$



If Δ represents the sum of the distances from the $(r+1)$ th value X_{r+1} , then the distance of each of the r values $X_1, X_2, \dots, X_r$, will increase by $(X_{r+1} - A) = c$ (say) and



the distance of each of the $(n-r)$ values $X_{r+1}, \dots, X_n$ will diminish by c , so that ultimately

$$\Delta' = \Delta + rc - (n-r)c = \Delta - (n-2r)c$$

Since c is positive,

$$\Delta' < \Delta, \text{ when } r < n/2$$

$$\Delta' = \Delta, \text{ when } r = n/2$$

$$\Delta' > \Delta, \text{ when } r > n/2$$

These relations imply that, if the origin is shifted to the next higher value, the sum of the distances from A .

- (i) will reduce further, so long as less than half the total number of observations is smaller than A .
- (ii) will remain constant, when $r = n/2$, i.e. exactly half the total number of observations is smaller than A (and naturally the same number greater than A).
- (iii) will increase, if more than half the total number of observations is smaller than A .

By (i) and (iii) we find that the minimum value of Δ will be attained when A lies between the $\frac{n}{2}$ th and the $\left(\frac{n}{2} + 1\right)$ th values. If n is odd, say $n = 2m + 1$, Δ will therefore be minimum, when A coincides with the $(m + 1)$ th value, i.e. $A = \text{Median}$. Again if n is even, say $n = 2m$, any value of A lying between the m th and $(m + 1)$ th given values (in order of magnitude) will minimise Δ , and by (ii), this Δ will remain a constant. Conventionally, we take $\text{Median} = \text{A.M. of the } m\text{th and the } (m + 1)\text{th values}$. Hence, in this case also, Δ will be minimum when $A = \text{Median}$.

This proves that $\sum |x_i - A|$ is the least, when $A = \text{Median}$.

5.14 MODE

Mode of a given set of observations is that value which occurs with the maximum frequency. It is the most typical or prevalent value, and at times represents the true characteristic of the distribution as a measure of central tendency. The modal wage of a group of workers is that wage which the largest number of workers receive, and as such this wage may be considered as the representative wage of the group. Mode is used in business, because it is most likely to occur. Meteorological forecasts are, in fact, based on mode.

5.15 CALCULATION OF MODE

From a simple series, mode can be determined by locating that value, which occurs the maximum number of times [Example 5.2, *Illustration 14(i)*].

From a simple frequency distribution, mode can be determined by inspection only. It is that value of the variable which corresponds to the largest frequency (Example 5.57).

From a grouped frequency distribution, it is very difficult to find the mode accurately. However, if all classes are of equal width, mode is usually calculated by the formula

$$\text{Mode} = l_1 + \frac{d_1}{d_1 + d_2} \times c \quad (5.15.1)$$



where

l_1 = lower boundary of the *modal class* (i.e the class containing the largest frequency);

d_1 = difference of the largest frequency and the frequency of class just preceding the modal class;

d_2 = difference of the largest frequency and the frequency of class just following the modal class;

c = common width of classes.

If f_0, f_{-1}, f_1 represent the frequencies of the modal class, the class just preceding and the class just following it, then

$$d_1 = f_0 - f_{-1} \quad \text{and} \quad d_2 = f_0 - f_1,$$

so that (5.15.1) may also be written in the form (see Example 5.61)

$$\text{Mode} = l_1 + \frac{f_0 - f_{-1}}{2f_0 - f_{-1} - f_1} \times c \quad (5.15.1a)$$

Formula (5.15.1) or (5.15.1a) is based on the assumption that the distances of mode from the two boundaries of modal class are proportional to the differences in frequencies of the modal class and its two adjoining classes, viz. d_1 and d_2 (Fig. 5.4, Page 154).

If however the frequency distribution has classes of unequal width formula (5.15.1) of (5.15.1a) cannot be applied. In this case, an approximate value of mode is usually obtained by the relation (5.17.1), when the mean and median are known.

5.16 ADVANTAGES AND DISADVANTAGES OF MODE

Advantages

1. From a simple frequency distribution, mode can be obtained only by inspection. Also, for a simple series with a small number of observations, mode can often be determined without any calculation.
2. It is unaffected by the presence of extreme values.
3. Unlike A. M., it can be calculated from frequency distributions, with open-end classes. In fact, if the modal class and its two adjoining classes together with their class frequencies are available, mode can be determined, provided it is known that all classes are of equal width.

Disadvantages

1. Mode has no significance unless a large number of observations is available.
2. It is a peculiar measure of central tendency. For any given set of observations, it is always possible to find the values of A.M., G.M. H.M., or median. But mode may not exist. When all values occur with equal frequency, there is no mode. On the other hand if two or more values have the same maximum frequency, there is more than one mode (A distribution is called *unimodal*, *bimodal* or *multimodal*, when it has one, two or more than two modes). See *Illustration 14*, Example 5.2 (p. 106)
3. From a grouped frequency distribution, it is difficult to locate the mode accurately. An approximate value of mode is obtained by the formula (5.15.1)



or (5.15.1a), only when all classes are of equal width, and the class with the largest frequency is preceded and followed by two other classes. Even then, if class frequencies do not show any gradual tendency to maximum concentration in a class, or the largest frequency occurs in two or more classes, the formula is inapplicable.

4. For the calculation of mode, the data must be arranged in the form of a frequency distribution.
5. Mode cannot be treated algebraically.

Example 5.57 The numbers of telephone calls received in 245 successive one minute intervals at an exchange are shown in the following frequency distribution:

No. of Calls	0	1	2	3	4	5	6	7	Total
Frequency	14	21	25	43	51	40	39	12	245

Evaluate the mean, median and mode.

[LC.W.A., June '74]

Solution [Hint: (i) $\Sigma fx = 922$, $N = 245$, $\bar{x} = \Sigma fx/N = 3.8$

(ii) Proceeding as in Example 5.46 (page, 166)

Median = Value corresponding to cumulative frequency $(N + 1)/2$, i.e. $123 = 4$

(iii) Mode is the value of the variable corresponding to the highest frequency 51; i.e. Mode = 4]

Ans. Mean = 3.8. Median = 4, Mode = 4.



Example 5.58 Find the modal wage from the following:

Wages (Rs)	50.00–59.99	60.00–69.99	70.00–79.99	80.00–89.99
No. of Employees	8	10	16	14
	90.00–99.99	100.00–109.99	110.00–119.99	
	10	5	2	

[B.U., B.A.(Econ) '72]

Solution The class 70.00–79.99, having the maximum frequency 16, is the 'modal class'; its lower boundary is 69.995. Since the classes are of equal width, formula (5.15.1a) can be applied: $l_1 = 69.995$, $f_0 = 16$, $f_{-1} = 10$, $f_1 = 14$, $c = 10$.

$$\text{Mode} = 69.995 + \frac{16 - 10}{2 \times 16 - 10 - 14} \times 10 = 77.50 \text{ Rs.}$$



Example 5.59 The monthly profits in rupees of 100 shops are distributed as follows:

Profits per Shop	0–100	100–200	200–300	300–400	400–500	500–600
No. of Shops	12	18	27	20	17	6

Draw the histogram to the data and hence find the modal value. Check this value by direct calculation.

[I.C.W.A., Jan. '64]

Solution The histogram has been drawn (Fig. 5.4)

In the histogram, the top right corner of the highest rectangle is joined by a straight line to the top right corner of the preceding rectangle. Similarly, the top left corner of the highest rectangle is joined to the top left corner of the following rectangle. From the point of intersection



of these two lines (dotted lines in Fig. 5.4) a perpendicular is drawn on the horizontal axis. The foot of the perpendicular indicates the Mode. This is read from the horizontal scale and the modal value is found to be 256 (in Rs) approximately.

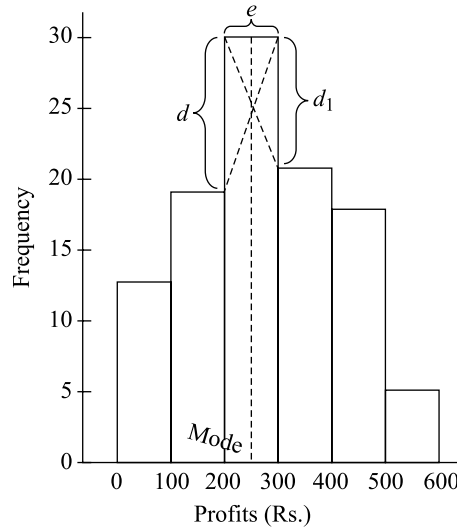


Fig. 5.4 Histogram for Distribution of Profits

In order to calculate the Mode directly, we find that the largest class frequency, viz, 27, lies in the class 200–300, and hence this is the *modal class*. Therefore, in formula (5.15.1), $l_1 = 200$, $d_1 = 27 - 18 = 9$, $d_2 = 27 - 20 = 7$, $c = 300 - 200 = 100$.

$$\text{Mode} = 200 + \frac{9}{9+7} \times 100 = 256.25 \text{ Rs.}$$

Ans. Rs 256.25 P.



Example 5.60 Calculate the Median and Mode from the following data:

Value	Frequency
Less than 10	4
Less than 20	16
Less than 30	40
Less than 40	76
Less than 50	96
Less than 60	112
Less than 70	120
Less than 80	125

[C.A., Nov. '67]

Solution We have to construct a cumulative frequency distribution for the calculation of Median and a grouped frequency distribution for the calculation of Mode (see Example 4.9, page 82). Using simple interpolation in Table 4.22, we can now find the value corresponding to cumulative frequency $N/2 = 125/2 = 62.5$, giving the median.



$$\frac{\text{Median} - 30}{40 - 30} = \frac{62.5 - 40}{76 - 40}$$

Solving this, we get Median = 36.25.

In order to find the mode, we apply formula (5.15.1). It will be found in Table 4.23 that the largest class frequency 36 corresponds to the class 30-40, and hence this is the 'modal class' (note that Table 4.23 shows class boundaries). Here $l_1 = 30$, $d_1 = 36 - 24 = 12$, $d_2 = 36 - 20 = 16$, $c = 40 - 30 = 10$.

$$\therefore \text{Mode} = 30 + \frac{12}{12 + 16} \times 10 = 34.29$$

Ans. Median = 36.25, Mode = 34.29.



Example 5.61 In a grouped distribution the class interval with the highest frequency has boundaries l_1 and l_2 . The frequencies in the two adjoining classes, preceding and following the modal class, are f_{-1} and f_1 respectively. If the frequency of the modal class is f_0 and if the classes are of equal width c , show that mode is given by

$$M_0 = l_1 + c \cdot \frac{f_0 - f_{-1}}{2f_0 - f_{-1} - f_1}$$

[B.U.B.A, (Econ) '71]

Solution (Refer to the histogram at Fig. 5.4, p, 155), Since all classes are of equal width, the frequencies f_{-1} , f_0 , f_1 , represent the heights of the 3 adjacent rectangles (starting from the left) with the tallest at the middle. The vertical lines forming boundaries of the tallest rectangle meet the x -axis at l_1 and l_2 , so that the Mode (M_0) lies in between them, i.e. $l_1 < M_0 < l_2$ (in the Fig. $l_1 = 200$, $l_2 = 300$) and $c = l_2 - l_1$.

Let d_1 and d_2 represent the differences in heights of the tallest rectangle from the two adjacent ones (see figure), i.e. $d_1 = f_0 - f_{-1}$, and $d_2 = f_0 - f_1$. If d_1 and d_2 were equal, the mode would lie exactly midway between l_1 and l_2 . In other words, distances $M_0 - l_1$, and $l_2 - M_0$ would be equal. The distance of M_0 from l_1 , is expected to be larger (smaller) than its distance from l_2 as d_1 is more (less) than d_2 . If we assume that the distance of the mode from the lower (upper) boundary is proportional to the corresponding difference in the heights of rectangles to the left (right), we have

$$\frac{M_0 - l_1}{l_2 - M_0} = \frac{d_1}{d_2}$$

$$\text{or, } d_2 (M_0 - l_1) = d_1 (l_2 - M_0)$$

$$\begin{aligned} \text{or, } M_0(d_1 + d_2) &= d_1 l_2 + d_2 l_1 \\ &= d_1(l_1 + c) + d_2 l_1, \quad \text{since } l_2 - l_1 = c \\ &= l_1(d_1 + d_2) + d_1 c \end{aligned}$$

Dividing both sides by $(d_1 + d_2)$, we get (5.15.1), viz.

$$M_0 = l_1 + \frac{d_1}{d_1 + d_2} \cdot c$$

Now substituting the values $d_1 = f_0 - f_{-1}$ and $d_2 = f_0 - f_1$, the required result follows.





5.17 RELATION BETWEEN MEAN, MEDIAN, MODE

For unimodal distributions of moderate skewness (Section 7.7) the following approximate relation has been found to hold:

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median}) \quad (5.17.1)$$

Sometimes this relation may be utilised for the calculation of mode. When the distribution is symmetrical, mean, median and mode coincide. In particular, for the “normal distribution” (Section 12.6) mean, median and mode are all equal.

In most frequency distributions, it has been observed that the three measures of central tendency, viz. Mean, Median and Mode, obey the approximate relation (5.17.1), provided the distribution is not very skew. This relation is therefore applied to estimate one of them when the values of the other two are known.

Example 5.62 *Arithmetic Mean = 26.8, Median = 27.9. What is the value of Mode ? Explain the principle involved.* [C.A. Nov. '72]

Solution Using (5.17.1), $26.8 - \text{Mode} = 3(26.8 - 27.9)$
or, $26.8 - \text{Mode} = -3.3$; $\therefore \text{Mode} = 30.1$



5.18 PARTITION VALUES—QUARTILES, DECILES, PERCENTILES

Just as median divides the total number of observations into two equal parts, there are similar other measures which are used to divide, or partition, the observations into a fixed number of parts, say 4, 10 or 100. These are collectively known as *partition values* or *quantiles* or *fractiles*. Some of the important types of partition values are

- (a) Median, (b) Quartiles, (c) Deciles, (d) Percentiles.

We know that Median is the middle-most value of a set of observations, i.e. it divides the total number of observations into 2 equal parts. The number of observations smaller than median is the same as the number larger than it. For data of continuous type, exactly one-half of the observations are smaller than median, i.e. median is the value of the variable corresponding to cumulative frequency $N/2$. These ideas are extended to Quartiles, Deciles and Percentiles.

Quartiles are such values which divide the total number of observations into 4 equal parts. Obviously, there are 3 quartiles—

- (i) *First quartile* (or *Lower quartile*) : Q_1
- (ii) *Second quartile*, (or *Middle quartile*) : Q_2
- (iii) *Third quartile* (or *Upper quartile*) : Q_3

The number of observations smaller than Q_1 , is the same as the number lying between Q_1 and Q_2 , or between Q_2 and Q_3 , or larger than Q_3 . For data of continuous type, one-quarter of the observations is smaller than Q_1 , two-quarters are smaller than Q_2 , and three-quarters are smaller than Q_3 . This means that Q_1 , Q_2 , Q_3 are values of the variable corresponding to ‘less-than’ cumulative frequencies $N/4$, $2N/4$, $3N/4$ respectively. Since, $2N/4 = N/2$, it is evident that the second quartile Q_2 is the same as median.



$$Q_1 < Q_2 < Q_3; \quad Q_3 = \text{Median.} \quad (5.18.1)$$

Quartiles are used for measuring central tendency, dispersion and skewness. For instance, the second quartile Q_2 is itself taken as a measure of central tendency, where it is known as *Median*. The lower and the upper quartiles, viz. Q_1 and Q_3 , are used to define *Quartile Deviation* (6.3.1), which is a measure of dispersion. In *Bowley's formula* (7.7.3) for skewness all the three quartiles are used.

$$\text{Median} = Q_2$$

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

$$\text{Skewness} = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

[**Note:** (i) Students must not carry a false impression that there is a common difference between the successive quartiles, or that $Q_3 - Q_2 = Q_2 - Q_1$. Only the number of observations' (i.e. frequency) between them is the same. In fact,

Less than Q_1 , the frequency = $N/4$

Between Q_1 and Q_2 , the frequency = $N/4$

Between Q_2 and Q_3 , the frequency = $N/4$

Above Q_3 , the frequency = $N/4$

(ii) Again, when asked to define quartiles, many students often write $Q_2 = N/4$, $Q_2 = 2N/4$, $Q_2 = 3N/4$. This is incorrect. The correct position is

Corresponding to Q_1 , the cumulative frequency = $N/4$

Corresponding to Q_2 , the cumulative frequency = $2N/4$

Corresponding to Q_3 , the cumulative frequency = $3N/4$

Deciles are such values which divide the total number of observations into 10 equal parts. There are 9 deciles $D_1, D_2, \dots, D_9$ called the *first decile*, the *second decile*, etc. The number of observations smaller than D_1 , or between two successive deciles, or larger than D_9 is the same. For data of continuous type, $D_1, D_2, \dots, D_9$ correspond to cumulative frequencies $N/10, 2N/10, \dots, 9N/10$ respectively.

$$D_1 < D_2 < \dots < D_9; \quad D_5 = Q_2 = \text{Median.} \quad (5.18.2)$$

Percentiles are such values which divide the total number of observations into 100 equal parts. There are 99 percentiles $P_1, P_2, \dots, P_{99}$, called the *first percentile*, the *second percentile*, and so on. The k -th percentile (P_k) is, therefore, that value of the variable upto which lie exactly $k\%$ of the total number of observations. Hence, P_k corresponds to 'less-than' cumulative frequency $kN/100$. In particular,

$$\begin{aligned} P_{10} &= D_1, & P_{20} &= D_2, \dots, & P_{90} &= D_9 \\ P_{25} &= Q_1, & P_{50} &= D_5 = Q_2 = \text{Median}, & P_{75} &= Q_3 \\ P_1 &< P_2 < \dots < P_{99} \end{aligned} \quad (5.18.3)$$

5.19 CALCULATION OF PARTITION VALUES

The method of calculation of quartiles, deciles, percentiles is exactly the same as that employed for median, using simple interpolation (page 77) in a cumulative frequency distribution. These may also be obtained graphically from ogive. For example, the 7th decile D_7 is the abscissa of that point on the ogive whose ordinate is $7N/10$.



(a) *From simple series*—The given data are arranged in increasing order of magnitude and a number showing the rank is attached to each observation. The smallest value is given rank 1, the next higher value rank 2, etc. and the largest value is given rank n . The ranks of partition values are as follows:

$$\begin{aligned}
 \text{Rank of Median} &= \frac{1}{2}(n+1) \\
 \text{Rank of } Q_1 &= \frac{1}{4}(n+1) \\
 \text{Rank of } Q_3 &= \frac{3}{4}(n+1) \\
 \text{Rank of } D_k &= \frac{k}{10}(n+1) \\
 \text{Rank of } P_k &= \frac{k}{100}(n+1)
 \end{aligned} \tag{5.19.1}$$

Using simple interpolation, the value of the variable corresponding to the appropriate rank is determined, giving the partition value (see Example 5.63)

$$\begin{aligned}
 \text{Median} &= \text{Value corresponding to rank } \frac{1}{2}(n+1) \\
 Q_1 &= \text{Value corresponding to rank } \frac{1}{4}(n+1) \\
 Q_3 &= \text{Value corresponding to rank } \frac{3}{4}(n+1) \\
 D_k &= \text{Value corresponding to rank } \frac{k}{10}(n+1) \\
 P_k &= \text{Value corresponding to rank } \frac{k}{100}(n+1)
 \end{aligned} \tag{5.19.2}$$

(b) *From simple frequency distribution*—The cumulative frequency corresponding to each distinct value of the variable is calculated (see Table 5.25, p. 142). If the total frequency be N .

$$\begin{aligned}
 \text{Median} &= \text{Value corresponding to cumulative frequency } \frac{1}{2}(N+1) \\
 Q_1 &= \text{Value corresponding to cumulative frequency } \frac{1}{4}(N+1) \\
 Q_3 &= \text{Value corresponding to cumulative frequency } \frac{3}{4}(N+1) \\
 D_k &= \text{Value corresponding to cumulative frequency } \frac{k}{10}(N+1) \\
 P_k &= \text{Value corresponding to cumulative frequency } \frac{k}{100}(N+1)
 \end{aligned} \tag{5.19.3}$$

(c) *From grouped frequency distribution*—(i) *By application of simple interpolation*—A cumulative frequency distribution is constructed showing the class boundaries



and the corresponding cumulative frequencies ('less-than' type). Using simple interpolation, we now find (see Examples 5.64, 5.65, 5.66)

$$\begin{aligned}
 \text{Median} &= \text{Value corresponding to cumulative frequency } \frac{1}{2}N \\
 Q_1 &= \text{Value corresponding to cumulative frequency } \frac{1}{4}N \\
 Q_3 &= \text{Value corresponding to cumulative frequency } \frac{3}{4}N \\
 D_k &= \text{Value corresponding to cumulative frequency } \frac{k}{10}N \\
 P_k &= \text{Value corresponding to cumulative frequency } \frac{k}{100}N
 \end{aligned} \tag{5.19.4}$$

If the cumulative frequencies are viewed as ranks, some authors define median as the value corresponding to rank $N/2$, Q_1 corresponding to rank $N/4$, etc. (see Example 5.68).

(ii) *Graphical method*—An ogive ('less-than' type) is drawn. From this ogive (see Example 5.66)

$$\begin{aligned}
 \text{Median} &= \text{Abscissa corresponding to ordinate } \frac{1}{2}N \\
 Q_1 &= \text{Abscissa corresponding to ordinate } \frac{1}{4}N \\
 Q_3 &= \text{Abscissa corresponding to ordinate } \frac{3}{4}N \\
 D_k &= \text{Abscissa corresponding to ordinate } \frac{k}{10}N \\
 P_k &= \text{Abscissa corresponding to ordinate } \frac{k}{100}N
 \end{aligned} \tag{5.19.5}$$

Example 5.63 Obtain the values of Median and the two Quartiles: 391 384 591 407 672 522 777 773 2488 1490

[C.A. May '69]

Solution The given values are arranged in increasing order of magnitude and ranked (Table 5.31). Here, $n = 10$; so using (5.19.1)

$$\begin{aligned}
 \text{Rank of Median} &= \frac{1}{2}(10 + 1) = 5.5 \\
 \text{Rank of } Q_1 &= \frac{1}{4}(10 + 1) = 2.75 \\
 \text{Rank of } Q_3 &= \frac{3}{4}(10 + 1) = 8.25
 \end{aligned}$$

**Table 5.31** Ordered Values and Their Ranks

Rank	Value
1	384
2	391
2.75 →	← Q_1
3	407
4	522
5	591
5.5 →	← Median
6	672
7	733
8	777
8.25 →	← Q_3
9	1490
10	2488

Using simple interpolation, we have

$$\frac{5.5 - 5}{6 - 5} = \frac{\text{Median} - 591}{672 - 591}; \quad \frac{2.75 - 2}{3 - 2} = \frac{Q_1 - 391}{407 - 391}$$

$$\frac{8.25 - 8}{9 - 8} = \frac{Q_3 - 777}{1490 - 777}$$

Solving the three equations,

$$\text{Median} = 631.5; \quad Q_1 = 403; \quad Q_3 = 955.25$$

[**Note:** Since the second quartile Q_3 is the same as Median, for practical purposes we speak of two quartiles only, viz. the lower quartile Q_1 , and the upper quartile Q_3].

Example 5.64 Determine Q_1 , Q_3 and the median (Q_2) from the following frequency table:

Marks in English	10–19	20–29	30–39	40–49	50–59	60–69	Total
Frequency	8	11	15	17	12	7	70

[B.U.,B.A. (Econ) '73]

Solution In a grouped frequency distribution, Q_1 , Q_2 and Q_3 are values of the variable corresponding to cumulative frequencies $N/4$, $2N/4$ and $3N/4$ respectively.

Table 5.32 Cumulative Frequency Distribution

Marks (Class boundary)	Cumulative Frequency (less-than)
9.5	0
19.5	8
$Q_1 \rightarrow$	← $N/4 = 17.5$
29.5	19
39.5	34
$Q_2 \rightarrow$	← $2N/4 = 35$
49.5	51
$Q_3 \rightarrow$	← $3N/4 = 52.5$
59.5	63
69.5	70 = N



[**Note:** The values of $N/4$, $2N/4$, $3N/4$ are first obtained numerically, and then their positions are indicated by arrow marks in the column of cumulative frequencies. Q_1 , Q_2 , Q_3 are then shown in the corresponding spaces under the first column. Here, since $N/4 = 70/4 = 17.5$ exceeds cumulative frequency 8, but is smaller than the next cumulative frequency 19, $N/4$ is shown in the space between 8 and 19. Q_1 therefore must lie in between the corresponding values of the variable, viz. 19.5 and 29.5. Similarly, $2N/4 = 35$ lies between the cumulative frequencies 34 and 51, and hence the position of Q_2 is shown between the corresponding values 39.5 and 49.5. Similarly for Q_3]

Using simple interpolation (3.16.1), we have

$$\frac{Q_1 - 19.5}{29.5 - 19.5} = \frac{17.5 - 8}{19 - 8}; \quad \frac{Q_2 - 39.5}{49.5 - 39.5} = \frac{35 - 34}{51 - 34}$$

$$\frac{Q_3 - 49.5}{59.5 - 49.5} = \frac{52.5 - 51}{63 - 51}$$

Solving these equations, we get $Q_1 = 28$, $Q_2 = 40$, $Q_3 = 51$



Example 5.65 From the following data calculate the first quartile and 4th decile :

X	0-5	5-10	10-15	15-25	25-35	35-60	60-80
Freq.	12	30	51	84	66	50	7

[C.U., B.Com. (Hons) '69]

Solution (Note that here all classes do not have the same width ; but this does not cause any difficulty in the calculation of partition values). The total frequency is $N = 300$. Hence, using (5.19.4), the first quartile Q_1 and the fourth decile D_4 are those values of X which correspond to

cumulative frequencies $\frac{1}{4} \times 300 = 75$ and $\frac{4}{10} \times 300 = 120$ respectively.

Table 5.33 Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency (less-than)
0	0
5	12
10	42
$Q_1 \rightarrow$	$\leftarrow N/4 = 75$
15	93
$D_4 \rightarrow$	$\leftarrow 4/N \cdot 10 = 120$
25	177
35	243
60	293
80	$300 = N$

Using simple interpolation,

$$\frac{Q_1 - 10}{15 - 10} = \frac{75 - 42}{93 - 42}, \text{ hence we get } Q_1 = 13.24$$

$$\frac{D_4 - 15}{25 - 15} = \frac{120 - 93}{177 - 93}, \text{ hence we get } D_4 = 18.21 \quad \text{Ans. } Q_1 = 13.24, D_4 = 18.21$$



**Example 5.66**

Height in inches	57.5–	60.0–	62.5–	65.0–	67.5–	70.0–	72.5
Number ...	6	26	190	281	412	7	38

Compute the third decile and the quartiles of the distribution given above. Check your results by graphical means. [I.C.W.A., July '65]

Solution By (5.19.4), D_3 , Q_1 , Q_2 , and Q_3 are values of the variable corresponding to

cumulative frequencies $\frac{3}{10} \times 1080 = 324$, $\frac{1}{4} \times 1080 = 270$, $\frac{2}{4} \times 1080 = 540$, and $\frac{3}{4} \times 1080 =$

810 respectively. Their positions are shown by arrow marks in the table below.

(Note that here the upper ends of classes have not been stated explicitly. But the class boundaries can be obtained; see Table 4.17 at p. 76 and Example 4.3 (h) at p. 77)

Table 5.34 Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency (less-than)
57.5	0
60.0	6
62.5	32
65.0	222
$Q_1 \rightarrow$	$\leftarrow N/4 = 270$
$D_2 \rightarrow$	$\leftarrow 3N/10 = 324$
67.5	503
$Q_2 \rightarrow$	$\leftarrow 2N/4 = 540$
$Q_3 \rightarrow$	$\leftarrow 3N/4 = 810$
70.0	915
72.5	1042
75.0	1080 = N

See Example 4.3(h), page 77

Now, using simple interpolation

$$(a) \quad \frac{Q_1 - 65.0}{67.5 - 65.0} = \frac{270 - 222}{503 - 222}; \quad \text{or, } Q_1 = 65.0 + \frac{48}{281} \times 2.5$$

$$= 65.43 \text{ inches}$$

$$(b) \quad \frac{D_3 - 65.0}{67.5 - 65.0} = \frac{324 - 222}{503 - 222}; \quad \text{or, } D_3 = 65.0 + \frac{102}{281} \times 2.5$$

$$= 65.91 \text{ inches}$$

$$(c) \quad \frac{Q_2 - 67.5}{70.0 - 67.5} = \frac{540 - 503}{915 - 503}; \quad \text{or, } Q_2 = 67.5 + \frac{37}{412} \times 2.5$$

$$= 67.72 \text{ inches}$$

$$(d) \quad \frac{Q_3 - 67.5}{70.0 - 67.5} = \frac{810 - 503}{915 - 503}; \quad \text{or, } Q_3 = 67.5 + \frac{307}{412} \times 2.5$$

$$= 69.36 \text{ inches}$$

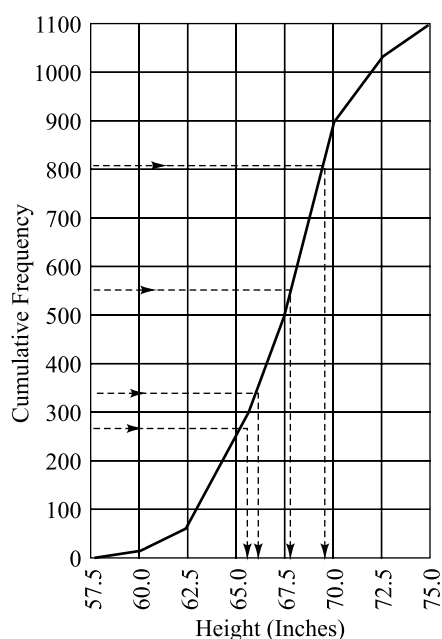


Fig. 5.5 Ogive and Location of Partition Values

We draw the ogive (less-than type) from the data in Table 5.34 (Fig. 5.5). Horizontal lines are drawn from cumulative frequencies 270, 324, 540, 810 on the vertical scale meeting the ogive. The abscissae of the points of intersection are read approximately from the horizontal scale, giving the values of Q_1 , D_3 , Q_2 , Q_3 respectively. (Students are advised to draw the ogive and check how far the graphical values agree with those obtained by direct calculation. The more accurate the drawing, the better will be the agreement. Graphical values are only approximate, and are hardly expected to agree exactly with those obtained by arithmetic interpolation).



Example 5.67 In a group of 500 wage earners the weekly wages of 4% were under Rs 30 and those of 15% were under Rs 32.50. The median and quartile wages were Rs 52.25, Rs 42.75 and Rs 60.50; the fourth and sixth decile wages were Rs 48.75 and Rs 55.25 respectively.

Put the above information in the form of a frequency distribution and calculate the second and third deciles. [C.U., B.A. (Econ) '74]

Solution Here the variable is 'weekly wage (Rs)'. Let us first compile the given information:

- (i) Total frequency $N = 500$,
- (ii) 4% of total frequency lie under 30,
- (iii) 15% of total frequency lie under 32.50,
- (iv) Median = 52.25, $Q_1 = 42.75$, $Q_3 = 60.50$ (note that of the two quartiles Q_1 is the smaller); i.e. 50%, 25% and 75% of total frequency lie under 52.25, 42.75 and 60.50 respectively.
- (v) $D_4 = 48.75$, $D_6 = 55.25$; i.e. 40% and 60% of total frequency lie under 48.75 and 55.25 respectively.



This information can be re-written as follows:

- (a) Under 30.00 the frequency is 4% of 500, i.e. 20
- (b) Under 32.50 the frequency is 15% of 500, i.e. 75
- (c) Under 52.25 the frequency is 50% of 500, i.e. 250
- (d) Under 42.75 the frequency is 25% of 500, i.e. 125
- (e) Under 60.50 the frequency is 75% of 500, i.e. 375
- (f) Under 48.75 the frequency is 40% of 500, i.e. 200
- (g) Under 55.25 the frequency is 60% of 500, i.e. 300

Now arranging the results at (a) to (g) above in increasing order of values of the variable, we have the following cumulative frequency distribution. The second and third deciles, viz. D_2 and D_3 , are values of the variable corresponding to cumulative frequencies $2N/10 = 100$ and $3N/10 = 150$ respectively. These may be obtained from Table 5.35, using simple interpolation.

Table 5.35 Cumulative Frequency Distribution

Weekly Wage (Rs)	Cumulative Frequency
30.00	20
32.50	75
$D_2 \rightarrow$	$\leftarrow 2N/10 = 100$
42.75	125
$D_3 \rightarrow$	$\leftarrow 3N/10 = 150$
48.75	200
52.25	250
55.25	300
60.50	375

$$\frac{D_2 - 32.50}{42.75 - 32.50} = \frac{100 - 75}{125 - 75}; \quad \frac{D_3 - 42.75}{48.75 - 42.75} = \frac{150 - 125}{200 - 125}$$

Solving these equations, we get $D_2 = 37.625$ Rs and $D_3 = 44.75$ Rs

The frequency distribution (grouped) is now constructed from Table 5.35 (as in Example 4.9).

Table 5.36 Frequency Distribution of Wages

Weekly Wages (Rs)	Frequency
Under 30.00	20
30.00–32.50	55
32.50–42.75	50
42.75–48.75	75
48.75–52.25	50
52.25–55.25	50
55.25–60.50	75
60.50 & above	125
Total	500

Ans. $D_2 = \text{Rs } 37.62$ P, $D_3 = \text{Rs } 44.75$ P



EXERCISES



- Find the mean and the median of: 88, 72, 33, 29, 70, 54, 86, 91, 57, 61.
[C.U., B.Com. '73]
- Find the mean, median and mode of the following numbers: 7, 4, 3, 5, 6, 3, 3, 2, 4, 3, 4, 3, 3, 4, 4, 3, 2, 2, 4, 3, 5, 4, 3, 4, 3, 4, 3, 1, 2, 3. [C.U., B.Com. '71]
- Evaluate the arithmetic mean, median, and mode for the following distribution of 'number of telephone calls received per one-minute interval':

No. of Calls	0	1	2	3	4	5	6	7	8
Frequency	5	22	31	43	51	40	35	15	3

[B.U., B.A.(Econ) '72]

- Calculate the simple and weighted average from the following and account for the difference between the two:

Price per ton (Rs P.)	45.60	50.70	42.45
Tons Purchased	135	40	25

[C.A., Nov. '72]

- The numbers 3.2, 5.8, 7.9 and 4.5 have frequencies x , $(x + 2)$, $(x - 3)$ and $(x + 6)$ respectively. If the arithmetic mean is 4.876, find the value of x .

[C.U., M.Com. '73]

- Calculate the arithmetic mean from the following data:

(i) Class Interval	50-59	60-69	70-79	80-89	90-99	100-109	110-119
Frequency	14	38	44	54	45	30	25
(ii) Height in Inches	57.5	-60.0	-62.5	-65.0	-67.5	-70.0	-72.5-
Number of Men	6	26	190	281	412	127	38
(iii) Weight in lbs.	137.5-147.5	147.5-157.5	157.5-167.5				
Number of Men	2	5	4				
	167.5-177.5	177.5-187.5	187.5-197.5	197.5-217.5	217.5-247.5		
	5	7	5	3	1		
(iv) X	20-30	30-50	50-100	100-200	200-350	350-550	
Freq.	2	9	11	52	10	3	

- From the data in the following table calculate the average marks of the M.Com. examinees in Statistics at a class test:

Marks	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No. of Examinees	2	3	11	20	32	25	7

[C.U., M.Com. '72]

- The following table gives the rise in price of 300 commodities between two dates. Calculate the mean rise in price:

% Increase	0-	5-	10-	15-	25-	35-	45-	60-80
Frequency	12	30	51	84	66	35	15	7

[Dip. Soc. Welfare, '71]



9. The following are the monthly salaries (in Rs) of 30 employees in a firm:

140	139	126	114	100	88	62	77	99	103
108	129	144	148	134	63	69	148	132	118
142	116	123	104	95	80	85	106	123	133

The firm gave bonus of Rs 10, 15, 20, 25, 30, 35 for individuals in the respective salary groups: 'exceeding Rs 60 but not exceeding Rs 75'; 'exceeding Rs 75 but not exceeding Rs 90'; and so on upto 'exceeding Rs 135 but not exceeding Rs 150'. Find the average bonus paid per worker.

[I.C.W.A., Dec. '76-old]

10. For the variable x , taking the values $0, 1, 2, \dots, k$, the cumulative frequencies of more-than type are $F_0, F_1, F_2, \dots, F_k$. Show that $\bar{x} = \sum_{i=1}^k F_i / n$, where n is the total frequency.

[B.U., B.A. (Econ) '73]

11. (a) The arithmetic mean calculated from the following frequency distribution is known to be 67.45 inches. Find the value of f_3 .

Height (inches)	60–62	63–65	66–68	69–71	72–74
Frequency	15	54	f_3	81	24

[I.C.W.A., July '71]

- (b) The expenditure of 1000 families is given below:

Expenditure (Rs):	40–59	60–79	80–99	100–119	120–139
No. of Families:	50	?	500	?	50

The median and mean for the distribution are both Rs 87.50 P. Calculate the missing frequencies.

[I.C.W.A., June '78]

12. Find out the missing frequencies of the following data, given that A.M. is 67.45 inches.

Height (inches)	60–62	63–65	66–68	69–71	72–74	Total
No. of Students	5	18	f_3	f_4	8	100

[Dip. Management, '72]

13. (i) The average marks obtained in an examination by two groups of students was found to be 75 and 85 respectively. Determine the ratio of students in the two groups, if the average mark for all students was 80.

[B.U., B.A. (Econ) '69]

- (ii) On a certain examination, the average grade of all students in class A is 68.4 and all students in class B is 71.2. If the average of both classes combined is 70.0, what is the ratio of the number of students in class A to the number in class B?

[C.U., M.Com. '62]

- (iii) The mean age of a combined group of men and women is 30 years. If the mean age of the group of men is 32 and that of the group of women is 27, find out the percentage of men and women in the group.

[C.A., Nov. '65]



14. Out of the total population in a certain town in South Africa, 60% belonged to the Black Race and the rest belonged to the White Race. It was estimated that their mean incomes were respectively 2,000 and 5,000 pounds. Find the average income of the entire town. [C.A., May '68]
15. (i) A factory has five sections employing 105, 184, 130, 93 & 125 workers. The mean earnings in a certain week per worker are Rs 13.80, Rs 15.00, Rs 15.20, Rs 18.20 and Rs 14.20 for the 5 sections. Determine the mean earning per worker of the whole factory. [Dip. Management, '70]
- (ii) In a survey of locality the following figures regarding the income of the people in different occupations were received. Find out the average per capita income:

Occupation	Average Income (in Rs)	Number of People
Business	500	700
Labour	300	300
Craftmanship	200	200
Other	400	100

[D.S.W., Nov. '70]

16. The following shows some data collected for three regions of a country:

Region	No. of Inhabitants (million)	Percentage of Literates	Average Annual Income per Person (Rs)
A	10	52	850
B	5	68	620
C	18	39	730

Obtain the over-all figures for the three regions taken together.

[C.U., B.A. (Econ) '77]

17. The population of India in 1951 and in 1961 were 361 and 439 million respectively. (i) What was the average percentage increase per year during the period? (ii) If the average rate of increase from 1961 to 1971 remains the same, what would be the population in 1971? [I.C.W.A., July '71]
18. A man gets three successive annual increments in salary of 20%, 30% and 25%, each percentage being reckoned on his salary at the end of the previous year. How much better or worse off would he have been if he had been given 3 annual increments of 25% each, reckoned in the same way? [I.C.W.A., Dec. '74-Old]
19. A machine is assumed to depreciate 40% in value in the first year, 25% in the second year and 10% per annum for the next 3 years, each percentage being calculated on the diminishing value. What is the average percentage depreciation, reckoned on the diminishing value, for the 5 years? [B.U., B.A. (Econ) '66; C.U., M.Com. '73]
20. The G.M. of 4 observations is 47, and the G.M. of 6 others is 40. Find the G.M. of all the 10 observations.
21. The geometric mean of six numbers is 75. If the geometric mean of four of them is 67, what is the geometric mean of the other two? [B.U., B.A. (Econ) '71]



22. You fly to a place X in a Boeing at a speed of 500 miles per hour and come back from X , following the same route, at a speed of 160 m.p.h. What is your average speed for the to-and-fro journey? [C.U., M.Com. '72]
23. An aeroplane flies around a square the sides of which measure 100 Kms. each. The aeroplane covers at a speed of 100 Kms. per hour the first side, at 200 Kms per hour the second side, at 300 Kms. per hour the third side, and at 400 Kms per hour the fourth side. Use the correct mean to find the average speed round the square. [I.C.W.A., June '78]
24. If two grades of oranges sell at 10 for Re. 1 and 20 for Re. 1 respectively, calculate the average price per orange, stating your assumptions explicitly. [B.U., B.A. (Econ) '69]
25. (a) The weights (in lb.) of 8 persons are: 138, 143, 141, 139, 152, 148, 160 and 267. Find the average weight using a suitable form of average. Give reasons for your choice.
- (b) What is the suitable form of average in each of the following cases? (i) The length of a rod is measured by a tape 10 times. You are to estimate the length of the rod by averaging these 10 determinations. (ii) A person purchases 5 rupees worth of eggs from 10 different markets. You are to find the average number of eggs per rupee for all the markets taken together. (iii) You are given the population of India for the censuses of 1961 and 1971. You are to find the population of India at the middle of the period by averaging these population figures, assuming a constant rate of increase of population. [B.U., B.A. (Econ) '72]
26. Find the mean and the median for the following data, and comment on the shape of the distribution:

Weight in Kg	36–40	41–45	46–50	51–55	56–60	61–65	66–70
No. of Persons	14	26	40	53	50	37	25

[I.C.W.A., June '75-old]

27. The G.M., H.M. and A.M. of three observations are 3.63, 3.27 and 4 respectively. Find the observations. [C.U., M.Com. '75]
28. Using a suitable formula calculate the median value from the following data:

Midvalue	115	125	135	145	155	165	175	185	195	Total
Frequency	6	25	48	72	116	60	38	22	3	390

[C.A., Nov. '66]

29. In a group of 1000 wage earners the monthly wages of 4% are below Rs 60 and those of 15% are under Rs 62.50. 15% earned Rs 95 and over, and 5% got Rs 100 and over. Find the median wage. [B.U., B.A. (Econ) '70]
30. The table below gives the frequency distribution of weights of 80 apples:

Weight (gms)	110–119	120–129	130–139	140–149	150–159	160–169	170–179	180–189
Frequency	5	7	12	20	16	10	7	3

Draw the cumulative frequency diagram and hence determine the median weight of an apple. [I.C.W.A., Dec. '76]



31. Draw the less-than Ogive and estimate the value of median on the basis of the data given below:

<i>Mid-Point</i>	18	25	32	39	46	53	60	
<i>Frequency</i>	10	15	32	42	26	12	9	$N = 146$

Verify the above result by calculation. [C.A., May '74]

32. An incomplete frequency distribution is given below:

<i>Height (inches)</i>	5.1–6.0	6.1–7.0	7.1–8.0	8.1–9.0	9.1–10.0	10.1–11.0	11.1–12.0
<i>No. of Plants</i>	3	8	27	?	17	11	9

It is known that the median height of the plant is 8.53 inches. Calculate the missing frequency. [I.C.W.A., Jan. '72]

33. In the following data two class frequencies are missing:

<i>C.I.</i>	100–110	110–120	120–130	130–140	140–150	150–160
<i>Frequency</i>	4	7	15	?	40	?
	160–170	170–180	180–190	190–200		
	16	10	6	3		

However it was possible to ascertain that the total number of frequencies was 150 and that the median has been correctly found out as 146.25. Find the two missing frequencies. [C.A., May '73]

34. In each of the following cases, explain whether the description applies to the mean, median or both:
- can be calculated from a frequency distribution with open end intervals.
 - the values of all items are taken into consideration in the calculation,
 - the values of extreme items do not influence the average,
 - in a distribution with a single peak and moderate skewness to the right, it is closer to the concentration of the distribution. [C.A., Nov. '65]

35. Calculate the value of the mode by the usual formula (after grouping if necessary):

x	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100	100–110
f	4	6	5	10	20	22	24	6	2	1

[C.A., May '74]

36. From the following distribution of weekly earnings, calculate (i) the most usual wage, and (ii) the percentage earning more than Rs. 31.50.

<i>Weekly Earnings (Rs):</i>	25–	26–	27–	28–	29–	30–	31–	32–	
<i>No. of Persons:</i>	25	70	210	275	430	550	340	130	
					33–	34–	35–	36	Total
					90	55	25		2200

[I.C.W.A., Dec. '73]



37. Find the mean and mode for the following:

<i>Year under</i>	10	20	30	40	50	60
<i>Number of Persons</i>	15	32	51	78	97	109

[B.U., B.A. (Econ) '73]

38. From the following cumulative frequency distribution of marks obtained by 22 students, calculate (a) Arithmetic mean, (b) Median, and (c) Mode.

<i>Marks</i>	<i>No. of Students</i>
Below 10	3
Below 20	8
Below 30	17
Below 40	20
Below 50	22

[I.C.W.A., June '77]

39. The table below gives the numbers (f) of candidates obtaining marks x or higher in a certain examination (all marks are given in whole numbers):

x	10	20	30	40	50	60	70	80	90	100
f	140	133	118	100	75	45	25	9	2	0

Calculate the mean and the median marks obtained by the candidates.

[I.C.W.A. June '75]

40. Calculate the values of (i) mean, (ii) median, and (iii) the two quartiles:

Income

(Rs '000): Under	1	1-2	2-3	3-5	5-10	10-25	25-50	50-100	100-1000
<i>No. of Persons:</i>	13	90	81	117	66	27	6	2	2

[C.U., M.Com. '74]

41. In a moderately asymmetrical distribution, the mean and the median are respectively 25.6 and 26.1 inches. What is the mode of the distribution?

[I.C.W.A., Jan. '71]

42. In a moderately skewed distribution, A. Mean = 24.6, and the mode = 26.1. Find the value of the Median and explain the reason for the method employed.

[C.A., Nov. '67]

43. Calculate arithmetic mean, and median of the frequency distribution given below. Hence calculate the mode using the empirical relation between the three.

<i>Class Limits:</i>	130-134	135-139	140-144	145-149	150-154	155-159	160-164
<i>Frequency:</i>	5	15	28	24	17	10	1

[I.C.W.A., Dec. '74]

44. Compute the median and the upper quartile of the following:

<i>Intelligence Quotient (IQ):</i>									
55-64	65-74	75-84	85-94	95-104	105-114	115-124	125-134	135-144	
<i>No. of Students:</i>									
2	20	79	184	302	207	82	24	4	

[C.U., B.A. (Econ) '76]



45. The weekly wages earned by the hundred workers of a factory are set out in the following table:

Weekly Wages (Rs):	12.5–17.5	17.5–22.5	22.5–27.5	27.5–32.5
No. of Workers:	12	16	25	14
	32.5–37.5	37.5–42.5	42.5–47.5	47.5–52.5
	13	10	6	3
				1

Calculate the three quantiles of the above distribution, taking $n/4$, $2n/4$, and $3n/4$, as their respective ranks. [C.A., Nov. '63]

46. The following table shows the age distribution of heads of families in a certain country during the year 1957. Find the median, the third quartile and the second decile of the distribution. Check your results by the graphical method:

Age of Head of Family (Yrs):	Under 25	25–29	30–34	35–44	45–54
Number (million):	2.3	4.1	5.3	10.6	9.7
	55–64	65–74	Above 74	Total	
	6.8	4.4	1.8	45.0	

[I.C.W.A., Jan. '73]

47. For an income distribution of a group of men 20 per cent of men have income below Rs 30, 35 per cent below Rs 70, 60 per cent below Rs 150 and 80 per cent below Rs 250. The first and third quartiles are Rs 50 and Rs 170.

Put the above information in a cumulative frequency distribution and find the median. [C.U., M.Com. '66]

48. "For a group of 5000 workers the weekly wages vary from Rs 20 to Rs 80. The wages of 4 per cent of the workers are under Rs 25 and those of 10 per cent are under Rs 30; 15 per cent of the workers earn Rs 60 and over, and 5 per cent of them get Rs 70 and over. The quartile wages are Rs 40 and Rs 54, and the sixth decile is Rs 50". Put the above information in the form of a frequency distribution and find the mean wage therefrom. [I.C.W.A., Jan. '71]

49. For a certain group of 'Saree' weavers of Varanasi, the median and quartile of earnings per week are Rs 44.30, Rs 43.00 and Rs 45.90 respectively. Ten per cent of the group earn under Rs 42 per week and 13% earn Rs 47 and over, and 6% Rs 48 and over. The range of earnings per week is Rs 40–Rs 50. Put the data into a frequency distribution. [C.U., B.A. (Econ) '70]

50. Comment on the following statements:

(a) "The median of a distribution is $N/2$, the lower quartile is $N/4$ and the upper quartile is $3N/4$ ". (Here N denotes the total frequency).

[I.C.W.A. Dec. '74]

(b) "If Q_1 , Q_2 , Q_3 be respectively the lower quartile, median and the upper quartile of a distribution, then $Q_2 - Q_1 = Q_3 - Q_2$ ". [I.C.W.A. June '75]

51. Given below is the frequency distribution of carbon content (present) in 150 determinations of a certain mixed powder.

Per cent Carbon:	4.0–4.1	4.2–4.3	4.4–4.5	4.6–4.7	4.8–4.9	5.0–5.1
Frequency:	1	2	7	20	25	30
				5.2–5.3	5.4–5.5	5.6–5.7
				10	25	30

Compute the arithmetic mean and median.

[I.C.W.A., Dec. '78]



52. Write True or False: (i) The mean, median and mode are located at the same point in a frequency distribution.
 (ii) Extreme values in the series affect the utility of the mean but not the median or mode. [D.S.W., '78]

53. Compute the arithmetic mean, median and mode of the following distribution and explain their relationships:

Monthly income

(Rs):	0-75	75-150	150-225	225-300	300-375	375-450
Frequency:	15	200	250	225	10	5

[C.U., M.Com. '76]

54. (a) If $z_1 = x_1 + y_1, z_2 = x_2 + y_2, \dots, z_n = x_n + y_n$, then prove that $\bar{z} = \bar{x} + \bar{y}$, where the symbols have their usual meaning.
 (b) Prove that the logarithm of geometric mean of observations is the arithmetic mean of logarithms of the observations. [D.M., '73]
55. The following are the population figures (in thousands) of 10 cities. Find the median: 2488, 1490, 777, 733, 522, 672, 591, 407, 387 and 391. [D.M., '78]



ANSWERS



1. 64.1, 65.5
2. 3.47, 3, 3
3. 3.90, 4, 4
4. Rs 46.25, Rs 46.23
5. 5
6. (i) 85.22, (ii) 67.45 inches, (iii) 176.72 lbs. (iv) 150.98
7. 72.5
8. 24.1%
9. Rs 24.50
10. (see Example 5.13)
11. (a) 126, (b) 250, 150
12. $f_3 = 42, f_4 = 27$
13. (i) 1 : 1, (ii) 3 : 4 (iii) Men 60%, Women 40%
14. £3200
15. (i) Rs 15.15, (ii) Rs 400
16. 33 million, 47.3%, Rs 749.70
17. 2%, 533.9 million
18. He would have received $R/320$ more (R = starting salary)
19. 20%
20. 42.67
21. 94.0
22. 242.4 m.p.h.
23. 192 k m.p.h.
24. 7.5 Paise (assuming equal numbers of the two grades are sold)
25. (a) Median = 145.5 lbs. (Since there is one extremely large value, A.M. will be unsuitable. Mode does not exist; moreover it will be inappropriate with only 8 observations). (b) (i) A.M., (ii) H.M., (iii) G.M.



26. Mean 54.3 Kg., Median 54.5 Kg. Since Mean is less than Median, distribution has negative skewness; i.e. the longer tail of the frequency curve lies to the left (see pages 114 and 272)
27. 2, 4, 6
28. 153.8
29. Rs 78.75
30. 147.5 gms.
31. 38.2
32. 25
33. 24, 25
34. (i) Median, (ii) Mean, (iii) Median, (iv) Median.
35. Since the frequency of the class immediately following the modal class has declined abruptly, re-group the given classes in two's giving 10–30, 30–50, 50–70, 70–90, 90–110 with frequencies 10 15, 42, 30, 3. Mode = 63.8
36. (i) Mode Rs 30.36, (ii) 21.4%
37. 29.95, 35.00 (years)
38. 22.7, 22.8, 23.5
39. 50.7, 51.2
40. (i) 8.06, (ii) 3.31, (iii) 1.98, 5.15 (all in Rs '000)
41. 27.1 inches
42. 25.1
43. 145.35, 144.92, 144.06
44. 100.03, 108.90
45. Quartiles 21.56, 26.90, 35.38 (Rs)
46. 44.7, 57.1, 32.0 years
47. Rs 118

48. <i>Weekly Wage (Rs)</i>	20–25	25–30	30–40	40–50	50–54	54–60
<i>Number of Workers</i>	200	300	750	1750	750	500
	60–70	70–80	Total			
	500	250	5000	Mean = Rs 4730		

49. <i>Weekly Earnings (Rs.)</i>	40.00–	42.00–	43.00–	44.30–
<i>Percentage of Weavers</i>	10	15	25	25
	45.90–	47.00–	48.00–50.00	Total
	12	7	6	100

50. (a) The statement is not correct. Median is the value of the $N/2$ -th item when the observations are arranged in order of magnitude, and not $N/2$ itself. In fact, $N/2$ is the *rank* of median. Similarly for the lower and the upper quartiles.
- (b) The statement is not generally correct. The number of observations (i.e. frequency) lying between Q_1 and Q_2 is the same as the number lying between Q_2 and Q_3 . The differences $(Q_2 - Q_1)$ and $(Q_3 - Q_2)$ may not be equal.
51. 5.118%, 5.083%
52. (i) False, (ii) True
53. (Rs) 190.69, 191.25, 200.00.
55. 631.5 thousand.

6 MEASURES OF DISPERSION

6.1 MEANING AND NECESSITY OF 'MEASURES OF DISPERSION'

The word *dispersion* is used to denote the 'degree of heterogeneity' in the data. It is an important characteristic indicating the extent to which observations vary among themselves. The dispersion of a given set of observations will be zero, only when all of them are equal. The wider the discrepancy from one observation to another, the larger will be the dispersion.

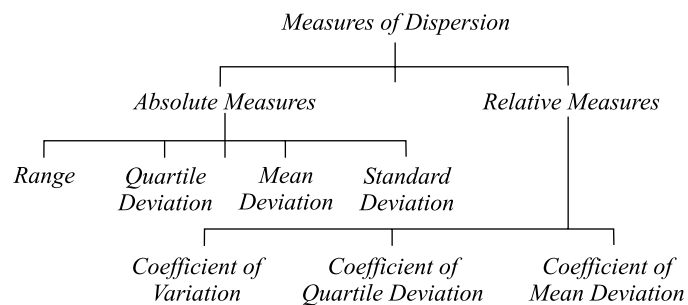
It may be noted that a measure of central tendency exhibits quite a different characteristic of data, viz. the size of a typical value representing a whole set of observations. Two series of observations may have the same average, but unequal dispersion. For example, consider the following:

Series A— 30, 32, 36, 38, 39, Total 175

Series B— 4, 19, 36, 53, 63, Total 175

These series with equal number of observations have the same mean and also the same median, so that when we consider their averages the two series are not different. However, it may be noticed that observations in Series A are close to one another, while those in Series B are widely dissimilar. We say that the *dispersion* (or *variability*) of Series B is more than that of Series A. Thus averages alone are not sufficient to reveal all the characteristics of data and hence the necessity of measures of dispersion.

A *measure of dispersion* is designed to state numerically the extent to which individual observations vary on the average. There are several measures of dispersion.





Example 6.1 *State the requisites of a satisfactory measure of dispersion and examine in their light any two common measures of dispersion.*

[I.C.W.A., Dec.'74]

Solution According to G.U. Yule, a good measure of dispersion should obey conditions similar to those (Example 5.1) for a satisfactory average:

- (i) It should be rigidly defined.
- (ii) It should be based on all observations.
- (iii) It should be readily comprehensible.
- (iv) It should be fairly easily calculated.
- (v) It should be affected as little as possible by fluctuations of sampling; and
- (vi) It should readily lend itself to algebraic treatment.

Standard Deviation (S.D.) obeys the majority of these requisites. It is rigidly defined, based on all observations, calculated fairly easily, and can be treated algebraically. Given the mean, the standard deviation and the number of observations in each of several groups, S.D. of the composite group can be determined. No other measure of dispersion has this property which makes S.D. the most suitable among the different measures. S.D. is also the least affected by sampling fluctuations. However, the significance of S.D. is not easy to understand. It is a round-about measure obtained by squaring deviations from mean, finding the average of these squares, and then taking the square-root (called 'Root-Mean-Square Deviation from mean').

Quartile Deviation is also rigidly defined and based on all observations. It has two definite advantages over S.D., viz. that it can be calculated with great ease and has a very simple meaning. It is half the difference between the two quartiles Q_1 and Q_3 , which include the central 50% of observations. It can also be calculated from frequency distributions with open-end classes. However, it has no simple algebraic properties and its behaviour as regards sampling fluctuations is difficult to decide.



Example 6.2 *What are the different measures of variability of observations?*

[C.U., B.A.(Econ) '72; M.Com. '71]

Solution The measures of variability (or dispersion) in common use are:

- (1) Range,
- (2) Quartile Deviation (or Semi-interquartile Range),
- (3) Mean Deviation, (or Mean Absolute Deviation),
- (4) Standard Deviation,
- (5) Coefficient of Variation.

Two other measures of dispersion, rarely used, are

- (6) Coefficient of Quartile Deviation,
- (7) Coefficient of Mean Deviation.

The measures (1) to (4) are known as 'absolute measures', and the remaining three, viz. the coefficients of dispersion, are called 'relative measures.'

Range is the difference between the largest and the smallest observations. It is simple to understand and easy to calculate, but is not generally used because of its many disadvantages.

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$



Illustration 1. For the observations (in Rs) 6, 4, 1, 6, 5, 10, 3, the maximum and the minimum values are 10 and 1. Therefore, $\text{Range} = 10 - 1 = 9$ Rs.

Quartile Deviation (also called *Semi-interquartile Range*) is defined as half the difference between the lower and the upper quartiles. This is also not very much used, because it does not take into account the variability of all observations.



$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

Illustration 2. The data in Illustration 1, arranged in order of magnitude, are 1, 3, 4, 5, 6, 6, 10, so that $Q_1 = 3$, $Q_3 = 6$. Therefore, Quartile Deviation = $\frac{1}{2}(6 - 3) = 1.5$ Rs.

Mean Deviation (also called *Mean Absolute Deviation*) is the arithmetic mean of absolute deviations (i.e. differences) from mean or any other specified value. Although it depends on all observations, the procedure of neglecting the signs and taking absolute deviations only makes algebraic treatment difficult.

$$\text{Mean Deviation} = \frac{1}{n} \sum |x_i - \bar{x}|$$

Illustration 3. For the data in Illustration 1, mean = $35/7 = 5$ Rs and deviations from mean are 1, -1, -4, 1, 0, 5, -2, so that absolute deviations (i.e. deviations without regard to signs) are 1, 1, 4, 1, 0, 5, 2. The arithmetic mean of these absolute deviations is Mean Deviation = $(1 + 1 + 4 + 1 + 0 + 5 + 2)/7 = 14/7 = 2$ Rs.

Standard Deviation is the square-root of the arithmetic mean of squares of deviations from arithmetic mean. It is the most useful among all measures of dispersion because of many advantage.

$$\text{Standard Deviation} = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$$

Illustration 4. In Illustration 3, the deviations from mean have been found. Therefore, Standard Deviation

$$\begin{aligned} \text{S.D.} &= \sqrt{\{1^2 + (-1)^2 + (-4)^2 + 1^2 + 0^2 + 5^2 + (-2)^2\}/7} \\ &= \sqrt{48/7} = \sqrt{686} = 2.6 \text{ Rs} \end{aligned}$$

Absolute measures of dispersion are expressed in the same unit as the observations, and therefore, cannot be used for comparing the variability of two or more series given in different units. The relative measures, defined below, are pure numbers and are employed for the purpose.

$$\text{Coefficient of Variation} = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

$$\text{Coefficient of Quartile Deviation} = \frac{\text{Quartile Deviation}}{\text{Median}} \times 100$$

$$\begin{aligned} \text{Coefficient of Mean Deviation} \\ &= \frac{\text{Mean Deviation about mean (or median)}}{\text{Mean (or median)}} \times 100 \end{aligned}$$

Illustration 5. For the data in illustration 1, Mean = 5 Rs, Standard Deviation = 2.6 Rs Therefore,

$$\text{Coefficient of Variation} = \frac{2.6 \text{ Rs}}{5 \text{ Rs}} \times 100 = 52.$$

Median = 5 Rs, and Q.D. = 1.5 Rs. Therefore,



$$\text{Coefficient of Quartile Deviation} = \frac{1.5 \text{ Rs}}{5 \text{ Rs}} \times 100 = 30$$

Mean = 5 Rs and Mean Deviation = 2 Rs. Therefore,

$$\text{Coefficient of Mean Deviation} = \frac{2 \text{ Rs}}{5 \text{ Rs}} \times 100 = 40.$$

[**Note:** All the relative measures of dispersion are defined as the ratio of an absolute measure of dispersion to the corresponding measure of central tendency, and the ratio is expressed as a percentage.]

Example 6.3 *Explain the advantages and disadvantages of the different measures of dispersion.* [C.U., M.Com. '72 ; B.A. (Econ) '72, '74]

Solution The advantages of Range are that it is very easy to understand and simple to calculate. However, Range has many disadvantages, which make it virtually useless as a reliable measure of dispersion. It is highly affected by the presence of even one extremely high or low value (Example 6.6). It does not depend on all observations. Lastly, Range cannot be calculated from frequency distributions with open-end classes.

Unlike Range, Quartile Deviation is dependent only on the smallest and the largest of the central 50% observations, and does not take into account the variability of the highest 25% and the lowest 25% observations. It can be calculated from frequency distributions with open-end classes (Example 6.8). However, it does not depend on the actual magnitudes of all observations.

Mean Absolute Deviation is based on all observations and hence takes into account the variability of each of them. However, the practice of neglecting signs and taking absolute deviations makes it difficult to be treated algebraically.

Standard Deviation is comparatively more difficult to calculate and not easy to understand. But, it has many advantages, for which it is most extensively used as a measure of dispersion. It is based on all observations, and is suitable for algebraic treatment. Given the mean, standard deviation and number of observations in each of several groups, standard deviation of the composite group can be calculated (Examples 6.39, 6.40). It is also the least affected by sampling fluctuations (see (4) of Example 6.12).

All absolute measures of dispersion suffer from the disadvantage that they are expressed in the same unit as original observations, and hence are unsuitable for comparing the dispersion of two or more sets of observations given in different units, e.g. heights in inches, and weights in lbs. However, relative measures are pure numbers and therefore may be used for this purpose (Example 6.54). Again, there are cases where it is found that although the unit of measurement is the same, the averages differ widely and in such cases a relative measure, e.g. Coefficient of Variation, provides a more useful measure of variability than standard deviation itself (Examples 6.48, 6.55, 6.56). Coefficient of Quartile Deviation and Coefficient of Mean Deviation are not generally used.



Example 6.4 *Distinguish between absolute and relative measures of dispersion.*

[C.A., Nov. '71; I.C.W.A., Jan. '73]

Solution Measures of dispersion are of two types—Absolute and Relative. (1) Absolute measures of dispersion are: Range, Quartile deviation, Mean deviation, and Standard deviation. Relative measures are: Coefficient of Variation, Coefficient of Quartile deviation, and Coefficient of Mean deviation.



(2) Absolute measures are expressed in the same unit in which the observations are given. Relative measures are obtained by expressing an absolute measure as percentage of a measure of central tendency and hence relative measures are independent of the units of measurement.

(3) Usually the absolute measures are employed for measuring dispersion. But for purposes of comparing the dispersion in different series, relative measures are used. Again when two sets of data are given in dissimilar units, there is no other alternative but to use a relative measure for comparison (Example 6.54).

(4) Relative measures may also be used to compare the relative accuracy of data. But absolute measures cannot be so used (Example 6.46).



6.2 RANGE

Range of a set of observations is the difference between the maximum and the minimum values.

$$\text{Range} = \text{Maximum value} - \text{Minimum value} \quad (6.2.1)$$

Example 6.5 Discuss the advantages and limitations of 'range' as a measure of variability of observations. [B.U., B.A.(Econ) '77]

Solution Range is the simplest of all measures of dispersion.

Advantages—(i) It is easy to understand. Range is the difference between the two extremes, viz. the largest and the smallest values, and as such represents the maximum possible difference between any two observations.

(ii) Range is also simple to calculate. This simplicity in the calculation of Range has led to its extensive application in Statistical Quality Control (Chapter 17). Here, 'Control Chart for Range' and estimates of variability based on Range are preferred.

However, Range has many limitations.

Limitations—(i) It does not depend on all observations, and is based on only the largest and the smallest among them. The values of intermediate observations, are not at all necessary for its calculation.

(ii) It is highly affected by extreme values. The presence of even one very high or low value is sufficient to increase Range many times.

(iii) It does not take into account the form of the distribution. The same value of Range might be obtained from a symmetrical or a J-shaped distribution, although the nature of dispersion in the two are completely different.

(iv) The greatest disadvantage of Range is that it cannot be calculated from frequency distributions with open-end classes (Example 6.8).



Example 6.6 The weight of 11 forty-year old men were 148, 154, 158, 160, 161, 162, 166, 170, 182, 195 and 236 pounds. If the heaviest man is omitted, what is the percentage change in the range? [C.U., M.Com '68]

Solution In the given data, maximum value = 236, minimum value = 148. Hence,

$$\text{Range} = 236 - 148 = 88 \text{ pounds.}$$

If the heaviest man is omitted, the maximum value 236 is cancelled, and among the remaining 10 observations, maximum = 195, minimum = 148.

$$\text{Range (revised)} = 195 - 148 = 47 \text{ pounds}$$

$$\text{Change in range} = 88 - 47 = 41, \text{ out of the original 88 pounds.}$$



$$\therefore \text{Percentage change in range} = \frac{41}{88} \times 100 = 47$$

Ans. 47%



6.3 QUARTILE DEVIATION (OR SEMI-INTERQUARTILE RANGE)

Quartile Deviation is defined as half the difference between the upper and the lower quartiles.

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2} \quad (6.3.1)$$

The difference $Q_3 - Q_1$ being the distance between the two quartiles, may be called *interquartile range*, and half of this is *Semi-interquartile Range* ('Semi' denotes 'half', e.g. semi-circle). Thus, the name 'Semi-interquartile Range' itself gives the definition of Quartile Deviation.

Quartile Deviation (Q.D.) is dependent on the two quartiles and does not take into account the variability of the largest 25% or the smallest 25% of observations. It is therefore unaffected by extreme values. Since in most cases the central 50% of observations tend to be fairly typical, Q.D. affords a convenient measure of dispersion. It can be calculated from frequency distributions with open-end classes (Example 6.8). Q.D. is thus superior to Range in many ways. Its unpopularity lies in the fact that Q.D. does not depend on the magnitudes of all observations. The calculation of Q.D. only depends on that of the two quartiles, Q_1 and Q_3 , which can be found from a cumulative frequency distribution using simple interpolation (Example 6.7).

Example 6.7 Calculate the quartile deviation from the following:

Class Interval	10-15	15-20	20-25	25-30	30-40	40-50	50-60	60-70	Total
Frequency	4	12	16	22	10	8	6	4	82

[C.A., Nov. '76]

Solution In order to compute Quartile Deviation, we have to find Q_1 and Q_3 , i.e. values of the variable corresponding to cumulative frequencies $N/4$ and $3N/4$. Here, total frequency $N = 82$. Therefore, $N/4 = 20.5$ and $3N/4 = 61.5$.

TABLE 6.1 Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency (less-than)
10	0
15	4
20	16
$Q_1 \rightarrow$	$\leftarrow N/4 = 20.5$
25	32
30	54
$Q_3 \rightarrow$	$\leftarrow 3N/4 = 61.5$
40	64
50	72
60	78
70	$82 = N$



Applying simple interpolation (page 77),

$$\frac{Q_1 - 20}{25 - 20} = \frac{20.5 - 16}{32 - 16}$$

$$\text{or, } \frac{Q_1 - 20}{5} = \frac{4.5}{16}$$

$$\text{or, } Q_1 - 20 = \frac{4.5}{16} \times 5 = 1.4$$

$$\therefore Q_1 = 20 + 1.4 = 21.4$$

$$\text{Similarly, } \frac{Q_3 - 30}{40 - 30} = \frac{61.5 - 54}{64 - 54}$$

$$\text{or, } \frac{Q_3 - 30}{10} = \frac{7.5}{10}$$

$$\text{or, } Q_3 - 30 = \frac{7.5}{10} \times 10 = 7.5$$

$$\therefore Q_3 = 30 + 7.5 = 37.5$$

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2} = \frac{37.5 - 21.4}{2} = 8.0$$



Example 6.8 Calculate the appropriate measure of dispersion from the following data:

Wages in Rupees per Week	No. of Wage Earners
less than 35	14
35-37	62
38-40	99
41-43	18
over 43	7

[I.C.W.A., Jan.'64]

Solution Since there are open-end classes in the frequency distribution. Quartile Deviation would be the most appropriate measure of dispersion for the data. So, we have to determine the quartiles Q_1 and Q_3 , which can be done from a cumulative frequency distribution using simple interpolation, as in Example 6.7.

TABLE 6.2 Cumulative Frequency Distribution

Wages (Rs) per Week	Cumulative Frequency
34.5	14
$Q_1 \rightarrow$	$\leftarrow N/4 = 50$
37.5	76
$Q_3 \rightarrow$	$\leftarrow 3N/4 = 150$
40.5	175
43.5	193
...	200 = N



Applying simple interpolation,

$$\frac{Q_1 - 34.5}{37.5 - 34.5} = \frac{50 - 14}{76 - 14}, \frac{Q_3 - 37.5}{40.5 - 37.5} = \frac{150 - 76}{175 - 76}$$

Solving $Q_1 = 36.24$ Rs., $Q_3 = 39.74$ Rs. Therefore,
 Quartile Deviation = $(39.74 - 36.24) / 2 = 1.75$ Rs.

[Note: (i) Because of the presence of open-end classes in the frequency distribution, Range cannot be determined. Mean Deviation and Standard Deviation also cannot be calculated, as mid-values for the open end classes cannot be obtained.

(2) Quartile deviation, and in fact all absolute measures of dispersion, must be expressed in the same unit in which observations are given].

6.4 MEAN DEVIATION (OR MEAN ABSOLUTE DEVIATION)

Mean Deviation (also called *Mean Absolute Deviation*) of a set of observations is the arithmetic mean of absolute deviations from mean or any other specified value (The term 'absolute deviation' used here refers to deviation without regard to sign, and is commonly known as 'difference').

Given the observations $x_1, x_2, \dots, x_n$, in order to 'find 'Mean Deviation about A', we first obtain the deviations $(x_1 - A), (x_2 - A), \dots, (x_n - A)$. Some of these deviations may be positive and some negative. If we write $|x_i - A|$ to denote the positive value of $(x_i - A)$, whatever be the actual sign, the sum of these 'absolute deviations' is

$$|x_1 - A| + |x_2 - A| + \dots + |x_n - A| = \sum |x_i - A|$$

and A.M. of the absolute deviations is

$$\text{Mean Deviation about } A = \frac{1}{n} \sum |x_i - A| \quad (6.4.1)$$

Mean Deviation (M.D.) is usually calculated about arithmetic mean ($\bar{x}$), and hence 'Mean Deviation' only refers to M.D. about mean.

For simple series,

$$\text{Mean Deviation} = \frac{1}{n} \sum |x_i - \bar{x}| \quad (6.4.2)$$

For frequency distribution,

$$\text{Mean Deviation} = \frac{1}{N} \sum f_i |x_i - \bar{x}| \quad (6.4.3)$$

An important property of M.D. is that it has the minimum value when deviations are taken from median, i.e. M.D. about median is the least (see Examples 6.17, 6.56).

Example 6.9 Calculate the Mean Deviation of the following values about the median: 8, 15, 53, 49, 19, 62, 7, 15, 95, 77 [C.U., B.Com. '73]

Solution Since there are an even number of observations, viz. 10, the median is the average of the two middlemost observations, when arranged in order of magnitude: 7, 8, 15, 15, (19, 49), 53, 62, 77, 95.

$\therefore$

$$\text{Median} = (19 + 49)/2 = 34$$

TABLE 6.3 Calculations for Mean Deviation

x	$ x - \text{median} $ i.e. difference from median
8	26
15	19
53	19
49	15
19	15
62	28
7	27
15	19
95	61
77	43
Total	272

Applying (6.4.1)

Mean Deviation about median

$$= \frac{1}{n} \sum |x - \text{median}|$$

$$= \frac{1}{10} \times 272$$

$$= 27.2$$

Example 6.10 Find the mean deviation of the following series:

x	10	11	12	13	14	Total
Frequency	3	12	18	12	3	48

[C.A., May '63]

*Solution***TABLE 6.4** Calculations for Mean Deviation

x	f	fx	$ x - \bar{x} $	$f x - \bar{x} $
10	3	30	2	6
11	12	132	1	12
12	18	216	0	0
13	12	156	1	12
14	3	42	2	6
Total	48	576	—	36

$$\bar{x} = \sum fx / N = 576 / 48 = 12$$

Applying (6.4.3), Mean Deviation = $36 / 48 = 0.75$

Ans. 0.75



Example 6.11 Measurements of the lengths in feet of 50 iron rods are distributed as follows:

Class Boundary	Frequency
2.35–2.45	1
2.45–2.55	4
2.55–2.65	7
2.65–2.75	15
2.75–2.85	11
2.85–2.95	10
2.95–3.05	2

Find, to two decimal places, the value of the mean deviation.

[C.U., M.Com. '65]

Solution Let $x_1, x_2, \dots, x_n$ represent mid-values of classes with frequencies $f_1, f_2, \dots, f_n$ respectively and $\bar{x}$ be their arithmetic mean. Then

$$\text{Mean Deviation} = \frac{1}{N} \sum f |x - \bar{x}|$$

where $N = \sum f$

TABLE 6.5 Calculations for Mean Deviation

Class Boundaries (ft.)	Frequency f	Mid-value x	$y = \frac{x - 2.70}{0.10}$	fy	$x - \bar{x}$	$f x - \bar{x} $
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2.35–2.45	1	2.40	–3	–3	–0.338	0.338
2.45–2.55	4	2.50	–2	–8	–0.238	0.952
2.55–2.65	7	2.60	–1	–7	–0.138	0.966
2.65–2.75	15	2.70	0	0	–0.038	0.570
2.75–2.85	11	2.80	1	11	0.062	0.682
2.85–2.95	10	2.90	2	20	0.162	1.620
2.95–3.05	2	3.00	3	6	0.262	0.524
Total	50	–	–	19	–	5.652

By (5.3.4), $\bar{x} = c + d\bar{y} = 2.70 + 0.10 \times \frac{19}{50} = 2.738$ ft. Using this, cols. (6) and (7) are then calculated in the above Table.

Mean Deviation = $5.652/50 = 0.11$ ft. (upto 2 decimals).



6.5 STANDARD DEVIATION (S.D.)

Standard Deviation of a set of observations is the square-root of the arithmetic mean of squares of deviations from arithmetic mean. In short, S.D. may be defined as “Root-Mean-Square-Deviation from mean”. It is usually denoted by the Greek small letter σ (sigma). If $x_1, x_2, \dots, x_n$ be a set of observations and $\bar{x}$ their A.M. then,



Deviations from mean: $(x_1 - \bar{x}), (x_2 - \bar{x}), \dots, (x_n - \bar{x})$
 Square-Deviations from mean: $(x_1 - \bar{x})^2, (x_2 - \bar{x})^2, \dots, (x_n - \bar{x})^2$
 Mean-Square-Deviation from mean:

$$\frac{1}{n} [(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2] = \frac{1}{n} \Sigma (x_i - \bar{x})^2$$

Root-Mean Square-Deviation from mean, i.e.

$$\text{Standard Deviation } (\sigma) = \sqrt{\frac{1}{n} \Sigma (x_i - \bar{x})^2} \quad (6.5.1)$$

The square of standard deviation, i.e. σ^2 , is known as *variance*.

Variance = (S.D.)²

For simple series,

$$\sigma^2 = \frac{1}{n} \Sigma (x_i - \bar{x})^2 \quad (6.5.2)$$

For frequency distribution,

$$\sigma^2 = \frac{1}{N} \Sigma f_i (x_i - \bar{x})^2 \quad (6.5.3)$$

S.D. is always considered as positive. Thus, S.D. is the positive square-root of Variance.

6.6 IMPORTANT PROPERTIES OF S.D.

(a) S.D. is independent of the change of origin ; i.e. if $y = x - c$, where c is a constant, then

$$\text{S.D. of } x = \text{S.D. of } y$$

In symbols,

$$\sigma_x = \sigma_y \quad (6.6.1)$$

This implies that the same S.D. will be obtained if each of the observations is increased or decreased by a constant.

(b) If two variables x and z are so related that $z = ax + b$ for each $x = x_i$, where a and b are constants, then

$$\sigma_z = |a| \cdot \sigma_x \quad (6.6.2)$$

where $|a|$ denotes the positive value of a .

In particular, if $y = (x - c)/d$, where c and d are constants (d positive), then

$$\sigma_x = d \cdot \sigma_y \quad (6.6.3)$$

This implies that S.D. does not depend on origin, but depends on scale of measurement. If each observation is multiplied or divided by a constant, S.D. will also be similarly affected.

(c) If a group of n_1 observations has mean $\bar{x}_1$ and S.D. σ_1 , and another group of n_2 observations has mean $\bar{x}_2$ and S.D. σ_2 , then S.D. (σ) of the composite group of $n_1 + n_2$ ($= N$, say) observations can be obtained by the formula

$$N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2) + (n_1d_1^2 + n_2d_2^2) \quad (6.6.4)$$

where $d_1 = \bar{x}_1 - \bar{x}$, $d_2 = \bar{x}_2 - \bar{x}$, and $N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$.

Relation (6.6.4) may be extended to any number of groups:

$$N\sigma^2 = \Sigma n_i\sigma_i^2 + \Sigma n_id_i^2 \quad (6.6.5)$$



where $d_i = \bar{x}_i - \bar{x}$, $N = \sum n_i$, and $\bar{x}$ is the mean of composite group, given by $N\bar{x} = \sum n_i \bar{x}_i$ (see 5.3.6).

(d) S.D. is the minimum root-mean-square-deviation, i.e.

$$\sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2} \leq \sqrt{\frac{1}{n} \sum (x_i - A)^2} \quad (6.6.6)$$

whatever be the value of A (see Example 6.16).

Example 6.12 Explain why the standard deviation is regarded as superior to other measures of dispersion. What is its chief defect?

[C.U., B.A.(Econ) '77; I.C.W.A, Dec. '75, June '76, B.U., B.A. '72, '74; '76]

Solution There are four absolute measures of dispersion—Range, Quartile Deviation, Mean Deviation, and Standard Deviation. Among these, S.D. is considered superior, because it possesses almost all the requisites of a good measure of dispersion (see Example 6.1, page 176).

(1) It is rigidly defined. S.D. of a set of observations is the square-root of the mean of squared deviations from mean.

(2) It is based on all observations. Even if one of the observations is changed, S.D. changes. However, Range and Quartile Deviation do not possess this property. The whole set of observations intermediate between the largest and the smallest values may be changed without affecting Range. Similarly, Quartile Deviation is based only on the two extremes of the central 50% of observations, and is independent of the magnitudes of other observations, so long as the two quartiles do not change.

(3) Although not so simple as Range or Quartile Deviation, the calculation of S.D. is not very difficult, and does not necessitate any special technique for computation. A change of origin does not affect S.D. As such, observations may be reduced conveniently by subtracting a constant, and then the formula applied to the reduced values (Examples 6.29, 6.28). If possible, the scale may also be changed to simplify the calculations further; but in that case, the S.D. obtained must be multiplied later (Example 6.30).

(4) S.D. is the least affected by fluctuations of sampling. If several independent samples are drawn from the same statistical population and each time all the four absolute measures of dispersion calculated, it will be found that S.D. fluctuates less from sample to sample than any other measure of dispersion.

(5) The unique property which makes S.D. superior to other measures of dispersion is that it is amenable to algebraic treatment. Given the number of observations, mean and S.D. for each of several groups, S.D. of the composite group can be calculated (Section 6.8).

Because of these advantages, S.D. is almost exclusively used unless there are definite reasons for using any other measure.

The chief defect of S.D. is that it cannot be used for comparing the dispersion of two or more series of observations given in different units. S.D. is also difficult to understand, and the process of squaring deviations from mean, and then taking the square-root of the mean of these squared deviations seems to be a complicated affair.



Example 6.13 Prove that the standard deviation is independent of change of origin.
[I.C.W.A June '76-old]

Solution Let $x_1, x_2, \dots, x_n$ be a set of n observations and let $y_1, y_2, \dots, y_n$ be the quantities derived from them by shifting the origin to an arbitrary constant c , i.e. $y_i = x_i - c$ ($i = 1, 2, \dots, n$). We have to prove that $\sigma_x = \sigma_y$. By definition (6.5.2),



$$\sigma_x^2 = \Sigma(x_i - \bar{x})^2/n, \quad \text{where } \bar{x} = \Sigma x_i/n.$$

Now, $y_i = x_i - c$, so that $\bar{x} = c + \bar{y}$ (see Example 5.10, p. 112)

Therefore, $x_i - \bar{x} = (c + y_i) - (c + \bar{y}) = y_i - \bar{y}$

Substituting this $\sigma_x^2 = \Sigma(x_i - \bar{x})^2/n = \Sigma(y_i - \bar{y})^2/n = \sigma_y^2$

or, $\sigma_x = \sigma_y$

This shows that σ_x is exactly equal to σ_y , proving thereby that S.D. has not changed by the choice of new origin c .



Example 6.14 Prove that the standard deviation is independent of any change of origin, but is dependent on the change of scale.

[I.C.W.A., July '71, Dec. '76; Dip. Management, May '72]

Solution Let $x_1, x_2, \dots, x_n$ be a set of observations. If we change the origin of x to c and the scale to d (positive), i.e. write

$$y_i = \frac{x_i - c}{d}, \quad \text{then} \quad x_i = c + dy_i$$

Hence, $\bar{x} = c + d\bar{y}$ (Example 5.11, page 113)

or, $x_i - \bar{x} = (c + dy_i) - (c + d\bar{y}) = d(y_i - \bar{y})$

$\therefore \sigma_x^2 = \Sigma(x_i - \bar{x})^2/n = \Sigma\{d(y_i - \bar{y})\}^2/n = d^2 \Sigma(y_i - \bar{y})^2/n$

or, $\sigma_x^2 = d^2 \sigma_y^2$, so that $\sigma_x = d \sigma_y$.

It is observed from the result that on the right hand side, the new origin c is absent, but the scale d is present. This proves that S.D. is unaffected by any change of origin, but depends on scale.



Example 6.15 Prove that the standard deviation calculated from two values x_1 and x_2 of a variable x is equal to half their difference. [I.C.W.A., Jan. '68]

Solution By definition (6.5.2),

$$\sigma^2 = \frac{1}{n} \Sigma(x_i - \bar{x})^2 = \frac{1}{2} \{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2\}.$$

Since there are only two values x_1 and x_2 , $\bar{x} = \frac{1}{2}(x_1 + x_2)$.

$$\therefore x_1 - \bar{x} = x_1 - \frac{1}{2}(x_1 + x_2) = \frac{x_1 - x_2}{2}$$

$$x_2 - \bar{x} = x_2 - \frac{1}{2}(x_1 + x_2) = \frac{x_2 - x_1}{2}$$

$$\therefore \sigma^2 = \frac{1}{2} \left\{ \left(\frac{x_1 - x_2}{2} \right)^2 + \left(\frac{x_2 - x_1}{2} \right)^2 \right\}$$

$$= \frac{1}{2} \left\{ \left(\frac{x_1 - x_2}{2} \right)^2 + \left(\frac{x_1 - x_2}{2} \right)^2 \right\}$$

$$= \frac{1}{4} (x_1 - x_2)^2$$

$$\therefore \sigma = \frac{1}{2} |x_1 - x_2|$$



[**Note:** It is necessary to take the positive value of $(x_1 - x_2)$, i.e. the difference of x_1 and x_2 , since S.D. is always positive.]



Example 6.16 Show that the standard deviation is the minimum root-mean-square deviation. [B.U., B.A. (Econ) '72]

Solution Let $x_1, x_2, \dots, x_n$ be a given set of observations. By definition (see page 184),

S.D. = Root-Mean-Square deviation from mean

$$= \sqrt{\Sigma(x_i - \bar{x})^2 / n} \quad \dots (i)$$

If instead of taking deviations from mean, we consider deviations from an arbitrary constant A , then

$$\text{Root-Mean-Square deviation from } A = \sqrt{\Sigma(x_i - A)^2 / n} \quad \dots (ii)$$

We have to show that (ii) cannot be smaller than (i) whatever be the value of A .

Now, it has been shown that (see Example 5.12, page 113)

$$\Sigma(x_i - A)^2 = \Sigma(x_i - \bar{x})^2 + n(\bar{x} - A)^2$$

Since the second term on the right can never be negative

$$\Sigma(x_i - A)^2 \geq \Sigma(x_i - \bar{x})^2$$

Dividing both sides by n and then taking the square-root, we have

$$\sqrt{\Sigma(x_i - A)^2 / n} \geq \sqrt{\Sigma(x_i - \bar{x})^2 / n} \quad (\text{shown})$$



Example 6.17 Show that the mean absolute deviation is least when the deviations are taken from median, but $\Sigma(x_i - A)^2 / n$ is least when $A = \bar{x}$, the arithmetic mean of $x_1, x_2, \dots, x_n$. [C.U., B.A. (Econ) '67, '73, '75; B.U., B.A. (Econ) '67; D.M. '77]

Solution (First Part) In Example 5.56 (page 150), it has been proved that $\Sigma|x_i - A|$ is minimum when $A = \text{median}$.

Hence, Mean Absolute Deviation about A , i.e. $\Sigma|x_i - A|/n$, is the least, when $A = \text{median}$.

(Second Part) see Example 5.12 (page 113).



Example 6.18 Show that $\bar{x}$ if is the arithmetic mean of the quantities $x_1, x_2, \dots, x_n$, then

$$\sum_1^n (x_i - \bar{x})^2 = \sum_1^n x_i^2 - n\bar{x}^2$$

[C.U., M.Com. '63; I.C.W.A., Jan. '69; D.M. '76]

Solution

(First method) By definition, $\bar{x} = \frac{1}{n} \sum_1^n x_i$

$$\text{so that} \quad n\bar{x} = \sum_1^n x_i \quad \dots (i)$$

$$\text{Therefore,} \quad \sum_1^n (x_i - \bar{x})^2 = \sum_1^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2)$$



$$\begin{aligned}
 &= \sum_1^n x_i^2 - \sum_1^n 2x_i\bar{x} + \sum_1^n \bar{x}^2 && \text{by (3.15.2)} \\
 &= \sum_1^n x_i^2 - 2\bar{x} \sum_1^n x_i + n\bar{x}^2, && \text{by (3.15.3) \& (3.15.4)} \\
 &= \sum_1^n x_i^2 - 2\bar{x} \cdot n\bar{x} + n\bar{x}^2, && \text{by (i) above} \\
 &= \sum_1^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\
 &= \sum_1^n x_i^2 - n\bar{x}^2.
 \end{aligned}$$

(Second method)

$$\begin{aligned}
 \sum_1^n (x_i - \bar{x})^2 &= (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2 \\
 &= (x_1^2 - 2x_1\bar{x} + \bar{x}^2) + (x_2^2 - 2x_2\bar{x} + \bar{x}^2) + \dots + (x_n^2 - 2x_n\bar{x} + \bar{x}^2). \\
 &= (x_1^2 + x_2^2 + \dots + x_n^2) - 2x_1\bar{x} - 2x_2\bar{x} - \dots - 2x_n\bar{x} + (\bar{x}^2 + \bar{x}^2 + \dots + \bar{x}^2). \\
 &= \sum_1^n x_i^2 - 2\bar{x}(x_1 + x_2 + \dots + x_n) + n\bar{x}^2 \\
 &= \sum_1^n x_i^2 - 2\bar{x} \cdot n\bar{x} + n\bar{x}^2 \\
 &= \sum_1^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\
 &= \sum_1^n x_i^2 - n\bar{x}^2. && \text{(Shown)}
 \end{aligned}$$

Example 6.19 Show that if $\bar{x}$ is the arithmetic mean of the values x_i ($i = 1, 2, \dots, n$), then

$$\sum_{i=1}^n f_i(x_i - \bar{x})^2 = \sum_{i=1}^n f_i x_i^2 - N\bar{x}^2, \quad \text{where } N = \sum_{i=1}^n f_i.$$

[C.U., M.Com. '64]

Solution (First method) If f_i represents the frequency of the value x_i ($i = 1, 2, \dots, n$), then by definition, the arithmetic mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i \quad \text{where } N = \sum_{i=1}^n f_i$$

By cross multiplication, $N\bar{x} = \sum_{i=1}^n f_i x_i \quad \dots \quad (i)$

Therefore,
$$\begin{aligned}
 \sum_{i=1}^n f_i(x_i - \bar{x})^2 \\
 &= \sum_{i=1}^n f_i(x_i^2 - 2x_i\bar{x} + \bar{x}^2)
 \end{aligned}$$



$$\begin{aligned}
 &= \sum_{i=1}^n (f_i x_i^2 - 2f_i x_i \bar{x} + f_i \bar{x}^2) \\
 &= \sum_{i=1}^n f_i x_i^2 - \sum_{i=1}^n 2f_i x_i \bar{x} + \sum_{i=1}^n f_i \bar{x}^2 && \text{by (3.15.2)} \\
 &= \sum_{i=1}^n f_i x_i^2 - 2\bar{x} \sum_{i=1}^n f_i x_i + \bar{x}^2 \sum_{i=1}^n f_i && \text{by (3.15.3)} \\
 &= \sum_{i=1}^n f_i x_i^2 - 2\bar{x} \cdot N\bar{x} + \bar{x}^2 \cdot N && \text{by (i) above} \\
 &= \sum_{i=1}^n f_i x_i^2 - 2N\bar{x}^2 + N\bar{x}^2 \\
 &= \sum_{i=1}^n f_i x_i^2 - N\bar{x}^2 && \text{(Shown)}
 \end{aligned}$$

(Second method) The arithmetic mean $\bar{x}$ of the values $x_1, x_2, \dots, x_n$ (having frequencies $f_1, f_2, \dots, f_n$ respectively) is given by

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{N}, \quad \text{where } N = \sum_{i=1}^n f_i$$

By cross multiplication, $N\bar{x} = f_1 x_1 + f_2 x_2 + \dots + f_n x_n$ (ii)

$$\begin{aligned}
 \text{Hence, } \sum_{i=1}^n f_i (x_i - \bar{x})^2 &= f_1(x_1 - \bar{x})^2 + f_2(x_2 - \bar{x})^2 + \dots + f_n(x_n - \bar{x})^2 \\
 &= f_1(x_1^2 - 2x_1\bar{x} + \bar{x}^2) + f_2(x_2^2 - 2x_2\bar{x} + \bar{x}^2) + \dots + f_n(x_n^2 - 2x_n\bar{x} + \bar{x}^2) \\
 &= (f_1 x_1^2 - 2f_1 x_1 \bar{x} + f_1 \bar{x}^2) + (f_2 x_2^2 - 2f_2 x_2 \bar{x} + f_2 \bar{x}^2) + \dots + (f_n x_n^2 - 2f_n x_n \bar{x} + f_n \bar{x}^2) \\
 &= f_1 x_1^2 + f_2 x_2^2 + \dots + f_n x_n^2 - 2f_1 x_1 \bar{x} - 2f_2 x_2 \bar{x} - \dots - 2f_n x_n \bar{x} + (f_1 \bar{x}^2 + f_2 \bar{x}^2 + \dots + f_n \bar{x}^2) \\
 &= \sum_{i=1}^n f_i x_i^2 - 2\bar{x}(f_1 x_1 + f_2 x_2 + \dots + f_n x_n) + \bar{x}^2(f_1 + f_2 + \dots + f_n) \\
 &= \sum_{i=1}^n f_i x_i^2 - 2\bar{x} \cdot N\bar{x} + \bar{x}^2 \sum_{i=1}^n f_i && \text{by (ii)} \\
 &= \sum_{i=1}^n f_i x_i^2 - 2N\bar{x}^2 + \bar{x}^2 N \\
 &= \sum_{i=1}^n f_i x_i^2 - 2N\bar{x}^2 + N\bar{x}^2 \\
 &= \sum_{i=1}^n f_i x_i^2 - N\bar{x}^2 && \text{(Shown)}
 \end{aligned}$$



Example 6.20 Show that the formula for standard deviation (σ) of $x_1, x_2, \dots, x_n$ can be expressed as

$$\begin{aligned}
 \sigma^2 &= \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 && \text{for simple series} \\
 \sigma^2 &= \frac{\sum f x^2}{N} - \left(\frac{\sum f x}{N} \right)^2 && \text{for frequency distributions}
 \end{aligned}$$



Solution For simple series

$$\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2, \quad \text{by definition (6.5.2)}$$

$$= \frac{1}{n} (\sum x_i^2 - n\bar{x}^2), \quad \text{by Example 6.18}$$

$$= \frac{\sum x_i^2}{n} - \bar{x}^2$$

$$= \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2, \quad \text{since } \bar{x} = \frac{\sum x}{n}.$$

For frequency distributions,

$$\sigma^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2, \quad \text{by definition (6.5.3)}$$

$$= \frac{1}{N} (\sum f_i x_i^2 - N\bar{x}^2), \quad \text{by Example 6.19}$$

$$= \frac{\sum f_i x_i^2}{N} - \bar{x}^2$$

$$= \frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N} \right)^2, \quad \text{since } \bar{x} = \frac{\sum fx}{N}.$$



Example 6.21 The standard deviation calculated from a set of 32 observations is 5. If the sum of the observations is 80, what is the sum of the squares of these observations? [I.C.W.A., Jan. '71]

Solution We are given $n = 32$, $\sigma = 5$, $\sum x = 80$. It is required to find the value of $\sum x^2$. Now from Example 6.20)

$$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2$$

$$\text{Substituting the values,} \quad 25 = \frac{\sum x^2}{32} - \left(\frac{80}{32} \right)^2$$

$$\text{Solving this, we get} \quad \sum x^2 = 1000.$$

Ans. 1000.



Example 6.22 Find the mean and the standard deviation of the first n natural numbers. [Dip. Management, May '67]

Solution The first n natural numbers are 1, 2, 3, ... n . The sum and the sum of squares of these n numbers are (see page 55)

$$\sum x = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\sum x^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\therefore \text{Mean } (\bar{x}) = \frac{\sum x}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}$$



$$\begin{aligned}
 \sigma^2 &= \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n} \right)^2, && \text{by Example 6.20} \\
 &= \frac{n(n+1)(2n+1)}{6} - \left\{ \frac{n(n+1)}{2n} \right\}^2 \\
 &= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} \\
 &= \frac{2(n+1)(2n+1) - 3(n+1)^2}{12} \\
 &= \frac{(n+1)(4n+2-3n-3)}{12} = \frac{(n+1)(n-1)}{12} = \frac{n^2-1}{12}
 \end{aligned}$$

Therefore, Standard deviation (σ) = $\sqrt{(n^2-1)/12}$

$$\text{Ans. Mean} = (n+1)/2, \quad \text{S.D.} = \sqrt{(n^2-1)/12}.$$



Example 6.23 The arithmetic mean and variance of n values $x_1, x_2, \dots, x_n$ are 0 and σ^2 respectively, and y is a variable defined by $y = x^2$. Prove that $\bar{y} = \sigma^2$.

[C.U., B.A. (Econ) '72; D.M. '78]

Solution We are given $\bar{x} = 0$, i.e. $\Sigma x/n = 0$, so that $\Sigma x = 0$.

$$\begin{aligned}
 \text{Again, } \sigma_x^2 &= \sigma^2; && \text{or, } \Sigma x^2/n - (\Sigma x/n)^2 = \sigma^2 \\
 \text{or, } &&& \Sigma x^2/n - (0/n)^2 = \sigma^2 \\
 \text{or, } &&& \Sigma x^2/n = \sigma^2
 \end{aligned}$$

Since, $y = x^2$, we have $\bar{y} = \Sigma y/n = \Sigma x^2/n = \sigma^2$

(Proved)



Example 6.24 The mean and S.D. calculated from 20 observations are 15 and 10 respectively. If an additional observation 5, left out through oversight, be included in the calculations, find the corrected mean and S.D. [I.C.W.A., Jan. '69]

Solution The sum of the 20 observations is $\Sigma x = 20 \times 15 = 300$. The sum of the squares of these 20 observations can be obtained from

$$\sigma^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n} \right)^2; \quad \text{or, } 10^2 = \frac{\Sigma x^2}{20} - \left(\frac{300}{20} \right)^2$$

so that $\Sigma x^2 = 6500$. When the additional observation 5 is included, for the 21 observations, the corrected values are $\Sigma x = 300 + 5 = 305$; $\Sigma x^2 = 6500 + 5^2 = 6525$. Therefore, Mean = $305/21 = 14.52$;

$$(\text{S.D.})^2 = \frac{6525}{21} - \left(\frac{305}{21} \right)^2 = \frac{44000}{441};$$

$\therefore$

$$\text{S.D.} = \sqrt{44000/441} = 9.99$$

Ans. Mean = 14.52; S.D. = 9.99





Example 6.25 The mean income per month of a friendly society of 25 members is Rs 350 and the standard deviation is Rs 50. Five more members are admitted to the society and their incomes in Rs per month are 260, 300, 320, 490 and 590. Find the mean and standard deviation of income for the new group of 30 members.

[I.C.W.A., July '69]

Solution Let us suppose that the income figures of 25 members form Group I and those of 5 new members form Group II. We have to find the mean and S.D. of the composite group of 30 observations.

For group I, (as in Example 6.24) $\Sigma x = 25 \times 350 = 8,750$

and $50^2 = \Sigma x^2/25 - (350)^2$.

Solving we get $\Sigma x^2 = 3,125,000$.

For group II, $\Sigma x = 260 + 300 + 320 + 490 + 590 = 1,960$

$\Sigma x^2 = 260^2 + 300^2 + 320^2 + 490^2 + 590^2 = 848,200$

Hence, for the composite group of $n = 30$ observations,

$\Sigma x = 8,750 + 1,960 = 10,710$

$\Sigma x^2 = 3,125,000 + 848,200 = 3,973,200$

Therefore,

$\bar{x} = 10710/30 = 357$ Rs

$$\sigma^2 = \frac{3973200}{30} - \left(\frac{10710}{30} \right)^2 = 4991; \sigma = \sqrt{4991} = 70.65 \text{ Rs}$$

Ans. Mean = 357 Rs; S.D. = 70.65 Rs



Example 6.26 The mean and standard deviation of 20 items is found to be 10 and 2 respectively. At the time of checking it was found that one item 8 was incorrect. Calculate the mean and standard deviation, if (i) the wrong item is omitted, and (ii) it is replaced by 12.

[I.C.W.A., Dec. '77]

Solution (As in Example 6.24) We have for the 20 items (including the incorrect item 8)

$\Sigma x = 20 \times 10 = 200$ and $2^2 = \Sigma x^2/20 - (10)^2$, giving

$\Sigma x = 200$ and $\Sigma x^2 = 2080$

(i) When the incorrect item 8 is omitted, we have for the remaining 19 items, $\Sigma x = 200 - 8 = 192$, $\Sigma x^2 = 2080 - 8^2 = 2016$. Now applying the formulae for mean and variance,

Mean = $192/19 = 10.11$

$(\text{S.D.})^2 = 2016/19 - (192/19)^2 = 3.9889$

$\therefore$ S.D. = $\sqrt{3.9889} = 1.997$

(ii) When the incorrect item 8 is replaced by 12, then for the 20 correct items $\Sigma x = 192 + 12 = 204$, $\Sigma x^2 = 2016 + 12^2 = 2160$. Hence, Mean = $204/20 = 10.2$

$(\text{S.D.})^2 = 2160/20 - (204/20)^2 = 3.96$

$\therefore$ S.D. = $\sqrt{3.96} = 1.990$

Ans. (i) 10.11, 1.997; (ii) 10.2, 1.990



Example 6.27 For a distribution of 280 observations, mean = 54 and standard deviation = 3. On checking it was discovered that observations which should correctly read as 62 and 82 had been wrongly recorded as 64 and 80 respectively. Calculate the correct values of mean and standard deviation.

[C.U., B.A. (Econ) '73, '75]



Solution For the 280 observations (which include the incorrect values 64 and 80), using the formulae for mean and variance, viz.

$$\begin{aligned}\bar{x} &= \Sigma x/n, & \sigma^2 &= \Sigma x^2/n - (\Sigma x/n)^2 \\ \text{we have } 54 &= \Sigma x/280, & \text{so that } \Sigma x &= 280 \times 54 \\ 3^2 &= \Sigma x^2/280 - (54)^2. & \text{Hence, } \Sigma x^2 &= 280 \times 2925\end{aligned}$$

When 64 and 80 are replaced by 62 and 82, the corrected values are

$$\begin{aligned}\Sigma x &= 280 \times 54 - (64 + 80) + (62 + 82) = 280 \times 54 \\ \Sigma x^2 &= 280 \times 2925 - (64^2 + 80^2) + (62^2 + 82^2) = 819072\end{aligned}$$

Using these in the formulae for mean and variance,

$$\begin{aligned}\text{Mean} &= (280 \times 54)/280 = 54 \\ (\text{S.D.})^2 &= 819072/280 - (54)^2 = 9.257 \\ \therefore \text{S.D.} &= \sqrt{9.257} = 3.04\end{aligned}$$

Ans. Mean = 54, S.D. = 3.04



6.7 CALCULATION OF STANDARD DEVIATION (σ)

If the observations are small, S.D. can be calculated by using the following relations (see Example 6.20).

For simple series

$$\sigma^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n} \right)^2 \quad (6.7.1)$$

For frequency distributions

$$\sigma^2 = \frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N} \right)^2 \quad (6.7.2)$$

The calculations can, however, be simplified based on the following results (see Examples 6.13, 6.14, at p. 186, 187)

(I) if $y_1, y_2, \dots, y_n$ represent the deviations of $x_1, x_2, \dots, x_n$ from an arbitrary constant c , then

$$\text{S.D. of } x = \text{S.D. of } y$$

In symbols, if $y = x - c$, then $\sigma_x = \sigma_y$. (6.7.3)

This result implies that in order to find the S.D. we can always reduce the given observations by subtracting a convenient number. S.D. of the reduced values gives the S.D. of the original values of x .

(II) If $y_1, y_2, \dots, y_n$ represent the deviations of $x_1, x_2, \dots, x_n$ from an arbitrary constant c , in units of another constant d , then

$$\text{S.D. of } x = d (\text{S.D. of } y).$$

In symbols, if $y = \frac{x - c}{d}$, then $\sigma_x = d \cdot \sigma_y$ (6.7.4)

This implies that, if after taking deviations of x from a constant c , it is possible to reduce these deviations on division by a constant d , then the S.D. of these reduced values when multiplied by d , will give the S.D. of the original values of x . The result is particularly suitable when the consecutive values of x have a common difference. In such cases the values of y can be reduced to 0, 1, 2, 3, ... and $-1, -2, -3, \dots$, whatever be the values of x .



σ_y used in (6.7.3) or (6.7.4) is calculated by using formulae similar to (6.7.1) or (6.7.2), replacing x by y , i.e.

$$\sigma_y^2 = \frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n} \right)^2 \quad \text{for simple series}$$

$$\sigma_y^2 = \frac{\Sigma fy^2}{N} - \left(\frac{\Sigma fy}{N} \right)^2 \quad \text{for frequency distributions}$$

For the calculation of S.D. from a grouped frequency distribution, see Example 6.32.

Example 6.28 Find the standard deviation from the following data: 49, 63, 46, 59, 65, 52, 60, 54. [C.U., B.Com. '72]

Solution Since the standard deviation remains unaffected by change of origin, we may reduce each observation by 50 (suppose) and write $y = x - 50$. By (6.7.3),

$$\sigma_x = \sigma_y \quad \text{where} \quad \sigma_y^2 = \frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n} \right)^2$$

Table 6.6 Calculations for S.D.

x	$y = x - 50$	y^2
49	-1	1
63	13	169
46	-4	16
59	9	81
65	15	225
52	2	4
60	10	100
54	4	16
Total	48	612

Here, $n = 8$, $\Sigma y = 48$, $\Sigma y^2 = 612$

$$\begin{aligned} \therefore \sigma_y^2 &= \frac{612}{8} - \left(\frac{48}{8} \right)^2 \\ &= 76.5 - (6)^2 = 40.5 \\ \sigma_y &= \sqrt{40.5} = 6.36 \end{aligned}$$

Hence, $\sigma_x = \sigma_y = 6.36$

Ans. 6.36

[Note: Students are advised to try this problem by taking different origins. For instance, if the origin is taken at 55, i.e. $y = x - 55$, we have the following table.

**Table 6.7** Calculations for S.D.

x	$y = x - 55$	y^2
49	-6	36
63	8	64
46	-9	81
59	4	16
65	10	100
52	-3	9
60	5	25
54	-1	1
Total	8	332

$$\sigma_y^2 = \frac{332}{8} - \left(\frac{8}{8}\right)^2$$

$$= 41.5 - (1)^2 = 40.5$$

$$\therefore \sigma_y = \sqrt{40.5} = 6.36$$

$$\text{Hence, } \sigma_x = 6.36$$

We get the same result as before. This verifies that S.D. does not depend on the origin. However, numerical computations will be small, when a value close to the mean is taken as origin.



Example 6.29 Compute the standard deviation of household size from the following frequency distribution of 500 households:

Household Size	1	2	3	4	5	6	7	8	9
No. of Households	92	49	52	82	102	60	35	24	4

[C.U., B.A. (Econ) '77]

Solution Since the S.D. is unaffected by change of origin, we write $y = x - 4$; then $\sigma_x = \sigma_y$.

Table 6.8 Calculations for S.D.

x	f	$y = x - 4$	fy	fy^2
1	92	-3	-276	828
2	49	-2	-98	196
3	52	-1	-52	52
4	82	0	0	0
5	102	1	102	102
6	60	2	120	240
7	35	3	105	315
8	24	4	96	384
9	4	5	20	100
Total	500	—	17	2217

$$\sigma_y^2 = \frac{\Sigma fy^2}{N} - \left(\frac{\Sigma fy}{N}\right)^2 = \frac{2217}{500} - \left(\frac{17}{500}\right)^2$$

$$= 4.434 - .0012 = 4.4328$$

$$\sigma_y = \sqrt{4.4328} = 2.11;$$

$\therefore$

$$\sigma_x = \sigma_y = 2.11$$

Ans. 2.11

Example 6.30 Find out the S.D. for the following table:

x	5	15	25	35	45	55	65	75
f	3	7	9	23	15	8	6	4

[D.S.W., Nov. '70]

Solution Since the successive values of x have a common difference, we use formula (6.7.4).

Table 6.9 Calculations for S.D.

x	f	$y = \frac{x - 35}{10}$	fy	fy^2
5	3	-3	-9	27
15	7	-2	-14	28
25	9	-1	-9	9
35	23	0	0	0
45	15	1	15	15
55	8	2	16	32
65	6	3	18	54
75	4	4	16	64
Total	75	-	33	229

$$\sigma_y^2 = \frac{\Sigma fy^2}{N} - \left(\frac{\Sigma fy}{N} \right)^2 = \frac{229}{75} - \left(\frac{33}{75} \right)^2 = 3.053 - (0.44)^2 = 2.859$$

$$\therefore \sigma_y = \sqrt{2.859} = 1.69. \quad \sigma_x = d \cdot \sigma_y = 10 \times 1.69 = 16.9$$

[**Note:** For the calculation of S.D. only an extra column fy^2 is needed beyond what is necessary for the calculation of A.M.]

Example 6.31 How do you calculate the Standard Deviation of a grouped variate?

[C.U., M.Com.'61]

Solution For finding the standard deviation from a grouped frequency distribution, the mid-values are taken as representatives of all observations falling in the respective classes (see Example 5.16). If $x_1, x_2, \dots, x_n$ be the mid-values of classes with frequencies $f_1, f_2, \dots, f_n$ respectively, and $y_1, y_2, \dots, y_n$ represent the deviations of these mid-values from an arbitrary origin c , in units of another constant d , then $\sigma_x = d \cdot \sigma_y$, where

$$\sigma_y^2 = \frac{\Sigma fy^2}{N} - \left(\frac{\Sigma fy}{N} \right)^2 \text{ and } N = \Sigma f.$$

Calculations are done in the following tabular form:

**Table 6.10** Calculations for S.D.

Class Interval	Frequency f	Mid-value x	Deviation $= x - c$	$y = \frac{x - c}{d}$	fy	fy^2
(1)	(2)	(3)	(4)	(5)	(6)	(7)
	f_1	x_1		y_1		
	f_2	x_2		y_2		
	$\vdots$	$\vdots$		$\vdots$		
	$\vdots$	$\vdots$		$\vdots$		
	f_n	x_n		y_n		
Total	$N = \Sigma f$	—	—	—	Σfy	Σfy^2

The steps in the calculation are as follows:

- (i) From the given class intervals, we find the mid-values in col. (3). This is done by taking the arithmetic mean of class limits or class boundaries, as given in the frequency distribution.
- (ii) One of the mid-values is chosen as the arbitrary origin c , preferably the mid-value corresponding to the maximum frequency, if it lies near the middle of the distribution. Deviation of each mid-value from c is calculated, as shown in col. (4).
- (iii) These deviations are divided by their highest common factor (d) and the results, called *step deviations* (y), are shown in col. (5). If all class intervals have the same width, d will be equal to this width. (In such a case, it will be found that we get 0 against the mid-value which has been taken as origin, and $-1, -2, -3, \dots$ successively as we move away towards the smaller mid-values, and $+1, +2, +3, \dots$ towards the higher mid-values).
- (iv) Cols. (2) and (5) are multiplied to get col. (6).
- (v) Cols. (5) and (6) are multiplied to complete col. (7).
- (vi) Total of cols. (2), (6) and (7) are obtained, and these values are substituted in the formula to obtain the required S.D.



Example 6.32 Compute the standard deviation from the following distribution of marks obtained by 90 students:

Marks	20–29	30–39	40–49	50–59	60–69	70–79	80–89	90–99
No. of Students	5	12	15	20	18	10	6	4

[I.C.W.A., June '76]

Solution

Table 6.11 Calculations for S.D.

Class Interval	Frequency f	Mid-value x	$y = \frac{x - 54.5}{10}$	fy	fy^2
(1)	(2)	(3)	(4)	(5)	(6)
20–29	5	24.5	–3	–15	45
30–39	12	34.5	–2	–24	48
40–49	15	44.5	–1	–15	15
50–59	20	54.5	0	0	0

(Contd.)



Class Interval	Frequency f	Mid-value x	$y = \frac{x - 54.5}{10}$	fy	fy^2
60–69	18	64.5	1	18	18
70–79	10	74.5	2	20	40
80–89	6	84.5	3	18	54
90–99	4	94.5	4	16	64
Total	90	–	–	18	284

If $y = \frac{x - c}{d}$, then $\sigma_x = d \cdot \sigma_y$. Here in col. (4), $y = \frac{x - 54.5}{10}$; hence $c = 54.5$, $d = 10$.

$$\begin{aligned}\sigma_y^2 &= \frac{\Sigma fy^2}{N} - \left(\frac{\Sigma fy}{N} \right)^2 = \frac{284}{90} - \left(\frac{18}{90} \right)^2 \\ &= 3.156 - .04 = 3.116\end{aligned}$$

$\therefore$

$$\begin{aligned}\sigma_y &= \sqrt{3.116} = 1.77; \\ \sigma_x &= 10 \times 1.77 = 17.7\end{aligned}$$

Ans. 17.7



Example 6.33 Find the standard deviation from the following frequency distribution:

Height in Inches	No. of Students
Over 60 but not more than 62	35
Over 62 but not more than 64	27
Over 64 but not more than 66	20
Over 66 but not more than 68	13
Over 68 but not more than 70	5
	<u>100</u>

[C.U., B.Com. '76]

Solution

Table 6.12 Calculations for S.D.

Height* (inch)	Frequency f	Mid-value x	$y = \frac{x - 65}{2}$	fy	fy^2
(1)	(2)	(3)	(4)	(5)	(6)
60–62	35	61	–2	–70	140
62–64	27	63	–1	–27	27
64–66	20	65	0	0	0
66–68	13	67	1	13	13
68–70	5	69	2	10	20
Total	100	–	–	–74	200

* See Table 4.18(D) and Example 4.3(1) at p, 76–77.



If $y = (x - c)/d$, then $\sigma_x = d \cdot \sigma_y$

$$\begin{aligned}\sigma_y^2 &= \frac{\sum fy^2}{N} - \left(\frac{\sum fy}{N} \right)^2 = \frac{200}{100} - \left(\frac{-74}{100} \right)^2 \\ &= 2.00 - (-0.74)^2 = 2.00 - 0.5476 = 1.4524 \\ \sigma_y &= \sqrt{1.4524} = 1.21; \sigma_x = 2 \times 1.21 = 24 \text{ inches.}\end{aligned}$$



Example 6.34 Calculate the standard deviation for the following frequency distribution. Find out the numbers which will lie between $\bar{x} \pm s$ and $\bar{x} \pm 2s$.

Values	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	2	5	7	13	21	16	8	3

[D.S.W., Nov. '78]

Solution (Note that here s denotes the standard deviation).

Table 6.13 Calculations for Mean and S.D.

Class Interval	Frequency (f)	Mid-value (x)	$y = \frac{x - 22.5}{5}$	fy	fy ²
(1)	(2)	(3)	(4)	(5)	(6)
0-5	2	2.5	-4	-8	32
5-10	5	7.5	-3	-15	45
10-15	7	12.5	-2	-14	28
15-20	13	17.5	-1	-13	13
20-25	21	22.5	0	0	0
25-30	16	27.5	1	16	16
30-35	8	32.5	2	16	32
35-40	3	37.5	3	9	27
Total	75	—	—	-9	193

If $y = (x - c)/d$, then (i) $\bar{x} = c + d\bar{y}$, and (ii) $s_x = d \cdot s_y$

Here in col. (4), $y = (x - 22.5)/5$; so, $c = 22.5$, $d = 5$.

$$\bar{y} = \sum fy/N = -9/75 = -0.12$$

$$\bar{x} = 22.5 + 5(-0.12) = 22.5 - 0.6 = 21.9$$

$$\begin{aligned}\text{Again, } s_y^2 &= \sum fy^2/N - (\sum fy/N)^2 = 193/75 - (-9/75)^2 \\ &= 2.5733 - .0144 = 2.5589\end{aligned}$$

$\therefore s_y = \sqrt{2.5589} = 1.60$; $s_x = 5 \times 1.60 = 8.0$. Thus, for the given frequency distribution, we have $\bar{x} = 21.9$, $s = 8.0$.

In order to find the number of observations lying between $\bar{x} \pm s$, we use simple interpolation in a cumulative frequency distribution (see Example 4.12).

$$\bar{x} \pm s = 21.9 \pm 8.0 = 21.9 + 8.0 \text{ and } 21.9 - 8.0 = 29.9 \text{ and } 13.9$$

**Table 6.14** Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency
0	0
5	2
5.9→	←C
10	7
13.9→	←A
15	14
20	27
25	48
29.9→	←B
30	64
35	72
37.9→	←D
40	75

If A and B denote cumulative frequencies corresponding to 13.9 and 29.9 respectively, then

$$\frac{13.9 - 10}{15 - 10} = \frac{A - 7}{14 - 7}, \quad \frac{29.9 - 25}{30 - 25} = \frac{B - 48}{64 - 48}$$

Solving, $A = 12.46$, $B = 63.68$

Hence, the number of observations lying between $\bar{x} \pm s$, i.e. between 13.9 and 29.9, is $B - A = 63.68 - 12.46 = 51$

Similarly, $\bar{x} \pm 2s = 21.9 \pm 2 \times 8.0 = 21.9 \pm 16.0 = 37.9$ and 5.9

If C and D be the cumulative frequencies of 5.9 and 37.9 respectively, then

$$\frac{5.9 - 5}{10 - 5} = \frac{C - 2}{7 - 2}, \quad \frac{37.9 - 35}{40 - 35} = \frac{D - 72}{75 - 72}$$

whence $C = 2.9$ and $D = 73.74$. Hence the frequency lying between $\bar{x} \pm 2s$ is $D - C = 73.74 - 2.9 = 71$.

Ans. 8.0; 51 and 71.



Example 6.35 Find the standard deviation of the following distribution:

X	0-500	500-1000	1000-1500	1500-2000	2000-3000
Frequency	90	218	86	41	15

[C.U., B.A.(Econ) '74]

Solution

Table 6.15 Calculations for S.D.

X	Frequency (f)	Mid-value (x)	$x - 750$	$y = \frac{x - 750}{250}$	fy	fy^2
0-500	90	250	-500	-2	-180	360
500-1000	218	750	0	0	0	0
1000-1500	86	1250	500	2	172	344
1500-2000	41	1750	1000	4	164	656
2000-3000	15	2500	1750	7	105	735
Total	450	—	—	—	261	2095



$$\sigma_y^2 = \frac{\sum fy^2}{n} - \left(\frac{\sum fy}{N} \right)^2 = \frac{2095}{450} - \left(\frac{261}{450} \right)^2 = 4.6556 - (0.58)^2 = 4.3192$$

$$\sigma_x = d \cdot \sigma_y = 250 \times \sqrt{4.3192} = 250 \times 2.0783 = 519.6$$



Example 6.36 Calculate the mean and the S.D. from the data in Example 6.34 by the **Method of Summation**.

Solution *Method of Summation* (similar to that shown in the Table at p. 115) may be used to obtain the mean and the S.D. from a frequency distribution, if the class intervals are of equal width.

Table 6.16 Method of Summation for Mean and S.D.

Class Interval	Frequency (f)	1st Cumulation	2nd Cumulation	
(1)	(2)	(3)	(4)	
0–5	2	2	2	
5–10	5	7	9	
10–15	7	14	23	
15–20	13	27	50	
20–25	21	48	98	
25–30	16	64	162	
30–35	8	72	234	
35–40	3	–	–	
Total	75	234	578	

Note: Col. (3) shows cumulative totals of col. (2), i.e. cumulation frequencies.

Col. (4) shows cumulative totals of col. (3).

Let x_n denote the mid-value of the last class interval, and h the common width of classes (here $x_n = 37.5$ and $h = 5$).

$$F_1 = \frac{\text{Total of 1st cumulation}}{N} = \frac{234}{75} = 3.12$$

$$F_2 = \frac{\text{Total of 2nd cumulation}}{N} = \frac{578}{75} = 7.7067$$

Then, Mean = $x_n - h \cdot F_1 = 37.5 - 5(3.12) = 21.9$

$$\begin{aligned} \text{S.D.} &= h \sqrt{2F_2 - F_1 - F_1^2} = 5 \sqrt{15.4134 - 3.12 - 9.7344} \\ &= 5 \sqrt{2.5590} = 8.0 \end{aligned}$$

The results agree with those obtained in Example 6.34.



Example 6.37 A variable X assumes two distinct values, 0 and 1. Of the total number of observations N , the fraction p of N are ones and the fraction q of N are zeros. Find the mean and standard deviation of the N observations.

[C.U., B.Com.(Hons) '64]



Solution The variable X has two distinct values 0 and 1 with frequencies qN and pN respectively, the total frequency being N . Hence, $qN + pN = N$; or, $N(q + p) = N$;
 $\therefore q + p = 1$.

Table 6.17 Calculations for Mean and S.D.

X	Frequency f	fX	fX^2
0	qN	0	0
1	pN	pN	pN
Total	N	pN	pN

Therefore, $\bar{x} = \frac{\sum fX}{\sum f} = \frac{pN}{N} = p$

$$\sigma^2 = \frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N} \right)^2 = \frac{pN}{N} - \left(\frac{pN}{N} \right)^2$$

$$= p - p^2 = p(1 - p) = pq \text{ (since } q + p = 1\text{)}$$

$\therefore \sigma = \sqrt{pq}$ Ans. Mean = p , S.D. = $\sqrt{pq}$.



6.8 S.D. OF COMPOSITE GROUP

If two groups contain n_1 and n_2 observations with means $\bar{x}_1$ and $\bar{x}_2$, and standard deviations σ_1 and σ_2 respectively, then the S.D. (σ) of the composite group is given by

$$N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2) + (n_1d_1^2 + n_2d_2^2) \quad (6.8.1)$$

where $d_1 = \bar{x}_1 - \bar{x}$, $d_2 = \bar{x}_2 - \bar{x}$, and $\bar{x}$ is the mean of composite group, given by (5.5.1), viz. $N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$. (see Example 6.43)

It will usually be found convenient to arrange the data (see Table 5.18, page 125) in the following tabular form:

Table 6.18 Data for S.D. of Composite Group

Characteristics	Groups		Composite Group
	I	II	
No. of Observations	n_1	n_2	N
Mean	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}$
Standard Deviation	σ_1	σ_2	σ

In general, if there are several group, the i -th group containing n_i observations with mean $\bar{x}_i$ and S.D. σ_i , then the S.D. (σ) of the composite group is given by

$$N\sigma^2 = \sum n_i\sigma_i^2 + \sum n_id_i^2 \quad (6.8.2)$$

where $N\bar{x} = \sum n_i\bar{x}_i$, $d_i = \bar{x}_i - \bar{x}$, and $N = \sum n_i$.

Example 6.38 For a group of 50 boys the mean score and the standard deviation of scores on a test are 59.5 and 8.38. For a group of 40 girls the same results are 54.0 and 8.23. Find the mean and S.D. of the combined group of 90 children.

[C.U., M.Com. '72]



Solution

Table 6.19 Data for Mean and S.D. of Combined Group

Characteristics	Group		Combined Group
	I (Boys)	II (Girls)	
No. of Observations	$n_1 = 50$	$n_2 = 40$	$N = 90$
Mean	$\bar{x}_1 = 59.5$	$\bar{x}_2 = 54.0$	$\bar{x} = ?$
Standard Deviation	$\sigma_1 = 8.38$	$\sigma_2 = 8.23$	$\sigma = ?$

The formulae for mean and S.D. of combined group are

$$(1) \quad N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$$

$$(2) \quad N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2) + (n_1d_1^2 + n_2d_2^2)$$

Using (1), $90\bar{x} = 50 \times 59.5 + 40 \times 54.0 = 5135.0$

$\therefore \bar{x} = 5135.0/90 = 57.06$

Hence, $d_1 = 59.5 - 57.06 = 2.44$, $d_2 = 54.0 - 57.06 = -3.06$

Using (2), $90\sigma^2 = \{50 \times (8.38)^2 + 40 \times (8.23)^2\} + \{50 \times (2.44)^2 + 40 \times (-3.06)^2\}$
 $= (3511.22 + 2709.32) + (297.68 + 374.54) = 6892.76$

$$\sigma^2 = 6892.76/90 = 76.59, \quad \sigma = \sqrt{76.59} = 8.75$$

Ans. 57.06, 8.75



Example 6.39 Find the standard deviation of the composite set from the means and standard deviations of its component sets as given in the following table:

	First Set	Second Set	Third Set
Number of Observations	200	250	300
Arithmetic Mean	25	10	15
Standard Deviation	3	4	5

[B.U., B.A.(Econ) '70]

Solution

Table 6.20 Mean and S.D. of Component Sets

Characteristics	Set I	Set II	Set III	Composite Set
No. of Observations	$n_1 = 200$	$n_2 = 250$	$n_3 = 300$	$N = 750$
Mean	$\bar{x}_1 = 25$	$\bar{x}_2 = 10$	$\bar{x}_3 = 15$	$\bar{x} = ?$
Standard Deviation	$\sigma_1 = 3$	$\sigma_2 = 4$	$\sigma_3 = 5$	$\sigma = ?$

The required formulae are

$$(1) \quad N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3 \quad (\text{where } N = n_1 + n_2 + n_3)$$

$$(2) \quad N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2 + n_3\sigma_3^2) + (n_1d_1^2 + n_2d_2^2 + n_3d_3^2)$$

Putting the values in (1),

$$750\bar{x} = 200 \times 25 + 250 \times 10 + 300 \times 15 = 12,000$$

$$\bar{x} = 12000/750 = 16$$

So, $d_1 = \bar{x}_1 - \bar{x} = 9$, $d_2 = \bar{x}_2 - \bar{x} = -6$, $d_3 = \bar{x}_3 - \bar{x} = -1$.



Now using (2),

$$750\bar{\sigma}^2 = (200 \times 3^2 + 250 \times 4^2 + 300 \times 5^2) + \{200 \times 9^2 + 250(-6)^2 + 300(-1)^2\}$$

$$= (1800 + 4000 + 7500) + (16200 + 9000 + 300) = 38800$$

$$\sigma^2 = 38800/750 = 51.73; \quad \sigma = \sqrt{51.73} = 7.19$$

Example 6.40 The mean and the variance of a group of 100 observations are 6.5 and 3.0 respectively. 55 of these observations have mean 6.6 and standard deviation 1.5. Find the mean and the S.D. of the remaining 45 observations.

[I.C.W.A., Dec. '75-old]

Solution We assume that the 100 observations have been split up into two groups—Group I containing 55 observations with mean $(\bar{x}_1) = 6.6$ and standard deviation $(\sigma_1) = 1.5$, and Group II containing 45 observations. The composite group of 100 observations has mean $(\bar{x}) = 6.5$ and variance $(\sigma^2) = 3.0$. It is required to find the mean $(\bar{x}_2)$ and s.d. (σ_2) of Group II.

Table 6.21 Data for Mean and S.D. of Composite Group

Characteristics	I	Groups	II	Composite Group
No. of Observations	$n_1 = 55$		$n_2 = 45$	$N = 100$
Mean	$\bar{x}_1 = 6.6$		$\bar{x}_2 = ?$	$\bar{x} = 6.5$
Standard Deviation	$\sigma_1 = 1.5$		$\sigma_2 = ?$	$\sigma = \sqrt{3.0}$

Formulae for Mean and S.D. of composite group are

(i) $N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$

(ii) $N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2) + (n_1d_1^2 + n_2d_2^2)$

Substituting the values in (i), $100 \times 6.5 = 55 \times 6.6 + 45\bar{x}_2$; hence,

$$\bar{x}_2 = 287/45 = 6.38$$

Therefore, $d_1 = \bar{x}_1 - \bar{x} = 6.6 - 6.5 = 0.1$

$$d_2 = \bar{x}_2 - \bar{x} = 6.38 - 6.5 = -0.12$$

Now, applying (ii),

$$100 \times (\sqrt{3.0})^2 = 55 \times (1.5)^2 + 45\sigma_2^2 + 55 \times (0.1)^2 + 45(-0.12)^2$$

$$\text{or, } 300 = 123.75 + 45\sigma_2^2 + 0.55 + 0.648$$

$$\text{or, } 45\sigma_2^2 = 300 - 123.75 - 0.55 - 0.648 = 175.052$$

$$\therefore \sigma_2^2 = 175.052 \div 45 = 3.89$$

$$\sigma_2 = \sqrt{3.89} = 1.97$$

Ans. 6.38, 1.97



Example 6.41 In a batch of 10 children, the I.Q. of a dull boy is 36 below the average I.Q. of the other children. Show that the standard deviation of I.Q. for all the children cannot be less than 10.8. If this standard deviation is actually 11.4, determine what the s.d. will be, when the dull boy is left out. [C.U., M.Com. '75]

Solution Let a denote the average I.Q. of the 9 children (excluding the dull boy). Then, I.Q. of the dull boy is $a - 36$. We separate the 10 children into two groups—Group I consisting of the 9 children and Group II including only the dull boy.

**Table 6.22** Mean and S.D. of Composite Group

Characteristics	Group		Composite Group
	I	II	
No. of Observations	$n_1 = 9$	$n_2 = 1$	$N = 10$
Mean	$\bar{x}_1 = a$	$\bar{x}_2 = a - 36$	$\bar{x}$
Standard Deviation	σ_1	$\sigma_2 = 0$	σ

Since the s.d. of one observation is zero, therefore $\sigma_2 = 0$.

$$\begin{aligned}
 (1) \quad N\bar{x} &= n_1\bar{x}_1 + n_2\bar{x}_2 \\
 \text{or,} \quad 10\bar{x} &= 9a + 1(a - 36) = 10a - 36 \\
 \therefore \quad \bar{x} &= a - 3.6 \\
 d_1 &= \bar{x}_1 - \bar{x} = a - (a - 3.6) = 3.6 \\
 d_2 &= \bar{x}_2 - \bar{x} = (a - 36) - (a - 3.6) = -32.4 \\
 (2) \quad N\sigma^2 &= (n_1\sigma_1^2 + n_2\sigma_2^2) + (n_1d_1^2 + n_2d_2^2) \\
 \text{or,} \quad 10\sigma^2 &= (9\sigma_1^2 + 1 \times 0) + \{9(3.6)^2 + 1 \times (-32.4)^2\} \\
 \text{or,} \quad 10\sigma^2 &= 9\sigma_1^2 + 1166.4 \quad \dots (i) \\
 \text{or,} \quad \sigma^2 &= 0.9\sigma_1^2 + 116.64 \\
 \therefore \quad \sigma^2 &\geq 116.64 \text{ (since } \sigma_1^2 \text{ cannot be negative)} \\
 \text{or,} \quad \sigma &\geq \sqrt{116.64}, \text{ i.e. } 10.8 \\
 \text{If } \sigma &= 11.4, \text{ then from (i), } 10 \times (11.4)^2 = 9\sigma_1^2 + 1166.4 \\
 \text{Solving, } \sigma &= \sqrt{14.8} = 3.85
 \end{aligned}$$

Example 6.42 Particulars regarding income of two villages are given below:

	Village X	Village Y
Number of People	600	500
Average Income in Rs	175	186
Variance of Income in Rs	100	81

(a) In which village is the variation in income greater? (b) What is the total income of both villages put together? (c) What is the average income of the people in X and Y villages put together? (d) What is the combined Standard Deviation of the villages X and Y put together? [C.A., May '74]

Solution (a) Standard Deviation (S.D.) provides a measure of the variation or discrepancy among observations. Since $S.D. = \sqrt{\text{Variance}}$, the S.D.s of villages X and Y are respectively $\sigma_1 = \sqrt{100} = 10$, $\sigma_2 = \sqrt{81} = 9$. Since σ_1 is larger than σ_2 , we conclude that the variation in income is greater in village X.

(b) Total incomes of villages X and Y are respectively $600 \times 175 = 105,000$ Rs and $500 \times 186 = 93,000$ Rs. Hence the combined total income of both villages put together is $105,000 + 93,000 = 198,000$ Rs.

(c) The total number of people $N = n_1 + n_2 = 600 + 500 = 1100$, whose total income is Rs 198,000. Average income $= 198,000 \div 1100 = 180$ Rs (We could also apply the formula $N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$).

(d) We use the formula $N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2) + (n_1d_1^2 + n_2d_2^2)$.
 Here $d_1 = \bar{x}_1 - \bar{x} = 175 - 180 = -5$, $d_2 = \bar{x}_2 - \bar{x} = 186 - 180 = 6$.
 $\therefore 1100\sigma^2 = (600 \times 100 + 500 \times 81) + (600 \times 25 + 500 \times 36) = 133,500$
 $\sigma^2 = 133500/1100 = 121.36$; $\sigma = \sqrt{121.36} = 11.02$ Rs

Ans. (a) Village X, (b) Rs 198,000, (c) Rs 180, (d) Rs 11.02



Example 6.43 If n_1 , σ_1 and $\bar{x}_1$ be the number of observations, standard deviation and mean of a set of observations, and n_2 , σ_2 , and $\bar{x}_2$ be those for a second set of observations, prove that the standard deviation σ of the combined set of $(n_1 + n_2)$ observations is given by

$$(n_1 + n_2)\sigma^2 = n_1\sigma_1^2 + n_2\sigma_2^2 + n_1d_1^2 + n_2d_2^2$$

where $d_1 = \bar{x}_1 - m$, $d_2 = \bar{x}_2 - m$, and $m = (n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$.
[C.U., B.A.(Econ) '68, '75]

Solution Let x_i ($i = 1, 2, \dots, n_1$) and x'_j ($j = 1, 2, \dots, n_2$) be the observations in the 1st set and 2nd set respectively. By definition

$$\begin{aligned}\bar{x}_1 &= \sum_i x_i / n_1 & \sigma_1^2 &= \sum_i (x_i - \bar{x}_1)^2 / n_1 \\ \bar{x}_2 &= \sum_j x'_j / n_2 & \sigma_2^2 &= \sum_j (x'_j - \bar{x}_2)^2 / n_2\end{aligned}$$

By cross-multiplication, we get

$$\left. \begin{aligned}\sum_i x_i &= n_1\bar{x}_1 & \sum_i (x_i - \bar{x}_1)^2 &= n_1\sigma_1^2 \\ \sum_j x'_j &= n_2\bar{x}_2 & \sum_j (x'_j - \bar{x}_2)^2 &= n_2\sigma_2^2\end{aligned}\right\} \dots (1)$$

Now considering the combined set of $(n_1 + n_2)$ observations x_i and x'_j , the mean (m) and standard deviation (σ) are given by

$$\begin{aligned}m &= \left(\sum_i x_i + \sum_j x'_j \right) / (n_1 + n_2) \\ &= (n_1\bar{x}_1 + n_2\bar{x}_2) / (n_1 + n_2), \text{ by (1)}\end{aligned}$$

and

$$\sigma^2 = \left\{ \sum_i (x_i - m)^2 + \sum_j (x'_j - m)^2 \right\} / (n_1 + n_2) \dots (2)$$

But,

$$x_i - m = (x_i - \bar{x}_1) + (\bar{x}_1 - m) = (x_i - \bar{x}_1) + d_1, \text{ where } d_1 = \bar{x}_1 - m$$

$\therefore$

$$\begin{aligned}\sum_i (x_i - m)^2 &= \sum_i \{(x_i - \bar{x}_1) + d_1\}^2, \\ &= \sum_i \{(x_i - \bar{x}_1)^2 + d_1^2 + 2(x_i - \bar{x}_1)d_1\} \\ &= \sum_i (x_i - \bar{x}_1)^2 + \sum_i d_1^2 + \sum_i 2(x_i - \bar{x}_1)d_1; \quad \text{by (3.15.2)} \\ &= \sum_i (x_i - \bar{x}_1)^2 + n_1d_1^2 + 2d_1 \sum_i (x_i - \bar{x}_1), \\ &\quad \text{by (3.15.4) and (3.15.3)} \\ &= n_1\sigma_1^2 + n_1d_1^2 + 2d_1 \times 0 \\ &= n_1\sigma_1^2 + n_1d_1^2\end{aligned}$$

because, $\sum_i (x_i - \bar{x}_1) = \sum_i x_i - \sum_i \bar{x}_1 = n_1\bar{x}_1 - n_1\bar{x}_1 = 0.$

Similarly, $\sum_j (x'_j - m)^2 = \sum_j \{(x'_j - \bar{x}_2) + d_2\}^2 = n_2\sigma_2^2 + n_2d_2^2.$

Substituting in (2),

$$\begin{aligned}\sigma^2 &= \{(n_1\sigma_1^2 + n_1d_1^2) + (n_2\sigma_2^2 + n_2d_2^2)\} / (n_1 + n_2) \\ \text{i.e. } (n_1 + n_2)\sigma^2 &= n_1\sigma_1^2 + n_2\sigma_2^2 + n_1d_1^2 + n_2d_2^2\end{aligned}$$



Example 6.44 Show that the combined standard deviation of two distributions pooled together is given by the expression

$$NS^2 = n_1s_1^2 + n_2s_2^2 + \frac{n_1n_2}{N} (\bar{x}_1 - \bar{x}_2)^2$$

Where the symbols have their usual meaning.

[C.U., B.Com.(Hons) '67; B.A.(Econ) '73]

Solution Here n_1, s_1 and $\bar{x}_1$ represent the number of observations, S.D. and mean of the first distribution; $n_2, s_2, \bar{x}_2$ those of the second distribution; and N, S represent the total number of observations and S.D. of the combined distribution respectively. Obviously, $N = n_1 + n_2$.

From Example 6.43, we have (changing the notations)

$$NS^2 = n_1s_1^2 + n_2s_2^2 + n_1d_1^2 + n_2d_2^2$$

$$\text{But, } d_1 = \bar{x}_1 - m = \bar{x}_1 - \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{n_2(\bar{x}_1 - \bar{x}_2)}{n_1 + n_2}$$

$$\text{and similarly, } d_2 = \frac{n_1(\bar{x}_2 - \bar{x}_1)}{n_1 + n_2}$$

$$\begin{aligned} \therefore n_1d_1^2 + n_2d_2^2 &= n_1 \frac{n_2^2(\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2} + n_2 \frac{n_1^2(\bar{x}_2 - \bar{x}_1)^2}{(n_1 + n_2)^2} \\ &= \frac{n_1n_2^2(\bar{x}_1 - \bar{x}_2)^2 + n_1^2n_2(\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2} \\ &= \frac{n_1n_2(\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2} (n_2 + n_1) \\ &= \frac{n_1n_2(\bar{x}_1 - \bar{x}_2)^2}{n_1 + n_2} = \frac{n_1n_2}{N} (\bar{x}_1 - \bar{x}_2)^2 \end{aligned}$$

Hence, the result.



6.9 RELATION BETWEEN S.D. AND OTHER MEASURES

Many of the distributions occurring in practice have frequency curves of the symmetrical bell shaped type. Such frequency curves may be regarded as approximations to an important curve, known as "Normal" curve (Chapter 12). Some of the properties of this curve therefore hold approximately for observed frequency distributions. For instance, about 99.7% of the area under normal curve lies between $mean \pm 3\sigma$, so that the effective range is 6σ . Hence, for any distribution the S.D. may be taken to be about one-sixth of the range.

Again, approximately 68% and 95% of area under normal curve are included between $(mean \pm \sigma)$ and $(mean \pm 2\sigma)$ respectively. For moderately asymmetrical distributions, therefore, about two-thirds of the total number of observations are expected to lie between $mean \pm s.d.$ (see Example 6.34).

It can be proved (Example 6.45) that for any given set of observations S.D. is never less than M.D.

$$S.D. \geq M.D. \quad (6.9.1)$$



For the normal distribution, Q.D. = 0.67σ and M.D. = 0.80σ (appr.). These results form the very basis of the following approximate relations which hold for any distribution, not far from normal:

$$\text{Range} = 6\sigma, \quad \text{Q.D.} = \frac{2}{3}\sigma, \quad \text{M.D.} = \frac{4}{5}\sigma \quad (6.9.2)$$

Example 6.45 Show that $\sum_{i=1}^n x_i^2 \geq \left(\sum_{i=1}^n x_i \right)^2 / n$, where $x_1, x_2, \dots, x_n$ are any given

values of the variable x . Hence, or otherwise, prove that the mean deviation about the mean cannot exceed the standard deviation. When will the two be equal?

[B.U., B.A.(Econ) '74]

Solution We have shown (Example 6.18) that $\Sigma(x_i - \bar{x})^2 = \Sigma x_i^2 - n\bar{x}^2$. But the L.H.S. being the sum of the squares cannot be negative; i.e. $\Sigma(x_i - \bar{x})^2 \geq 0$. Hence

$$\Sigma x_i^2 \geq n\bar{x}^2, \quad \text{or,} \quad \Sigma x_i^2 \geq (\Sigma x_i)^2 / n$$

The two sides are equal, only when all x_i 's are equal.

The above inequality holds for any given set of real numbers $x_1, x_2, \dots, x_n$. In particular, if we replace x_i by $|x_i - \bar{x}|$, then

$$\begin{aligned} \Sigma\{|x_i - \bar{x}|\}^2 &\geq \{\Sigma|x_i - \bar{x}|\}^2 / n \\ \text{or,} \quad \Sigma(x_i - \bar{x})^2 &\geq \{\Sigma|x_i - \bar{x}|\}^2 / n \\ \text{or,} \quad \Sigma(x_i - \bar{x})^2 / n &\geq \{\Sigma|x_i - \bar{x}| / n\}^2 \\ \text{or,} \quad (\text{S.D.})^2 &\geq (\text{M.D.})^2 \\ \text{or,} \quad \text{S.D.} &\geq \text{M.D.} \end{aligned}$$

The equality sign holds only when all $|x_i - \bar{x}|$ are equal, i.e. all the observations x_i are equal (then S.D. = M.D. = 0)

6.10 RELATIVE MEASURES OF DISPERSION

The four measures—Range, Quartile Deviation, Mean Deviation and Standard Deviation, are expressed in the same units as the original observations, and are called *Absolute measures* of variability. So, they cannot be used for comparing the variability of two or more distributions given in different units. In order to meet such situations, the *Relative measures* of variability have been introduced which are independent of the units of measurement. There are 3 such measures:

(i) *Coefficient of Variation*

$$= 100 \times \frac{\text{Standard Deviation}}{\text{Mean}}$$

(ii) *Coefficient of Quartile Deviation*

$$= 100 \times \frac{\text{Quartile Deviation}}{\text{Median}}$$

(iii) *Coefficient of Mean Deviation*

$$= 100 \times \frac{\text{Mean Deviation}}{\text{Mean or Median}}$$

Among these, coefficient of variation is the most important and is used in almost all cases.



The relative measures of variability may also be used to measure the precision of observations, although given in the same unit.



Example 6.46 Define 'coefficient of variation'. What are the special uses of this measure? [B.U., B.A. (Econ) '76; D.M. '73; D.S.W. '77; M.B.A. '79]

Solution Coefficient of Variation is defined by the formula:

$$\text{Coefficient of Variation} = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

It is a relative measure of dispersion and is independent of the unit of measurement.

Uses—(i) The absolute measures cannot be used for comparing the dispersion of two or more distributions given in different units. The coefficient of variation is a pure number and as such may be employed for the purpose (Example 6.54).

(ii) Again, where the means of distributions are widely different, although given in the same unit, a relative measure will be more logical than an absolute measure of dispersion (Examples 6.48, 6.55, 6.56).

(iii) It may also be used to compare the precision of two or more sets of measurements. If repeated measurements of the length of a 10 ft. rod and also of a 100 ft. long road (the exact lengths not given) are taken, it will be found that the observations will vary among themselves. If both sets of measurements are taken with the same care, we expect that S.D. of the former will be less than that of the latter. The same S.D. in both sets will imply that the first set of measurements were less precise than the second. The coefficient of variation will however reveal the correct position.



Example 6.47 If the record of amounts spent by you on various evenings gives a coefficient of variation of 50 per cent and a standard deviation of Rs 4.00, what was your average expenditure? [D.S.W., Nov '73]

Solution By definition, Coefficient of Variation = $\frac{\text{S.D.}}{\text{Mean}} \times 100$

$$\text{or, } 50 = \frac{4.00 \text{ Rs}}{\text{Mean}} \times 100; \quad \text{Solving, Mean} = 8.00 \text{ Rs}$$



Example 6.48 From some financial statistics, it is found that the monthly average Electricity Charges was Rs 2,460 and S.D. Rs 120. The monthly average Direct Wages was Rs 42,000 and S.D. Rs 1,200. State which is the more variable with proper reasons. [C.A., Nov. '67]

Solution The given data are (in Rs)

	Electricity Charges	Direct Wages
Mean	2,460	42,000
S.D.	120	1,200

Since the means in the two cases differ widely, although given in the same unit, viz. Rs, it will be inappropriate to employ S.D. for comparing the variability. The coefficient of variation (C.V.) will be more logical here.



$$\text{C.V. (for Electricity Charges)} = \frac{\text{Rs } 120}{\text{Rs } 2460} \times 100 = 4.9$$

$$\text{C.V. (for Direct Wages)} = \frac{\text{Rs } 1200}{\text{Rs } 42000} \times 100 = 2.9$$

Since the coefficient of variation for Electricity Charges is the larger, Electricity Charges are more variable than Direct Wages.



Example 6.49 In an industrial establishment, the coefficients of variation of wages of male and female workers were 55% and 70% respectively. The standard deviations were Rs 22.00 and Rs 15.40 respectively. Calculate the combined average wages for all the workers, if 80% of the workers were males. [C.A., Nov. '75]

Solution Let $\bar{x}_1$ and $\bar{x}_2$ denote the mean wages of male and female workers respectively. Then, as in Example 6.47,

$$55 = \frac{\text{Rs } 22.00}{\bar{x}_1} \times 100 \quad \text{and} \quad 70 = \frac{\text{Rs } 15.40}{\bar{x}_2} \times 100.$$

Solving we get $\bar{x}_1 = \text{Rs } 40.00$, $\bar{x}_2 = \text{Rs } 22.00$.

If the total number of workers be 100, the numbers of male and female workers are $n_1 = 80$ and $n_2 = 20$. Now applying formula (5.5.1) for mean of a composite group (p. 146), $100\bar{x} = 80 \times 40.00 + 20 \times 22.00$, from which $\bar{x} = 36.40$ Rs

Ans. Rs 36.40



Example 6.50 Find the coefficient of variation from the following:

Weight (gm)	110–119	120–129	130–139	140–149	150–159
Frequency	5	7	12	20	16
			160–169	170–179	180–189
			10	7	3

[I.C.W.A., Dec. '76]

Solution In order to find the coefficient of variation, we have to calculate both the mean and the standard deviation.

Table 6.23 Calculations for Mean and S.D.

Weight (gm)	Frequency (f)	Mid-value (x)	$y = \frac{x - 144.5}{10}$	fy	fy ²
110–119	5	114.5	–3	–15	45
120–129	7	124.5	–2	–14	28
130–139	12	134.5	–1	–12	12
140–149	20	144.5	0	0	0
150–159	16	154.5	1	16	16
160–169	10	164.5	2	20	40
170–179	7	174.5	3	21	63
180–189	3	184.5	4	12	48
Total	80	–	–	28	252



Proceeding as in Example 6.34,

$$\text{Mean } (\bar{x}) = 144.5 + 10 (28/80) = 148.0 \text{ gm.}$$

$$\sigma_y^2 = 252/80 - (28/80)^2 = 3.15 - (0.35)^2 = 3.0275$$

$$\therefore \sigma_y = \sqrt{3.0275} = 1.74; \quad \sigma_x = 10 \times 1.74 = 17.4 \text{ gm.}$$

$$\text{Coefficient of variation} = (17.4 + 148.0) \times 100 = 12$$



Example 6.51 The scores of two batsmen, A and B, in ten innings during a certain season, are as under:

A	32	28	47	63	71	39	10	60	96	14
B	19	31	48	53	67	90	10	62	40	80

Find which of the batsmen is more consistent in scoring. [I.C.W.A., Jan. '70]

Solution We use Coefficient of Variation for measuring dispersion. The batsman with the smaller dispersion is more consistent. The mean and S.D. for simple series are obtained by using formulae (5.4.1) and (6.7.3): If $y = x - c$, then (i) $\bar{x} = c + \bar{y}$, (ii) $\sigma_x = \sigma_y$.

Table 6.24 Calculations for Mean and S.D.

Cricketer A			Cricketer B		
x	$y = x - 50$	y^2	x	$y = x - 50$	y^2
32	-18	324	19	-31	961
28	-22	484	31	-19	361
47	-3	9	48	-2	4
63	13	169	53	3	9
71	21	441	67	17	289
39	-11	121	90	40	1600
10	-40	1600	10	-40	1600
60	10	100	62	12	144
96	46	2116	40	-10	100
14	-36	1296	80	30	900
Total	-40	6660	Total	0	5968

For Cricketer A:

$$\text{Mean} = 50 + (-40/10) = 46$$

$$\text{S.D.} = \sqrt{\frac{6660}{10} - \left(\frac{-40}{10}\right)^2} = \sqrt{650} = 25.5$$

$$\therefore \text{C.V.} = \frac{25.5}{46} \times 100 = 55$$

For Cricketer B:

$$\text{Mean} = 50 + (0/10) = 50$$

$$\text{S.D.} = \sqrt{\frac{5968}{10} - \left(\frac{0}{10}\right)^2} = \sqrt{596.8} = 24.4$$

$$\therefore \text{C.V.} = \frac{24.4}{50} \times 100 = 49$$



Since for Cricketer B, the coefficient of variation is smaller, he is more consistent.

Ans. B.



Example 6.52 From the data given below, state which series is more variable (Use standard deviation):

Variable	Series A	Series B
10–20	10	18
20–30	18	22
30–40	32	40
40–50	40	32
50–60	22	18
60–70	18	10

[C.A., Nov. '71]

Solution Means and standard deviations for the two series are (calculations have not been shown here) as follows:

Series A: Mean = 42.14, S.D. = 14.06

Series B: Mean = 37.86, S.D. = 14.06

If we use standard deviation directly, it is not possible to decide which series is more variable; because the standard deviations are equal. However, since the means of the two Series are different, a relative measure of dispersion (using standard deviation), viz. Coefficient of Variation, would be appropriate.

$$\text{C.V. (for Series A)} = \frac{14.06}{42.14} \times 100 = 33$$

$$\text{C.V. (for Series B)} = \frac{14.06}{37.86} \times 100 = 37$$

Since the Coefficient of Variation for Series B is larger than that for Series A, the former is more variable.

Ans. Series B



Example 6.53 Given the following results relating to two groups containing 20 and 30 observations, calculate the coefficient of variation of all the 50 observations by combining both the groups:

	Groups	
	I	II
Σx	45	55
Σx^2	118	132

[I.C.W.A., Jan. '68]

Solution For the combined group,

$$n = 20 + 30 = 50;$$

$$\Sigma x = 45 + 55 = 100;$$

$$\Sigma x^2 = 118 + 132 = 250;$$

$$\therefore \text{Mean} = \Sigma x/n = 100/50 = 2.$$

$$\sigma^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n} \right)^2 = \frac{250}{50} - \left(\frac{100}{50} \right)^2 = 5 - 4 = 1. \quad \therefore \sigma = \sqrt{1} = 1.$$



$$\therefore \text{Coefficient of Variation} = \frac{\text{S.D.}}{\text{Mean}} \times 100 = \frac{1}{2} \times 100 = 50.$$

Ans. 50.



Example 6.54 The following frequency distributions have been constructed from measurements of heights (in inches) and weights (in lbs.) of the same group of adult persons. Which is the more variable, height or weight?

Mid-point (height)	Frequency	Mid-point (weight)	Frequency
60	10	84	10
62	10	94	10
64	30	104	15
66	30	114	20
68	15	124	15
70	5	134	10
		144	10
	100	154	
			100

[I.C.W.A., July '68]

Solution The variability of the two distributions, given in different units, can only be compared by means of a relative measure. We calculate their coefficients of variation, and therefore have to find their means and S.Ds.

Table 6.25 Calculations for Mean and S.D. (Height)

Mid-point (X)	f	$x = \frac{X - 64}{2}$	fx	fx ²
60	10	-2	-20	40
62	10	-1	-10	10
64	30	0	0	0
66	30	1	30	30
68	15	2	30	60
70	5	3	15	45
Total	100	—	45	185

$$\bar{X} = c + d\bar{x} = 64 + 2 \times \frac{45}{100} = 64.9 \text{ inches}$$

$$\sigma_x = d \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} = 2 \times \sqrt{\frac{185}{100} - \left(\frac{45}{100}\right)^2} = 2.57 \text{ inches.}$$

**Table 6.26** Calculations for Mean and S.D. (Weight)

Mid-point (Y)	f	$y = \frac{Y - 114}{10}$	fy	fy ²
80	10	-3	-30	90
94	10	-2	-20	40
104	15	-1	-15	15
114	20	0	0	0
124	15	1	15	15
134	10	2	20	40
144	10	3	30	90
154	10	4	40	160
Total	100	—	40	450

$$\bar{Y} = c + d\bar{y} = 114 + \frac{40}{100} \times 10 = 118 \text{ lbs.}$$

$$\sigma_y = d \sqrt{\frac{\sum fy^2}{N} - \left(\frac{\sum fy}{N}\right)^2} = 10 \times \sqrt{\frac{450}{100} - \left(\frac{40}{100}\right)^2} = 20.83 \text{ lbs.}$$

$$\text{C.V. (for height)} = \frac{2.57 \text{ inches}}{64.9 \text{ inches}} \times 100 = 4$$

$$\text{C.V. (for weight)} = \frac{20.83 \text{ lbs.}}{118 \text{ lbs.}} \times 100 = 18$$

Since, the coefficient of variation for weight is greater than that for height, weight is the more variable.

Example 6.55 You are given two variables A and B. Using Quartile Deviation, state which is more variable.

A.	Mid point	15	20	25	30	35	40	45	
	Frequency	15	33	56	103	40	32	10	
B.	Mid point	100	150	200	250	300	350	400	450
	Frequency	340	492	890	1420	620	360	187	140

[C.A., Nov. '70]

Solution It is seen that averages of the two distributions differ widely (Averages being measures of central tendency will lie near the middle of the distributions, viz. near about 30 and 250 respectively). It will, therefore, be inappropriate to compare the variability of distributions based on an absolute measure of dispersion. A relative measure will be more logical here. Using Quartile Deviation, the relative measure is

$$\begin{aligned} \text{Coefficient of Quartile Deviation} &= 100 \times \frac{\text{Quartile Deviation}}{\text{Median}} \\ &= 100 \times \frac{Q_3 - Q_1}{2(\text{Median})} \end{aligned}$$

Medians and quartiles (Q_1 and Q_3) for the two distributions are (Calculations have not been shown here. Note that here mid-points are given, and therefore it is necessary to obtain the class boundaries for the calculation):



For A: $Q_1 = 24.67$ Median = 29.47 $Q_3 = 33.72$
 For B: $Q_1 = 190.74$ Median = 242.69 $Q_3 = 290.71$

Coefficient of Quartile Deviation (for A)

$$= 100 \times \frac{33.72 - 24.67}{2(29.47)} = 15$$

Coefficient of Quartile Deviation (for B)

$$= 100 \times \frac{290.71 - 190.74}{2(242.69)} = 21$$

Since the Coefficient of Quartile Deviation for B is larger than that for A, B is more variable (i.e. has larger variability) than A.

Example 6.56 With median as the base, calculate mean deviation and compare the variability of the two series A and B.

Series A	3484	4572	4124	3682	5624	4388	3680	4308
Series B	487	508	620	382	408	266	186	218

[C.A., May '73]

Solution (As in the previous Example, the averages of the two series differ widely, as is evident from the sizes of items. Hence, for comparing variability, we have to employ a relative measure). Using *Mean Deviation from median*, the relative measure of dispersion is

$$\text{Coefficient of Mean Deviation} = \frac{\text{M.D. about median}}{\text{Median}} \times 100$$

For the determination of Median, we arrange the values of each series in order of magnitude. Since there are 8 observations in each series, the average of the 4th and the 5th items gives the median. The ordered values are 3484, 3680, 3682, 4124, 4308, 4388, 4572, 5624 for series A, and 186, 218, 266, 382, 408, 487, 508, 620 for series B.

$$\text{Median for series A} = (4124 + 4308)/2 = 4216$$

$$\text{Median for series B} = (382 + 408)/2 = 395$$

Table 6.27 Calculations for M.D. from Median

Series A		Series B	
x	$ x - 4216 $	x	$ x - 395 $
3484	732	487	92
4572	356	508	113
4124	92	620	225
3682	534	382	13
5624	1408	408	13
4388	172	266	129
3680	536	186	209
4308	92	218	177
Total	3922	Total	971

Using formula (6.4.1), Mean Deviations about median are $3922/8 = 490.25$ for Series A, and $971/8 = 121.375$ for Series B.

The Coefficients of Mean Deviation are

$$\frac{490.25}{4216} \times 100 = 12 \text{ (for Series A)}$$

$$\frac{121.375}{395} \times 100 = 31 \text{ (for Series B)}$$

Since the coefficient of M.D. for Series B, viz. 31, is the larger, we conclude that Series B has greater variability.

6.11 LORENZ CURVE

Lorenz Curve is a diagram for showing the dispersion of a group. It is, in effect, a cumulative percentage curve, combining the percentage of items under review with the percentage of the factor (say, wealth distribution) among the items. If wealth were equally distributed among the people, the curve would be the straight line *ACB*, connecting the two extremes of the scales. In practice, however, curves like *ADB* are obtained. The less the area between the Lorenz curve *ADB* and the diagonal straight line *ACB*, the greater is the homogeneity in the distribution of wealth, i.e. less is the dispersion. On the other hand, the larger the area, the larger is the percentage of poor people and greater is the concentration of wealth in the hands of a few, Lorenz curve does not yield a numerical measure. It is, in this respect, inferior to the familiar measures of dispersion, e.g. Range, Standard deviation, etc. But the advantage is that it affords a picture of the dispersion at a glance. Lorenz curve is useful in such studies as the distribution of land, wages and income among the population of a country or the distribution of profits over different groups in business.

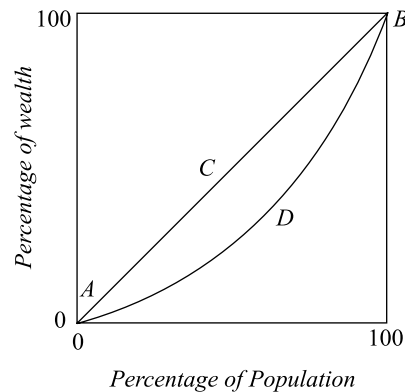


Fig. 6.1 Lorenz Curve

EXERCISES

- If each item is reduced by 10, what effect would this have on (i) the arithmetic mean, (ii) the range, and (iii) the standard deviation? [C.A., May '64]
- If the variables are increased or decreased (i) by the same amount, (ii) by the same proportion, what will be the effect on standard deviation? [B.U., B.A.(Econ) '68; I.C.W.A., Dec. '76-old]
- (i) If the first quartile is 142 and the semi-interquartile range is 18, what is the third quartile?
(ii) The coefficient of variation is 40 and the mean is 30; find the standard deviation. [C.U., M.Com. '69]
- Find out the range of the following data:

Height (inches)	60-62	63-65	66-68	69-71	72-74
No. of Students	8	27	42	18	5

[D.S.W., Nov. '73]



5. Calculate the quartile deviation and its coefficient from the following:

Cl. Interval	10–15	15–20	20–25	25–30	30–40	40–50	50–60	60–70	Total
Frequency	4	12	16	22	10	8	6	4	82

[C.A., Nov. '76]

6. The following table shows the distribution of the maximum loads supported by certain cables produced by a company:

Maximum load (Short tons)	9.3–9.7	9.8–10.2	10.3–10.7	10.8–11.2	11.3–11.7
No. of Cables	2	5	12	17	14
			11.8–12.2	12.3–12.7	12.8–13.2
			6	3	1

Find the semi-interquartile range.

[D.S.W., '68]

7. Define Mean Deviation. Find the mean deviation about the arithmetic mean of the numbers 31, 35, 29, 63, 55, 72, 37. [B.U., B.Com. '76]
8. Calculate the mean deviation of the following: 13, 84, 68, 24, 96, 139, 84, 27, about the median. [B.U., B.Com. '77]
9. Find the mean deviation about median from the following data: 46, 79, 26, 85, 39, 65, 99, 29, 56, 72. [C.U., B.Com. '77]

10. Find mean deviation for the following frequency distribution:

Variable	3	5	7	9	11	13
Frequency	2	7	10	9	5	1

[D.M.(Suppl.) '77]

11. Calculate the mean deviation from the following data, relating to heights (to the nearest inch) of 100 children:

Height (inches)	60	61	62	63	64	65	66	67	68
No. of Children	2	0	15	29	25	12	10	4	3

[I.C.W.A., Jan. '73]

12. Calculate mean deviation from median from the following:

Class Intervals	2–4	4–6	6–8	8–10
Frequencies	3	4	2	1

[I.C.W.A., Dec. '77]

13. In a certain distribution of $N = 25$ measurements it was found that $\bar{x} = 56$ inches and S.D. = 2 inches. After these results were computed it was discovered that a mistake had been made in one of the measurements which was recorded as 64 inches. Find the mean and standard deviation, if the incorrect measurement is omitted. [C.U., M.Com. '62]

14. The mean and S.D. of a group of 25 observations were found to be 30 and 3 respectively. After the calculations were made, it was found that two of the observations were incorrect, which were recorded as 29 and 31. Find the mean and S.D. if the incorrect observations are excluded.

[C.U., B.Com.(Hons) '68]

15. The mean and standard deviation of a group of 100 observations were found to be 20 and 3 respectively. After the calculations were made it was found that



three of the observations were incorrect which were recorded as 21, 21 and 18. Find the mean and s.d. if the incorrect observations are omitted.

[C.U., B.A.(Econ) '65]

16. The mean and the standard deviation of a sample of size 10 were found to be 9.5 and 2.5 respectively. Later on, an additional observation became available. This was 15.0 and was included in the original sample. Find the mean and the standard deviation of the 11 observations.

[I.C.W.A., June '75]

17. The mean and the s.d. of a sample of 100 observations were calculated as 40 and 5.1 respectively, by a student who by mistake took one observation as 50 instead of 40. Calculate the correct s.d.

[I.C.W.A., Dec. '76-old]

18. For a distribution of 280 observations mean and standard deviation were found to be 54 and 3 respectively. On checking it was discovered that two observations, which should correctly read as 62 and 82, had been wrongly recorded as 64 and 80 respectively. Calculate the correct values of mean and s.d.

[C.U., B.A.(Econ) '75]

19. $\bar{X}$ is the mean of X_1, X_2 and X_3 . If x_1, x_2, x_3 , are the deviations of X_1, X_2, X_3 , from $\bar{X}$ respectively, prove that

$$x_1^2 + x_2^2 + x_3^2 = X_1^2 + X_2^2 + X_3^2 - 3\bar{X}^2. \quad [C.U., B.A (Econ) '69]$$

20. Let $x_1, x_2, \dots, x_n$ be a set of observations. Suppose we compute $y_i = a + bx_i$ ($i = 1, 2, \dots, n$), where a and b are constants. Express the s.d. of the y 's in terms of the s.d. of the x 's and comment on the relation between the two.

[C.U., B.A.(Econ) '78]

21. 'If the mean and the standard deviation of n observations $x_1, x_2, \dots, x_n$ be $\bar{x}$ and σ respectively, then the mean and the standard deviation of $-x_1, -x_2, \dots, -x_n$ will be $-\bar{x}$ and $-\sigma$ respectively'—Comment.

[I.C.W.A., June '75]

22. If d^2 = mean square deviation about A , σ = standard deviation, and $A - \bar{x} = a$, then show that $d^2 = \sigma^2 + a^2$.

[D.S.W., '71]

23. Calculate the standard deviation from the following series:

20, 85, 120, 60, 40.

[B.U., B.Com. '71]

24. Find the standard deviation of weights (to the nearest pound) of 15 students given below: 138, 156, 147, 115, 145, 132, 163, 158, 130, 123, 103, 109, 100, 105, 106.

[B.U., B.A.(Econ) '72]

25. Calculate the s.d. of the following observations:

240.12, 240.13, 240.15, 240.12, 240.17,

240.15, 240.17, 240.16, 240.22, 240.21. [I.C.W.A., June '76]

26. Find the standard deviation for the distribution given below:

x	1	2	3	4	5	6	7
Frequency	10	20	30	35	14	10	2

[Dip. Management, '67]

27. Find the s.d. from the following table giving the age distribution of 540 members of a Parliament:

Age in Years	30	40	50	60	70
No. of Members	64	132	153	140	51

[C.U., B.Com. '78]



28. Find the s.d. from the following frequency distribution:

Wt. (lbs.)	120–124	125–129	130–134	135–139	140–144	145–149	Total
No. of Boys	12	25	28	15	12	8	100

[B.U., B.Com, '74]

29. The weights of a certain product produced in a factory are given below from a sample of 121 articles :

Weight (oz.)	3.0–3.1	3.1–3.2	3.2–3.3	3.3–3.4	3.4–3.5	3.5–3.6
Frequency	5	10	12	20	25	18
	3.6–3.7	3.7–3.8	3.8–3.9	3.9–4.0		
	10	8	8	5		

Calculate the arithmetic mean and standard deviation.

[C.A., May '72]

30. Compute the standard deviation of the following data:

Weekly Wages in Rs	Number of Men
30 and under 40	8
40 and under 50	12
50 and under 60	6
60 and under 70	4
70 and under 80	10

[B.U., B.Com. '73]

31. Find the standard deviation of the following distribution:

Turnover (Rs '000 p.a.)	50–100	100–150	150–200	200–250	250–300	300–350	350–400
No. of Firms	5	8	9	12	18	23	17

[C.U., B.Com. '74]

32. Compute the arithmetic mean, standard deviation and mean deviation about the mean for the following data:

Scores	4–5	6–7	8–9	10–11	12–13	14–15	Total
f	4	10	20	15	8	3	60

[I.C.W.A., Dec. '78]

33. Compute the S.D. of income from the following:

Income (Rs)	Below 200	200–399	400–599	600–799	800–999	1000–1199
No. of Earners	25	72	47	22	13	7

[C.U., B.A.(Econ) '78]

34. Find the mean and the S.D. from the following frequency distribution:

Weight (lb.)	131–140	141–150	151–160	161–170	171–180
No. of Persons	2	5	4	9	7
	181–190	191–210	211–240		
	5	3	1		

[I.C.W.A., Jan'71]

35. Calculate the proportion of farms in which costs of production are within the range $A.M. \pm S.D.$ in the following distribution:



<i>Costs of Production</i> (Rs per 5 litres)	4-6	6-8	8-10	10-12	12-14	14-16	Total
<i>No. of Dairy Farms</i>	13	111	182	105	19	7	437

[I.C.W.A., Dec. '73]

36. Out of 400 observations, 100 observations have the value one and the rest of the observations are zero. Find the mean and s.d. of 400 observations together.

[B.U., B.A.(Econ) '66]

37. Two samples of sizes 60 and 90 have 52 and 48 as the respective arithmetic means, and 9 and 12 as the respective standard deviations. Find the arithmetic mean and the standard deviation of the combined sample of size 150.

[I.C.W.A., July '70]

38. The means of two samples of sizes 50 and 100 respectively are 54.4 and 50.3 and the standard deviations are 8 and 7. Obtain the mean and standard deviation of the sample of size 150 obtained by combining the two samples. (Give answers correct to one decimal place.)

[D.M. '72; I.C.W.A., Dec. '77 & June '78]

39. An analysis of monthly wages paid to workers in two firms A and B, belonging to the same industry, gives the following results:

	<i>Firm A</i>	<i>Firm B</i>
<i>Number of Wage Earners</i>	550	650
<i>Average Monthly Wages</i>	Rs 50	Rs 45
<i>S.D. of the Distribution of Wages</i>	Rs ($\sqrt{90}$)	Rs ($\sqrt{120}$)

Answer the following questions with proper justifications:

- (a) Which firm A or B pays out larger amount as monthly wages?
 (b) In which firm A or B is there greater variability in individual wages?
 (c) What are the measures of (i) average monthly wages, and (ii) standard deviation in the distribution of individual wages of all workers in the two firms taken together?

[I.C.W.A., June '77]

40. A company has three establishments E_1 , E_2 and E_3 in three cities. Analysis of the monthly salaries paid to the employees in the three establishments is given below:

	E_1	E_2	E_3
<i>Number of Employees</i>	20	25	40
<i>Average Monthly Salary (Rs)</i>	305	300	340
<i>S.D. of Monthly Salaries (Rs)</i>	50	40	45

Find the average and the standard deviation of the monthly salaries of all the 85 employees in the company.

[I.C.W.A., Dec. '76]

41. Three sets of values of the variable x have means 26.3, 27.0 and 28.5 and standard deviations 4.5, 3.9 and 4.8. If the three sets have respectively 50, 60 and 55 values, what would be the mean and variance of x , if the three sets are taken together?

[B.U., B.A.(Econ) '77]

42. The mean and the variance calculated from a group of 80 observations are 63.2 and 25.93 respectively. If 60 of these observations have mean 64.8 and S.D. 4. find the mean and the s.d. of the remaining 20 observations.

[I.C.W.A., July '71]



43. A group has the following measurements:

$$\bar{x} = 10, \quad \sigma^2 = 4 \text{ and } n = 60.$$

A subgroup of the above has $\bar{x}_1 = 11$, $\sigma_1^2 = 2.25$ and $n_1 = 40$. Find the mean and standard deviation of the other subgroup.

[I.C.W.A., June '73; M.B.A. '78]

44. The following data refer to the dividend (%) paid by two companies A and B over the last 7 years.

A	4	8	4	15	10	11	9
B	12	8	3	15	6	4	10

Calculate the coefficients of variation and comment.

[I.C.W.A., June '75-old]

45. From the prices of shares x and y below find out which is more stable in value:

x	35	54	52	53	56	58	52	50	51	49
y	108	107	105	105	106	107	104	103	104	101

[I.C.W.A., Dec. '76-old]

46. Calculate the coefficient of variation from the following data, showing Grades of 100 students in M.A. Mathematics:

Grades	30–39	40–49	50–59	60–69	70–79	80–89	90–99
Frequency	2	3	11	20	32	25	7

[C.U., M.Com. '73]

47. The mean life in days and standard deviation for two types of electric bulbs are given below:

	Mean Life in days	Standard Deviation in days
Type I	310	9
Type II	260	14

Compare the relative variability of life of the two types of bulbs.

[B.U., B.A.(Econ) '65]

48. You are given the distribution of wages in two factories X and Y.

Wages (Rs)	50–100	100–150	150–200	200–250	250–300	300–350
No. of X	2	9	29	54	11	5
Workers Y	6	11	18	32	27	11

State in which factory the wages are more variable (Use Standard Deviation and Mean.) [C.A., May '75]

49. Calculate a suitable measure of dispersion for the following distribution:

Cotton Consumed in Thousand Candles	0–2	2–4	4–6	6–8	8–10	10–12	12–14
No. of Mills	5	13	12	11	8	4	1
				14–16	16–18	18–20	20–22
				3	1	1	2

How does this dispersion compare with a S.D. of 3 lbs for the weight



of yarns per spindle) among mills producing yarns of average weight 20 lbs. per spindle? [I.C.W.A., June '74]

50. In a small town, a survey was conducted in respect of profit made by retail shops. The following results were obtained:

Profit or Loss (Rs '000)	-4 to -3	-3 to -2	-2 to -1	-1 to 0	0 to 1
No. of Shops	4	10	22	28	38
	1 to 2	2 to 3	3 to 4	4 to 5	5 to 6
	56	40	24	18	10

- Calculate: (i) the average profit made by a retail shop;
(ii) total profit made by all the shops;
(iii) the coefficient of variation of earnings.,

[C.A., Nov. '77]



ANSWERS



1. (i) A.M. reduces by 10; (ii) & (iii) Range and S.D. Unchanged.
2. (i) Unchanged, (ii) Changes by the same proportion.
3. (i) 178, (ii) 12
4. 15 inches (= 74.5 - 59.5)
5. 8.05, 30
6. 0.49 short ton.
7. 14.9
8. 33.9
9. 20.4
10. 2.02
11. 1.24 inches
12. 1.48
13. 55.67, 1.18 (inches)
14. 30, 3.1
15. 20, 3.04
16. 10, 2.86
17. 5.0
18. 54, 3.04
19. (see Example 6.18, for $n = 3$)
20. $\sigma_y = |b| \cdot \sigma_x$
21. Mean = $-\bar{x}$, S.D. = σ [see (5.3.3) & (6.6.2); $z = -x$, when $a = -1$ & $b = 0$]
22. (see Example 5.12)
23. 34.9
24. 21 lbs.
25. 0.033
26. 1.4
28. 11.64 years
28. 7.18 lbs.
29. 3.47, 0.23 (oz.)
40. Rs. 14.80
31. 88.4 (Rs '000)
32. 9.23, 2.48, 2.03
33. Rs. 251.40
34. 169.25, 19.5 (lbs.)
35. 0.65
36. $1/4, \sqrt{3/4}$
37. 49.6, 11.1
38. 51.7, 7.6
39. B, B; 47.29 & 10.60 (Rs)
40. 320.00, 48.69 (Rs)
41. 27.29 Rs, 20.16 (Rs)²
42. 58.4, 5.0
43. 8, 1.22
44. 41.5, 48.6; Dividend paid by B less stable.
45. Share y; (C.V. 11.6 & 1.9)
46. 18.1
47. Type II more variable (C.V. 2.9 & 5.4)
48. Y(C.V. 23 & 30)
49. $\sigma = 4.78$ units; (C.V. 67 & 45). Dispersion for cotton consumption more.
50. 1.348 & 337 (Rs '000); 153.

7 MOMENTS, SKEWNESS AND KURTOSIS

7.1 MOMENTS

Given n observations $x_1, x_2, \dots, x_n$, and an arbitrary constant A ,

$\frac{1}{n} \Sigma(x - A)$ is called the *1st moment about A*,

$\frac{1}{n} \Sigma(x - A)^2$ is called the *2nd moment about A*, (7.1.1)

$\frac{1}{n} \Sigma(x - A)^3$ is called the *3rd moment about A*,

and so on. Let us denote these moments successively by m_1', m_2', m_3' , etc. (Sometimes, these are also represented by μ_1', μ_2', μ_3' , etc.)

$$\begin{aligned} \text{Then } m_1' &= \Sigma(x - A)/n = (\Sigma x - \Sigma A)/n = (\Sigma x - nA)/n \\ &= \bar{x} - A \end{aligned} \quad (7.1.2)$$

i.e. the 1st moment about A equals $(\bar{x} - A)$.

Moments about zero (i.e. when $A = 0$), and moments about mean (i.e. when $A = \bar{x}$) are particularly important.

(1) Moments about zero (or Raw moments)

$$\text{1st moment about zero} = \frac{1}{n} \Sigma x = \bar{x}$$

$$\text{2nd moment about zero} = \frac{1}{n} \Sigma x^2 \quad (7.1.3)$$

$$\text{3rd moment about zero} = \frac{1}{n} \Sigma x^3$$

and so on. Note that the 1st moment about zero is the mean $\bar{x}$.

$$m_1' = \bar{x} \quad (7.1.4)$$

(2) Moments about mean (or Central moments)

$$\text{1st moment about mean} = \frac{1}{n} \Sigma(x - \bar{x}) = 0$$



$$\text{2nd moment about mean} = \frac{1}{n} \sum (x - \bar{x})^2 = \sigma^2$$

$$\text{3rd moment about mean} = \frac{1}{n} \sum (x - \bar{x})^3$$

$$\text{4th moment about mean} = \frac{1}{n} \sum (x - \bar{x})^4 \quad (7.1.5)$$

and so on. These are usually denoted by m_1, m_2, m_3, m_4 , etc. (Sometimes, these are represented by $\mu_1, \mu_2, \mu_3, \mu_4$, etc.) Note that the 1st central moment is always zero, and the 2nd central moment is the variance σ^2 . By (5.3.2) and (6.5.2),

$$m_1 = 0, \quad m_2 = \sigma^2 \quad (7.1.6)$$

From the second relation, we find that the standard deviation is the square-root of the second central moment m_2 .

The 3rd central moment m_3 is used to measure skewness (Section 7.7) and the 4th central moment m_4 to measure kurtosis (Section 7.8). Higher order moments m_5, m_6 etc. are seldom used.

In general, given n observations $x_1, x_2, \dots, x_n$, the r -th order, moments ($r = 0, 1, 2, \dots$) are defined as follows:

$$r\text{-th moment about } A: \quad m'_r = \frac{1}{n} \sum (x - A)^r$$

$$r\text{-th raw moment:} \quad m'_r = \frac{1}{n} \sum x^r$$

$$r\text{-th central moment:} \quad m_r = \frac{1}{n} \sum (x - \bar{x})^r \quad (7.1.7)$$

For a frequency distribution,

$$r\text{-th moment about } A: \quad m'_r = \frac{1}{N} \sum f(x - A)^r$$

$$r\text{-th raw moment:} \quad m'_r = \frac{1}{N} \sum f x^r$$

$$r\text{-th central moment:} \quad m_r = \frac{1}{N} \sum f (x - \bar{x})^r \quad (7.1.8)$$

where $N = \sum f$. [Note that moments about mean are written without dashes ('), but moments about any other origin, i.e. non-central moments, with dashes.]

There are important relations between central and non-central moments. For example, if the non-central moments (m'_1, m'_2, m'_3 etc.) about any arbitrary origin A are known, the central moments can be obtained by using the relations (7.2.2), viz.

$$\begin{aligned} m_2 &= m'_2 - m_1'^2 \\ m_3 &= m'_3 - 3m'_2m'_1 + 2m_1'^3 \\ m_4 &= m'_4 - 4m'_3m'_1 + 6m'_2m_1'^2 - 3m_1'^4 \end{aligned} \quad (7.1.9)$$

In particular, using the first two moments m'_1 and m'_2 , about an arbitrary origin A , the mean and variance may be obtained:

$$\bar{x} = m'_1 + A, \quad \sigma^2 = m'_2 - m_1'^2 \quad (7.1.10)$$



Example 7.1 The first two moments of a distribution about the value 5 of the variable are 2 and 20. Find the mean and the variance. [I.C.W.A. June '77]

Solution Using formulae (7.1.10), we have here

$$m'_1 = 2, m'_2 = 20, \text{ and } A = 5.$$

$\therefore$

$$\bar{x} = 2 + 5 = 7,$$

$$\sigma^2 = 20 - 2^2 = 16.$$

Ans. 7, 16.



Example 7.2 The first four moments of a distribution about the value 3 are 2, 10, 40 and 218. Find the moments about the origin and mean.

Solution Given

$$\Sigma(x-3)/n = 2,$$

$$\Sigma(x-3)^2/n = 10,$$

$$\Sigma(x-3)^3/n = 40,$$

$$\Sigma(x-3)^4/n = 218,$$

we have to find (a) $\Sigma x/n$, $\Sigma x^2/n$, $\Sigma x^3/n$, $\Sigma x^4/n$; and (b) m_1 , m_2 , m_3 , m_4 .

$$(a) \quad \Sigma(x-3)/n = 2;$$

$$\text{or,} \quad (\Sigma x - 3n)/n = 2$$

$$\text{or,} \quad \Sigma x/n - 3 = 2$$

$$\therefore \quad \Sigma x/n = 2 + 3 = 5 \quad \dots (i)$$

$$\Sigma(x-3)^2/n = 10$$

$$\text{or,} \quad \Sigma(x^2 - 6x + 9)/n = 10$$

$$\text{or,} \quad \Sigma x^2/n - 6 \Sigma x/n + 9 = 10$$

$$\text{or,} \quad \Sigma x^2/n - 6 \times 5 + 9 = 10; \text{ putting } \Sigma x/n = 5 \text{ from (i)}$$

$$\therefore \quad \Sigma x^2/n = 10 + 30 - 9 = 31 \quad \dots (ii)$$

$$\text{Again,} \quad \Sigma(x-3)^3/n = 40$$

Using the binomial $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ we have

$$\Sigma(x^3 - 3x^2 \cdot 3 + 3x \cdot 3^2 - 3^3)/n = 40$$

$$\text{or,} \quad \Sigma x^3/n - 9 \Sigma x^2/n + 27 \Sigma x/n - 27 = 40$$

$$\text{or,} \quad \Sigma x^3/n - 9 \times 31 + 27 \times 5 - 27 = 40; \text{ from (i) and (ii).}$$

$$\therefore \quad \Sigma x^3/n = 40 + 279 - 135 + 27 = 211 \quad \dots (iii)$$

$$\text{Also,} \quad \Sigma(x-3)^4/n = 218$$

Using the binomial expansion $(a-b)^4 = a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4$,

$$\Sigma(x^4 - 4x^3 \cdot 3 + 6x^2 \cdot 3^2 - 4x \cdot 3^3 + 3^4)/n = 218$$

$$\text{or,} \quad \Sigma x^4/n - 12 \Sigma x^3/n + 54 \Sigma x^2/n - 108 \Sigma x/n + 81 = 218$$

$$\text{or,} \quad \Sigma x^4/n - 12 \times 211 + 54 \times 31 - 108 \times 5 + 81 = 218;$$

from (i), (ii) & (iii)

$$\therefore \quad \Sigma x^4/n = 218 + 2532 - 1674 + 540 - 81 = 1535 \quad \dots (iv)$$

(b) The moments about 3 are given as

$$m'_1 = 2, m'_2 = 10, m'_3 = 40, m'_4 = 218$$

Hence, using relations (7.1.9), the moments about mean are

$$m_1 = 0 \text{ (in all cases)}$$

$$m_2 = m'_2 - m_1'^2 = 10 - 2^2 = 6$$

$$m_3 = m'_3 - 3m'_2m'_1 + 2m_1'^3$$

$$= 40 - 3 \times 10 \times 2 + 2 \times 2^3 = -4$$

$$m_4 = m'_4 - 4m'_3m'_1 + 6m'_2m_1'^2 - 3m_1'^4$$

$$= 218 - 4 \times 40 \times 2 + 6 \times 10 \times 2^2 - 3 \times 2^4$$

$$= 218 - 320 + 240 - 48 = 90$$

[Note that in relations (7.1.9), the moments about any arbitrary origin may be used to obtain the central moments. Hence, we may also use the moments about origin, obtained earlier, viz. 5, 31, 211, 1535, to find the central moments.

$$m_2 = 31 - 5^2 = 6$$

$$m_3 = 211 - 3 \times 31 \times 5 + 2 \times 5^3 = 211 - 465 + 250 = -4$$



$$\begin{aligned} m_4 &= 1535 - 4 \times 211 \times 5 + 6 \times 31 \times 5^2 - 3 \times 5^4 \\ &= 1535 - 4220 + 4650 - 1875 = 90 \end{aligned}$$

Ans. 5, 31, 211, 1535; 0, 6, -4, 90



Example 7.3 The arithmetic mean of a certain distribution is 5. The second and the third moments about the mean are 20 and 140 respectively. Find the third moment of the distribution about 10. [I.C.W.A., June '74]

Solution Given $\Sigma(x-5)^2/n = 20$ and $\Sigma(x-5)^3/n = 140$, we have to find the value of $\Sigma(x-10)^3/n$. Applying the binomial expansion $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$, we have

$$\begin{aligned} (x-10)^3 &= \{(x-5)-5\}^3 \\ &= (x-5)^3 - 15(x-5)^2 + 75(x-5) - 125 \end{aligned}$$

Summing over all values of x and then dividing both sides by n ,

$$\Sigma(x-10)^3/n = \Sigma(x-5)^3/n - 15\Sigma(x-5)^2/n + 75\Sigma(x-5)/n - 125$$

But $\Sigma(x-5)/n = (\Sigma x - \Sigma 5)/n = (\bar{x} - 5) = 0$ (since A.M. = 5). Substituting the values,

$$\begin{aligned} \Sigma(x-10)^3/n &= 140 - 15 \times 20 + 75 \times 0 - 125 \\ &= 140 - 300 + 0 - 125 = -285 \end{aligned}$$

Ans. -285



Example 7.4 If m_2' and m_2 are respectively the second moment about an arbitrary origin a and that about $\bar{x}$, then show that $m_2' = m_2 + d^2$, where $d = \bar{x} - a$. [C.U., M.Com. '76]

Solution By definition

$$m_2' = \Sigma(x-a)^2/n, \quad m_2 = \Sigma(x-\bar{x})^2/n$$

We may write $(x-a)^2 = \{(x-\bar{x}) + (\bar{x}-a)\}^2$

$$= (x-\bar{x})^2 + (\bar{x}-a)^2 + 2(x-\bar{x})(\bar{x}-a)$$

$$\therefore \Sigma(x-a)^2/n = \Sigma(x-\bar{x})^2/n + \Sigma(\bar{x}-a)^2/n + 2(\bar{x}-a)\Sigma(x-\bar{x})/n$$

$$\text{or,} \quad m_2' = m_2 + n(\bar{x}-a)^2/n + 2(\bar{x}-a).0/n$$

$$\text{or,} \quad m_2' = m_2 + (\bar{x}-a)^2$$

$$\text{or,} \quad m_2' = m_2 + d^2$$

(Shown)



Example 7.5 Show how moments are used to describe the characteristics of a distribution, viz. central tendency, dispersion, skewness, and kurtosis.

[I.C.W.A., Dec. '73, June '75]

Solution All the important characteristics of a frequency distribution can be described by moments. A measure of central tendency is given by the first moment about zero, which is the arithmetic mean. By (7.1.4)

$$\bar{x} = m_1'$$

A measure of dispersion is given by the second central moment, which is the square of standard deviation. By (7.1.6)

$$\sigma = \sqrt{m_2}$$

Measures of skewness and kurtosis are provided by the 3rd and the 4th central moments. By (7.7.4) and (7.8.1)

$$\text{Skewness } (\gamma_1) = \frac{m_3}{(\sqrt{m_2})^3}; \quad \text{Kurtosis } (\gamma_2) = \frac{m_4}{m_2^2} - 3.$$





7.2 RELATION BETWEEN CENTRAL AND NON-CENTRAL MOMENTS

(I) Formula for m_r in terms of m'_r and moments of lower order:

$$m_r = \Sigma(x_i - \bar{x})^r/n \quad m'_r = \Sigma(x_i - A)^r/n$$

Let us write

$$x_i - \bar{x} = (x_i - A) - (\bar{x} - A) = \{(x_i - A) - d\}, \text{ (suppose)}$$

Using the binomial expansion

$$(x_i - \bar{x})^r = (x_i - A)^r - {}^rC_1(x_i - A)^{r-1}d + {}^rC_2(x_i - A)^{r-2}d^2 - \dots + (-1)^r d^r$$

Summing over all values of $i = 1, 2, \dots, n$

$$\Sigma(x_i - \bar{x})^r = \Sigma(x_i - A)^r - {}^rC_1 d \Sigma(x_i - A)^{r-1} + {}^rC_2 d^2 \Sigma(x_i - A)^{r-2} - \dots + (-1)^r n d^r.$$

Now dividing both sides by n ,

$$m_r = m'_r - {}^rC_1 m'_{r-1} d + {}^rC_2 m'_{r-2} d^2 - \dots + (-1)^r d^r \quad (7.2.1)$$

where $d = \bar{x} - A = m'_1$ (see 7.1.2).

In particular, putting $r = 1, 2, 3, 4$, we get

$$\begin{aligned} m_1 &= m'_1 - d \\ m_2 &= m'_2 - 2m'_1 d + d^2 \\ m_3 &= m'_3 - 3m'_2 d + 3m'_1 d^2 - d^3 \\ m_4 &= m'_4 - 4m'_3 d + 6m'_2 d^2 - 4m'_1 d^3 + d^4 \end{aligned}$$

Writing $d = m'_1$ and simplifying, the central moments (m_r) when expressed in terms of the moments (m'_r) about any origin are

$$\begin{aligned} m_1 &= 0, & \text{as it should be, by (7.1.6)} \\ m_2 &= m'_2 - m_1'^2 \\ m_3 &= m'_3 - 3m'_2 m'_1 + 2m_1'^3 \\ m_4 &= m'_4 - 4m'_3 m'_1 + 6m'_2 m_1'^2 - 3m_1'^4 \end{aligned} \quad (7.2.2)$$

(II) Formula for m'_r in terms of m_r and moments of lower order

We write $x_i - A = (x_i - \bar{x}) + (\bar{x} - A) = (x_i - \bar{x}) + d$, (suppose)

Therefore, using the binomial expansion

$$(x_i - A)^r = (x_i - \bar{x})^r + {}^rC_1(x_i - \bar{x})^{r-1}d + {}^rC_2(x_i - \bar{x})^{r-2}d^2 + \dots + d^r$$

Summing over all values of $i = 1, 2, \dots, n$

$$\Sigma(x_i - A)^r = \Sigma(x_i - \bar{x})^r + {}^rC_1 d \Sigma(x_i - \bar{x})^{r-1} + {}^rC_2 d^2 \Sigma(x_i - \bar{x})^{r-2} + \dots + n d^r.$$

Dividing both sides by n ,

$$m'_r = m_r + {}^rC_1 m_{r-1} d + {}^rC_2 m_{r-2} d^2 + \dots + d^r \quad (7.2.3)$$

where $d = \bar{x} - A = m'_1$ is the first moment about A .

In particular, putting $r = 1, 2, 3, 4$ we get

$$\begin{aligned} m'_1 &= m_1 + d \\ m'_2 &= m_2 + 2m_1 d + d^2 \\ m'_3 &= m_3 + 3m_2 d + 3m_1 d^2 + d^3 \\ m'_4 &= m_4 + 4m_3 d + 6m_2 d^2 + 4m_1 d^3 + d^4 \end{aligned}$$

Since $m_1 = 0$ and $d = \bar{x} - A = m'_1$, we have

$$\begin{aligned} m'_1 &= m'_1 \text{ (as expected)} \\ m'_2 &= m_2 + m_1'^2 \\ m'_3 &= m_3 + 3m_2 m'_1 + m_1'^3 \\ m'_4 &= m_4 + 4m_3 m'_1 + 6m_2 m_1'^2 + m_1'^4 \end{aligned} \quad (7.2.4)$$



Example 7.6 Let $a_r = \mu_r/\sigma^r$ where μ_r is the r th moment about mean and σ is the standard deviation. Show that $\alpha_1 = 0$ and $\alpha_2 = 1$. [C.U., B.Com. (Hons) '67]

Solution The r th moment about mean, is defined as

$$\mu_r = \Sigma(x_i - \bar{x})^r/n$$

Putting $r = 1$ and 2 successively, we have

$$\mu_1 = \Sigma(x_i - \bar{x})/n = 0/n = 0, \quad \text{by (5.3.2)}$$

$$\mu_2 = \Sigma(x_i - \bar{x})^2/n = \sigma^2. \quad \text{by (6.5.2)}$$

Also, we are given $a_r = \mu_r/\sigma^r$. Therefore,

$$\alpha_1 = \mu_1/\sigma = 0/\sigma = 0, \quad \alpha_2 = \mu_2/\sigma^2 = \sigma^2/\sigma^2 = 1$$



7.3 BETA-COEFFICIENTS AND GAMMA-COEFFICIENTS

'Beta-coefficients' are defined as follows:

$$\beta_1 = \frac{m_3^2}{m_2^3}; \quad \beta_2 = \frac{m_4}{m_2^2} \quad (7.3.1)$$

(Note that β is the Greek small letter 'beta').

The beta-coefficients can never be negative, and are pure numbers, independent of origin and scale of observations. Since for a symmetrical distribution all odd order moments are zero, $m_3 = 0$ (see Example 7.17); consequently $\beta_1 = 0$, when the distribution is symmetrical.

Frequency distributions are classified as *leptokurtic*, *platykurtic*, or *mesokurtic*, according as the value of β_2 is greater than, less than, or equal to 3. Beta-coefficients are used for measuring skewness and kurtosis. (Sections 7.7 and 7.8)

'Gamma-coefficients' are defined as follows:

$$\gamma_1 = \sqrt{\beta_1}; \quad \gamma_2 = \beta_2 - 3 \quad (7.3.2)$$

(Note that γ is the Greek small letter 'gamma'). γ_1 must have the same sign as m_3 . The gamma-coefficients may be positive, negative, or zero; but are pure numbers like the beta-coefficients. These are used as measures of 'skewness' and 'kurtosis'.

$$\text{Skewness } (\gamma_1) = \sqrt{\beta_1} = \frac{m_3}{\sigma^3} \quad (7.3.3)$$

$$\text{Kurtosis } (\gamma_2) = \beta_2 - 3 = \frac{m_4}{\sigma^4} - 3$$

Distributions are said to be 'positively skew', 'negatively skew', or 'symmetrical', according as γ_1 is positive, negative, or zero. Similarly positive, negative, or zero values of γ_2 are associated with 'leptokurtic', 'platykurtic', or 'mesokurtic' distributions.

From the results of Example 7.2, we have

$$\beta_1 = \frac{(-4)^2}{6^3} = .074; \quad \beta_2 = \frac{90}{6^2} = 2.5$$

Hence, $\gamma_1 = \sqrt{.074} = -0.27$ (since $m_3 = -4$ is negative), and $\gamma_2 = 2.5 - 3 = -0.5$. The distribution is therefore negatively skew and platykurtic.



For the 'Normal distribution' (Chapter 12), $\beta_1 = 0$ and $\beta_2 = 3$; i.e. $\gamma_1 = 0$ and $\gamma_2 = 0$. If, therefore, for any given distribution γ_1 and γ_2 do not significantly differ from 0, the data may be assumed to belong to a 'normal' distribution.

7.4 STANDARDIZED VARIABLE

In statistical theories, it is often found convenient to express deviations from mean ($\bar{x}$) in terms of standard deviation (σ) as unit.

$$z = \frac{x - \bar{x}}{\sigma} = \frac{\text{Variable} - \text{Mean}}{\text{S.D.}} \quad (7.4.1)$$

is called the *Standardized variable* or *Standardized form of x*. The standardized variable z is independent of origin and unit of measurement. If the variable x represents length in ft., its mean $\bar{x}$ and S.D. σ will both be expressed in ft., so that the numerator and the denominator of z will be in ft., and hence z is a pure number.

Mean of the standardized variable z is zero, and its S.D. is 1.

$$\bar{z} = 0, \quad \sigma_z = 1 \quad (7.4.2)$$

Since the mean is zero, the r th central moment of z is the same as the r th raw moment of z , and is usually denoted by α_r

$$\alpha_r = m_r / \sigma^r \quad (7.4.3)$$

Note that $\alpha_1 = 0$, $\alpha_2 = 1$, $\alpha_3 = \sqrt{\beta_1} = \gamma_1$, $\alpha_4 = \beta_2 = \gamma_2 + 3$.

Example 7.7 If m_r and α_r denote the r th central moments of x and of the standardized variable $z = (x - \bar{x})/\sigma$ respectively, where σ is the S.D. of x , then show that $\alpha_r = m_r/\sigma^r$.

Solution By definition, $m_r = \Sigma(x_i - \bar{x})^r/n$, $\alpha_r = \Sigma(z_i - \bar{z})^r/n$

But $z_i = (x_i - \bar{x})/\sigma$; hence by (5.3.2)

$$\bar{z} = \frac{1}{n} \Sigma z_i = \frac{1}{n} \Sigma \left(\frac{x_i - \bar{x}}{\sigma} \right) = \frac{1}{n\sigma} \Sigma(x_i - \bar{x}) = 0$$

$$\therefore \alpha_r = \Sigma(z_i - 0)^r/n = \Sigma z_i^r/n$$

$$= \frac{1}{n} \Sigma \left(\frac{x_i - \bar{x}}{\sigma} \right)^r = \frac{1}{n} \Sigma \frac{(x_i - \bar{x})^r}{\sigma^r} = \frac{1}{\sigma^r} \Sigma \frac{(x_i - \bar{x})^r}{n} = \frac{m_r}{\sigma^r}$$



7.5 MOMENTS OF FREQUENCY DISTRIBUTIONS

If $x_1, x_2, \dots, x_n$ have frequencies $f_1, f_2, \dots, f_n$ respectively, the r -th moment about A is defined as

$$m'_r = \frac{1}{N} \Sigma f(x - A)^r \quad (7.5.1)$$

where $N = \Sigma f$.

The r -th central moment is similarly defined as

$$m_r = \frac{1}{N} \Sigma f(x - \bar{x})^r \quad (7.5.2)$$

where $\bar{x} = \Sigma fx/N$.



In the case of grouped frequency distributions, the midvalues are taken as representatives of the respective classes and $x_1, x_2, \dots, x_n$ denote these midvalues. If the successive mid values have a common difference, the calculation of central moments can be simplified.

If $y = (x - c)/d$, where c and d are constants, the r -th central moment of x is equal to d^r times the r -th central moment of y .

$$m_{r(x)} = d^r \cdot m_{r(y)} \quad (7.5.3)$$

Since the values of y are usually small, $m_{r(y)}$ can be calculated easily from the raw moments of y . Finally, $m_{r(y)}$ is multiplied by d^r to give the required central moments of x . In particular, for the standardized variable $z = (x - \bar{x})/\sigma$, we have $c = \bar{x}$ and $d = \sigma$, so that $m_{r(x)} = \sigma^r \cdot m_{r(z)}$; or, $m_{r(z)} = m_{r(x)}/\sigma^r$; i.e. $\alpha_r = m_r/\sigma^r$, the result derived in Example 7.7.

Example 7.8 Show that the central moments are invariant under change of origin, but not under change of scale. [I.C.W.A. Dec. '75]

Solution Let $y = (x - c)/d$. Then, by (5.3.4), $\bar{x} = c + d\bar{y}$.

$$\therefore x - \bar{x} = (c + dy) - (c + d\bar{y}) = d(y - \bar{y})$$

$$\begin{aligned} \text{Hence, } m_{r(x)} &= \Sigma(x - \bar{x})^r/N = \Sigma fd^r(y - \bar{y})^r/N \\ &= d^r \Sigma f(y - \bar{y})^r/N = d^r \cdot m_{r(y)} \end{aligned}$$

Since the expression on the right hand side is independent of c , but contains d , the result follows.



Example 7.9 Find the second, the third and the fourth central moments of the frequency distribution given below. Hence, find a measure of skewness (γ_1) and a measure of kurtosis (γ_2).

Class Limits	110.0–114.9	115.0–119.9	120.0–124.9	125.0–129.9
Frequency	5	15	20	35
	130.0–134.9	135.0–139.9	140.0–144.9	Total
	10	10	5	100

[I.C.W.A., June '76]

Solution

Table 7.1 Calculations for Moments

Mid-value x	f	$y = \frac{x - 127.45}{5}$	fy	fy^2	fy^3	fy^4	$f(y + 1)^4$
112.45	5	-3	-15	45	-135	405	80
117.45	15	-2	-30	60	-120	240	15
122.45	20	-1	-20	20	-20	20	0
127.45	35	0	0	0	0	0	35
132.45	10	1	10	10	10	10	160
137.45	10	2	20	40	80	160	810
142.45	5	3	15	45	135	405	1280
Totals	100	—	-20	220	-50	1240	2380

**Charlier's check:**

C.V.L. Charlier gave a simple and effective check on the calculations for moments from a grouped frequency distribution. Expanding $(y + 1)^4$ by binomial theorem, multiplying by f , and then summing, we have the identity

$$\Sigma f(y + 1)^4 = \Sigma fy^4 + 4\Sigma fy^3 + 6\Sigma fy^2 + 4\Sigma fy + N \quad (7.5.4)$$

Along with the columns for fy, fy^2, fy^3, fy^4 needed for the calculation of moments upto the 4th order, an extra column for $f(y + 1)^4$ is also calculated. If it is found that the numerical values obtained from the table satisfy the identity (7.5.4), then we are sure that the calculations are correct.

Applying Charlier's check on the calculations in Table 7.1,

$$\begin{aligned} 2380 &= 1240 + 4(-50) + 6(220) + 4(-20) + 100 \\ &= 1240 - 200 + 1320 - 80 + 100 \\ &= 2660 - 280 \\ &= 2380 \end{aligned}$$

Hence, the calculations in Table 7.1 are correct.

Raw moments of y :

$$\begin{aligned} m_1' &= \Sigma fy/N = -20/100 = -0.2 \\ m_2' &= \Sigma fy^2/N = 220/100 = 2.2 \\ m_3' &= \Sigma fy^3/N = 50/100 = -0.5 \\ m_4' &= \Sigma fy^4/N = 1240/100 = 12.4 \end{aligned}$$

Central moments of y : Applying (7.2.2), we have

$$\begin{aligned} m_2 &= m_2' - (m_1')^2 = 2.2 - (-0.2)^2 = 2.16 \\ m_3 &= m_3' - 3m_2'm_1' + 2(m_1')^3 \\ &= -0.5 - 3(2.2)(-0.2) + 2(-0.2)^3 \\ &= -0.5 + 1.32 - .016 \\ &= 0.804 \\ m_4 &= m_4' - 4m_3'm_1' + 6m_2'(m_1')^2 - 3(m_1')^4 \\ &= 12.4 - 4(-0.5)(-0.2) + 6(2.2)(-0.2)^2 - 3(-0.2)^4 \\ &= 12.4 - 0.4 + 0.528 - .0048 \\ &= 12.5232 \end{aligned}$$

Central moments of x : Using (7.5.3), we have

$$\begin{aligned} m_{2(x)} &= d^2.m_{2(y)} = 5^2 \times 2.16 = 54.0 \\ m_{3(x)} &= d^3.m_{3(y)} = 5^3 \times 0.804 = 100.5 \\ m_{4(x)} &= d^4.m_{4(y)} = 5^4 \times 12.5232 = 7827.0 \end{aligned}$$

Also,

$$\bar{x} = c + d\bar{y} = 127.45 + 5(-0.2) = 126.45$$

Since the beta-coefficients (7.3.1) and gamma-coefficients (7.3.2) are independent of origin and scale, we may use the moments of y :

$$\beta_1 = \frac{(0.804)^2}{(2.16)^3} = 0.064; \quad \beta_2 = \frac{12.5232}{(2.16)^2} = 2.68$$

$$\therefore \gamma_1 = \sqrt{0.064} = 0.25; \quad \gamma_2 = 2.68 - 3 = -0.32$$

The distribution is positively skew and platykurtic.

Ans. 54.0, 100.5, 7827.0; 0.25, -0.32.

**7.6****SHEPPARD'S CORRECTION FOR ERRORS DUE TO GROUPING**

For the calculation of mean, variance and other moments from a grouped frequency distribution, the mid-value of a class interval is taken as representative of all



observations falling within that class interval. This, however, introduces some error in the calculated values, known as *error due to grouping*. The presence of error will be evident from the fact that moments calculated from ungrouped data will, in general, be different from those calculated from the grouped frequency distribution. To minimize such errors, the class intervals may be made smaller, or the class limits so chosen that the mid-values coincide with the majority of observations falling within a class interval. Even then, the differences can not be removed altogether. However, under certain conditions it is possible to make corrections for these discrepancies by using certain formulae due to W.F. Sheppard; which are known as *Sheppard's corrections for errors due to grouping*. So far as the first four moments are concerned:

- (i) No correction is necessary for the mean m_1' and the third central moment m_3 .
- (ii) Corrections for the 2nd and the 4th central moments are m_2 (corrected) = m_2

$$(\text{uncorrected}) - \frac{c^2}{12}$$

$$m_4(\text{corrected}) = m_4(\text{uncorrected}) - \frac{c^2}{2} m_2(\text{uncorrected}) + \frac{7c^4}{240}$$

where c is the common width of class intervals.

Sheppard's corrections should be applied only when:

- (a) the distribution relates to a continuous variable, and is of moderate asymmetry;
- (b) the frequency becomes smaller and smaller approaching zero at each end of the distribution.

These corrections are not applicable to J- or U-shaped distributions or to very skew distributions. Moreover, unless the total frequency is fairly large, the corrections will be of little practical importance.

Example 7.10 Apply Sheppard's corrections to the central moments calculated in Example 7.9.

Solution The moments calculated from the grouped data are $m_2 = 54.0$; $m_3 = 100.5$; $m_4 = 7827.0$; and $c = 5$

$$\therefore \begin{aligned} \text{corrected } m_2 &= 54.0 - 5^2/12 = 54.0 - 2.1 = 51.9 \\ \text{corrected } m_3 &= 100.5 \end{aligned}$$

$$\begin{aligned} \text{corrected } m_4 &= 7827.0 - \frac{1}{2} (5^2)(54.0) + 7(5^4)/240 \\ &= 7827.0 - 675.0 + 18.2 = 7170.2 \end{aligned}$$



7.7 SKEWNESS

A frequency distribution is said to be 'symmetrical', if the frequencies are symmetrically distributed about mean, i.e. when values of the variable equidistant from mean have equal frequencies.

Illustration 1. Symmetrical distributions:

(i) x :	10	15	20	25	30
f :	3	7	16	7	3



(ii) x :	10	15	20	25	30	35
f :	3	7	16	16	7	3
(iii) x :	5-8	9-12	13-16	17-20	21-24	
f :	7	18	23	18	7	
(iv) x :	5-8	9-12	13-16	17-20	21-24	25-28
f :	7	18	23	23	18	7

Note that in the above distributions, the means are respectively 20, 22.5, 14.5 and 16.5. The median and the mode for each also have the same value. In fact, for any symmetrical distribution mean, median and mode are equal.

In general, however, frequency distributions are not symmetrical; some are slightly asymmetrical and some others may be highly asymmetrical.

Illustration 2. Asymmetrical (or Skew) distributions:

(i) x :	5-8	9-12	13-16	17-20	21-24
f :	7	18	23	16	7
(ii) x :	5-8	9-12	13-16	17-20	21-24
f :	7	18	23	10	3

Here the frequencies are not symmetrically distributed; in distribution (i) the extent of asymmetry is small, while in (ii) it is comparatively larger.

The word “*skewness*” is used to denote the ‘extent of asymmetry’ in the data. When the frequency distribution is not symmetrical, it is said to be ‘*skew*’. The word ‘skewness’ literally denotes ‘asymmetry, or ‘lack of symmetry’, and ‘skew’ denotes ‘asymmetrical’. A symmetrical distribution has therefore zero skewness. Skewness may also be positive or negative.

Skewness is measured by the following formulae:

(1) *Pearsons’ first measure*—

$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}} \quad (7.7.1)$$

(2) *Pearson’s second measure*—

$$\text{Skewness} = \frac{3 (\text{Mean} - \text{Median})}{\text{Standard Deviation}} \quad (7.7.2)$$

(3) *Bowley’s measure*—

$$\begin{aligned} \text{Skewness} &= \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)} \\ &= \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1} \end{aligned} \quad (7.7.3)$$

where Q_1, Q_2, Q_3 denote the first, second and third quartiles of the distribution.

(4) *Moment measure*—

$$\text{Skewness } (\gamma_1) = \frac{m_3}{\sigma^3} = \frac{m_3}{(\sqrt{m_2})^3} \quad (7.7.4)$$

where m_2 and m_3 are the second and third central moments, and σ denotes the S.D. [Note that this is the third moment α_3 of the standardized variable z , i.e. $\alpha_3 = m_3/\sigma^3 = \sqrt{\beta_1} = \gamma_1$; see (7.3.3) and (7.4.3)].



It should be noted that all the measures of skewness are pure numbers (which do not depend upon the units of measurement), and have the value zero when the distribution is symmetrical.

(**Note:** The measures of skewness (7.7.1) to (7.7.4) are also sometimes known as ‘coefficients of skewness’)

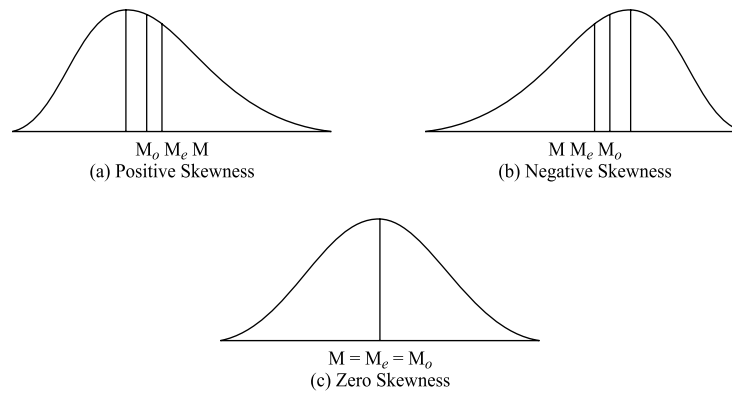


Fig. 7.1 Positions of Mean (M), Median (M_e), Mode (M_o) for Different Types of Skewness

Discussion on different measures—For a symmetrical distribution Mean, Median and Mode are equal. The more the asymmetry in the data, the larger is the discrepancy between them. So the difference between Mean and Mode may be taken as a measure of skewness. This difference, when judged relative to S.D., gives Pearson’s measure (7.7.1). If Mode is not known accurately, the approximate relation $Mean - Mode = 3 (Mean - Median)$ is utilised in (7.7.2). Note that skewness measured by (7.7.1) and (7.7.2) is positive when Mean is larger than Median and Mode, and negative when Mean is less.

Bowley’s measure of skewness (7.7.3) has been developed from the following viewpoint. For a symmetrical distribution, Q_2 (i.e. Median) lies exactly midway between Q_1 and Q_3 ; for a positively skew distribution (i.e. when the longer “tail” of the frequency curve lies towards the right, Fig. 7.1a) Q_3 will be wider away from Q_2 than Q_1 , and for a negatively skew distribution (Fig. 7.1b) the reverse will be the case. The difference of the midpoint of Q_1 and Q_3 from the Median Q_2 , taken relative to Quartile Deviation, gives (7.7.3).

$$\text{Skewness} = \frac{\frac{1}{2}(Q_1 + Q_3) - Q_2}{\frac{1}{2}(Q_3 - Q_1)} = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

As regards the moment measure of skewness (7.7.4), it may be noted that in a symmetrical distribution, for each positive value of $(x_i - \bar{x})$ there is a corresponding negative value. When these deviation $(x_i - \bar{x})$ are cubed, positive values retain their



positive sign and negative values the negative sign, so that $m_3 = \sum f_i(x_i - \bar{x})^3/N$ will be zero. For positively skew distributions (Fig. 7.1a), large positive values of $(x_i - \bar{x})$ are magnified considerably when cubed, and ultimately the sum of positive cubed deviations outweigh the negative cubes, making m_3 positive. In a similar manner, for negatively skew distributions (Fig. 7.1b), m_3 becomes negative.

Limits of different measures—Bowley's measure (7.7.3) lies between -1 and $+1$. There are no theoretical limits to Pearson's measure (7.7.1); but in practice the value is rarely very high. The measure of skewness (7.7.2) lies between -3 and $+3$, although these limits are very seldom attained. No theoretical limits can be set for the moment measure of skewness (7.7.4).

Example 7.11 Find a suitable measure of skewness for the following distribution:

Annual Sales (Rs '000)	0–20	20–50	50–100	100–250	250–500	500–1000
No. of Firms	20	50	69	30	25	19

[CU., M.Com. '72]

Solution Since the class-intervals are of unequal width, Bowley's measure of skewness (7.7.3), which is based on quartiles, will be appropriate here (Note that unequal widths of class intervals do not cause any difficulty in the calculation of partition values; however, other measures of skewness will involve laborious calculations for mean, s.d., moments, etc.). Using simple interpolation in a cumulative frequency distribution (as in Example 5.64), we find that (in Rs '000) $Q_1 = 39.95$, $Q_2 = 76.45$, $Q_3 = 203.75$ (Calculations are not shown here). Putting these values in formula (7.7.3):

$$\text{Skewness} = \frac{203.75 - 2 \times 76.45 + 39.95}{203.75 - 39.95} = \frac{90.80}{163.80} = +0.55$$



Example 7.12 Calculate the coefficient of skewness based on quartiles from the following:

More than	0	5474	More than	60	2718
More than	10	5426	More than	70	1406
More than	20	5259	More than	80	764
More than	30	5023	More than	90	370
More than	40	4475	More than	100	160
More than	50	3712	More than	110	39

[C.A., Nov. '74]

Solution The measure of skewness based on quartiles is given by Bowley's formula (7.7.3). We have to find the quartiles Q_1 , Q_2 , Q_3 , by using simple interpolation (page 64) in the following table showing the cumulative frequencies of more-than type.

**Table 7.2** Cumulative Frequencies

x	Cum. Freq. (more-than)
0	5474 = N
10	5426
20	5259
30	5023
40	4475
$Q_1 \rightarrow$	$\leftarrow 3N/4 = 4105.5$
50	3712
$Q_2 \rightarrow$	$\leftarrow 2N/4 = 2737$
60	2718
70	1406
$Q_3 \rightarrow$	$\leftarrow N/4 = 1368.5$
80	764
90	370
100	160
110	39

Since Q_1, Q_2, Q_3 are values of x corresponding to (less-than) cumulative frequencies $N/4, 2N/4, 3N/4$ respectively, they also correspond to more-than cumulative frequencies $3N/4, 2N/4, N/4$.

Using simple interpolation,

$$\frac{Q_1 - 40}{50 - 40} = \frac{4105.5 - 4475}{3712 - 4475}$$

$$\frac{Q_2 - 50}{60 - 50} = \frac{2737 - 3712}{2718 - 3712}$$

$$\frac{Q_3 - 70}{80 - 70} = \frac{1368.5 - 1406}{764 - 1406}$$

Solving, we get $Q_1 = 44.84, Q_2 = 59.81, Q_3 = 70.58$

Substituting these values in the formula

$$\text{Skewness} = \frac{70.58 - 2 \times 59.81 + 44.84}{70.58 - 44.84} = \frac{-4.20}{25.74} = -0.16$$



Example 7.13 A frequency distribution gives the following results: (i) Coefficient of variation = 5, (ii) Variance = 4, (iii) Karl Pearson's coefficient of skewness = 0.5. Find the mean and the mode of the distribution. [I.C.W.A., Dec. '75 & '76-old]

Solution From (ii), S.D. = $\sqrt{\text{Variance}} = \sqrt{4} = 2$.

Substituting the values of S.D. and C.V. in the formula

$$\text{C.V.} = \frac{\text{S.D.}}{\text{Mean}} \times 100, \text{ we get } 5 = \frac{2}{\text{Mean}} \times 100$$

Solving this, Mean = 40. Now, putting the values of Mean and Skewness in formula (7.7.1), viz.



$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}}, \quad 0.5 = \frac{40 - \text{Mode}}{2}$$

we get Mode = 39.

Ans. 40, 39



Example 7.14 From the data given below, calculate the coefficient of variation: Pearson's measure of skewness = 0.42; Arithmetic mean = 86; Median = 80.

[C.A., Nov. '75-old]

Solution Using A.M. and Median, Pearson's measure of skewness is

$$\text{Skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}}; \quad \text{or,} \quad 0.42 = \frac{3(86 - 80)}{\text{S.D.}}$$

Solving this, S.D. = $18/0.42 = 43$. Hence

$$\text{C.V.} = \frac{\text{S.D.}}{\text{Mean}} \times 100 = \frac{43}{86} \times 100 = 50.$$

Ans. 50



Example 7.15 If the first quartile is 142 and the semi-interquartile range is 18, find the median (assuming the distribution to be symmetrical about mean or median).

[I.C.W.A., Dec. '78]

Solution Semi-interquartile range is given by the formula $(Q_3 - Q_1)/2$. Hence, $18 = (Q_3 - 142)/2$. Solving, $Q_3 = 178$.

For a symmetrical distribution, Skewness = 0. Using quartiles,

$$\text{Skewness} = \frac{Q_3 - 2.Q_2 + Q_1}{Q_3 - Q_1} = 0;$$

$$\text{i.e.} \quad Q_3 - 2.Q_2 + Q_1 = 0 \quad \text{or,} \quad 178 - 2.Q_2 + 142 = 0.$$

$$\text{Solving this,} \quad Q_2 = 160.$$

Ans. Median = 160



Example 7.16 The following facts were gathered from a firm before and after an industrial dispute:

		Before Dispute	After Dispute
Mean Wages	... (Rs)	185	190
Median Wages	... (Rs)	182	180
Modal Wages	... (Rs)	176	160
Quartiles	... (Rs)	175 and 192	175 and 195
S.D.	... (Rs)	13	19
Number Employed	...	600	550

Compare the position of the firm before and after the dispute by making use of the above data as fully as possible. [C.A., May '76-old]

Solution (a) Number of workers has decreased by 50, from 600 to 550 as a result of the dispute.

(b) Although the mean wage has slightly increased, the firm saves Rs 6,500 (after dispute) in respect of the monthly salary bill:



Total wages (before dispute) = $600 \times 185 = 1,11,000$ Rs

Total wages (after dispute) = $550 \times 190 = 1,04,500$ Rs

Difference = 6,500 Rs

(c) The median and modal wages have decreased. Before the dispute, 50% of the workers used to get Rs 182 and above, but after the dispute, workers in this category are less than 50%. Similarly, most of the workers are being paid around Rs 160 (after dispute) as against Rs 180 (before dispute).

(d) The first quartile Q_1 has not changed. The second quartile Q_2 (i.e. Median) has decreased slightly, but the third quartile Q_3 has increased. The significance of this information is as shown below:

Category of Workers	Wages (Rs)	
	Before Dispute	After Dispute
(A) Lowest-paid 25%	Upto 175	Upto 175
(B) Next higher group of 25%	175–182	175–180
(C) Next higher group of 25%	182–192	180–195
(D) Highest-paid 25%	Above 192	Above 195

Category (A) workers are not affected. The next higher category (B) workers are now confined to a narrower range of salary. But the highest-paid categories (C) and (D) are now generally paid more after the dispute.

(e) S.D. has increased from Rs 13 to Rs 19, implying thereby that the variability in individual wages has increased after dispute.

C.V. (before dispute) = $(13 \div 185) \times 100 = 7$

C.V. (after dispute) = $(19 \div 190) \times 100 = 10$

The variability relative to mean has also increased.

(f) Measures of skewness are:

	Before dispute	After dispute
Pearson's measure	$\frac{185-176}{13} = 0.69$	$\frac{190-160}{19} = 1.58$
Bowley's measure	$\frac{192-2(182)+175}{192-175} = 0.18$	$\frac{195-2(180)+175}{195-175} = 0.50$

Skewness, i.e. the asymmetry in the distribution of wages has increased considerably after dispute.



Example 7.17 Show that the measure of skewness based on third central moment is a pure number and is zero for symmetrical distribution.

[C.U., B.Com. (Hons) '65]

Solution The measure of skewness based on the third central moment is

$$\gamma_1 = \frac{m_3}{\sigma^3} = \frac{\sum f(x - \bar{x})^3}{N\sigma^3}$$

where m_3 denotes the third central moment of the distribution, and σ is the standard deviation.

Let us suppose that the variable x represents length in ft. Then $\bar{x}$ and σ will be in ft., so that $(x - \bar{x})$ will be in ft. and $(x - \bar{x})^3$ will have "cubic ft." as unit. So, m_3 will be in cubic ft. and $\gamma_1 = m_3/\sigma^3$ is a ratio whose numerator is in cubic ft. and denominator also in cubic ft. Hence, the ratio is a pure number, independent of the unit of measurement.

If the values of x are symmetrically distributed about the mean, then for every positive value of $(x - \bar{x})$ there is a corresponding negative value. When these deviations $(x - \bar{x})$ are cubed, the negative values retain the negative sign and positive values the positive sign. Hence



in $\sum f(x - \bar{x})^3$, positive and negative values cancel each other and become zero. Thus, γ_1 is zero for a symmetrical distribution. If the frequency curve has a longer tail towards the right, positive values will outweigh the negative values and m_3 will be positive, so that γ_1 will be positive. The case for a negatively skew distribution is just the opposite.



Example 7.18 *The frequency distribution of a large number of balls when classified according to their radius is symmetrical. Show that if we classify the balls according to their volume, the resulting frequency distribution will be skewed.*

[C.U., B.A. (Econ) '66]

Solution Let q_1 , q_2 , and q_3 represent the first, second and third quartiles of the frequency distribution of radius R . Then 25% of the balls have radius upto q_1 , 50% have radius upto q_2 , and 75% upto q_3 . It is given that the frequency distribution of radius is symmetrical; hence the first quartile q_1 and the third quartile q_3 are equidistant from the second quartile q_2 , on either side. Let

$$q_1 = q_2 - h, \quad q_3 = q_2 + h, \quad (h \text{ positive})$$

If now Q_1 , Q_2 and Q_3 represent the corresponding quartiles of the frequency distribution of volume V , then 25% of the same balls have volume upto Q_1 , 50% have volume upto Q_2 , and 75% upto Q_3 . Again, since the volume V of a ball of radius R is given by

$$V = \frac{4}{3} \pi R^3$$

and 25% of the balls have radius upto q_1 , the same balls have volume upto Q_1 . Substituting Q_1 and q_1 for V and R respectively

$$Q_1 = \frac{4}{3} \pi q_1^3 = \frac{4}{3} \pi (q_2 - h)^3$$

Similarly,

$$Q_2 = \frac{4}{3} \pi q_2^3$$

$$Q_3 = \frac{4}{3} \pi q_3^3 = \frac{4}{3} \pi (q_2 + h)^3$$

Using Bowley's formula (7.7.3) for the frequency distribution of volume V

$$\begin{aligned} \text{Skewness} &= \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1} = \frac{\frac{4}{3} \pi \{(q_2 + h)^3 - 2q_2^3 + (q_2 - h)^3\}}{\frac{4}{3} \pi \{(q_2 + h)^3 - (q_2 - h)^3\}} \\ &= \frac{6q_2 h^2}{2(3q_2^2 h + h^3)} = \frac{3q_2 h}{3q_2^2 + h^2} \end{aligned}$$

This is positive (and *not zero*), because both q_2 and h are positive. Thus, the frequency distribution of volume is positively skew.



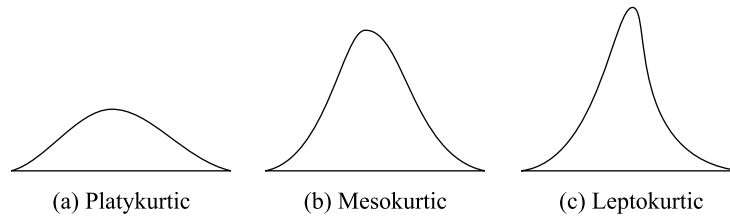
7.8 KURTOSIS

Kurtosis refers to the degree of "peakedness" of the frequency curve. Two distributions may have the same average, dispersion and skewness; yet, in one there may be high concentration of values near the mode, showing a sharper peak in the frequency curve than in the other. This characteristic of the frequency distribution is known as "kurtosis". The only measure of kurtosis is based on moments, viz.



$$\text{Kurtosis } (\gamma_2) = \frac{m_4}{\sigma^4} - 3 = \beta_2 - 3 \quad (7.8.1)$$

where m_4 and σ denote the fourth central moment and S.D. respectively. [Note that m_4/σ^4 is the fourth moment α_4 of the standardized variable z , i.e. $\alpha_4 = m_4/\sigma^4 = \beta_2$; see (7.3.3) and (7.4.3)].



Fig, 7.2 Different Types of Kurtosis

$\beta_2 < 3$, for Platykurtic distribution

$\beta_2 = 3$, for Mesokurtic distribution

$\beta_2 > 3$, for Leptokurtic distribution

For the 'Normal Distribution' (Chapter 12), which is neither very peaked nor flat-topped, $\beta_2 = 3$. This important distribution is taken as a standard for measuring kurtosis, and it has become customary to use $\gamma_2 = \beta_2 - 3$ as a measure of kurtosis.

A distribution is said to be "platykurtic", when γ_2 is negative; it is said to be "mesokurtic" when $\gamma_2 = 0$, and "leptokurtic" when γ_2 is positive.

The frequency curve for a platykurtic distribution is relatively flat-topped, and for a leptokurtic distribution it has a relatively high peak. A mesokurtic distribution (e.g. Normal distribution) is of moderate peakedness.



EXERCISES



- What is meant by 'moment' of a distribution? What are the 'raw' and the 'central' moments? [I.C.W.A., June & Dec. '75]
- The first two moments of a distribution about the value 4 are -1.5 and 2.7 .
(a) Find the moments about zero. (b) Also calculate the mean and S.D.
- The first three moments of a distribution about the value 3 of the variable are 2, 10 and 30 respectively. Obtain the first three moments about zero. Show also that the variance of the distribution is 6. [I.C.W.A., Jan. '64]
- The first four moments about the value 1 are 2.6, 10.2, 43.4 and 192.6 respectively. Find the A.M. and the first four moments about 4.
- The first three moments of a distribution about the value 7, calculated from a set of 9 observations, are 0.2, 19.4 and -41.0 . Find the measures of central tendency, dispersion, and also the third moment about the origin. [I.C.W.A., Dec. '75]
- Find the first, the second and the third central moments of the frequency distribution of expenditure (Rs per month) given below:

Expenditure	3-6	6-9	9-12	12-15	15-18	18-21	21-24	Total
No. of Families	28	292	389	212	59	18	2	1000

[I.C.W.A., June '78]



7. Find the first four moments and the values of β_1 and β_2 from the following frequency distribution:

x	21–24	25–28	29–32	33–36	37–40	41–44
f	40	90	190	110	50	20

Also, find the measures of skewness and kurtosis.

8. Establish the relation between the central moments and the raw moments of a frequency distribution, and hence express the first four moments in terms of the raw moments. *[I.C.W.A., June '77]*
9. Explain the terms 'Skewness' and 'Kurtosis' used in connection with the frequency distribution of a continuous variable. Give the measures of skewness (any three) and kurtosis. *[I.C.W.A., June '77]*
10. Find the coefficient of skewness = (Mean – Mode)/S.D.:

Marks	55–58	58–61	61–64	64–67	67–70
Frequency	12	17	23	18	11

[D.M., '78]

11. Calculate Pearson's measure of skewness on the basis of Mean, Mode and Standard deviation–

x	14.5	15.5	16.5	17.5	18.5	19.5	20.5	21.5
f	35	40	48	100	125	87	43	22

[C.A., May '75]

12. Calculate from the undernoted table the measure of skewness based on Mean, Median and Standard Deviation:

X	100–200	200–300	300–400	400–500	500–600	600–700
F	45	88	146	206	79	52
					700–800	800–900
					30	14

[C.A., Nov. '73]

13. Calculate the measure of skewness based on quartiles and median from the following data:

Variable	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	358	2417	976	129	62	18	10

[C.A., May '72]

14. Calculate the coefficient of skewness, based on quartiles:

Class Limits	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
Frequency	5	9	14	20	25	15	8	4

[I.C.W.A., Dec. '76]

15. Calculate Bowley's measure of skewness from the following:

x	10–14	15–19	20–29	30–39	40–49	50–59
f	786	924	320	172	96	32

[C.A., Nov. '75-old]



16. Compute Bowley's and Pearson's measures of skewness:

Monthly Income (Rs)	0–75	75–150	150–225	225–300	300–375	375–450
Frequency	15	200	250	225	10	5

[C.U., M.Com. '76]

17. Calculate the quartile measure of skewness for the distribution of time taken by 100 workers to complete a job:

Time (seconds)	–12	13–15	16–18	19–21	22–24	25–27	28–
No. of Workers	4	16	22	28	15	9	6

[I.C.W.A., Dec. '74-old & D.M., '77]

18. Find the appropriate measure of skewness from the following:

Age (years)	Below 20	20–25	25–30	30–35	35–40
No. of Employees	13	29	46	60	112
			40–45	45–55	55 and above
			94	45	21

[I.C.W.A., Dec. '75]

19. Compute an appropriate measure of skewness from:

Marks (per cent)	Under 30	30–40	40–50	50–60	60–70	70–80	80–90
No. of Students	45	40	24	12	9	3	2

[C.U., M.Com. '75]

20. Calculate with the use of quartiles the coefficient of skewness for the following frequency distribution:

Under Years	10	20	30	40	50	60
No. of Persons	15	32	51	78	97	109

[C.U., M.Com. '61]

21. Coefficient of skewness = -0.375 , Mean = 62, Median = 65. Find the value of standard deviation.

[C.A., Nov. '72]

22. The measure of skewness for a certain distribution is -0.8 . If the lower and the upper quartiles are 44.1 and 56.6 respectively, find the median.

[I.C.W.A., Jan. '71]

23. The median, mode and coefficient of skewness for a certain distribution are respectively 17.4, 15.3 and 0.35. Calculate the coefficient of variation.

[I.C.W.A., Jan. '73]

24. The mean, median and the coefficient of variation of the weekly wages of a group of workers are respectively Rs 45, Rs 42 and 40. Find the (i) mode, (ii) variance, and (iii) coefficient of skewness, for the distribution of wages.

[I.C.W.A., Jan. '72]

25. You are given: Mean = 50, C.V. = 40%, Sk. = - 0.4. Find the S.D., Mode and Median. [C.A., Nov. '73]
26. The first two moments of a distribution about the value 4 of the variable are - 1.5 and 2.7. It is also known that the median of the distribution is 2.1. Comment on the shape of the distribution. [I.C.W.A., June '74-old]
27. Define moments and quartiles. Show how they provide measures of dispersion and skewness. [I.C.W.A., June '73]
28. The following facts were gathered before and after an industrial dispute:

	<i>Before Dispute</i>	<i>After Dispute</i>
<i>No. of Workers Employed</i>	516	508
<i>Mean Wages (Rs)</i>	49.50	51.75
<i>Median Wages (Rs)</i>	52.70	50.00
<i>Variance of Wages (Rs²)</i>	100.00	121.00

Compare the position before and after the dispute in respect of (a) total wages, (b) modal wages, (c) standard deviation, (d) coefficient of variation, (e) skewness. [C.A., Nov. '70]

— ANSWERS

2. (a) 2.5, 6.7; (b) 2.5, 0.671
3. 5, 31, 201
4. 3.6; -0.4, 3.6, -5.2, 22.8
5. Mean = 72, S.D. = 4.4; 738.8
6. 0, 9.4 Rs², 17.4 Rs³
7. mean = 31.3 Rs, $m_2 = 23.04 \text{ Rs}^2$, $m_3 = 26.11 \text{ Rs}^3$, $m_4 = 1496.68 \text{ Rs}^4$; $\beta_1 = .056$,
 $\beta_2 = 2.82$; Skewness (γ_1) = + 0.24, Kurtosis (γ_2) = - 0.18
10. (62.46 - 62.64)/3.76 = -0.048
11. (18.07 - 18.40)/1.77 = - 0.19
12. 3(430.6 - 424.8)/158.6 = + 0.11
13. + 0.13 (quartiles 22.63, 26.73, 32.07)
14. - 0.103 (quartiles 37.36, 50.30, 60.83)
15. + 0.105 (quartiles 13.21, 16.55, 20.67)
16. - .0070 (quartiles 135.47, 191.25, 246.25 Rs) and - 0.14 (mean = 190.69,
mode = 200, S.D. = 67.36 Rs).
17. - .0063 (quartiles 16.18, 19.36, 22.50 sec.)
18. Bowley's measure = - 0.103 (quartiles 31.42, 37.77, 42.93 yr.)
19. Bowley's measure = - .081 (quartiles 22.50, 35.62, 46.77 p.c.).
20. - 0.14 (quartiles 17.2, 31.3, 42.0 years)
21. 24 22. 55.35 23. 49
24. (i) Rs 36, (ii) 324 Rs², (iii) 0.5. 25. 20, 58, 52.67
26. Sk. = + 1.8 (Frequency curve is asymmetrical and has the longer tail on the
right hand side.
28. (a) 25,542 and 26,289 Rs, i.e. 3% increase; (b) 59.10 and 46.50 Rs; (c) 10
and 11 Rs; (d) 20.2 and 21.3; (e) - 0.96 and + 0.48.

8 CURVE FITTING AND METHOD OF LEAST SQUARES

8.1 CURVE FITTING

When observations in respect of two variables are available, very often a relation is found to exist between them. For example, height and weight of persons are interdependent, expenditure depends on income, yield of a crop depends on the amount of rainfall, production depends on price, etc. Frequency, it is found desirable to express this relationship between variables by means of some mathematical equation, representing a certain geometrical curve. The process of finding such a curve or its equation on the basis of a given set of observations is called *curve fitting*.

We list below the equations of some common types of curves:

- | | |
|---------------------------------------|-----------------------------------|
| 1. $y = a + bx$ | <i>Straight line</i> |
| 2. $y = a + bx + cx^2$ | <i>Parabola</i> |
| 3. $y = a + bx + cx^2 + dx^3$ | <i>Cubic curve</i> |
| 4. $y = a + bx + cx^2 + \dots + px^n$ | <i>Polynomial of n-th degree</i> |
| 5. $y = ab^x$ | <i>Exponential curve</i> |
| 6. $y = a + bc^x$ | <i>Modified exponential curve</i> |
| 7. $\frac{1}{y} = a + bc^x$ | <i>Logistic curve</i> |
| 8. $\log y = a + bc^x$ | <i>Gompertz curve</i> |
| 9. $y = ax^b$ | <i>Geometric curve</i> |

The variables x and y are often referred to as *independent variable* and *dependent variable* respectively. (All letters $a, b, c, d, \dots, p$, except x and y , appearing in the above equations represent constants.)

The choice of the appropriate equation to be fitted to a given set of observations is often facilitated by the following rules:

- (i) Plot the corresponding observations of x and y as points on a graph paper. If the pattern of points shows approximately a linear path, use the straight line. (or, if the successive differences in y corresponding to equidistant values of x are approximately equal, use the straight line).
- (ii) If $\log y$ when plotted against x shows a linear path, use the exponential curve.
- (iii) If $\log y$ when plotted against $\log x$ show a linear path, use the geometric curve.



8.2 STRAIGHT LINE AND PARABOLA

Straight Line

Straight line is the geometrical representation of an equation of the form

$$Ax + By + C = 0 \quad (8.2.1)$$

where A, B, C are constants. When B is not zero, i.e. the equation contains a term in y , the equation of the straight line can be solved for y , giving $y = (-A/B)x + (-C/B)$, which is of the form

$$y = a + bx \quad (8.2.2)$$

where a and b are new constants. This is the form we shall generally use to represent a straight line. Sometimes, the form $y = mx + c$ is also used. [Note that the equations $y = a + bx$ and $y = mx + c$ are of the same form. The left hand side contains only y and the right hand side contains a constant and another term involving x .]

Illustration 1. The following equations represent straight lines:

- (i) $y = 3 + 2x$
- (ii) $y = 2x$
- (iii) $y = 5$
- (iv) $5x + 2y - 6 = 0$
- (v) $ax + by + c = 0$
- (vi) $4y + 7 = 0$
- (vii) $3x - 8 = 0$
- (viii) $y = 1600 + 200t$
- (ix) $y - 3 = 2.8(x - 5)$
- (x) $x - 4 = -5(y + 8)$

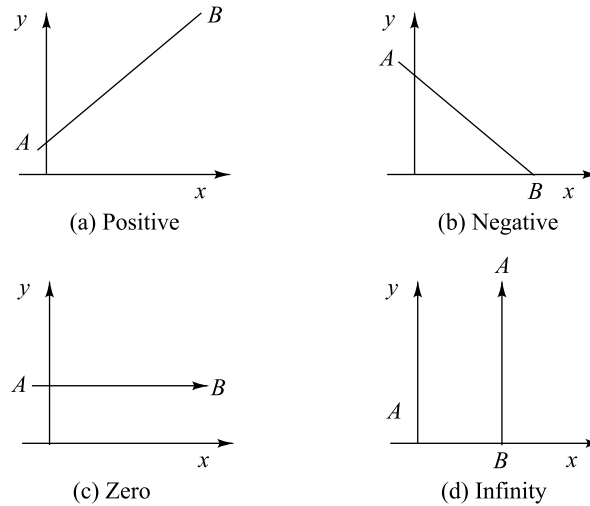
[Note that the equation of a straight line contains terms involving only single power of one or both the variables and a constant term. There must not be any term involving (say) x^2 , xy , $\sqrt{x}$, $1/x$, etc. Here, equations (iv) to (vii) are of the form $Ax + By + C = 0$, and (i), (ii), (iii), (viii) belong to the type $y = a + bx$. Equations (ix) and (x) may also be brought to any of these two forms.]

Slope of Straight Line

In the equation $y = a + bx$, the coefficient of x on the right, viz. b , is called the 'slope' of the straight line. The slope may be positive, negative, or zero. If the equation is given in this form, the slope can be obtained simply by inspection. If however the equation is given in the form $Ax + By + C = 0$, it should be reduced to the form $y = a + bx$ and then the slope easily obtained. If the equation of the straight line does not contain a term in y , the slope cannot be determined.

Illustration 2. (i) The slopes of equations (i), (ii), (iii) and (viii) in Illustration 1 above, are respectively 2, 2, 0 and 200.

(ii) To find the slope of the straight line $5x + 2y - 6 = 0$, let us write the equation as $2y = 6 - 5x$. Dividing both sides by 2, we get $y = 3 - \frac{5}{2}x$ which is of the form $y = a + bx$. The slope is seen to be $-\frac{5}{2}$.

**Fig. 8.1** Slopes of Straight Lines

(iii) The equation $y = 5$, can be written as $y = 5 + 0.x$. Thus, the slope of the straight line is zero.

(iv) The slope of the straight line $3x - 8 = 0$ is indeterminate.

When the slope is positive, y increases as x increases; when the slope is negative, y decreases as x increases; when the slope is zero, y remains a constant whatever be the value of x . Slope represents the amount of change (increase or decrease) in the value of y for a unit increase in the value of x . Conversely, if it is found that the change in y per unit increase in x is always the same, then the relation between x and y can be given by an equation of the form $y = a + bx$.

Illustration 3. (i) In the equation $y = 3 + 2x$, for $x = 0, 1, 2, 3$, the corresponding values are $y = 3, 5, 7, 9$. It is seen that the values of y *increase*, as x increases; because the slope is positive. Also the increase in y for successive values of x is always 2, which is the slope of the straight line.

(ii) In the equation $y = 6 - 2x$, for $x = 0, 1, 2, 3$, the values of y are 6, 4, 2, 0. That is, the values of y *decrease*, as x increases, the successive changes being -2 , which is the slope.

Geometrically, the slope depends on the inclination of the straight line with the x -axis. Two parallel straight lines have the same slope. When the slope is zero, the straight line is parallel to the x -axis. As the slope increases, the inclination also increases. When the slope is positive, the straight line is inclined towards the right; when the slope is negative, the line is inclined towards the left (Fig. 8.1).

Some useful information about straight lines are:

- The constant term a on the right of the equation $y = a + bx$ is called the *y-intercept*, i.e. the value of y at the point where the line crosses the vertical axis ($x = 0$). The quantity b is called the *slope*.
- If a straight line $y = a + bx$ passes through the point (x_0, y_0) , the co-ordinates must satisfy the equation, i.e. $y_0 = a + bx_0$.



- (c) If a straight line passes through the points (x_1, y_1) and (x_2, y_2) , its slope is $b = \frac{y_1 - y_2}{x_1 - x_2}$. If a straight line passes through the point (x_0, y_0) and has slope b , its equation is

$$y - y_0 = b(x - x_0) \quad (8.2.3)$$

Conversely, if the equation of a straight line can be written in this form we are certain that the line passes through the point (x_0, y_0) and has slope b .

- (d) The co-ordinates of the point of intersection of two straight lines is obtained by solving the two equations.

Parabola

Parabola is the geometrical representation of an equation of the form

$$y = a + bx + cx^2 \quad (8.2.4)$$

where a, b, c are constants (the term in x^2 must be present, i.e. $c \neq 0$). The parabola is a special type of curve. In fact, the curved path followed by a flying projectile (e.g. a cricket ball thrown from a distance), is a parabola. The following equations represent parabolas:

- | | |
|-------------------------|-------------------------------|
| (i) $y = 3 - 2x + 7x^2$ | (ii) $y = 6 - 4x^2$ |
| (iii) $y = 0.5x^2$ | (iv) $y = a + bx + cx(x - 1)$ |

Example 8.1 (a) Find the slopes of the following straight lines:

- (i) $y - 3 = 2.8(x - 5)$; (ii) $x - 4 = -5(y + 8)$
 (b) Show that the straight line $3x - 2y + 7 = 0$ passes through the point $(5, 11)$.
 (c) Find the slopes of the straight lines passing through the points (i) $(3, 8)$ and $(5, 11)$, (ii) $(-2, 8)$ and $(4, 12)$, (iii) $(8, 5)$ and $(12, 2)$.
 (d) Find the equation of the straight line passing through the points $(3, 8)$ and $(5, 11)$.
 (e) Show that the straight line $y - 3 = 2.8(x - 5)$ passes through the point $(5, 3)$ and has slope 2.8.
 (f) Find the slopes of the straight lines (i) $y - \bar{y} = b(x - \bar{x})$ and (ii) $x - \bar{x} = b'(y - \bar{y})$, where $\bar{x}, \bar{y}, b$ and b' are constants.
 (g) Show that the lines $2x - 3y - 1 = 0$ and $3x + 4y - 10 = 0$ intersect at the point $(2, 1)$.

Solution (a) (i) The equation can be written as $y = 3 + 2.8(x - 5)$. It is seen that the left-hand side (L.H.S.) contains y only, and on the right-hand side (R.H.S.) the coefficient of x is clearly 2.8, which is the slope.

(ii) Dividing both sides by -5 , we get $\frac{x - 4}{-5} = y + 8$, which is the same as $y + 8 = -\frac{1}{5}(x - 4)$. As discussed in (i), the slope is $-\frac{1}{5}$.

(b) Put $x = 5$ and $y = 11$, and we see that L.H.S. $= 3x - 2y + 7 = 3 \times 5 - 2 \times 11 + 7 = 15 - 22 + 7 = 0 =$ R.H.S. Thus the equation $3x - 2y + 7 = 0$ is satisfied. Hence the result.

(c) (i) Slope $= \frac{8 - 11}{3 - 5} = \frac{-3}{-2} = 3/2$. (ii) $2/3$, (iii) $-3/4$.

(d) As in (c), the slope is $3/2$. Since the straight line passes through the point $(3, 8)$ and has slope $3/2$, by (8.2.3), its equation is $y - 8 = (3/2)(x - 3)$; simplifying, we get $3x - 2y + 7 = 0$.



(e) Put $x = 5, y = 8$, and we find that each side is zero; i.e. L.H.S. = R.H.S. Hence the straight line passes through $(5, 8)$. See solution (a) (i) above.

(f) (i) This is of the form (8.2.3). Hence the straight line passes through the point $(\bar{x}, \bar{y})$ and has slope b . See solution (e) above.

(ii) Dividing both sides by b' , we get $\frac{x - \bar{x}}{b'} = y - \bar{y}$, which is the same as $y - \bar{y} = b'(x - \bar{x})$. This is of the form (8.2.3), indicating that the straight line passes through the point $(\bar{x}, \bar{y})$ and has slope $1/b'$.

[Note: It is interesting to observe that since both the lines $y - \bar{y} = b(x - \bar{x})$ and $x - \bar{x} = b'(y - \bar{y})$ pass through the same point $(\bar{x}, \bar{y})$, they intersect at the point].

(g) Solving the two equations, we get $x = 2, y = 1$. Hence the point of intersection is $(2, 1)$.



8.3 FREE-HAND METHOD OF CURVE FITTING

When the given data are plotted as points on a graph paper, it is often possible to draw a smooth curve through the cluster of points, which appears best to represent their pattern. The smooth curve so drawn is called a *free-hand curve*. It may be noted that the free-hand curve depends entirely on individual judgement, and may either be a straight line or a curved line. If the pattern of points is linear, the equation of a straight line of the form $y = a + bx$ is obtained by choosing two points on the line. Let $(1, 5)$ and $(4, 11)$ be two such points. Substituting for x and y in the equation $y = a + bx$, we find the relation $5 = a + b$ and $11 = a + 4b$, solving which we get $a = 3$ and $b = 2$. Hence $y = 3 + 2x$ is the equation of the fitted free-hand curve, which may be used to estimate the value of y for any given value of x . Similarly, for other types of curves, it is necessary to choose as many points on the smooth curve as there are constants in the equation. The free-hand method has the disadvantage that different individuals will get different curves and equations.

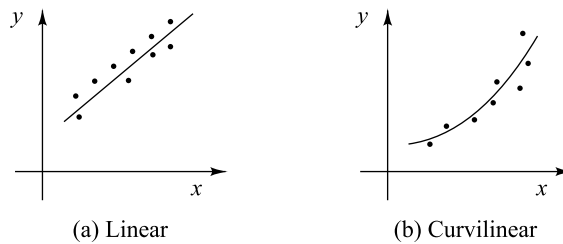


Fig. 8.2 Free-hand Curves

8.4 METHOD OF LEAST SQUARES

Method of Least Squares is a device for finding the equation of a specified type of curve, which best fits a given set of observations. The method depends upon the *Principle of Least Squares*, which suggests that for the “best-fitting” curve, the *sum*



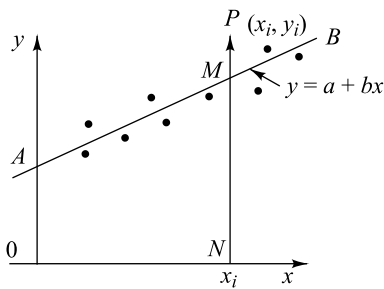
of the squares of differences between the observed and the corresponding estimated values should be the minimum possible.

Suppose, we are given n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, and it is required to fit a straight line to these data. The general equation of a straight line $y = a + bx$ is taken, where a and b are constants. Any values for a and b would give a straight line, and once these values are obtained, an estimate of y can be had by substituting the values of x . That is to say, the estimated values of y when $x = x_1, x_2, \dots, x_n$ would be more $a + bx_1, a + bx_2, \dots, a + bx_n$ respectively. In order that the equation $y = a + bx$ gives a good representation of the relationship between x and y , it is desirable that the estimated values $a + bx_1, a + bx_2, \dots, a + bx_n$ are, on the whole, close enough to the corresponding observed values $y_1, y_2, \dots, y_n$. For the best fitting straight line, therefore, our problem is only to choose such values of a and b for the equation $y = a + bx$ which will provide estimates of y as close as possible to the observed values. This can be done in different ways. However, according to the Principle of Least Squares, the “best-fitting” equation is interpreted as that which minimises the sum of the squares of differences $\Sigma(y_i - a - bx_i)^2$, i.e.

$$(y_1 - a - bx_1)^2 + (y_2 - a - bx_2)^2 + \dots + (y_n - a - bx_n)^2$$

Principle of Fitting Straight Line by Least Squares Method

x (1)	Observed y (2)	Estimated $y = a + bx$ (3)	Difference = (2) – (3) (4)	(Difference) ² (5)
x_1	y_1	$a + bx_1$	$y_1 - a - bx_1$	$(y_1 - a - bx_1)^2$
x_2	y_2	$a + bx_2$	$y_2 - a - bx_2$	$(y_2 - a - bx_2)^2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	y_n	$a + bx_n$	$y_n - a - bx_n$	$(y_n - a - bx_n)^2$
Total		–	–	$\Sigma(y_i - a - bx_i)^2$



(Geometrical Interpretation)

Fig. 8.3 Method of Least Squares

In the geometric sense, the problem of finding the best-fitting straight line is as follows:

If the pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ are plotted as points on a graph paper, and all possible straight lines are drawn on it (Fig. 8.3), that straight line



will be considered to be the “best-fitting” for which the *sum of the squares of vertical distances* PM between the plotted points P and the line AB is the least.

Fortunately, we do not have to find the values of a and b (or geometrically select the straight line) by trial. The problem is tackled by mathematical methods. This leads to a set of equations, called *normal equations*, solving which we get the desired values of a and b . [See Mathematical Note at the end of this Chapter]

Similarly, the method of least squares can be used to fit other types of curves, e.g. parabola, exponential curve, geometric curve, etc. Method of Least Squares is applied to find regression lines (Section 9.12) and also in the determination of trend in time series (Section 14.4).

8.5 FITTING STRAIGHT LINE

Let $y = a + bx$ (8.5.1)

be the equation of the straight line to be fitted to a given set of n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. Applying the method of least squares, the values of a and b are so determined as to minimise $\sum_{i=1}^n (y_i - a - bx_i)^2$. This leads to the *normal equations*

$$\begin{aligned}\Sigma y &= an + b\Sigma x \\ \Sigma xy &= a\Sigma x + b\Sigma x^2\end{aligned}\quad (8.5.2)$$

Here, the values of n , Σx , Σy , Σx^2 and Σxy are substituted on the basis of the given data. We have then two equations involving a and b , solving which the values of a and b are obtained.

[**Note:** The normal equation (8.5.2) can be easily remembered in the following way: Multiply both sides of (8.5.1) by the coefficients of a and b , viz. 1 and x , respectively; and then sum, using Σ notation. Thus

$$\Sigma y = \Sigma(a + bx) = \Sigma a + \Sigma bx = an + b\Sigma x$$

giving the first equation, and

$$\Sigma xy = \Sigma x(a + bx) = \Sigma(ax + bx^2) = \Sigma ax + \Sigma bx^2 = a\Sigma x + b\Sigma x^2$$

giving the second equation. However, it should be borne in mind that this is not a derivation of the normal equations, but only a means of remembering them].

8.6 SIMPLIFIED CALCULATIONS

The labour in computation can often be minimised by a transformation of one or both the variables, either by change of origin only or by changes of origin and scale both.

1. If we change the origin of x only, i.e. write $X = x - c$, where c is an arbitrary constant, then x is replaced by X in the equations (8.5.2), giving

$$\begin{aligned}\Sigma y &= an + b\Sigma X \\ \Sigma Xy &= a\Sigma X + b\Sigma X^2\end{aligned}\quad (8.6.1)$$

2. If we change the origins of both x and y , i.e. write $X = x - c$ and $Y = y - c'$, where c and c' are arbitrary constants, then x and y are replaced by X and Y in the equation (8.5.2), so that



$$\Sigma Y = an + b\Sigma X$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2 \quad (8.6.2)$$

3. In case the successive values of the independent variable x are found to have a common difference, two special transformations are available for the cases, when (i) n is odd, and (ii) n is even.

(i) when n is odd, write

$$u = \frac{x - (\text{central value of } x)}{\text{common difference}} \quad (8.6.3)$$

The values of u will be 0 at the middle; 1, 2, 3, 4, ... for the successive values; and $-1, -2, -3, -4, \dots$ for the preceding values (see Tables 8.5, 8.8, 8.11, 8.12)

(ii) when n is even, write

$$u = \frac{x - (\text{mean of two central values of } x)}{\frac{1}{2}(\text{common difference})} \quad (8.6.4)$$

The values of u will be successively 1, 3, 5, 7, ... for the lower half, and $-1, -3, -5, -7, \dots$ for the upper half, starting from the middle (see Tables 8.6, 8.7).

If we write the equation of straight line in the form $y = a + bu$, the normal equations will be

$$\Sigma y = an + b\Sigma u$$

$$\Sigma uy = a\Sigma u + b\Sigma u^2 \quad (8.6.5)$$

But since by the above transformation $\Sigma u = 0$, these equations take very simple forms

$$\Sigma y = an$$

$$\Sigma uy = b\Sigma u^2 \quad (8.6.6)$$

from which the values of a and b can easily be determined.

However, in all cases, finally we have to rewrite the transformed variables X, Y, u in terms of the original variables x and y , and then simplify the result, to get the equation of the fitted straight line.

Example 8.2 Determine the equation of a straight line which best fits the following data:

x	10	12	13	16	17	20	25
y	19	22	24	27	29	33	37

[C.U., M.Com., '62; D.S.W., '69]

Solution (First method) Let $y = a + bx$ be the equation of the best fitting straight line by method of least squares. The constants a and b are obtained by solving the normal equations (8.5.2),

$$\Sigma y = an + b\Sigma x, \quad \Sigma xy = a\Sigma x + b\Sigma x^2$$

where n is the number of pairs of observations.

**Table 8.1** Calculations for Fitting Straight Line

x	y	x^2	xy
10	19	100	190
12	22	144	264
13	24	169	312
16	27	256	432
17	29	289	493
20	33	400	660
25	37	625	925
Total	113	1983	3276

Putting $\Sigma y = 191$, $n = 7$, $\Sigma x = 113$, $\Sigma xy = 3276$, $\Sigma x^2 = 1983$ in the normal equations, we have

$$191 = 7a + 113b \quad \dots(i)$$

$$3276 = 113a + 1983b \quad \dots(ii)$$

Multiplying (i) by 113 and (ii) by 7, and subtracting

$$\begin{array}{rcl} 21,583 & = & 791a + 12,769b \\ 22,932 & = & 791a + 13,881b \\ \hline -1,349 & = & -1,112b \end{array} \quad \therefore b = \frac{-1349}{-1112} = 1.21$$

Putting the value of b in (i) we have

$$7a = 191 - 113 \times 1.21 = 54.27, \text{ so that } a = 54.27/7 = 7.75$$

Now, substituting the values of a and b in $y = a + bx$, the equation of the fitted straight line is

$$y = 7.75 + 1.21x \quad \dots(iii)$$

[**Note:** The accuracy of equation (iii) may be verified by observing that its both sides will be equal, when x and y are replaced by their means $\bar{x} = 113/7 = 16.14$ and $\bar{y} = 191/7 = 27.29$. Here $\bar{y} = 27.29$ and $7.75 + 1.21 \times 16.14 = 27.28$. The discrepancy of 1 in the last place is due to rounding error].

$$\text{Ans. } y = 7.75 + 1.21x$$

(*Second method*) Let $y = a + bX$ be the equation of the best fitting straight line, where we write $X = x - 16$. The normal equations for determining the values of a and b are (8.6.1), viz.

$$\Sigma y = an + b\Sigma X, \quad \Sigma Xy = a\Sigma X + b\Sigma X^2$$

Table 8.2 Calculations for Fitting Straight Line

x	y	$X = x - 16$	X^2	Xy
10	19	-6	36	-114
12	22	-4	16	-88
13	24	-3	9	-72
16	27	0	0	0
17	29	1	1	29
20	33	4	16	132
25	37	9	81	333
Total	191	1	159	220



Substituting the values $\Sigma y = 191$, $\Sigma X = 1$, $\Sigma X^2 = 159$, $\Sigma XY = 220$ and $n = 7$ in the normal equations, we get

$$191 = 7a + b \quad \dots(\text{iv})$$

$$220 = a + 159b \quad \dots(\text{v})$$

Multiplying (v) by 7 and then subtracting from (iv), we have

$$\begin{array}{rcl} 191 & = & 7a + b \\ 1540 & = & 7a + 1113b \\ \hline -1349 & = & -1112b \end{array} \quad \therefore b = \frac{-1349}{-1112} = 1.21$$

Substituting this value in (iv) we get

$$7a = 191 - 1.21 = 189.79, \quad \therefore a = 189.79/7 = 27.11$$

Hence, the equation of the best-fitting straight line is

$$\begin{aligned} y &= a + bX \\ \text{or, } y &= 27.11 + 1.21(x - 16) \\ &= 27.11 + 1.21x - 19.36 \\ &= 7.75 + 1.21x \end{aligned}$$

(Third method) Let $Y = a + bX$ be the equation of the best fitting straight line, where $X = x - 16$ and $Y = y - 26$ (Deviations should be taken from values close to the respective means; here $\bar{x} = 113/7 = 16.1$ and $\bar{y} = 191/7 = 27.3$). The normal equations are (8.6.2):

$$\Sigma Y = an + b\Sigma X, \quad \Sigma XY = a\Sigma X + b\Sigma X^2$$

Table 8.3 Calculations for Fitting Straight Line

x	y	$X = x - 16$	$Y = y - 26$	X^2	XY
10	19	-6	-7	36	42
12	22	-4	-4	16	16
13	24	-3	-2	9	6
16	27	0	1	0	0
17	29	1	3	1	3
20	33	4	7	16	28
25	37	9	11	81	99
Total	—	1	9	159	194

Substituting the values in the normal equations, we get

$$9 = 7a + b \quad \dots(\text{vi})$$

$$194 = a + 159b \quad \dots(\text{vii})$$

Multiplying (vii) by 7 and subtracting from (vi),

$$\begin{array}{rcl} 9 & = & 7a + b \\ 1358 & = & 7a + 1113b \\ \hline -1349 & = & -1112b \end{array} \quad \therefore b = \frac{-1349}{-1112} = 1.21$$

Putting the value in (vi), $7a = 7.79$; $a = 7.79/7 = 1.11$

Therefore, $Y = a + bX$

$$\text{or, } y - 26 = 1.11 + 1.21(x - 16)$$

$$\text{or, } y = 26 + 1.11 + 1.21x - 19.36$$

$$\text{or, } y = 7.75 + 1.21x$$

[Note: The First method should be employed when the values of x and y are small, say x not exceeding 10 and y not exceeding 100. The Second method will be found convenient when the



values of x exceed 10 and those of y are not very large. When the values of x and y are both large, the *Third method* should be employed. In case, the values of the independent variable x are found to have a common difference, the special methods shown in Examples 8.4 and 8.5 are the most convenient, requiring the minimum possible calculations].



Example 8.3 Fit a straight line by the method of least squares to the following data:

Age	21	42	38	64	53	61	47
Absence (no. of days)	4	14	10	38	19	34	17

Estimate the probable number of days absent when age is 40 years.

[C.U., B.A. (Econ)' 70]

Solution Let x denote age (in years) and y denote the number of days absent, and $y = a + bX$ be the equation of the fitted straight line, where $X = x - 47$ (**Note:** The origin of X should preferably be taken near the mean of the given values of x). Using the method of the squares, the normal equations for determining the values of a and b are

$$\Sigma y = an + b\Sigma X \quad \Sigma Xy = a\Sigma X + b\Sigma X^2$$

Table 8.4 Calculations for Fitting Straight Line

Age (x)	Absence (y)	$X = x - 47$	X^2	Xy
21	4	-26	676	-104
42	14	-5	25	-70
38	10	-9	81	-90
64	38	17	289	646
53	19	6	36	114
61	34	14	196	476
47	17	0	0	0
Total 326	136	-3	1303	972

Putting the values from the table in the normal equations, we have

$$136 = 7a - 3b, \quad 972 = -3a + 1303b$$

Solving these two equations we get $a = 19.77$ and $b = 0.791$. The equation of the fitted straight line is therefore

$$y = a + bX$$

$$\text{or, } y = 19.77 + 0.791(x - 47)$$

$$\text{i.e. } y = 0.791x - 17.41$$

When $x = 40$, $y = 0.791 \times 40 - 17.41 = 14$ days (approx.)

Ans. $y = 0.791x - 17.41$; 14 days



Example 8.4 Fit a straight line to the following data by the method of least squares:

x	15	20	25	30	35
y	12	14	18	25	31



Solution [Note: Here, the successive values of x have a common difference and n is odd; so, the transformation (8.6.3) will be the most suitable.]

Let us write the equation of the straight line in the form

$$y = a + bu \quad \dots(i)$$

where $u = \frac{(x-25)}{5}$. The normal equations are

$$\Sigma y = an + b\Sigma u, \quad \Sigma uy = a\Sigma u + b\Sigma u^2$$

Table 8.5 Calculations for Fitting Straight Line

x	y	$u = \frac{(x-25)}{5}$	u^2	uy
15	12	-2	4	-24
20	14	-1	1	-14
25	18	0	0	0
30	25	1	1	25
35	31	2	4	62
Total	100	0	10	49

Putting the values from Table 8.5 in the normal equations,

$$\begin{cases} 100 = a(5) + b(0) \\ 49 = a(0) + b(10) \end{cases}$$

or,

$$\begin{cases} 100 = 5a \\ 49 = 10b \end{cases}$$

Hence $a = 100/5 = 20$ and $b = 49/10 = 4.9$. Substituting these values in (i), and rewriting y is terms of x ,

$$y = 20 + 4.9 \left(\frac{x-25}{5} \right)$$

or

$$y = 20 + (0.98x - 24.5) = -4.5 + 0.98x$$

Thus, the required equation of the straight line is

$$y = -4.5 + 0.98x$$

...(ii)

$$\text{Ans. } y = -4.5 + 0.98x$$



Example 8.5 Fit a straight line to the following data by the method of least squares:

x	15	20	25	30	35	40
y	12	14	18	25	31	44

Solution [Note: As in the previous Example, here also successive values of x have a common difference; but n is even. So the transformation (8.6.4) will be used.]

Let the equation of the straight line be $y = a + bu$, where $u = \frac{(x-27.5)}{2.5}$. The normal equations are the same as (8.6.5)

**Table 8.6** Calculations for Fitting Straight Line

x	y	$u = \frac{(x - 27.5)}{2.5}$	u^2	uy
15	12	-5	25	-60
20	14	-3	9	-42
25	18	-1	1	-18
30	25	1	1	25
35	31	3	9	93
40	44	5	25	220
Total	114	0	70	218

Substituting the values from Table 8.6 in the normal equations,

$$\begin{cases} 144 = a(6) + b(0) \\ 218 = a(0) + b(70) \end{cases}$$

or

$$\begin{cases} 6a = 144 \\ 70b = 218 \end{cases}$$

Hence $a = 144/6 = 24$, and $b = 218/70 = 3.114$.

Putting the values of a and b and rewriting u in terms of x ,

$$y = 24 + 3.114 \left(\frac{x - 27.5}{2.5} \right)$$

i.e.

$$y = -10.25 + 1.246x$$



Example 8.6 The data below show the lengths (y) in cm. attained by a coiled spring corresponding to various weights (x) in gm. Fit a straight line of the form $y = a + bx$.

x (gm.)	100	200	300	400	500	600
y (cm.)	90.2	92.3	94.2	96.3	98.2	100.3

[I.C.W. A., July '71]

Solution Let us write the equation of the straight line as $Y = A + BX$, where $X = (x - 350)/50$ and $Y = y - 90$. Using the method of least squares, the normal equations for determining the values of A and B are $\Sigma Y = An + B\Sigma X$, $\Sigma XY = A\Sigma X + B\Sigma X^2$

Table 8.7 Calculations for Fitting Straight Line

x	y	$X = \frac{x - 350}{50}$	$Y = y - 90$	X^2	XY
100	90.2	-5	0.2	25	-1.0
200	92.3	-3	2.3	9	-6.9
300	94.2	-1	4.2	1	-4.2
400	96.3	1	6.3	1	6.3
500	98.2	3	8.2	9	24.6
600	100.3	5	10.3	25	51.5
Total	—	0	31.5	70	70.3



Substituting the values from the table in the normal equations,

$$\begin{cases} 31.5 = A \times 6 + B \times 0 \\ 70.3 = A \times 0 + B \times 70 \end{cases} \quad \begin{cases} 6A = 31.5 \\ 70B = 70.3 \end{cases}$$

Solving, $A = 5.25$, $B = 1.004$. The equation of the straight line is therefore,

$$Y = A + BX$$

or, $y - 90 = 5.25 + 1.004 \left(\frac{x - 350}{50} \right)$

Simplifying we get $y = 88.22 + 0.0201x$ as the equation of the fitted straight line.

Ans. $y = 88.22 + .0201x$.



Example 8.7 Fit straight line $y = mx + c$ to the following data and plot the observed and fitted values on the same graph paper:

x	1	3	5	7	9	11	13	15	17
y	10	15	20	27	31	35	30	35	40

[C.U., B.A. (Econ) '74]

Solution (First method) [Note: There are $n = 9$ (odd) pairs of observations, and the values of x have a common difference 2. So we may use the special transformation as in Example 8.4]

Let us write the equation $y = mx + c$ in the form $y = a + bu$, where $u = (x - 9)/2$. The normal equations are,

$$\Sigma y = an + b \Sigma u, \quad \Sigma uy = a \Sigma u + b \Sigma u^2.$$

Table 8.8 Fitting Straight Line

x	Observed Values (y)	$u = \frac{x-9}{2}$	u^2	uy	bu	Fitted Values $a + bu$
1	2	3	4	5	6	7
1	10	-4	16	-40	-13.88	13.12 = 13.1
3	15	-3	9	-45	-10.41	16.59 = 16.6
5	20	-2	4	-40	-6.94	20.06 = 20.1
7	27	-1	1	-27	-3.47	23.53 = 23.5
9	31	0	0	0	0	27.00 = 27.0
11	35	1	1	35	3.47	30.47 = 30.5
13	30	2	4	60	6.94	33.94 = 33.9
15	35	3	9	105	10.41	37.41 = 37.4
17	40	4	16	160	13.88	40.88 = 40.9
Total	243	0	60	208	—	243.0

Substituting the values in the normal equations, we get (since $\Sigma u = 0$), $243 = 9a$, $208 = 60b$. Therefore, $a = 243/9 = 27$, $b = 208/60 = 3.467$. The equation of the fitted line is therefore $y = a + bu$; i.e.

$$y = 27 + 3.467u$$

or, $y = 27 + 3.467 \left(\frac{x-9}{2} \right)$

On simplification, we get $y = 11.40 + 1.73x$



The fitted values of y , corresponding to $x = 1, 3, 5, \dots$, i.e. $u = -4, -3, -2, \dots$, may be obtained from the equation $y = 27 + 3.467u$, and are shown in col. (7), Table 8.8.

The observed values are plotted on a graph paper as points with co-ordinates (1, 10), (3, 15), (5, 20), etc. The fitted values are also plotted with co-ordinates (1, 13.1), (3, 16.6), etc., and they lie on a straight line.

[**Note:** (i) The total of the observed values in col. (2) will be the same as the total of the fitted values in col. (7).

(ii) Since the successive values of u in col. (3) increase by 1, the values in col. (7) will increase by $b = 3.47$. When $x = 1$, i.e. $x = -4$, the fitted value is $27 + 3.47(-4) = 27 - 13.88 = 13.12$. The successive values may then be obtained by repeatedly adding 3.47; e.g. $13.12 + 3.47 = 16.59$; $16.59 + 3.47 = 20.06$; etc.]

(*Second method*) If we use the direct method with the equation as $y = mx + c$, then the normal equations will be

$$\Sigma y = m \Sigma x + cn, \quad \Sigma xy = m \Sigma x^2 + c \Sigma x$$

Table 8.9 Fitting Straight Line

x	Observed Values y	x^2	xy	mx	Fitted Values $mx + c$
1	10	1	10	1.73	$13.13 = 13.1$
3	15	9	45	5.19	$16.59 = 16.6$
5	20	25	100	8.65	$20.05 = 20.0$
7	27	49	189	12.11	$23.51 = 23.5$
9	31	81	279	15.57	$26.97 = 27.0$
11	35	121	385	19.03	$30.43 = 30.4$
13	30	169	390	22.49	$33.89 = 33.9$
15	35	225	525	25.95	$37.35 = 37.4$
17	40	289	680	29.41	$40.81 = 40.8$
Total 81	243	969	2603	—	—

Putting the values, the normal equations are $243 = 81m + 9c$ and $2603 = 969m + 81c$. Solving, $m = 1.73$, $c = 11.40$. Hence, the equation of the fitted line is $y = 1.73x + 11.40$.

Putting $x = 1, 3, 5$, etc. successively, we get the fitted values, as shown in the last column (Table 8.9). Note that some fitted values in Tables 8.8 and 8.9 differ slightly in the last place due to rounding error. Those in Table 8.8 are correct to one decimal.



8.7 FITTING PARABOLA

Let $y = a + bx + cx^2$ (8.7.1)

be the equation of the parabola to be fitted to a given set of n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. Using the method of least squares, the constants a, b, c of the best fitting parabola are obtained by solving the normal equations

$$\left. \begin{aligned} \Sigma n &= an + b \Sigma x + c \Sigma x^2 \\ \Sigma xy &= a \Sigma x + b \Sigma x^2 + c \Sigma x^3 \\ \Sigma x^2 y &= a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4 \end{aligned} \right\} \quad (8.7.2)$$

[**Note:** (1) we have 3 equations in order to find the 3 constants a, b, c .



(2) As stated in the Note, p. 290, it will be found convenient to remember the normal equations in the following way: Multiply both sides of (8.7.1) by the coefficients of a , b , c , viz. 1, x , x^2 respectively, and then sum, using Σ notation.

Thus

$$\Sigma y = \Sigma(a + bx + cx^2) = \Sigma a + \Sigma bx + \Sigma cx^2 = an + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = \Sigma x(a + bx + cx^2) = \Sigma ax + \Sigma bx^2 + \Sigma cx^3 = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2 y = \Sigma x^2(a + bx + cx^2) = \Sigma ax^2 + \Sigma bx^3 + \Sigma cx^4 = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4]$$

In practical problems, simplifications in calculation are often introduced either by change of origin of the independent variable x , or of both the variables x and y , as stated in (1) and (2) of Section 8.6 (p. 251 – 252). When the successive values of x have a common difference, the transformations suggested at (3) of Section 8.6 would, of course, be the most convenient.

Example 8.8 The profits y (Rs lakh) of a certain company in the x th year of its life are given by

x	1	2	3	4	5
y	2.18	2.44	2.78	3.25	3.83

Fit a second degree parabola $y = a + bx + cx^2$ to the data.

[I.C.W.A., Jan '68]

Solution (First method) The constants a , b , c appearing in the equation

$$y = a + bx + cx^2 \quad \dots(i)$$

are obtained by solving the normal equations (8.7.2).

Table 8.10 Calculations for Fitting Parabola

x	y	x^2	x^3	x^4	xy	x^2y
1	2.18	1	1	1	2.18	2.18
2	2.44	4	8	16	4.88	9.76
3	2.78	9	27	81	8.34	25.02
4	3.25	16	64	256	13.00	52.00
5	3.83	25	125	625	19.15	95.75
Total	15	14.48	55	225	47.55	184.71

Substituting the values from Table 8.10 (here $n = 5$)

$$14.48 = 5a + 15b + 55c \quad \dots(ii)$$

$$47.55 = 15a + 55b + 225c \quad \dots(iii)$$

$$184.71 = 55a + 225b + 979c \quad \dots(iv)$$

Multiplying (ii) by 3 and subtracting from (iii),

$$47.55 = 15a + 55b + 225c$$

$$43.44 = 15a + 45b + 165c$$

$$4.11 = 10b + 60c$$

$$\text{i.e. } 10b + 60c = 4.11 \quad \dots(v)$$

Again multiplying (ii) by 11 and subtracting from (iv),

$$184.71 = 55a + 225b + 979c$$

$$159.28 = 55a + 165b + 605c$$

$$25.43 = 60b + 374c$$



i.e. $60b + 374c = 25.43$... (vi)

Solving (v) and (vi), we get $b = .081$ and $c = .055$. Putting these values in (ii), we have $a = 2.048$. Now putting the values of a , b and c in (i), the required equation of the parabola is

$$y = 2.048 + .081x + .005x^2 \quad \dots \text{(vii)}$$

(Second method)

[**Note:** Since the successive values of x have a common difference, and n is odd, we use the transformed variable u , defined by (8.6.3)].

Let $y = a + bu + cu^2$... (viii)

be the equation of the parabola, where $u = x - 3$. Applying the method of least squares, this normal equations for determining the constants a , b , c are

$$\Sigma y = an + b\Sigma u + c\Sigma u^2$$

$$\Sigma uy = a\Sigma u + b\Sigma u^2 + c\Sigma u^3$$

$$\Sigma u^2 y = a\Sigma u^2 + b\Sigma u^3 + c\Sigma u^4 \quad \dots \text{(ix)}$$

where n is the number of pairs of observations given.

Table 8.11 Calculations for the Fitting Parabola

x	y	$u = x - 3$	u^2	u^3	u^4	uy	u^2y
1	2.18	-2	4	-8	16	-4.36	8.72
2	2.44	-1	1	-1	1	-2.44	2.44
3	2.78	0	0	0	0	0	0
4	3.25	1	1	1	1	3.25	3.25
5	3.83	2	4	8	16	7.66	15.32
15	14.48	0	10	0	34	4.11	29.73

Putting the values from Table 8.11 in the normal equations

$$\begin{cases} 14.48 = a(5) + b(0) + c(10) \\ 4.11 = a(0) + b(10) + c(0) \\ 29.73 = a(10) + b(0) + c(34) \end{cases}$$

i.e. $\begin{cases} 14.48 = 5a + 10c \\ 4.11 = 10b \\ 29.73 = 10a + 34c \end{cases}$

From the second equation, $b = 0.411$. Also solving the other two equations, $a = 2.786$ and $c = .055$. Now putting the values of a , b , c in (viii), and rewriting u in terms of x , i.e. $u = x - 3$,

$$\begin{aligned} y &= 2.786 + 0.411(x - 3) + .055(x - 3)^2 \\ &= 2.786 + 0.411(x - 3) + .055(x^2 - 6x + 9) \end{aligned}$$

Simplifying, we get

$$y = 2.048 + .081x + .055x^2$$

[**Check:** The accuracy of the fitted equation (vii) may be verified by observing that both sides of the equation will have the same value when y , x , x^2 are replaced by their means. In the present case,

$$\text{Mean of } y = \Sigma y/n = 14.48/5 = 2.896$$

$$\text{Mean of } x = \Sigma x/n = 15/5 = 3$$

$$\text{Mean of } x^2 = \Sigma x^2/n = 55/5 = 11$$

Putting these values in (vii).

$$2.896 = 2.048 + .081 \times 3 + 0.55 \times 11 = 2.048 + 0.243 + 0.605 = 2.896].$$





Example 8.9 Fit a second degree parabola to the given data, using the method of least squares:

Length (in.)	7.5	10.0	12.5	15.0	17.5	20.0	22.5
Weight (oz.)	1.9	4.5	10.1	17.6	27.8	40.8	56.9

Solution Let x and y represent the length and the weight respectively. The equation of the parabola may be written as

$$y = a + bu + cu^2 \quad \dots(i)$$

where

$$u = \frac{x - 15.0}{2.5}$$

[**Note:** If we use the direct method, the calculations would be large. However, since the values of x have a common difference, viz. 2.5, and n is odd, we use the transformed variable (8.6.3)].

Using the method of least squares, the normal equations for determining the constants a, b, c are

$$\begin{aligned} \Sigma y &= an + b\Sigma u + c\Sigma u^2 \\ \Sigma uy &= a\Sigma v + b\Sigma u^2 + c\Sigma u^3 \\ \Sigma u^2y &= a\Sigma u^2 + b\Sigma u^3 + c\Sigma u^4 \end{aligned}$$

Table 8.12 Calculations for Fitting Parabola

x	y	u	u^2	u^3	u^4	uy	u^2y
7.5	1.9	-3	9	-27	81	-5.7	17.1
10.0	4.5	-2	4	-8	16	-9.0	18.0
12.5	10.1	-1	1	-1	1	-10.1	10.1
15.0	17.6	0	0	0	0	0	0
17.5	27.8	1	1	1	1	27.8	27.8
20.0	40.8	2	4	8	16	81.6	163.2
22.5	56.9	3	9	27	81	170.7	512.1
Total	159.6	0	28	0	196	255.3	748.3

Substituting for $\Sigma y, \Sigma u, \Sigma u^2$ etc. in the normal equations (here $n = 7$), we get

$$159.6 = 7a + 28c, 255.3 = 28b, 748.3 = 28a + 196c$$

Solving

$$a = 17.6, b = 9.1 \text{ and } c = 1.3$$

Putting these values in (i), the equation of the parabola is

$$y = 17.6 + 9.1u + 1.3u^2, \text{ where } u = (x - 15)/2.5 = (0.4x - 6).$$

Substituting for u , we have

$$\begin{aligned} y &= 17.6 + 9.1(0.4x - 6) + 1.3(0.4x - 6)^2 \\ &= 17.6 + 3.64x - 54.6 + 1.3(0.16x^2 - 4.8x + 36) \\ &= -37.0 + 3.64x + 0.208x^2 - 6.24x + 46.8 \end{aligned}$$

or,

$$y = 9.8 - 2.60x + 0.208x^2 \quad \dots(ii)$$

This is the required equation of the parabola.

$$\text{Ans. } y = 9.8 - 2.60x + 0.208x^2$$





8.8 FITTING EXPONENTIAL AND GEOMETRIC CURVES

In order to fit curves with equations of the form $y = ab^x$ and $y = ax^b$, the procedure is to take logarithms of both sides and then form normal equations. For example, to fit the *Exponential Curve* $y = ab^x$, we take logarithms of both sides, obtaining

$$\log y = (\log a) + x(\log b).$$

This can be written as

$$Y = A + Bx$$

where $Y = \log y$, $A = \log a$, $B = \log b$. The normal equations are then

$$\Sigma Y = An + B\Sigma x, \quad \Sigma xY = A\Sigma x + B\Sigma x^2$$

These equations are solved for A and B , and then taking antilog, we find the values of a and b .

Similarly, the *Geometric Curve* $y = ax^b$ may be written as

$$\log y = (\log a) + b(\log x)$$

i.e.

$$Y = A + bX$$

where $Y = \log y$, $A = \log a$, $X = \log x$. The normal equations are

$$\Sigma Y = An + b\Sigma X, \quad \Sigma XY = A\Sigma X + b\Sigma X^2$$

These are solved for A and b , and then $a = \text{antilog } A$ is obtained.

Example 8.10 Fit an equation of the form $y = ab^x$ to the following data:

x	2	3	4	5	6
y	144	172.8	207.4	248.8	298.5

[B.U., B.A. (Econ) '69]

Solution Taking logarithms of both sides in the equation $y = ab^x$, we have $\log y = \log a + x(\log b)$, i.e.

$$Y = A + Bx$$

where $Y = \log y$, $A = \log a$ and $B = \log b$. Applying the method of least squares, the normal equations are

$$\Sigma Y = An + B\Sigma x, \quad \Sigma xY = A\Sigma x + B\Sigma x^2$$

Table 8.13 Fitting Exponential Curve

x	y	$Y = \log y$	x^2	xY
2	144	2.1584	4	4.3168
3	172.8	2.2375	9	6.7125
4	207.4	2.3168	16	9.2672
5	248.8	2.3959	25	11.9795
6	298.5	2.4749	36	14.8494
Total 20	—	11.5835	90	47.1254

Substituting the values in the normal equations.

$$11.5835 = 5A + 20B$$

$$47.1254 = 20A + 90B$$

Solving these equations we get $A = 2.00014$ and $B = 0.07914$; i.e. upto 4 decimals $A = 2.0001$, $B = 0.0791$. Thus



$$\begin{cases} \log a = 2.0001 \\ \log b = 0.0791 \end{cases} \quad \text{or,} \quad \begin{cases} a = \text{antilog } 2.0001 = 100.0 = 100 \text{ approx.} \\ b = \text{antilog } 0.0791 = 1.200 = 1.2 \text{ approx.} \end{cases}$$

Therefore, the equation of the fitted curve is $y = 100(1.2)^x$.



Example 8.11 Determine the constants of the curve $y = ax^n$ which best fits the data given below:

x :	4	5	6	7	8
y :	8	12.5	18	24.5	32

Solution Taking logarithms of both sides in the equation $y = ax^n$, we have $\log y = (\log a) + n(\log x)$;

$$\text{i.e.} \quad Y = A + nX \quad \dots(i)$$

where $Y = \log y$, $A = \log a$, $X = \log x$. The normal equations for determining the constants A and n in (i) are

$$\Sigma Y = Ar + n\Sigma X, \quad \Sigma XY = A\Sigma X + n\Sigma X^2$$

where r is the number of pairs of observations given (here, $r = 5$).

Table 8.14 Fitting Geometric Curve

x	y	$x = \log x$ from log-table	Rounded	$Y = \log y$ from log-table	rounded	X^2	XY
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
4	8	0.6021	0.60	0.9031	0.90	0.3600	0.5400
5	12.5	0.6990	0.70	1.0969	1.10	0.4900	0.7700
6	18	0.7782	0.78	1.2553	1.26	0.6084	0.9828
7	24.5	0.8451	0.85	1.3892	1.39	0.7225	1.1815
8	32	0.9031	0.90	1.5051	1.51	0.8100	1.3590
Total	—	—	3.83	—	6.16	2.9909	4.8333

***Note:** Values of $\log x$ and $\log y$, obtained from 4-figure log tables, are rounded to two demicals for simplifying the calculations.

Putting the values from Table 8.14, we have

$$6.16 = 5A + 3.83n \quad \dots(ii)$$

$$4.8333 = 3.83A + 2.9909n \quad \dots(iii)$$

Dividing both sides of (ii) by 5 and both sides of (iii) 3.83,

$$1.232 = A + 0.766n \quad \dots(iv)$$

$$1.262 = A + 0.781n \quad \dots(v)$$

$$\text{Subtracting} \quad -0.030 = -0.015n; \quad \therefore n = 2$$

$$\text{Putting } n = 2 \text{ in (iv), } A = 1.232 - 2(0.766) = -0.3$$

$$\text{i.e.} \quad \log a = -0.3 = 1.7; \quad \text{or,} \quad a = \text{antilog } (1.7) = 0.5$$

Thus, the constants of the curve $y = ax^n$ are $a = 0.5$ and $n = 2$, and the equation of the curve is $y = 0.5x^2$ (note that this is the equation of a parabola).

Ans. $a = 0.5$, $n = 2$

**EXERCISES**

1. Explain the method of least squares for fitting a straight line $y = mx + c$ to a given set of n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. Indicate some uses of this method in business statistics. [I.C.W.A., Dec '74 (old)]
2. Fit a straight line of the form $y = a + bx$ to each of the following sets of data:

- (i) $\begin{cases} x & 2 & 5 & 6 & 8 & 9 \\ y & 8 & 14 & 19 & 20 & 31 \end{cases}$
- (ii) $\begin{cases} x & 1 & 3 & 5 & 7 & 9 & 11 \\ y & 16 & 12 & 10 & 7 & 5 & 4 \end{cases}$
- (iii) $\begin{cases} x & 4 & 6 & 8 & 12 & 15 & 17 & 22 \\ y & 46 & 42 & 40 & 36 & 30 & 25 & 19 \end{cases}$
- (iv) $\begin{cases} x & 1.0 & 1.5 & 2.0 & 2.5 & 3.0 & 3.5 & 4.0 \\ y & 5.3 & 5.7 & 6.3 & 7.2 & 8.2 & 8.7 & 8.4 \end{cases}$
- (v) $\begin{cases} x & 70 & 80 & 90 & 100 & 110 & 120 \\ y & 553 & 547 & 539 & 533 & 527 & 520 \end{cases}$
- (vi) $\begin{cases} x & 5 & 7 & 9 & 11 & 13 & 15 \\ y & 1.7 & 2.4 & 2.8 & 3.4 & 3.7 & 4.4 \end{cases}$

3. Given the following data on quantities exchanged and prices, fit a linear demand curve $y = a + bx$.

Price (x)	16	10	8	9	5	4	3
Quantity (y)	70	85	100	115	120	124	130

[C.U., M.Com. (Business Econ) '70]

4. Apply the principle of least squares to fit a straight line $y = a + bx$ to the following data:

x :	2	4	6	8	10	12	14
y :	10	14	15	16	15	17	18

[C.U., B.Sc. (Math. Hons) '68]

5. Fit a straight line to the following data and estimate the most probable yield of rice for 40 inches of water.

Water x (inches):	12	18	24	30	36	42	48
Yield y (tons):	5.27	5.68	6.25	7.21	8.02	8.71	8.42

[C.U., M.Com. '64]

6. Calculate the values of m and k for the equation $y = mx + k$ to show the regression of profit per unit of output on output.



Output x ('000)	5	7	9	11	13	15
Profit per unit of Output y (Rs)	1.7	2.4	2.8	3.4	3.7	4.4

Estimate the profit per unit of output when there is an output of 10,500.

[I.C.W.A., June '73]

7. The following data relate to results of a fertiliser experiment on crop yields:

Units of Fertiliser used (X):	0	2	4	6	8	10
Units of Yield (Y):	110	113	118	119	120	118

Fit a straight line to the above data and estimate the amounts of yield when units of fertiliser used are 3 and 7 respectively. [C.U., M.Com. (Old) '69]

8. The weights (in lbs.) of a calf taken at weekly intervals are given below. Fit a straight line, and calculate the average rate of growth per week.

Age (x)	1	2	3	4	5	6	7	8	9	10
Weight (y)	52.5	58.7	65.0	70.2	75.4	81.1	87.2	95.5	102.2	106.4

[Hint: Since n even, take origin of x at 5.5 and $1/2$ as unit. i.e. $y = a + bx$, where $u = (x - 5.5)/(1/2) = 2x - 11$. The values of u will be as usual $-9, -7, -5, -3, -1, 1, 3, 5, 7, 9$].

9. In the following table, S is the weight of potassium bromide which will dissolve in 100 gms of water at $T^\circ\text{C}$. Fit an equation of the form $S = mT + b$ by the method of least squares. Use this relation of estimate S , when $T = 50^\circ$

T	0	20	40	60	80
S	54	65	75	85	96

[I.C.W.A., Jan. '65]

10. Fit a curve of the form $y = ab^x$ to the following data:

x :	2	3	4	5	6	7
y :	640	512	410	328	262	210

11. Fit a curve of the form $y = ax^b$ to the following data:

x :	1	2	3	4	5
y :	5.0	6.3	7.2	7.9	8.5

12. Estimate the constants of the Pareto Curve $n = Ax^{-a}$ which fits the data below:
1945 – 46: Number of Net Incomes more than Rs x after Tax

Income (Rs x)	Number (n)
150	14,000,000
500	825,000
1000	173,000
2000	35,500

[I.C.W.A., Dec. '73]

[Hint: Use logarithm and reduce the curve to straight line form; i.e. $\log n = (\log A) + (-a)(\log x)$; or, $Y = a' + b'X$]



13. Fit a second degree parabola ($y = a + bx + cx^2$), to the following data:

x	0	1	2	3	4
y	1	5	10	22	38

[C.U., M.Com. '66]

14. Fit a parabola of the second degree to the following data taking x as the independent variable ($y = a + bx + cx^2$), by the method of least squares.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

Find out the difference between the actual value of y and the value of y obtained from the fitted curve when $x = 2$.

[I.C.W.A. July '65]

15. The profits (in '000 Rs) of a company in the x -th year of its life are given below. Fit a parabola and estimate its profits in the sixth year.

x	1	2	3	4	5
y	1250	1400	1650	1950	2300

[Hint: Write the equation in the form $v = a + bu + cu^2$, where $u = x - 3$ and $v = (y - 1650)/50$].



ANSWERS



2. (i) $y = 1.0 + 2.9x$ (ii) $y = 16.2 - 1.2x$
- (iii) $y = 52 - 1.5x$ (iv) $y = 4.04 + 1.23x$
- (v) $y = 599.2 - 0.66x$ (vi) $y = 0.50 + 0.257x$
3. $y = 143.0 - 4.667x$ 4. $y = 10.72 + 0.536x$
5. $y = 3.99 + 0.103x$; 8.11 tons 6. $y = 0.50 + 0.257x$; 3.20 Rs
7. $Y = 111.88 + 0.886X$; 114.5 and 118.1 units.
8. $y = 46.13 + 6.053x$; 6.053 lbs. 9. $S = 0.52T + 54.2$; 80.2 units
10. $y = 1000(0.8)^x$ 11. $y = 5.0x^{0.33}$
12. $\log A = 12.22$, $a = -2.33$ 13. $y = 1.42 + 0.26x + 2.21x^2$
14. $y = 1.42 - 1.07x + 0.55x^2$; 0.18 15. $y = 1140 + 72.1x + 32.2x^2$; Rs 2732

MATHEMATICAL NOTE

How the Normal Equations are Derived

- (1) Fitting straight line $y = a + bx$ (page 290):

Applying the Method of Least Squares, the constants a and b are so chosen as

to minimise $\sum_{i=1}^n (y_i - a - bx_i)^2$. This is a problem of Differential Calculus.

Taking partial derivatives with respect to a and b , and equating them to zero, we get

$$\frac{\partial}{\partial a} \Sigma(y - a - bx)^2 = 0, \quad \frac{\partial}{\partial b} \Sigma(y - a - bx)^2 = 0$$



$$\text{i.e. } \begin{cases} \Sigma(y - a - bx)(-2) = 0 \\ \Sigma(y - a - bx)(-2x) = 0 \end{cases} \quad \text{or, } \begin{cases} \Sigma(y - a - bx) = 0 \\ \Sigma x(y - a - bx) = 0 \end{cases}$$

These give the normal equations (8.5.2).

(2) Fitting parabola $y = a + bx + cx^2$ (page 299):

Here, the constants a, b, c are so chosen as to minimise $\sum_{i=1}^n (y_i - a - bx_i - cx_i^2)^2$.

Taking partial derivatives with respect to a, b, c , and equating them to zero,

$$\frac{\partial}{\partial a} \Sigma(y - a - bx - cx^2)^2 = 0; \quad \frac{\partial}{\partial b} \Sigma(y - a - bx - cx^2)^2 = 0;$$

$$\frac{\partial}{\partial c} \Sigma(y - a - bx - cx^2)^2 = 0$$

These lead to the normal equation (8.7.2).

9 CORRELATION AND REGRESSION

9.1 CONCEPTS OF 'CORRELATION' AND 'REGRESSION'

So far we have studied some characteristics of one variable only, e.g. mean of the distribution of height, standard deviation of weight, skewness of the distribution of income. But, many situations arise in which we may have to study two variables simultaneously, say x and y . For example, the variables may be (i) the amount of rainfall and yield of a certain crop, (ii) the height and weight of a group of children, (iii) income and expenditure of several families, (iv) ages of husband and wife; etc. There are two main problems involved in such studies :

Firstly, the data may reveal some association between x and y , and we may be interested to measure numerically the strength of this association between the variables. Such a measure will determine how well a linear or other equation explains the relationship between the variables. This is the problem of *Correlation*.

Secondly, there may be one variable of particular interest, and the other variable, regarded as an auxiliary variable, may be studied for its possible aid in throwing some light on the former. In such a case, one is then interested in using a mathematical equation for making estimates or predictions regarding the principal variable. This equation is known as a *Regression Equation*, and the problem of making predictions on the basis of the equation is called the problem of *Regression*.

In short, correlation is concerned with the measurement of the 'strength of association' between variable; while regression is concerned with the 'prediction' of the most likely value of one variable when the value of the other variable is known.

In *simple correlation* and *simple regression* (also called *linear correlation* and *linear regression*), we consider the simplest kind of relationship, viz. a linear relationship, as the regression equation. Simple correlation is, therefore, concerned with the strength of linear trpe of relationship between the variables.

9.2 BIVARIATE DATA

The word '*bivariate*' is used to describe situations in which two characters are measured on each individual or item, the characters being represented by two variables. Statistical data relating to the simultaneous measurement of two variables are called *bivariate data*. (Data relating to one variable only are called *univariate data*). The observations on each individual are then paired, one for each variable— $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, suppose.



Table 9.1 Bivariate Data Showing Height and Weight of 15 Individuals

Height (inches)	Weight (lbs.)	Height (Inches)	Weight (lbs.)
68	135	66	131
64	127	70	149
69	149	70	130
62	133	59	118
67	145	72	139
63	114	71	136
67	152	67	136
68	157		

The above data show that height (in inches) and weight (in lbs.) for the 1st individual are (68, 135), for the 2nd individual (64, 127), ..., for the last individual (67, 136).

9.3 BIVARIATE FREQUENCY DISTRIBUTION

Example 9.1 Explain clearly the method of constructing a bivariate frequency distribution and explain the meaning of marginal distribution, conditional distribution and conditional mean values with reference to such a table.

[C.U. B.Com.(Hons.)' 65]

Solution In bivariate data, when a large number of pairs of observations is available, it becomes necessary to condense the data in the form of a two-way frequency table, called *Bivariate Frequency Table*, or *Bivariate Frequency Distribution*, or *Correlation Table*.

The method of constructing a bivariate frequency distribution is similar to that of a univariate frequency distribution (p. 91). Suppose, we are given N pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$ relating to two variables x and y . At first we find the maximum and the minimum values in each of the x - and y -series, based on which the number of class intervals and the class limits are fixed. If there are m class intervals for the x -series and n for the y -series, we construct a table with m rows and n columns, thus giving $m \times n$ rectangular spaces, called *cells*. The class limits of the x -series and y -series are shown as row-headings and column-headings respectively. Each pair of observations is now represented by a tally mark (/) placed side by side, in the appropriate cell. Every fifth tally mark, in any cell is however placed across the preceding four to form a group of 5 tally marks (page 93). When all the N pairs of observations have been shown in the two-way table by tally marks, these marks are counted and the number of observations falling in each cell, i.e. the *cell frequencies*, are obtained. A two-way table like Table 9.2 is finally prepared showing the class intervals for x and y and the corresponding frequencies shown in the cells, instead of the tally marks. This is the Bivariate Frequency Distribution.

Table 9.2 Bivariate Frequency Distribution

		Weight (lbs.)					
Height (inches)	$\begin{matrix} y \\ \diagdown \\ x \end{matrix}$	111–120	121–130	131–140	141–150	151–160	Total
	58–61	1					1
	62–65	1	1	1			3
	66–69			3	2	2	7
	70–73		1	2	1		4
	Total	2	2	6	3	2	15 = N

Source: Data of Table 9.1



A univariate frequency distribution, derived from the bivariate frequency distribution, irrespective of the values assumed by the other variable, is called a *Marginal Distribution*. Totals of all frequencies shown in the last row and the last column are called *marginal totals* or *marginal frequencies*. The marginal totals in the last column represent class frequencies of the distribution of height, called the 'marginal distribution of height' whose class intervals are shown in the first column. Similarly, the marginal totals in the last row represent class frequencies of the 'marginal distribution of weight', whose class intervals are shown in the first row.

Table 9.3 Marginal Distribution of Height

Height (inches)	Frequency
58–61	1
62–65	3
66–69	7
70–73	4
Total	15

Table 9.4 Marginal Distribution of Weight

Height (lbs.)	Frequency
111–120	2
121–130	2
131–140	6
141–150	3
151–160	2
Total	15

A univariate frequency distribution, derived from the bivariate frequency distribution, for a specified value (or class interval) of the other variable, is called a *Conditional Distribution*.

Table 9.5 Conditional Distribution of Height
(when weight 131–140 lbs.)

Height (inches)	Frequency
58–61	0
62–65	1
66–69	3
70–73	2
Total	6

Table 9.6 Conditional Distribution of Weight
(when height 70–73 inches)

Weight (lbs.)	Frequency
111–120	0
121–130	1
131–140	2
141–150	1
151–160	0
Total	4

There is only one marginal distribution of x and one marginal distribution of y . But there is one conditional distribution of x corresponding to each class interval of y ; and similarly, corresponding to each class interval of x , there is one conditional distribution of y .

The *Conditional Mean Value* is the arithmetic mean calculated from the conditional distribution.



Example 9.2 The ages of twenty husbands and wives are given below : From a two-way frequency table showing the relationship between the ages of husbands and wives with class intervals 20–25, 25–30 etc.

S. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Age of Husband	28	37	42	25	29	47	37	35	23	41	27	39	23	33	36	32	22	29	38	48
Age of Wife	23	30	40	26	25	41	35	25	21	38	24	34	20	31	29	35	23	27	34	47

[C.A., May '70]

**Solution**

[**Note:** For the construction of a frequency distribution, we must use *class limits*. But in the question, the class intervals have been specified by means of class boundaries, viz. 20–25, 25–30, etc. Our class limits should therefore be interpreted either as “20 and above but below 25; 25 and above but below 30; etc.”, or as “Above 20 but not exceeding 25; Above 25 but not exceeding 30; etc.” (see Example 4.3 (k, 1), p. 77). The latter type is not applicable here, as the age 20 of wife at Sl. No. 13 cannot be accommodated. The given class intervals 20–25, 25–30, etc. must therefore be interpreted in the sense “20 and above but below 25, 25 and above but below 30, etc.”]

Table 9.7 Two-way Frequency Table showing Ages of Husbands and Wives
Age of Wife (years)

	20–25*	25–30	30–35	35–40	40–45	45–50	Total
Age of Husband (years)							
20–25*	3						3
25–30	2	3					5
30–35			1	1			2
35–40		2	3	1			6
40–45				1	1		2
45–50					1	1	2
Total	5	5	4	3	2	1	20

*Class intervals 20–25, 25–30 etc. represent “20 and above but below 25, 25 and above but below 30, etc.”

**Example 9.3** Find the conditional mean values of y when $x = 1, 3$ and 5 :

$x \backslash y$	0	3	6	9	12
1	1	2	1	0	1
3	2	3	4	1	0
5	3	1	3	1	6

[C.U., B.Com. (Hons.)’ 68]

Solution**Table 9.8** Calculations for Conditional Mean Values of y (a) when $x = 1$

y	f	fy
0	1	0
3	2	6
6	1	6
9	0	0
12	1	12
Total	5	24

(b) when $x = 3$

y	f	fy
0	2	0
3	3	9
6	4	24
9	1	9
12	0	0
Total	10	42

(c) when $x = 5$

y	f	fy
0	3	0
3	1	3
6	3	18
9	1	9
12	6	72
Total	14	102

$$\bar{y} = \frac{\Sigma fy}{N} = \frac{24}{5} = 4.8$$

$$\bar{y} = \frac{42}{10} = 4.2$$

$$\bar{y} = \frac{102}{14} = 7.3$$

The conditional mean values of y when $x = 1, 3, 5$ are respectively 4.8, 4.2, 7.3.

9.4 SCATTER DIAGRAM

Example 9.4 What is a Scatter Diagram? Explain how this can be used to indicate the degree and type of association between two variables.

[D.S.W., Nov., '70; I.C.W.A., Jan. '67, July '71, Dec. '73, '74;
C.U., M.Com. '63, '67; B.A. (Econ) '71, '73, '75; B.U., B.A. '77]

Solution When statistical data relating to the simultaneous measurement on two variables are available, each pair of observations can be geometrically represented by a point on the graph paper—the values of one variable being shown along the X -axis and those of the other variable along Y -axis. If there are n pairs of observations, finally the graph paper will contain n points. This diagrammatic representation of bivariate data is known as *Scatter Diagram* (Figs 9.1 and 9.2).

A scatter diagram indicates the nature of association between the two variables, i.e., the type of correlation between them. If the pattern of points (or dots) on the scatter diagram shows a linear path diagonally across the graph paper from the the bottom left-hand corner to the top right, correlation will be positive (Fig. 9.1a). In other words, association between the variables is direct, indicating thereby that high values of one variable are in general, associated with high values of the other variable, and low values are associated with low values.

On the other hand, if the pattern of dots be such as to indicate a straight line path from the upper left-hand corner to the bottom right, correlation is negative, i.e. the association is indirect, high values of one variable being associated with low values of the other (Fig. 9.1b).

When the dots do not indicate any straight line tendency, but a swarm (Fig. 9.1c), or concentration around a curved line, correlation is small (Fig. 9.1d). In fact, if no straight line tendency is noticed, correlation will be zero.

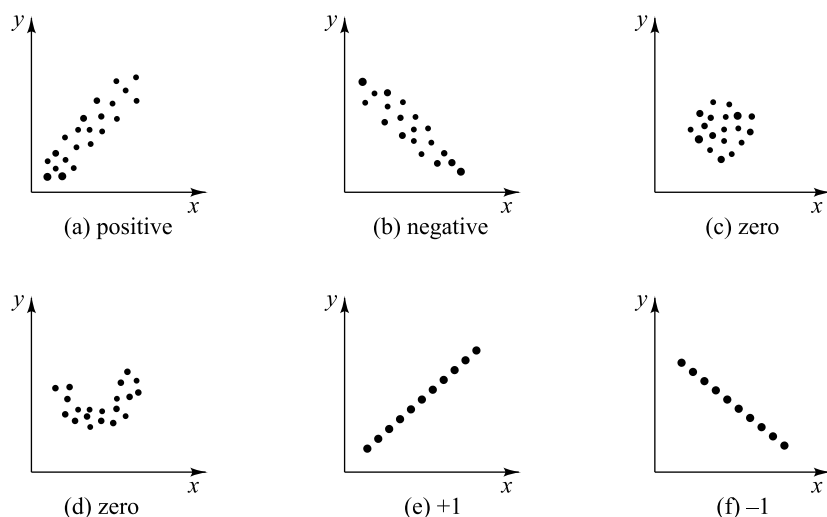


Fig. 9.1 Scatter Diagrams showing Different Types and Degrees of Correlation



When the dots lie exactly on a straight line, correlation is perfect—the correlation coefficient being $+1$ or -1 , according as the slope of the straight line is positive or negative (Figs 9.1e, 9.1f).

The scatter diagram also gives an indication of the degree of linear correlation between the variables, i.e. whether correlation is high or low. If the plotted points on the scatter diagram lie approximately on, or near about, a straight line (Figs 9.1e, 9.1f, 9.2), correlation coefficient will be nearly one, numerically. The more scattered the points are around a straight line, the less is the correlation coefficient (Figs 9.1a, 9.1b). A scatter diagram like Fig. 9.1c will indicate almost zero correlation.



Example 9.5 Construct the scatter diagram of the data given below. Draw a free hand straight line through the group of points and from your diagram discuss the probable amount of correlation:

	Average Values in Lakhs							
	1946	1947	1948	1949	1950	1951	1952	1953
Raw Cotton Imports:	47	64	100	97	126	203	171	115
Cotton Manufacturers' Exports:	70	85	100	103	111	139	133	115

[C.U., M.Com. '69, 72]

Solution The variables here are Raw Cotton Imports (x , say) and Cotton Manufacturers' Exports (y). The 8 pairs of observations (47, 70), (64, 85), ..., (115, 115) are plotted as points on a graph paper, giving the scatter diagram (Fig. 9.2). A free hand straight line has been drawn. It is found that out of 8 points, 4 lie almost on the straight line and the remaining 4 are also very close to it. This indicates that the correlation coefficient between the variables x and y is numerically very high, say about 0.95. Again since the slope of the straight line is positive, the correlation coefficient is also positive. The amount of correlation is thus estimated to be about $+0.95$ (The calculated value of the correlation coefficient for the data is $+0.98$).

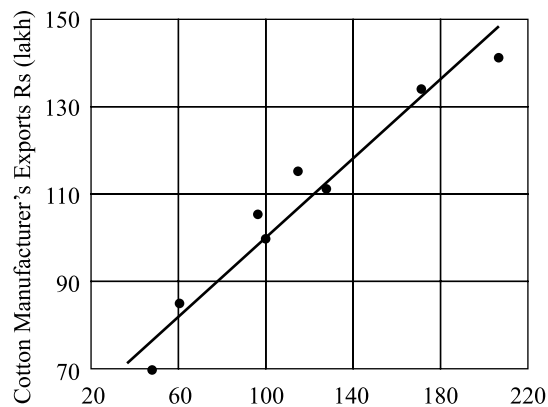


Fig. 9.2 Scatter Diagram





9.5 CORRELATION

The word “correlation” is used to denote the *degree of association between variables*. If two variables x and y are so related that variations in the magnitude of one variable tend to be accompanied by variations in the magnitude of the other variable, they are said to be *correlated*. If y tends to increase as x increases, the variables are said to be *positively correlated*. If y tends to decrease as x increases, the variables are *negatively correlated*. If the values of y are not affected by changes in the values of x , the variables are said to be *uncorrelated* (see Example 9.6).

Correlation may also be *linear* or *non-linear*. If the amount of change in one variable tends to bear a constant ratio to the amount of change in the other variable, then correlation is said to be ‘linear’; because the scatter diagram would show a linear path. Here, we shall be concerned with *linear correlation* or *simple correlation* only. This is measured by ‘*Correlation Coefficient*’ (Section 9.7).

9.6 COVARIANCE

Given a set of n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ relating to two variables x and y , the *Covariance of x and y* , usually represented by $\text{cov}(x, y)$, is defined as

$$\text{cov}(x, y) = \frac{1}{n} \sum (x - \bar{x})(y - \bar{y}) \quad (9.6.1)$$

Expanding the expansion on the right, it can be shown that

$$\text{cov}(x, y) = \frac{\sum xy}{n} - \left(\frac{\sum x}{n} \right) \left(\frac{\sum y}{n} \right) \quad (9.6.2)$$

This form is generally used for calculations. Covariance has properties similar to those of variance, i.e. the square of S.D.

(i) If $X = x - c$ and $Y = y - c'$, where c, c' are constants, then

$$\text{cov}(x, y) = \text{cov}(X, Y) \quad (9.6.3)$$

(ii) If $u = (x - c)/d$ and $v = (y - c')/d'$, where c, c', d, d' are constants, then

$$\text{cov}(x, y) = dd' \cdot \text{cov}(u, v) \quad (9.6.4)$$

However, while variance must be always positive, covariance may be positive, negative or zero.

[Note: By definition,

$$\text{Variance of } x = \sigma_x^2 = \frac{1}{n} \sum (x - \bar{x})^2 = \frac{1}{n} \sum (x - \bar{x})(x - \bar{x})$$

$$\text{Variance of } y = \sigma_y^2 = \frac{1}{n} \sum (y - \bar{y})^2 = \frac{1}{n} \sum (y - \bar{y})(y - \bar{y})$$

$$\text{Covariance of } x \text{ and } y = \frac{1}{n} \sum (x - \bar{x})(y - \bar{y})$$

For the variances, both factors in the expression on the right are same, either both $(x - \bar{x})$, or both $(y - \bar{y})$; both for the covariance one factor is $(x - \bar{x})$ and another $(y - \bar{y})$. Again, the working formulae are



$$\text{Variance of } x = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n} \right)^2 = \frac{\Sigma x \cdot x}{n} - \left(\frac{\Sigma x}{n} \right) \left(\frac{\Sigma x}{n} \right)$$

$$\text{Variance of } y = \frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n} \right)^2 = \frac{\Sigma y \cdot y}{n} - \left(\frac{\Sigma y}{n} \right) \left(\frac{\Sigma y}{n} \right)$$

$$\text{Covariance of } x \text{ and } y = \frac{\Sigma x \cdot y}{n} - \left(\frac{\Sigma x}{n} \right) \left(\frac{\Sigma y}{n} \right)$$

For the variances, the first term on the right contains $x \cdot x$ or $y \cdot y$ and for the covariance it is $x \cdot y$, i.e. one is x and the other y ; in the second term similarly instead of both factors same, $\Sigma x/n$ or $\Sigma y/n$, we have one from x and another from y , i.e. $(\Sigma x/n)(\Sigma y/n)$.

It is thus seen that covariance is a variance like quantity, but obtained by the combination of two variables. Covariance may, therefore, be looked upon as a 'conjoint variance'.

It is also interesting to note that covariance of a variable and the same variable is the variance itself. Replacing y by x in (9.6.1) or (9.6.2), we see that $\text{cov.}(x, x) = \sigma_x^2$; similarly, $\text{cov.}(y, y) = \sigma_y^2$

9.7 CORRELATION COEFFICIENT (r)

Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be a given set of n pairs of observations on two variables x and y . The *Correlation Coefficient*, or *Coefficient of Correlation*, between x and y (denoted by the symbol r) is then defined as

$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \quad (9.7.1)$$

where σ_x and σ_y are the standard derivations of x and y respectively, and $\text{cov}(x, y)$ denotes the covariance of x and y (Section 9.6). This expression is known as *Pearson's product-moment formula*, and is used as a measure of linear correlation between x and y .

The formula for r may be written in various other forms. Putting the explicit expressions for $\text{cov}(x, y)$, σ_x and σ_y in (9.7.1), and multiplying both the numerator and the denominator by n , we have

$$r = \frac{\Sigma (x - \bar{x})(y - \bar{y})}{\sqrt{[\Sigma (x - \bar{x})^2 \cdot \Sigma (y - \bar{y})^2]}} \quad (9.7.2)$$

Now, expanding the expressions

$$r = \frac{\Sigma xy - n\bar{x}\bar{y}}{\sqrt{[(\Sigma x^2 - n\bar{x}^2) \cdot (\Sigma y^2 - n\bar{y}^2)]}} \quad (9.7.3)$$

Multiplying the numerator and the denominator by n again, and since $n\bar{x} = \Sigma x$ and $n\bar{y} = \Sigma y$, we may write

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2] \cdot [n\Sigma y^2 - (\Sigma y)^2]}} \quad (9.7.4)$$

[**Note:** In all the forms shown above, the denominators contains two factors under the square-root. They may be obtained on replacing y by x , and x by y , in the numerator.]



9.8 PROPERTIES OF CORRELATION COEFFICIENT

- (i) The correlation coefficient r is independent of the choice of both origin and scale of observations. This means that if

$$u = \frac{x - c}{d} \quad \text{and} \quad v = \frac{y - c'}{d'}$$

where c, c', d, d' are arbitrary constants (d and d' positive), then

$$r_{xy} = r_{uv} \quad (9.8.1)$$

i.e. *Correlation coefficient between x and y*

= Correlation coefficient between u and v

In general, if $X = a + bx, Y = a' + b'y$, then

$$r_{XY} = \pm r_{xy} \quad (9.8.2)$$

according as b and b' have the same sign, or opposite signs.

- (ii) The correlation coefficient r is a pure number and is independent of the units of measurement. This means that if, for example, x represents height in inches and y weight in lbs., then the correlation coefficient between x and y will neither be in inches nor in lbs. or any other unit, but only a number.
- (iii) The correlation coefficient r lies between -1 and $+1$; i.e. r cannot exceed 1 numerically.

$$-1 \leq r \leq +1 \quad (9.8.3)$$

9.9 CALCULATION OF r

Correlation coefficient (r) is unaffected by the choice of origin and scale of one or both the variables (Property (i), Section 9.8). Therefore, it can be calculated from a given set of n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ as follows:

- (I) If $X = x - c$ and $Y = y - c'$, (here c, c' are constants), then

$$r_{xy} = r_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \quad (9.9.1)$$

$$\text{where } \sigma_X^2 = \frac{\sum X^2}{n} - \left(\frac{\sum X}{n} \right)^2, \quad \sigma_Y^2 = \frac{\sum Y^2}{n} - \left(\frac{\sum Y}{n} \right)^2$$

$$\text{cov}(X, Y) = \frac{\sum XY}{n} - \left(\frac{\sum X}{n} \right) \left(\frac{\sum Y}{n} \right) \quad (9.9.2)$$

Thus, we can always reduce the given values of x and y on subtracting convenient numbers c and c' , and obtain deviations $X = x - c, Y = y - c'$. From these reduced values X and Y , the two standard deviations and the covariance, viz. σ_X, σ_Y and $\text{cov}(X, Y)$, are now calculated, and finally the correlation coefficient r_{XY} between them.

This will be exactly equal to the correlation coefficient r_{xy} between the original values of x and y (see Example 9.9).

- (II) If $u = \frac{x - c}{d}$ and $v = \frac{y - c'}{d'}$, (here c, c', d, d' are constants and d, d' are positive), then



$$r_{xy} = r_{uv} = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v} \quad (9.9.3)$$

$$\text{where } \sigma_u^2 = \frac{\sum u^2}{n} - \left(\frac{\sum u}{n} \right)^2, \quad \sigma_v^2 = \frac{\sum v^2}{n} - \left(\frac{\sum v}{n} \right)^2$$

$$\text{cov}(u, v) = \frac{\sum uv}{n} - \left(\frac{\sum u}{n} \right) \left(\frac{\sum v}{n} \right) \quad (9.9.4)$$

In some cases, the given values of x and y may be such that it is further possible to reduce the deviations $x - c$ and $y - c'$ on division by constant factors d and d' , i.e. $u = (x - c)/d$, $v = (y - c')/d'$. From these values of u and v , the two standard deviations σ_u , σ_v and the covariance $\text{cov}(u, v)$ are calculated, and finally the correlation coefficient r_{uv} between u and v is obtained. This will be exactly equal to the correlation coefficient r_{xy} between the original values of x and y (see Example 9.11).

Example 9.6 State in each case whether you would expect to find a positive correlation, a negative correlation or no correlation:

- (i) The ages of husband and wives,
- (ii) Shoe size and intelligence,
- (iii) Insurance companies' profit and the number of claims they have to pay,
- (iv) Years of education and income,
- (v) Amount of rainfall and yield of crop.

Solution (i) Positive, (ii) No correlation, (iii) Negative,
(iv) Positive, (v) Positive.



Example 9.7 Calculate the correlation coefficient and its probable error from the following results:

$$\sum_{i=1}^{100} x_i = 280, \quad \sum_{i=1}^{100} y_i = 60, \quad \sum_{i=1}^{100} x_i^2 = 2384, \quad \sum_{i=1}^{100} y_i^2 = 117, \quad \sum_{i=1}^{100} x_i y_i = 438$$

[I.C.W.A., July '71]

Solution

$$\text{Correlation coefficient } r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$$\text{Here } \sigma_x^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 = \frac{2384}{100} - \left(\frac{280}{100} \right)^2 = 16$$

$$\sigma_y^2 = \frac{\sum y^2}{n} - \left(\frac{\sum y}{n} \right)^2 = \frac{117}{100} - \left(\frac{60}{100} \right)^2 = 0.81$$

$$\text{cov}(x, y) = \frac{\sum xy}{n} - \left(\frac{\sum x}{n} \right) \left(\frac{\sum y}{n} \right) = \frac{438}{100} - \frac{280}{100} \times \frac{60}{100} = 2.7$$

Substituting the values,

$$r = \frac{2.7}{\sqrt{16} \sqrt{0.81}} = \frac{2.7}{4 \times 0.9} = 0.75$$



The Probable Error (P. E.) of correlation coefficient is

$$\begin{aligned} \text{P.E.} &= 0.6745 \frac{1-r^2}{\sqrt{n}} \\ &= 0.6745 \times \frac{1-(0.75)^2}{\sqrt{100}} = 0.0295. \end{aligned}$$

Ans. $r = 0.75$, P.E. = 0.0295



Example 9.8 While calculating the coefficient of correlation between two variables x and y , the following results were obtained : $n = 25$, $\Sigma x = 125$, $\Sigma y = 100$, $\Sigma x^2 = 650$, $\Sigma y^2 = 460$, $\Sigma xy = 508$. It was however later discovered at the time of checking that two pairs of observations (x, y) were copied $(6, 14)$ and $(8, 6)$, while the correct values were $(8, 12)$ and $(6, 8)$ respectively. Determine the correct value of the coefficient of correlation.

[I.C.W.A., June '77, '78]

Solution For the two incorrect pairs of observations

$$\begin{aligned} x : 6, \quad 8; \quad \Sigma x &= 6 + 8 = 14; \quad \Sigma x^2 = 6^2 + 8^2 = 100; \\ y : 14, \quad 6; \quad \Sigma y &= 14 + 6 = 20; \quad \Sigma y^2 = 14^2 + 6^2 = 232; \\ \Sigma xy &= 6 \times 14 + 8 \times 6 = 132. \end{aligned}$$

For the two correct pairs

$$\begin{aligned} x : 8, \quad 6; \quad \Sigma x &= 8 + 6 = 14; \quad \Sigma x^2 = 8^2 + 6^2 = 100; \\ y : 12, \quad 8; \quad \Sigma y &= 12 + 8 = 20; \quad \Sigma y^2 = 12^2 + 8^2 = 208; \\ \Sigma xy &= 8 \times 12 + 6 \times 8 = 144. \end{aligned}$$

When the incorrect pairs of observations are replaced by the correct pairs, the revised results for the 25 pairs of observations are:

$$\begin{aligned} \Sigma x &= 125 - 14 + 14 = 125; & \Sigma y &= 100 - 20 + 20 = 100; \\ \Sigma x^2 &= 650 - 100 + 100 = 650; & \Sigma y^2 &= 460 - 232 + 208 = 436; \\ \Sigma xy &= 508 - 132 + 144 = 520 \end{aligned}$$

Using these results,

$$\begin{aligned} \sigma_x^2 &= 650/25 - (125/25)^2 = 26 - 25 = 1; & \sigma_x &= 1 \\ \sigma_y^2 &= 436/25 - (100/25)^2 = 17.44 - 16 = 1.44; & \sigma_y &= 1.2 \\ \text{cov}(x, y) &= 520/25 - (125/25)(100/25) = 20.8 - 20 = 0.8 \\ r &= \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{0.8}{1 \times 1.2} = \frac{2}{3} \end{aligned}$$

Ans. $2/3$.



Example 9.9 Find the coefficient of correlation from the following data:

X	65	63	67	64	68	62	70	66
Y	68	66	68	65	69	66	68	65

[C.U., B.A. (Econ) '69]

Solution Since the correlation coefficient is unaffected by change of origin (and also scale), let us change the origins of X and Y to 65 and 67 respectively, i.e. write $x = X - 65$, $y = Y - 67$.

**Table 9.9** Calculations for Correlation Coefficient

X	Y	$x = X - 65$	$y = Y - 67$	x^2	y^2	xy
(1)	(2)	(3)	(4)	(5)	(6)	(7)
65	68	0	1	0	1	0
63	66	-2	-1	4	1	2
67	68	2	1	4	1	2
64	65	-1	-2	1	4	2
68	69	3	2	9	4	6
62	66	-3	-1	9	1	3
70	68	5	1	25	1	5
66	65	1	-2	1	4	-2
525	535	5	-1	53	17	18

$$\sigma_x^2 = \frac{53}{8} - \left(\frac{5}{8}\right)^2 = \frac{399}{64}$$

$$\sigma_y^2 = \frac{17}{8} - \left(\frac{-1}{8}\right)^2 = \frac{135}{64}$$

$$\text{cov}(x, y) = \frac{18}{8} - \left(\frac{5}{8}\right)\left(\frac{-1}{8}\right) = \frac{149}{64}$$

$$r_{xy} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{\frac{149}{64}}{\sqrt{\frac{399}{64}} \sqrt{\frac{135}{64}}} = \frac{149}{\sqrt{(399 \times 135)}} = 0.64$$

Therefore, the required correlation coefficient is 0.64.

[**Note:** (i) The origins should be changed to values near the means. Here, the means of X and Y are $525/8 = 65.625$ and $535/8 = 66.875$ respectively.

(ii) The following check may be introduced for the accuracy of figures in cols. (3) and (4), on which the calculations in the subsequent columns depend. (3) must be $525 - 8 \times 65 = 525 - 520 = 5$. Similarly the total of col. (2) may be used to check the total of col. (4), viz. $535 - 8 \times 67 = 535 - 536 = -1$.]



Example 9.10 Marks of 10 students in Mathematics and Statistics are given below:

Mathematics (X)	32	38	48	43	40	22	41	69	35	64
Statistics (Y)	30	31	38	43	33	11	27	76	40	59

Calculate (i) product-moment correlation coefficient, and (ii) its standard error.
[I.C.W.A., June '74]

Solution Since the correlation coefficient is unaffected by changes of origin, we write $x = X - 43$, $y = Y - 38$, and construct a table like Table 9.9. It will be seen that

$$n = 10, \Sigma x = 2, \Sigma y = 8, \Sigma x^2 = 1806, \Sigma y^2 = 2902, \Sigma xy = 2140.$$

Using these results, $\sigma_x^2 = 180.56$, $\sigma_y^2 = 289.56$, $\text{cov}(x, y) = 213.84$.

$$r = \frac{213.84}{\sqrt{(180.56 \times 289.56)}} = 0.94$$

Standard Error (S.E.) of correlation coefficient is

$$S.E. = \frac{1 - r^2}{\sqrt{n}} = \frac{1 - (0.94)^2}{\sqrt{10}} = 0.04 \quad \text{Ans. (i) 0.94, (ii) .04}$$

Example 9.11 Calculate the coefficient of correlation from the following data :

<i>x</i>	2.52	2.49	2.49	2.45	2.43	2.42	2.41	2.40
<i>y</i>	730	710	770	890	970	1020	970	1040

[C.U., B.A. (Econ) '73 : M.Com. '76]

Solution Since the coefficient of correlation is unaffected by changes of origin and scale, let us write

$$u = \frac{x - 2.45}{.01}, \quad v = \frac{y - 900}{10}$$

Table 9.10 Calculations for Correlation Coefficient

<i>x</i>	<i>y</i>	$u = \frac{x - 2.45}{.01}$	$v = \frac{y - 900}{10}$	u^2	v^2	uv
2.52	730	7	-17	49	289	-119
2.49	710	4	-19	16	361	-76
2.49	770	4	-13	16	169	-52
2.45	890	0	-1	0	1	0
2.43	970	-2	7	4	49	-14
2.42	1020	-3	12	9	144	-36
2.41	970	-4	7	16	49	-28
2.40	1040	-5	14	25	196	-70
Total 19.61	7100	1	-10	135	1258	-395

[Note: (i) For the calculation of correlation coefficient, a change of origin is always possible, and must be made; a change of scale, wherever possible, should also be made to simplify the calculations. In the present case, the given values of *y* are divisible by 10, indicating that changes of both origin and scale are possible. Also, since the values of *x* are given in two decimals, division by .01 (i.e. multiplication by 100) will change them to whole numbers, and consequently worries about fixing the decimal point can be avoided.

(ii) Another important point to note here is how to effect changes of origin and scale both in one step. When data are given in two decimals the origin should be changed to a convenient two-decimal number near the mean, and the scale shall be changed to 01. Here, $u = (x - 2.45)/.01 = 100x - 245$. This shows that for obtaining the values of *u*, each *x* is to be multiplied by 100 (which can be achieved simply by ignoring the decimal point) and then 245 subtracted. Thus, 2.52 changes to 252, and $252 - 245 = 7$; 2.49 changes to 249 and $249 - 245 = 4$; etc. (For data given in one-decimal figures, the origin should be changed to a one-decimal number and scale to 0.1).



Again, $v = (y - 900)/10 = y/10 - 90$. This shows that for obtaining the values of v , each y is to be divided by 10 (i.e. ignore the last zero) and then subtract 90. For example, 730 changes to 73 and $73 - 90 = -17$; 710 changes to 71 and $71 - 90 = -19$; etc.]

Now using formula (9.9.3),

$$\sigma_u^2 = \Sigma u^2/n - (\Sigma u/n)^2 = 135/8 - (1/8)^2 = 1079/64$$

$$\sigma_v^2 = \Sigma v^2/n - (\Sigma v/n)^2 = 1258/8 - (-10/8)^2 = 9964/64$$

$$\text{cov}(u, v) = \Sigma uv/n - (\Sigma u/n)(\Sigma v/n) = -395/8 - (1/8)(-10/8) = -3150/64$$

$$r = \frac{\frac{-3150}{64}}{\sqrt{\frac{1079}{64}} \sqrt{\frac{9964}{64}}} = \frac{-3150}{\sqrt{1079 \times 9964}} = -0.96$$



Example 9.12 The following table gives the index numbers of industrial production in a country and the number of registered unemployed persons in the same country during the eight consecutive years. Calculate the coefficient of correlation and comment on the result.

Year	1954	1955	1956	1957	1958	1959	1960	1961
Index of industrial production	100	102	103	105	106	104	103	98
No. of registered unemployed (in thousands)	10.5	11.4	13.0	11.5	12.0	12.5	15.6	20.8

[I.C.W.A., July '64; C.U., B.A. (Econ)' 65]

Solution Let x denote the index of industrial production the y the number of registered unemployed (in thousands). Since the correlation coefficient remains unaffected by changes of origin and scale, we write $u = x - 100$ and $v = (y - 13.0)/0.1$

$$\sigma_u^2 = \Sigma u^2/n - (\Sigma u/n)^2 = 103/8 - (21/8)^2 = 6.0$$

$$\sigma_v^2 = \Sigma v^2/n - (\Sigma v/n)^2 = 7991/8 - (33/8)^2 = 982$$

$$\text{cov}(u, v) = \Sigma uv/n - (\Sigma u/n)(\Sigma v/n) = -265/8 - (21/8)(33/8) = -44$$

Table 9.11 Calculations for Correlation Coefficient

x (1)	y (2)	$u = x - 100$ (3)	$v = (y - 13.0)/0.1$ (4)	u^2 (5)	v^2 (6)	uv (7)
100	10.5	0	-25	0	625	0
102	11.4	2	-16	4	256	-32
103	13.0	3	0	9	0	0
105	11.5	5	-15	25	225	-75
106	12.0	6	-10	36	100	-60
104	12.5	4	-5	16	25	-20
103	15.6	3	26	9	676	78
98	20.8	-2	78	4	6084	-156
Total 821	107.3	21	33	103	7991	-265



$$\therefore r_{xy} = r_{uv} = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v} = \frac{-44}{\sqrt{(6.0 \times 982)}} = -0.57$$

Comments—The scatter diagram (not shown here) does not indicate any straight line path and hence $r = -0.57$ does not really serve as a useful measure of association between the variables. In fact, if the data for 1961 are excluded, we find $r = +0.21$, a result which differs much from the former. The observed correlation coefficient $r = -0.57$ may be called *non-sense correlation* (Page 286).



Example 9.13 Prove that the correlation coefficient does not depend on the origin or scale of the observations.

[C.U., B.A. (Econ) '66, M. Com. '76; B.U., B.A. (Econ) '68; I.C.W.A., July '70, Jan. '72, June '75]

Solution Let x and y be the variables, and $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ denote n pairs of observations. Let us change the origins of x and y to a and c , and the units of measurement (i.e. scales) to b and d respectively, and write

$$u_i = \frac{x_i - a}{b}, \quad v_i = \frac{y_i - c}{d} \quad (i)$$

where a, b, c, d are arbitrary constants (b and d positive). We have to prove that the correlation coefficient between x and y is the same as that between u and v ; i.e. $r_{xy} = r_{uv}$.

From (i), $x_i = a + bu_i, \quad y_i = c + dv_i$

Therefore, $\bar{x} = \Sigma x_i / n = \Sigma(a + bu_i) / n = (na + b \Sigma u_i) / n = a + b \bar{u}$

Similarly, $\bar{y} = c + d \bar{v}$

Hence, $\sigma_x^2 = \Sigma(x_i - \bar{x})^2 / n = \Sigma(a + bu_i - a - b\bar{u})^2 / n$
 $= b^2 \Sigma(u_i - \bar{u})^2 / n = b^2 \cdot \sigma_u^2$

$\therefore \sigma_x = b \cdot \sigma_u$ (since b is positive)

Similarly, $\sigma_y = d \cdot \sigma_v$ (since d is positive) (ii)

Again, $\text{cov}(x, y) = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / n$
 $= \Sigma\{(a + bu_i - a - b\bar{u})(c + dv_i - c - d\bar{v})\} / n$
 $= bd \Sigma(u_i - \bar{u})(v_i - \bar{v}) / n$
 $= bd \cdot \text{cov}(u, v)$ (iii)

Substituting the values from (ii) and (iii),

$$r_{xy} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{bd \cdot \text{cov}(u, v)}{(b \cdot \sigma_u)(d \cdot \sigma_v)} = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v} = r_{uv}$$

i.e. Correlation coefficient between x and y

= Correlation coefficient between u and v .



Example 9.14 If $X_i = (x_i - a)/b, Y_i = (y_i - c)/d, (i = 1, 2, \dots, n)$ where a, b, c, d are arbitrary constants, prove that $r_{xy} = r_{XY}$, if b and d are of the same sign, and $r_{xy} = -r_{XY}$ if they are of opposite signs.

[C.U., B.A. (Econ) '68, '70; M. Com. '74; B.U., B.A., (Econ) '72]



Solution As in Example 9.13, $\sigma_x^2 = b^2 \cdot \sigma_X^2$, $\sigma_y^2 = d^2 \cdot \sigma_Y^2$, and $\text{cov}(x, y) = bd \cdot \text{cov}(X, Y)$. These results are true whatever be the signs of b and d . But since the standard deviation is always positive, we must have $\sigma_x = |b| \cdot \sigma_X$, $\sigma_y = |d| \cdot \sigma_Y$ (see 6.6.2, page 185)

$$\begin{aligned} r_{xy} &= \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{bd \cdot \text{cov}(X, Y)}{|b| \cdot \sigma_X |d| \cdot \sigma_Y} \\ &= \frac{bd}{|b| \cdot |d|} \cdot \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{bd}{|b| \cdot |d|} r_{XY} \end{aligned}$$

If b and d are of the same sign (i.e. either both positive or both negative), the product bd will be positive, and exactly equal to $|b| \cdot |d|$, which is always positive. Therefore,

$$\frac{bd}{|b| \cdot |d|} = +1, \text{ so that } r_{xy} = r_{XY}.$$

If b and d are of opposite signs (i.e. one of them positive and the other negative), the product bd will be negative, but numerically equal to $|b| \cdot |d|$. Therefore,

$$\frac{bd}{|b| \cdot |d|} = -1, \text{ so that } r_{xy} = -r_{XY}. \quad (\text{Proved})$$



Example 9.15 If the correlation coefficient between x and y is 0.5, what would be the correlation coefficient between $5x$ and $-3y$?

Solution Let us write $X = 5x$ and $Y = -3y$. Then by formula (9.8.2), we have (here $a = 0$, $b = 5$, $a' = 0$, $b' = -3$),

$$r_{XY} = -r_{xy} = -0.5$$

since the coefficients of x and y (viz. 5 and -3) have opposite signs.



Example 9.16 Prove that the correlation coefficient r lies between -1 and $+1$.

[C.U., B.A.(Econ) '67, '74; M.Com. '74; B.U., B.A. (Econ) '65, '68, '72, '73]

Solution Let x and y be the variables, and $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ denote n pairs of observations, with means $\bar{x}$, $\bar{y}$ and standard deviations σ_x , σ_y respectively. If we write

$$u_i = \frac{x_i - \bar{x}}{\sigma_x} \quad \text{and} \quad v_i = \frac{y_i - \bar{y}}{\sigma_y}$$

$$\text{then} \quad \sum u_i^2 = \sum \left(\frac{x_i - \bar{x}}{\sigma_x} \right)^2 = \frac{\sum (x_i - \bar{x})^2}{\sigma_x^2} = \frac{n \sigma_x^2}{\sigma_x^2} = n$$

$$\text{Similarly,} \quad \sum v_i^2 = n$$

$$\begin{aligned} \text{Again,} \quad \sum u_i v_i &= \sum \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right) = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sigma_x \sigma_y} \\ &= \frac{n \cdot \text{cov}(x, y)}{\sigma_x \sigma_y} = nr \end{aligned}$$

where r denotes the correlation coefficient between x and y .



Now, $(u_i + v_i)^2$ can never be negative, because it is a perfect square. Hence, the sum of all such squares for $i = 1, 2, \dots, n$, cannot be negative; i.e.

$$\Sigma (u_i + v_i)^2 \geq 0$$

$$\text{or, } \Sigma u_i^2 + \Sigma v_i^2 + 2\Sigma u_i v_i \geq 0$$

$$\text{or, } n + n + 2nr \geq 0$$

$$\text{or, } 2n + 2nr \geq 0$$

$$\text{or, } 2n(1 + r) \geq 0$$

$$\text{or, } 1 + r \geq 0$$

$$\text{or, } r \geq -1; \text{ i.e. } -1 \leq r$$

Similarly, since $(u_i - v_i)^2$ cannot be negative,

$$\Sigma (u_i - v_i)^2 \geq 0$$

$$\text{or, } \Sigma u_i^2 + \Sigma v_i^2 - 2\Sigma u_i v_i \geq 0$$

$$\text{or, } n + n - 2nr \geq 0$$

$$\text{or, } 2n(1 - r) \geq 0$$

$$\text{or, } 1 - r \geq 0$$

$$\text{or, } 1 \geq r; \text{ i.e. } r \leq 1$$

Combining the results $-1 \leq r$ and $r \leq 1$, we have $-1 \leq r \leq +1$. This proves that the correlation coefficient lies between -1 and $+1$.



9.10 INTERPRETATION AND USE OF r

Example 9.17 Discuss the various uses and limitations of coefficient of correlation, indicating the difficulties involved in the interpretation of the coefficient of correlation.

[C.U., B.A.(Econ) '66; M.Com '68]

Solution Uses—

(i) The coefficient of correlation (r) is a measure of the degree of association between two variables. For comparing two series of observations, it is sometimes necessary to determine whether they are associated or not, and to establish relations of cause and effect. The coefficient of correlation furnishes a method of determining numerically the existence of such causal connection between them, e.g. whether weights of school boys are connected with their parents salaries. When the pattern of dots in the scatter diagram is linear, the correlation coefficient (r) can be considered as a useful measure of such relationship. A positive value of r indicates that high values of one variable are in general associated with high values of the other, and low values with low values. When r is negative, high values of one variable are in general associated with low values of the other.

(ii) Again, the proportion of variation explained by regression is equal to the square of correlation coefficient (Section 9.14), i.e.

$$r^2 = \text{Proportion of variation explained by regression}$$

The value of r^2 , therefore, enables us to state the relative amount of variation in the dependent variable which can be explained by the regression equation.

(iii) The coefficient of correlation also helps in estimating the value of the dependent variable, when the value of the independent variable is known. Thus, for a given x , the estimated value of y is obtained from the regression equation of y on x (page 291)



$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

When $r = 0$, neither y nor x can be estimated by a linear function of the other variable. Again, when $r = \pm 1$, the two regression lines coincide, and the value of one variable can be accurately predicted by the linear regression equation.

(iv) In educational and psychological measurements, coefficients of correlation is used in problems of *reliability and validity* of tests.

Limitations—

(i) In linear correlation, it is assumed that there is a straight line relationship between the variables. A small value of r therefore indicates only a poor *linear type* of relationship between the variables. This however, does not rule out the possibility that the association is very close, but the relationship is non-linear (Example 9.21). Before using r as a measure of the degree of association between the variables, it is, therefore, advisable to draw a scatter diagram and see whether the pattern of points is linear or not.

(ii) Again, a high value of r also does not imply that there is a direct cause-and-effect relationship between the variables. The high value of r may be generated solely due to the influence of a third variable affecting both. In this cause, the effect of the third variable should be eliminated from the first two and then the 'partial' correlation coefficient between them found out (Section 9.17).

(iii) Sometimes, it may happen that two series of observations show a high correlation coefficient even though there is no logical basis for any relationship between them. For example, a renowned statistician observed a high correlation between the stork population in Oslo, Norway, over a period of several years and the number of babies born there each year. Yet it is hard to develop a theory as to why this should be. Such correlation is said to be *Spurious correlation* or *Non-sense correlation*. One should apply common-sense in deciding whether the association indicated by the value of r is real or spurious (Example 9.12).

(iv) If the data are not reasonably homogeneous the coefficient of correlation may give a misleading picture of the extent of association. For example, if the scatter diagram shows the points in separate clusters or groups, the correlation coefficient based on all the groups taken together may be very high; yet if separate values of r are computed for each group, they may be close to zero. If some reasonable basis can be found for separating the data into groups, it is desirable to compute values of r for each group.



9.11 VARIANCE OF THE SUM (DIFFERENCE) OF TWO SERIES

Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be a set of n pairs of observations on the variables x and y , and a new series $(x + y)$ is formed by combining the corresponding values of the two series, giving $x_1 + y_1, x_2 + y_2, \dots, x_n + y_n$. If we denote the variances of x -series, y -series and $(x + y)$ -series by $\text{var}(x)$, $\text{var}(y)$ and $\text{var}(x + y)$ respectively, and $\text{cov}(x, y)$ denotes the covariance between x and y , then

$$\text{var}(x + y) = \text{var}(x) + \text{var}(y) + 2 \cdot \text{cov}(x, y) \quad (9.11.1)$$

Similarly, if another series $(x - y)$ is formed from the given data, then

$$\text{var}(x - y) = \text{var}(x) + \text{var}(y) - 2 \cdot \text{cov}(x, y) \quad (9.11.2)$$



In the usual notations, if we write σ_x , σ_y , σ_{x+y} , σ_{x-y} to denote the standard deviations of x , y , $x + y$, $x - y$ respectively, the above formulae may be written as

$$\sigma_{x+y}^2 = \sigma_x^2 + \sigma_y^2 + 2r\sigma_x\sigma_y \quad (9.11.3)$$

$$\sigma_{x-y}^2 = \sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y \quad (9.11.4)$$

When x and y are uncorrelated i.e. $r = 0$, then $\text{cov}(x, y) = 0$, and the last terms on the right of the above formulae vanish. Hence

$$\begin{aligned} \text{var}(x + y) &= \text{var}(x - y) = \text{var}(x) + \text{var}(y) \\ \text{i.e.} \quad \sigma_{x+y}^2 &= \sigma_{x-y}^2 = \sigma_x^2 + \sigma_y^2 \end{aligned} \quad (9.11.5)$$

This implies that the variance of the sum (or difference) of uncorrelated variables is equal to the *sum* of their variances.

In general, if $z = ax + by$, where a and b are constants, then

$$\begin{aligned} \text{var}(ax + by) &= a^2 \cdot \text{var}(x) + b^2 \cdot \text{var}(y) + 2ab \cdot \text{cov}(x, y) \\ \text{i.e.} \quad \sigma_z^2 &= a^2 \sigma_x^2 + b^2 \sigma_y^2 + 2ab r \sigma_x \sigma_y \end{aligned} \quad (9.11.6)$$

Note that formulae (9.11.3) and (9.11.4) may be obtained from this by putting $a = 1$, $b = 1$, and $a = 1$, $b = -1$ respectively.

Again, solving (9.11.3) and (9.11.4) for r , we see that

$$r = \frac{\sigma_{x+y}^2 - \sigma_x^2 - \sigma_y^2}{2\sigma_x\sigma_y}, \quad r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y} \quad (9.11.7)$$

Thus, the standard deviations of x , y and $x + y$ (or $x - y$) may be used to find the correlation coefficient between x and y .

Example 9.18 n pairs of values of two variables x and y are given. The variances of x , y and $(x - y)$ are σ_x^2 , σ_y^2 and σ_{x-y}^2 respectively. Show that the correlation coefficient r_{xy} between x and y is given by

$$\frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

[I.C.W.A., July '66, '72; C.U., B.A.(Econ) '72]

Solution Let (x_i, y_i) , $i = 1, 2, \dots, n$, denote n pairs of observations with means $\bar{x}$, $\bar{y}$ and standard deviations σ_x , σ_y for the x -series and y -series respectively. If we write $u_i = x_i - y_i$, then $\bar{u} = \bar{x} - \bar{y}$.

$$u_i - \bar{u} = (x_i - y_i) - (\bar{x} - \bar{y}) = (x_i - \bar{x}) - (y_i - \bar{y})$$

$$\begin{aligned} \therefore \quad \sigma_u^2 &= \Sigma(u_i - \bar{u})^2/n \\ &= \Sigma\{(x_i - \bar{x}) - (y_i - \bar{y})\}^2/n \\ &= \Sigma(x_i - \bar{x})^2/n - 2\Sigma(x_i - \bar{x})(y_i - \bar{y})/n + \Sigma(y_i - \bar{y})^2/n \\ &= \sigma_x^2 - 2 \cdot \text{cov}(x, y) + \sigma_y^2 \\ &= \sigma_x^2 - 2r_{xy} \sigma_x \sigma_y + \sigma_y^2, \end{aligned}$$

$$\text{because by definition } r_{xy} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \text{ and hence } \text{cov}(x, y) = r_{xy} \sigma_x \sigma_y$$



$$\text{i.e.} \quad \sigma_{x-y}^2 = \sigma_x^2 - 2r_{xy} \sigma_x \sigma_y + \sigma_y^2$$

$$\text{or,} \quad 2r_{xy} \sigma_x \sigma_y = \sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2$$

$$\text{or,} \quad r_{xy} = (\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2) / (2\sigma_x \sigma_y) \quad (\text{Shown})$$



Example 9.19 The differences ($y_i - x_i$), $i = 1, 2, \dots, 15$, are given for a set of 15 pairs of observations below :

15, 15, 13, 14, 11, 13, 12, 12, 15, 15, 15, 19, 19, 19, 14.

If $\bar{x} = 49.34$, $\bar{y} = 64.07$, $\sigma_x = 3.53$, $\sigma_y = 4.30$, find the coefficient of correlation between x and y .

[C.U., B.Sc. '64; B.A. (Econ) '67]

Solution We apply the result of the previous Example. If $u_i = y_i - x_i$, then $\bar{u} = \bar{y} - \bar{x}$. In the present case, the mean of the given differences is $(15 + 15 + 13 + 14 + \dots + 19 + 14)/15 = 221/15 = 14.73$, which agrees with $\bar{y} - \bar{x} = 64.07 - 49.34 = 14.73$. For the calculation of σ_{y-x}^2 , we use deviations from 15 (Table 9.12).

Table 9.12 Calculations for S.D.

Deviation from 15

$y - x$	(z)	z^2
15	0	0
15	0	0
13	-2	4
14	-1	1
11	-4	16
13	-2	4
12	-3	9
12	-3	9
15	0	0
15	0	0
15	0	0
19	4	16
19	4	16
19	4	16
14	-1	1
Total	-4	92

$$\begin{aligned} \sigma_{y-x}^2 &= \Sigma z^2/n - (\Sigma z/n)^2 \\ &= 92/15 - (-4/15)^2 = 6.06 \end{aligned}$$

$$\therefore \sigma_{x-y}^2 = \sigma_{y-x}^2 = 6.06$$

Now, putting the values

$$r_{xy} = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x \sigma_y}$$

$$\begin{aligned}
 &= \frac{(3.53)^2 + (4.30)^2 - 6.06}{2 \times 3.53 \times 4.30} \\
 &= \frac{24.8909}{30.3580} = 0.82.
 \end{aligned}$$

Ans. 0.82.

Example 9.20 If three uncorrelated variables x_1, x_2 and x_3 have the same standard deviation, find the coefficient of correlation between $x_1 + x_2$ and $x_2 + x_3$.

[B.U., B.A. (Econ)' 71]

Solution Let us write $u = x_1 + x_2, v = x_2 + x_3$. It is required to find r_{uv} . Let $\sigma_1, \sigma_2, \sigma_3$, denote the standard deviations of the variables x_1, x_2, x_3 respectively; and $\text{cov}(x_1, x_2), \text{cov}(x_2, x_3), \text{cov}(x_1, x_3)$ denote the covariances between pairs of variables. Since x_1, x_2, x_3 are uncorrelated,

$$\text{cov}(x_1, x_2) = 0, \quad \text{cov}(x_2, x_3) = 0, \quad \text{cov}(x_1, x_3) = 0$$

Also since the variables are assumed to have the same s.d.,

$$\sigma_1 = \sigma_2 = \sigma_3 = \sigma \text{ (suppose)}$$

Using the formula (9.11.5) for uncorrelated variables

$$\sigma_u^2 = \sigma_1^2 + \sigma_2^2 = 2\sigma^2; \quad \sigma_v^2 = \sigma_2^2 + \sigma_3^2 = 2\sigma^2$$

If $\bar{x}_1, \bar{x}_2, \bar{x}_3$ denote the means, then $\bar{u} = \bar{x}_1 + \bar{x}_2$ and $\bar{v} = \bar{x}_2 + \bar{x}_3$.

$$\begin{aligned}
 \text{cov}(u, v) &= \Sigma(u - \bar{u})(v - \bar{v})/n \\
 &= \Sigma \{[(x_1 - \bar{x}_1) + (x_2 - \bar{x}_2)] [(x_2 - \bar{x}_2) + (x_3 - \bar{x}_3)]\}/n \\
 &= \Sigma(x_1 - \bar{x}_1)(x_2 - \bar{x}_2)/n + \Sigma(x_1 - \bar{x}_1)(x_3 - \bar{x}_3)/n + \Sigma(x_2 - \bar{x}_2)^2/n \\
 &\quad + \Sigma(x_2 - \bar{x}_2)(x_3 - \bar{x}_3)/n \\
 &= \text{cov}(x_1, x_2) + \text{cov}(x_1, x_3) + \sigma_2^2 + \text{cov}(x_2, x_3) \\
 &= 0 + 0 + \sigma^2 + 0 = \sigma^2
 \end{aligned}$$

$$\text{Therefore, } r_{uv} = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v} = \frac{\sigma^2}{2\sigma^2} = 0.5.$$

Example 9.21 If two variates are independent, their correlation coefficient is zero. Is the converse true? Explain by means of an example.

[I.C.W.A., June '64, '77]

Solution If two variates are independent, their correlation coefficient is zero. But the converse is *not* true. A zero correlation coefficient does not necessarily signify that the variables are independent. It only implies that there is no linear relationship between the variables. However, the possible existence of a non-linear relationship cannot be ruled out altogether. For example, let us consider the following data:

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

Here $\Sigma x = 0, \Sigma y = 28, \Sigma xy = 0, n = 7$. Therefore, using (9.6.2.),

$$\text{cov}(x, y) = \frac{\Sigma xy}{n} - \left(\frac{\Sigma x}{n} \right) \left(\frac{\Sigma y}{n} \right) = \frac{0}{7} - \left(\frac{0}{7} \right) \left(\frac{28}{7} \right) = 0.$$

Hence, $r_{xy} = \text{cov}(x, y)/(\sigma_x \sigma_y) = 0/(\sigma_x \sigma_y) = 0$

i.e. the correlation coefficient between x and y is zero. But it may be noticed that x and y are bound by the relation $y = x^2$. So, x and y are not independent. Thus the correlation coefficient may be zero, even when the variables are not independent.



9.12 REGRESSION

The word “regression” is used to denote *estimation* or *prediction* of the average value of one variable for a specified value of the other variable. The estimation is done by means of suitable equations, derived on the basis of available bivariate data. Such an equation is known as a *Regression equation* and its geometrical representation is called a *Regression curve*.

In *linear regression* (or *simple regression*) the relationship between the variables is assumed to be linear. The estimate of y (say, y') is obtained from an equation of the form

$$y' - \bar{y} = b_{yx}(x - \bar{x}) \quad (i)$$

and the estimate of x (say, x') from another equation (usually different from the former) of the form

$$x' - \bar{x} = b_{xy}(y - \bar{y}) \quad (ii)$$

Equation (i) is known as *Regression equation of y on x* , and equation (ii) as *Regression equation of x on y* . The coefficient b_{yx} appearing in the regression equation of y on x is known as the *Regression coefficient of y on x* . Similarly, b_{xy} is called the *Regression coefficient of x on y* . The geometrical representation of linear regression equations (i) and (ii) are known as *Regression lines*. These lines are “best fitting” straight lines obtained by the Method of Least Squares (page 249).

Example 9.22 Derive the regression equations of y on x and x on y .

[C.U., B.A. (Econ) '71, '73, '78]

Or,

Obtain the equations of the two lines of regression for a bivariate distribution.

[I.C.W.A., July '70]

Solution (1) *Regression Equation of y on x*

The regression equation of y on x is the equation of the best-fitting straight line in the form $y = a + bx$, obtained by the method of least squares (Section 8.5, page 251).

Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be a set of n pairs of observations, and let us fit a straight line of the form

$$y = a + bx \quad (i)$$

to these data (here, x is considered to be the independent variable and y and dependent variable, i.e. a value of y is obtained when a value of x is given). Applying the method of least squares, the constants a and b are obtained by solving the normal equations (8.5.2), viz.

$$\Sigma y = an + b \Sigma x \quad (ii)$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 \quad (iii)$$

Dividing both sides of (ii) by n , we get $\bar{y} = a + b\bar{x}$ so that $a = \bar{y} - b\bar{x}$.

Substituting this in equation (i),

$$y - \bar{y} = b(x - \bar{x}) \quad (iv)$$

Again, multiplying (ii) by Σx and (iii) by n , we have

$$(\Sigma x)(\Sigma y) = na(\Sigma x) + b(\Sigma x)^2$$

$$n(\Sigma xy) = na(\Sigma x) + nb(\Sigma x^2)$$

Subtracting the first from the second,

$$n\Sigma xy - (\Sigma x)(\Sigma y) = b\{n\Sigma x^2 - (\Sigma x)^2\}$$



$$\therefore b = \frac{n \Sigma xy - (\Sigma x)(\Sigma y)}{n \Sigma x^2 - (\Sigma x)^2}$$

Dividing both numerator and denominator by n^2 ,

$$b = \frac{\Sigma xy/n - (\Sigma x/n)(\Sigma y/n)}{n \Sigma x^2/n - (\Sigma x/n)^2} = \frac{\text{cov}(x, y)}{\sigma_x^2} \quad (\text{v})$$

Writing b with the usual subscripts, we have from (iv),

$$y - \bar{y} = b_{yx}(x - \bar{x}) \quad (\text{vi})$$

where $b_{yx} = \text{cov}(x, y)/\sigma_x^2$. This is the required regression equation of y on x . Since $r = \text{cov}(x, y)/(\sigma_x \sigma_y)$, we see that $\text{cov}(x, y) = r \sigma_x \sigma_y$.

Substituting this

$$b_{yx} = \frac{\text{cov}(x, y)}{\sigma_x^2} = r \frac{\sigma_y}{\sigma_x} \quad (\text{vii})$$

(2) *Regression Equation of x on y*

If we fit an equation of the form $x = a' + b' y$ (assuming x to be the dependent variable and y the independent variable), we obtain the regression equation of x on y . Applying the method of least squares, the normal equations for determined a' and b' are

$$\Sigma x = a' n + b' \Sigma y; \quad \Sigma xy = a' \Sigma y + b' \Sigma y^2 \quad (\text{viii})$$

Proceeding the same way as before, the regression equation of x on y is found to be

$$x - \bar{x} = b_{xy}(y - \bar{y}) \quad (\text{ix})$$

where
$$b_{xy} = \frac{\text{cov}(x, y)}{\sigma_y^2} = r \frac{\sigma_x}{\sigma_y} \quad (\text{x})$$



9.13 PROPERTIES OF LINEAR REGRESSION

The regression equation of y on x is

$$y - \bar{y} = b_{yx}(x - \bar{x}) \quad (9.13.1)$$

where $b_{yx} = \frac{\text{cov}(x, y)}{\sigma_x^2} = r \frac{\sigma_y}{\sigma_x}$. This equation is used to estimate y , when the value of x is known.

The regression equation of x on y is

$$x - \bar{x} = b_{xy}(y - \bar{y}) \quad (9.13.2)$$

where $b_{xy} = \frac{\text{cov}(x, y)}{\sigma_y^2} = r \frac{\sigma_x}{\sigma_y}$. This equation is used to estimate x , when the value of y is known.

Substituting the values of b_{yx} and b_{xy} , the regression equations (9.13.1) and (9.13.2) may also be written as

$$\frac{y - \bar{y}}{\sigma_y} = r \left(\frac{x - \bar{x}}{\sigma_x} \right) \quad \text{and} \quad \frac{x - \bar{x}}{\sigma_x} = r \left(\frac{y - \bar{y}}{\sigma_y} \right) \quad (9.13.3)$$

respectively.



It will be noticed that both regression equations are satisfied (i.e. the value of the right-hand side is equal to the value of the left hand side), when we put $x = \bar{x}$ and $y = \bar{y}$. Geometrically, this implies that both the regression lines pass through, and hence intersect at, the point $(\bar{x}, \bar{y})$. The slopes of the regression lines are respectively b_{yx} and $1/b_{xy}$ [Example 8.1 (f) page 248].

The two regression lines are usually different. However, since they must always intersect at the point $(\bar{x}, \bar{y})$, the line will coincide when their slopes are equal, i.e. $b_{yx} = 1/b_{xy}$; or $b_{yx} \cdot b_{xy} = 1$; i.e. $r^2 = 1$; or, $r = \pm 1$.

When $r = +1$, we find from (9.13.3) that both the regression equation reduce to $\frac{y - \bar{y}}{\sigma_y} = \frac{x - \bar{x}}{\sigma_x}$. It is evident that the common regression line represented by this equation passes through the point $(\bar{x}, \bar{y})$ with a positive slope σ_y/σ_x .

When $r = -1$, as in the previous case, the regression equations reduce to $\frac{y - \bar{y}}{\sigma_y} = -\left(\frac{x - \bar{x}}{\sigma_x}\right)$. This common regression line passes through the point $(\bar{x}, \bar{y})$ with a negative slope $(-\sigma_y/\sigma_x)$.

When $r = 0$, from (9.13.3) we find that regression equations reduce to $\frac{y - \bar{y}}{\sigma_y} = 0$ and $\frac{x - \bar{x}}{\sigma_x} = 0$ respectively. That is, the regression equations become $y = \bar{y}$ and $x = \bar{x}$. The regression line of y on x , viz. $y = \bar{y}$, is now a straight line parallel to the x -axis; and the regression line of x on y , viz. $x = \bar{x}$, is a straight line parallel to the y -axis. The regression lines are thus perpendicular to each other, nevertheless intersecting at $(\bar{x}, \bar{y})$ as usual. Since now the regression equation of y on x does not contain x , it cannot be used to estimate y as a linear function of x . Similarly, the regression equation of x on y cannot be used to estimate x on the basis of y .

Properties of Linear Regression

- (1) There are two linear regression equations—

- (i) Regression equation of y on x :

$$y - \bar{y} = b_{yx}(x - \bar{x}) \quad (9.13.4)$$

- (ii) Regression equation of x on y :

$$x - \bar{x} = b_{xy}(y - \bar{y}) \quad (9.13.5)$$

where b_{yx} and b_{xy} are respectively the regression coefficient of y on x and the regression coefficient of x on y .

$$b_{yx} = \frac{\text{cov}(x, y)}{\sigma_x^2} = r \frac{\sigma_y}{\sigma_x} \quad (9.13.6)$$

$$b_{xy} = \frac{\text{cov}(x, y)}{\sigma_y^2} = r \frac{\sigma_x}{\sigma_y} \quad (9.13.7)$$

- (2) The product of the two regression coefficients is equal to the square of correlation coefficient.

$$b_{yx} \cdot b_{xy} = r^2 \quad (9.13.8)$$



- (3) r , b_{yx} and b_{xy} , all have the same sign. If the correlation coefficient r is zero, the regression coefficients b_{yx} and b_{xy} are also zero.
- (4) The regression lines always intersect at the point $(\bar{x}, \bar{y})$. The slopes of the regression line of y on x and the regression line of x on y are respectively b_{yx} and $1/b_{xy}$.
- (5) The angle between the two regression lines depends on the correlation coefficient r . When $r = 0$, the two lines are perpendicular to each other; when $r = +1$, or $r = -1$, they coincide. As r increases numerically from 0 to 1, the angle between the regression lines diminishes from 90° to 0° .
- (6) The two regression equations are usually different. However, when $r = \pm 1$, they become identical; and in this case, there is an exact linear relationship between the variables. When $r = 0$, the regression equations reduce to $y = \bar{y}$ and $x = \bar{x}$, and neither y nor x can be estimated from linear regression equations.

Example 9.23 Find the regression of x on y from the following data :

$$\begin{array}{lll} \Sigma x = 24 & \Sigma y = 44 & \Sigma xy = 306 \\ \Sigma x^2 = 164 & \Sigma y^2 = 574 & n = 4 \end{array}$$

Find the estimated value of x , when $y = 6$. [C.U., B.A.(Econ) '65]

Solution The regression equation of x on y is

$$x - \bar{x} = b_{xy}(y - \bar{y}), \text{ where } b_{xy} = \text{cov}(x, y)/\sigma_y^2.$$

$$\text{Here, } \bar{x} = \Sigma x/n = 24/4 = 6, \quad \bar{y} = \Sigma y/n = 44/4 = 11.$$

$$\text{By (9.6.2), } \text{cov}(x, y) = 306/4 - (24/4)(44/4) = 10.5$$

$$\text{Also, } \sigma_y^2 = 574/4 - (44/4)^2 = 22.5; \quad \therefore b_{xy} = 10.5/22.5 = 0.467$$

The regression equation of x on y is then

$$x - 6 = 0.467(y - 11); \text{ or, } x = 0.467y + 0.86$$

$$\text{When } y = 6, \text{ we have } x = 0.467 \times 6 + 0.86 = 3.7$$

$$\text{Ans. } x = 0.467y + 0.86; 3.7$$



Example 9.24 The following data show the maximum and the minimum temperatures (degrees centigrade) on a certain day at ten important centres located throughout India :

Max. temp. (x)	29	23	25	15	27	29	24	31	22	28
Min. temp. (y)	8	3	7	5	8	19	10	7	5	11

It is known that for the given data $\Sigma x^2 = 6595$, $\Sigma y^2 = 867$, $\Sigma xy = 2193$. Draw a scatter diagram and show regression line of y on x thereon. [I.C.W.A., July '72]

Solution With the given values of (x, y) , viz. (29, 8), (23, 3), ..., points are plotted on a graph paper treating the pairs of values as co-ordinates. This will be the scatter diagram.

From the given data, find $n = 10$, $\Sigma x = 253$, $\Sigma y = 83$. Also given $\Sigma x^2 = 6595$, $\Sigma y^2 = 867$, $\Sigma xy = 2193$. With these results it is possible to find the regression equation of y on x , viz. $y - \bar{y} = b_{yx}(x - \bar{x})$. Proceeding as in the previous example, $\bar{x} = 25.3$, $\bar{y} = 8.3$, $\text{cov}(x, y) = 9.31$, $\sigma_x^2 = 19.41$. Hence $b_{yx} = 9.31/19.41 = 0.48$, and the regression equation is $y - 8.3 = 0.48(x - 25.3)$; or, $y = 0.48x - 3.84$.

In order to show this line on the scatter diagram, any two points satisfying this equation are taken. For example, when $x = 20$, $y = 5.76$; when $x = 30$, $y = 10.56$. The straight line joining the points (20, 5.76) and (30, 10.56) will be the required regression line.





Example 9.25 From the following results, obtain the two regression equations and estimate the yield of crops when the rainfall is 22 cms, and the rainfall when the yield is 600 kg:

	y (Yield in kg.)	x (Rainfall in cm.)
Mean	508.4	26.7
S.D.	36.8	4.6

Coefficient of correlation between yield and rainfall = 0.52

[C.A., May '76]

Solution For estimating yield (y) we have to use the regression equation of y on x, and for rainfall (x) the regression equation of x on y. Given $\bar{x} = 26.6$, $\bar{y} = 508.4$, $\sigma_x = 4.6$, $\sigma_y = 36.8$, $r = 0.52$,

$$b_{yx} = 0.52 \times 36.8/4.6 = 4.16,$$

$$b_{xy} = 0.52 \times 4.6/36.8 = .065$$

The regression equations are therefore $y - 508.4 = 4.16 (x - 26.7)$ and $x - 26.7 = .065(y - 508.4)$ respectively, which simplify to

$$y = 4.16x + 397.33, \text{ and } x = .065y - 6.346$$

When $x = 22$, $y = 4.16 \times 22 + 397.33 = 488.8$ kg.

When $y = 600$, $x = .065 \times 600 - 6.346 = 32.7$ cm.



Example 9.26 The following results were obtained from records of age (x) and systolic blood pressure (y) of a group of 10 women:

	x	y
Mean ...	53	142
Variance ...	130	165

$$\Sigma(x - \bar{x})(y - \bar{y}) = 1220$$

Find the appropriate regression equation and use it to estimate the blood pressure of a women whose age is 45. [I.C.W.A., Jan. '73]

Solution The appropriate equation is the regression equation of y on x, viz. $y - \bar{y} = b_{yx}(x - \bar{x})$, where $b_{yx} = \text{cov}(x, y)/\sigma_x^2$.

But $\text{cov}(x, y) = \Sigma(x - \bar{x})(y - \bar{y})/n = 1220/10 = 122$,
so that $b_{yx} = 122/130 = 0.94$.

The regression equation is therefore $y - 142 = 0.94 (x - 53)$; or, $y = 0.94x + 92.18$. When $x = 45$, $y = 0.94 \times 92.18 = 134.5$. The estimated blood pressure is 134.5.

Ans. $y = 0.94x + 92.18$; 134.5



Example 9.27 Marks obtained by 12 students in the college test (x) and the university test (y) are as follows:

x	41	45	50	68	47	77	90	100	80	100	40	43
y	60	63	60	48	85	56	53	91	74	98	65	43

What is your estimate of the marks a student could have obtained in the university test if he obtained 60 in the college test but was ill at the time of the university test?

[B.U., B.A. (Econ) '70]



Solution For estimating y , we use the regression equation of y on x , viz. $y - \bar{y} = b_{yx}(x - \bar{x})$, where $b_{yx} = \text{cov}(x, y)/\sigma_x^2$. Therefore, we have to find $\bar{x}$, $\bar{y}$, $\text{cov}(x, y)$ and σ_x^2 .

Table 9.13 Calculations for Regression

x	y	$X = x - 60$	$Y = y - 60$	X^2	XY
41	60	-19	0	361	0
45	63	-15	3	225	-45
50	60	-10	0	100	0
68	48	8	-12	64	-96
47	85	-13	25	169	-325
77	56	17	-4	289	-68
90	53	30	-7	900	-210
100	91	40	31	1,600	1,240
80	74	20	14	400	280
100	98	40	38	1,600	1,520
40	65	-20	5	400	-100
43	43	-17	-17	289	289
Total 781	796	61	76	6,397	2,485

$$\bar{x} = 781/12 = 65.08, \quad \bar{y} = 796/12 = 66.33$$

Since the variance and covariance are unaffected by changes of origin,

$$\begin{aligned} \text{cov}(x, y) &= \text{cov}(X, Y) = \Sigma XY/n - (\Sigma X/n)(\Sigma Y/n) \\ &= 2485/12 - (61/12)(76/12) = 25184/144 \\ \sigma_x^2 &= \Sigma X^2/n - (\Sigma X/n)^2 = 6397/12 - (61/12)^2 = 73043/144 \end{aligned}$$

$$\text{Therefore, } b_{yx} = \frac{25184/144}{73043/144} = \frac{25184}{73043} = 0.345$$

The regression equation is $y - 66.33 = 0.345(x - 65.08)$;

$$\text{i.e. } y = 43.88 + 0.345x$$

When $x = 60$, $y = 43.88 + 0.345 \times 60 = 64.58 \approx 65$ (approx.)

$$\text{Ans. } y = 43.88 + 0.345x; 65$$

[**Note:** In Examples 9.26 and 9.27, the use of formula $b_{yx} = r\sigma_y/\sigma_x$ would involve laborious calculations for r , and must be avoided.]



Example 9.28 Derive the regression line which you consider more important from the following series of observations :

Output (thousand)	5	7	9	11	13	15
Profit per unit of Output (Rs)	1.70	2.40	2.80	3.40	3.70	4.40

[B.U., B.A.(Econ) '66; C.U., B.A. (Econ) '71; I.C.W.A., June '73]

Solution Let x represent 'Output (thousand)' and y represent 'Profit per unit of output (Rs.)'. It is considered that the regression line of y on x is the more important; because, using the equation it will be possible to find the expected profit based on output of the firm.

[**Note:** Here the successive values of x are equidistant. Also, since the number of pairs of observations is even, viz. 6, we apply the transformation (8.6.4) at p. 252. for simplifying the calculations]

**Table 9.14** Calculations for Regression

x	y	$u = x - 10$	u^2	uy
5	1.70	-5	25	-8.50
7	2.40	-3	9	-7.20
9	2.80	-1	1	-2.80
11	3.40	1	1	3.40
13	3.70	3	9	11.10
15	4.40	5	25	22.00
Total 60	18.40	0	70	18.00

Proceeding as in Example 9.27, $\bar{x} = 60/6 = 10$, $\bar{y} = 18.406 = 3.07$

$$\text{cov}(x, y) = \text{cov}(u, y) = \Sigma uy/n - (\Sigma u/n)(\Sigma y/n)$$

$$= 18.00/6 - (0/6)(18.40/6) = 3.00$$

$$\sigma_x^2 = \sigma_u^2 = \Sigma u^2/n - (\Sigma u/n)^2 = 70/6 - (0/6)^2 = 70/6$$

$$b_{yx} = 18.00/70 = 0.257$$

The regression equation of y on x is $y - \bar{y} = b_{yx}(x - \bar{x})$;

i.e., $y - 3.07 = 0.257(x - 10)$; or, $y = 0.257x + 0.50$



Example 9.29 Find the equation of the line of regression of x on y for the following data:

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	5.3	5.7	6.3	7.2	8.2	8.7	8.4

[C.U., B.A. (Econ) '72]

Solution The regression equation of x on y is $x - \bar{x} = b_{xy}(y - \bar{y})$, where $b_{xy} = \text{cov}(x, y)/\sigma_y^2$. For simplifying the calculations, let us make a change of origin and scale for both the variables

$$u = \frac{x - 2.5}{0.5}, \quad v = \frac{y - 7.0}{0.1}$$

[**Note:** Here the values of x are equidistant and n is odd. So, we use the transformation (8.6.3), page 252].

$$\bar{x} = 17.5/7 = 2.5, \quad \bar{y} = 49.8/7 = 7.11$$

$$\text{cov}(u, v) = \Sigma uv/n - (\Sigma u/n)(\Sigma v/n) = 172/7 - (0/7)(8/7) = 172/7$$

$$\sigma_v^2 = \Sigma v^2/n - (\Sigma v/n)^2 = 1140/7 - (8/7)^2 = 7916/49$$

Using (9.6.4), $\text{cov}(x, y) = d.d.' \text{cov}(u, v) = (0.5)(0.1)(172/7)$

Table 9.15 Calculations for Regression

x	y	u	v	v^2	uv
1.0	5.3	-3	-17	289	51
1.5	5.7	-2	-13	169	26
2.0	6.3	-1	-7	49	7
2.5	7.2	0	2	4	0
3.0	8.2	1	12	144	12
3.5	8.7	2	17	289	34
4.0	8.4	3	14	196	42
Total 17.5	49.8	0	8	1,140	172

Using (6.7.4), $\sigma_y^2 = d^2 \cdot \sigma_v^2 = (0.1)^2 (7916/49)$

$$\text{Hence, } b_{xy} = \frac{\text{cov}(x, y)}{\sigma_y^2} = \frac{0.5 \times 0.1 \times (172/2)}{(0.1)^2 \times (7916/49)} = 0.76$$

Therefore, the regression equation of x on y is

$$x - 2.5 = 0.76(y - 7.11); \text{ i.e. } x = 0.76y - 2.90.$$

Example 9.30 From the following bivariate frequency distribution, calculate (i) the coefficient of correlation and (ii) the regression equation of marks (y) on age (x)—
Age (years)

	15–20	20–25	25–30	30–35	35–40	Total
0–19	4	2				6
20–39	6	5	3	1		15
40–59		9	4	2	1	16
60–79		7	4	1		12
80–99			1			1
Total	10	23	12	4	1	50

Solution Using the results from Table 9.16, we have

$$\sigma_u^2 = \frac{\sum f_1 u^2}{N} - \left(\frac{\sum f_1 u}{N} \right)^2 = \frac{71}{50} - \left(\frac{-37}{50} \right)^2 = 0.8724$$

Table 9.16 Calculations for Correlation and Regression

x_u y_v	17.5 – 2	22.5 – 1	27.5 0	32.5 1	37.5 2	Totals f_2	$f_2 v$	$f_2 v^2$	fuv
9.5–2	4 (16)	2 (4)				6	– 12	24	20
29.5–1	6 (12)	5 (5)	3 (0)	1 (– 1)		15	– 15	15	16
49.5–0		9 (0)	4 (0)	2 (0)	1 (0)	16	0	0	0
69.5–1		7 (– 7)	4 (0)	1 (1)		12	12	12	– 6
89.5–2			1 (0)			1	2	4	0
Totals									
f_1	10	23	12	4	1	50 = N	– 13	55	30
$f_1 u$	– 20	– 23	0	4	2	– 37			
$f_1 u^2$	40	23	0	4	4	71			
fuv	28	2	0	0	0	30			

Explanation of symbols used in Table 9.16:

- (i) x and y represent mid-values of the class intervals of age and marks respectively.
- (ii) f_1 and f_2 represent marginal frequencies of the distribution of x and y respectively.



$$(iii) \quad u = \frac{x - 27.5}{5} \text{ and } v = \frac{y - 49.5}{20}.$$

(iv) Number within brackets in each cell is the product of that cell frequency and the corresponding values of u and v .

(v) f_{uv} represent totals of the numbers within brackets, mentioned at (iv) above, in any row or column.

$$\begin{aligned}\sigma_v^2 &= \frac{\sum f_2 v^2}{N} - \left(\frac{\sum f_2 v}{N} \right)^2 = \frac{55}{50} - \left(\frac{-13}{50} \right)^2 = 1.0324 \\ \text{cov}(u, v) &= \frac{\sum f_{uv}}{N} - \left(\frac{\sum f_1 u}{N} \right) \left(\frac{\sum f_2 v}{N} \right) = \frac{30}{50} - \left(\frac{-37}{50} \right) \left(\frac{-13}{50} \right) = 0.4076 \\ \sigma_u &= \sqrt{0.8724} = 0.93, \quad \sigma_v = \sqrt{1.0324} = 1.02\end{aligned}$$

$$\therefore \quad r_{xy} = r_{uv} = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v} = \frac{0.4076}{0.93 \times 1.02} = 0.43$$

Again, since $u = (x - 27.5)/5$ and $v = (y - 49.5)/20$,

$$\bar{x} = 27.5 + 5(-37/50) = 23.8$$

$$\bar{y} = 49.5 + 20(-13/50) = 44.3$$

Applying (9.6.4) and (6.6.3),

$$\text{cov}(x, y) = 5 \times 20 \times \text{cov}(u, v) = 100 \times 0.4076$$

$$\sigma_x^2 = 5^2 \times \sigma_u^2 = .25 \times 0.8724$$

$$\therefore \quad b_{yx} = \frac{\text{cov}(x, y)}{\sigma_x^2} = \frac{100 \times 0.4076}{25 \times 0.8724} = 1.87$$

The regression equation of y on x is $y - \bar{y} = b_{yx}(x - \bar{x})$

$$\text{i.e.} \quad y - 44.3 = 1.87(x - 23.8)$$

$$\text{or,} \quad y = 1.87x - 0.21$$

$$\text{Ans. } r = +0.43, y = 1.87x - 0.21$$



Example 9.31 Let the lines of regression concerning two variables x and y be given by $y = 32 - x$ and $x = 13 - 0.25y$. Obtain the values of the means and the correlation coefficient. [C.U., B.Sc. '71]

Solution Since the regression lines intersect at $(\bar{x}, \bar{y})$, the means will be obtained by solving the two equations. Solving $y = 32 - x$ and $x = 13 - 0.25y$, we get $x = 6.7$ and $y = 25.3$. So, $\bar{x} = 6.7$, $\bar{y} = 25.3$.

$y = 32 - x$ is the regression equation of y on x ; therefore the 'regression coefficient of y on x ' is the coefficient of x on the right; i.e. $b_{yx} = -1$. Similarly, $x = 13 - 0.25y$ being the regression equation of x on y , the 'regression coefficient of x on y ' is the coefficient of y on the right; i.e. $b_{xy} = -0.25$.

$$r^2 = b_{yx} \times b_{xy} = (-1)(-0.25) = 0.25;$$

$$r = \sqrt{0.25} = \pm 0.5$$

But since the regression coefficients are negative (note that both must have the same sign), the correlation coefficient also must be negative, i.e. $r = -0.5$.

$$\text{Ans. } \bar{x} = 6.7, \bar{y} = 25.3, r = -0.5.$$





Example 9.32 If x and y satisfy the relationship $y = -5 + 6x$, what is the product-moment correlation? [B.U., B.A. (Econ) '73]

Solution Since the relationship between x and y is exact and *linear*, there is perfect correlation between the variables; i.e. $r = \pm 1$. But the slope of the straight line (viz. $+6$) is positive. Hence r must be positive, and $r = +1$.

[**Note:** We speak of two types of relationship between the variables—(i) 'exact' or 'functional' relationship, and (ii) 'statistical' relationship. If to any specified value of one variable there corresponds some definite value or values of another variable, the relationship is exact. However, in many practical situations, no such relationship exists, e.g. between Height and Weight, or Income and Expenditure, etc. We then try to find the best possible estimates of one variable by means of an equation connecting them. This equation, known as 'Regression Equation', shows the statistical relationship between the variables.]



Example 9.33 Regression of savings(s) of a family on income (y) may be expressed as $s = a + \frac{y}{m}$, where a and m are constants. In a random sample of 100 families the variance of savings is one-quarter of the variance of incomes and the correlation is found to be 0.4. Obtain the estimate of m .

[I.C.W.A., June '74-old]

Solution The given regression equation of s on y can be written as $s = a + (1/m)y$, which is the equation of a straight line. Therefore the regression coefficient of s on y is the coefficient of y on the right;

$$\text{i.e.} \quad b_{sy} = 1/m \quad (i)$$

$$\text{But} \quad b_{sy} = r \frac{\sigma_s}{\sigma_y} \quad (ii)$$

However, we are given that $\sigma_s^2 = (1/4) \sigma_y^2$ and $r = 0.4$, so that $\frac{\sigma_s}{\sigma_y} = \frac{1}{2}$

$$\text{Hence, using (ii),} \quad b_{sy} = (0.4) \times \frac{1}{2} = 0.2$$

$$\text{Now, from (i),} \quad 0.2 = 1/m; \quad \therefore m = 1/0.2 = 5$$



Example 9.34 You are given that the variance of x is 9. The regression equations are $8x - 10y + 66 = 0$ and $40x - 18y = 214$. Find (i) Average values of x and y , (ii) Correlation coefficient between the two variables, (iii) Standard deviation of y .

[C.A., Nov. '77; I.C.W.A., June '77]

Solution

[**Note:** The regression equations of y on x and of x on y are usually shown in the forms $y = a + bx$ and $x = a' + b'y$ respectively, and in such cases it is not difficult to find the regression coefficients (Example 9.31). If however the equations are shown otherwise, they should be brought to the usual forms and then the regression coefficients can be easily determined. In the present problem, it is not known which one is the regression equation of y on x and which one the regression equation of x on y . A method has been shown for identifying the appropriate equations.]

Since the regression lines always intersect at the point $(\bar{x}, \bar{y})$, the averages of x and y will be obtained by solving the two equations [see Example 8.1 (g), p. 248]. Multiplying the first equation by 5 and then subtracting the result from the second equation, we have $32y = 544$, so



that $y = 17$. Now putting this value in any of the given equations and solving we get $x = 13$. Therefore, the averages of x and y are 13 and 17 respectively.

Let us assume that $8x - 10y + 66 = 0$ represents the regression equation of x on y , and $40x - 18y = 214$ the regression equation of y on x . These equations can be re-written as $x = (5/4)y - (33/4)$ and $y = (20/9)x - (107/9)$ respectively. The regression coefficients should then be $b_{xy} = 5/4$ and $b_{yx} = 20/9$.

$$\therefore r^2 = b_{xy} \cdot b_{yx} = (5/4) (20/9) = 25/9$$

This result is impossible, because r cannot exceed 1 numerically. So, our assumptions must be wrong.

The correct position must be the other way round; namely. $8x - 10y + 66 = 0$ must be the regression equation of y on x , and $40x - 18y = 214$ the regression equation of x on y . The two equations when expressed in the usual forms give $y = (4/5)x + (33/5)$ and $x = (9/20)y + (107/20)$. The correct values of the regression coefficients are $b_{yx} = 4/5$, $b_{xy} = 9/20$.

$$\text{i.e., } r = \sqrt{(4/5)(9/20)} = +0.6$$

(The positive value of the square-root is taken, since the regression coefficients are positive).

Again, given $\sigma_x^2 = 9$, i.e. $\sigma_x = 3$, using the relation $b_{yx} = 4/5$,

$$\text{we have } r \frac{\sigma_y}{\sigma_x} = 0.8; \text{ or } 0.6 \times \frac{\sigma_y}{3} = 0.8$$

This gives $\sigma_y = 4$.

Ans. 13, 17; +0.6; 4



Example 9.35 Prove that the coefficient of correlation is the geometric mean of the coefficients of regression.

[I.C.W.A., Jan. '65, '69, June '74, Dec. '76; B.U., B.A., (Econ) '66; C.U., M.Com. '64, '68, '71]

Solution The regression coefficients b_{yx} and b_{xy} may be expressed in terms of the correlation coefficient (r) and the standard deviations of x and y (viz. σ_x and σ_y) by the relations

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}, \quad b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

$$\therefore b_{yx} \times b_{xy} = \left(r \frac{\sigma_y}{\sigma_x} \right) \left(r \frac{\sigma_x}{\sigma_y} \right) = r^2$$

$$\text{or, } r^2 = b_{yx} \times b_{xy}; \quad \therefore r = \sqrt{b_{yx} \times b_{xy}}$$

Thus, the correlation coefficient r is the geometric mean of the two regression coefficients b_{yx} and b_{xy} .



Example 9.36 What are regression lines? Explain why we have two regression lines and why these two lines are identical if r the correlation coefficient is $+1$ or -1 .

[C.U., B.A. (Econ) '63; M.Com. '66; I.C.W.A., Jan. '73, Dec. '77]

Solution If bivariate data are plotted as points on a graph paper, it will be found that the concentration of points follows a certain pattern showing the relationship between the variables. When the trend of points is found to be linear, we determine the best-fitting straight line by the Method of Least Squares. Such straight lines which are used to obtain best estimates of one variable for given values of the other, are called *regression lines*. If x is considered as the independent variable and y the dependent variable, a linear equation of the form $y = a + bx$ is fitted to the pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ leading to the equation



$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \quad (i)$$

This is known as the *regression line of y on x*. If however, the variable x is considered to be dependent on y , a linear equation of the form $x = a' + b'y$ is fitted, yielding

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y}) \quad (ii)$$

This is known as the *regression line of x on y*. Equation (i) is used to obtain best estimates of y for given values of x , and equation (ii) to obtain best estimates of x for given values of y .

There are two regression lines, which are separately used to obtain best estimates of each variable. The same equation can not serve both the purposes; because the two equations (i) and (ii) are derived under two different assumptions. In deriving equation (i) it is assumed that the values of x are known exactly and those of y are subject to error. In deriving equation (ii) the assumption is just the reverse, i.e. the values of y are known exactly and those of x only are subject to error.

Since the two regression lines (i) and (ii) both pass through $(\bar{x}, \bar{y})$, they will be identical if their slopes are equal, i.e. if $b_{yx} = 1/b_{xy}$

$$\text{or, } r \frac{\sigma_y}{\sigma_x} = \frac{1}{r} \frac{\sigma_y}{\sigma_x}; \text{ simplifying, we get } r = \pm 1.$$



9.14 EXPLAINED VARIATION AND UNEXPLAINED VARIATION

If y'_i represents the *estimated* value of y from the regression equation of y on x (note that y_i denotes the *observed* value) when $x = x_i$, i.e., $y'_i - \bar{y} = b_{yx}(x_i - \bar{x})$, then it can be shown that

$$\Sigma(y_i - \bar{y})^2 = \Sigma(y_i - y'_i)^2 + \Sigma(y'_i - \bar{y})^2$$

$\Sigma(y_i - \bar{y})^2$ is called *Total variation* of the observed values of y . $\Sigma(y_i - y'_i)^2$ is the sum of the squares of vertical distances of the points on the scatter diagram around the regression line of y on x , and as such is called *Variation around the regression line* or *Unexplained variation* or *Residual variation*. $\Sigma(y'_i - \bar{y})^2$ is called *Variation due to regression* or *Explained variation*.

It can be shown that

$$\frac{\text{Explained variation}}{\text{Total variation}} = r^2$$

i.e. *Proportion of Total variation explained by regression* = r^2

We have, thus, a new interpretation of the correlation coefficient, viz. the square of the correlation coefficient is equal to the proportion of total variation explained by regression.

When $r^2 = 1$, the whole of the total variation is explained by regression, so that the unexplained variation is zero and each $y_i = y'_i$. All the points on the scatter diagram lie on the regression line of y on x (the two regression lines now coincide), and there is *perfect linear dependence* between the variables. For a given value of x , we have a fixed value of y .



When $r^2 = 0$, the explained variation is zero and each $y_i' = \bar{y}$. That is, the estimated value of y is a constant $\bar{y}$, whatever be the value of x . In this case there is *no linear dependence* between the variables.

r^2 may, therefore, be used as a measure of the degree of linear dependence between the variables. It may be noted that the sign of the correlation coefficient r only indicates whether x and y move in the same direction or opposite directions; but r^2 is always positive and gives a better measure (than r) of the degree of linear dependence.

Example 9.37 For a set of pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, prove that the total variation $\Sigma(y_i - \bar{y})^2$ can be written as

$$\Sigma(y_i - \bar{y})^2 = \Sigma(y_i - y_i')^2 + \Sigma(y_i' - \bar{y})^2$$

where y_i' represents the values of y for given values of $x = x_1, x_2, \dots, x_n$ obtained from the regression equation of y on x .

Explain the significance of the identity given above with reference to regression equations.

[C.U., B.A. (Econ) '67 '69; I.C.W.A., Dec. '76]

Solution The regression equation of y on x is $y - \bar{y} = b_{yx}(x - \bar{x})$ where $b_{yx} = \text{cov}(x, y)/\sigma_x^2$. If y_i' represents the estimated values of y when $x = x_i$, then $y_i' - \bar{y} = b_{yx}(x_i - \bar{x})$;

$$\text{i.e., } y_i' = \bar{y} + b_{yx}(x_i - \bar{x}) \quad (i)$$

$$\begin{aligned} \text{Now, } (y_i - \bar{y})^2 &= \{(y_i - y_i') + (y_i' - \bar{y})\}^2 \\ &= (y_i - y_i')^2 + (y_i' - \bar{y})^2 + 2(y_i - y_i')(y_i' - \bar{y}) \end{aligned}$$

Summing both sides over all values of $i = 1, 2, \dots, n$.

$$\Sigma(y_i - \bar{y})^2 = \Sigma(y_i - y_i')^2 + \Sigma(y_i' - \bar{y})^2 + 2\Sigma(y_i - y_i')(y_i' - \bar{y}) \quad (ii)$$

Considering the last term on the right and substituting for y_i' from (i),

$$\begin{aligned} \Sigma(y_i - y_i')(y_i' - \bar{y}) &= \Sigma\{(y_i - \bar{y}) - b_{yx}(x_i - \bar{x})\}\{b_{yx}(x_i - \bar{x})\} \\ &= \Sigma[b_{yx}(x_i - \bar{x})(y_i - \bar{y}) - b_{yx}^2(x_i - \bar{x})^2] \\ &= b_{yx}\Sigma(x_i - \bar{x})(y_i - \bar{y}) - b_{yx}^2\Sigma(x_i - \bar{x})^2 \\ &= b_{yx}[n.\text{cov}(x, y) - b_{yx}.n\sigma_x^2], \text{ because} \\ \text{cov}(x, y) &= \Sigma(x_i - \bar{x})(y_i - \bar{y})/n \quad \text{and} \quad \sigma_x^2 = \Sigma(x_i - \bar{x})^2/n \\ &= b_{yx}\left[n.\text{cov}(x, y) - \frac{\text{cov}(x, y)}{\sigma_x^2} \cdot n\sigma_x^2\right] \\ &= b_{yx}[n.\text{cov}(x, y) - n.\text{cov}(x, y)] = b_{yx} \times 0 = 0. \end{aligned}$$

Hence, from (ii)

$$\Sigma(y_i - \bar{y})^2 = \Sigma(y_i - y_i')^2 + \Sigma(y_i' - \bar{y})^2 \quad (iii)$$

Let us suppose that x represents the height and y the weight of several persons. Then the estimated weight y_i' of the person with height x_i is given by (i). It can not be expected that the actual weight y_i will be equal to its estimate y_i' obtained on the basis of his height x_i , i.e. the regression value. But y_i' provides *some* estimate of the weight of the person, i.e. a part of the actual weight y_i can be estimated or *explained* by the regression value y_i' . The difference $(y_i - y_i')$, called the *residual*, is the *unexplained* part of the weight.

The significance of the identity (iii) is that it is possible to split up the total variation of the observed values of y into two parts—a part (the second term on the right) explained by regression, i.e. which is entirely due to variation in the values of x , and another part which is due to other factors, unexplained by regression. This signifies that

$$\text{Total variation} = \text{Unexplained variation} + \text{Explained variation}.$$

9.15 REGRESSION CURVE IN BIVARIATE FREQUENCY DISTRIBUTION

With reference to a bivariate frequency distribution, the *Regression Curve of y on x* may be defined as that curve which shows the relationship between the conditional mean values of y and the corresponding specified values (or class marks) of x . If this curve happens to be a straight line, we say that the ‘regression of y on x is linear’, if it is a parabola, we come to the case of ‘parabolic regression’.

Similarly, the *Regression Curve of x on y* may be defined as that curve which shows the relationship between the conditional mean values of x and the corresponding values of y ; if this curve happens to be a straight line, we come to the case of linear regression, and say that the ‘regression of x on y is linear’. It is possible for either regression to be linear or both.

Example 9.38 In the table below verify that the means of the x -arrays are collinear and also those of the y -arrays and hence show that the correlation coefficient is -0.535 .

$y \backslash x$	0	1	2	3	4
0	—	—	—	4	3
1	—	—	18	36	9
2	—	12	54	36	3
3	1	12	18	4	—

[I.C.W.A., June '73]

Solution (Note that the “means of the x -arrays” refers to the ‘conditional mean values of x , and “means of the y -arrays” denotes the ‘conditional mean values of y ’; see Example 9.3, page 272). We may proceed as follows—

x	Total of y -array	Marginal Frequency	Mean of y -array
0	$1 \times 3 = 3$	1	$3/1 = 3$
1	$12 \times 2 + 12 \times 3 = 60$	24	$60/24 = 2.5$
2	$18 \times 1 + 54 \times 2 + 18 \times 3 = 180$	90	$180/90 = 2$
3	$4 \times 0 + 36 \times 1 + 36 \times 2 + 4 \times 3 = 120$	80	$120/80 = 1.5$
4	$3 \times 0 + 9 \times 1 + 3 \times 2 = 15$	15	$15/15 = 1$

The means of y -arrays corresponding to $x = 0, 1, 2, 3, 4$ are 3, 2.5, 2, 1.5, 1 respectively. Since, for unit increments in x these means change by equal amounts, namely -0.5 , they are collinear. This implies that if the means of y -arrays are plotted as points on a graph paper against the corresponding values of x , these points will lie on a straight line, which is the



regression line of y on x . The regression coefficient of y on x , which measures the change in the value of y for unit increment in x , is thus found to be $b_{yx} = -0.5$.

y	Total of x -array	Marginal Frequency	Mean of x -array
0	$4 \times 3 + 3 \times 4 = 24$	7	$24/7$
1	$18 \times 2 + 36 \times 3 + 9 \times 4 = 180$	63	$180/63 = 20/7$
2	$12 \times 1 + 54 \times 2 + 36 \times 3 + 3 \times 4 = 240$	105	$240/105 = 16/7$
3	$1 \times 0 + 12 \times 1 + 18 \times 2 + 4 \times 3 = 60$	35	$60/35 = 12/7$

Here again it is seen that the means of x -arrays change by equal amounts, namely $-4/7$, for unit increments in y , and so these means are collinear. The regression of x on y is linear, and the regression coefficient of x on y is $b_{xy} = -4/7$.

Since the correlation coefficient can be obtained from the relation $r = \sqrt{b_{yx} \cdot b_{xy}}$, with a sign common to both b_{yx} and b_{xy} ,

$$r = \sqrt{(-0.5)(-4/7)} = \sqrt{2/7} = -0.535$$



9.16 RANK CORRELATION

The product-moment correlation coefficient (r) is calculated by using 'values' of the variables. But many situations arise in which either precise measurements are not available, or the characters cannot be measured at all. For example, in order to find the extent of association between 'intelligence' and 'efficiency in salesmanship' for a group of salesmen, we could attempt to measure the two qualities by allotting marks to each salesman. But this method is open to many objections, and an exact measurement of the two qualities is not at all possible.

Difficulties of this type however disappear if we arrange the individuals in order of merit or proficiency in the possession of the qualities, using numbers 1, 2, 3, The individuals are then said to be *ranked* and the number allotted to a particular candidate is called his *rank*. For each individual, we then have a pair of ranks, one in each character. The correlation coefficient between the two series of ranks is called '*Rank Correlation Coefficient*'. It is given by the formula

$$R = 1 - \frac{6\sum d^2}{n^3 - n} \quad (9.16.1)$$

where d represents the difference of the ranks of an individual in the two characters, and n is the number of individuals. This formula is also known as *Spearman's formula* for rank correlation coefficient.

The rank correlation coefficient lies between -1 and $+1$.

$$-1 \leq R \leq +1$$

It has the maximum value $+1$, when the ranks in the two characters are equal, i.e. individuals with ranks 1, 2, 3, ... n in any character have exactly the same ranks 1, 2, 3, ... n respectively in the other. Again, R has the minimum value -1 , when the ranks are just the opposite, i.e. individuals with ranks 1, 2, 3, ... n in one character have ranks $n, n-1, \dots, 3, 2, 1$ in the other.

In the calculation of rank correlation coefficient from given scores, if several individuals have the same score in any character, they must be allotted the same ranks and we are then concerned with what are known as '*tied ranks*'. In dealing with such



cases, the usual way is to allot the average rank to each of these individuals, and then calculate the product-moment correlation coefficient from these ranks (see Example 9.42). However in such cases one way of correcting formula (9.16.1) is to increase Σd^2 by $(t^3 - t)/12$ in respect of each tie, where t denotes the number of individuals involved in a tie whether in the first or second series. The modified formula for rank correlation coefficient, when there are ties, is then

$$R' = 1 - \frac{6\{\Sigma d^2 + \Sigma(t^3 - t)/12\}}{n^2 - n} \quad (9.16.2)$$

Uses—

- (1) The rank correlation coefficient R is used as a measure of the 'degree of association between two attributes', where measurements on the characters are not available, but it is possible to rank the individuals in some order without difficulty (Note that the product-moment correlation coefficient r is a measure of the degree of association between two variables). It is therefore applied in vocational and applied psychology for finding the degree of correspondence between abilities.
- (2) It is also used in cases where exact or reliable measurements are not available. In educational tests it is known to all that the usual method of allotting marks to the candidates is open to the objection that different examiners would probably give different marks to the same candidate. The product-moment correlation coefficient r between scores would not therefore accurately reflect the relation ship between the abilities. However, if the candidates are ranked, the discrepancies in marks given by examiners would not seriously affect the value of rank correlation coefficient.
- (3) Even when exact measurements are available, the rank correlation coefficient R provides a quick estimate of the degree of association between the variables. The laborious calculations for the product-moment correlation coefficient r are replaced by ranking the individuals and then using formula (9.16.1) or (9.16.2).

Example 9.39 In a contest, two judges ranked seven candidates in order of their preference as in the following table:

Candidates		A	B	C	D	E	F	G
Ranks by Judge I	I	2	1	4	5	3	7	6
Ranks by Judge II	II	3	4	2	5	1	6	7

Calculate the rank correlation coefficient.

Solution

Table 9.17 Calculations for Rank Correlation Coefficient

Candidates	Ranks by		$d = x - y$	d^2
	Judge I x	Judge II y		
A	2	3	-1	1
B	1	4	-3	9

(Contd.)



Candidates	Ranks by		$d = x - y$	d^2
	Judge I x	Judge II y		
C	4	2	2	4
D	5	5	0	0
E	3	1	2	4
F	7	6	1	1
G	6	7	-1	1
Total	—	—	0	20

Here, $n = 7$, $\Sigma d^2 = 20$,

$$\therefore R = 1 - \frac{6 \Sigma d^2}{(n^3 - n)} = 1 - \frac{6 \times 20}{(7^3 - 7)} = 0.64. \quad \text{Ans. } R = 0.64$$



Example 9.40 In a certain examination 10 students obtained the following marks in Mathematics and Physics. Find Spearman's rank correlation coefficient.

Student (Roll No.)	1	2	3	4	5	6	7	8	9	10
Marks in Mathematics	90	30	82	45	32	65	40	88	73	66
Marks in Physics	85	42	75	68	45	63	60	90	62	58

[I.C.W.A., June '76-old]

Solution We have to obtain the ranks of students in the two subjects and calculate the rank correlation coefficient. For example, in Mathematics the student with Roll No. 1 gets the highest marks 90 and is given rank 1. Roll No. 8 with 88 marks has rank 2, and so on (see col. (3), Table 9.18.)

Table 9.18 Calculations for Rank Correlation Coefficient

Roll No. (1)	Mathematics		Physics		$d = x - y$ (6)	d^2 (7)
	Marks (X) (2)	Rank (x) (3)	Marks (Y) (4)	Rank (y) (5)		
1	90	1	85	2	-1	1
2	30	10	42	10	0	0
3	82	3	75	3	0	0
4	45	7	68	4	3	9
5	32	9	45	9	0	0
6	65	6	63	5	1	1
7	40	8	60	7	1	1
8	88	2	90	1	1	1
9	73	4	62	6	-2	4
10	66	5	58	8	-3	9
Total	—	—	—	—	0	26

Applying formula (9.6.1),

$$R = 1 - \frac{6 \times 26}{10^3 - 10} = 1 - \frac{26}{165} = +0.84$$



[**Note:** It is interesting to note that Spearman's rank correlation coefficient $R = 0.84$, calculated above, is in good agreement with Pearson's product-moment correlation coefficient $r = 0.86$ calculated from actual marks.]



Example 9.41 Deduce Spearman's formula for rank correlation coefficient.

[B.U., B.A. (Econ) '73]

Solution Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ denote the ranks of n individuals in the two characters. Spearman's rank correlation coefficient is the product-moment correlation coefficient between these ranks, treating them as values of variables.

It may be noted that $x_1, x_2, \dots, x_n$ being ranks (and not the actual values) are only the numbers $1, 2, \dots, n$ arranged in some order. Similarly $y_1, y_2, \dots, y_n$ are also the same numbers, but possibly in a different order [see Table 9.18, cols (3) and (5)].

$$\Sigma x = 1 + 2 + 3 + \dots + n = n(n+1)/2$$

(see p 67)

$$\Sigma x^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = n(n+1)(2n+1)/6$$

$$\therefore \bar{x} = \Sigma x/n = (n+1)/2$$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2 = \frac{n(n+1)(2n+1)}{6n} - \frac{(n+1)^2}{4} = \frac{n^2-1}{12}$$

Since the x -series and the y -series comprise the same numbers, their means and variances will be equal. Therefore

$$\bar{x} = \bar{y} = \frac{n+1}{2}; \quad \sigma_x^2 = \sigma_y^2 = \frac{n^2-1}{12}$$

In order to find $\text{cov}(x, y)$, we proceed as follows. Let us write

$$d_i = x_i - y_i$$

Since $\bar{x}$ and $\bar{y}$ are equal, we may write $d_i = (x_i - \bar{x}) - (y_i - \bar{y})$.

$$\therefore \Sigma d_i^2/n = \Sigma (x_i - \bar{x})^2/n + \Sigma (y_i - \bar{y})^2/n - 2\Sigma (x_i - \bar{x})(y_i - \bar{y})/n$$

$$= \sigma_x^2 + \sigma_y^2 - 2 \text{cov}(x, y) = \frac{n^2-1}{6} - 2 \text{cov}(x, y)$$

Transposing and dividing by 2, we find that

$$\text{cov}(x, y) = \frac{n^2-1}{12} - \frac{\Sigma d^2}{2n}$$

$$\therefore R = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{\frac{n^2-1}{12} - \frac{\Sigma d^2}{2n}}{\sqrt{\frac{n^2-1}{12}} \sqrt{\frac{n^2-1}{2n}}} = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$



Example 9.42 In the following table are recorded data showing the test scores made by 10 salesmen on an intelligence test and their weekly sales:

Salesmen	1	2	3	4	5	6	7	8	9	10
Test Scores	50	70	50	60	80	50	90	50	60	60
Sales ('000 Rs)	25	60	45	50	45	20	55	30	45	30

Calculate the rank correlation coefficient between intelligence and efficiency in salesmanship.

Note: It will be noticed that in Test Scores, 3 salesmen (viz. 4, 9 and 10) with score 60 each tie for the fourth place, and hence are given the average rank $(4 + 5 + 5)/3$



= 5; again, 4 salesmen (viz. 1, 3, 6 and 8) tie for the seventh place with score 50 each and are given the average rank $(7 + 8 + 9 + 10)/4 = 8.5$. Similarly in Sales, 3 salesmen tie for the fourth place and 2 salesmen tie for the seventh place, where average ranks are given].

Solution

Table 9.19 Calculations for Rank Correlation

Salesmen	Intelligence		Sales		$d = x - y$	d^2
	Test score	Rank (x)	Amount ('000 Rs)	Rank (y)		
1	50	8.5	25	9	-0.5	0.25
2	70	3	60	1	2	4
3	50	8.5	45	5	3.5	12.25
4	60	5	50	3	2	4
5	80	2	45	5	-3	9
6	50	8.5	20	10	-1.5	2.25
7	90	1	55	2	-1	1
8	50	8.5	30	7.5	1	1
9	60	5	45	5	0	0
10	60	5	30	7.5	-2.5	6.25
Total	—	—	—	—	0	40

In the two series together, there are in all 4 ties, viz. one tie involving 4 individuals, 2 ties involving 3 individuals each, and 1 tie with 2 individuals. Hence,

$$\frac{\Sigma(t^3 - t)}{12} = \frac{4^3 - 4}{12} + \frac{3^3 - 3}{12} + \frac{3^3 - 3}{12} + \frac{2^3 - 2}{12}$$

$$= 5 + 2 + 2 + 0.5 = 9.5$$

Substituting the values in formula (9.16.2) we get

$$R' = 1 - \frac{6(40 + 9.5)}{10^3 - 10} = 0.70$$



9.17 MULTIPLE CORRELATION AND PARTIAL CORRELATION

In bivariate distributions, we know that when the values of one variable are influenced by those of another, the simple correlation coefficient r provides a measure of the degree of association between these two variables, and estimates of the values of a variable are obtained from the appropriate linear regression equation. But, very often it happens that the values of a variable are influenced not only by another single variable, but by several others. For example, (i) the yield of crop depends on rainfall, temperature, amount of fertiliser applied, etc.; (ii) the weight of a person depends on his height as well as his chest measurement; (iii) the yield of dry fibre from jute plants



depends on the height of plant, the diameter of stem, etc. In each of these cases are concerned with the joint distribution of three or more variables, called *Multivariate Distribution*.

The problems in multivariate distribution are of three types:

- To estimate the value of one variable from those of several others; this is the problem of *Multiple Regression*.
- To measure the extent of *combined influence* of a group of variables upon another variable; this is the problem of *Multiple Correlation*.
- To examine the influence of one variable upon another, *after eliminating the effects* of all other variables; this is the problem of *Partial Correlation*.

Let us consider the cases of three variables, denoted by x_1, x_2, x_3 . The simplest kind of relationship which can be used for estimating x_1 in terms of x_2 and x_3 is an equation of the form

$$x_1 = a + bx_2 + cx_3,$$

where a, b, c are constants. Applying the method of least squares, the constants are determined. The equation is usually written as

$$x_1 - \bar{x}_1 = b_{12.3}(x_2 - \bar{x}_2) + b_{13.2}(x_3 - \bar{x}_3) \quad (9.17.1)$$

where $\bar{x}_1, \bar{x}_2, \bar{x}_3$ are the means of variables x_1, x_2, x_3 respectively, and $b_{12.3}, b_{13.2}$ are certain constants. This is known as *regression equation of x_1 on x_2 and x_3* (sometimes called *multile regression equation*). The constants $b_{12.3}$ and $b_{13.2}$ are called *partial regression coefficients*, and are given by

$$b_{12.3} = \frac{\sigma_1}{\sigma_2} \left(\frac{r_{12} - r_{13}r_{23}}{1 - r_{23}^2} \right); b_{13.2} = \frac{\sigma_1}{\sigma_3} \left(\frac{r_{13} - r_{12}r_{23}}{1 - r_{23}^2} \right) \quad (9.17.2)$$

where $\sigma_1, \sigma_2, \sigma_3$ are the standard deviation of the variables x_1, x_2, x_3 , and r_{12}, r_{13}, r_{23} denote the product-moment correlation coefficients between x_1 and x_2 , between x_1 and x_3 , and between x_2 and x_3 respectively.

The product-moment correlation coefficient between the observed values of x_1 and the estimated values from the regression equation (9.17.1) is called the *Multiple Correlation Coefficient* of x_1 on x_2 and x_3 , and is given by

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}} \quad (9.17.3)$$

It is always considered as positive.

$$0 \leq R_{1.23} \leq 1. \quad (9.17.4)$$

In order to understand the significance of *Partial Correlation Coefficient*, let us consider the regressions of x_1 on x_3 and of x_2 on x_3 . viz. $x_1 = a + b_{13}x_3$ and $x_2 = a' + b_{23}x_3$. We do not expect that $a + b_{13}x_3$ gives a very good estimate of x_1 , but it shows only that part of x_1 which can be estimated as a linear function of x_3 alone.

The difference

$$x_1 - (a + b_{13}x_3) = x_{1.3} \text{ (suppose)}$$

is the part of x_1 that remains, after the effect of x_3 has been eliminated. Similarly,

$$x_2 - (a' + b_{23}x_3) = x_{2.3} \text{ (suppose)}$$

is the part of x_2 that remains, after the effect of x_3 has been eliminated.

The product-moment correlation coefficient between the residuals $x_{1.3}$ and $x_{2.3}$ is called *Partial Correlation Coefficient between x_1 and x_2* , and is given by



$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}} \quad (9.17.5)$$

The partial correlation coefficient lies between -1 and $+1$.

$$-1 \leq r_{12.3} \leq 1 \quad (9.17.6)$$

[Note:

- (1) r_{12} is the correlation coefficient between x_1 and x_2 , *ignoring* altogether any influences of x_3 . But $r_{12.3}$ is the correlation coefficient between x_1 and x_2 , after eliminating the influence of x_3 from both. r_{12} is sometimes called *Total Correlation Coefficient*, in order to distinguish it from *Partial Correlation Coefficient* $r_{12.3}$.
- (2) Just as r_{12} is the same as r_{21} , similarly $r_{12.3}$ is the same as $r_{21.3}$. The multiple correlation coefficient $R_{1.23}$ is the same $R_{1.32}$. But just as the *Total Regression Coefficients* b_{12} and b_{21} are different, similarly the partial regression coefficients $b_{12.3}$ and $b_{21.3}$ are also different. $b_{12.3}$ is the coefficient of x_2 in the multiple regression equation for estimating x_1 as a linear function of x_2 and x_3 ; but $b_{21.3}$ is the coefficient of x_1 in the multiple regression equation for estimating x_2 as a linear function of x_1 and x_3 .
- (3) The regression equation of x_2 and x_1 and x_3 may be obtained from (9.17.1) by interchanging the subscripts 1 and 2. The multiple correlation coefficient $R_{2.13}$ may also be obtained from (9.17.3) similarly. The partial correlation $r_{23.1} = r_{32.1}$ may be obtained from (9.17.5) by interchanging the subscripts 1 and 3.]

Important Results

$$(i) \quad 1 - R_{1.23}^2 = (1 - r_{12}^2)(1 - r_{13.2}^2) \\ = (1 - r_{13}^2)(1 - r_{12.1}^2) \quad (9.17.7)$$

$$(ii) \quad R_{1.23} \geq |r_{12}| \text{ and } R_{1.23} \geq |r_{13}| \quad (9.17.8)$$

$$(iii) \quad \text{If } R_{1.23} = 0, \text{ then } r_{12} = r_{13} = 0 \quad (9.17.9)$$

- (iv) Just as the product of total regression coefficients b_{12} and b_{21} is the square of total correlation coefficient r_{12} , i.e. $b_{12} \times b_{21} = r_{12}^2$, similarly the product of partial regression coefficients $b_{12.3}$ and $b_{21.3}$ is the square of partial correlation coefficient $r_{12.3}$, i.e.

$$b_{12.3} \times b_{21.3} = r_{12.3}^2 \quad (9.17.10)$$

This is, probably, the reason why the regression coefficients $b_{12.3}$ etc. are called 'partial' regression coefficients, and not 'multiple' regression coefficients.

$$(v) \quad b_{12.3} = \frac{b_{12} - b_{13}b_{32}}{1 - b_{23}b_{32}} \quad (9.17.11)$$

Example 9.43 Given $r_{12} = +0.65$, $r_{13} = +0.60$ and $r_{23} = +0.90$, calculate the values of $r_{12.3}$ and $R_{1.23}$ (the symbols have their usual significance).

Solution Using formula (9.17.5) and (9.17.3), we have

$$r_{12.3} = \frac{0.65 - 0.6 \times 0.9}{\sqrt{(1 - 0.6^2)(1 - 0.9^2)}} = \frac{0.11}{0.349} = 0.32$$

$$R_{1.32} = \sqrt{\frac{0.65^2 + 0.6^2 - 2 \times 0.65 \times 0.6 \times 0.9}{1 - 0.9^2}} = \sqrt{\frac{0.0805}{0.19}} = 0.651.$$



EXERCISES



1. The data given below relate to the heights and weights of 20 persons. You are required to form a two-way frequency table with class-intervals 62" to 64", 64" to 66" and so on, and 115 to 125 lbs., 125 to 135 lbs., and so on.

S. No.	1	2	3	4	5	6	7	8	9	10	11	12	13
Height	70	65	65	64	69	63	65	70	71	62	70	67	63
Weight	170	135	136	137	148	124	117	128	143	129	163	139	122
							14	15	16	17	18	19	20
							68	67	69	66	68	67	67
							134	140	132	120	148	129	152

[C.A., May '66]

2. Calculate r from the following given results: $n = 10$; $\Sigma x = 125$; $\Sigma x^2 = 1585$; $\Sigma y = 80$; $\Sigma y^2 = 650$; $\Sigma xy = 1007$.
3. Find the coefficient of correlation from the following results:

$$\sum_{i=1}^8 X = 42.2, \sum_{i=1}^8 Y = 46.4, \sum_{i=1}^8 X^2 = 291.20,$$

$$\sum_{i=1}^8 Y^2 = 290.52, \sum_{i=1}^8 XY = 230.42$$

4. Obtain the correlation coefficient from the following:

$x -$	6	2	10	4	8
$y -$	9	11	5	8	7

[D.S.W., '77]

5. Calculate the coefficient of correlation for the ages of husband and wife :

Age of Husband	23	27	28	29	30	31	33	35	36	39
Age of Wife	18	22	23	24	25	26	28	29	30	32

[I.C.W.A., Jan. '70]

6. Calculate the correlation coefficient r_{xy} from the following:

x	65	66	67	67	68	69	70	72
y	67	68	65	68	72	72	69	71

[D.M., '78]

7. Calculate the coefficient of correlation between x and y :

x	155	157	153	151	159	162	158
y	118	129	125	124	129	133	127

[C.U., B.A. (Econ) '75]

8. Calculate Pearson's coefficient of correlation from the following data using 44 and 26 as the origins of X and Y respectively :

X	43	44	46	40	44	42	45	42	38	40	42	57
Y	29	31	19	18	19	27	27	29	41	30	26	10

[C.A., May '78]



9. Specimens of similarly treated alloy steel containing various percentages of nickel are tested for toughness with the following results:

<i>Toughness (arbitrary units)</i>	47	50	52	52	54	56	58	59
<i>Percentage of Nickel</i>	2.7	2.7	2.8	2.8	2.9	3.2	3.2	3.3
			60	60	62	64	65	66
			3.4	3.5	3.5	3.6	3.7	3.8

Find the coefficient of correlation between 'toughness' as measured by the test, and 'percentage content of Nickel' in the alloy steel.

[C.U., M. Com. '65; D.S.W., May '71]

10. Calculate the coefficient of correlation from the following data:

<i>Export of Raw Cotton (Rs crores)</i>	42	44	58	55	89	98	66
<i>Import of Manufactured Goods (Rs crores)</i>	56	49	53	58	65	76	58

Calculate also the standard error of the coefficient of correlation.

[I.C.W.A., July '65]

11. Determine the correlation coefficient between x and y :

x	5	7	9	11	13	15
y	1.7	2.4	2.8	3.4	3.7	4.4

[Dip. Management, '67]

12. From the following figures, calculate the coefficient of correlation between the income and the general level of prices:

<i>Income (X)</i>	360	420	500	550	600	640	680	720	750
<i>General Level of Prices (Y)</i>	100	104	115	160	180	290	300	320	330

[C.U., M. Com. '68]

13. The following data give the hardness (x) and tensile strength (y) for some specimens of a material, in certain units. Find the correlation coefficient and calculate its probable error:

x	23.3	17.5	17.8	20.7	18.1	20.9	22.9	20.8
y	4.2	3.8	4.6	3.2	5.2	4.7	4.4	5.6

[I.C.W. A., Jan. '72]

14. The following table gives the saving-bank deposits in billions of dollars and strikes and lock-outs in thousands over a number of years. Compute the correlation coefficient and comment on the results.

<i>Saving Deposits</i>	5.1	5.4	5.5	5.9	6.5	6.0	7.2
<i>Strikes and Lock-outs</i>	3.8	4.4	3.3	3.6	3.3	2.3	1.0

[I.C.W.A., Jan. '64]

15. Two positively correlated variables x_1 and x_2 have variances σ_1^2 and σ_2^2 respectively. Determine the value of the constant a such that $x_1 + ax_2$ and

$x_1 + \frac{\sigma_1}{\sigma_2} x_2$ are uncorrelated.

[B.U., B.A.(Econ) '72]



16. State whether the correlation will be positive or negative in the following cases:
- Age and income;
 - Speed of an automobile and the distance required to stop the car after applying brakes;
 - Sale of woollen garments and day-temperature;
 - Sale of cold-drinks and day-temperature;
 - Production and price per unit.
17. Given $\Sigma x = 56$, $\Sigma y = 40$, $\Sigma x^2 = 524$
 $\Sigma y^2 = 256$, $\Sigma xy = 364$, $n = 8$,
 find (i) the correlation coefficient and (ii) the regression equation of x on y .
[I.C.W.A., July '67]
18. The following sums have been obtained from 100 observations-pairs:
 $\Sigma x = 12,500$, $\Sigma y = 8,000$, $\Sigma x^2 = 1,585,000$,
 $\Sigma y^2 = 648,100$, $\Sigma xy = 1,007,425$.
 (i) Find the regression of y on x , and estimate the value of y when $x = 130$.
 (ii) Compute the correlation coefficient (r) between x and y and state what you learn from the value of r obtained by you.
[C.U., B.A. (Econ) '76]
19. Given the following totals for 10 pairs of observations on two characters x and y , obtain the two regression equations and hence calculate the correlation coefficient:
 $\Sigma x = 12$, $\Sigma y = 4$, $\Sigma x^2 = 16.20$, $\Sigma y^2 = 1.96$, $\Sigma xy = 5.2$
[M.B.A. '79, D.M. '77]
20. Estimate from the information given below, the probable crop yield, when rainfall is 29 inches:
- | | Mean | S.D. |
|-------------------------|------|------|
| Rainfall in inches | 25 | 3 |
| Yield in units per acre | 40 | 6 |
- Coefficient of correlation between the variables: 0.65
[C.U., B.SC. '73]
21. The correlation coefficient between two variates x and y is $r = 0.60$. If $\sigma_x = 1.50$, $\sigma_y = 2.00$, $\bar{x} = 10$, $\bar{y} = 20$, find the equations of the regression lines of
 (i) y on x , (ii) x on y .
[I.C.W.A., Dec. '77]
22. The following data pertain to the marks in two subjects, say A and B .
 Mean marks in $A = 39.5$, Mean marks in $B = 47.5$
 S.D. of marks in $A = 10.8$, S.D. of marks in $B = 16.8$
 Coefficient of correlation between marks in A and $B = 0.42$
 Obtain the equations of the two regression lines and then estimate the marks in B for candidates who secured 50 marks in A .
[I.C.W.A., June '78]
23. Given the following results of the height and weight of 1000 men students:
 $\bar{x} = 68$ inches, $\bar{y} = 150$ lbs., $r = 0.60$, $\sigma_x = 2.50$ inches, $\sigma_y = 20.000$ lbs. John



Doe weights 200 lbs., Richard Roe is five feet tall. Estimate the height of Doe from his weight, and weight of Roe from his height.

[C.U., M. Com. '72, '76]

24. From the following data find the coefficient of linear correlation between X and Y . Determine also the regression line of Y on X , and then make an estimate of the value of Y when $X = 12$.

X	1	3	4	6	8	9	11	14
Y	1	2	4	4	5	7	8	9

[I.C.W.A., June '75]

25. Obtain the lines of regression for the following data:

(X)	1	2	3	4	5	6	7	8	9
(Y)	9	8	10	12	11	13	14	16	15

[C.U., M.Com. '68; D.S.W. '78; I.C.W.A., Dec. '78]

26. Find the two lines of regression from the following data:

Age of Husband (x)	25	22	28	26	35	20	22	40	20	18
Age of Wife (y)	18	15	20	17	22	14	16	21	15	14

Hence, estimate (i) the age of husband when the age of wife is 19, (ii) the age of wife when the age of husband is 30.

[C.U., M.Com. '70]

27. From the following data, obtain the two regression equations:

Sales	91	97	108	121	67	124	51	73	111	57
Purchases	71	75	69	97	70	91	39	61	80	47

[C.A., May '77]

28. Obtain the equation of the line of regression of yield of rice (y) on water (x) from the data given in the following table:

Water in Inches (x)	12	18	24	30	36	42	48
Yield in Tons (y)	5.27	5.68	6.25	7.21	8.02	8.71	8.42

Estimate the most probable yield of rice for 40 inches of water.

[C.U., M.Com. '64; I.C.W. A., Dec. '76]

29. If the regression equation of y on x be $Y = 0.57x + 6.93$ and the regression equation of x on y be $X = 1.12y - 2.46$, find the correlation coefficient between x and y .

[B.U., B.A. (Econ) '72]

30. For some bivariate data the following results were obtained. The mean value of $X = 53.2$, the mean value of $Y = 27.9$, the regression coefficient of Y on $X = -1.5$, and the regression coefficient of X on $Y = -0.2$. Find the (i) most probable value of Y when $X = 60$, (ii) r the coefficient of correlation between X and Y .

[C.U., M.Com '74]

31. The regression equation calculated from a given set of observations are $x = -0.2y + 4.2$, $y = -0.8x + 8.4$. Calculate (i) x and y , (ii) r , (iii) the estimated value of y when $x = 4$.

[I.C.W.A., Jan. '68; C.U., B Com. (Hons) '69]



32. The two regression lines involving the two variables x and y are $Y = 5.6 + 1.2x$ and $X = 12.5 + 0.6y$. Find the means of x and y and their correlation coefficient.

[W.B.H.S., '78]

33. Two variates have the least squares regression lines $x + 4y + 3 = 0$ and $4x + 9y + 5 = 0$. Find their mean values and the correlation coefficient.

[I.C.W.A., July '70; M.B.A. '78]

34. Two lines of regression are given by $x = 2y = 5$ and $2x + 3y = 8$, and $\sigma_x^2 = 12$. Calculate the values of $\bar{x}$, $\bar{y}$, σ_y and r .

[I.C.W.A., Dec '76-old]

35. In order to find the correlation coefficient between two variates x and y from 12 pairs of observations, the following calculations were made:

$$\Sigma x = 30, \quad \Sigma y = 5, \quad \Sigma x^2 = 670, \quad \Sigma y^2 = 285, \quad \Sigma xy = 334.$$

On subsequent verification it was found that the pair ($x = 11$, $y = 4$) was copied wrongly, the correct values being ($x = 10$, $y = 14$). Find the correct value of correlation coefficient.

[I.C.W.A., June '75-old]

36. Obtain the linear regression equation that you consider more relevant for the following set of paired observations and given reasons why you consider it to be so:

Age	56	42	72	36	63	47	55	49	38	42	68	60
Blood Pressure	147	125	160	118	149	128	150	145	115	140	152	155

Also estimate the blood pressure of a person whose age is 45.

[C.U., M.Com., '73]

37. For the variables x and y the equations of two regression lines are $4x - 5y + 33 = 0$ and $20x - 9y = 107$. Identify the regression line of y on x and that of x on y . What is the estimated value of y , when $x = 10$? If this estimate is denoted by y_0 , find the estimated value of x when $y = y_0$.

[C.U., B.A. (Econ) '75]

38. State the meaning of the terms *explained variation* and *unexplained variation*, used in the theory of regression. If the coefficient of correlation between two variables X and Y be 0.83, what percentage of total variation remains unexplained by the regression equation?

[I.C.W.A., Dec. '75]

39. Given: Unexplained variation = 19.70, and Explained variation = 19.22, determine the coefficient of correlation.

[I.C.W.A., Dec. '76]

40. In a contest, two judges ranked eight candidates A, B, C, D, E, F, G and H in order of their preference, as shown in the following table. Find the rank correlation coefficient.

	A	B	C	D	E	F	G	H
First Judge	5	2	8	1	4	6	3	7
Second Judge	4	5	7	3	2	8	1	6

[I.C.W.A., June '75]



41. Compute the correlation coefficient of the following ranks of a group of students in two examinations. What conclusion do you draw from the result?

Roll Nos.	1	2	3	4	5	6	7	8	9	10
Ranks in B.Com. Exam.	1	5	8	6	7	4	2	3	9	10
Ranks in M.Com. Exam.	2	1	5	7	6	3	4	8	10	9

[C.U., M.Com. '75]

42. Ten competitors in a musical contest were ranked by 3 judges A, B, C in the following order :

Ranks by A	1	6	5	10	3	2	4	9	7	8
Ranks by B	3	5	8	4	7	10	2	1	6	9
Ranks by C	6	4	9	8	1	2	3	10	5	7

Using Rank Correlation method, discuss which pair of judges has the nearest approach to common likings in music.

[I.C.W.A., Dec. '78]

43. Ten students obtained the following marks in Mathematics and Statistics. Calculate the rank correlation coefficient.

Student (Roll No.)	1	2	3	4	5	6	7	8	9	10
Marks in Mathematics	78	36	98	25	75	82	90	62	65	39
Marks in Statistics	84	51	91	60	68	62	86	58	53	47

44. Given the following coefficients : $r_{12} = 0.41$, $r_{13} = 0.71$, $r_{23} = 0.5$ Find $r_{12.3}$, $r_{13.2}$ and $R_{1.23}$, where the symbols have their usual significance.

[C.U., M.Com. '74]

45. In a three-variate multiple correlation analysis, the following results were found:

$$\begin{array}{lll} \bar{x}_1 = 60 & x_2 = 70 & \bar{x}_3 = 100 \\ s_1 = 3 & s_2 = 4 & s_3 = 5 \\ r_{12} = 0.7 & r_{13} = 0.6 & r_{23} = 0.4 \end{array}$$

the symbols having their usual significance. Find the regression of x_1 on x_2 and x_3 , and the multiple correlation coefficient $R_{1.23}$.

[B.U., M.A.(Econ) '68]



ANSWERS



1. (See Example 9.2)
2. + 0.47
3. - 0.37
4. - 0.92
5. + 0.996
6. 0.60
7. + 0.71
8. - 0.73
9. + 0.98
10. + 0.90, 0.072
11. + 0.995
12. + 0.94
13. - 0.072, 0.237
14. - 0.82
15. $-\sigma_1/\sigma_2$
16. (i) positive, (ii) positive, (iii) negative, (iv) positive, (v) negative
17. + 0.98; $x = 1.5y - 0.5$
18. $y = 0.33x + 38.75$; 81.65; 0.55
19. $y = 0.222x + 0.133$; $x = 1.11y + 0.756$; 0.50



20. 45.2 units per acre
21. $y = 0.8x + 12$; $x = 0.45y + 1$
22. $y = 0.65x + 21.82$; $x = 0.27y + 26.68$; 54
23. 71.75 inches; 111.60 lbs.
24. $+0.98$; $y = 0.64x + 0.52$; 8.2
25. $X = 0.95Y - 6.4$; $Y = 0.95X + 7.25$
26. $x = 2.23y - 12.76$; $y = 0.39x + 7.22$; (i) 30, (ii) 19.
27. $y = 0.613x + 14.83$; $x = 1.360y - 5.2$
28. $y = 3.99 + 0.103x$; 8.11 tons.
29. $r = \sqrt{0.57 \times 1.12} = +0.80$
30. (i) 17.7, (ii) -0.55
31. (i) 3 and 6, (ii) -0.4 , (iii) 5.2
32. 56.64, 73.57; $+0.85$
33. $\bar{x} = 1$, $\bar{y} = -1$, $r = -0.75$
34. $\bar{x} = 1$, $\bar{y} = 2$, $\sigma_y = 2$, $r = -\sqrt{3/2}$
35. $+0.77$
36. Regression equation of blood pressure (y) on age (x) is
 $y = 1.14x + 80.67$; 132
37. First equation regression line of y on x ; $y_0 = 14.6$; 11.92
38. 31%
39. $|r| = 0.70$
40. $2/3$
41. $+0.64$
42. Rank corr. coeffs. between A & B , A & C and B & C respectively are -0.21 , $+0.64$, -0.30 . Since $+0.64$ is the largest, Judges A and C have the nearest approach to common likings in music.
43. $+0.82$
44. $+0.09$, $+0.64$, $+0.71$
45. $x_1 = 8.3 + 0.41x_2 + 0.23x_3$; $R_{1,23} = 0.78$

10 INTERPOLATION

10.1 INTRODUCTION

'Interpolation' has been defined as the 'art of reading between the lines of a table', and the term usually denotes the process of finding the intermediate value of a function from a set of given values of that function. For example, using the following given values of x and y

x :	1	3	5	7	9	11
y :	1	15	65	175	369	671

we may be required to find, i.e. *interpolate*, the value of y when $x = 4$. Since the mathematical relationship between x and y is not known, we have necessarily to base our calculations on the given data.

In *Simple Interpolation* (page 63) we use only two pairs of observations on either side of the unknown value, and assume a constant rate of change (i.e. linear relationship) in this interval.

x	y	Using (3.16.1), $\frac{4-3}{5-3} = \frac{y-15}{65-15}$. Solving, we get $y = 40$.
3	15	
4 $\rightarrow$	$\leftarrow ?$	
5	65	

However, the mathematical relationship between the values of x and y given above is $y = \frac{1}{2}(x^3 + x)$, so that when $x = 4$, the value of y is 34. The discrepancy of the interpolated value $y = 40$ from the true value $y = 34$ is due to the fact that we have not utilised the whole of the given data, but only two pairs of observations. Applying formulae for interpolation, stated in the following pages, it is possible (see Example 10.12) in some cases, to obtain the value of y exactly, and if not to obtain a close approximation to the true value.

10.2 FINITE DIFFERENCES: Δ AND E OPERATORS

In problems of interpolation, the independent variable x is often known as 'argument', and the dependent variable or the function $y = f(x)$ is known as 'entry'. Let $x_0, x_1, x_2, \dots, x_n$ denote a set of equidistant values of the argument, i.e.



$$x_1 - x_0 = x_2 - x_1 = \dots = x_n - x_{n-1} = h$$

where h is a constant; and $y_0, y_1, y_2, \dots, y_n$ denote the corresponding values of the entry. Differences of the successive values of y , viz.

$$(y_1 - y_0), (y_2 - y_1), (y_3 - y_2), \dots, (y_n - y_{n-1})$$

are called *finite differences of the first order* (or simply *first differences*), and are denoted by

$$\Delta y_0, \Delta y_1, \Delta y_2, \dots, \Delta y_{n-1}$$

respectively. The differences of the successive first order differences Δy , namely

$$(\Delta y_1 - \Delta y_0), (\Delta y_2 - \Delta y_1), \dots, (\Delta y_{n-1} - \Delta y_{n-2})$$

are known as *finite differences of the second order* (or simply *second differences*) and are denoted by

$$\Delta^2 y_0, \Delta^2 y_1, \dots, \Delta^2 y_{n-2}$$

respectively. Similarly, the *third differences* $\Delta^3 y$, the *fourth differences* $\Delta^4 y$, and differences of higher order may be defined.

Argument (x)	Entry (y)	First Differences (Δy)	Second Differences ($\Delta^2 y$)
x_0	y_0		
x_1	y_1	$y_1 - y_0 = \Delta y_0$	$\Delta y_1 - \Delta y_0 = \Delta^2 y_0$
x_2	y_2	$y_2 - y_1 = \Delta y_1$	$\Delta y_2 - \Delta y_1 = \Delta^2 y_1$
x_3	y_3	$y_3 - y_2 = \Delta y_2$	$\Delta y_3 - \Delta y_2 = \Delta^2 y_2$
x_4	y_4	$y_4 - y_3 = \Delta y_3$	

A table which shows the finite differences is known as *Difference Table*.

Table 10.1 Difference Table

Argument x	Entry y	First Differences Δy	second Differences $\Delta^2 y$	Third Differences $\Delta^3 y$	Fourth Differences $\Delta^4 y$
$x_0 = 1$	$y_0 = 1$	14			
$x_1 = 3$	$y_1 = 15$	50	36		
$x_2 = 5$	$y_2 = 65$	110	60	24	0
$x_3 = 7$	$y_3 = 175$	194	84	24	0
$x_4 = 9$	$y_4 = 369$	302	108		
$x_5 = 11$	$y_5 = 671$				



In symbols, the difference table may be shown as follows:

Table 10.2 Difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
x_0	y_0	Δy_0				
x_1	y_1	Δy_1	$\Delta^2 y_0$	$\Delta^3 y_0$		
x_2	y_2	Δy_2	$\Delta^2 y_1$	$\Delta^3 y_1$	$\Delta^4 y_0$	
x_3	y_3	Δy_3	$\Delta^2 y_2$	$\Delta^3 y_2$	$\Delta^4 y_1$	$\Delta^5 y_0$
x_4	y_4	Δy_4	$\Delta^2 y_3$			
x_5	y_5					

The arrangement of differences should be noted. Each difference is written between the two quantities which are used to calculate it.

The initial term y_0 of the entry is called the *leading term* and the initial terms in the difference columns, viz. Δy_0 , $\Delta^2 y_0$, $\Delta^3 y_0$ etc. are called *leading differences*.

[**Note:** Δ (called *delta*) is not a quantity multiplied to y^2 s, but an ‘operator’ like the root-sign $\sqrt{\quad}$ or log. These operators separately have no meaning, but when operating on a number, e.g. $\sqrt{9}$ or $\log 7$, have a real significance. Again, Δ^2 does not imply the square of Δ , but the repetition of operation by Δ twice].

Another symbolic operator E is defined as follows:

$$Ey_0 = y_1, \quad Ey_1 = y_2, \quad Ey_2 = y_3, \quad \dots$$

Both the operators Δ and E can be applied repeatedly, the repeated operations being indicated by Δ^2 , Δ^3 , ... and E^2 , E^3 , etc. Thus

$$\begin{aligned} \Delta^2 y_0 &= \Delta(\Delta y_0) = \Delta y_1 - \Delta y_0 \\ &= (y_2 - y_1) - (y_1 - y_0) = y_2 - 2y_1 + y_0 \\ \Delta^3 y_0 &= \Delta(\Delta^2 y_0) = \Delta^2 y_1 - \Delta^2 y_0 \\ &= (y_3 - 2y_2 + y_1) - (y_2 - 2y_1 + y_0) \\ &= y_3 - 3y_2 + 3y_1 - y_0 \end{aligned}$$

$$\text{Similarly, } E^2 y_0 = E(Ey_0) = E(y_1) = y_2$$

$$E^3 y_0 = E(E^2 y_0) = E(y_2) = y_3; \quad \text{etc.}$$

From the definitions, we may in general write

$$\Delta y_r = y_{r+1} - y_r \quad (10.2.1)$$

$$Ey_r = y_{r+1} \quad (10.2.2)$$

These operators may thus be interpolated in the following manner:

- Δ when prefixed to y_r implies that y_r is to be subtracted from the next value of the entry y_{r+1} .
- E when prefixed to y_r denotes the next value of the entry y_{r+1} .



From (10.2.1) and (10.2.2) we find that

$$\begin{aligned} Ey_r &= y_r + \Delta y_r \\ \text{or, } Ey_r &= (1 + \Delta) y_r \quad (\text{suppose}) \end{aligned} \quad (10.2.3)$$

Omitting y_r from both sides, we find that the operators E and Δ are connected by the symbolic relation

$$E = 1 + \Delta \quad (10.2.4)$$

This does not mean that 1 when added to Δ gives E , but that the operation by E is equivalent to the operation by $(1 + \Delta)$. It may be shown that the above relation follows certain algebraic rules.

As shown earlier, we have

$$\begin{aligned} \Delta y_0 &= y_1 - y_0 \\ \Delta^2 y_0 &= y_2 - 2y_1 + y_0 \\ \Delta^3 y_0 &= y_3 - 3y_2 + 3y_1 - y_0 \\ \Delta^4 y_0 &= y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 \end{aligned} \quad (10.2.5)$$

Alternatively, we may write

$$\begin{aligned} \Delta y_0 &= Ey_0 - y_0 \\ \Delta^2 y_0 &= E^2 y_0 - 2Ey_0 + y_0 \\ \Delta^3 y_0 &= E^3 y_0 - 3E^2 y_0 + 3Ey_0 - y_0 \\ \Delta^4 y_0 &= E^4 y_0 - 4E^3 y_0 + 6E^2 y_0 - 4Ey_0 + y_0 \end{aligned}$$

With the operators only (removing y_0 's from both sides)

$$\begin{aligned} \Delta &= E - 1 \\ \Delta^2 &= E^2 - 2E + 1 = (E - 1)^2, \\ \Delta^3 &= E^3 - 3E^2 + 3E - 1 = (E - 1)^3 \\ \Delta^4 &= E^4 - 4E^3 + 6E^2 - 4E + 1 = (E - 1)^4 \end{aligned}$$

Thus, we have developed a convenient method of expressing the finite difference of any order in terms of the entries.

(see Example 10.1)

Example 10.1 Express $\Delta^5 y_0$ in terms of $y_0, y_1, y_2, \dots$

Solution Using (10.2.4), we replace Δ by $E - 1$ and expand $(E - 1)^5$ by the binomial series (see pages 68-69).

$$\begin{aligned} \Delta^5 y_0 &= (E - 1)^5 y_0 \\ &= (E^5 - 5E^4 + 10E^3 - 10E^2 + 5E - 1) y_0 \quad [\text{Example 3.27 (iii)}] \\ &= E^5 y_0 - 5E^4 y_0 + 10E^3 y_0 - 10E^2 y_0 + 5E y_0 - y_0 \\ &= y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0. \end{aligned}$$



Example 10.2 If $y = c_0 + c_1 x + \frac{c_2}{2!} x(x - 1)$ passes through the points $(0, y_0), (1, y_1), (2, y_2)$ calculate c_0, c_1, c_2 in terms of differences of y^2 s.

[I.C.W.A., July '68]

Solution The statement that the curve passes through the points $(0, y_0), (1, y_1)$ and $(2, y_2)$ implies that when $x = 0, 1, 2$, the values of y are respectively y_0, y_1, y_2 .

Putting $x = 0$,

$$\begin{aligned} y_0 &= c_0 + c_1 \times 0 + \frac{c_2}{2!} \times 0(0 - 1) \\ \text{i.e. } y_0 &= c_0 \end{aligned} \quad \dots(i)$$



Putting $x = 1$,

$$y_1 = c_0 + c_1 \times 1 + \frac{c_2}{2!} \times 1(1-1)$$

or, $y_1 = y_0 + c_1 + 0$ (since $c_0 = y_0$)

i.e. $c_1 = y_1 - y_0 = \Delta y_0$... (ii)

Putting $x = 2$,

$$y_2 = c_0 + c_1 \times 2 + \frac{c_2}{2!} \times 2(2-1)$$

or, $y_2 = y_0 + 2(y_1 - y_0) + c_2$, from (i) and (ii)

i.e. $c_2 = y_2 - y_0 - 2(y_1 - y_0)$... (iii)

$$= y_2 - 2y_1 + y_0$$

$$= E^2 y_0 - 2E y_0 + y_0$$

$$= (E^2 - 2E + 1) y_0 = (E - 1)^2 y_0$$

$$= \Delta^2 y_0$$

Thus, $c_0 = y_0$, $c_1 = \Delta y_0$, $c_2 = \Delta^2 y_0$.



10.3 DIFFERENCES OF A POLYNOMIAL FUNCTION

Theorem: If y is a polynomial of the n -th degree [Example 3.29 (f), page 58], then the n -th differences are constants, and differences of higher order are all zero.

The third differences in Table 10.1 are found to be all equal and fourth differences are zero; because $y = \frac{1}{2}(x^3 + x)$ is a polynomial of the third degree.

Example 10.3 Given $u_1 = (12 - x)(4 + x)$, $u_2 = (5 - x)(4 - x)$, $u_3 = (x + 18)(x + 18)$ ($x + 6$), $u_4 = 94$, obtain a value of x such that second differences of u are constant.

[B.U., B.A. (Econ) '72]

Solution If the 2nd differences are constant, the 3rd difference (and all higher order differences) will be zero. In particular,

$$\Delta^3 u_1 = 0; \quad \text{i.e.} \quad (E - 1)^3 u_1 = 0$$

Expanding as in the previous example, and substituting the values

$$u_4 - 3 \cdot u_3 + 3 \cdot u_2 - u_1 = 0$$

or, $94 - 3(x + 18)(x + 6) + 3(5 - x)(4 - x) - (12 - x)(4 + x) = 0$.

Simplifying, $x^2 - 107x - 218 = 0$; or, $(x + 2)(x - 109) = 0$.

$\therefore x = -2$, or 109 .



Example 10.4 Estimate U_2 from the following table:

$x:$	1	2	3	4	5
$U_x:$	7	*	13	21	37

State the necessary assumptions made.

[I.C.W.A., June '77]

Solution Since only 4 values of the function are given, we assume that U_x is a 3rd degree polynomial in x . So the 4th differences may be regarded as zero; i.e. $\Delta^4 U_x = 0$. In particular, $\Delta^4 U_1 = 0$; or, $(E - 1)^4 U_1 = 0$;

or, $(E^4 - 4E^3 + 6E^2 - 4E + 1) U_1 = 0$



$$\text{or, } U_5 - 4U_4 + 6U_3 - 4U_2 + U_1 = 0$$

$$\text{or, } 37 - 4(21) + 6(13) - 4U_2 + 7 = 0$$

$$\text{or, } 38 - 4U_2 = 0. \quad \therefore U_2 = 9.5$$

[**Note:** It is always possible to pass a straight line, i.e. first degree polynomial, through any 2 given points; a parabola, i.e. second degree polynomial through any 3 given points; a third degree polynomial through any 4 given points; etc.]



Example 10.5 The monthly average number of deaths of infants (under age 1 year) in Calcutta are as shown below. Find estimates for the missing terms.

Year ...	1950	1951	1952	1953	1954	1955
Number of deaths (Monthly average)	939	?	907	841	?	773

Solution Since only 4 values are given, we assume a third degree polynomial, so that the fourth differences are zero. Denoting the entries corresponding to the years 1950, 1951, 1952 etc. by $y_0, y_1, y_2, \dots$, we have then

$$\Delta^4 y_0 = 0, \quad \Delta^4 y_1 = 0$$

$$\text{or, } (E-1)^4 y_0 = 0, \quad (E-1)^4 y_1 = 0$$

$$\text{Expanding, we get } y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$$

$$y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0$$

$$\text{Putting } y_0 = 939, y_2 = 907, y_3 = 841, y_5 = 773, \text{ we have}$$

$$y_4 - 4(841) + 6(907) - 4y_1 + 939 = 0$$

$$773 - 4y_4 + 6(841) - 4(907) + y_1 = 0$$

$$\text{or, } y_4 - 4y_1 = -3017 \quad \text{and} \quad y_1 - 4y_4 = -2191$$

$$\text{Solving these two equations, we get } y_1 = 951, y_4 = 785$$

Thus, the missing terms are 951 for 1951 and 785 for the year 1954.



Example 10.6 The values of a function $f(x)$ are given for certain values of x :

x	4	5	6	7	8
$f(x)$	3.11	2.96	2.85	—	2.70

Obtain the best approximation of $f(7)$.

[I.C.W.A., June '74-old]

Solution Since only 4 values are known, we assume a third degree polynomial for $f(x)$, so that 4th and higher order differences are zero. In particular $\Delta^4 f(4) = 0$;

$$\text{or, } (E-1)^4 f(4) = 0$$

$$\text{i.e. } f(8) - 4f(7) + 6f(6) - 4f(5) + f(4) = 0$$

$$\text{or, } 2.70 - 4f(7) + 6 \times 2.85 - 4 \times 2.96 + 3.11 = 0$$

$$f(7) = 2.7675.$$

[**Note:** This method can be applied only when, the values of the argument, including those for which the function is to be determined (e.g. 4, 5, 6, 7, 8) are found to have a common difference. The method will not work if, for example, it is required to find $f(5.5)$ or $f(7.3)$ from the above data. Lagrange's formula is however applicable in all cases; see Example 10.15]



Example 10.7 Show that third differences of a cubic function are constant.

[I.C.W.A., June '74; C.U., B.A. (Econ) '76]



Solution Let us write the cubic function as

$$y = f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

where a_0, a_1, a_2, a_3 are arbitrary constants. When the successive values of the argument x increase by a constant h , then the next higher value of the function y will be obtained on replacing x by $x + h$, i.e.

$$f(x + h) = a_0 + a_1(x + h) + a_2(x + h)^2 + a_3(x + h)^3$$

Subtracting $f(x)$ from the next value of the entry, viz. $f(x + h)$,

$$\begin{aligned}\Delta y &= \Delta f(x) = f(x + h) - f(x), \\ &= a_1\{(x + h) - x\} + a_2\{(x + h)^2 - x^2\} + a_3\{(x + h)^3 - x^3\} \\ &= a_1(x + h - x) + a_2(x^2 + 2xh + h^2 - x^2) + a_3(x^3 + 3x^2h + 3xh^2 + h^3 - x^3) \\ &= (a_1h + a_2h^2 + a_3h^3) + (2a_2h + 3a_3h^2)x + (3a_3h)x^2 \\ &= b_0 + b_1x + b_2x^2 \quad (\text{suppose})\end{aligned}$$

where b_0, b_1, b_2 are also constants. Applying Δ operator again,

$$\begin{aligned}\Delta^2 y &= \Delta^2 f(x) = \Delta[\Delta f(x)] \\ &= \Delta f(x + h) - \Delta f(x) \\ &= b_0 + b_1(x + h) + b_2(x + h)^2 - b_0 - b_1x - b_2x^2 \\ &= b_1(x + h - x) + b_2(x^2 + 2xh + h^2 - x^2) \\ &= (b_1h + b_2h^2) + (2b_2h)x \\ &= c_0 + c_1x \quad (\text{suppose})\end{aligned}$$

Repeating the operation by Δ ,

$$\begin{aligned}\Delta^3 y &= \Delta^3 f(x) = \Delta[\Delta^2 f(x)] = \Delta^2 f(x + h) - \Delta^2 f(x) \\ &= c_0 + c_1(x + h) - c_0 - c_1x = c_1h\end{aligned}$$

which is independent of x and hence a constant. This shows that the third differences of a cubic function are constants.



Example 10.8 Find the polynomial function $f(x)$ for which it is known that $f(2) = f(3) = 27$, $f(4) = 77$ and $f(5) = 165$.

Solution (First method) Since only 4 values of the function are known, we may assume that the polynomial is of 3rd degree.

$$f(x) = a + bx + cx^2 + dx^3$$

where a, b, c, d are certain constants to be determined. Putting $x = 2, 3, 4, 5$ successively,

$$\begin{aligned}f(2) &= a + b \times 2 + c \times 2^2 + d \times 2^3 \quad \text{i.e.} \quad 27 = a + 2b + 4c + 8d \\ f(3) &= a + b \times 3 + c \times 3^2 + d \times 3^3 \quad 27 = a + 3b + 9c + 27d \\ f(4) &= a + b \times 4 + c \times 4^2 + d \times 4^3 \quad 77 = a + 4b + 16c + 64d \\ f(5) &= a + b \times 5 + c \times 5^2 + d \times 5^3 \quad 165 = a + 5b + 25c + 125d\end{aligned}$$

Solving these 4 equations, we get the four constants, viz. $a = 225$, $b = -177$, $c = 43$, $d = -2$. Hence the required polynomial is

$$f(x) = 225 - 177x + 43x^2 - 2x^3$$

(Second method) The difficulty in solving 5 equations in the previous method can be avoided if we write the 3rd degree polynomial in the form

$$f(x) = A + B(x - 2) + C(x - 2)(x - 3) + D(x - 2)(x - 3)(x - 4)$$

where A, B, C, D are constants to be determined.

$$\text{Putting } x = 2, \quad f(2) = A + 0 + 0 + 0$$

$$\text{or,} \quad 27 = A, \quad \text{i.e.} \quad A = 27 \quad \dots(i)$$

$$\text{Putting } x = 3, \quad f(3) = A + B + 0 + 0$$

$$\text{or,} \quad 27 = 27 + B \quad (\text{since } A = 27) \quad \dots(ii)$$

$$\text{i.e.} \quad B = 0$$



Putting $x = 4$, $f(4) = A + 2B + 2C + 0$

or, $77 = 27 + 0 + 2C$ (since $A = 27, B = 0$)

i.e. $C = 25$

...(iii)

Putting $x = 5$, $f(5) = A + 3B + 6C + 6D$

or, $165 = 27 + 0 + 150 + 6D$

i.e. $D = -2$

...(iv)

Substituting the values of A, B, C, D , we get

$$f(x) = 27 + 25(x-2)(x-3) - 2(x-2)(x-3)(x-4)$$

or, $f(x) = 225 - 117x + 43x^2 - 2x^3$

(Third method) Since the function is given for equidistant values of x (viz. 2, 3, 4, 5), we can construct a difference table:

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	27			
3	27	0		
4	77	50	50	
5	165	88	38	-12

$$f(x) = E^{x-2} f(2) = (1 + \Delta)^{x-2} f(2)$$

$$= [1 + (x-2)\Delta + \frac{(x-2)(x-3)}{1 \times 2} \Delta^2 + \frac{(x-2)(x-3)(x-4)}{1 \times 2 \times 3} \Delta^3] f(2)$$

$$= f(2) + (x-2) \cdot \Delta f(2) + \frac{(x-2)(x-3)}{2} \Delta^2 f(2) + \frac{(x-2)(x-3)(x-4)}{6} \Delta^3 f(2).$$

$$= 27 + 0 + 25(x-2)(x-3) - 2(x-2)(x-3)(x-4)$$

This is of the same form as obtained in the 2nd method [Note that it is unnecessary to continue the expansion beyond 3rd differences, since higher order differences are zero.]



Example 10.9 Given for a function $f(x)$,

$$f(0) = 1; f(1) + f(2) = 10; f(3) + f(4) + f(5) = 65; \text{ find } f(4). \quad [\text{I.C.W.A., Dec. '73}]$$

Solution Since only 3 values are known, we may assume a 2nd degree polynomial for the function, and write $f(x) = a + bx + cx^2$.

Putting $x = 0$, $f(0) = a + b \times 0 + c \times 0^2 = a$

Putting $x = 1$ and 2, and adding

$$f(1) + f(2) = (a + b + c) + (a + 2b + 4c) = 2a + 3b + 5c$$

Putting $x = 3, 4, 5$, and adding

$$\begin{aligned} f(3) + f(4) + f(5) &= (a + 3b + 9c) + (a + 4b + 16c) + (a + 5b + 25c) \\ &= 3a + 12b + 50c \end{aligned}$$

Thus, from the given results, we have

$$a = 1; 2a + 3b + 5c = 10; 3a + 12b + 50c = 65.$$

Solving these equations we get $a = 1, b = 1, c = 1$. Hence $f(x) = 1 + x + x^2$, so that $f(4) = 1 + 4 + 4^2 = 21$.



Example 10.10 If u is any function of x , and given that

$$u_0 + u_8 = 56 \quad u_2 + u_6 = 32$$

$$u_1 + u_7 = 42 \quad u_3 + u_5 = 26$$

find the value of u_4 .



Solution Since only 4 values are available, we assume a 3rd degree polynomial for u , so that the 4th and higher order differences are zero. In particular, $\Delta^4 u_0 = 0$; or, $(E - 1)^4 u_0 = 0$. Expanding,

$$\begin{aligned} u_8 - 8u_7 + 28u_6 - 56u_5 + 70u_4 - 56u_3 + 28u_2 - 8u_1 + u_0 &= 0 \\ \text{or, } (u_8 + u_0) - 8(u_7 + u_1) + 28(u_6 + u_2) - 56(u_5 + u_3) + 70u_4 &= 0 \\ \text{Substituting the values and solving, we get } u_4 &= 12. \end{aligned}$$



10.4 NEWTON'S FORWARD INTERPOLATION FORMULA

Let $y_0, y_1, y_2, \dots, y_n$ be some tabulated values of a function $y = f(x)$ corresponding to the equidistant values $x = x_0, x_1, x_2, \dots, x_n$.

$$x_1 - x_0 = x_2 - x_1 = x_3 - x_2 = \dots = x_n - x_{n-1} = h \text{ (say).}$$

It is required to find the value of y corresponding to an intermediate value of x lying near the beginning of the tabulated values. This is given by *Newton's Forward Interpolation Formula*:

$$\begin{aligned} y = y_0 + u\Delta y_0 + \frac{u(u-1)}{1 \times 2} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{1 \times 2 \times 3} \Delta^3 y_0 + \dots \\ \dots + \frac{u(u-1)(u-2) \dots (u-n+1)}{1 \times 2 \times 3 \times \dots \times n} \Delta^n y_0. \end{aligned} \quad (10.4.1)$$

where $u = \frac{x - x_0}{h}$

Derivation of Newton's Forward Formula

All interpolation formulae are based on the following principle. Within the range of $n + 1$ given values, the tabulated function $y = f(x)$ can be replaced by an n -th degree polynomial $\phi(x)$, which coincides with $f(x)$ when $x = x_0, x_1, x_2, \dots, x_n$; i.e.

$$\phi(x_0) = y_0, \phi(x_1) = y_1, \phi(x_2) = y_2, \dots, \phi(x_n) = y_n \quad \dots(1)$$

Let us write the n -th degree polynomial $\phi(x)$ in the form

$$\begin{aligned} \phi(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + a_3(x - x_0)(x - x_1)(x - x_2) \\ + \dots + a_n(x - x_0)(x - x_1) \dots (x - x_{n-1}). \end{aligned} \quad \dots(2)$$

The $n + 1$ constants $a_0, a_1, a_2, \dots, a_n$ are so determined as to satisfy the $n + 1$ relations (1).

Putting $x = x_0$ in (2), we have

$$\phi(x_0) = a_0$$

because all other terms will contain $(x_0 - x_0)$ and therefore vanish.

or, $y_0 = a_0$; i.e. $a_0 = y_0$

Now putting $x = x_1$ in (2)

$$\phi(x_1) = a_0 + a_1(x_1 - x_0) \text{ because the other terms vanish.}$$

or, $y_1 = y_0 + a_1h$, since $x_1 - x_0 = h$ (given)

or, $a_1h = y_1 - y_0$

$$\therefore a_1 = \frac{y_1 - y_0}{h} = \frac{\Delta y_0}{h}$$

Again, putting $x = x_2$ in (2),



$$\phi(x_2) = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1)$$

$$\text{or, } y_2 = y_0 + \frac{y_1 - y_0}{h} (2h) + a_2(2h^2) (h)$$

$$\text{or, } y_2 = y_0 + 2(y_1 - y_0) + a_2(2h^2)$$

Solving, we get

$$a_2 = \frac{y_2 - 2y_1 + y_0}{2h^2} = \frac{\Delta^2 y_0}{2h^2}$$

Now putting $x = x_3$ in (2),

$$\phi(x_3) = a_0 + a_1(x_3 - x_0) + a_2(x_3 - x_0)(x_3 - x_1) + a_3(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)$$

$$\text{or, } y_3 = y_0 + \frac{y_1 - y_0}{h} (3h) + \frac{y_2 - 2y_1 + y_0}{2h^2} (3h)(2h) + a_3(3h)(2h)(h)$$

$$\text{or, } y_3 = y_0 + 3(y_1 - y_0) + 3(y_2 - 2y_1 + y_0) + a_3(1 \times 2 \times 3) h^3$$

$$\text{Solving, } a_3 = \frac{y_3 - 3y_2 + 3y_1 - y_0}{(1 \times 2 \times 3)h^3} = \frac{\Delta^3 y_0}{(1 \times 2 \times 3)h^3}$$

Similarly, putting $x = x_4, \dots, x_n$ successively in (2), we find that

$$a_4 = \frac{\Delta^4 y_0}{(1 \times 2 \times 3 \times 4)h^4}, \dots a_n = \frac{\Delta^n y_0}{(1 \times 2 \times \dots \times n) h^n}$$

Substituting the values of $a_0, a_1, a_2, \dots, a_n$ in (2),

$$\begin{aligned} y = \phi(x) &= y_0 + \frac{\Delta y_0}{h} (x - x_0) + \frac{\Delta^2 y_0}{(1 \times 2)h^2} (x - x_0)(x - x_1) \\ &+ \frac{\Delta^3 y_0}{(1 \times 2 \times 3)h^3} (x - x_0)(x - x_1)(x - x_2) + \dots \\ &+ \frac{\Delta^n y_0}{(1 \times 2 \times \dots \times n)h^n} (x - x_0)(x - x_1) \dots (x - x_{n-1}) \quad \dots(3) \end{aligned}$$

If we write $\frac{x - x_0}{h} = u$, i.e. $x = x_0 + hu$, then

$$x - x_0 = hu,$$

$$x - x_1 = (x_0 + hu) - (x_0 + h) = h(u - 1),$$

$$x - x_2 = (x_0 + hu) - (x_0 + 2h) = h(u - 2),$$

$$\dots \dots \dots$$

$$x - x_{n-1} = (x_0 + hu) - \{x_0 + (n - 1)h\} = h(u - n + 1)$$

Substituting these values in (3), we have

$$\begin{aligned} y &= y_0 + \Delta y_0 \cdot u + \frac{\Delta^2 y_0}{1 \times 2} u(u - 1) + \frac{\Delta^3 y_0}{1 \times 2 \times 3} u(u - 1)(u - 2) \\ &+ \dots \dots + \frac{\Delta^n y_0}{1 \times 2 \times \dots \times n} u(u - 1)(u - 2) \dots (u - n + 1). \end{aligned}$$

Rearranging the terms on the right, we get Newton's forward interpolation formula (10.4.1).



Uses—Newton's forward interpolation formula can be used only when the values of the argument are equidistant. Since the formula contains only the leading term y_0 and the leading differences Δy_0 , $\Delta^2 y_0$, etc., it is most suitable for interpolation near the beginning of the tabulated values. It may also be used to find the value of the function corresponding to values slightly before the first tabulated value of argument (The process of finding the value of the function outside the range of tabulated values is known as *Extrapolation*).

Example 10.11 From the following table, find by interpolation the value of $f(x)$ for $x = 24$:

x	20	25	30	35	40
$f(x)$	30.5	34.5	40	47.75	59.25

Solution Since the given values of x are equidistant and the value $x = 24$ lies near the beginning of these values, we use Newton's forward interpolation formula (10.4.1).

Table 10.3 Difference Table

x	$y = f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
20	30.5				
		4.0			
25	34.5		1.5		
		5.5		0.75	
30	40		2.25		0.75
		7.75		1.50	
35	47.75		3.75		
		11.50			
40	59.25				

Hence, $u = \frac{x - x_0}{h} = \frac{24 - 20}{5} = 0.8$. Applying (10.4.1),

$$\begin{aligned}
 y &= 30.5 + 0.8(4.0) + \frac{0.8(0.8 - 1)}{2} (1.5) + \frac{0.8(0.8 - 1)(0.8 - 2)}{6} (0.75) \\
 &\quad + \frac{0.8(0.8 - 1)(0.8 - 2)(0.8 - 3)}{24} (0.75) \\
 &= 30.5 + 3.2 - 0.12 + 0.024 - 0.0132 \\
 &= 33.5908 = 33.59
 \end{aligned}$$

Ans. 33.59



Example 10.12 Given the following table, find $(3.24)^3$:

x	31	32	33	34	35
x^3	29791	32768	35937	39304	42875

Solution Since the successive values of x have a common difference, we shall use Newton's forward interpolation formula and find $y = (32.4)^3$, from which the value of $(3.24)^3$ can be easily obtained by shifting the decimal point three places to the left.

**Table 10.4** Difference Table

x	$y = x^3$	Δy	$\Delta^2 y$	$\Delta^3 y$
31	29,791			
32	32,768	2,977		
33	35,937	3,169	192	6
34	39,304	3,367	198	6
35	42,875	3,571	204	

Here $u = (x - x_0)/h = (32.4 - 31)/1 = 1.4$. Using (10.4.1),

$$y = (32.4)^3 = 29791 + 1.4(2977) + \frac{1.4(1.4-1)}{2} (192) + \frac{1.4(1.4-1)(1.4-2)}{6} (6) \quad (6)$$

$$= 29791 + 4167.8 + 53.76 - 0.336 = 34012.224$$

Hence, $(3.24)^3 = 34.012,224$. This result is exact.



10.5 NEWTON'S BACKWARD INTERPOLATION FORMULA

As in Section 10.4, let $x_0, x_1, x_2, \dots, x_n$ be some equidistant values of the argument and $y_0, y_1, y_2, \dots, y_n$ the corresponding entries. It is required to find the value of y corresponding to a specified value of x lying near the end of the tabulated values. This is given by *Newton's Background Interpolation Formula*:

$$y = y_n + v\Delta y_{n-1} + \frac{v(v+1)}{1 \times 2} \Delta^2 y_{n-2} + \frac{v(v+1)(v+2)}{1 \times 2 \times 3} \Delta^3 y_{n-3} + \dots$$

$$\dots + \frac{v(v+1)(v+2) \dots (v+n-1)}{1 \times 2 \times 3 \times \dots \times n} \Delta^n y_0. \quad (10.5.1)$$

where $v = \frac{x - x_n}{h}$.

Derivation of Newton's backward formula

As before, within the range of the given values, the tabulated function $y = f(x)$ is replaced by an n -th degree polynomial $\phi(x)$, which coincides with $f(x)$ when $x = x_0, x_1, \dots, x_n$; i.e.

$$\phi(x_0) = y_0, \phi(x_1) = y_1, \phi(x_2) = y_2, \dots, \phi(x_n) = y_n \quad \dots(1)$$

We write the n -th degree polynomial $\phi(x)$ in the form

$$\phi(x) = a_0 + a_1(x - x_n) + a_2(x - x_n)(x - x_{n-1}) + a_3(x - x_n)(x - x_{n-1})(x - x_{n-2})$$

$$+ \dots + a_n(x - x_n)(x - x_{n-1}) \dots (x - x_1). \quad \dots(2)$$

The $n + 1$ constants $a_0, a_1, a_2, \dots, a_n$ are so determined as to satisfy the $n + 1$ relations (1).

Putting $x = x_n$ in (2), we have

$$\phi(x_n) = a_0$$

because all other terms contain the factor $(x_n - x_n)$ and therefore vanish.

or, $y_n = a_0$; i.e. $a_0 = y_n$



Now putting $x = x_{n-1}$ in (2),

$$\phi(x_{n-1}) = a_0 + a_1(x_{n-1} - x_n), \text{ because the other terms vanish;}$$

$$\text{or, } y_{n-1} = y_n + a_1(-h), \text{ since } x_n - x_{n-1} = h$$

$$\text{or, } a_1 h = y_n - y_{n-1} = \Delta y_{n-1}$$

$$\therefore a_1 = \frac{y_n - y_{n-1}}{h} = \frac{\Delta y_{n-1}}{h}$$

Again, putting $x = x_{n-2}$ in (2),

$$\phi(x_{n-2}) = a_0 + a_1(x_{n-2} - x_n) + a_2(x_{n-2} - x_n)(x_{n-2} - x_{n-1})$$

$$\text{or, } y_{n-2} = y_n + \frac{y_n - y_{n-1}}{h}(-2h) + a_2(-2h)(-h)$$

$$\text{or, } y_{n-2} = y_n - 2(y_n - y_{n-1}) + a_2(2h^2)$$

$$\text{Solving, } a_2 = \frac{y_n - 2y_{n-1} + y_{n-2}}{2h^2} = \frac{\Delta^2 y_{n-2}}{2h^2}$$

Similarly, putting $x = x_{n-3}, x_{n-4}, \dots, x_0$ successively in (2), we get

$$a_3 = \frac{\Delta^3 y_{n-3}}{(1 \times 2 \times 3)h^3},$$

$$a_4 = \frac{\Delta^4 y_{n-4}}{(1 \times 2 \times 3 \times 4)h^4}, \dots a_n = \frac{\Delta^n y_0}{(1 \times 2 \times \dots \times n)h^n}$$

Substituting the values of $a_0, a_1, a_2, \dots, a_n$ in (2),

$$\begin{aligned} y &= y_n + \frac{\Delta y_{n-1}}{h} (x - x_n) + \frac{\Delta^2 y_{n-2}}{(1 \times 2)h^2} (x - x_n)(x - x_{n-1}) \\ &+ \frac{\Delta^3 y_{n-3}}{(1 \times 2 \times 3)h^3} (x - x_n)(x - x_{n-1})(x - x_{n-2}) + \dots \\ &\dots + \frac{\Delta^n y_0}{(1 \times 2 \times \dots \times n)h^n} (x - x_n)(x - x_{n-1}) \dots (x - x_1) \end{aligned} \quad \dots(3)$$

If we write $\frac{x - x_n}{h} = v$, i.e. $x = x_n + hv$, then

$$x - x_n = hv$$

$$x - x_{n-1} = (x_n + hv) - (x_n - h) = h(v + 1)$$

$$x - x_{n-2} = (x_n + hv) - (x_n - 2h) = h(v + 2)$$

...

$$x - x_1 = (x_n + hv) - \{x_n - (n - 1)h\} = h(v + n - 1)$$

When these values are substituted in (3), we have

$$\begin{aligned} y &= y_n + \Delta y_{n-1} v + \frac{\Delta^2 y_{n-2}}{1 \times 2} v(v + 1) + \frac{\Delta^3 y_{n-3}}{1 \times 2 \times 3} v(v + 1)(v + 2) \\ &+ \dots + \frac{\Delta^n y_0}{1 \times 2 \times \dots \times n} v(v + 1)(v + 2) \dots (v + n - 1). \end{aligned}$$



Rearranging the terms on the right, we get Newton's backward interpolation formula (10.5.1).

Uses—Newton's backward interpolation formula can be used only when the values of the argument are equidistant. Since the formula contains only the terms at the bottom of the difference table (Table 10.2), it is most suitable for interpolation near the end of the tabulated values. It may also be used to extrapolate the value of the function corresponding to an argument slightly beyond the last tabulated value.

[**Note:** (i) Usually, each term in formulae (10.4.1) and (10.5.1) is less than the preceding. Hence, in numerical problems it is not necessary to calculate all the terms given in the formula. Only the first few terms are sufficient to obtain the required estimate of y to the same degree of accuracy as the given entries.

(ii) If all the terms in formulae (10.4.1) and (10.5.1) are used, Newton's forward formula and backward formula would give exactly the same result.]

Example 10.13 Given the following table, estimate y when $x = 0.35$, by using Newton's backward interpolation formula.

x :	0	0.1	0.2	0.3	0.4
y :	1	1.095	1.179	1.251	1.310

(Give answer correct to 3 decimal places).

[I.C.W.A., June '78]

Solution (For simplicity, we shall ignore the decimal point while computing the differences).

Table 10.5 Difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	1	95			
0.1	1.095	84	– 11		
0.2	1.179	72	– 12	– 1	
0.3	1.251	59	– 13	– 1	0
0.4	1.310				

Here $v = (x - x_n)/h = (0.35 - 0.4)/0.1 = -0.5$

Using 3 decimals,

$$\begin{aligned}
 y &= 1310 + (-0.5) \times 59 + \frac{(-0.5)(-0.5+1)}{2} \times (-13) \\
 &\quad + \frac{(-0.5)(-0.5+1)(-0.5+2)}{6} \times (-1) \\
 &= 1310 - 29.5 + 1.6 + 0.1 = 1282.2 \\
 &= 1282
 \end{aligned}$$

Adjusting the decimal point, the required value is 1.282.





10.6 CENTRAL DIFFERENCE FORMULAE-STIRLING'S AND BESSEL'S

When the successive values of the argument are equidistant, Newton's forward and backward formulae are generally applied in all cases of interpolation. However, for interpolation near the middle of the set of tabulated values, *Central Difference* formulae, which utilize differences near the central part of the difference table, are found more useful; because the successive terms converge more rapidly than in Newton's forward and backward formulae.

Let the function $y = f(x)$ be tabulated for some equidistant values of the argument

$$(x_0 - nh), \dots (x_0 - 2h), (x_0 - h), x_0, (x_0 + h), (x_0 + 2h), \dots (x_0 + nh)$$

the common difference being h ; and let the corresponding entries be denoted by

$$y_{-n}, \dots y_{-2}, y_{-1}, y_0, y_1, y_2, \dots y_n.$$

It is required to interpolate the value of the function $y = f(x)$ for a value of x in the interval $(x_0 - h)$ to $(x_0 + h)$.

1. Stirling's Interpolation Formula

$$y = y_0 + u \cdot \frac{(\Delta y_0 + \Delta y_{-1})}{2} + \frac{u^2}{2!} \cdot \Delta^2 y_{-1} + \frac{u(u^2 - 1^2)}{3!} \cdot \frac{(\Delta^3 y_{-1} + \Delta^3 y_{-2})}{2} \\ + \frac{u^2(u^2 - 1^2)}{4!} \cdot \Delta^4 y_{-2} + \frac{u(u^2 - 1^2)(u^2 - 2^2)(\Delta^5 y_{-2} + \Delta^5 y_{-3})}{5!} + \dots \quad (10.6.1)$$

where $u = (x - x_0)/h$.

Stirling's formula is appropriate when the value of u lies in the interval -0.25 to $+0.25$. It uses y_0 and even order differences on the horizontal line through y_0 , but the average of odd order differences above and below that line.

2. Bessel's Interpolation Formula

$$y = \frac{(y_0 + y_1)}{2} + v \cdot \Delta y_0 + \frac{\left(v^2 - \frac{1}{4}\right)}{2!} \cdot \frac{(\Delta^2 y_0 + \Delta^2 y_{-1})}{2} + \frac{v\left(v^2 - \frac{1}{4}\right)}{3!} \cdot \Delta^3 y_{-1} \\ + \frac{\left(v^2 - \frac{1}{4}\right)\left(v^2 - \frac{9}{4}\right)}{4!} \cdot \frac{(\Delta^4 y_{-1} + \Delta^4 y_{-2})}{2} + \frac{v\left(v^2 - \frac{1}{4}\right)\left(v^2 - \frac{9}{4}\right)}{5!} \cdot \Delta^5 y_{-2} + \dots \quad (10.6.2)$$

where $u = (x - x_0)/h$ and $v = u - \frac{1}{2}$.

Bessel's formula is appropriate for values of u lying in the interval $+0.25$ to $+0.75$, i.e. for interpolation in the middle-half of two tabulated values. It uses the average of y_0 and y_1 , and odd order differences on the horizontal line mid-way between them, but the average of even order differences above and below that line.

Bessel's formula is especially found useful for interpolating the value of the function *exactly* at the middle of two given values, i.e. when $u = \frac{1}{2}$ so that $v = 0$. Every alternate term in Bessel's formula (10.6.2) then vanish. This special case of Bessel's formula is known as "*Formula for Interpolating to Halves*".



Example 10.14 Mantissae of 4 numbers are given below:

Numbers	4,200	4,210	4,220	4,230
Mantissae	6,232,493	6,242,821,	6,253,125,	6,263,404

Find the mantissa of the logarithm of 4,213.

Solution Since the values of the argument are equidistant, and 4213 lies near the middle of the tabulated values, we use a *central difference formula*. Let us take $x_0 = 4210$; then

$$u = (x - x_0)/h = (4213 - 4210)/10 = 0.3$$

Bessel's interpolation formula will be appropriate here.

Table 10.6 Difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^2 y$
$x_{-1} = 4200$	$y_{-1} = 6,232,493$			
$x_0 = 4210$	$y_0 = 6,242,821$	$\Delta y_{-1} = 10,328$		
$x_1 = 4220$	$y_1 = 6,253,125$	$\Delta y_0 = 10,304$	$\Delta^2 y_{-1} = -24$	
$x_2 = 4230$	$y_2 = 6,263,404$	$\Delta y_1 = 10,279$	$\Delta^2 y_0 = -25$	$\Delta^2 y_{-1} = -1$

Here, $v = u - \frac{1}{2} = 0.3 - 0.5 = -0.2$. Substituting the values from Table 10.7 in formula (10.6.2),

$$\begin{aligned}
 y &= \frac{(6,242,821 + 6,253,125)}{2} + (-0.2)(10,304) \\
 &\quad + \frac{(0.04 - 0.25)}{2} \cdot \frac{(-24 - 25)}{2} + \frac{(-0.2)(0.04 - 0.25)}{6} (-1) \\
 &= 6,247,973 - 2,060.8 + \frac{(-0.21)(-49)}{4} + \frac{(-0.2)(-0.21)(-1)}{6} \\
 &= 6,247,973 - 2,060.8 + 2.6 - 0.0 \\
 &= 6,245,914.8 \\
 &= 6,245,915.
 \end{aligned}$$

This result is correct to the last place.



10.7 LAGRANGE'S INTERPOLATION FORMULA

Let $y_0, y_1, y_2, \dots, y_n$ denote the tabulated values of a function $y = f(x)$ corresponding to the values of the argument $x_0, x_1, x_2, \dots, x_n$ (which may *not* be equidistant). It is required to find the value of y corresponding to a specified value of x lying in between the given values. This is obtained by using Lagrange's Interpolation Formula:

$$\begin{aligned}
 y &= \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 + \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1 \\
 &\quad + \dots \dots \dots \dots \\
 &\quad + \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} y_n
 \end{aligned} \tag{10.7.1}$$



Derivation of Lagrange's formula

As before, the tabulated function $y = f(x)$ is replaced by an n -th degree polynomial $\phi(x)$, which coincides with $f(x)$ when $x = x_0, x_1, \dots, x_n$; i.e.

$$\phi(x_0) = y_0, \phi(x_1) = y_1, \phi(x_2) = y_2, \dots, \phi(x_n) = y_n \quad \dots(1)$$

We write the n -th degree polynomial $\phi(x)$ in the form

$$\begin{aligned} \phi(x) = & a_0(x-x_1)(x-x_2)(x-x_3)\dots(x-x_n) \\ & + a_1(x-x_0)(x-x_2)(x-x_3)\dots(x-x_n) \\ & + a_2(x-x_0)(x-x_1)(x-x_3)\dots(x-x_n) \\ & + \dots \dots \dots \\ & + a_n(x-x_0)(x-x_1)(x-x_2)\dots(x-x_{n-1}). \end{aligned} \quad \dots(2)$$

[Note that each term on the right is a polynomial of degree n ; $(x-x_0)$ is absent from the 1st term, $(x-x_1)$ from the 2nd term, and so on, finally $(x-x_n)$ is absent from the last term]. Then $n+1$ constants $a_0, a_1, a_2, \dots, a_n$ are so determined as to satisfy the $n+1$ relations (1).

Putting $x = x_0$ in (2), we have

$$y_0 = a_0(x_0-x_1)(x_0-x_2)(x_0-x_3)\dots(x_0-x_n),$$

because all other terms contain the factor (x_0-x_0) and vanish identically. Hence

$$a_0 = \frac{y_0}{(x_0-x_1)(x_0-x_2)(x_0-x_3)\dots(x_0-x_n)}$$

Similarly, putting $x = x_1, x_2, \dots, x_n$ successively in (2), we find that

$$a_1 = \frac{y_1}{(x_1-x_0)(x_1-x_2)(x_1-x_3)\dots(x_1-x_n)}$$

$$a_2 = \frac{y_2}{(x_2-x_0)(x_2-x_1)(x_2-x_3)\dots(x_2-x_n)}$$

$$\dots \quad \dots \quad \dots$$

$$a_n = \frac{y_n}{(x_n-x_0)(x_n-x_1)(x_n-x_2)\dots(x_n-x_{n-1})}$$

Substituting in the values of $a_0, a_1, a_2, \dots, a_n$ in (2), we get Lagrange's interpolation formula (10.7.1).

Uses—Lagrange's interpolation formula is generally used when the values of the argument are not equidistant, though it can be used for equidistant arguments also. It is applicable in any part of the tabulated values, and does not require the construction of a difference table. Lagrange's formula may also be used to find the value of the argument for a given value of the function, i.e. to find the value of x for a given value of y (This process is known as '*Inverse Interpolation*'; page 335).

[**Note:** (i) Lagrange's formula may be used in all cases, whether the values of x differ by a constant or not; or whether the value of x , corresponding to which the value of y is to be found, lies near the beginning or end or middle of the tabulated values. However, if the given values of x are equidistant, it is preferable to use Newton's Forward or Backward Formula, because the application of Lagrange's formula would involve more laborious computations.

(ii) When the values of the argument are equidistant, Lagrange's formula may be reduced to Newton's Forward or Backward formula.]



Example 10.15 The values of a function $f(x)$ are given for certain values of x :

x	4	5	6	8
$f(x)$	3.11	2.96	2.85	2.70

Obtain the best approximation of $f(7)$.

[I.C.W.A., June '74-old]

Solution Since the successive values of x for the tabulated function $f(x)$ are not equidistant, Newton's forward or backward formula cannot be applied. Applying Lagrange's formula,

$$\begin{aligned}
 y = f(x) &= \frac{(7-5)(7-6)(7-8)}{(4-5)(4-6)(4-8)} \times 3.11 + \frac{(7-4)(7-6)(7-8)}{(5-4)(5-6)(5-8)} \times 2.96 \\
 &+ \frac{(7-4)(7-5)(7-8)}{(6-4)(6-5)(6-8)} \times 2.85 + \frac{(7-4)(7-5)(7-6)}{(8-4)(8-5)(8-6)} \times 2.70 \\
 &= \frac{2 \times 1 \times (-1)}{(-1)(-2)(-4)} \times 3.11 + \frac{3 \times 1 \times (-1)}{1 \times (-1)(-3)} \times 2.96 \\
 &+ \frac{3 \times 3 \times (-1)}{2 \times 1 \times (-2)} \times 2.85 + \frac{3 \times 2 \times 1}{4 \times 3 \times 2} \times 2.70 \\
 &= \frac{1}{4} \times (3.11) + (-1) \times (2.96) + \frac{3}{2} \times (2.85) + \frac{1}{4} \times (2.70) \\
 &= 0.7775 - 2.9600 + 4.2750 + 0.6750 \\
 &= 2.7675 \text{ (see Example 10.6)}
 \end{aligned}$$

[**Note:** In numerical calculations with Lagrange's formula, the sum of the coefficients of the values of y is always 1. Here

$$\frac{1}{4} + (-1) + \frac{3}{2} + \frac{1}{4} = 1.$$

This gives a check on the accuracy of calculations]



10.8 INVERSE INTERPOLATION

Given a set of tabulated values of a function $y = f(x)$ corresponding to some values of the argument x , the process of finding the value of the argument for an intermediate value of the function is called '*Inverse Interpolation*'.

Let $y_0, y_1, \dots, y_n$ be a set of tabulated values of a function $y = f(x)$ corresponding to some given values (usually equidistant) of the argument $x = x_0, x_1, \dots, x_n$. In some cases it may be required to determine the value of x for another value of y lying in between $y_0, y_1, \dots, y_n$. The process of finding such a value of x is known as 'inverse interpolation'. If y is a function of x , say $y = f(x)$, in many cases it is possible to treat x also as a function of y , say $x = g(y)$. Since Lagrange's interpolation formula is applicable for unequal intervals, we can use this formula for inverse interpolation, interchanging the role of the argument x and the function y .

$$\begin{aligned}
 x &= \frac{(y - y_1)(y - y_2)(y - y_3) \dots (y - y_n)}{(y_0 - y_1)(y_0 - y_2)(y_0 - y_3) \dots (y_0 - y_n)} x_0 \\
 &+ \frac{(y - y_0)(y - y_2)(y - y_3) \dots (y - y_n)}{(y_1 - y_0)(y_1 - y_2)(y_1 - y_3) \dots (y_1 - y_n)} x_1
 \end{aligned}$$

$$\begin{aligned}
& + \frac{(y - y_0)(y - y_1)(y - y_3) \dots (y - y_n)}{(y_2 - y_0)(y_2 - y_1)(y_2 - y_3) \dots (y_2 - y_n)} x_2 \\
& + \dots \dots \dots \\
& + \frac{(y - y_0)(y - y_1)(y - y_2) \dots (y - y_{n-1})}{(y_n - y_0)(y_n - y_1)(y_n - y_2) \dots (y_n - y_{n-1})} x_n
\end{aligned} \tag{10.8.1}$$

Example 10.16 The values of a function $f(x)$ are given for certain values of x :

x	4	5	6	8
$f(x)$	3.11	2.96	2.85	2.70

Find the value of x for which $f(x) = 3.00$.

Solution Applying the formula (10.8.1),

$$\begin{aligned} x &= \frac{(3.00 - 2.96)(3.00 - 2.85)(3.00 - 2.70)}{(3.11 - 2.96)(3.11 - 2.85)(3.11 - 2.70)} \times 4 \\ &\quad + \frac{(3.00 - 3.11)(3.00 - 2.85)(3.00 - 2.70)}{(2.96 - 3.11)(2.96 - 2.85)(2.96 - 2.70)} \times 5 \\ &\quad + \frac{(3.00 - 3.11)(3.00 - 2.96)(3.00 - 2.70)}{(2.85 - 3.11)(2.85 - 2.96)(2.85 - 2.70)} \times 6 \\ &\quad + \frac{(3.00 - 3.11)(3.00 - 2.96)(3.00 - 2.85)}{(2.70 - 3.11)(2.70 - 2.96)(2.70 - 2.85)} \times 8 \\ &= \frac{.04 \times .15 \times .30}{.15 \times .26 \times .41} \times 4 + \frac{(-.11) \times .15 \times .30}{(-.15) \times .11 \times .26} \times 5 \\ &\quad + \frac{(-.11) \times .04 \times .30}{(-.26) (-.11) (.15)} \times 6 + \frac{(-.11) (.04) (.15)}{(-.41) (-.26) (-.15)} \times 8 \\ &= \frac{60}{533} \times (4) + \frac{15}{13} \times (5) + \frac{-4}{13} \times (6) + \frac{22}{533} \times (8) \\ &\qquad\qquad\qquad [\text{Check: } \frac{60}{533} + \frac{15}{13} + \frac{(-4)}{13} + \frac{22}{533} = 1] \\ &= 0.450 + 5.769 - 1.846 + 0.330 = 4.70 \end{aligned}$$



Example 10.17 The mode of a certain frequency curve $y = f(x)$ is attained at $x = 9.1$ and the value of the frequency function $f(x)$ for $x = 8.9, 9.0$ and 9.3 are respectively equal to $0.30, 0.35$ and 0.25 . Calculate the approximate value of $f(x)$ at the mode.

[I.C.W.A., Dec. '78]

Solution We are given the following information:

x	8.9	9.0	9.3
$y = f(x)$	0.30	0.35	0.25

It is required to find the value of y when $x = 9.1$. Using Lagrange's interpolation formula (10.7.1),



$$\begin{aligned}
 y &= \frac{(9.1 - 9.0)(9.1 - 9.3)}{(8.9 - 9.0)(8.9 - 9.3)} \times 0.30 + \frac{(9.1 - 8.9)(9.1 - 9.3)}{(9.0 - 8.9)(9.0 - 9.3)} \times 0.35 \\
 &\quad + \frac{(9.1 - 8.9)(9.1 - 9.0)}{(9.3 - 8.9)(9.3 - 9.0)} \times 0.25 \\
 &= \frac{0.1(-0.2)}{(-0.1)(-0.4)} \times 0.30 + \frac{0.2(-0.2)}{0.1(-0.3)} \times 0.35 + \frac{0.2 \times 0.1}{0.4 \times 0.3} \times 0.25 \\
 &= \left(\frac{-1}{2} \right) \times 0.30 + \left(\frac{4}{3} \right) \times 0.35 + \left(\frac{1}{6} \right) \times 0.25 \\
 &= \frac{1}{6} \times (-0.90 + 2.80 + 0.25) = 0.358
 \end{aligned}$$

Note that this result is larger than any of the given values of y , because $y = f(x)$ is the maximum at the Mode.



Example 10.18 Using three pairs of observations (x_0, y_0) , (x_1, y_1) , (x_2, y_2) , show that Lagrange's interpolation formula reduces to Newton's forward formula, when x_0, x_1, x_2 are equidistant.

Solution Using Lagrange's formula (10.7.1), we have

$$y = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

Here, since $x_1 - x_0 = x_2 - x_1 = h$, therefore

$$x_1 = x_0 + h; \quad x_2 = x_0 + 2h$$

Also, writing $u = (x - x_0)/h$, we have $x = x_0 + hu$

$$\begin{aligned}
 \therefore \quad x - x_0 &= (x_0 + hu) - x_0 = hu \\
 x - x_1 &= (x_0 + hu) - (x_0 + h) = h(u - 1) \\
 x - x_2 &= (x_0 + hu) - (x_0 + 2h) = h(u - 2)
 \end{aligned}$$

Substituting these values,

$$\begin{aligned}
 y &= \frac{h(u-1) \cdot h(u-2)}{(-h)(-2h)} y_0 + \frac{hu \cdot h(u-2)}{(h)(-h)} y_1 + \frac{hu \cdot h(u-1)}{(2h)(h)} y_2 \\
 &= \frac{(u-1)(u-2)}{2} y_0 - \frac{u(u-2)}{1} y_1 + \frac{u(u-1)}{1} y_2 \\
 &= \frac{(u-1)(u-2)}{2} y_0 - \frac{u(u-2)}{1} (1 + \Delta)y_0 + \frac{u(u-1)}{2} (1 + \Delta)^2 y_0 \\
 &= \frac{(u-1)(u-2)}{2} y_0 - \frac{u(u-2)}{1} (y_0 + \Delta y_0) + \frac{u(u-1)}{2} (y_0 + 2\Delta y_0 + \Delta^2 y_0) \\
 &= y_0 \left\{ \frac{(u-1)(u-2)}{2} - \frac{u(u-2)}{1} + \frac{u(u-1)}{2} \right\} \\
 &\quad + \Delta y_0 \left\{ -\frac{u(u-2)}{1} + \frac{u(u-2)}{2} (2) \right\} + \Delta^2 y_0 \left\{ \frac{u(u-1)}{2} \right\}
 \end{aligned}$$

$$\text{i.e. } y = y_0 + y \Delta y_0 + \frac{u(u-1)}{2} \Delta^2 y_0$$

Thus, Lagrange's interpolation formula reduces to Newton's forward interpolation formula, for equidistant values of the argument.





EXERCISES



1. The following data show the monthly average number of deaths under one year in a certain large city. Find the missing term:

Year	1960	1961	1962	1963	1964
Number of Deaths (monthly average)	940	?	907	843	798

[I.C.W.A., Jan. '72]

2. The following gives the amount y of cement in thousands of tons manufactured in India in the year x . Find the missing term.

x	1946	1948	1950	1952	1954	1956
y	39	85	?	151	264	388

[I.C.W.A., July '69]

3. The growth of population in India, according to the decennial census, is shown below:

Year	1901	1911	1921	1931	1941	1951
Population (lakh)	2384	2552	2514	2791	...	3613

The census figure for 1941 is not given here. Give an estimate of the actual population for 1941.

[C.U., B.Sc. '71]

4. Below are given the values of a function U_x for certain values of x :

x	0	1	2	3	4
U_x	1	0	5	22	57

Construct the table of differences. What does the table suggest? Use this table to find U_5 .

[I.C.W.A., Dec. '76]

5. Form a difference table and find the values of y_3 and y_9 from the following:
 $y_4 = 135$, $y_5 = 122$, $y_6 = 1015$, $y_7 = 2016$, $y_8 = 3591$.

6. (a) Given $y_0 = 8$, $y_1 = 6$, $y_2 = 4$, $y_4 = 24$, find the value of y_3 .
 (b) Given $y_2 = 5$, $y_5 = 122$, $y_6 = 193$, find the values of y_3 and y_4 .
 (c) Given $u_5 = 3$, $u_{10} = 0$, $u_{20} = -24$, $u_{25} = -33$, estimate u_{15} under suitable assumption.

7. Find the missing term:

x	0	1	2	3	4
$f(x)$	1	3	9	?	81

8. Find y for $x = 2$, from the following table:

x	...	0	1	3	4	5
y	...	39	85	151	264	388

[I.C.W.A., Jan. '69]

9. Find $f(5)$ from the following data:

$$f(3) = 4, \quad f(4) = 13, \quad f(6) = 43$$



10. Find the polynomial function $f(x)$ from the following values:

$$f(3) = -1, \quad f(4) = 5, \quad f(5) = 15$$

11. Given the following table, find the function $f(x)$, assuming it to be a polynomial of the 3rd degree in x :

x	0	1	2	3
$f(x)$	1	2	11	34

[I.C.W.A., June '75]

12. u_x is a polynomial in x . Given the following table, find u_x .

x	0	1	2	5
u_x	2	3	12	147

13. Below are given the values of a function $f(x)$ for certain values of x . Find $f(2)$, stating your assumption.

x :	0	1	3	4
$f(x)$:	5	6	50	105

[I.C.W.A., Dec. '75]

14. Find $f(x)$, given that

$$f(0) = -3, \quad f(1) = 6, \quad f(2) = 8, \quad f(3) = 12.$$

(State your assumptions, if any). Hence find $f(6)$. [I.C.W.A., June '76]

15. Using any algebraic method, find the value of y when $x = 6$,

x :	3	7	9	10
y :	16.8	12.0	7.2	6.3

[I.C.W.A., Dec. '75-old]

16. For a certain polynomial function y_x it is known that

$$y_1 = -1, \quad y_2 + y_3 = -1, \quad y_4 + y_5 + y_6 = 61.$$

Find y_x and hence the value of y_3 .

17. Given $u_0 + u_6 = -107$, $u_1 + u_5 = -36$, $u_2 + u_4 = -3$, find the value of u_3 .

18. From the following table estimate by interpolation the number of units of a commodity supplied when the price is Rs 4:

Price in Rs	1	3	5	7	9
No. of units Supplied	256	625	935	1201	1433

19. The following table gives the expectation of life e_x° at age x . Calculate the expectation of life at age 12 by using Newton's forward interpolation formula.

x	10	15	20	25	30	35
e_x°	35.4	32.2	29.1	26.0	23.1	20.4

[I.C.W.A., Dec. '77]

20. The following shows the values of a function $y = f(x)$ for a number of values of x :

x :	0.5	0.6	0.7	0.8	0.9
y :	0.35207	0.33322	0.31225	0.28969	0.26609

Obtain the value of y when $x = 0.58$, using a suitable interpolation formula.

[C.U., B.A. (Econ) '76]



21. The table below gives the average number of years of life remaining to persons who survive to exact age x , for male African population of Belgian Congo:

x	0	5	10	15	20
${}^{\circ}e_x$	37.64	44.04	41.40	37.78	34.41

Obtain ${}^{\circ}e_2$ approximately.

[I.C.W.A., Dec. '73]

22. State Newton's Forward Interpolation formula, and use it to find $\sqrt{5.5}$, given that $\sqrt{5} = 2.236$, $\sqrt{6} = 2.449$, $\sqrt{7} = 2.646$ and $\sqrt{8} = 2.828$.

[I.C.W.A., June '74]

23. The following table shows the number of earners earning incomes exceeding different amounts during a certain period:

Income (Rs)	50,000	75,000	100,00	125,000	150,000
No. of Earners	412	304	225	147	88

Find the number of earners earning more than Rs 60,000 by linear interpolation and also by using Newton's forward formula.

[C.U., B.A. (Econ) '77]

24. Using suitable interpolation formulae, calculate the values of y , when (i) $x = 10$, and (ii) $x = 25$.

x	7	11	15	19	23	27
y	20,256	20,625	21,296	22,407	24,098	26,511

25. Find the value of $\sin 48^\circ$ from the following table:

x (degrees)	30	35	40	45	50
$\sin x$	.5000	.5736	.6428	.7071	.7660

26. Using Newton's interpolation formula, find the number of factories earning less than Rs 65,000 as profits, from the following data:

Profits (Rs '000)	30-40	40-50	50-60	60-70	70-80
No. of Factories	34	43	56	39	29

[I.C.W.A., Dec. '75]

27. Apply the appropriate interpolation formula to find $\log 3.146$, given

$$\begin{aligned} \log 3.141 &= 0.497, 0679 & \log 3.144 &= 0.497, 4825 \\ \log 3.142 &= 0.497, 2062 & \log 3.145 &= 0.497, 62.05 \\ \log 3.143 &= 0.497, 3444 \end{aligned}$$

(Find correct upto 7 decimal places).

[I.C.W.A., Dec. '78]

28. Use Stirling's interpolation formula to find the value of the probability integral (P) when $X = 1.52$:

$X...$	1.3	1.4	1.5	1.6	1.7
$P = \text{prob. integral}$	.90320	.91924	.93319	.94520	.95543

29. Given the following cube-roots, find by Bessel's interpolation formula, the cube root of 102.5:

Number	101	102	103	104
Cube-root	4.657,0095	4.672,3287	4.687,5481	4.702,6694

[B.U., B.A. (Econ) '65]



30. Find by using Bessel's interpolation formula, the expectation of life at age 22 from the following data:

Age (x)	10	15	20	25	30	35
Exp. of Life (y)	35.4	32.2	29.1	26.0	23.1	20.4

31. The following table gives the normal weight of a baby during the first six months of life:

Age in Months	0	2	3	5	6
Weight in lbs.	5	7	8	10	12

Estimate the weight of a baby at the age of 4 months.

[I.C.W.A., Jan. '70]

32. State Lagrange's interpolation formula. Use it to find $f(x)$ when $x = 0$, given

x	-1	-2	2	4
$f(x)$	-1	-9	11	69

[I.C.W.A., Dec. '74]

33. State Lagrange's interpolation formula. Use it to find the value of U_4 of a function U_x , given that

$$U_1 = 10, \quad U_2 = 15, \quad U_5 = 42.$$

[I.C.W.A., Dec. '76]

34. Using Lagrange's formula or otherwise, obtain the value of $\log 95$ approximately from the following table:

x	95	97	98	99
$\log x$	1.977,7236	1.986,7717	1.991,2261	1.995,6352

[C.U., B.Com. (Hons) '66, '69]

35. Given $\log_{10} 654 = 2.8156$, $\log_{10} 658 = 2.8182$
 $\log_{10} 659 = 2.8189$, $\log_{10} 661 = 2.8202$,
 find by Lagrange's interpolation formula $\log_{10} 656$ (Retain 4 decimal places in your answer).
 [I.C.W.A., June '78]

36. Find the value of x for which $y = 40$:

x	10	12	15	20
y	25	32	35	45



ANSWERS



- 952
- 96.4
- 3322 lakh
- 3rd differences constant; hence, U_x may be given by a 3rd degree polynomial.
 $U_5 = 116$.
- 16; 5920
- (a) 8; (b) 28, 67; (c) -11
- 31
- 96.4
- 26
- $2x^2 - 8x + 5$
- $x^3 + x^2 - x + 1$
- $x^3 + x^2 - x + 2$
- 19; assuming $f(x)$ to be a 3rd degree polynomial.



14. 126 : assuming $f(x)$ to be a 3rd degree polynomial.

$$f(x) = \frac{1}{2}(3x^3 - 16x^2 + 31x - 6)$$

15. 14.7; (use *Second method*, Example 10.8).

16. $2x^2 - 7x + 4$; 1

17. 3.2

18. 786

19. 34.1

20. 0.33718

21. 42.08

22. 2.345

23. 369; 362

24. 20,510; 25,205

25. 0.7431

26. 156 (use Cumulative frequency distribution)

27. 0.497, 7584

28. 0.93574

29. 4.679, 9508

30. 27.85

31. $8\frac{8}{9}$ lbs.

32. 1

33. 31

34. 1.9822, 7115 (see Examples 10.6 and 10.15)

35. 2.8168

36. 19.56

THEORETICAL QUESTIONS

1. Distinguish between a *discrete* variable and a *continuous* variable.
[W.B.H.S. '84]
2. State which of the following represent discrete data and which represent continuous data:
 - (i) Number of inches of rainfall in a city during various months of the year ;
 - (ii) Speed of a car in miles per hour;
 - (iii) Number of ten-rupee notes circulating in India at any time;
 - (iv) Annual incomes of college professors. [C.U., B.Sc. (Econ) '86]
3. Define *primary* data and *secondary* data. Discuss the different methods of collecting primary data. [I.C.W.A., June '86]
4. Enumerate the important sources of secondary data. What precautions one should take before using such data? [C.A., Nov. '84]
5. Write short explanatory notes (with illustrations) on:— Population and Sample. [C.U., B.Com(H) '86]
6. Distinguish between the 'census' and 'sampling' methods of collecting data and compare their merits and demerits.
7. What are the advantages of sampling over census as a method of investigation?
8. Explain the different parts of a table. [W.B.H.S. '83, '87]
9. Define graphical representation and tabular representation of data and distinguish between them, and write with examples where and why they are suitable. [W.B.H.S. '88]
10. Discuss briefly the different types of diagrams generally used to represent numerical data and point out the relative merits and demerits of diagrammatic representation compared to other method used for the purpose. [C.U., B.Com.(H) '81]
11. Distinguish between the *natural* scale and *logarithmic* scale used in graphical presentation of data.
12. Explain what is meant by semi-logarithmic chart. Discuss the uses of such a chart. [C.U., B.Sc.(Econ) '87; W.B.H.S. '87]
13. What do you mean by *bar diagram*? Describe the different types of bar diagrams, and the situations and techniques of applying them. [W.B.H.S. '86]



14. What is a Pie diagram? Describe how it is drawn.
15. Discuss the different considerations to be kept in view in drawing up a frequency distribution from raw data. [C.U., B.Sc.(Econ) '85]
16. Explain clearly with illustrations the following term: Class limits, Class boundaries, Frequency density; and discuss their uses with reference to frequency distribution. [C.U., B.Sc.(Econ) '88]
17. Explain clearly the distinction between (i) class limits and class boundaries; (ii) relative frequency and density; (iii) histogram and frequency polygon; (iv) less-than and more-than cumulative frequencies.
18. Discuss how would draw a histogram to represent the frequency distribution of a continuous variable. [C.U., B.Com.(H) '88]
19. Explain the construction of histogram with class intervals of unequal width. [I.C.W.A.; Dec. '84]
20. What do you mean by ogives? Describe how they are drawn. [W.B.H.S. '86]
21. The point of intersection of the less-than ogive and greater-than ogive corresponds to (fill up the blank). [C.U., B.Com.(H) '83]
22. Explain what is meant by 'central tendency' of data. [I.C.W.A., Dec. '78, '80 & June '83 ; W.B.H.S. '79, '85]
23. Define 'weighted arithmetic mean' of a set of observations. Show that the weighted arithmetic mean is unaffected if all the weights are multiplied by a common factor. [C.U., B.A.(Econ) '77]
24. Prove that the sum of the deviations $x_1, x_2, \dots, x_n$ from their mean $\bar{x}$ is equal to zero. [I.C.W.A. June '83]
25. Let $x_1, x_2, \dots, x_n$ be the value of a variable with frequencies $f_1, f_2, \dots, f_n$ respectively, Prove that $\sum f_i (x_i - \bar{x}) = 0$. [C.U., B.Com.(H) '80]
26. Prove that the sum of squares of deviations is minimum when deviations are measured from the mean. [W.B.H.S. '78, '82 ; C.U., B.Com.(H) '81, '83 ; B.Sc.(Econ) '87]
27. Show that if $\bar{x}_1$ and $\bar{x}_2$ are the arithmetic means of two sets of values of x , the mean $\bar{x}$ of the combined set will lie between $\bar{x}_1$ and $\bar{x}_2$. [C.U., B.Sc. (Econ) '85]
28. Define A.M., G.M. and H.M. of n positive quantities $x_1, x_2, \dots, x_n$. Of the three means which has the greatest value? [I.C.W.A., Dec. '81; C.U., B.Com.(H) '83]
29. Define geometric mean. Show that the geometric mean of n observations $x_1, x_2, \dots, x_n$ with frequencies $f_1, f_2, \dots, f_n$ is the antilogarithm of the weighted arithmetic mean of the logarithms of the set of x 's. [C.U., B.Com.(H) '82]
30. If A , G and H be the arithmetic, geometric and harmonic mean respectively between x and y , show that $A.H = G^2$. [C.U., B.Com.(H) '85, '88]
31. Prove that the geometric mean of two positive numbers is equal to the geometric mean of their arithmetic and harmonic means. [I.C.W.A., Dec; '81, '83, '85]
32. If a variety assumes n values $a, ar, ar^2, \dots, ar^{n-1}$ with ($r < 1$) equal (constant) frequencies then find the arithmetic mean (A), geometric mean (G) and harmonic mean (H), and show that $AH = G^2$. [W.B.H.S. '86]



33. Prove that for any two positive quantities.

$$A.M. \geq G.M. \geq H.M.$$

[I.C.W.A., Dec. '82, '84]
34. If x and y are two variables such that $y = a + bx$ and if M is the median of x , then show that the median of y is $a + bM$.
35. What is meant by mode of a distribution? Does it always exist? Is it unique? Give examples to illustrate your answer. [C.U., B.Com.(H) '82]
36. If mean and median of a frequency distribution are known, how do you estimate the mode?
37. Compare mean, median and mode as measures of central tendency.

[W.B.H.S. '79, '81, '83, '87, '88]
38. Indicate the merits and shortcomings of the different measures of central tendency as well as situations where to use each. [C.U., B.Com.(H) '81]
39. What are 'quartiles' of a distribution? How do you use them for measuring dispersion and skewness? [C.U., B.Com.(H) '80]
40. Explain how by graphically estimating appropriate measures from the ogive of a given frequency distribution on a continuous variable, one can throw light on its central tendency, dispersion and skewness.

[C.U., B.Com.(H) '88]
41. Explain what you mean by the dispersion of a frequency distribution.

[W.B.H.S. '78, '80, '82, '86]
42. Show that mean deviation is least when calculated about median.

[W.B.H.S. '79, '81, '84, '87]
43. Obtain the standard deviation of the first n natural numbers.

[C.U., B.Sc.(Econ) '85]
44. Prove that the standard deviation calculated from two values x_1 and x_2 of a variable x is equal to half their difference.

[W.B.H.S. '82, '86 ; I.C.W.A., June '80, Dec., '82]
45. Prove that the standard deviation is independent of any change of origin but is dependent on change of scale.

[W.B.H.S. '82 ; C.U., B.Com.(H) '88 ; I.C.W.A., June '82, '86]
46. Show that root-mean-square deviation is minimum when measured about the arithmetic mean. [W.B.H.S. '80]
47. Find the A.M. and S.D. of y when $y_i = (x_i - \bar{x})/S_x$, where $\bar{x}$ and S_x have usual meaning. [C.U., B.A.(Econ.) '79]
48. Let $x_1, x_2, \dots, x_n$ be a set of observations. Suppose we compute $y_i = a + bx_i$ ($i = 1, 2, \dots, n$) where a and b are constants. Express the s.d. of y 's in terms of the s.d. of x 's and comment on the relation between the two.

[C.U., B.A.(Econ.) '78]
49. $\bar{x}$ is the mean of x_1, x_2 and x_3 . If d_1, d_2 and d_3 are the deviations of x_1, x_2 and x_3 from $\bar{x}$ respectively, prove that

$$d_1^2 + d_2^2 + d_3^2 = x_1^2 + x_2^2 + x_3^2 - 3\bar{x}^2$$

[C.U., BSc. (Econ.) '88]
50. Compare range, mean deviation and standard deviation as measures of dispersion. [W.B.H.S. '78, '80]
51. Define 'coefficient of variation'. Point out its importance in statistical data analysis. [I.C.W.A., Dec. '84]



52. Write short notes on:—Relative merits and demerits of various measures of dispersion. [C.U., B.Com.(H) '82, '84]
53. Discuss how different measures of variability are affected when all variety values are changed (i) by the same amount, (ii) in the same proportion. [C.U., B.Com.(H) '87]
54. Define 'moments' of a distribution. What are raw and central moments? [C.U., B.Com.(H) '86]
55. Express the second, third and fourth central moments in terms of raw moments. [I.C.W.A., Dec. '76]
56. Show that central moments are invariant under change of origin but not under change of scale. [I.C.W.A., Dec. '75]
57. Let $x_1, x_2, \dots, x_n$ be n observations with mean $\bar{x}$ and standard deviation σ . Let m_r denote the r -th central moment of the above observations and let α_r denote the r -th central moment of $z_1, z_2, \dots, z_n$ where $z_i = (x_i - \bar{x})/\sigma$. Prove that $\alpha_r = m_r/\sigma^r$. [I.C.W.A. Dec. '81, June '85]
58. Explain the terms 'skewness' and 'kurtosis' of a frequency distribution. Write down the different measures of skewness and kurtosis. [C.U., B.Com.(H) '82, '87]
59. With the help of rough diagrams show the different types of skewness and kurtosis indicating their names. [C.U., B.Com.(H) '84, '85]
60. Show by means of sketches the relative positions of mean, median and mode for frequency curves which are (i) positively skew, (ii) negatively skew, and (iii) symmetrical.
61. What do you understand by bivariate data and a two-way table? [C.A., May '82]
62. Define the term 'correlation' between two variables x and y . [I.C.W.A., June '85, C.U., B.Com.(H) '84]
63. Write down Karl Pearson's *product moment* formula for measuring correlation between two variables. [C.U., B.Com.(H) '80, '87; I.C.W.A., June '82, '85; W.B.H.S. '88]
64. Prove that the product moment correlation coefficient (r) lies between -1 and $+1$. [I.C.W.A., Dec. '81, '83; W.B.H.S. '81, '84, '88; C.U., B.com.(H) '81, 87]
65. Prove that $-1 \leq r \leq +1$. Also interpret the cases when $r = -1$, $r = 0$ and $r = 1$. [I.C.W.A. Dec. '85]
66. Prove that the correlation coefficient is independent of change of origin and scale. [I.C.W.A. June '86]
67. If $u = ax + b$ and $v = cy + d$, where a, b, c and d are constants, express r_{xy} in terms of r_{xy} . [C.U., B.Com.(H) '88]
68. Mention the various properties of correlation coefficient. [C.A., Nov. '78]
69. If x and y are independent variates, then prove that they are uncorrected. Is the converse true? Explain your answer with the help of an example. [W.B.H.S. '80; I.C.W.A., June '80, Dec. '81]
70. Show that $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \cdot \text{Cov}(X, Y)$, where Var denotes variance and Cov denotes covariance. [C.U., B.Sc.(Econ) '85]



71. If $s_y^2 = s_x^2$, show that $(x + y)$ and $(x - y)$ are uncorrected.
[C.U., B.Com.(H) '87]
72. If X_1 , X_2 and X_3 are uncorrected variables each having the same standard deviation, obtain the correlation coefficient between $X_1 + X_2$ and $X_2 + X_3$.
[W.B.H.S. '84]
73. Describe the uses and limitations of correlation coefficient.
[I.C.W.A. Dec. '83 ; C.U., B.Com.(H) '87]
74. What is a scatter diagram? Indicate by mean of suitable scatter diagrams the different types of correlation that may exist between the variables in bivariate data.
[C.U., B.Com.(H) '83 ; I.C.W.A., June '82]
75. Write short notes on—Least squares theory and its uses.
[W.B.H.S. '78, '84, '86, '88]
76. Define the term 'regression'. What is meant by 'regression of x on y '?
[I.C.W.A., Dec. '82, '84, '85; C.U., B.Com.(H) '88]
77. What is the difference between 'correlation' and 'regression'?
[I.C.W.A., Dec. '79]
78. In bivariate data, what are regression lines?
[W.B.H.S. '82, '83]
79. Explain how you will find the linear regression equation of one variable on another.
[C.U., B.Com.(H) '87]
80. Explain the concept of regression and write down the equations of regression lines.
[C.A., May '81]
81. Show that the two regression lines meet at the point . [C.U., B.Com.(H) '88]
82. Show that $r^2 = b_{yz} \cdot b_{zy}$, where the symbols have their usual meaning.
[I.C.W.A., Dec. '76]
83. Prove that the arithmetic mean of the two regression coefficients is greater than the correlation coefficient (r) for positive r .
[W.B.H.S. '83]
84. If the two coefficients of regression are negative, does it mean that their correlation would be positive ?
[I.C.W.A., Dec., '79]
85. If the two regression lines coincide, show that the correlation coefficient $r = \pm 1$. If $r = 0$, show that the two regression lines are at right angles.
[I.C.W.A., Dec. '79, '84, '85]
86. If the two variables are independent, show that the two regression lines are at right angles.
[I.C.W.A., Dec. '77]
87. What can you say about the angle between the regression lines when (i) $r = 0$, (ii) $r = \pm 1$, (iii) r increases from 0 to 1 ?
[I.C.W.A., Dec. '76]
88. When are the regression lines (i) perpendicular to each other, and (ii) coincide ?
[I.C.W.A., June; C.A., Nov. '79]
89. Deduce the expression of regression coefficients by the method of least squares. Point out the important properties of these estimates. [I.C.W.A., June '83]
90. If $u = ax + b$ and $v = cy + d$, where a , b , c and d are constants, express b_{uv} in terms of b_{xy} , and b_{vu} in terms of b_{yz} .
91. Why should there be two regression lines?
[C.A., Nov. '79; I.C.W.A., June '84 and Dec. '84, '85; C.U., B.Com.(H) '88]
92. If x and y are two correlated variables with the same variance and the correlation coefficient is r , find the regression coefficient of x on $(x + y)$ and that of $(x + y)$ on x . Hence find the correlation coefficient between x and $(x + y)$.
[W.B.H.S. '85]



93. Are the following statements valid? Justify your answers.
- Positive value of correlation coefficient between x and y implies that if x decreases, y tends to increase.
 - Correlation coefficient is independent of the origin of reference.
 - Correlation coefficient between x and y turned out to be 1.02. [I.C.W.A., Dec. '80]
94. Give reasons to justify that the following statements cannot be true:
- $b_{yx} = 0.8$ and $b_{xy} = -0.3$;
 - $b_{yx} = 0.8$ and $b_{xy} = 2.4$. [C.U., B.Com.(H) '84]
95. Is it possible for two regression lines to be as follows:
 $Y = -1.5X + 7$ and $X = 0.6Y + 9$? Give reasons. [C.U., B.Com. (H) '87]
96. For a set of pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ prove that
- $$\sum_i (y_i - \bar{y})^2 - \sum_i (y_i - y_i)^2 + \sum_i (y_i - \bar{y})^2$$
- where y_i represents the values of y for given values of
 $x = x_1, x_2, \dots, x_n$
 obtained from the regression equation of y on x . [C.U., B.Sc.(Econ) '88]
97. Define the term 'rank correlation'. Write down the formula for Spearman's 'rank correlation coefficient'.
[C.A., May '82; C.U., B.Com.(H) '83; I.C.W.A., Dec., '84]
98. If the difference in ranks in each pair is zero, show that the rank correlation coefficient is + 1. [I.C.W.A., Dec. '84]
99. Derive the formula for Spearman's rank correlation coefficient when there are no ties. [I.C.W.A., Dec. '84; W.B.H.S. '79, '82, '86]
100. Prove that $-1 \leq \rho \leq +1$, where ρ is Spearman's rank correlation coefficient in case of no ties. [W.B.H.S. '82]
101. Explain the significance of the results; (i) $\rho = +1$, (ii) $\rho = -1$, where ρ Denotes the rank correlation coefficient.
102. Explain the concept of 'multiple regression'. [I.C.W.A., June '81]
103. In the case of three variables x_1, x_2 and x_3 , write down the formulae for
- Multiple regression equation of x_1 on x_2 and x_3 ;
 - Partial correlation coefficient $r_{12.3}$;
 - Multiple correlation coefficient $R_{1.23}$;
 - Partial regression coefficient $b_{12.3}$.
104. What are the limits for (i) Partial correlation coefficient, (ii) Multiple correlation coefficient?
105. What is meant by 'interpolation'? [C.U., B.Com.(H) '82, '84, '86]
106. Define the two operators Δ and E , and show that $E = 1 + \Delta$. If $U_x = x_3$, evaluate $\Delta^2 U$. [C.U., B.Com.(H) '83]
107. State Newton's 'forward' and 'backward' interpolation formulae, and comment on their uses. [C.U., B.Com.(H) '82, '84, '85, '86]
108. State Newton's Backward Interpolation formula with assumptions. [I.C.E.A., June '83, '84]
109. What is 'inverse interpolation'? [I.C.W.A., June '81, Dec. '82]
110. State Lagrange's interpolation formula. [I.C.W.A., June '82, Dec. '83]



111. Show that the weighted A.M. of first n natural numbers whose weights are equal to the corresponding numbers is equal to $(2n + 1)/3$.

[I.C.W.A., June '81]

112. Let x be a variable assuming the values $1, 2, \dots, n$ and let $F_1, F_2, \dots, F_n$ be the corresponding cumulative frequencies of the more-than type. Prove that the A.M. of x is given by

$$\bar{x} = (F_1 + F_2 + \dots + F_n)/N$$

where N is the total frequency.

113. Given a set of n values $x_1, x_2, \dots, x_n$, show that the square of their arithmetic mean cannot exceed the arithmetic mean of their squares.
114. Show that if a variable takes the values $0, 1, 2, \dots, n$ with frequencies $1, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ respectively, then the mean square deviation about the origin is $n(n + 1)/4$. Also show that the mean and variance are $n/2$ and $n/4$ respectively.

[I.C.W.A., Dec. '79]

115. Suppose that a variable x assumes the values a and b and $(n - 2)$ other values all equal to $(a + b)/2$. Find
- the mean deviation about mean,
 - the standard deviation.

[B.U., B.A. (Econ.) '82]

[Ans. $|a - b|/n, |a - b|/\sqrt{2n}$].

116. If the observations $x_1, x_2, \dots, x_n$ all lie between a and b ($a < b$), show that their s.d. cannot be larger than $(b - a)/2$.
117. For a given set of observations with mean $\bar{x}$, prove that

$$S^2 = \sigma^2 + d^2$$

where S^2 is the mean square deviation from a constant A , σ is the standard deviation and $d = \bar{x} - A$.

118. If in a series of measurements we obtain m_1 values of magnitude x_1 , m_2 values of magnitude x_2 and so on, and if $\bar{x}$ is the mean of all the measurements, prove that the s.d. is

$$\sqrt{\left[\frac{\sum m_r (k - x_r)^2}{\sum m_r} - \bar{x}^2 \right]}$$

where $x = k + \delta$, and k is a constant.

[I.C.W.A. June '86]

119. Given n pairs of observations (x_i, y_i) for $i = 1, 2, \dots, n$, let $z_i = x_i + y_i$. Prove that
- $$\sigma_z^2 = \sigma_x^2 + \sigma_y^2 + 2r_{xy} \sigma_x \sigma_y$$

120. Show that $R_{1,23}^2 = r_{12}^2 + r_{13}^2$, if $r_{23} = 0$

[I.C.W.A., June, '87]

SUPPLEMENT

CHAPTER 1

COLLECTION OF DATA, CLASSIFICATION AND TABULATION

(*Summary*)

STATISTICS

The word “*Statistics*” has two different meanings. In the plural sense, it is used to denote ‘facts and figures’ on a particular topic. In the singular sense, Statistics refers to the subject of study which deals with the theories and methods of collection, analysis and interpretation of such numerical figures.

VARIABLE AND ATTRIBUTE

Variable is a quantitative character of an individual or item. *Attribute* is a qualitative character. Two types of variables—*discrete* (or *discontinuous*) and *continuous*. ‘Height’ and ‘temperature’ are continuous variables; ‘family size’ is a discrete variable. ‘Mother-tongue’ of an individual is an attribute.

PRIMARY AND SECONDARY DATA

Statistical data which are collected for a specific purpose directly from the field of enquiry are called *primary data*. “Reserve Bank of India Bulletin” published by Reserve Bank of India contains primary data regarding banking and currency in India.

Secondary data are those which have previously been collected by one agency for some purpose and merely compiled from that source for use in a different connection. “Statistical Abstract of the Indian Union” published by Central Statistical Organisation, Government of India, is a source of secondary data.

Methods of Collecting Primary Data—(1) Direct personal observation, (2) Indirect oral investigation, (3) Questionnaires sent through mail, (4) Interview method with schedules.



POPULATION AND SAMPLE

Statistical information on a particular character in respect of each and every unit covered by an enquiry is called *population*. A selected part of the population which is analysed with a view to ascertaining the properties of the population is called a *sample*.

CENSUS AND SAMPLE SURVEY

Collection of statistical data from each and every member of the population is called *census*; e.g. census of industries in West Bengal. Collection of data from only a selected part of the population (i.e. a sample) is called *sample survey*.

Sample survey method has definite advantages over census method of collecting data—(1) It takes less time, labour and money. (2) In sample survey the magnitude of error can be estimated; but in census method this is not possible. (3) In some cases census method is not possible, and sample survey is the only alternative.

QUESTIONNAIRE

This is a proforma containing a sequence of questions relevant to statistical enquiry, and is used for collection of primary data from individuals through their replies to the questions.

Essentials—(1) The questionnaire should be short. (2) Questions should be simple, unambiguous and precise. (3) Questions should be logically arranged. (4) Questions affecting pride and sentiments should not be asked.

CLASSIFICATION AND TABULATION

Classification is the process of arranging the collected information under different categories or classes according to some common characteristic of the individual members.

Tabulation is the logical and systematic organisation of data in rows and columns.

Different Parts of a Table: (1) Title, (2) Stub, (3) Caption and Box-head, (4) Body, (5) Footnote.



ADDITIONAL EXERCISES 1



- Classify each of the following characters are either a discrete variable or a continuous variable: (i) Monthly income of a person, (ii) Heights of students of a particular school in the year 1978. [W.B.H.S. '78]
- State which of the following represent discrete data and which represent continuous data: (i) Number of inches of rainfall in a city during various months of the year, (ii) Speed of a car in miles per hour, (iii) Number of ten-rupee notes circulating in India at any time, (iv) Annual incomes of college professors. [C.U., B.Sc.(Econ.) '86]
- Distinguish between primary data and secondary data. [W.B.H.S. '83 ; C.A., May '84]
- What do you mean by Primary data? How can they be collected? [W.B.H.S. '85]
- Distinguish between the interview method and the mail questionnaire method of collecting data, state their relative merits and demerits. [B.U., B.A. (Econ.) '79]



6. Distinguish between a population and a sample. [C.U., B.Sc. (Econ.) '83]
7. Describe the characteristics of a good questionnaire. [C.A., Nov. '82]
8. Form a blank table to show the total number of students with males and females reading in 1st year, 2nd year and 3rd year in the streams Arts, Science, and Commerce in 1982 of a particular College under Burdwan University. [W.B.H.S. '86]
9. Prepare a model table with different columns and subcolumns to present the statistical data, giving the values of exports and imports through the ports of Calcutta and Bombay in 1971 and 1972 with respect to foodgrains, cash crops, and industrial products. Make provisions for the following in the table: (i) Totals for different columns and subcolumns, (ii) % increase or decrease in trade in 1972 with respect to 1971. [W.B.H.S. '81]
10. Present the following data in a table:
The Government grants to a University in the years 1946–47, 1947–48 and 1948–49 are respectively Rs 5,59,685, Rs 7,77,892 and Rs 7,48,000. The receipts by way of fees, endowments and donations during the year 1946–47 are respectively Rs 2,52,220, Rs 2,71,636 and Rs 4,83,934. The receipts for the years 1947–48 and 1948–49 in respect of fees are Rs 3,21,286 and Rs 3,83,153. In 1947–48 the University received by way of donation Rs 3,71,971 and in 1948–49 the University received by way of endowments Rs 3,350. The total receipts in the years 1947–48 and 1948–49 are respectively Rs 14,80,362 and Rs 14,39,025. [I.C.W.A., June '86 (old)]
11. Present the following information in a tabular form:
In 1965 out of a total of 2000 workers of a factory, 1500 workers were members of a trade union. The number of women employed was 150 of which 128 do not belong to a trade union. In 1970 the number of union workers increased to 1620 of which 1582 were men. On the other hand the number of non-union workers fell down to 448 of which 318 were men. [W.B.H.S. '85]
12. Present the following data in a tabular form:
The Orissa Government has taken up a scheme of providing drinking water for each village. During the first four years of the sixth plan the Government has installed 39,664 tubewells. Whereas out of the funds earmarked for natural calamities the Government has sunk 14,072 tubewells during the first four years of the sixth plan.
Thus out of the sixth plan fund 9,245 and 8,630 tubewells were sunk respectively in 1980–81 and 1981–82. Whereas out of natural calamities fund the number of tubewells installed in 1980–81 and 1981–82 were 4,511 and 637 respectively. The investment for 1980–81 and 1981–82 were Rs 863.41 lakhs and 1185.65 lakhs respectively.
The number of tubewells installed in 1982–83 were 16,740, out of which 4,800 were installed out of natural calamities fund and the expenditure in respect of sinking of tubewells during 1982–83 were Rs 1411.17 lakhs.
The number of tubewells installed in 1983–84 were 13,973, out of which 9,849 tubewells were sunk out of fund for sixth plan and the total expenditure during the first four years were Rs 5443.05 lakhs. [I.C.W.A., Dec, '84]

ANSWERS

1. (i) Discrete, (ii) Continuous
2. Continuous—(i), (ii); Discrete—(iii), (iv)

8. Number of Students in a College under Burdwan University, 1982

Class	Arts	Science	Commerce	Total
1st year:				
Males				
Females				
Total				
2nd year:				
Males				
Females				
Total				
3rd year:				
Males				
Females				
Total				
All Classes:				
Males				
Females				
Total				

9. Exports & Imports through Calcutta and Bombay Ports (1971-72)

Values (Rs Crores)

Ports/ Items	Imports			Exports		
	1971	1972	%	1971	1972	%
	increase (+)			increase (+)		
	decrease (–)			decrease (–)		
CALCUTTA:						
1. Foodgrains						
2. Cash crops						
3. Industrial products						
Total						
BOMBAY:						
1. Foodgrains						
2. Cash crops						
3. Industrial products						
Total						
Grand Total						



10. Annual Receipts of a University (1946–49)

(Rupees)

Source	1946–47	1947–48	1948–49
1. Govt. Grant	5,59,685	7,77,892	7,48,000
2. Fees	2,52,220	3,21,286	3,83,153
3. Endowments	2,71,636	9,213	3,350
4. Donations	4,83,934	3,71,971	3,04,522
Total	15,67,475	14,80,362	14,39,025

11. Number of Trade Union Members among Workers in a Factory (Years 1965 & 1970)

Trade Union	1965			1970		
	Men	Women	Total	Men	Women	Total
Member	1478	22	1500	1582	38	1620
Non-member	372	128	500	318	130	448
Total	1850	150	2000	1900	168	2068

12. Tubewells for Drinking Water in Orissa Villages during Sixth Plan

Year	Number of Tubewells sunk			Expenditure (Rs lakh)
	6th Plan Fund	Natural Calamities Fund	Total	
1980–81	9,245	4,511	13,756	863.41
1981–82	8,630	637	9,267	1,185.65
1982–83	11,940	4,800	16,740	1,411.17
1983–84	9,849	4,124	13,973	1,982.82
Total	39,664	14,072	53,736	5,443.05

CHAPTER 2

CHARTS AND DIAGRAMS

(*Summary*)

CHARTS AND DIAGRAMS

- (1) Line Diagram (or Graph),
- (2) Bar Diagram,
- (3) Pie Diagram,
- (4) Pictogram,
- (5) Histogram, Frequency Polygon and Ogives.

LINE DIAGRAM

'Natural scale' or 'Arithmetic scale' shows equal distances for equal amounts of change. 'Ratio scale' or 'Logarithmic scale' shows equal distances for equal ratios of change.

Ordinary graph shows natural scale in both horizontal and vertical directions. Semi-logarithmic graph shows natural scale in the horizontal direction, but ratio scale in the vertical direction.

BAR DIAGRAM

Bar diagram consists of a group of equispaced rectangular diagrams each of equal width. The rectangles start from a common base line and their lengths are proportional to the statistical data.

Bar diagrams may be 'vertical' or 'horizontal'. They may again be (i) Single bar, (ii) Grouped (or Multiple) bar, (iii) Component (or Subdivided) bar, (iv) Paired bar; etc.

PIE DIAGRAM

Pie diagram is a circular diagram whose area is divided proportionately into different sectors by straight lines drawn from the centre to the circumference of the circle. For drawing a Pie diagram, it is first necessary to express each value as a percentage of the total. The percentages are then multiplied by 3.6 each to give the angle of the sector at the centre of the circle. A compass and a protractor are necessary for drawing a Pie diagram.

■ ■ ■ ————— ADDITIONAL EXERCISES 2 ————— ■ ■ ■

1. Represent the following data by a suitable diagram:

Year	1970	1971	1972	1973	1974	1975
Sale of Steel (thousand tonnes)	8	8.8	9.2	10.2	7.6	12.5

[W.H.B.S. '82]

2. What do you mean by Bar Diagram? Describe the different types of bar diagrams, the situations and techniques of applying them. [W.B.H.S. '86]



3. Production of Rubber in 1968 in the principal rubber producing countries:

Country	Malaysia	Indonesia	Thailand	Sri Lanka
Production ('000 tonne)	1,042	740	254	146
	India	Combodia	Others	Total
	68	51	294	2,595

Give a proper diagrammatic representation of the above data.

[W.B.H.S. '81]

4. The percentage of literacy in West Bengal is shown below separately for males and females for 4 years:

Years	1941	1951	1961	1971
Males	27.4	34.1	40.1	44.8
Females	3.9	12.3	17.0	22.1

Represent the data in a suitable diagram.

[W.B.H.S. '80]

5. Represent the following data of revenue of two Governments in a particular year by a suitable chart:

Sources	Revenue (Rs Crores) Govt. I	Govt. II
Customs	170	110
Excise	480	380
Income Tax	350	320
Corporation Tax	110	140
Others	90	50
Total	1200	1000

[W.B.H.S. '85]

6. The population of India is to be classified Statewise and is to be shown separately for males and females. Suggest suitable methods of diagrammatic representation of the data.

[W.B.H.S. '80]

7. Draw a (rough) Pie Chart to represent the following data relating to the production cost of manufacture:

Cost of Materials	:	Rs 38,400
Cost of Labour	:	Rs 30,720
Direct Expenses	:	Rs 11,520
Overhead Expenses	:	Rs 15,360

[C.U., B.Com.(Hons) '86]

8. The following table gives the distribution of out-lay in a Five-year Plan of India under the major heads of development expenditure:

Head of Expenditure	Expenditure (Crores of Rs)
Agriculture and Community Development	8,000
Irrigation and Power	4,000
Industry and Mining	7,000
Transport and Communication	5,500
Miscellaneous	2,500
Total	27,000

Represent the above information by a Pie Chart.

[I.C.W.A., June '83]



9. Construct a Pie Diagram for the following data:

<i>Mines</i>	<i>Coal</i>	<i>Iron Ore</i>	<i>Copper</i>	<i>Manganese</i>	<i>Mica</i>	<i>Others</i>
<i>Average Daily Number of Workers Employed ('000)</i>	348	34	4	90	31	84

[C.U., M.Com., '83]



ANSWERS



1. Line diagram
3. Horizontal Bar diagram
4. Grouped Vertical Bar
5. Component Bar
6. Paired Bar (Horizontal)
7. **Hint:** Total 96,000; Percentages 40, 32, 12, 16;
Angle (degree) 144, 115, 43, 58.
8. **Hint:** Percent. 30, 15, 26, 20, 9;
Angle (degree.) 108, 54, 94, 72, 32.
9. **Hint:** Total 591; Percent. 59, 6, 1, 15, 5, 14;
Angle (degree.) 212, 22, 4, 54, 18, 50.

CHAPTER 4

FREQUENCY DISTRIBUTION (Solved Problems)

Problem 4.1 Construct the Grouped Frequency Distribution and Cumulative Frequency Distribution from the following:

(a)	Marks	No. of Students	(b)	Age	No. of Persons
	<i>Below 10</i>	15		<i>Under 10</i>	15
	<i>Below 20</i>	35		<i>Under 20</i>	30
	<i>Below 30</i>	60		<i>Under 30</i>	53
	<i>Below 40</i>	84		<i>Under 40</i>	75
	<i>Below 50</i>	106		<i>Under 50</i>	100
	<i>Below 60</i>	120		<i>Under 60</i>	110
	<i>Below 70</i>	125		<i>Under 70</i>	115
(Source: I.C.W.A., June '84)				<i>Under 80</i>	125
			(Source: I.C.W.A., June '80)		

Solution (a) [Hint: 'Marks' obtained by students is a *discrete* variable, which can take the values 0, 1, 2, 3, etc. So, mark "Below 10" can be only the values 0, 1, 2, ..., 9, i.e. upto 9; "Below 20" denotes upto 19, and so on. If marks "Below 10" are separated from those "Below 20", there remain only the values between 10 and 19. Hence, the class intervals are 0–9, 10–19, etc. The class boundaries between successive classes are then 9.5, 19.5, ..., 69.5. Note that the lower boundary of the first class interval is not stated in Table 2, because a negative class boundary (– 0.5) would be meaningless here. Also see No. 9, page 98.]

Table 1 Grouped Frequency Distribution

Class interval	Frequency
0–9	15
10–19	20
20–29	25
30–39	24
40–49	22
50–59	14
60–69	5
Total	125 = N

Table 2 Cumulative Frequency Distribution

Class	Cumulative
9.5	15
19.5	35
29.5	60
39.5	84
49.5	106
59.5	120
69.5	125 = N

(b) [Hint: Here the variable 'Age' is *continuous*, which can take any value, even fractional values. "Under 10" can be any value between 0 and a number very close to 10, say 9.9999; no definite upper limit can be stated. Hence, for all practical purposes, age "Under 10" may be assumed to be between 0 and 10, i.e. 0–10. Similarly, "Under 20" can go upto a number which is practically 20. Hence the class intervals are 0–10, 10–20, etc. It should be noted that we cannot speak of a person aged exactly 10 years; because time ticks off in split-seconds.]



Table 3 Grouped Frequency Distribution **Table 4** Cumulative Frequency Distribution

<i>Class Interval</i>	<i>Frequency</i>
0–10	15
10–20	15
20–30	23
30–40	22
40–50	25
50–60	10
60–70	5
70–80	10
Total	125 = N

<i>Class Boundary</i>	<i>Cumulative Frequency</i>
0	0
10	15
20	30
30	53
40	75
50	100
60	110
70	115
80	125 = N

[**Note:** For calculating Mean, Standard Deviation (S.D.), Mean Deviation and Moments from data shown as in Problem 4.1 (a) and (b), it is necessary to obtain *mid-values* of class intervals. Hence Grouped Frequency Distribution is to be constructed.

For calculating Median, Mode, Quartiles, Deciles etc. it is necessary to obtain *class boundaries*. This cannot be done correctly unless one has a clear idea about the nature of Grouped Frequency Distribution. If class intervals are shown by using class limits (see Table 1), we must calculate class boundaries therefrom (see Table 2). If however class intervals are shown by using class boundaries (see Table 3), these class boundaries are shown directly in the Cumulative Frequency Distribution (Table 4). Also see No. 38, Page 171.]



Problem 4.2 Construct the Grouped Frequency Distribution and Cumulative Frequency Distribution in each of the following cases:

(a)	<i>Value</i>	<i>Frequency</i>
	<i>Less than 10</i>	5
	<i>Less than 20</i>	15
	<i>Less than 30</i>	35
	<i>Less than 40</i>	81
	<i>Less than 50</i>	90
	<i>Less than 60</i>	118
	<i>Less than 70</i>	130
	<i>Less than 80</i>	140

(b)	<i>Annual Sales (Rs '000)</i>	<i>Frequency</i>
	<i>Less than 10</i>	4
	<i>Less than 20</i>	20
	<i>Less than 30</i>	35
	<i>Less than 40</i>	55
	<i>Less than 50</i>	62
	<i>Less than 60</i>	67

(Source : I.C.W.A., June '83)

(Source : I.C.W.A., Dec, '81)

Solution (a) [**Hint:** If nothing is known about the nature of the variable, whether continuous or discrete, it is assumed to be *continuous*. This should be treated as in Problem 4.1 (b), the class intervals being 0–10, 10–20, etc. and the class boundaries being 0, 10, 20, etc. Also see Example 4.9, page 81.]



(b) [Hint: Strictly speaking the variable 'Annual Sales' is *discrete*, because Sales "Less than 10" thousand rupees can be only upto Rs 9999.99 P. However, for practical purposes Sales (in Rs '000) are taken to be 0–10, 10–20, etc.; i.e. the variable is treated as *continuous*. See Problem 4.1 (b) and Problem 5.4]



Problem 4.3 Construct Cumulative Frequency Distribution and Grouped Frequency Distribution from the following data:

Mid-value	10	30	50	70	90
Frequency	40	51	64	38	7

(Source : I.C.W.A., June '85)

Solution [Hint: If a grouped frequency distribution has classes of *equal width*, then the successive

Mid-values have a common difference,

Class boundaries have a common difference,

Lower class limits have a common difference,

Upper class limits have a common difference,

this common difference being the common width of classes.

Hence, if the successive mid-values are found to have a common difference, the frequency distribution must have classes of equal width, and the class boundaries will lie exactly midway between the successive mid-values. The common width of classes will be equal to the common difference between mid-values.]

Table 5 Cumulative Frequency Distribution

Class Boundary	Cumulative Frequency
0	0
20	40
40	91
60	155
80	193
100	200 = N

Table 6 Grouped Frequency Distribution

Class Interval	Frequency
0–20	40
20–40	51
40–60	64
60–80	38
80–100	7
Total	200 = N

[Note: For calculating Mean, S. D., M. D. and Moments from data as shown in this problem, the mid-values may be used directly.

For calculating Median, Quartiles, Deciles and Percentiles, Cumulative Frequency Distribution (Table 5) is necessary.

For calculating Mode, Grouped Frequency Distribution (Table 6) is required.]

■ ■ ■ ————— **ADDITIONAL EXERCISES 4** ————— ■ ■ ■

1. Consider the following frequency distribution:

<i>Monthly Wages (Rs)</i>	50.00–59.99	60.00–69.99	70.00–79.99	80.00–89.99
<i>No. of Employees</i>	8	10	16	14
	90.00–99.99	100.00–109.99	110.00–119.99	Total
	10	5	2	65

- Determine (i) lower limit of the sixth class;
(ii) class-mark of the third class;
(iii) class boundaries of the fifth class; and
(iv) relative frequency of the third class.

[C.U., B.Sc. (Econ.) '83]

2. Find the class boundaries of the frequency distribution of weight in pounds of a group of boys, where the class limits used are 40.0–42.4, 42.5–44.9, 45.0–47.4, 47.5–49.9, 50.0–52.4, the measurement were made to the nearest $\frac{1}{10}$ th of a pound.

[C.U., B.Sc. (Econ) '86]

3. Construct (i) Grouped frequency distribution, and (ii) Cumulative frequency distribution, from the following data:

(a)		(b)		(c)	
<i>No. of Days Absent</i>	<i>No. of Students</i>	<i>Value</i>	<i>Frequency</i>	<i>Weekly Wages (Rs)</i>	<i>No. of Workers</i>
Less than 5	29	Below 10	2	Below 10	8
Less than 10	239	Below 20	21	Below 20	18
Less than 15	469	Below 30	56	Below 30	45
Less than 20	584	Below 40	133	Below 40	90
Less than 25	634	Below 50	238	Below 50	113
Less than 30	644	Below 60	303	Below 60	120
Less than 35	650	Below 70	333		
Less than 40	655	Below 80	351		
		Below 90	359		
		Below 100	360		
(Source: I.C.W.A. Dec., '83)		(Source: I.C.W.A. June '85)		(Source: C.U., B.Com. (H) '80)	

4. From the following table, find (i) the number of men whose heights are less than 67.2 inches, and (ii) the number of men whose heights lie between 58.5 and 67.2 inches:

<i>Height (inches)</i>	56–57	58–59	60–61	62–63	64–65	66–67	68–69
<i>Frequency</i>	2	4	9	30	23	20	9
	70–71	72–73					
	2	1					

[C.U., B.Sc. (Econ) '86]



5. From the following table, find the number of wage earners receiving between Rs 37 and Rs 47 per week:

<i>Weekly Wages (Rs) :</i>	30–	32–	34–	36–	38–	40–	42–
<i>Frequency</i>	3	8	24	31	50	61	38
	44–	46–	48–50				
	21	12	2				

[I.C.W.A., June '82]

6. (a) From the frequency distribution of heights shown in Q. 4, prepare a table showing class boundaries, class marks and cumulative frequencies (less than and more than).
 (b) Draw the histogram for the height distribution. Shade that portion of the histogram which will represent the total number of men whose heights lie between 60.5 and 66.5 inches.
 [C.U., B.Com.(H) '84]
7. Draw histogram of the following frequency distribution and use it to find the total number of wage earners in the age group 19–32 years:

<i>Age Group</i>	14–15	16–17	18–20	21–24	25–29	30–34	35–39
<i>No. of Wage Earners</i>	60	140	150	110	110	100	90

[C.U., B.Com.(H) '81]

8. Draw histogram for the following frequency distribution of heights of students. Hence find the height above which 120 students have their heights.

<i>Height (cm)</i>	126–135	136–145	146–150	151–155	156–160
<i>Frequency</i>	11	42	33	41	36
	161–165	166–180			
	21	16			

[W.B.H.S. '86]

9. Construct (a) a frequency polygon and (b) a cumulative frequency polygon, from the frequency distribution shown in Q. 1 above.

[C.U., B.Com.(H) '85]

10. Draw a suitable cumulative frequency diagram for the following distribution of family size for 100 families:

<i>Family Size</i>	1	2	3	4	5	6	7	8	9	10
<i>Frequency</i>	2	7	12	21	26	17	9	3	2	1

[W.B.H.S. '79]

11. Draw cumulative frequency polygon (less than type) on the basis of the following data:

<i>Mid-value of the Class</i>	7	14	21	28	35	42
<i>Frequency</i>	8	18	32	28	18	6

[W.B.H.S. '85]

ANSWERS

1. (i) Rs 100.00, (ii) Rs 74.995, (iii) Rs 89.995–99.995, (iv) $16/65 = 0.246$.
2. (in pound) 39.95–42.45, 42.45–44.95, 44.95–47.45, 47.45–49.95, 49.95–52.45.
3. (a) **Hint:** “No. of days absent” is a discrete variable. See Problem 4.1 (a). (b) **Hint:** See Problem 4.2(a).
- (c) **Hint:** “Weekly wages” is a discrete variable, although it may be treated as continuous. See Problem 4.2(b).

as continuous. See Problem 42(b).									
(i) <i>Class Interval</i>		0–9.99		10–19.99		20–29.99		30–39.99	
<i>Frequency</i>		8		10		27		45	
		40–49.99		50–59.99		Total			
		23		7		120			
(ii) <i>Cl. Boundary</i>		9.995	19.995	29.995	39.995	49.995	59.995		
<i>Cum. Freq.</i>		8	18	45	90	113	120		

4. (i) 85, (ii) 81 (see Example 4.12, p. 86)
5. 192 (See p. 86)
6. (a) Table showing Class boundaries, Class marks etc.

<i>Class</i>	<i>Boundaries</i>	<i>Class mark</i>	<i>Cumulative</i>	<i>Frequencies</i>
<i>Lower</i>	<i>Upper</i>		<i>Less than</i>	<i>More than</i>
55.5	57.5	56.5	2	100
57.5	59.5	58.5	6	98
59.5	61.5	60.5	15	94
61.5	63.5	62.5	45	85
63.5	65.5	64.5	68	55
65.5	67.5	66.5	88	32
67.5	69.5	68.5	97	12
69.5	71.5	70.5	99	3
71.5	73.5	72.5	100	1

Note: ‘Less than’ cumulative frequency corresponds to ‘Upper’ class boundary and ‘More than’ corresponds to ‘Lower’ class boundary.

- (b) **Hint:** Since all classes are of equal width, the class frequency is taken as height of each rectangle in the histogram. Area under the histogram between the vertical lines drawn at 60.5 and 66.5 (inches) on the horizontal scale, is shaded.
7. **Hint:** The classes are of unequal width. See Example 4.15, page 92. 345 wage earners.
8. Note the unequal width of classes. See Example 4.15, page 92.
9. **Hint:** See Example 4.13, page 89.
10. **Hint:** See Fig. 4.8 page 95.
11. **Hint:** See Problem 4.3 and Fig. 4.4 (page 91).

<i>Class Boundary</i>	3.5	10.5	17.5	24.5	31.5	38.5	45.5
<i>Cum. Frequency</i>	0	8	26	58	86	104	110

CHAPTER 5

MEASURES OF CENTRAL TENDENCY

(Solved Problems)

Problem 5.1 The average weight of the following frequency distribution is 117 lbs. Find the value of x .

Weight (in lbs.) :	100	110	120	$x + 25$	140	Total
No. of Men :	1	4	2	2	1	10

[C.U., B.Com (P) '83]

Solution Mean $\Sigma fX/N$.

$$\begin{aligned} \text{Here, } \Sigma fX &= (100 \times 1) + (110 \times 4) + (120 \times 2) + (x + 25) \times 2 + (140 \times 1) \\ &= 100 + 400 + 240 + (2x + 50) + 140 \\ &= 2x + 970 \end{aligned}$$

Mean = 117 (given)

$$\therefore 117 = \frac{2x + 970}{10};$$

$$\text{i.e. } 1170 = 2x + 970$$

$$\text{or, } 2x = 1170 - 970 = 200$$

$$\therefore x = 100.$$

Ans. 100.



Problem 5.2 The weighted arithmetic mean of the four numbers 26, 18, 12 and 4 is 10.4. If the weight of the first three numbers are 1, 3 and 4 respectively, find the weight of the fourth number.

Solution Let the required weight be w .

x	f	fx
26	1	26
18	3	54
12	4	48
4	w	$4w$
Total	$8 + w$	$128 + 4w$

$$\bar{x} = \Sigma fx/N$$

$$\text{or, } 10.4 = \frac{128 + 4w}{8 + w} \quad \text{or, } \frac{104}{10} = \frac{128 + 4w}{8 + w}$$

$$\text{or, } 832 + 104w = 1280 + 40w$$

$$\text{or, } 64w = 448$$

$$\therefore w = 448/64 = 7$$

Ans. 7.





Problem 5.3 Prove that for any two real quantities,
 $A.M. \geq G.M. \geq H.M.$

[I.C.W.A., June '79 ; Dec. '82, '84]

Solution Let a and b be two real quantities (both positive).

Then $A.M. = \frac{a+b}{2}$, $G.M. = \sqrt{ab}$, $H.M. = \frac{2ab}{a+b}$

Since $(a-b)^2$ is a perfect square, it cannot be negative.

$$(a-b)^2 \geq 0$$

or, $(a-b)^2 + 4ab \geq 0 + 4ab$

or, $(a+b)^2 \geq 4ab$

or, $\left(\frac{a+b}{2}\right)^2 \geq ab$

or, $\frac{a+b}{2} \geq \sqrt{ab}$... (i)

or, $A.M. \geq G.M.$... (ii)

Again, dividing both sides of (i) by ab ,

$$\frac{a+b}{2ab} \geq \frac{\sqrt{ab}}{ab}$$

or, $\frac{a+b}{2ab} \geq \frac{1}{\sqrt{ab}}$

or, $(a+b) \sqrt{ab} \geq 2ab$

or, $\sqrt{ab} \geq \frac{2ab}{a+b}$

or, $G.M. \geq H.M.$... (iii)

From (ii) and (iii), we get $A.M. \geq G.M. \geq H.M.$



Problem 5.4 Calculate Median and Mode of the following:

Annual Sales (Rs '000)	Frequency
less than 10	4
less than 20	20
less than 30	35
less than 40	55
less than 50	62
less than 60	67

Is it possible to calculate the arithmetic mean? If possible calculate it.

[I.C.W.A., June '83]

Solution [Note: Median is calculated from the Cumulative Frequency Distribution and Mode from the Grouped Frequency Distribution. See Problem 4.2 (b), page 359.]

**Table 7** Cumulative Frequency Distribution

<i>Class Boundary</i>	<i>Cumulative Frequency</i>
10	4
20	20
Median →	← $N/2 = 33.5$
30	35
40	55
50	62
60	$67 = N$

Using simple interpolation,

$$\frac{\text{Median} - 20}{30 - 20} = \frac{33.5 - 20}{35 - 20}$$

or,
$$\frac{\text{Median} - 20}{10} = \frac{13.5}{15}$$

or,
$$\text{Median} - 20 = \frac{13.5}{15} \times 10 = 9$$

∴
$$\text{Median} = 20 + 9 = 29 \text{ (Rs '000)}$$

Table 8 Grouped Frequency Distribution

<i>Class Interval</i>	<i>Frequency</i>
upto 10	4
10–20	16
20–30	15
(30–40)	20
40–50	7
50–60	5
Total	67

The maximum frequency 20 lies in the class interval (30–40), which is the *modal class*.

$$\begin{aligned} \text{Mode} &= l_1 + \frac{d_1}{d_1 + d_2} \times c = 30 + \frac{(20 - 15)}{(20 - 15) + (20 - 7)} \times 10 \\ &= 30 + \frac{5}{18} \times 10 = 30 + 2.78 \\ &= 32.78 \text{ (Rs '000)} \end{aligned}$$

It is possible to calculate the Mean if we assume that the class intervals are of equal width, and hence the first class interval is 0–10. (Note that the grouped frequency distribution contains an open-end class.)

**Table 9** Calculations for Mean

Class Interval	Frequency (f)	Mid-value (x)	$y = \frac{x - 25}{10}$	fy
0–10	4	5	–2	–8
10–20	16	15	–1	–16
20–30	15	25	0	0
30–40	20	35	1	20
40–50	7	45	2	14
50–60	5	55	3	15
Total	67	–	–	25

If $y = (x - c)/d$, then $\bar{x} = c + d\bar{y}$

$$\text{Mean} = 25 + 10(25/67) = 25 + 3.73 = 28.73 \text{ (Rs '000)}$$

Ans. 29.00, 32.78, 28.73 (Rs '000)



Problem 5.5 In the following frequency distribution, two class frequencies are missing:

Intelligence Quotient	No. of Students
55–64	2
65–74	19
75–84	78
85–94	?
95–104	301

Intelligence Quotient	No. of Students
105–114	?
115–124	92
125–134	14
135–144	4

It is however known that the total frequency is 900 and the Median 100.048. Find the two missing frequencies. [C.U., B.Com. (H) '81]

Solution Let us denote the missing frequencies in the fourth and the sixth classes by f_4 and f_6 respectively.

Class Boundary	Cumulative Frequency
54.5	0
64.5	2
74.5	21
84.5	99
94.5	$99 + f_4$
Median = 100.048 →	$\leftarrow N/2 = 450$
104.5	$400 + f_4$
114.5	$400 + f_4 + f_6$
124.5	$492 + f_4 + f_6$
134.5	$506 + f_4 + f_6$
144.5	$510 + f_4 + f_6 = N = 900 \text{ (given)}$



[**Note:** Since Median = 100.048 lies between 94.5 and 104.5, $N/2 = 450$ must lie in the corresponding interval.]

Using simple interpolation,

$$\frac{100.048 - 94.5}{104.5 - 94.5} = \frac{450 - (99 + f_4)}{(400 + f_4) - (99 + f_4)}$$

$$\text{or, } \frac{5.548}{10} = \frac{450 - 99 - f_4}{400 + f_4 - 99 - f_4}$$

$$\text{or, } 0.5548 = \frac{351 - f_4}{301}$$

$$\text{or, } 0.5548 \times 301 = 351 - f_4$$

$$\text{or, } 167 = 351 - f_4$$

$$\text{or, } f_4 = 351 - 167 = 184 \quad \dots (1)$$

Again, Total frequency = 900 (given)

$$\text{or, } 2 + 19 + 78 + f_4 + 301 + f_6 + 92 + 14 + 4 = 900$$

$$\text{or, } 510 + f_4 + f_6 = 900$$

$$\text{or, } f_4 + f_6 = 390$$

$$\text{Using (1), we get } f_6 = 390 - 184 = 206.$$

$$\text{Ans. } f_4 = 184, f_6 = 206.$$

(Also see Example 5.55, page 149).



Problem 5.6 The median and mode of the following frequency distribution are known to be 27 and 26 respectively. Find the values of a and b .

Values	0–10	10–20	20–30	30–40	40–50
Frequency	3	a	20	12	b

[I.C.W.A., Dec. '85]

Solution

x	Cum. Freq.
0	0
10	3
20	$3 + a$
Median = 27 →	← $N/2$
30	$23 + a$
40	$35 + a$
50	$35 + a + b = N$

(i) Median = 27 (given) lies between 20 and 30, and so $N/2$ lies between the corresponding cumulative frequencies $3 + a$ and $23 + a$, as indicated by arrow marks. Now using simple interpolation,



$$\frac{27 - 20}{30 - 20} = \frac{N/2 - (3 + a)}{(23 + a) - (3 + a)}$$

$$\text{or, } \frac{7}{10} = \frac{N/2 - 3 - a}{20} = \frac{N - 6 - 2a}{40}$$

$$\text{or, } N - 6 - 2a = \frac{7}{10} \times 40 = 28$$

Putting $N = 35 + a + b$, we have $35 + a + b - 6 - 2a = 28$

$$\text{or, } a - b = 1 \quad \dots (1)$$

(ii) Mode = 26 (given) lies in the class (20 – 30). This is the modal class. Applying formula for mode,

$$\text{Mode} = l_1 + \frac{d_1}{d_1 + d_2} \times C$$

$$\text{or, } 26 = 20 + \frac{(20 - a)}{(20 - a) + (20 - 12)} \times 10$$

$$\text{or, } 6 = \frac{20 - a}{28 - a} \times 10; \quad \text{Simplifying } a = 8 \quad \dots (2)$$

Now using (1) and (2), we get $a = 8$, $b = 7$



Problem 5.7 If $y = a + bx$ and M_o is the mode of x , then show that the mode of y must be $a + bM_o$. [C.U., B.Sc.(Econ.) '85 ; M.Com. '75]

Solution [Hint: We begin with an example. Let 6, 8, 9, 10 be the values of x with frequencies 2, 5, 4, 1 respectively and $y = 3 + 2x$.

x	f	$y = 3 + 2x$
6	2	15
8	5	19
9	4	21
10	1	23

Since 5 is the largest frequency, mode of x is the corresponding value of x ; i.e. $M_o = 8$.

If $y = 3 + 2x$, the values of y are 15, 19, 21, 23 with frequencies 2, 5, 4, 1 respectively. Thus $y = 19$ has the largest frequency. This means that Mode of y is 19. But $19 = 3 + 2 \times 8$; i.e. Mode of $y = 3 + 2$ (Mode of x).]

Let $x_1, x_2, \dots, x_n$ be the values of x and $f_1, f_2, \dots, f_n$ be the respective frequencies. If the maximum frequency is f_k , then the

x	f	$y = a + bx$
x_1	f_1	$y_1 = a + bx_1$
x_2	f_2	$y_2 = a + bx_2$
$\vdots$	$\vdots$	$\vdots$
x_n	f_k	$y_k = a + bx_k$
$\vdots$	$\vdots$	$\vdots$
x_n	f_n	$y_n = a + bx_n$



mode of x is the corresponding value of x , viz. x_k ; i.e.

$$M_o = x_k \quad \dots (i)$$

If $y = a + bx$, then the values of y , viz. $y_1, y_2, \dots, y_n$ and obtained by putting $x = x_1, x_2, \dots, x_n$. These values of y must be all different; because $y = a + bx$ will give different values of x . Hence that value of y has the largest frequency for which $x = x_k$. That is, the mode of y is y_k . But $y_k = a + bx_k$; i.e.

$$\text{Mode} = y = a + bM_o \quad \text{from (i)}$$



SHORT-ANSWER TYPE QUESTIONS



Problem 5.8 Find the A.M. of the numbers 3, 5, 7, ..., 47.

Solution [Hint : A.M. of a series of numbers having a common difference, is equal to the A.M. of the 1st and the last numbers]. A.M. = $(3 + 47)/2 = 25$.



Problem 5.9 (i) If a variable assumes the values 1, 2, 3, ..., 10 with frequencies 1, 2, 3, ..., 10 respectively, find the A.M. [W.B.H.S. '80]

(ii) The weighted A.M. of 3, 5 and 10 with weights x , y and z respectively is known to be 8. What is the weighted A.M. if the weights are changed to $2x$, $2y$ and $2z$ respectively?

(iii) A variable x takes the values 10 and 20 with equal frequencies. Find the A.M. [W.B.H.S. '78]

(iv) A variable x takes all integral values from 1 to 10 with equal frequencies. Find its A.M. [W.B.H.S. '79]

$$\text{Solution} \quad (i) \text{ A.M.} = \frac{1 \times 1 + 2 \times 2 + 3 \times 3 + \dots + 10 \times 10}{1 + 2 + 3 + \dots + 10} = \frac{385}{55} = 7$$

(ii) If all the weights are multiplied by a constant, the A.M. does not change. A.M. = 8.

(iii) Since the frequencies are equal,

$$\text{Weighted A.M.} = \text{Simple A.M.} = (10 + 20)/2 = 15$$

(iv) Since the frequencies are equal,

$$\text{Weighted A.M.} = \text{Simple A.M.}$$

$$= \text{A.M. of } 1, 2, 3, \dots, 10$$

$$= (1 + 10)/2, \text{ since there is common difference.}$$

$$= 11/2 = 5.5$$



Problem 5.10 If the A.M. of 3, 5, x , 12, 17 be 9, find the value of x .

Solution $(3 + 5 + x + 12 + 17)/5 = 9$ (given)

$$\text{or,} \quad 37 + x = 45, \quad \text{i.e.} \quad x = 8$$



Problem 5.11 The algebraic sum of the deviations of 25 observations measured from 45 is -55 . Find the A.M. of the observations. [C.U., B.Com.(Pass) '85]



Solution [Hint: If $y = x - c$, then $\bar{x} = c + \bar{y}$; p. 136]

Here $y = x - 45 = \text{Deviation from 45}$; $n = 25$, $\Sigma y = -55$

$$\therefore \bar{x} = 45 + (-55)/25 = 45 - 2.2 = 42.8$$



Problem 5.12 (i) Find the geometric mean of 3, 12 and 48.

[C.U., B.Com(Pass) '83]

(ii) If a variable increase from 20 to 2000 at the same ratio in 10 years, find its value after 5 years.

[W.B.H.S. '80]

Solution (i) $G.M. = \sqrt[3]{3 \times 12 \times 48}$

$$= \sqrt[3]{3 \times (3 \times 4) \times (3 \times 4 \times 4)} = 12$$

(ii) [Hint: (1) If there is a *common difference* in a series, Middle-most value = A.M. of the 1st and the last values.

(2) If there is *common ratio* in a series, Middle-most value = G.M. of the 1st and the last values.

Since there is a common ratio, and the values after 5 years is the middle-most item,

Required value = G.M. of 20 and 2000

$$= \sqrt{20 \times 2000} = \sqrt{40000} = 200$$



Problem 5.13 (i) A piece of work was done in 2, 4 and 3 days by 3 men. Find (with the help of harmonic mean) the average rate of working.

[C.U., B.Com(Hons) '84]

(ii) Find the H.M. of 1, 1/2, 1/3 and 1/4.

Solution (i) Average Rate of working

= H.M. of the 3 observations 2, 4, 3

$$= \frac{3}{\frac{1}{2} + \frac{1}{4} + \frac{1}{3}} = \frac{36}{13} = 2.77 \text{ days.}$$

$$(ii) H.M. = \frac{4}{\frac{1}{1} + \frac{1}{\frac{1}{2}} + \frac{1}{\frac{1}{3}} + \frac{1}{\frac{1}{4}}} = \frac{4}{1 + 2 + 3 + 4} = 0.4$$



Problem 5.14 (i) Find A.M., G.M. and H.M. of 2, 9, 12; and show that A.M. > G.M. > H.M.

(ii) The A.M. of two observations is 36 and their G.M. is 24. Find the H.M.

(iii) Find A.M., G.M. and H.M. of the following: 2, 9, 2, 12, 9, 12

(iv) Find A.M., G.M. and H.M. of the following:

(a) 3, 3, 3, 3;

(b) 3, -2, 4, 0, 5.



Solution (i) $A.M. = \frac{1}{3}(2 + 9 + 12) = \frac{1}{3} \times 23 = 7.67$

$$G.M. = \sqrt[3]{2 \times 9 \times 12} = \sqrt[3]{216} = 6$$

$$H.M. = \frac{1}{\frac{1}{2} + \frac{1}{9} + \frac{1}{12}} = \frac{3}{\frac{18 + 4 + 3}{36}} = \frac{108}{25} = 4.32$$

From these results we see that $A.M. > G.M. > H.M.$

(ii) For two observations, $A.M. \times H.M. = (G.M.)^2$

$$36 \times H.M. = 24^2; \quad \therefore H.M. = 16$$

(iii) Since 2, 9 and 12 occur with equal frequency, therefore

Weighted A.M. = Simple A.M. = 7.67; see (i)

Weighted G.M. = Simple G.M. = 6

Weighted H.M. = Simple H.M. = 4.32

(iv) (a) Since all observations are equal,

$$A.M. = G.M. = H.M. = 3$$

(b) $A.M. = \frac{3 - 2 + 4 + 0 + 5}{5} = \frac{10}{5} = 2$

G.M. and H.M. cannot be determined, since one of the observations is zero. Moreover, G.M. and H.M. become meaningless, when all observations do not have the same sign.



Problem 5.15 Find median and mode of the following data:

4, 7, 10, 7, 9, 15, 12, 7, 9, 6

Solution Arranging: 4, 6, 7, 7, 7, 9, 9, 10, 12, 15.

There are two middlemost observations 5th and 6th. viz. 7 and 9.

$$\text{Median} = (5\text{th item} + 6\text{th item})/2 = (7 + 9)/2 = 8$$

Amongst the given observations, 7 occurs the largest number of times.

Mode = 7.



Problem 5.16 Find the mode of the following:

(i) 2, 7, 4, 3, 6

(ii) 2, 7, 4, 3, 6, 7, 4

Solution (i) Mode does not exist; because all values occur with equal frequency.

(ii) There are two modes: 4 and 7; because both of them have the highest frequency.



Problem 5.17 If mean and median of a frequency distribution are 20 and 17 respectively, find the mode.



Solution Mean – Mode = 3(Mean – Median)
 or, $20 - \text{Mode} = 3(20 - 17)$
 Solving, Mode = 11.



Problem 5.18 (a) Suppose, the highest value (97.5) of a set of 100 observations is changed to 197.5. what would happen to its mean, median and mode?

(b) For the following observations regarding monthly income for 6 persons, which measure of central tendency, you think, would be appropriate? Justify your answer.

Rs 500, 525, 475, 480, 520, 2593

(c) In a set of values of a variable x , the smallest value is 0.3 and the highest is 1.1. Which of the following is a possible value of the mean? Give reason in support of your answer:

5.2, 1.0, 0.2, – 0.9,

[C.U., M.Com. '80]

Solution (a) Mean increases by 1; because the total of 100 observations increases by 100 (from 97.5 to 197.5).

Median does not change; because with the change in the highest value, the middlemost observation remains unaffected.

Mode does not change; because usually mode lies near the middle of the given values.

(b) Since one observation is extremely large compared to the others, Median is the appropriate average. Arranging the data, Median = $(500 + 520)/2 = 510$ Rs.

(c) 1.0; because all measures of central tendency must lie between the smallest and the largest values.



Problem 5.19 (a) The arithmetic mean of a variable x is 100. Find the mean of the variable $5x - 10$. [W.B.H.S. '78]

(b) Two variables x and y are given $y = 2x - 11$. If the median of x be 30, find the median of y . [W.B.H.S. '78]

(c) If the relation between two variables x and y is $3y - 2x + 5 = 0$, and median of y is 40, find the median of x . [W.B.H.S. '83]

(d) If the relation between two variable x and y is $2y - 6x = 6$ and mode of x is 21, find the mode of y . [W.B.H.S. '82]

Solution [Note: If $y = a + bx$... (i)

then A.M. of $y = a + b(\text{A.M. of } x)$

Median of $y = a + b(\text{Median of } x)$

Mode of $y = a + b(\text{Mode of } x)$

In general, if two variables x and y are related by an equation of the form

$$Ax + By + C = 0 \quad \dots (ii)$$

where A, B, C are constants, then

$$A(\text{A.M. of } x) + B(\text{A.M. of } y) + C = 0$$

$$A(\text{Median of } x) + B(\text{Median of } y) + C = 0$$

$$A(\text{Mode of } x) + B(\text{Mode of } y) + C = 0$$

Such relations do not however hold for G.M. and H.M. (unless $C = 0$). Note that (i) and (ii) are both equations of straight lines.]



- (a) Let $y = 5x - 10$.
Then $\bar{y} = 5\bar{x} - 10 = 5 \times 100 - 10 = 490$.
- (b) Since $y = 2x - 11$
 $\therefore$ Median of $y = 2 \cdot (\text{Median of } x) - 11 = 49$
- (c) $3 \cdot (\text{Median of } y) - 2 \cdot (\text{Median of } x) + 5 = 0$
or, $3 \times 40 - 2 \cdot (\text{Median of } x) + 5 = 0$
Solving, Median of $x = 125/2 = 62.5$
- (d) Since $2y - 6x = 6$
 $\therefore 2 \cdot (\text{Mode of } y) - 6 \cdot (\text{Mode of } x) = 6$
Putting Mode of $x = 21$ and solving, we get Mode of $y = 66$



Problem 5.20 If $2u = 5x$, G.M. of x is 1 and H.M. of x is 0.4, find G.M. and H.M. of u . [W.B.H.S. '86]

Solution [Hint: If $y = kx$, where k is a constant, the (Average of y) = $k \cdot$ (Average of x)
This is true for the Averages, viz. A.M., G.M., H.M., Median and Mode.] Here $2u = 5x$, so that $u = (2.5)x$

$$\text{G.M. of } u = 2.5 \times (\text{G.M. of } x) = 2.5 \times 1 = 2.5$$

$$\text{H.M. of } u = 2.5 \times (\text{H.M. of } x) = 2.5 \times 0.4 = 1$$



Problem 5.21 (a) Find the 1st and the 3rd quartiles for the following data: 0, 1, -1, -2, 6, 4, 5, 8, 12, 10, 11, [I.C.W.A., June '86]

(b) If for a symmetrical distribution $Q_1 = 20$ and $Q_3 = 30$, find Median.

(c) How many deciles lie below the first quartile? [C.A., Nov. '82]

Solution (a) The data are arranged in increasing order and serially numbered (i.e. ranked):

Values	-2	-1	0	1	4	5	6	8	10	11	12	
Ranks	1	2	3	4	5	6	7	8	9	10	11	($n = 11$)

$$Q_1 = \text{Value of } (n + 1)/4 \text{th item, i.e. 3rd item} = 0$$

$$Q_3 = \text{Value of } 3(n + 1)/4 \text{th item, i.e. 9th item} = 10$$

(b) For a symmetrical distribution,

$$\text{Median} = (Q_1 + Q_3)/2 = (20 + 30)/2 = 25$$

(c) First decile $D_1 = P_{10}$ and second decile $D_2 = P_{20}$ are the only two deciles which lie below 1st quartile $Q_1 = P_{25}$.



ADDITIONAL EXERCISES 5



- The weighted arithmetic mean of 3.2, 3.4, 3.8 and 4.6 is known to be 4.656. If the weights of the first observations are respectively 12, 20 and 8, find the weight of 4.6. [K.U., B.Com.(Pass) '81]



2. From the following table, calculate the value of mean:

Monthly Wages (Rs)	0–10	10–20	20–30	40–50	50–60	60–70	70–80	80–90
No. of Servants	1	4	10	30	35	10	7	1

[C.U., B.Com.(Hons) '84]

3. A machine depreciates at a rate of 20% for the first year, 10% for the next two years and 5% for the next three years. What is the average rate of depreciation ?
[C.U., M.B.A. '81]
4. Prove that the logarithm of the geometric mean of n positive values is the arithmetic mean of the logarithms of the values.
[I.C.W.A., June '86]
5. The total interest paid on each of three different sums of money (for a given period of time) yielding 3%, 4% and 5% simple interest per annum respectively is the same. Which of the three means will you use to find the average yield per cent of the total sum invested? Calculate this average.
[C.U., B.Com.(Hons) '83]
6. One person purchased 4 types of tea at the prices of Rs 16, Rs 20, Rs 32 and Rs 80 each per kilogram by spending equal amounts of money for each type of tea and mixed them together to sell at a profit of Re 1.00 per kilogram. Find the selling price of the mixed tea.
[W.B.H.S. '83]
7. Find the arithmetic mean, geometric mean and harmonic mean of the elements of the set $\{1, 2, 3, 4, 5\}$, and verify the relation $A > G > H$.

[C.U., B.Com.(Hons) '85]

8. Using the formula, find median of the following:

Class Interval	130–134	135–139	140–144	145–149	150–154	155–159	160–164
Frequency	5	15	28	24	17	10	1

[C.U., B.Com.(Hons) '86]

9. Draw Ogive (less than type) from the following data and find Median from the ogive:

Weekly Wages (Rs)	51–60	61–70	71–80	81–90	91–100
No. of Workers	10	15	34	23	11

[K.U., B.Com.(Pass) '83]

10. (a) What is meant by Mode of a distribution? Does it always exist? Is it always unique? Give examples to illustrate your answer.
(Hint: Illustration 14, page 124.)

- (b) Calculate mode of the following frequency distribution:

Marks (x)	5–10	10–15	15–20	20–25	25–30	30–35	35–40	40–45
No. of (f) Students	5	6	15	10	5	4	2	2

[C.U., B.Com.(Hons) '82]



11. Runs scored by individual English batsmen in the 1st innings of the 3rd test match (India vs. England, 1986-series) are as follows: 0, 21, 0, 49, 183 (not out), 44, 38, 17, 18, 8, 0. Find mean, median and mode.
12. Calculate Arithmetic Mean and Median of the frequency distribution given below. Hence calculate Mode using the empirical relation between the three.

Class Limits	1–10	11–20	21–30	31–40	41–50	51–60	61–70
Frequency	8	15	25	20	16	10	6

[C.U., B.Com.(Pass) '82]

13. Calculate Mean and Median from the following data:

Weekly Wages (Rs)	Number of Workers
Below 10	8
Below 20	18
Below 30	45
Below 40	90
Below 50	113
Below 60	120

[C.U., B.Com.(Hons) '80]

(Hint: No. 3 (c), page 361 and Note on p. 359.)

14. For a certain frequency table with total frequency 150, the mean was found to be 56.47. But while copying out the table, a typist left out two of the class frequencies, say f_3 and f_5 , so that the table is given to you in the following form. Find the values of f_3 and f_5 .

Weekly Wages in Rs (mid value)	45	50	55	60	65	70	75	Total
Frequency	5	48	f_3	30	f_5	8	6	150

[I.C.W.A., Dec. '83]

15. If the mode of the following frequency distribution which represents the number of minutes spent per week by a group of teenagers in going to movies is 339.5 minutes per week, find the missing frequency. Then find the median of the complete frequency distribution.

Number of minutes/week	0–99	100–199	200–299	300–399	400–499	500 and above
Number of Teenagers	26	32	65	?	60	42

[W.B.H.S. '84]

16. Calculate the quartiles from the following:

Weight (kg)	14–18	18–22	22–26	26–30	30–34	Total
Frequency	15	40	25	13	7	100

[K.U., B.Com.(Pass) '85]



17. If a variate assumes n values $a, ar, \dots, ar^{n-1}$ with ($r > 1$) equal (constant) frequencies then find the arithmetic mean (A), geometric mean (G) and harmonic mean (H) and show that $AH = G^2$. [W.B.H.S. '82]

■ ■ ■ ——— **SHORT ANSWER TYPE QUESTIONS** ——— ■ ■ ■

18. Find the A.M., G.M. and H.M. of the four number 3, 6, 24 and 48.
[K.U., B.Com.(Pass) '84]
19. Find the arithmetic mean of two observations if their geometric mean and harmonic mean are 15 and 9 respectively. [W.B.H.S. '85]
20. Find mean, median and mode of the following data: 44, 60, 33, 63, 48, 61, 51, 48.
[C.U., B.Com.(Pass) '85]
21. For some symmetrical distribution $Q_1 = 24$ and $Q_3 = 42$. Find Median.
[C.U., B.Com.(Pass) '85]
22. If the relation between two variables x and y is $2x + 3y = 7$ and the median of y is 2, find the median of x . [W.B.H.S. '85]
23. Examine the validity of the following statements:
- The geometric mean can never be calculated when one of the observations is a negative one.
 - One can calculate the harmonic mean for a set of observations which includes 0 (zero) as one of its observations.
 - The median of a set of 31 values is the 11th value in the arrangement of the values in an increasing order (or in a decreasing order) of magnitude.
 - The mode of a variable is the highest frequency.
 - The relation: $\bar{x} - \text{Mode} = 3(\bar{x} - \text{Median})$ holds only when the distribution is moderately skewed. [C.U., M.Com. '82]
24. Examine the validity of the following:
- A.M. is unit free.
 - When all observations are equal, $A.M. = G.M. = H.M.$
 - Median can be calculated even for a frequency distribution with open-end classes.
 - When all observations are equal, $A.M. \geq G.M. \geq H.M.$
 - For a set of observations, one obtains the following:
 $\sum |x_i - 10|/n = 19.35$, $\sum |x_i - 5|/n = 14.35$,
where Median (x) = 10.
 - $Q_2 = 2\text{nd Quartile} = N/2$, where N = Total number of observations. [C.U., M.Com '83]
25. Fill up the missing parts:
- A.M. of a set of observations is defined as
 - G.M. of a set of n positive observations is defined as
 - H.M. of a set of n positive observations is defined as
 - Median of a set of observations is defined as
 - Mode of a set of observations is defined as



26. Fill up the missing parts:
- Simple and weighted arithmetic means are equal only when
 - The relations between A.M., G.M. and H.M. are
 - When all observations are equal, the relations between A.M., G.M. and H.M. are
 - For two observations a and b , the expressions for A.M., G.M. and H.M. are respectively
 - Mean, Median and Mode calculated from two observations x_1 and x_2 are respectively
 - For two observations only, if A.M. and G.M. are known, then H.M. may be obtained by the relation
 - The empirical relation connecting Mean, Median and Mode is
27. State the appropriate type of average to use in each of the following cases:
- Marks obtained in Mathematics by the students in an examination.
 - The size of shoes sold in a shop.
 - A frequency distribution having open-end classes.
 - Different rates of depreciation charged on diminishing balance.
 - Speeds in several journeys when the distance covered is the same in each journey.
 - Speeds in several journeys when the time taken is the same in each journey.
28. Comment on the following statements:
- Mean is independent of origin, but depends on scale.
 - Median is unaffected by the presence of extreme values.
 - Mode is the value of the largest observation in the series.
 - The lower quartile of a distribution is $N/4$ and the upper quartile is $3N/4$.
 - The quartiles of a distribution are connected by the relation $Q_3 - Q_2 = Q_2 - Q_1$.



ANSWERS



- 10
- Rs 46.83
- 9.3%
- H.M. = 3.83%
- Rs 26.60 per kg.
- 3.00, 2.61, 2.19
- 144.92
- Rs 76.82
- 18.21
- 34.4, 18, 0
- 33.0, 31.5, 28.5
- (Rs) 32.16, 33.33.
- 41, 12
- 75, 335.5 min/week
- (kg) 19.0, 21.5, 25.2
- $A = \frac{a(r^n - 1)}{n(r - 1)}, G = ar^{(n-1)/2}, H = \frac{na(r-1)r^{n-1}}{r^n - 1}$
- 20.25, 12, 7.11
- 25
- 51, 49.5, 48
- 33



22. 0.5
23. True—(a), (c), (e); False—(b), (d).
24. True (b), (c); False (a), (d), (e) since M.D. is minimum when deviations are taken from median, and (f).
25. (a) the sum of the observations, divided by their number.
 (b) the n -th root of their product.
 (c) the reciprocal of the A.M. of their reciprocals.
 (d) the middlemost observation, when all items are arranged in order of magnitude.
 (e) that observation(s) which has (have) the highest frequency.
26. (a) all weights are equal;
 (b) $A.M. \geq G.M. \geq H.M.$;
 (c) $A.M. = G.M. = H.M.$
 (d) $(a + b)/2$, $\sqrt{ab}$, $2ab/(a + b)$;
 (e) $(x_1 + x_2)/2$, $(x_1 + x_2)/2$, no mode;
 (f) $A.M. \times H.M. = (G.M.)^2$;
 (g) $\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$.
27. (a) A.M., (b) Mode, (c) Median, (d) G.M.,
 (e) H.M., (f) A.M.
28. (a) False; Mean depends on the both origin and scale.
 (b) True.
 (c) False; Mode is the value of the observation which has the largest frequency.
 (d) False; Lower quartile and upper quartile are those values which correspond to cumulative frequencies $N/4$ and $3N/4$ respectively.
 (e) False; Quartiles are connected by $Q_1 \leq Q_2 \leq Q_3$. The given relation holds only for a symmetrical distribution.

CHAPTER 6

MEASURES OF DISPERSION

(Solved Problems)

Problem 6.1 Find Range from the following data:

(a) 0, 1, -1, -2, 6, 4, 5, 8, 12, 10, 11.

[I.C.W.A., June '86]

(b)

x	f
3	5
4	9
6	10
7	4
11	2
Total	30

(c)

Length (cm)	Frequency
20–24	3
25–29	6
30–34	13
35–44	8
46–59	2
Total	32

Solution (a) Maximum value = 12, Minimum value = -2

$\therefore$ Range = $12 - (-2) = 14$

(b) Maximum value = 11, Minimum value = 3

$\therefore$ Range = $11 - 3 = 8$

(c) Max. value = Upper boundary of the class 45 – 59 = 59.5

Min. value = Lower boundary of the class 20 – 24 = 19.5

$\therefore$ Range = $59.5 - 19.5 = 40.0$ cm.

(Note: All absolute measures of dispersion are expressed in the same unit as the observations.)



Problem 6.2 Find Semi-interquartile Range from the following:

(a) 0, 1, -1, -2, 6, 4, 5, 8, 12, 10, 11.

[I.C.W.A. June '86]

(b) 391, 384, 591, 407, 672, 522, 777, 733, 2488, 1490.

(c) x : 0 1 2 3 4 5 6 7 Total

f : 14 21 25 43 51 40 39 12 245

Solution (a) The data are arranged in ascending order of magnitude and ranked:

Value	-2	-1	0	1	4	5	6	8	10	11	12
Rank	1	2	3	4	5	6	7	8	9	10	11

Rank of $Q_1 = \frac{1}{4} (11 + 1) = 3$; Rank of $Q_3 = \frac{3}{4} (11 + 1) = 9$

Q_1 = Value with rank 3 = 0

Q_3 = Value with rank 9 = 10

$$\therefore \text{Semi-interquartile Range} = \frac{1}{2} (10 - 0) = 5$$

(b) [Hint: see Example 5.63, p. 160]. $Q_1 = 403$, $Q_3 = 955.25$.

$$\text{Semi-interquartile Range} = \frac{1}{2} (955.25 - 403) = 276.125$$

(c) [Ref.: Example 5.57, p. 154] Here $N = 245$

x	f	$Cum. f$
0	14	14
1	21	35
2	25	60
3	43	103
4	51	154
5	40	194
6	39	233
7	12	245 = N
Total	245 = N	

$$\text{Rank of } Q_1 = \frac{1}{4} (N + 1) = \frac{1}{4} (245 + 1) = 61.5$$

$$\text{Rank of } Q_3 = \frac{3}{4} (N + 1) = \frac{3}{4} (245 + 1) = 184.5$$

$$Q_1 = 61.5\text{th value} = 3 \quad (\text{see Note, page 167})$$

$$Q_3 = 184.5\text{th value} = 5$$

$$\therefore \text{Quartile Deviation} = \frac{1}{2} (Q_3 - Q_1) = \frac{1}{2} (5 - 3) = 1.$$



Problem 6.3 Calculate Mean Deviation about mode in respect of marks (out of 25) obtained by 9 students given below:

Students :	A	B	C	D	E	F	G	H	I
Marks :	4	7	10	7	9	15	12	7	9

[I.C.W.A., June '86]

Solution Mode = 7 (because it occurs the largest number of times).

Student	Marks	Differences from Mode (= 7)
A	4	3
B	7	0
C	10	3
D	7	0
E	9	2
F	15	8
G	12	5
H	7	0
I	9	2
Total		23



$$\begin{aligned}\text{M.D. about mode} &= \text{A.M. of differences from mode} \\ &= 23/9 = 2.56\end{aligned}$$



Problem 6.4 After shift of origin and change of scale a frequency distribution of a continuous variable takes the form as under:

Step Deviation (u) :	-3	-2	-1	0	1	2	3
Frequency (f) :	5	7	18	3	124	12	3

If the mean and variance of the distribution are 46 and 189 respectively, find the original frequency distribution. [W.B.H.S.'85]

Solution If the origin of x is shifted to A and the scale changed to h , we write $u = (x - A)/h$, so that $x = A + hu$.

Therefore (i) $\bar{x} = A + h\bar{u}$

(ii) $\sigma_x^2 = h \cdot \sigma_u^2$ i.e. $\sigma_x^2 = h^2 \cdot \sigma_u^2$

We calculate mean (u) and variance (σ_u^2) from the given data and use relations (i) and (ii) to obtain the values of A and h .

u	f	fu	fu^2
-3	5	-15	45
-2	7	-14	28
-1	18	-18	18
0	31	0	0
1	24	24	24
2	12	24	24
3	3	9	27
Total	100	10	190

Hence

$$\bar{u} = \Sigma fu/N = 10/100 = 0.1$$

$$\begin{aligned}\sigma_u^2 &= \Sigma fu^2/N - (\Sigma fu/N)^2 = 190/100 - (10/100)^2 \\ &= 1.9 - 0.01 = 1.89\end{aligned}$$

We are given $\bar{x} = 46$, $\sigma_x^2 = 189$.

Putting these results in (i) and (ii) above,

$$46 = A + h(0.1) \text{ and } 189 = h^2(1.89)$$

Solving, $A = 45$ and $h = 10$.

Hence, $u = (x - 45)/10$; i.e. $x = 45 + 10u$

When $u = -3, -2, -1, \dots, 3$, the values of x are respectively 15, 25, 35, 45,, 75. These are midvalues of class intervals 10–20, 20–30, 30–40,, 70–80. (see Problem 4.3, Page 360). The original frequency distribution is as follows:

Class Interval	10–20	20–30	30–40	40–50	50–60	60–70	70–80	Total
Frequency	5	7	18	31	24	12	3	100





Problem 6.5 Find the missing figures from the following table:

	Group I	Group II	Group III	Combined
Number	50	?	90	200
Mean	113	?	115	116
Standard Deviation	6	7	?	7.746

[C.U., B.Com. (Hons.) '86]

Solution From 3 groups, Mean and S.D. of Composite Group are given by the formulae (see Example 6.39, p. 204)

$$(1) N = n_1 + n_2 + n_3$$

$$(2) N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3$$

$$(3) N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2 + n_3\sigma_3^2) + (n_1d_1^2 + n_2d_2^2 + n_3d_3^2)$$

$$\text{Using (1), } 200 = 50 + n_2 + 90. \quad \therefore n_2 = 60$$

$$\text{Using (2), } 200 \times 116 = 50 \times 113 + 60\bar{x}_2 + 90 \times 115$$

$$\text{or, } 23200 = 5650 + 60\bar{x}_2 + 10350. \quad \therefore \bar{x}_2 = 120$$

$$\text{Using (3), } d_1 = (113 - 116) = -3, d_2 = (120 - 116) = 4,$$

$$d_3 = (115 - 116) = -1$$

$$200 \times (7.746)^2 = (50 \times 6^2 + 60 \times 7^2 + 90\sigma_3^2) + (50 \times 9 + 60 \times 16 + 90 \times 1)$$

$$\text{or, } 12000 = (18000 + 2940 + 90\sigma_3^2) + (450 + 960 + 90)$$

$$\text{Solving } \sigma_3^2 = 64. \quad \therefore \sigma_3 = 8$$

Ans. Number = 60, mean = 120, s.d. = 8



Problem 6.6 The mean of 5 observations is 4.4 and the variance is 8.24. If 3 of the observations are 4, 6, 9, find the other two. [I.C.W.A., June' 85]

Solution Let the two other observations be a and b . We use the formula for Mean and Variance, namely

$$\bar{x} = \Sigma x/n, \quad \sigma^2 = \Sigma x^2/n - \bar{x}^2$$

$$4.4 = (4 + 6 + 9 + a + b)/5. \quad \therefore a + b = 3 \quad \dots (i)$$

$$8.24 = (4^2 + 6^2 + 9^2 + a^2 + b^2)/5 - (4.4)^2. \quad \therefore a^2 + b^2 = 5 \quad \dots (ii)$$

Solving (i) and (ii), we get a and b .

$$(a + b)^2 - (a^2 + b^2) = 3^2 - 5 = 4.$$

$$\text{or, } 2ab = 4$$

$$(a - b)^2 = (a^2 + b^2) - 2ab = 5 - 4 = 1 \quad \therefore a - b = \pm 1 \quad \dots (iii)$$

From (i) and (iii), either $a = 2, b = 1$ or, $a = 1, b = 2$

Ans. 1 and 2



Problem 6.7 Show that M.D. about mean for a set of values of a variable cannot exceed S.D. [B.U., B.A. (Econ) '82]



Solution [Hint: Also see Example 6.45, p. 209]

Let $x_1, x_2, \dots, x_n$ be a set of values and $\bar{x}$ their arithmetic mean.

$$\text{M.D.} = \Sigma |x_i - \bar{x}| / n, \text{ S.D.} = \sqrt{\Sigma (x_i - \bar{x})^2 / n}$$

To prove that $\text{M.D.} \leq \text{S.D.}$, we use Cauchy-Schwarz inequality (page 70):

$$\Sigma a_i^2 \Sigma b_i^2 \geq (\Sigma a_i b_i)^2$$

Putting $a_i = |x_i - \bar{x}|$, $b_i = 1$ ($i = 1, 2, \dots, n$)

$$\Sigma a_i^2 = \Sigma (|x_i - \bar{x}|)^2 = \Sigma (x_i - \bar{x})^2$$

$$\Sigma a_i^2 = \Sigma 1^2 = n$$

$$\Sigma a_i b_i = \Sigma |x_i - \bar{x}|$$

$$\text{Hence, } \Sigma (x_i - \bar{x})^2 \cdot n \geq \{\Sigma |x_i - \bar{x}|\}^2$$

Dividing both sides by n^2 ,

$$\Sigma (x_i - \bar{x})^2 / n \geq \{\Sigma |x_i - \bar{x}| / n\}^2$$

or,

$$(\text{S.D.})^2 \geq (\text{M.D.})^2$$

i.e.

$$\text{S.D.} \geq \text{M.D.}$$

(Proved)



Problem 6.8 For the values $x_1, x_2, \dots, x_n$ with respective frequencies $f_1, f_2, \dots, f_n$, show that

$$\frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 \geq \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})$$

where $\bar{x}$ is the arithmetic mean and $N = \Sigma f_i$.

[C.U., B.Com.(Hons) '82]

Solution We use Cauchy-Schwarz inequality:

$$\Sigma a_i^2 \Sigma b_i^2 \geq (\Sigma a_i b_i)^2$$

Putting $a_i = \sqrt{f_i} |x_i - \bar{x}|$, $b_i = \sqrt{f_i}$

$$\text{We have } \Sigma a_i^2 = \Sigma \{\sqrt{f_i} |x_i - \bar{x}|\}^2 = \Sigma f_i (x_i - \bar{x})^2$$

$$\Sigma b_i^2 = \Sigma f_i = N$$

$$(\Sigma a_i b_i)^2 = \{\Sigma f_i |x_i - \bar{x}|\}^2$$

$$\text{Therefore, } \Sigma f_i (x_i - \bar{x})^2 \cdot N \geq \{\Sigma f_i |x_i - \bar{x}|\}^2$$

Dividing both sides N^2 ,

$$\Sigma f_i (x_i - \bar{x})^2 / N \geq \{\Sigma f_i |x_i - \bar{x}| / N\}^2 \quad (\text{Proved})$$



SHORT-ANSWER TYPE QUESTIONS



Problem 6.9 The three quartiles of a variable x are 5, 12 and 17. Find its Quartile Deviation. [W.B.H.S. '79]

$$\text{Solution } \text{Q.D.} = \frac{1}{2} (Q_3 - Q_1) = \frac{1}{2} (17 - 5) = 6$$

Because Q_1 is the smallest and Q_3 is the largest among the 3 quartiles.





Problem 6.10 (a) Find the Standard Deviation, if in distribution $n = 10$, $\Sigma x = 120$, $\Sigma x^2 = 1690$. [C.U., B.Com. (Pass)'83]

(b) Calculate the Variance of the sample $\{2, 5, 8\}$.

[C.U., B.Com. (Pass)'82]

(c) Find the S.D. of 1, 2, 3, 4, ..., 10.

[C.U., B.Com. (Pass)'80]

Solution (a) $\sigma^2 = \Sigma x^2/n - (\Sigma x/n)^2 = 1690/10 - (120/10)^2$
 $= 169 - 144 = 25; \therefore \sigma = 5$

[Note: Do not write $\sigma = \pm 5$.]

(b) $\Sigma x = 2 + 5 + 8 = 15$

$\Sigma x^2 = 4 + 25 + 64 = 93$ and $n = 3$

Variance (σ^2) = $\Sigma x^2/n - (\Sigma x/n)^2 = 93/3 - (15/3)^2 = 6$

(c) Using formulae for the sum and the sum of squares of natural numbers (see Sec. 3.8, p.67)
 ; here $n = 10$,

$$\Sigma x = \frac{n(n+1)}{2} = \frac{10 \times 11}{2} = 55$$

$$\Sigma x^2 = \frac{n(n+1)(2n+1)}{6} = \frac{10 \times 11 \times 21}{6} = 385$$

$$\sigma^2 = 385/10 - (55/10)^2 = 38.5 - 30.25 = 8.25$$

$$\therefore \sigma = \sqrt{8.25} = 2.9$$



Problem 6.11 Find the S.D. of x from the following:

$$\Sigma f_i x_i^2 / \Sigma f_i = 25, \Sigma f_i x_i = 300, \Sigma f_i = 100$$

[C.U., M.Com.'83]

Solution For a frequency distribution

$$\sigma^2 = \Sigma f x^2 / N - (\Sigma f x / N)^2 = 25 - (300/100)^2 = 16$$

$$\therefore \sigma = 4$$



Problem 6.12 (a) A certain student obtained for a set of 25 values of the variable x , $\Sigma x_i = 12.4$, $\Sigma x_i^2 = 34.39$.

Would you take his computations to be correct? Give reasons for your answer.

[B.U., B.A.(Econ)'79]

(b) Examine whether the following results of a piece of computation are consistent or not:

$$\text{Total frequency} = 120, \Sigma fd = -125, \Sigma fd^2 = 128. \quad [\text{I.C.W.A., Dec. '82}]$$

Solution (a) $\sigma^2 = \Sigma (x - \bar{x})^2 / n$ can never be negative.

Again, $\sigma^2 = \Sigma x^2 / n - (\Sigma x / n)^2$ (Working formula)

Hence, the values of n , Σx^2 will be such that σ^2 does not give negative value.



$$\begin{aligned}\text{Here } \sigma^2 &= \frac{34.39}{25} - \left(\frac{12.4}{25}\right)^2 = \frac{34.39 \times 25 - (12.4)^2}{625} \\ &= \frac{859.75 - 153.76}{625} = \frac{705.99}{625} \quad (\text{which is positive})\end{aligned}$$

Thus, the computations may be correct.

(b) For a frequency distribution, the working formula is $\sigma^2 = \Sigma fd^2/N - (\Sigma fd/N)^2$, where $d = x - A$. The values of N , Σfd^2 must be such that σ^2 will not give negative value in any numerical computation.

Here, $\sigma^2 = \frac{128}{120} - \left(\frac{-125}{120}\right)^2 = \frac{128 \times 120 - (-125)^2}{14400} = \frac{-265}{14400}$ (negative). Thus, the given results are not consistent.



Problem 6.13 If the S.D. of x is 6, find the S.D. of

- | | |
|-----------------|-----------------|
| (i) $x + 7$, | (ii) $x - 5$ |
| (iii) $3x$ | (iv) $-3x$ |
| (v) $x/2$ | (vi) $-x/2$ |
| (vii) $3x + 10$ | (viii) $3x - 4$ |
| (ix) $10 - 3x$ | (x) $100 - 3x$ |
| (xi) $-x$. | |

Solution [Hint: Addition or Subtraction of a constant does not alter the S.D. of x .]

(i) S.D. of $(x + 7) =$ S.D. of $x = 6$

(ii) S.D. of $(x - 5) =$ S.D. of $x = 6$

[Hint: Multiplication or Division by a constant changes the S.D. similarly. The sign is ignored.]

- | | | |
|---------------------------|----------------------|---------------------|
| (iii) S.D. of $(3x)$ | $= 3$ (S.D. of x) | $= 3 \times 6 = 18$ |
| (iv) S.D. of $(-3x)$ | $= 3$ (S.D. of x) | $= 3 \times 6 = 18$ |
| (v) S.D. of $(x/2)$ | $=$ (S.D. of x)/2 | $= 6/2 = 3$ |
| (vi) S.D. of $(-x/2)$ | $=$ (S.D. of x)/2 | $= 6/2 = 3$ |
| (vii) S.D. of $(3x + 10)$ | $= 3$ (S.D. of x) | $= 3 \times 6 = 18$ |
| (viii) S.D. of $(3x - 4)$ | $= 3$ (S.D. of x) | $= 3 \times 6 = 18$ |
| (ix) S.D. of $(10 - 3x)$ | $= 3$ (S.D. of x) | $= 3 \times 6 = 18$ |
| (x) S.D. of $(100 - 3x)$ | $= 3$ (S.D. of x) | $= 3 \times 6 = 18$ |
| (xi) S.D. of $(-x)$ | $=$ S.D. of x | $= 6$ |

[Note: (1) If $y = a + bx$, where a and b are constants, then S.D.(y) = $|b| \cdot$ S.D.(x).

(2) This result holds also for Range, Q.D. and M.D.]



Problem 6.14 (a) Calculate S.D. of the sample 3, 5, 8, 7, 5, 7. Without any further calculations, write down the S.D. of the following two samples:

(i) 25, 27, 30, 29, 27, 29

(ii) 6, 10, 16, 14, 10, 14

[C.U., B. Com. (Hons) '83]



(b) The S.D. of a variable x is 10. Find the S.D. of the variable $(5x - 10)$.

[W.B.H.S. '78]

(c) Find the S.D. of the following data:

x : 0, 1, -1, -2, 6, 4, 5, 8, 12, 10, 11

Hence obtain the Variance of the following numbers:

y : -1, 1, -3, -5, 11, 7, 9, 15, 23, 19, 21

[I.C.W.A., June '86]

Solution Since S.D. is unaffected by change of origin, we subtract a constant 5 from each observation. $d = x - 5$.

x	d	d^2
3	-2	4
5	0	0
8	3	9
7	2	4
5	0	0
7	2	4
Total	5	21

$$\begin{aligned}\sigma^2 &= \Sigma d^2/n - (\Sigma d/n)^2 \\ &= 21/6 - (5/6)^2 = 101/36\end{aligned}$$

$$\therefore \sigma = \sqrt{101/6} = 10.05/6 = 1.675$$

(i) We see that these values are obtained by adding a constant 22 to each observation. Since S.D. is unaffected if each observation is increased or decreased by a constant, $\sigma = 1.675$.

(ii) We see that these values are obtained on multiplying each observation by a constant 2. Hence S.D. will also be multiplied by 2.

$$\sigma = 2 \times 1.675 = 3.35$$

(b) If $y = a + bx$, then $\sigma_y = |b| \cdot \sigma_x$

Hence if we write $y = 5x - 10$ (here $a = -10$, $b = 5$)

$$\sigma_y = 5 \cdot \sigma_x = 5 \times 10 = 50$$

(c) $n = 11$

$$\Sigma x = 0 + 1 - 1 - 2 + 6 + 4 + 5 + 8 + 12 + 10 + 11 = 54$$

$$\Sigma x^2 = 0 + 1 + 1 + 4 + 36 + 16 + 25 + 64 + 144 + 100 + 121 = 512$$

$$\begin{aligned}\sigma_x^2 &= \Sigma x^2/n - (\Sigma x/n)^2 \\ &= 512/11 - (54/11)^2 = 2716/121 = 22.45\end{aligned}$$

$$\therefore \sigma_x = \sqrt{22.45} = 4.74$$

It will be observed that the given values of y are obtained on multiplying x by 2 and then subtracting 1 from it.

$$y = 2x - 1; \therefore \text{S.D. (y)} = 2 \times \text{S.D. (x)}$$

$$\text{Squaring, Variance (y)} = 4 \times \text{Variance (x)} = 4 \times 22.45 = 89.8$$

Ans. 4.74, 89.8



Problem 6.15 (a) What is the S.D. of the two values 10 and 7?

(b) S.D. of 2 quantities is 3. If one of them is 7, find the other.

(c) Write down the S.D. of the six numbers: 5, 5, 5, 7, 7, 7. [W.B.H.S. '80]

(d) For 2 values of a variable x , the mean and S.D. are 10 and 2 respectively. What are the values of x ? [B.U., B.A. (Econ.) '80]

Solution (a) S.D. of two values = Half their difference

$$= \frac{1}{2} \times 3 = 1.5$$

(b) If the other value is a , then

$$\frac{1}{2} (\text{difference of } 7 \text{ and } a) = 3$$

$$\therefore \text{Difference of } 7 \text{ and } a = 6$$

(Note that a may be larger or smaller than 7)

$$\therefore a = 1 \text{ or } 13.$$

(c) There are 6 observations in which the 2 values 5 and 7 occur with equal frequency. Therefore

$$\text{S.D. of } 5, 5, 5, 7, 7, 7 = \text{S.D. of } 5 \text{ and } 7$$

$$= \frac{1}{2} (\text{Difference of } 5 \text{ and } 7)$$

$$= \frac{1}{2} \times 2 = 1$$

(d) Let the two values be a and b ($b \geq a$)

$$\text{Then Mean} = \frac{a+b}{2} = 10 \text{ (given)}$$

$$\text{S.D.} = \frac{b-a}{2} = 2 \text{ (given), since } b \text{ is larger, } b-a \text{ is positive.}$$

$$\therefore a+b=20, b-a=4$$

$$\text{Adding, } 2b=24. \text{ So, } b=12. \therefore a=8.$$

Ans. 8 and 12.

Problem 6.16 (a) Write down the algebraic expression for root-mean-square-deviation from 10.

(b) For a set of observations, variance of x turns out to be 5.95. What would be the minimum value of $\Sigma(x_i - A)^2/n$?

(c) When Variance (x) = 0, what can you say about the set of observations?

(d) What is the unit of variance ?

$$\text{Solution (a) } \sqrt{\Sigma(x_i - 10)^2/n}$$

(b) Minimum value = Variance = 5.95,

Because, S.D. = Minimum value of Root-Mean-Square-Deviation

$\therefore$ Variance = Minimum value of Mean-Square-Deviation.

(c) All the values of x are equal.



(d) Square of the unit in which the observations are given. For example, if observations are given in Rs, Variance will be in (Rs)².



Problem 6.17 (a) If the relation between two variables x and y is $5x - 7y = 11$ and range of x is 5, find the range of y . [W.B.H.S., '82]

(b) If the relation between 2 variables x and y is $2x + 3y = 7$, and quartile deviation of x is 3, then write down the Q.D. of y . [W.B.H.S. '83]

(c) If the relation between the variables y and v is $5v + 2y = 17$, and the mean deviation of y about its mean 6 is 5, write down the M.D. of v about its mean. [W.B.H.S. '86]

(d) The mean and S.D. of a variable x are known to be 38 and 6 respectively. What are the means and S.D.s of

(i) $100 - 2x$ (ii) $60 - x/2$ [B.U., B.A. (Econ) '82]

Solution [Note: If $Ax + By + C = 0$, then

$|A|$ Dispersion (x) = $|B|$ Dispersion (y) for any absolute measure of Dispersion.]

(a) $5 \times \text{Range } (x) = 7 \times \text{Range } (y)$

or, $5 \times 5 = 7 \times \text{Range } (y); \therefore \text{Range } (y) = 25/7$

(b) $2 \times \text{Q.D. } (x) = 3 \times \text{Q.D. } (y)$

or, $2 \times 3 = 3 \times \text{Q.D. } (y); \therefore \text{Q.D. } (y) = 2$

(c) $5 \times \text{M.D. } (v) = 2 \times \text{M.D. } (y)$ for M.D. about mean.

$= 2 \times 5 = 10. \therefore \text{M.D. } (v) = 2$

(d) If $y = a + bx$, then $\bar{y} = a + b\bar{x}$ and $\sigma_y = |b| \cdot \sigma_x$

Here $\bar{x} = 38, \sigma_x = 6$

(i) $y = 100 - 2x; \bar{y} = 100 - 2 \times 38 = 24$

$\sigma_y = 2 \times 6 = 12$

(ii) $y = 60 - \frac{1}{2}x;$

$y = 60 - \frac{1}{2} \times 38 = 41$

$\sigma_y = \frac{1}{2} \times 6 = 3$

(Also see Problem 6.13, p. 439)



Problem 6.18 Are the following statements consistent? Justify your answer.

(a) When all observations are same, any measure of dispersion would be 1.0 (one).

(b) If $x_i = 5 + y_i$ then $R(x) = 5 + R(y)$, where R is the Range of the variable concerned.

(c) Mean Absolute Deviation from median and mean are 5 and 4 respectively.

(d) $V(x) = \sum x_i^2/n - \bar{x}^2 = -30; x_i, \bar{x}$ and n are the usual notation.

(e) The unit of Variance is that of the observations. [C.U., M.Com., '82]



Solution (a) No; it is zero.

Range = 0, because maximum and minimum values are equal.

Q.D. = 0, because then $Q_3 = Q_1$ = common observation.

M.D. = 0; because mean is also the same as the common observation, so that $x_i - \bar{x} = 0$.

S.D. = 0; because $x_i - \bar{x} = 0$.

C.V. and all other relative measures contain an absolute measure of dispersion in the numerator, which is zero. So, any relative measure of dispersion = 0.

(b) No; $R(x) = R(y)$. Because Range (in fact, any absolute measure of dispersion) is unaffected by change of origin.

(c) No; because M.D. about median should be smaller than about mean (or any other value).

(d) No; because Variance cannot be negative.

(e) No; the unit of variance is the square of the unit in which observations are given.



Problem 6.19 The range, standard deviation and coefficient of variation in respect of a set of values are respectively 15, 4.38 and 7.78%. How would these results be affected if each value of the set be increased by 5? [I.C.W.A., Dec. '85]

Solution Original series: Range = 15, S.D. = 4.38

$$\text{C.V.} = 7.78\%; \quad \text{i.e. } \frac{4.38}{\text{mean}} \times 100 = 7.78;$$

$$\therefore \text{mean} = \frac{4.38 \times 100}{7.78} = 56.3$$

New Series: If each observation is increased by 5,

(i) Range and S.D. do not change (because absolute measures of dispersion are unaffected by change of origin).

$$\text{Range} = 15, \quad \text{S.D.} = 4.38$$

(ii) Mean increases by 5. Mean = $56.3 + 5 = 61.3$

$$\therefore \text{C.V.} = \frac{4.38}{61.3} \times 100 = 7.15 \quad \text{Ans. 15, 4.38, 7.15\%}$$



Problem 6.20 If the third quartile is 30 and median 22, find the coefficient of quartile deviation, assuming the distribution to be symmetrical.

[I.C.W.A., June '86]

Solution For a symmetrical distribution

$$\text{Median} = \frac{1}{2} (Q_1 + Q_3)$$

$$\text{or, } 22 = \frac{1}{2} (Q_1 + 30); \quad \therefore Q_1 = 14$$

$$\text{Hence, Q.D.} = \frac{1}{2} (Q_3 - Q_1) = \frac{1}{2} (30 - 14) = 8$$

$$\text{Coeff. of Q.D.} = \frac{\text{Q.D.}}{\text{Median}} \times 100 = \frac{8 \times 100}{22} = 36.4$$



ADDITIONAL EXERCISES 6

(Q.D. = Quartile Deviation, M.D. = Mean Deviation,
S.D. = Standard Deviation, C.V. = Coefficient of Variation)

1. Find Range and Q.D. for the following data:

<i>Variable</i>	3	5	7	9	11	13
<i>Frequency</i>	2	7	10	9	6	1

2. Calculate Q.D. from the following:

<i>Weight (kg.)</i>	14–18	18–22	22–26	26–30	30–34	Total
<i>Frequency</i>	15	40	25	13	7	100

[K.U., B.Com. (Pass) '85]

3. Calculate M.D. about median from the following data:

(a) (Rs) 46, 79, 26, 85, 39, 65, 29, 59, 73.

[C.U., B.Com. (Pass) '85]

(b) 79, 85, 99, 46, 56, 72, 65, 32, 26, 36.

[I.C.W.A., June '85]

4. Calculate Median and the M.D. about median from the following:

<i>x</i>	5	6	7	8	9
<i>y</i>	1	4	7	6	2

[K.U., B.Com.(Pass)'80]

5. Find Mean Absolute Deviation for the following:

<i>Height (inches)</i>	60–62	63–65	66–68	69–71	72–74
<i>Frequency</i>	5	18	42	27	8

[C.U., B.Sc. (Econ.) '83]

6. Find S.D. from the following data:

<i>x</i>	10	20	30	40	50	60	Total
<i>f</i>	9	18	25	27	14	7	100

[C.U., B.Com. (Pass) '84, M.Com. '82]

7. Calculate S.D. from the following distribution of weights (in gm) of Zinc coating:

<i>Cell Boundaries</i>	<i>Frequency</i>
12.75–13.25	3
13.25–13.75	7
13.75–14.25	15
14.25–14.75	11
14.75–15.25	8
15.25–15.75	4
15.75–16.25	2
Total	50

[C.U., B.Com. (Hons.) '83]



8. Calculate Mean and S.D. from the following data:

<i>Value</i> :	90–99	80–89	70–79	60–69	50–59	40–49	30–39	Total
<i>Frequency</i> :	2	12	22	20	14	2	1	75

[I.C.W.A., Dec. '81]

9. Calculate S.D. of the following frequency distribution showing Wages in a sample of factory workers in Calcutta, 1980:

<i>Weekly Wage (Rs)</i>	<i>No. of Employees</i>
30–34.99	3
35–39.99	9
40–44.99	15
45–49.99	27
50–54.99	18
55–59.99	12
60–64.99	9
65–69.99	7

[C.U., B.Sc. (Econ.) '82]

10. After change of origin and scale, a frequency distribution of a continuous variable takes the form as under:

<i>Step Deviation (u)</i> :	– 3	– 2	– 1	0	1	2	3
<i>Frequency (f)</i> :	10	15	25	25	10	10	5

If the mean and variance of the distribution are respectively 31 and 254, find the original frequency distribution.

[I.C.W.A., Dec. 85]

[Hint: Page 382].

11. Calculate S.D. stating the necessary assumption:

<i>Heights in Inches</i>	<i>No. of Men (hundred)</i>
Under 62	5
62 and under 63	10
63 and under 64	10
64 and under 65	30
65 and over	45

[I.C.W.A., June '86]

12. Calculate mean and S.D. from the following data:

<i>Age</i>	<i>No. of Persons Dying</i>	<i>Age</i>	<i>No. of Persons Dying</i>
Under 10	15	Under 50	100
Under 20	30	Under 60	110
Under 30	53	Under 70	115
Under 40	75	Under 80	125

[Hint: —Table 3, Page 359].

[I.C.W.A., June '80]



13. Find the number of items lying within $\text{Mean} \pm \sigma$ of following distribution:

<i>Class Interval</i>	<i>Frequency</i>	<i>Class Interval</i>	<i>Frequency</i>
11–12	5	21–22	395
13–14	426	23–24	38
15–16	720	25–26	8
17–18	741	27–28	5
19–20	665	29–30	7

[Hint: Example 6.34, p. 199],

[I.C.W.A., Dec. '82]

14. For a number of 21 observations, the mean and s.d. are 10 and 2 respectively. It was found after the calculations were made that one of the observations recorded as 7 was incorrect. Find the s.d. of 20 observations, if the incorrect observation is omitted.
[C.U., B.Sc. (Econ) '86]
15. For a number of $N = 10$ observations it was found that mean was 20 and S.D. $= \sqrt{35}$. After these computations were performed it was found that observations 5 and 15 were incorrect. Find the mean and S.D. if these two incorrect observations are omitted.
[C.U., M.Com. '82]
16. In a certain factory,
Number of employees = 100
Average wages per employee per month = 85 Rs
S.D. of the wages per employee per month = 4 Rs
Suppose the wages of an employee were wrongly noted as Rs 120 instead of Rs 100. What would be the correct S.D.? [W.B.H.S. '84]
17. For two values, say a and b ($a < b$), of a variable the mean and the s.d. are respectively 17 and 3. Find a and b . [B.U., B.A. (Econ) '82]
18. Find the missing figures from the following table:

	<i>No. of Observations</i>	<i>Mean</i>	<i>Variance</i>
Group I:	—	45	—
Group II:	150	—	144
Combined Group of I and II:	250	51	130

[I.C.W.A., Dec. '85]

19. The mean and variance of 200 values are 60 and 400 respectively. If at the time of calculation, two values were taken as 3 and 67 in place of 13 and 17, find the correct Coefficient of Variation.
[I.C.W.A., June '86]
20. Using the values of medians and quartiles given below for two distributions A and B , examine which is more variable:
For A : $Q_1 = 22.65$, Median = 29.45, $Q_3 = 31.70$
For B : $Q_1 = 192.72$, Median = 242.65, $Q_3 = 292.69$

[I.C.W.A., Dec. '85]



SHORT ANSWERS TYPE QUESTIONS



21. (a) Find Range of the daily wages of 10 persons:
(Rs) 24, 18, 25, 16, 20, 28, 22, 17, 21, 27
[C.U., B.Com(P) '84]
- (b) What would be the value of Range, if each observation is
(i) increased by 5, (ii) decreased by 3,
(iii) multiplied by 2, (iv) multiplied by -2,
(v) divided by 10, (vi) multiplied by 2 and then 3 added to it ?
22. Find Semi-interquartile Range from the following data:
4, -1, 8, 0, -3, 18, 7.
23. Find M.D. (i) about 7, (ii) about median, (iii) about mean, from the following data: 4, 0, 8, 6, -3.
24. Find Standard Deviation and Coefficient of Variation from the following:
41, 37, 45, 43, 34.
25. What is the S.D. of two observations (in cm) 53 and 57?
26. Calculate (i) Range, (ii) Quartile Deviation, (iii) Mean Deviation, (iv) Standard Deviation, and (v) Coefficient of Variation, from the following:
- | | | | | | |
|-----|----|----|----|----|-------|
| x | 40 | 50 | 60 | 70 | Total |
| f | 3 | 7 | 4 | 1 | 15 |
27. Calculate (i) Range, (ii) Quartile Deviation, (iii) Mean Deviation, (iv) Standard Deviation, and (v) Coefficient of Variation, from the following:
- | | | | | | |
|-------------|-----|-------|-------|-------|-------|
| Weight (kg) | 6-9 | 10-13 | 14-17 | 18-21 | Total |
| Frequency | 2 | 10 | 5 | 3 | 20 |
28. A student obtained the following results. Do you think that the computations are correct?
(a) $n = 10$, $\Sigma x = 100$, $\Sigma x^2 = 800$
(b) $\Sigma f = 200$, $\Sigma fx = 1000$, $\Sigma fx^2 = 4800$
29. Weekly average wage of workers in a factory increases from Rs 8.00 to Rs 12.00 and s.d. increases from Re 1 to Rs 5. Have the wages become less uniform now?
[I.C.W.A., Dec. '82]
30. Two samples A and B have the same s.d., but the mean of A is greater than that of B . The Coefficient of Variation of A is: (i) Greater than that of B , (ii) Less than that of B , (iii) Equal to that of B , (iv) None of these. Which is correct?
[C.A., Nov. '82]
31. If the third quartile is 178 and median is 160, find the coefficient of Quartile Deviation, assuming the distribution to be symmetrical. [I.C.W.A., June '85]
32. If the quartiles of a distribution are 28, 35 and 40, find Semi-interquartile Range.
33. In a certain distribution the sum of the two quartiles is 20 and Quartile Deviation is 4. Find the quartiles.



34. The mean and s.d. calculated from 9 observations are 17 and 3 respectively. If an additional observation 27 is included, find the mean and s.d. of all the 10 observations. [K.U., B.Com(P) '83]
35. The first of two groups has 100 items with mean 45 and variance 49. If the combined group has 250 items with mean 51 and variance 130, find the mean and s.d. of the second group. [I.C.W.A., June '84]
36. If $x_i = 99 + y_i$ and Range (x) = 99, find Range (y).
37. If the Quartile Deviation (Q.D.) of x is 3, what is the Q.D. of (i) $5 + 4x$, (ii) $5 - 4x$?
38. (a) If $2x + 3y + 9 = 0$ and range of x is 12, find range of y .
 (b) If two variables x and y are related by $7x - 2y + 5 = 0$ and M.D. (about mean) of x is 4, find M.D. of y .
39. (a) Two variables x and y are given by $y = 11 - 5x$. If the s.d. of x is 4, write down the s.d. of y . [W.B.H.S. '81]
 (b) If $x + 3y - 7 = 0$ and s.d. of y is 5, what is the s.d. of x ?
 (c) If $x = \frac{3}{5}(y - 10)$ and Variance (x) = $\frac{1}{5}$, find Variance (y).
40. If the s.d. of 1, 2, 3, 4, 5, 6, 7 is known to be 2, what is the s.d. of 2, 4, 6, 8, 10, 12, 14? Hence write down the s.d. of 1, 3, 5, 7, 9, 11, 13.
41. Weights of 10 students are recorded by a machine as under: (kg) 50, 55, 57, 49, 54, 61, 64, 59, 58, 56. If the weighing machine shows weight less by 5 kgs, find the correct values of Range, S.D. and C.V. without calculating the correct weights.
 (Hint: Range and S.D. do not change; but mean increases by 5 kg. See Problem 6.19, page 390) [I.C.W.A., June '84]
42. Examine each of the following statements and state whether it is True or False:
 (a) If each observation decreases by 10, Range decreases by 10.
 (b) Q. D. is independent of origin.
 (c) M.D. may be negative
 (d) S.D. does not depend on the scale of measurement.
 (e) Coefficient of Variation is unit-free.
 (f) If each observation is multiplied by 3, coefficient of Q.D. will not change.
43. Fill up the missing parts:
 (a) Range of a given set of observations is defined as
 (b) Q.D. is defined as
 (c) M.D. about an arbitrary origin A is defined as
 (d) S.D. of a given set of observations is defined as
 (e) Coefficient of Variation is defined as
 (f) Coefficient of Quartile Deviation is defined as



44. Fill up the missing parts:
- M.D. is minimum when deviations are taken about ...
 - Root-Mean-Square-Deviation is minimum when deviations are taken about ...
 - S.D. is ... than M.D. about mean.
 - S.D. of two observations is equal to ... their differences.
 - S.D. is ... when all observations are equal.
 - S.D. of one observations is ...
45. Fill up the blanks:
- The quickest measure of dispersion is ...
 - The appropriate measure of dispersion from a frequency distribution with open-end class is ...
 - The measure of dispersion which is based on all observation and also easy to understand is ...
 - The most useful measure of dispersion is ...
 - For comparing the variability of two or more distributions whose means differ widely, the appropriate measure of dispersion is ...
 - The most widely-used relative measure of dispersion is ...



ANSWERS



- 10, 2
- 3.1 kg
- Rs 18, 20.4
- 7, 0.8
- 2.265 inch
- 13.56
- 0.073 gm
- 68.1, 12.5
- Rs 8.76
- Class intervals 0 – 10, 10 – 20, ... 60 – 70
- 1.18 inch (assuming equal width for all classes)
- 35.16, 19.8
- 1929
- 1.93
- 22.5, 2.5
- Rs 2.44
- 14, 20
- 100, 55, 49
- 33.6
- B (Coeff. of Q.D. larger)
- (a) 12 Rs; (b) (Rs) 12, 12, 24, 24, 1.20, 24
- 4.5
- 4.4, 3.4, 3.6
- 4, 10
- 2 cm
- 30, 5, 6.67, 8.33, 16
- 16 kg, 2.6 kg, 2.96 kg, 3.46 kg, 26
- No (see Problem 6.12)
- Yes (use C.V.)
- (ii)
- 11.25
- 6
- 6, 14
- 18, 4.1
- 55, 12
- 99
- 12, 12
- 8, 14
- 20, 15, 5/9
- 4, 4
- 15 kg, 4.38 kg, 7.15
- True—(b), (e), (f); False—(a), (c), (d)



43. (a) the difference between the largest and the smallest items.
(b) half difference between the upper and lower quartiles.
(c) the mean of absolute deviations from A .
(d) the root-mean-square-deviation from mean; (i.e. the square-root of the mean of squared deviations from mean).
(e) the ratio of S.D., to A.M., multiplied by 100.
(f) the ratio of Q.D. to Median, multiplied by 100.
44. (a) median, (b) mean, (c) not smaller, (d) half, (e) and (f) zero.
45. (a) Range, (b) Q.D., (c) M.D., (d) S.D., (e) C.V., (f) C.V

CHAPTER 7

MOMENTS, SKEWNESS AND KURTOSIS

(Summary)

Moment Given n observations $x_1, x_2, \dots, x_n$ the “ r -th moment about an arbitrary origin A ” is defined by the formula

$$m_r = \frac{1}{n} \sum (x - A)^r; \quad r = 0, 1, 2, 3, \dots$$

Thus the r -th moment about A is the mean of the r -th powers of deviations from A .

Moments about zero are called “Raw Moments”, and moments about mean are called “Central moments”.

$$r\text{-th raw moment:} \quad m'_r = \frac{1}{n} \sum x^r$$

$$r\text{-th central moment:} \quad m_r = \frac{1}{n} \sum (x - \bar{x})^r$$

[**Note:** For a frequency distribution, the formulae can be written down easily: Change n to N , and bring f after $\sum$ sign.]

Important Results

(1) Mean $(\bar{x}) = 1\text{st raw moment } (m'_1)$

Variance $(\sigma^2) = 2\text{nd central moment } (m_2)$

(2) If the 1st and 2nd moments about any origin A are known,

$$\text{Mean } (\bar{x}) = A + m'_1$$

$$\text{S.D. } (\sigma) = \sqrt{m'_2 - m'^2_1}$$

(3) Relation between central moment and non-central moments:

$$m_1 = 0 \text{ (always)}$$

$$m_2 = m'_2 - m'^2_1$$

$$m_3 = m'_3 - 3m'_2m'_1 + 2m'^3_1$$

$$m_4 = m'_4 - 4m'_3m'_1 + 6m'_2m'^2_1 - 3m'^4_1$$

(4) Moments can be used to measure all the four characteristics of statistical data:

(i) Central tendency : Mean = 1st raw moment (m'_1)

(ii) Dispersion : S.D. = $\sqrt{m_2}$

(iii) Skewness : Skewness = $m_3 / (\sqrt{m_2})^3$

(iv) Kurtosis : Kurtosis = $m_4 / (m_2)^2 - 3$.

(5) Moments of odd order (e.g. 1st moment, 3rd moment) may be positive, negative or zero; but moments of even order (e.g. 2nd moment, 4th moment) can never be negative.



- (6) If observations are given in inch, then the 1st moment has unit 'inch', 2nd moment has unit 'square inch', 3rd moment has unit 'cubic inch', and so on.
- (7) The central moments m_2, m_3, m_4 etc. are independent of origin, but depend on scale. Thus, if $y = (x - c)/d$, then

$$m_2 = d^2 \cdot (\text{2nd central moment of } y),$$

$$m_3 = d^3 \cdot (\text{3rd central moment of } y); \text{ etc.}$$

Standardized Variable

If the variable x has mean $\bar{x}$ and standard deviation σ , then the variable z defined by

$$z = \frac{x - \bar{x}}{\sigma}$$

is called the 'Standardized variable'. The mean and s.d. of z are always 0 and 1 respectively.

$$r\text{-th moment of } z = m_r / \sigma^r$$

(Note that 3rd and 4th moments of z are used to measure "skewness" and "kurtosis" respectively.)

Skewness

"Skewness" denotes the degree of asymmetry (i.e. lack of symmetry) in the data. It is used to indicate the 'shape' of the frequency curve as regards its distortion from symmetry. Skewness may be positive, negative or zero.

Measures of Skewness

- (1) Pearson's 1st measure = $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$
- (2) Pearson's 2nd measure = $\frac{3(\text{Mean} - \text{Median})}{\text{S.D.}}$
- (3) Bowley's measure = $\frac{Q_3 - 2 \cdot Q_2 + Q_1}{Q_3 - Q_1}$
- (4) Moment measure = $\frac{m_3}{\sigma^3} = \frac{m_3}{(\sqrt{m_2})^3}$

When skewness is positive, distribution is called "positively skew"

When skewness is negative, distribution is called "negatively skew"

When skewness is zero, distribution is called "symmetrical"

Important Results

- (a) All measures of skewness are pure numbers (i.e. unit-free)
- (b) (i) For symmetrical distributions (i.e. skewness = 0)
Mean = Median = Mode; $Q_2 = (Q_1 + Q_3)/2$; $m_3 = 0$. The frequency curve is symmetrical.
- (ii) For positively skew distributions,
Mean > Median > Mode; m_3 positive.
The frequency curve has the longer tail towards the right.



(iii) For negatively skew distribution,

Mean < Median < Mode ; m_3 negative

The frequency curve has the longer tail towards the left.

(c) Bowley's measure always lies between -1 and $+1$.

Kurtosis

"Kurtosis" denotes the degree of 'peakedness' of the frequency curve. It is used to indicate the 'shape' of the frequency curve as regards the sharpness of its peak. Kurtosis may be positive, negative or zero.

The only measure of kurtosis is based on moments.

$$\text{Kurtosis} = \frac{m_4}{\sigma^4} - 3 = \frac{m_4}{(m_2)^2} - 3$$

When kurtosis is positive, distribution is called "leptokurtic"

When kurtosis is negative, distribution is called "platykurtic"

When kurtosis is zero, distribution is called "mesokurtic"

Important Results

(a) The measure of kurtosis is a pure number (i.e. unit-free).

(b) The frequency curve of

"leptokurtic" distribution has a sharp peak;

"platykurtic" distribution has a blunt peak;

"mesokurtic" distribution has a moderate peak

(c) "Normal distribution" (Chapter 12) is symmetrical and mesokurtic (i.e. skewness = 0, kurtosis = 0).



SOLVED PROBLEMS



Problem 7.1 Find the first, second and third moments about 4 of the set of numbers 2, 4, 6, 8. [I.C.W.A., June' 80]

Solution

x	$x - 4$	$(x - 4)^2$	$(x - 4)^3$
2	-2	4	-8
4	0	0	0
6	2	4	8
8	4	16	64
Total	4	24	64

Moments about 4:

$$m'_1 = \frac{\Sigma(x - 4)}{n} = \frac{4}{4} = 1$$

$$m'_2 = \frac{\Sigma(x - 4)^2}{n} = \frac{24}{4} = 6$$



$$m'_3 = \frac{\Sigma(x-4)^3}{n} = \frac{64}{4} = 16$$

Ans. 1, 6, 16.



Problem 7.2 Find the 1st, 2nd and 3rd raw moments of 2, 4, 6, 8.

Solution

x	x^2	x^3
2	4	8
4	16	64
6	36	216
8	64	512
20	120	800

Raw moments:

$$m'_1 = \frac{\Sigma x}{n} = \frac{20}{4} = 5$$

$$m'_2 = \frac{\Sigma x^2}{n} = \frac{120}{4} = 30$$

$$m'_3 = \frac{\Sigma x^3}{n} = \frac{800}{4} = 200$$

Ans. 5, 30, 200.



Problem 7.3 Find the first four moments about mean for the set of numbers 2, 4, 6, 8.
[I.C.W.A., June' 82, '83]

Solution

x	$x - \bar{x}$	$(x - \bar{x})^2$	$(x - \bar{x})^3$	$(x - \bar{x})^4$
2	-3	9	-27	81
4	-1	1	-1	1
6	1	1	1	1
8	3	9	27	81
20	0	20	0	164

Moments about mean:

$$m_1 = \frac{\Sigma(x - \bar{x})}{n} = \frac{0}{4} = 0$$

$$m_2 = \frac{\Sigma(x - \bar{x})^2}{n} = \frac{20}{4} = 5$$

$$m_3 = \frac{\Sigma(x - \bar{x})^3}{n} = \frac{0}{4} = 0$$



$$m_4 = \frac{\Sigma(x - \bar{x})^4}{n} = \frac{164}{4} = 41$$

Mean ($\bar{x}$) = $20/4 = 5$.

Ans. 0, 5, 0, 41.



Problem 7.4 The first three moments of a distribution about the value 2 of a variable x are 1, 16 and -40 . Find the mean, variance and the third central moment.

[C.U., B.Com. (Hons.) '84]

Solution Give, Moments about $A = 2$,

$$m'_1 = 1, m'_2 = 16, m'_3 = -40.$$

$$\text{Mean } (\bar{x}) = A + m'_1 = 2 + 1 = 3$$

$$\text{Variance } (m_2) = m'_2 - (m'_1)^2 = 16 - 1^2 = 15$$

$$\text{3rd Central Moment } (m_3) = m'_3 - 3m'_2 m'_1 + 2(m'_1)^3$$

$$= -40 - 3 \times 16 \times 1 + 2 \times 1^3$$

$$= -40 - 48 + 2$$

$$= -86.$$

Ans. 3, 15, -86 .



Problem 7.5 Find the first, second and third moments about 7 from the following data:

Values	2	4	6	8	10	12	Total
Frequency	6	9	7	15	7	6	50

Hence find the 1st, 2nd and 3rd central moments. [C.U., B.Com. (Pass) '81]

Solution

x	f	$d = x - 7$	fd	fd^2	fd^3
2	6	-5	-30	150	-750
4	9	-3	-27	81	-243
6	7	-1	-7	7	-7
8	15	1	15	15	15
10	7	3	21	63	189
12	6	5	30	150	750
Total	50	-	2	466	-46

Moments about 7: Writing $d = x - 7$,

$$m_1 = \frac{\Sigma fd}{N} = \frac{2}{50} = 0.04$$

$$m_2 = \frac{\Sigma fd^2}{N} = \frac{466}{50} = 9.32$$

$$m_3 = \frac{\Sigma fd^3}{N} = \frac{-46}{50} = -0.92$$



Central Moments:

$$\begin{aligned}
 m_1 &= 0 \text{ (always)} \\
 m_2 &= m_2' - (m_1')^2 \\
 &= 9.32 - (0.04)^2 = 9.32 - .0016 = 9.3184 \\
 m_3 &= m_3' - 3m_2m_1 + 2(m_1')^3 \\
 &= -0.92 - 3 \times 9.32 \times 0.04 + 2(0.04)^3 \\
 &= -0.92 - 1.1184 + .000128 \\
 &= -2.038,272
 \end{aligned}$$

[**Note:** Odd order moments like m_1' , m_3' , m_5' may be positive negative or zero; but even order moments like m_2' , m_4' , m_6' can never be negative.]



Problem 7.6 Find the first, second and third moments about 3 from the following distribution:

Class interval	0–2	2–4	4–6	6–8	Total
Frequency	3	8	5	4	20

Also, calculate the third central moment.

[K.U., B.Com. (Pass)' 85]

Solution

Midvalue (x)	frequency (f)	d = x – 3	fd	fd ²	fd ³
1	3	– 2	– 6	12	– 24
3	8	0	0	0	0
5	5	2	10	20	40
7	4	4	16	64	256
Total	20	–	20	96	272

Moment about 3:

$$\begin{aligned}
 m_1 &= 20/20 = 1 \\
 m_2' &= 96/20 = 4.8 \\
 m_3' &= 272/20 = 13.6
 \end{aligned}$$

Third Central Moment:

$$\begin{aligned}
 m_3 &= m_3' - 3m_2'm_1 + 2m_1'^3 \\
 &= 13.6 - 3 \times 4.8 \times 1 + 2 \times 1^3 \\
 &= 13.6 - 14.4 + 2 \\
 &= 1.2
 \end{aligned}$$

Ans. 1, 4.8, 13.6 ; 1.2



Problem 7.7 In a distribution the difference of two quartiles is 2.03; their sum is 72.67 and the median is 36.18. Find the coefficient of Skewness.

[I.C.W.A., June '81 ; C.U., B.Com (Hons) '85]

Solution Since the third quartile (Q_3) is larger than the first quartile (Q_1), we have $Q_3 - Q_1 = 2.03$. Also $Q_3 + Q_1 = 72.67$ and median (Q_2) = 36.18. Using Bowley's formula (7.7.3, p. 234)



$$\text{Skewness} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = \frac{72.67 - 2 \times 36.18}{2.03} = +0.15.$$



Problem 7.8 The central moments of a distribution are given by $m_2 = 140$, $m_3 = 148$, $m_4 = 6030$. Calculate the moment measures of skewness and kurtosis, and comment on the shape of the distribution. [I.C.W.A., Dec. '84]

Solution

$$\begin{aligned} \text{Skewness} &= \frac{m_3}{(\sqrt{m_2})^3} = \frac{148}{(140)^{3/2}} = \frac{148}{140\sqrt{140}} \\ &= \frac{148}{1656.5} = +0.089 \\ \text{Kurtosis} &= \frac{m_4}{m_2^2} - 3 = \frac{6030}{19600} - 3 = -2.692 \end{aligned}$$

The distribution is *positively skew* and *playkurtic* (kurtosis negative).



Problem 7.9 The first four moments of a distribution about origin are 1, 4, 10 and 46 respectively. Obtain the various characteristics of the distribution on the basis of the information given. Comment upon the nature of the distribution. [I.C.W.A., June '84]

Solution Given Moments about zero (i.e. Raw Moments):

$$m'_1 = 1, m'_2 = 4, m'_3 = 10, m'_4 = 46.$$

The various characteristics, namely (1) Central tendency, (2) Dispersion, (3) Skewness, and (4) Kurtosis, are measured by using 1st raw moment m'_1 and central moments m_2, m_3, m_4 .

$$\begin{aligned} m_2 &= m'_2 - m'_1{}^2 = 4 - 1^2 = 3 \\ m_3 &= m'_3 - 3m'_2 m'_1 + 2m'_1{}^3 \\ &= 10 - 3 \times 4 \times 1 + 2 \times 1^3 = 10 - 12 + 2 = 0 \\ m_4 &= m'_4 - 4m'_3 m'_1 + 6m'_2 m'_1{}^2 - 3m'_1{}^4 \\ &= 46 - 4 \times 10 \times 1 + 6 \times 4 \times 1^2 - 3 \times 1^4 \\ &= 46 - 40 + 24 - 3 = 27 \end{aligned}$$

$$\text{Mean} = \text{1st Raw Moment } (m'_1) = 1$$

$$\therefore \text{S.D. } (\sigma) = \sqrt{m_2} = \sqrt{3} = 1.732$$

$$\text{Skewness} = m_3/\sigma^3 = 0/(\sqrt{3})^3 = 0$$

$$\text{Kurtosis} = m_4/\sigma^4 - 3 = (27/9) - 3 = 0$$

Comment: The distribution is *symmetrical* (skewness = 0) and *mesokurtic* (kurtosis = 0).

Problem 7.10 The following values were obtained from a frequency distribution: $\Sigma f = 100$, $\Sigma fx = -100$, $\Sigma fx^2 = 5500$, $\Sigma fx^3 = -6250$ and $\Sigma fx^4 = 775000$. Comment on the nature of the distribution after calculating a measure of skewness and a measure of kurtosis. [I.C.W.A. (New), June '86]



Solution For a frequency distribution, the Raw Moments are

$$m'_1 = \Sigma fx/N = -100/100 = -1$$

$$m'_2 = \Sigma fx^2/N = 5500/100 = 55$$

$$m'_3 = \Sigma fx^3/N = -6250/100 = -62.5$$

$$m'_4 = \Sigma fx^4/N = 775000/100 = 7750$$

Using them, we find the Central Moments:

$$\begin{aligned} m_2 &= m'_2 - m_1'^2 \\ &= 55 - (-1)^2 = 55 - 1 = 54 \end{aligned}$$

$$\begin{aligned} m_3 &= m'_3 - 3m'_2 m'_1 + 2m_1'^3 \\ &= -62.5 - 3 \times 55 \times (-1) + 2(-1)^3 \\ &= -62.5 + 165 - 2 = 100.5 \end{aligned}$$

$$\begin{aligned} m_4 &= m'_4 - 4m'_3 m'_1 + 6m'_2 m_1'^2 - 3m_1'^4 \\ &= 7750 - 4(-62.5)(-1) + 6 \times 55(-1)^2 - 3(-1)^4 \\ &= 7750 - 250 + 330 - 3 = 7827 \end{aligned}$$

$$\text{Skewness} = \frac{m_3}{(\sqrt{m_2})^3} = \frac{100.5}{396.8} = 0.253$$

$$\text{Kurtosis} = \frac{m_4}{m_2^2} - 3 = \frac{7827}{2916} - 3 = -0.316$$

The distribution is “positively skew” and “platykurtic”.



Problem 7.11 Find the moment measures of skewness (γ_1) and kurtosis (γ_2) from the following distribution:

Class Interval	0–10	10–20	20–30	30–40	Total
Frequency	1	3	4	2	10

[I.C.W.A., Dec. '82, '84]

Solution

Mid-value (x)	f	$y = \frac{x-25}{10}$	fy	fy ²	fy ³	fy ⁴
5	1	-2	-2	4	-8	16
15	3	-1	-3	3	-3	3
25	4	0	0	0	0	0
35	2	1	2	2	2	2
Total	10	—	-3	9	-9	21

Raw Moments of y:

$$m'_1 = \Sigma fy/N = -3/10 = -0.3$$

$$m'_2 = \Sigma fy^2/N = 9/10 = 0.9$$

$$m'_3 = \Sigma fy^3/N = -9/10 = -0.9$$

$$m'_4 = \Sigma fy^4/N = 21/10 = 2.1$$



Central Moments of y :

$$m_2 = m_2' - (m_1')^2 \\ = 0.9 - (-0.3)^2 = 0.9 - 0.09 = 0.81;$$

$\therefore$

$$\sigma = \sqrt{0.81} = 0.9$$

$$m_3 = m_3' - 3m_2'm_1' + 2(m_1')^3 \\ = -0.9 - 3(0.9)(-0.3) + 2(-0.3)^3 \\ = -0.9 + 0.81 - 0.054 = -0.144$$

$$m_4 = m_4' - 4m_3'm_1' + 6m_2'(m_1')^2 - 3(m_1')^4 \\ = 2.1 - 4(-0.9)(-0.3) + 6(0.9)(-0.3)^2 - 3(-0.3)^4 \\ = 2.1 - 1.08 + 0.486 - 0.0243 \\ = 1.4817$$

Moment measure of Skewness and Kurtosis are unaffected by changes of origin and scale. Hence using the central moments of y ,

$$\text{Skewness} = m_3/\sigma^3 = -0.144/0.729 = -0.198$$

$$\text{Kurtosis} = (m_4/\sigma^4) - 3 = (1.4817/0.6561) - 3 = -0.742.$$



Problem 7.12 Derive the formula

$$m_3 = m_3' - 3m_2'm_1' + 2(m_1')^3$$

where m_1' , m_2' , m_3' denote moments about an arbitrary origin A .

Solution Let $x_1, x_2, \dots, x_n$ be the observations. Then

$$m_1' = \Sigma(x - A)/n = \bar{x} - A = d \text{ (suppose)} \quad \dots \text{ (i)}$$

$$m_2' = \Sigma(x - A)^2/n, \quad m_3' = \Sigma(x - A)^3/n \quad \dots \text{ (ii)}$$

$$\text{3rd central moment } m_3 = \Sigma(x - \bar{x})^3/n$$

$$\text{We can write } (x - \bar{x}) = (x - A) - (\bar{x} - A) \\ = (x - A) - d, \text{ from (i)}$$

$$\therefore (x - \bar{x})^3 = \Sigma(x - A)^3 - 3(x - A)^2d + 3(x - A)d^2 - d^3$$

Applying Σ operator on both sides and then dividing by n ,

$$\Sigma(x - \bar{x})^3/n = \Sigma(x - A)^3/n - 3d \Sigma(x - A)^2/n + 3d^2 \Sigma(x - A)/n - nd^3/n$$

or,

$$m_3 = m_3' - 3dm_2' + 3d^2m_1' - d^3, \quad \text{from (i) and (ii)} \\ = m_3' - 3m_1'm_2' + 3(m_1')^2m_1' - (m_1')^3, \quad \text{from (i)} \\ = m_3' - 3m_2'm_1' + 3(m_1')^3 - (m_1')^3 \\ = m_3' - 3m_2'm_1' + 2(m_1')^3.$$



ADDITIONAL EXERCISES 7



- Find the first, second and third moments about 10 from the following data:

6, 7, 12, 9, 15

- Find the first three (i) raw moments, and (ii) central moments, from the following:

2, 5, 8 9.



3. Find the first, second and third moments about 5 from the following:

x	1	3	5	7	9	Total
f	3	5	16	18	8	50

Hence, calculate 2nd moment about mean. [K.U., B. Com(Pass) '83]

4. Calculate the first four central moments:

<i>Class Interval</i>	12–13	14–15	16–17	18–19	Total
<i>Frequency</i>	3	10	5	2	20

5. The first two moments of a distribution about the value 3 of a variable are 1 and 10 respectively. Calculate the arithmetic mean and the standard deviation of the distribution. [C.U., M.B.A. '81]
6. For a given set of observations, the first three moments about 8 are 2, 17 and 42 respectively. Find (i) the 3rd raw moment, and (ii) 2nd central moment. (Hint: see Example 7.2, p. 226) [K.U., B.Com(Pass) '81]
7. If M_2' and M_2 are respectively the second moment about an arbitrary origin A and that about $\bar{x}$, then show that

$$M_2' = M_2 + d^2, \quad \text{where } d = \bar{x} - A. \quad [C.U., M.Com '83]$$

8. Compare the skewness of the following distributions:

	<i>Distr. I</i>	<i>Distr. II</i>
<i>Mean</i>	Rs 190	£ 210
<i>Mode</i>	Rs 175	£ 250
<i>S.D.</i>	Rs 25	£ 20

[C.U., M.Com '79]

9. The mean, median and the coefficient of variation based on 100 observations are found to be 90, 84 and 80. Find the coefficient skewness. [I.C.W.A., Dec. '83]
10. In a distribution, Mean = 65, Median = 70 and Coefficient of skewness = -0.6 . Find Mode and Coefficient of variation. [C.A., Nov. '78]
11. The following results were obtained:
 (i) Arithmetic mean = 40;
 (ii) Coefficient of variation = 5;
 (iii) Karl Pearson's measure of skewness = 0.5.
 Using them, find the mode and the s.d. [K.U., B.Com (Pass) '85]
12. Calculate with the use of quartiles the coefficient of skewness for the following distribution of age:

<i>Under Years</i>	10	20	30	40	50	60	70	80
<i>No. of Persons</i>	15	32	51	78	97	107	114	120

[Hint: Quartiles 18.82, 33.33, 46.32]

[C.U., M.Com. '78]



13. Calculate the coefficient of skewness for the following data by using quartiles.

<i>Marks Above</i>	0	15	30	45	60	75	90
<i>No. of Students</i>	180	160	130	100	65	20	5

[I.C.W.A., June '83]

14. Particulars relating to the wage distributions of two manufacturing firms are given below:

	<i>Firm A</i>	<i>Firm B</i>
<i>Median Wage (Rs)</i>	270	260
<i>Quartile Wages (Rs)</i>	260 and 290	250 and 290

Calculate a measure of skewness for both the firms and on that basis compare the two distributions.

[C.U., M.Com. '82]

15. In a certain distribution the sum of the two quartiles is 102 and their difference is 12. If the median is 49, find the coefficient of skewness,

[K.U., B.Com. (Pass) '86]

16. The coefficient of skewness for a certain distribution based upon the quartiles is -0.8 . If the sum of the upper and lower quartiles is 100.7 and the median is 55.35, find the upper and lower quartiles.

[I.C.W.A., June '84]

17. The first three moments of a distribution about 3 are respectively 4, 65 and 134. Find the arithmetic mean, s.d. and moment measure of skewness.

[C.U., B.Com. (Pass) '80]

18. The first three moments of a distribution about the value 1 are 2, 25 and 80. Find its mean, S.D. and the moment-measure of skewness.

[C.U., B.Com. (Pass) '83]

19. The first three moments of a distribution about the value 4 are -1.5 , 17, -30 . Find the values of mean, S.D. and the moment measure of skewness.

[I.C.W.A.(new), Dec. '85]

20. Comment on the shape of the following two distributions:

<i>Distribution</i>	<i>Mean</i>	<i>Variance</i>	M_3	M_4
I	3.0	4.0	16.0	48.0
II	3.0	4.0	12.0	56.0

where M_3 and M_4 are the 3rd and 4th central moments.

[C.U., M.Com. '83]

21. In a certain distribution, the first four moments about the point 4 are 1.5, 17, -30 and 108. Find Kurtosis of the frequency curve and comment on its shape.

[I.C.W.A., Dec '82]

22. Calculate Skewness and Kurtosis for the following distribution and comment on the nature of the distribution:

<i>Mid-value (x)</i>	34.5	44.5	54.5	64.5	74.5	84.5	94.5	Total
<i>Frequency</i>	2	3	11	20	32	25	7	100

[C.U., M.Com. '81]



23. Obtain suitable measure of skewness and kurtosis from the following data:

<i>Class Interval</i>	15–19	20–24	25–29	30–34	35–39
<i>Frequency</i>	1	3	7	12	29
<i>Class Interval</i>	40–44	45–49	50–54	55–59	Total
<i>Frequency</i>	20	17	9	2	100

[B.U., B.A.(Econ) '80]

24. In calculating the moments of a frequency distribution based on 100 observations the following results were obtained:

$$\text{Mean} = 9, \text{Variance} = 18, m_3^2/m_2^3 = 0.8$$

where m_2 and m_3 represent the 2nd and 3rd moment respectively. But later it was noticed that one observation 12 was misread as 21. Obtain the correct values of the first three moments about zero.

[I.C.W.A., Dec. '83]



ANSWERS



1. -0.2, 11, 8.2
2. 6, 43.5, 343.5; 0, 7.5, -7.5
3. 0.92, 5.36, 8.48; 4.51
4. 0, 2.84, 1.872, 21.2432
5. 4, 3
6. 1346, 13 (see Example 7.2, p. 226)
8. 0.6, -2.0; I is positively skew and II negatively skew. But skewness is more is distribution II.
9. 0.25 (**Hint:** Find s.d. from C.V. and use (7.7.2), p. 234)
10. 80, 38 (**Hint:** (i) Use (5.17.1), p. 127;
(ii) find s.d. from (7.7.2) and then C.V.)
11. 39, 2 (**Hint:** Find S.D. from (i) & (ii); use (7.7.1), p. 234)
12. -0.055 (Quartiles 18.82, 33.33, 46.32) [See—Table 4, p. 359]
13. -0.11 (Quartiles 28.00, 49.79, 67.17). [**Hint:** Table 2, p. 358 and Example 7.12, p. 236]
14. Bowley's measure 0.33, 0.50; wage distr. for B more skew.
15. 0.33
16. 56.6, 44.1
17. 7, 7, -1.51
18. 3, 4.58, -0.56
19. 2.5, 3.84, 0.70
20. Skewness 2 and 1.5; Kurtosis 0 and 0.5; I more asymmetrical with longer tail to the right and II has sharper peak.
21. -0.69; platykurtic (**Hint:** $m_2 = 14.75$, $m_4 = 502.3125$)
22. Skewness -0.50, Kurtosis +0.18; Negatively skew and Leptokurtic.
23. Moment measures -0.209, -0.115.
24. 8.91, 96.03, 1207.975.

CHAPTER 8

CURVE FITTING AND METHOD OF LEAST SQUARES

(Summary)

METHOD OF LEAST SQUARES

Method of Least Squares is a technique employed for finding the equation of a specified type of curve which best fits a given set of observations relating to the variables. This is based on the principle that for the “best-fitting” curve the sum of the squares of differences between the observed and the corresponding estimated values, as obtained from the equation, should be the minimum possible.

FITTING STARTING LINE

Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be the given set of observations on two variables x and y .

I. (When x is the independent variable) The equation $y = a + bx$ is taken, where a and b are constants. By the method of least squares, these constants are obtained by minimising $\Sigma(y_i - a - bx_i)^2$. This gives the normal equations, solving which we get a and b :

$$\begin{aligned}\Sigma y &= an + b \Sigma x \\ \Sigma xy &= a \Sigma x + b \Sigma x^2\end{aligned}$$

Geometrical interpretation: The observations are plotted as points on a graph paper. Then $y = a + bx$ shows that straight line (among all straight lines) for which the sum of the squares of *vertical distance* from the given points is the smallest possible. (see Fig. 8.3, page 289)

II. (When y is the independent variable) The equation of the straight line is now of the form $x = c + dy$, where c and d are constants. The values of these constants are obtained by minimising $\Sigma(x_i - c - dy_i)^2$, giving the normal equations

$$\begin{aligned}\Sigma x &= cn + d \Sigma y \\ \Sigma xy &= c \Sigma y + d \Sigma y^2\end{aligned}$$

Geometrically, this gives the straight line for which the sum of the squares of *horizontal distances* from the given points is the smallest possible.

Obviously, for the same set of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ the two best-fitting straight lines $y = a + bx$ and $x = c + dy$ are usually different.

$y = a + bx$ is used for estimating y when x is given.

$x = c + dy$ is used for estimating x when y is given.

FITTING PARABOLA

The equation is $y = a + bx + cx^2$, where x is the independent variable. The constants a, b, c are obtained by solving the normal equations

$$\begin{aligned}\Sigma y &= an + b \Sigma x + c \Sigma x^2 \\ \Sigma xy &= a \Sigma x + b \Sigma x^2 + c \Sigma x^3 \\ \Sigma x^2 y &= a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4\end{aligned}$$

FITTING EXPONENTIAL CURVE

The equation is $y = ab^x$, where x is the independent variable.



Taking logarithms of both sides, $\log y = (\log a) + (\log b) x$. This is of the form $Y = A + Bx$, where $Y = \log y$, $A = \log a$, $B = \log b$. The normal equations for finding A and B are

$$\begin{aligned}\Sigma Y &= nA + B \Sigma x \\ \Sigma xY &= A \Sigma x + B \Sigma x^2\end{aligned}$$

Hence we obtain $a = \text{antilog } A$, $b = \text{antilog } B$.

(Note: The computations can be considerably reduced by changes of origin and scale of the independent variable. In particular, if the successive values of the independent variable have a common difference special methods are available for the cases when n is odd or even (see Example 8.4, 8.5, p. 255, 6).

Uses—

- (1) Regression lines are straight lines of best-fit obtained by the method of least squares. The regression equation of y on x is a straight line of the form $y = a + bx$, where x is the independent variable. The regression equation of x on y is the straight line of the form $x = c + dy$, where y is the independent variable. (see pages 250, 251).
- (2) Method of least squares is also used in Time Series analysis (Chapter 16) for finding the equation of Trend line.

Problem 8.1 Fit a least square line to the data below, using (i) x as the independent variable, (ii) y as the independent variable. Find y when $x = 5$ and $x = 12$. Also find x when $y = 7$.

x	3	5	6	8	9	11
y	2	3	4	6	5	8

[C.U., B.Com.(Hons) '83]

Solution (i) When x is the independent variable, the equation of straight line is of the form $y = a + bx$. Normal equations for finding a and b are $\Sigma y = an + b \Sigma x$, $\Sigma xy = a \Sigma x + b \Sigma x^2$.

Fitting Straight Line

x	y	x^2	xy	y^2
3	2	9	6	4
5	3	25	15	9
6	4	36	24	16
8	6	64	48	36
9	5	81	45	25
11	8	121	88	64
42	28	336	226	154

Putting the values,

$$28 = 6a + 42b \quad \dots(1)$$

$$226 = 42a + 336b \quad \dots(2)$$

Multiplying (1) by 7 and subtracting from (2), we get $30 = 42b$.

$$\therefore b = 30/42 = 5/7$$

Using this in (1) and solving

$$a = -1/3$$

$$\therefore \text{The fitted line is } y = -1/3 + (5/7)x \quad \dots(3)$$



(ii) When y is the independent variable, the equation of straight line is of the form $x = c + dy$.
Normal equations for finding the constants c and d are

$$\Sigma x = cn + d\Sigma y \quad \text{or,} \quad 42 = 6c + 28d \quad \dots (4)$$

$$\Sigma xy = c\Sigma y + d\Sigma y^2 \quad 226 = 28c + 154d \quad \dots (5)$$

Multiplying (4) by 14 and (5) by 3 and subtracting, we get

$$70d = 90; \text{ i.e., } d = 9/7. \text{ Hence, } c = 1.$$

The fitted line is $x = 1 + (9/7)y$... (6)

To find y when x is given, we must use (3)

$$\text{When } x = 5, \quad y = -\frac{1}{3} + \frac{5}{7} \times 5 = \frac{68}{31} = 3.24$$

$$\text{When } x = 12, \quad y = -\frac{1}{3} + \frac{5}{7} \times 12 = \frac{178}{21} = 8.24$$

To find x when y is given, we must be (6).

$$\text{When } y = 5, \quad x = 1 + \frac{9}{7} \times 7 = 10$$

Ans. (i) $y = -1/3 + (5/7)x$, (ii) $x = 1 + (9/7)y$; 3.24, 8.24; 10.



Problem 8.2 Fit the equation $y = 2 + ab^x$ to the following data by the method of least squares:

x	2	3	4	5	6
y	14	25	50	98	194

[I.C.W.A., Dec. '85]

Solution (Note: This is a Modified Exponential Curve; see page 245). We may write,
 $y - 2 = ab^x$.

$$\therefore \log(y - 2) = (\log a) + x(\log b)$$

$$\text{or,} \quad Y = A + Bx$$

where $Y = \log(y - 2)$, $A = \log a$, $B = \log b$. Normal equations:

$$\Sigma Y = nA + B\Sigma x, \quad \Sigma xY = A\Sigma x + B\Sigma x^2$$

Calculations for Curve Fitting

x	y	$(y - 2)$	$\log(y - 2) = Y$	x^2	xY
2	14	12	1.0792	4	2.1584
3	25	24	1.3802	9	4.1406
4	50	48	1.6812	16	6.7248
5	98	96	1.9823	25	9.9115
6	194	192	2.2833	36	13.6998
20	—	—	8.4062	90	36.6351

$$\text{Normal equations:} \quad 8.4062 = 5A + 20B \quad \dots (1)$$

$$36.6351 = 20A + 90B \quad \dots (2)$$

Multiplying (1) by 4 and subtracting from (2)

$$10B = 3.0103; \quad B = 0.30103$$

Putting this value in (1) and solving, $A = 0.47712$

$$a = \text{antilog } A = \text{antilog } 0.47712 = 3$$

$$b = \text{antilog } B = \text{antilog } 0.30103 = 2$$

The fitted equation is $y = 2 + 3 \times 2^x$.



ADDITIONAL EXERCISES 8

1. Fit a straight line by the method of least squares to the following data. What is the value corresponding to $x = 5$?

x	1	2	3	4	5
y	3	4	6	9	10

[I.C.W.A. June '85]

2. Use the method of least squares to determine a and b in $y = a + bx$ to fit in the following data:

x	0	1	2	3	4
y	1.8	5.1	8.9	14.1	19.8

[I.C.W.A., June '82]

3. A firm has the following data on the price (x) of its product and the volume of sales (y):

Price per unit (Rs)	10	7	5	4
Volume of Sales ('000) units)	5	7	8	10

Assuming that a linear relationship exists between price and volume of sales, estimate the parameters of this relationship; and hence find the likely volume of sales when price rises to Rs 12 per unit. If price increases by one rupee, how much sales decreases?

[B.U., B.Com.(Hons) '82]

4. Given the following data on price and quantity demanded, fit a linear demand curve by the method of least squares:

Price (x)	15	12	9	8	7	6
Quantity (y)	75	85	93	115	120	130

[B.U., B.Com.(Hons) '85]

5. Based on the following data, find the linear regression equation of Y on X by the method of least squares. Predict value of Y if $X = 4$.

X	1	2	3	4	5	6
Y	6	4	3	5	4	2

[I.C.W.A., Dec. '82]

6. Fit a least square straight line to the following data taking x as independent variable and y as dependent variable:

x	0	5	10	15	20	25
y	12	15	17	22	24	30

[C.U., B.Com(H) '84]

7. Fit a curve of the $y = AB^x$ to the following data, by the method of the least squares:

x	0	1	2
y	2	6	18

[I.C.W.A., Dec. '86]



8. Given the following data on the sales of a company in different years, fit a linear trend to the data by the method of least squares and estimate the likely sales in the year 1989:

Year	1981	1982	1983	1984	1985
Sales ('000 Rs)	120	130	135	145	165

- (a) What is the annual increase in sales?
 (b) In what year will sales be Rs 1,80,000?



ANSWERS



- $y = 0.7 + 1.9x$; 10.2
- $a = 0.94$, $b = 4.5$
- $a = 12.45$, $b = -0.76$; 3.33 ('000 units); 760 units.
- $y = 160.66 - 6.07x$
- $Y = 5.8 - 0.514X$; 3.74 (Linear regression equation of Y on X is the best-fitting straight line of the form $Y = a + bX$; see page 290).
- $y = 11.3 + 0.7x$ (**Hint:** Example 8.5 and 8.6, p. 256-257)
- $y = 2 \times 3^x$ (**Hint:** Example 8.10, p. 263)
- $y = 139 + 10.5x$, origing in year 1983, x unit 1 year; 202 ('000 Rs); (a) 10,500;
 (b) Year 1987.
 (**Hint:** Example 16.17 and 16.18, pp. 290, Part II).

CHAPTER 9

CORRELATION AND REGRESSION

(*Summary*)

BIVARIATE DATA

Statistical data relating to the simultaneous measurement of two variables (say, x and y) are called 'bivariate data'. The observations are then paired— $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.

CORRELATION AND REGRESSION

Correlation denotes the strength of association between two variables; i.e. how closely the variables are related to each other.

Regression denotes prediction or estimation of the most likely value of one variable for a specified value of the other variable.

In *linear correlation* and *linear regression* it is assumed that there is a linear relationship between the variables.

SCATTER DIAGRAM

Scatter diagram is the graphical representation of bivariate data. The pairs of observations are plotted as points on a graph paper. This diagram which shows the observations as points on the graph paper is known as 'scatter diagram'. This is used to indicate (i) the type of correlation (positive, negative or zero) and (ii) the degree of correlation (high or low).

If the points have more or less a straight line pattern, the variables are correlated. The type of correlation is positive or negative, according as the slope of the straight line is positive or negative. If there is no linear pattern, the variables are 'uncorrelated' ($r = 0$). If all the points lie exactly on a straight line, the variables are 'perfectly correlated' ($r = \pm 1$). The more scattered the points are from the linear path, the less is the degree of correlation.

CORRELATION COEFFICIENT (r)

Karl Pearson's product-moment formula:

$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{[\sum (x - \bar{x})^2 \sum (y - \bar{y})^2]}}$$

PROPERTIES OF CORRELATION COEFFICIENT (r)

1. r is independent of origin and scale of the observations. In particular, if $u = ax + b$, $v = cy + d$, then $r_{uv} = \pm r_{xy}$ according as the coefficients of x and y have the same sign or opposite signs.
2. r is a pure number.
3. r lies between -1 and $+1$; i.e. $-1 \leq r \leq +1$.

VARIANCE OF THE SUM (DIFFERENCE)

$$\text{Var}(x + y) = \text{Var}(x) + \text{Var}(y) + 2.\text{Cov}(x, y)$$

$$\text{Var}(x - y) = \text{Var}(x) + \text{Var}(y) - 2.\text{Cov}(x, y)$$



If x and y are uncorrelated (i.e. $r = 0$), then $\text{cov}(x, y) = 0$. Hence

$$\text{Var}(x + y) = \text{Var}(x - y) = \text{Var}(x) + \text{Var}(y)$$

In general,

$$\text{Var}(ax + by) = a^2 \cdot \text{Var}(x) + b^2 \cdot \text{Var}(y) + 2ab \cdot \text{Cov}(x, y)$$

$$\text{Cov}(ax + by, cx + dy) = ac \cdot \text{Var}(x) + bd \cdot \text{Var}(y) + (ad + bc) \cdot \text{Cov}(x, y)$$

Note:

(i) To find $\text{Var}(ax + by)$, think of the square

$$(ax + by)^2 = a^2x^2 + b^2y^2 + 2ab \cdot xy.$$

Replace the square terms x^2, y^2 by $\text{Var}(x), \text{Var}(y)$ and the product term xy by $\text{Cov}(x, y)$.

(ii) To find $\text{Cov}(ax + by, cx + dy)$, think of the product $(ax + by)(cx + dy) = ac \cdot x^2 + bd \cdot y^2 + (ad + bc) \cdot xy$. As before, replace the square terms by Var and the product term by Cov .

For example, since $(2x - 3y)^2 = 4x^2 + 9y^2 - 12xy$.

$$\text{Var}(2x - 3y) = 4 \cdot \text{Var}(x) + 9 \cdot \text{Var}(y) - 12 \cdot \text{Cov}(x, y)$$

Similarly since $(2x - 3y)(x + 2y) = 2x^2 - 6y^2 + xy$,

$$\text{Cov}(2x - 3y, x + 2y) = 2 \cdot \text{Var}(x) - 6 \cdot \text{Var}(y) + \text{Cov}(x, y)$$

If x and y are uncorrelated, $\text{Cov}(x, y) = 0$; then

$$\text{Var}(2x - 3y) = 4 \cdot \text{Var}(x) + 9 \cdot \text{Var}(y)$$

$$\text{Cov}(2x - 3y, x + 2y) = 2 \cdot \text{Var}(x) - 6 \cdot \text{Var}(y)$$

LINEAR REGRESSION

The two regression equations are:

(1) Regression equation of y on x (used to estimate y):

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

(2) Regression equation of x on y (used to estimate x):

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$\text{where } b_{yx} = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = r \frac{\sigma_y}{\sigma_x} \text{ and } b_{xy} = \frac{\text{Cov}(x, y)}{\text{Var}(y)} = r \frac{\sigma_x}{\sigma_y}$$

Regression equations represent straight lines of best fit obtained by the method of least squares. Regression equation of y on x is of the form $y = a + bx$, where x is the independent variable and y is dependent. Regression equation of x on y is of another form $x = c + dy$, where y is independent and x is dependent (see page 75).

Geometrically, regression line of y on x is that straight line of the scatter diagram for which the sum of the squares of *vertical distances* of the points is minimum. For the regression line of x on y the sum of the squares of *horizontal distances* is minimum. So the two regression lines are usually different.

PROPERTIES OF LINEAR REGRESSION

(1) Correlation coefficient is the geometric mean of the two regression coefficients.

$$r = \sqrt{(b_{yx} \times b_{xy})}.$$

(2) Correlation coefficient and the two regression coefficients, all have the same sign.

(3) The regression lines always intersect at the point $(\bar{x}, \bar{y})$.



- (4) The angle θ between the regression lines is given by

$$\tan \theta = \pm \frac{1 - r^2}{r} \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

- (5) When $r = 0$, the regression lines are perpendicular to each other. When $r = \pm 1$, the regression lines coincide, the angle between them being zero.
 (6) When $r = 0$, neither x nor y can be estimated from linear regression equations. When $r = \pm 1$, there is an exact linear relationship between the variables.
 (7) Regression coefficients are independent of origin but depend on scale. In particular, if $u = ax + b$, $v = cy + d$, then

$$b_{uv} = \frac{a}{c} b_{xy}; b_{vu} = \frac{c}{a} b_{yx}$$

EXPLAINED VARIATION

Proportion of Total Variation explained by Regression = r^2

RANK CORRELATION COEFFICIENT (R)

Rank Correlation Coefficient is the product-moment correlation coefficient calculated from *ranks* (i.e. serial numbers indicating the order of preference/magnitude), and not from the actual magnitudes of observations.

Spearman's formula (when there are no ties):

$$R = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

Rank Correlation Coefficient (R) lies between -1 and $+1$.

$$-1 \leq R \leq +1$$

$R = +1$ when all the individuals have identical ranks in the two characters $R = -1$ when the individuals have opposite ranks.

Rank correlation coefficient is generally used as a measure of association between two attributes when measurements are either not available or not reliable, but the individuals can be arranged in order of preference. It is therefore applied in educational and psychological studies to find the degree of correspondence between abilities.

MULTIPLE AND PARTIAL CORRELATION (3 VARIABLES)

Multiple Correlation Coefficient of x_2 on x_2 and x_3 measures the strength of the combined influence of x_2 and x_3 on x_1 .

$$R_{1 \cdot 23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{32}^2}}$$

It lies between 0 and 1. $0 \leq R_{1 \cdot 23} \leq 1$

Partial Correlation Coefficient of x_1 and x_2 , the other variable being x_3 , measures the strength of association between x_1 and x_2 , after eliminating the effect of x_3 from both.

$$r_{1 \cdot 2 \cdot 3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

It lies between -1 and $+1$; $-1 \leq r_{1 \cdot 2 \cdot 3} \leq +1$



MULTIPLE REGRESSION EQUATION

Regression equation of x_1 on x_2 and x_3 (used to estimate x_1) is:

$$x_1 - \bar{x}_1 = b_{12.3} (x_2 - \bar{x}_2) + b_{13.2} (x_3 - \bar{x}_3)$$

$$\text{where } b_{12.3} = \frac{\sigma_1}{\sigma_2} \left(\frac{r_{12} - r_{13} r_{23}}{1 - r_{23}^2} \right); b_{13.2} = \frac{\sigma_1}{\sigma_2} \left(\frac{r_{13} - r_{12} r_{23}}{1 - r_{23}^2} \right)$$

It may be shown that $b_{12.3} \times b_{21.3} = r_{12.3}^2$

$b_{12.3}$, $b_{21.3}$, etc. are known as *Partial Regression Coefficients*.

Note:

- (1) $r_{12} = r_{21}$; $r_{12.3} = r_{21.3}$; but $r_{12.3} \neq r_{13.2}$
- (2) $R_{1.23} = R_{1.32}$; but $R_{1.23} \neq R_{2.13}$
- (3) $b_{12.3} \neq b_{21.3}$



SOLVED PROBLEMS



Problem 9.1 Calculate correlation coefficient from the following results: $n = 10$, $\Sigma x = 140$, $\Sigma y = 150$, $\Sigma(x - 10)^2 = 180$, $\Sigma(y - 15)^2 = 215$, $\Sigma(x - 10)(y - 15) = 60$.

[C.A., May '80]

Solution Let $u = x - 10$, $v = y - 15$. Then

$$\Sigma u = \Sigma(x - 10) = \Sigma x - 10n = 140 - 10 \times 10 = 40$$

$$\Sigma v = \Sigma(y - 15) = \Sigma y - 15n = 150 - 15 \times 10 = 0$$

Also given, $\Sigma u^2 = 180$, $\Sigma v^2 = 215$, $\Sigma uv = 60$

$$\therefore \sigma_u^2 = \Sigma u^2/n - (\Sigma u/n)^2 = 180/10 - (40/10)^2 = 2$$

$$\sigma_v^2 = \Sigma v^2/n - (\Sigma v/n)^2 = 215/10 - (0/10)^2 = 21.5$$

$$\text{cov}(u, v) = \Sigma uv/n - (\Sigma u/n)(\Sigma v/n) = 60/10 - (40/10)(0/10) = 6$$

$$\therefore r_{uv} = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v} = \frac{6}{\sqrt{(2 \times 21.5)}} = +0.91;$$

$$r_{xy} = r_{uv} = +0.91.$$

Ans. 0.91



Problem 9.2 Given $r = 0.8$, $\Sigma xy = 60$, $\sigma_y = 2.5$, $\Sigma x^2 = 90$, find the number of items.

[C.A., May '82]

Solution (**Note:** Here x and y denote deviations of X and Y from their means).

$$\text{cov}(X, Y) = \Sigma xy/n = 60/n; \sigma_X^2 = \Sigma x^2/n = 90/n$$

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}; \text{ or, } 0.8 = \frac{60/n}{\sqrt{(90/n) \times 2.5}}$$

$$\text{or, } 60/n = 2 \times \sqrt{(90/n)}; \text{ Squaring both sides and solving } n = 10.$$



Problem 9.3 This lines of regression of y on x on y are respectively $y = x + 5$ and $16x = 9y - 94$. Find the variance of x if the variance of y is 16. Also find the covariance of x and y .

[I.C.W.A., Dec. '86]



Solution Regression equations are $y = x + 5$ and

$$x = (9/16)y - (94/16). \text{ This shows that } b_{yx} = 1, b_{xy} = 9/16.$$

$$\therefore r^2 = b_{yx} b_{xy} = 1 \times \frac{9}{16} = \frac{9}{16}$$

$$\text{or } r = +\frac{3}{4} \text{ (positive; because, } b_{yx} \text{ and } b_{xy} \text{ are positive.)}$$

$$\text{But } b_{yx} = r \frac{\sigma_y}{\sigma_x}; \text{ or } 1 = \frac{3}{4} \times \frac{4}{\sigma_x}$$

$$\text{Solving, } \sigma_x = 3; \text{ or, } \text{var}(x) = 9$$

$$\therefore \text{cov}(x, y) = r \cdot \sigma_x \sigma_y = \frac{3}{4} \times 3 \times 4 = 9.$$

Ans. 9; 9



Problem 9.4 If x and y are two correlated variables with same variance and the correlation coefficient is r , find the regression coefficient of x on $(x + y)$ and that of $(x + y)$ on x . Hence find correlation coefficient between x and $(x + y)$. [W.B.H.S. '85]

Solution Let $\text{var}(x) = \text{var}(y) = \sigma^2$

$$\text{Since } r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}, \text{ by cross multiplication}$$

$$\text{cov}(x, y) = r \cdot \sigma_x \sigma_y = r \cdot \sigma \cdot \sigma = r\sigma^2$$

If we write $u = x + y$, we have to find

$$b_{xu} = \frac{\text{cov}(x, u)}{\text{var}(u)}; b_{ux} = \frac{\text{cov}(x, u)}{\text{var}(x)}$$

$$\begin{aligned} \text{Now, } \text{var}(u) &= \text{var}(x + y) \\ &= \text{var}(x) + \text{var}(y) + 2 \cdot \text{cov}(x, y); \text{ see p. 469} \\ &= \sigma^2 + \sigma^2 + 2r \cdot \sigma^2 \\ &= 2\sigma^2(1 + r) \end{aligned}$$

$$\begin{aligned} \text{Again, } \text{cov}(x, u) &= \text{cov}(x, x + y); \text{ see Note, p. 469} \\ &= \text{var}(x) + \text{cov}(x, y) \\ &= \sigma^2 + r \cdot \sigma^2 = \sigma^2(1 + r) \end{aligned}$$

$$\text{Thus, } \text{cov}(x, u) = \sigma^2(1 + r), \text{var}(u) = 2\sigma^2(1 + r), \text{var}(x) = \sigma^2$$

$$\therefore b_{xu} = \frac{\sigma^2(1 + r)}{2\sigma^2(1 + r)} = \frac{1}{2}; b_{ux} = \frac{\sigma^2(1 + r)}{\sigma^2} = 1 + r$$

Correlation coefficient between x and u is

$$r_{xu} = \sqrt{b_{xu} \cdot b_{ux}} = \sqrt{(1 + r) / 2} \quad \text{Ans. } 1/2, 1 + r; \sqrt{(1 + r) / 2}$$



Problem 9.5 The coefficient of rank correlation of the marks obtained by 10 students in Mathematics and Statistics was found to be 0.5. It was then detected that the difference in ranks in the two subjects for one particular student was wrongly taken to be 3 in place of 7. What should be the correct rank correlation coefficient?

[I.C.W.A., June '81; C.U., B.Com (Hons) '83]

Solution Putting $R = 0.5$ and $n = 10$ in the formula for Rank Correlation Coefficient (p. 304),



$$0.5 = 1 - \frac{6\Sigma d^2}{990}$$

Solving, $\Sigma d^2 = 82.5$. When one incorrect value $d = 3$ is replaced by 7,

$$\Sigma d^2 = 82.5 - 3^2 + 7^2 = 122.5.$$

$$\therefore R = 1 - \frac{6 \times 122.5}{990} = +0.26$$

Ans. + 0.26



Problem 9.6 x_1, x_2, x_3 are three variates measured from their means with $N = 10$, $\Sigma x_1^2 = 90$, $\Sigma x_2^2 = 160$, $\Sigma x_3^2 = 40$, $\Sigma x_1 x_2 = 60$, $\Sigma x_3 x_1 = 40$, $\Sigma x_2 x_3 = 60$. Calculate the partial correlation coefficient $r_{31 \cdot 2}$ and multiple correlation coefficient $R_{1 \cdot 23}$.

[I.C.W.A., June '86]

Solution Since x_1, x_2, x_3 are measured from their means,

$$\text{var}(x_1) = \Sigma x_1^2 / N = 90/10 = 9; \quad \therefore \sigma_1 = 3$$

$$\text{var}(x_2) = \Sigma x_2^2 / N = 160/10 = 16; \quad \sigma_2 = 4$$

$$\text{var}(x_3) = \Sigma x_3^2 / N = 40/10 = 4; \quad \sigma_3 = 2$$

$$\text{cov}(x_1, x_2) = \Sigma x_1 x_2 / N = 60/10 = 6$$

$$\text{cov}(x_2, x_3) = \Sigma x_2 x_3 / N = 60/10 = 6$$

$$\text{cov}(x_3, x_1) = \Sigma x_3 x_1 / N = 40/10 = 4$$

$$r_{12} = \frac{\text{cov}(x_1, x_2)}{\sigma_1 \sigma_2} = \frac{6}{3 \times 4} = \frac{1}{2}; \text{ Similarly } r_{23} = \frac{n}{4}, r_{31} = \frac{2}{3}$$

In the formula for $r_{12 \cdot 3}$ (page 308) if we write the subscripts 3, 1, 2 in place of 1, 2, 3 respectively, we get

$$r_{31 \cdot 2} = \frac{r_{31} - r_{23} r_{12}}{\sqrt{(1 - r_{23}^2)(1 - r_{12}^2)}} = \frac{\frac{2}{3} - \left(\frac{3}{4} \times \frac{1}{2}\right)}{\sqrt{\frac{7}{16} \times \frac{3}{4}}} = \frac{\sqrt{21}}{9} = 0.51$$

Using the formula (page 308)

$$R_{1 \cdot 23} = \sqrt{\frac{\frac{1}{5} + \frac{4}{9} - 2 \times \frac{1}{2} \times \frac{3}{4} \times \frac{2}{3}}{1 - \frac{9}{16}}} = \frac{2}{3} = 0.67 \quad \text{Ans. } 0.51, 0.67$$



(SHORT-ANSWER TYPE QUESTIONS)

Problem 9.7 (a) Given $\text{cov}(x, y) = -30$, $\sigma_x = 6$, $\sigma_y = 10$; find coefficient of correlation between x and y . [C.U., B.Com(P) '85]

(b) In studying a set of pairs of related variates, the following results are obtained: $N = 100$, $\Sigma x_i = 57$, $\Sigma y_i = 43$, $\Sigma(x_i - \bar{x})^2 = 169$, $\Sigma(y_i - \bar{y})^2 = 64$, $\Sigma(x_i - \bar{x})(y_i - \bar{y}) = 101$. Find the correlation coefficient. [C.U., B.Com(Hons) '84]

Solution (a) $r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{-30}{6 \times 10} = -0.5$

$$(b) r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\sqrt{[\Sigma(x - \bar{x})^2 \Sigma(y - \bar{y})^2]}} = \frac{101}{\sqrt{(169 \times 64)}} = 0.97 \quad \text{Ans. } -0.5; 0.97$$





Problem 9.8 (a) The corresponding values of two variables are:

x	2	3	5	8	9
y	4	6	10	16	18

What is the correlation coefficient between x and y ? [C.A., May '82]

(b) If $u + 3x = 5$, $2y - v = 7$ and the correlation coefficient of x and y is 0.12, find the correlation coefficient of u and v . [W.B.H.S., '83]

Solution

(a) Since each value of y is double the corresponding value of x , we have $y = 2x$. Thus x and y are bound by an exact linear relation with a positive slope. $r = +1$. (Also see Ex. 9.32, p. 342).

(b) $u = 5 - 3x$, $v = 2y - 7$, $r_{xy} = 0.12$. Thus u and v are linear functions of x and y respectively. So r_{uv} is numerically equal to r_{xy} . But since the coefficients of x and y have opposite signs $r_{uv} = -r_{xy} = -0.12$. (Also see Example 9.15, p. 284). Ans. +1; -0.12



Problem 9.9 (a) Karl Pearson's coefficient of correlation between x and y is 0.52, their covariance is +7.8. If the variance of x is 16, find the standard deviation of y . [C.U., B.Com(P) '83]

(b) Two uncorrelated random variables x and y have standard deviations 3 and 4 respectively. Find the s.d. of $(x + y)$.

Solution

(a) $r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$. Putting $r = 0.52$, $\text{cov}(x, y) = 7.8$, $\sigma_x = 4$, and solving $\sigma_y = 15/4 = 3.75$.

(b) If $r = 0$, $\text{var}(x + y) = \text{var}(x) + \text{var}(y)$
 $= 3^2 + 4^2 = 25$; $\therefore \sigma_{x+y} = 5$ Ans. 3.75; 5



Problem 9.10 Are the following statements valid? Justify your answers.

- Positive value of correlation coefficient between x and y implies that if x decreases, y tends to increase.
- Correlation coefficient is independent of the origin of reference but is dependent on the units of measurement.
- Correlation coefficient between x and y turned out to be 1.02.

[I.C.W.A., Dec. '80]

Solution No; the statements are not valid.

- Positive value of correlation coefficient implies that if x decreases, y also tends to decrease.
- Correlation coefficient is independent of both the origin of reference and the units of measurement.
- Correlation coefficient cannot be larger than 1.



Problem 9.11 (a) Regression equation of y on x is $3x + 2y = 26$, and regression equation of x on y is $6x + y = 31$. Find the regression coefficients b_{yx} and b_{xy} .

(b) The equations of two lines of regression are $2x + 3y - 8 = 0$ and $x + 2y - 5 = 0$. Find the means of x and y .

**Solution**

- (a) Regression equation of y on x may be written as $2y = 26 - 3x$; or, $y = 13 - \frac{3}{2}x$. This shows that

$$\text{Regression coefficient of } y \text{ on } x (b_{yx}) = -\frac{3}{2}$$

Regression equation of x on y may be written as $6x = 31 - y$; or, $x = \frac{31}{6} - \frac{1}{6}y$. This shows that

$$\text{Regression coefficient of } x \text{ on } y (b_{xy}) = -\frac{1}{6}$$

Note : Express the *regression equation of y on x* in the form $y = a + bx$; i.e. L.H.S. is only y and R.H.S. contains no y at all. The coefficient of x on R.H.S. gives b_{yx} . Similarly, express, the *regression equation of x on y* in the form $x = c + dy$, and coefficient of y on R.H.S. gives b_{xy} .]

- (b) Solving the two equations we get $x = 1$, $y = 2$. Since the regression lines intersect at $(\bar{x}, \bar{y})$, $\bar{x} = 1$, $\bar{y} = 2$.



Problem 9.12 (a) If $b_{yx} = -0.3$ and $b_{xy} = -1.2$, find the coefficient of correlation between x and y .

(b) If $u = 2x + 5$, $v = -3y + 1$, and the regression coefficient of y on x is -1.2 , find the regression coefficient of v on u .

Solution

- (a) $r^2 = b_{yx} \cdot b_{xy} = (-0.3)(-1.2) = +0.36$; hence $r = \pm 0.6$. But r , b_{yx} , b_{xy} all must have the same sign. $\therefore r = -0.6$

- (b) Given $u = 2x + 5$, $v = -3y + 1$, $b_{yx} = -1.2$.
 $\therefore b_{vu} = (-3/2)b_{yx} = (-3/2) \times (-1.2) = +1.8$.

[See Problem 9.14(b), Solution, p. 423]

Ans. -0.6 ; $+1.8$



Problem 9.13 (a) Give reasons to justify that the following statement cannot be true: (i) $b_{yx} = 0.8$ and $b_{xy} = -0.3$; (ii) $b_{yx} = 0.8$ and $b_{xy} = 2.4$.

[C.U., B.Com(Hons) '84]

(b) Criticise: The correlation coefficient between x and y (r_{xy}) is 0.9 and the regression coefficient (b_{xy}) is -1 .

[I.C.W.A., June '79]

(c) A student obtained the two regression lines as $2x - 5y - 7 = 0$ and $3x + 2y - 8 = 0$. Do you agree with him?

[I.C.W.A., Dec. '81]

Solution

- (a) (i) Both b_{yx} and b_{xy} must have the same sign,
 (ii) The product $b_{yx} \cdot b_{xy}$ must not exceed 1; because $b_{yx} \cdot b_{xy} = r^2$, and r^2 cannot be greater than 1.
 (b) The statement cannot be true; because r_{xy} , b_{yx} and b_{xy} all three must have the same sign.
 (c) No; because slopes of the two regression lines must be either both positive or both negative.





Problem 9.14 (a) With $b_{xy} = 0.5$, $r = 0.8$ and variance of $y = 16$, the standard deviation of x equals to 2.5, 6.4, 10.0, 25.6 —Which is correct? [C.A., Nov. '83]

(b) Pick out the correct answer: If $4u = 2x + 7$ and $6v = 2y - 15$, and the regression coefficient of y on x is 3, then regression coefficient of v on u is 2, 3, $9/2$, -2 , None of these. [W.B.H.S. '82]

Solution

(a) $b_{xy} = r \frac{\sigma_x}{\sigma_y}$. Putting the values $b_{xy} = 0.5$, $r = 0.8$, $\sigma_y = 4$ and solving $\sigma_x = 2.5$.

Ans. 2.5

(b) [Note: $\text{var}(ax + b) = a^2 \cdot \text{var}(x)$;
 $\text{cov}(ax + b, cy + d) = ac \cdot \text{cov}(x, y)$].

Here, $u = \frac{1}{2}x + \frac{7}{4}$, $v = \frac{1}{3}y - \frac{5}{2}$; $b_{yx} = 3$.

$$\begin{aligned} \text{Now, } b_{vu} &= \frac{\text{cov}(u, v)}{\text{var}(u)} = \frac{\frac{1}{2} \times \frac{1}{3} \times \text{cov}(x, y)}{\left(\frac{1}{2}\right)^2 \text{var}(x)} = \frac{2}{3} \times \frac{\text{cov}(x, y)}{\text{var}(x)} \\ &= \frac{2}{8} b_{yx} = \frac{2}{8} \times 3 = 2 \end{aligned}$$

Ans. 2



Problem 9.15 (a) If $\Sigma D^2 = 33$ and $N = 10$, find the value of the coefficient of rank correlation, where D represents the difference between the ranks of two series and N is the number of pairs of observations. [C.U., B.Com(P) '85]

(b) Mention the correct answer: The ranks according to two attributes in a sample are given below:

R_1	1	2	3	4	5
R_2	5	4	3	2	1

The rank correlation between them is 0, +1, -1, none of these. [C.A., Nov. '82]

Solution

(a) Putting the values in the formula for rank correlation coefficient

$$R = 1 - \frac{6 \times 33}{10 \times 99} = 0.8$$

Ans. 0.8

(b) Since the ranks are just opposite, $R = -1$.

Ans. -1



ADDITIONAL EXERCISES 9



- Find (a) the coefficient of correlation, and (b) the regression equation of x on y , from the following data:

$$\Sigma x = 60, \Sigma y = 40, \Sigma xy = 1150, \Sigma x^2 = 4160, \Sigma y^2 = 1720, N = 10$$

[C.A., Nov. '85]

- For 12 pairs of observations the results are: $\Sigma(x_i - \bar{x})^2 = 84.68$, $\Sigma(y_i - \bar{y})^2 =$



38.92, $\Sigma(x_i - \bar{x})(y_i - \bar{y}) = 40.34$. Find the value of product-moment correlation coefficient. [I.C.W.A., Dec. '84]

3. Calculate the coefficient of correlation from

x	1	2	3	4	5
y	6	8	11	8	12

[C.U., B.Com(P) '85]

4. Calculate Karl Pearson's coefficient of correlation from the following data, using 20 as the working mean for price, and 70 as the working mean for demand:

Price	14	16	17	18	19	20	21	22	23
Demand	84	78	70	75	66	67	62	58	60

[C.A., May '86]

5. The following data give the heights in inches (x) and the weights in lbs. (y) of 10 students of age 17 years:

x	61	66	68	64	65	70	63	62	64	67
y	112	123	130	115	110	125	100	113	116	126

Calculate the correlation coefficient. [C.U., B.Com(Hons) '81]

6. For some bivariate data, the following results were obtained:

Mean value of the variable $X = 53.2$

Mean value of the variable $Y = 27.9$

Regression coefficient of Y on $X = -1.5$

Regression coefficient of X on $Y = -0.2$

What is the most likely value of Y when $X = 60$? What is the Coefficient of correlation between X and Y ? [I.C.W.A., June '84; C.U., B.Com(Hons) '86]

7. For a bivariate data the mean value of X is 20 and the mean value of Y is 45.

The regression coefficient of Y on X is 4 and that of X on Y is $1/9$. Find

(i) the coefficient of correlation,

(ii) the s.d. of X if the s.d. of Y is 12,

(iii) the equations of regression lines. [C.A., Nov, '82]

8. To study the effect of rain on yield of wheat, the following results were obtained:

	Mean	s.d.
Yield in lbs. per acre	800	12
Rainfall in inches	50	2

$$r = 0.80$$

Estimate the yield when rainfall is 80 inches. [I.C.W.A., June '86]

9. From the following data find the two regression equations:

X	1	2	3	4	5
Y	2	3	5	4	6

Predict the value of Y when $X = 2.5$.

[C.U., B.Com(P) '83]



10. The following table shows the age (x) in years, of cars of a certain make and the annual maintenance cost (y) in hundred Rs.

x	1	2	3	4	5	6	7
y	20	19	22	23	21	25	27

Obtain the regression equation of y on x . Estimate the cost of maintaining a six-year old car of the same make. (Answer should be correct upto 2 places of decimal). [I.C.W.A. June '86 (old)]

11. The grades of 9 students at the College test (x) and that University examination (y) are as follows:

x	77	50	71	72	81	94	96	99	67
y	82	66	78	34	47	85	99	99	68

- Draw the scatter diagram for the data (rough sketch only).
- Calculate the value of r .
- Find a linear regression equation for the data and then estimate the grade of the University examination for a student who received 85 in the College test but was sick at the time of University examination.

[C.U., B.Com(Hons) '83]

12. In the estimation of regression equations of two variables X and Y the following results were obtained:

$\bar{x} = 90$, $\bar{y} = 70$, $N = 10$, $\Sigma x^2 = 6360$, $\Sigma y^2 = 2860$, $\Sigma xy = 3900$. Obtain the two equations. [C.A., May '83]

(Note that x and y denote deviations of X and Y from their means)

13. Given that the regression equations of y on x and of x on y are respectively $y = 2x$ and $6x - y = 4$. Find (i) Correlation coefficient between x and y , and (ii) the means of x and y . [I.C.W.A., June '86]

14. In a partially destroyed record the following data are available:

Variance of $x = 25$

Regression equation of x on y : $5x - y = 22$

Regression equation of y on x : $64x - 45y = 24$

Find

- the mean values of x and y .
 - the coefficient of correlation between x and y .
 - the standard deviation of y .
- [C.A., Nov. '81]

15. The equations of two regression lines between two variables are expressed as $2x - 3y = 0$ and $4y - 5x - 8 = 0$.

- Identify which of the two can be called regression of y on x and which of x on y .
- Find $\bar{x}$ and $\bar{y}$ and correlation coefficient (r) from the equations.

[I.C.W.A., June '83]

16. If the equations of two regression lines obtained in a correlation analysis are $3x + 12y = 19$ and $3y + 9x = 46$, determine which one of these is the regression equation of y on x and which one is the regression equation of x on y . Give reasons for your answer.



Find the means, correlation coefficient, and ratio of the variances of x and y .

[W.B.H.S. '83]

17. Two regression lines of a sample are $x + 6y = 6$ and $3x + 2y = 10$. Find the correlation coefficient.

[C.U., B.Com(Hons) '86]

18. Marks obtained in Statistics and Economics by 10 M.Com. students are given below:

Student	A	B	C	D	E	F	G	H	I	J
Statistics	61	38	49	57	42	34	93	87	46	35
Economics	31	38	28	35	19	42	45	48	22	25

Calculate Rank Correlation and interpret the result.

[C.U., M.Com. '83]

19. Ten students were given the following marks in Mathematics and Business Statistics:

Student No.	1	2	3	4	5	6	7	8	9	10
Mathematics (x)	40	20	10	48	6	37	14	8	7	18
B. Statistics (y)	30	35	25	32	15	45	20	3	10	27

Calculate

- the correlation coefficient between x and y .
 - the rank correlation coefficient between x and y .
 - Predict the value of y when $x = 41$.
- [I.C.W.A., Dec, '85 (old)]
20. In a trivariate distribution,
 $S_1 = 3, S_2 = S_3 = 5, r_{12} = 0.7, r_{23} = r_{31} = 0.6$.
 Find
- Partial correlation coefficient $r_{12.3}$,
 - Multiple correlation coefficient $R_{1.23}$,
 - Regression coefficient $b_{12.3}$.
- [I.C.W.A., Dec. '85]

(SHORT-ANSWER TYPE QUESTIONS)

21. Pick out the correct answer:
- If $u = 2x + 3$ and $v = 3y - 7$ and the correlation coefficient between x and y is 0.86, then the correlation coefficient between u and v is: 0.86, 0.43, -0.43, -0.86, none of these.
- [W.B.H.S. '81]
- If relation between two variables x and y and between other two variables u and v are $2x + 3y = 7$ and $3u + 4v = 5$ respectively, and correlation coefficient of x and u is 0.2, then correlation coefficient of y and v is: -0.2, 0.2, -0.4, none of these.
- [W.B.H.S. '85]
22. Pick out the correct answer:
- If $b_{xy} = 0.4, b_{yx} = 0.9$, then r equals: 0.36, 0.6, 1.3
- [C.A., May '81]
- If the relationship between x and u is $u + 2x = 0$ and other two variables y and v is $v = 6 + 3y$, and the regression coefficient of x on y is 2.4, then the regression coefficient of u on v is: -0.4, 0.4, 2.4, none of these.
- [W.B.H.S. '84]
23. If $N = 10, \Sigma d^2 = 280$, then the rank correlation coefficient is: +2.8, +0.28, -0.7. Which is correct?
- [C.A., May '81]
24. Which of the following two statements is correct? Give reasons for your answer.
- Two independent variables are always uncorrelated.



- (ii) Two uncorrelated variables must be always independent.
[I.C.W.A., June '85]
25. If two regression coefficients are negative, does it mean that their correlation would be positive?
[I.C.W.A., Dec. '79]
26. State whether 'true' or 'false':
(a) The correlation coefficient of $3x$ and $-2y$ is the same as the correlation coefficient of x and y .
(b) If the correlation coefficient of x and y is zero, then $\text{Var}(x + y) = \text{Var}(x - y)$.
[W.B.H.S. '81]
27. Find the equations of the lines of regression using the following data: $\bar{x} = 4$, $\bar{y} = 5$, $b_{yx} = 0.39$, $b_{xy} = 0.69$.
[C.U., B.Com(P) '82]
28. (a) If the two regression lines coincide, show that the correlation coefficient $r = \pm 1$.
(b) If $r = 0$, show that the two regression lines are at right angles.
[I.C.W.A., Dec. '85]
29. From the following data, calculate the coefficient of correlation between x and y :
(i) Number of item in x -series or y -series = 12.
(ii) Sum of the square of deviations from mean: 360 for x -series and 250 for y -series.
(iii) Sum of the products of deviations of the two series from their respective means = 225.
[I.C.W.A., June '83]
30. Write down the expression for $r_{12 \cdot 3}$ and $R_{1 \cdot 23}$ in terms of r_{12} , r_{23} and r_{31} . What are the limits between which $r_{12 \cdot 3}$ and $R_{1 \cdot 23}$ must lie? Also state the relation between $r_{12 \cdot 3}$, $b_{12 \cdot 3}$ and $b_{21 \cdot 2}$.
[I.C.W.A., Dec. '86]



ANSWERS



1. (a) 0.37, (b) $12x = 7y + 44$ (**Hints:** Example 9.7, p. 278 and Example 9.23, p. 294)
2. 0.70 (**Hint:** Use formula (9.7.2.), p. 276)
3. 0.77 (**Hint:** Find Σx , Σy , Σx^2 , Σy^2 , Σxy ; Example 9.7).
4. -0.95 (**Hint:** Example 9.9, p. 279).
5. 0.76.
6. 17.7 (**Hint:** formula (9.13.4.), p. 292; $Y = 107.7 - 1.5X$) -0.55 (**Hint:** formula (9.13.8), p. 292 and property (3))
7. (i) $2/3$ (**Hint:** formula (9.13.8). p. 292);
(ii) 2 (**Hint:** Use (9.13.6) or (9.13.7) p. 292)
(iii) $Y = 4X - 35$; $X = Y/9 + 15$ (**Hint:** (9.13.4 and 5).
8. 944 lbs. per acre (**Hint:** Example 9.25, p. 294)
9. $Y = 0.9X + 1.3$; $X = 0.9Y - 0.6$ (**Hint:** Obtain ΣX , ΣY , ΣX^2 , ΣY^2 , ΣXY ; Example 9.23, p. 293); 3.55 (**Hint:** Use Ist equations)
10. $7y = 8x + 125$; 24.71 (hundred Rs.). (**Hint:** Example 9.28, p. 296)
11. (i) **Hint:** see p. 283; (ii) 0.56 (**Hint:** Example 9.9, p. 279)
(iii) $y = 0.777x + 12.07$; 78 (**Hint:** Example 9.27, p. 295)



12. $Y = 0.6132X + 14.812$; $X = 1.3636Y - 5.452$. (**Hint:** $b_{YX} = \Sigma xy / \Sigma x^2$, $b_{XY} = \Sigma xy / \Sigma y^2$; use (9.13.4 and 5), p. 292).
13. (i) $1/\sqrt{3} = 0.58$, (ii) 1, 2 (**Hint:** In standard form $y = 2x$, $x = (1/6)y + 2/3$; see Example 9.31, p. 299)
14. (i) 6, 8; (ii) $8/15$; (iii) $40/3$ (**Hint:** Example 9.31 and 9.34)
15. (i) 1st, 2nd; (ii) $-24/7$, $-16/7$; (iii) 0.73 (**Hint:** Example 9.34, p. 300)
16. 1st, 2nd; because then $r^2 = 1/2$ does not exceed 1, as it must, 5, $1/3$; $r = -1/2\sqrt{3}$; 4.3 (**Hint:** Example 9.34, p. 300)
17. $-1/3$ (**Hint:** Example 9.34, p. 300)
18. 0.39 (**Hint:** Example 9.40, p. 308). On the whole, a student doing well in Statistics is also found good in Economics.
19. (i) 0.735 (**Hint:** Example 9.9, p. 279)
(ii) 0.842 (**Hint:** Example 9.40, p. 308)
(iii) 36.3 (**Hint:** Example 9.27, p. 295; $y = 11.72 + 0.60x$)
20. (i) 0.53125, (ii) 0.73527, (iii) 0.31875 (**Hint:** Use formulae (9.17.5), (9.17.3) and (9.17.2), p. 313)
21. (a) 0.86, (b) 0.2 (**Hint:** see formula (9.8.2), p. 277)
22. (a) 0.6 (**Hint:** (9.13.8), p. 292 & Problem 9.12a)
(b) None of these (**Hint:** Problem 9.12(b); correct -1.6)
23. -0.7 (**Hint:** formula (9.16.1), p. 304)
24. (i) is correct; because then Covariance is zero.
(**Hint:** Example 9.21, p. 289, Example 12.28, p. 128 of Part II).
25. No; is must be negative. (**Hint:** Property (3), p. 292)
26. (a) False (**Hint:** Example 9.15, p. 303)
(b) True (**Hint:** Formula (9.11.5), p. 306)
27. $y = 0.39x + 3.44$; $x = 0.69y + 0.55$ (**Hint:** use (9.13.4 & 5), p. 292)
28. **Hint:** (a) see p. 292 (last para); (b) see p. 292 (para 3).
29. 0.75 (**Hint:** Given $n = 12$, $\Sigma(x - \bar{x})^2 = 360$, $\Sigma(y - \bar{y})^2 = 250$, $\Sigma(x - \bar{x})(y - \bar{y}) = 225$; use formula (9.7.2), p. 276)
30. **Hint:** Formula (9.17.3) to (9.17.6) & (9.17.10), pp. 292–3.

CHAPTER 10
INTERPOLATION
(Solved Problems)

Problem 10.1 Construct a table of difference. Express y as a function of x and obtain the value of y when $x = 22$ from the following:

x	10	15	20	25	30	35
y	3	6	11	18	27	38

[I.C.W.A., June '86 (new)]

Solution [see Example 10.8, p. 324]

<i>Difference table</i>				
x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
10	3			
		3		
15	6		2	
		5		0
20	11		2	
		7		0
25	18		2	
		9		0
30	27		2	
		11		
35	38			

The second differences $\Delta^2 y$ are found to be constant. Hence we may assume that y is a 2nd degree polynomial.

$$\text{Let } y = a + b(x - 10) + c(x - 10)(x - 15) \quad \dots (1)$$

where a, b, c are constants,

Putting $x = 10$ in (1),

$$3 = a + b(10 - 10) + c(10 - 10)(10 - 15)$$

$$\text{or, } 3 = a + b(0) + c \times 0(-5)$$

$$\text{or, } 3 = a + 0 + 0; \quad \text{or, } a = 3 \quad \dots (2)$$

Putting $x = 15$ in (1),

$$6 = a + b(15 - 10) + c(15 - 10)(15 - 15);$$

$$\text{or, } 6 = 3 + b(5) + c(5)(0); \quad \text{since } a = 3$$

$$\text{or, } 5b = 3; \quad \text{or, } b = 3/5 = 0.6 \quad \dots (3)$$

Putting $x = 20$ in (1),

$$11 = a + b(20 - 10) + c(20 - 10)(20 - 15);$$

$$\text{or, } 11 = 3 + 6 + 50c; \quad \text{since } a = 3, b = 0.6$$

$$\text{or, } 50c = 2; \quad \text{or, } c = \frac{2}{50} = 0.04 \quad \dots (4)$$

Putting the values of a, b, c in (1),



$$\begin{aligned}
 y &= 3 + 0.6(x - 10) + 0.04(x - 10)(x - 15) \\
 &= 3 + 0.6x - 6 + 0.04(x^2 - 25x + 150) \\
 &= 3 + 0.6x - 6 + 0.04x^2 - x + 6 \\
 &= 3 - 0.4x + 0.04x^2 \quad \dots (5)
 \end{aligned}$$

When $x = 22$, from (5) we get

$$\begin{aligned}
 y &= 3 - 0.4 \times 22 + 0.04 \times 484 \\
 &= 3 - 8.8 + 19.36 = 13.56
 \end{aligned}$$

$$\text{Ans. } y = 3 - 0.4x + 0.04x^2; 13.56$$



Problem 10.2 With the help of the symbolic operator E , find the value of $\log_{10} 666$ from the following table:

$$\log_{10} 654 = 2.8156, \quad \log_{10} 658 = 2.8182, \quad \log_{10} 662 = 2.8209$$

[I.C.W.A., June '79]

Solution

x	$y = \log_{10} x$	x
654	2.8156	(y_0)
658	2.8182	(y_1)
662	2.8209	(y_2)
666	?	(y_3)

Let $y = \log_{10} x$. For *equidistant* values of $x = 654, 658, 662, 666$ let just denote the values of y by y_0, y_1, y_2, y_3 respectively. Since only 3 values of y are known, we assume that y is a 2nd degree polynomial. Hence the 3rd differences are zero. In particular

$$\Delta^3 y_0 = 0$$

$$\text{or, } (E - 1)^3 y_0 = 0$$

$$\text{or, } (E^3 - 3E^2 + 3E - 1)y_0 = 0$$

$$\text{or, } y_3 - 3y_2 + 3y_1 - y_0 = 0$$

$$\text{or, } y_3 - 3(2.8209) + 3(2.8182) - 2.8156 = 0$$

$$\text{or, } y_3 - 8.4627 + 8.4546 - 2.8156 = 0$$

$$\text{or, } y_3 - 2.8237 = 0$$

$$\therefore y_3 = 2.8237; \quad \text{i.e. } \log_{10} 666 = 2.8237$$

Problem 10.3 Using the following results, find $\sqrt[3]{26}$.

$$\sqrt[3]{27} = 3.0000, \quad \sqrt[3]{28} = 3.0366, \quad \sqrt[3]{29} = 3.0723, \quad \sqrt[3]{30} = 3.1072$$

Solution

x	$y = \sqrt[3]{x}$	
26	?	(y_0)
27	3.0000	(y_1)
28	3.0366	(y_2)
29	3.0723	(y_3)
30	3.1072	(y_4)

Let $y = \sqrt[3]{x}$. For *equidistant* values of $x = 26, 27, 28, 29, 30$, we denote the values of y by y_0, y_1, y_2, y_3, y_4 respectively. Since only 4 values of the function are given, it is assumed that y



is a 3rd degree polynomial. Hence the 4th differences $\Delta^4 y$ are zero. In particular $\Delta^4 y_0 = 0$.
Expanding $(E - 1)^4 y_0 = 0$

$$\text{or, } y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$$

$$\text{or, } 3.1072 - 4(3.0723) + 6(3.0366) - 4(3.0000) + y_0 = 0$$

Solving, $y_0 = 2.9624$. (Note: The true value is 2.9625).



Problem 10.4 Obtain estimates of the missing figures in the following table:

x	2.0	2.1	2.2	2.3	2.4	2.5	2.6
U_x	0.135	—	0.111	0.100	—	0.082	0.074

[I.C.W.A., Dec. '82, '85]

Solution For equidistant values of $x = 2.0, 2.1, \dots, 2.6$, let us denote the values of $y = U_x$ by $y_0, y_1, \dots, y_6$ respectively. Since only 5 values are given we assume a 4th degree polynomial. Hence 5th differences are zero.

$$\Delta^5 y_0 = 0 \qquad \Delta^5 y_1 = 0$$

$$\text{or, } (E - 1)^5 y_0 = 0 \qquad (E - 1)^5 y_1 = 0$$

Expanding we get (see Ex. 10.1, p. 367)

$$y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0 = 0 \qquad \dots (i)$$

$$y_6 - 5y_5 + 10y_4 - 10y_3 + 5y_2 - y_1 = 0 \qquad \dots (ii)$$

[Note that (ii) may be obtained from (i) by increasing each suffix by 1]

$$\text{or, } 0.082 - 5y_4 + 10(0.100) - 10(0.111) + 5y_1 - 0.135 = 0$$

$$0.074 - 5(0.082) + 10y_4 - 10(0.100) + 5(0.111) - y_1 = 0$$

$$\text{Simplifying, } -5y_4 + 5y_1 = 0.163 \qquad \dots (iii)$$

$$10y_4 - y_1 = 0.781$$

Multiplying (ii) by 2 and adding to (iv),

$$9y_1 = 1.107, \qquad \therefore y_1 = 0.123$$

$$\text{From (vi), } 10y_4 = 0.904 \qquad \therefore y_4 = 0.0904$$



Problem 10.5 If $U_x = x^3$, evaluate $\Delta^2 U_x$. [C.U., B.Com(Hons) '83]

Solution Assuming that the interval of differencing is *unity* (i.e. the successive values of x increase by 1),

$$\Delta U_x = U_{x+1} - U_x, \qquad \text{(by definition)}$$

$$= (x+1)^3 - x^3$$

[Note that U_{x+1} is obtained from

$U_x = x^3$ on replacing x by $(x+1)$.]

$$= (x^3 + 3x^2 + 3x + 1) - x^3$$

$$= 3x^2 + 3x + 1$$

$$\therefore \Delta^2 U_x = \Delta U_{x+1} - \Delta U_x, \text{ (by definition)}$$

$$= \{3(x+1)^2 + 3(x+1) + 1\} - \{3x^2 + 3x + 1\}$$

[$\because \Delta U_{x+1}$ is obtained from $\Delta U_x = 3x^2 + 3x + 1$ on replacing x by $(x+1)$.]

$$= 3(x+1)^2 + 3(x+1) + 1 - 3x^2 - 3x - 1$$

$$= 6(x+1), \text{ on simplification.}$$

Ans. $6(x+1)$





Problem 10.6 Using an appropriate formula for interpolation, estimate the number of students who obtained marks between 40 and 45 from the following table :

Marks	30–40	40–50	50–60	60–70	70–80
Number of Students	31	42	51	35	31

[C.U., B.Com.(Hons.) '82]

Solution Number of students obtaining marks between 40–45

$$= (\text{No. of students obtaining less than 45}) - (\text{No. of students obtaining less than 40})$$

$$= (\text{Cumulative frequency corresponding to 45}) - (\text{Cumulative frequency corresponding to 40})$$

Cumulative frequency corresponding to 40 is 31. Cumulative frequency corresponding to 45 may be obtained from a cumulative frequency table by using *Newton's forward interpolation* formula, because 'marks 45' lies near the beginning of the table.

Marks (x)	Cum. Freq. (y)	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
30	0					
		31				
40	31		11			
		42		-2		
50	73		9		-23	
		51		-25		60
60	124		-16		37	
		35		12		
70	159		-4			
		31				
80	190					

Newton's forward formula (see p. 326).

Here $x_0 = 30$, $y_0 = 0$, $x = 45$, $h = 10$.

$$\therefore u = (x - x_0)/h = (45 - 30)/10 = 3/2$$

$$\begin{aligned}
 y &= 0 + \frac{3}{2} \times 31 + \frac{3}{2} \left(\frac{3}{2} - 1 \right) \times \frac{11}{2!} + \frac{3}{2} \left(\frac{3}{2} - 1 \right) \left(\frac{3}{2} - 2 \right) \times \frac{(-2)}{3!} \\
 &\quad + \frac{3}{2} \left(\frac{3}{2} - 1 \right) \left(\frac{3}{2} - 2 \right) \left(\frac{3}{2} - 3 \right) \times \frac{(-23)}{4!} \\
 &\quad + \frac{3}{2} \left(\frac{3}{2} - 1 \right) \left(\frac{3}{2} - 2 \right) \left(\frac{3}{2} - 3 \right) \left(\frac{3}{2} - 4 \right) \times \frac{60}{5!} \\
 &= 0 + \frac{93}{2} + \frac{3}{2} \times \frac{1}{2} \times \frac{11}{2} + \frac{3}{2} \times \frac{1}{2} \times \left(-\frac{1}{2} \right) \left(-\frac{2}{8} \right) \\
 &\quad + \frac{3}{2} \times \frac{1}{2} \times \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) \left(-\frac{28}{24} \right) + \frac{3}{2} \times \frac{1}{2} \times \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) \left(-\frac{5}{2} \right) \times \frac{60}{120}
 \end{aligned}$$



$$\begin{aligned}
 &= 0 + \frac{93}{2} + \frac{33}{8} + \frac{1}{8} - \frac{69}{128} - \frac{45}{64} \\
 &= 0 + 46.5 + 4.125 + 0.125 - 0.539 - 0.703. \\
 &= 49.508 = 50 \text{ (rounded to the nearest whole number)}
 \end{aligned}$$

$$\therefore \text{ Required number of students} = 50 - 31 = 19$$



Problem 10.7 Using Newton's Backward formula find the annual premium at the age of 29 from the following data:

Age	24	28	32	36	40
Annual Premium (Rs)	28.06	30.19	32.75	34.94	40.00

[C.U., B.Com(Hons) '84]

Solution (For simplicity, we ignore the decimal point while computing the differences).

Age (x)	Premium (y)	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
24	28.06				
		213			
28	30.19		43		
		256		- 80	
32	32.75		- 37		404
		219		324	
36	34.94		287		
		506			
40	40.00				

Here $v = (x - x_n)/h = (29 - 40)/4 = -11/4$

Using 2 decimals, and the Backward formula (p. 330),

$$\begin{aligned}
 y &= 4000 + \left(-\frac{11}{4}\right) \times 506 + \left(-\frac{11}{4}\right) \left(-\frac{11}{4} + 1\right) \times \frac{287}{2!} \\
 &\quad + \left(-\frac{11}{4}\right) \left(-\frac{11}{4} + 1\right) \left(-\frac{11}{4} + 2\right) \times \frac{324}{3!} \\
 &\quad + \left(-\frac{11}{4}\right) \left(-\frac{11}{4} + 1\right) \left(-\frac{11}{4} + 2\right) \left(-\frac{11}{4} + 3\right) \times \frac{404}{4!} \\
 &= 4000 + \left(-\frac{11}{4}\right) \times 506 + \left(-\frac{11}{4}\right) \left(-\frac{7}{4}\right) \times \frac{287}{2} \\
 &\quad + \left(-\frac{11}{4}\right) \left(-\frac{7}{4}\right) \left(-\frac{3}{4}\right) + \left(-\frac{11}{4}\right) \left(-\frac{7}{4}\right) \left(-\frac{3}{4}\right) \left(\frac{1}{4}\right) \times \frac{404}{24} \\
 &= 4000 - \frac{2783}{2} + \frac{22099}{32} - \frac{6237}{32} - \frac{7777}{512}
 \end{aligned}$$



$$= 4000 - 1391.5 + 690.6 - 194.9 - 15.2$$

$$= 3089.0 = 3089 \text{ (rounded)}$$

Adjusting the decimal point, the required value is 30.89 (Rs)

Note: (i) If nothing is mentioned, we must use the most appropriate formula, here Newton's "forward".

(ii) Both Newton's forward and backward formulae applied to the same problem would give exactly the same result. However, by the appropriate formula the successive terms diminish rapidly, so that the first few terms together will give fairly accurate result.

(iii) If we use Newton's forward formula, here

$$u = (x - x_0)/h = (29 - 24)/4 = 5/4$$

Using 2 decimals,

$$y = 2806 + \frac{5}{4} \times 213 + \frac{5}{4} \left(\frac{5}{4} - 1 \right) \times \frac{43}{2!} + \frac{5}{4} \left(\frac{5}{4} - 1 \right) \left(\frac{5}{4} - 2 \right) \times \frac{-80}{3!}$$

$$+ \frac{5}{4} \left(\frac{5}{4} - 1 \right) \left(\frac{5}{4} - 2 \right) \left(\frac{5}{4} - 3 \right) \times \frac{404}{4!}$$

$$= 2806 + \frac{1065}{4} + \frac{5}{4} \cdot \frac{1}{4} \cdot \frac{43}{2} + \frac{5}{4} \cdot \frac{1}{4} \cdot \frac{-3}{4} \cdot \frac{-80}{6} + \frac{5}{4} \cdot \frac{1}{4} \cdot \frac{-3}{4} \cdot \frac{-7}{4} \cdot \frac{404}{24}$$

$$= 2806 + \frac{1065}{4} + \frac{215}{32} + \frac{25}{8} + \frac{3535}{512}$$

$$= 2806 + 266.2 + 6.7 + 3.1 + 6.9$$

$$= 3088.9 = 3089 \text{ (rounded)}$$

Adjusting the decimal point, the required value is 30.89 (Rs), which is exactly the same as obtained by Newton's backward formula.

The first 3 terms by the forward formula together give 30.79 (pretty close to 30.89), whereas by the backward formula we get only 32.99 (far away from 30.89); because, in this problem the appropriate formula is Newton's *Forward*.



ADDITIONAL EXERCISES 10



1. Given the following table, find the function $f(x)$ assuming it to be a polynomial of degree 4. Find also $f(5)$.

x	0	1	2	3	4
$f(x)$	1	5	31	121	341

[I.C.W.A., Dec. '79]

2. Obtain an estimate of the missing figure in the following table:

x	100	101	102	103	104
y_z	2.0000	2.0043	—	2.0128	2.0170

[I.C.W.A. Dec. '86]



3. Obtain an estimate of the missing figure in the following:

x	5.1	5.2	5.3	5.4	5.5
y	–	0.110	0.100	0.090	0.082

[I.C.W.A., Dec. '85]

4. Obtain an estimate of the missing figures in the following table:

x	3.0	3.1	3.2	3.3	3.4	3.5	3.6
$f(x)$	0.270	–	0.222	0.200	–	0.164	0.148

[I.C.W.A., Dec '83]

5. Given $U_{50} = 939$, $U_{52} = 907$, $U_{53} = 841$, $U_{55} = 773$, estimate U_{51} and U_{54} under suitable assumptions.

[I.C.W.A. Dec. '84]

6. From the following data, obtain the value of y when $x = 9$ by using Newton's Forward Difference Interpolation formula:

x	3	7	11	15	19
y	42	43	47	53	60

[I.C.W.A., Dec. '85]

7. The population of a city for five census years are given below :

Year (x)	1941	1951	1961	1971	1981
Population in Lakhs (y)	46	66	81	93	101

Estimate by using Newton's forward interpolation formula the population of the city for the year 1945.

[C.U., B.Com (P) '82]

8. Use Newton's Forward Interpolation formula to find y when $x = 12$ from the following table:

x	10	15	20	25	30	35
y	35.4	32.2	29.1	26.0	23.1	20.4

[C.U., B.Com. (Hon) '86]

9. The following table gives the values of an unknown function $f(x)$ for certain equidistant values of x . Use a suitable interpolation formula to find the value of the function at $x = 24$:

x	18	22	26	30	34
$f(x)$	11.725	14.221	16.958	19.954	23.222

[C.U., B.Com.(Hons) '81]

10. The values of y are 7, 21, 43 and 73 respectively, when the values of x are 2, 4, 6 and 8. Find by a suitable method of interpolation the value of y when x is equal to 7.

[C.U., B.Com. (P) '80]

11. Given $\sqrt[3]{5} = 1.710$, $\sqrt[3]{7} = 1.913$, $\sqrt[3]{9} = 2.080$, $\sqrt[3]{11} = 2.224$, find $\sqrt[3]{10}$.

[K.U., B.Com, (P) '83]



12. Use a suitable interpolation formula to find y when $x = 33$ from the following data:

x	24	28	32	36	40
y	25.06	27.19	29.75	31.94	37.00

[C.U., B.Com.(Hons.) '85]

13. Given $f(45) = 0.7071$, $f(50) = 0.7660$, $f(55) = 0.8192$, and $f(60) = 0.8693$, find $f(59)$ correct to 4 places decimals.

[I.C.W.A., June '79]

14. Using an appropriate interpolation formula, estimate the number of students who secured marks upto 45 and hence obtain the number so students securing marks between 45 and 50 from the following:

Marks Upto	40	50	60	70	80
No. of Students	31	73	124	159	190

[I.C.W.A., June '86 (old)]

15. Use Newton's Backward formula for interpolation to find the number of workers earning Rs 36 or more both less than Rs 40 per day, from the following table:

Earning per day less than (Rs)	20	25	30	35	40
Number of Workers	20	45	115	210	225

[I.C.W.A., June '83]

16. Given the following table of values, find $f(1.75)$:

x	1.40	1.60	1.70	1.80
$f(x)$	0.9855	0.9995	0.9917	0.9739

[I.C.W.A., Dec. '84]

17. Given $\log_{10} 654 = 2.8156$, $\log_{10} 658 = 2.8182$, $\log_{10} 659 = 2.8189$, $\log_{10} 661 = 2.8202$, find $\log_{10} 656$ using the interpolation formula available for observations at unequal intervals, say Lagrange's formula.

[I.C.W.A., June '83]

18. Use Lagrange's interpolation formula or otherwise to find $f(5)$ from the following data: $f(3) = 8.71940$, $f(7) = 9.08914$, $f(9) = 9.19433$.

[I.C.W.A., June '83]

19. Given the following table, find the value of x corresponding to $y = 2.8198$:

x	654	658	659	661
y	2.8156	2.8182	2.8189	2.8202

[I.C.W.A., June '81]

20. What is the inverse interpolation? Given the following table, find the value of x corresponding to $y = 1.2852$:

x	0.736	0.737	0.738	0.739
y	1.2833	1.2841	1.2849	1.2857

[I.C.W.A., Dec. '82]



ANSWERS



1. $f(x) = x^4 + x^3 + x^2 + x + 1$; 781.
[Hint: Example 10.8 (Second Method), p. 324 and Problem 10.1; write $f(x) = a + bx + cx(x-1) + dx(x-1)(x-2) + ex(x-1)(x-2)(x-3)$.]
2. 2.0086 **[Hint:** Example 10.6, p. 323]
3. 0.118 **[Hint:** Problem 10.3, p. 430]
4. 0.246, 0.1808 **[Hint:** Problem 10.4, p. 431]
5. 950.6, 785.4 **[Hint:** Example 10.5; p. 323]
6. 44.6875 **[Hint:** Example 10.11, p. 326]
7. 58.85 lakh **[Hint:** Example 10.11, p. 326].
8. 34.07 **[Hints:** Example 10.11, p. 326]
9. 15.558 **[Hint:** Newton's forward formula; ignore decimals in computing differences, see Problem 10.7]
10. 57 **[Hint:** Newton's backward formula Example 10.13, p. 329]
11. 2.154 **[Hint:** see Problem 10.3, $y = Nx$; use Newton's backward formula, Example 10.13, p. 329]
12. 30.27 **[Hint:** Newton's backward; Example 10.13, p. 329]
13. 0.8594 **[Hint:** Newton's backward; ignore decimals in difference table.]
14. 48, 25 **[Hint:** Problem 10.6]
15. 16 **[Hint:** Problem 10.6; use Newton's backward]
16. 0.9841 **[Hint:** Lagrange's formula; Example 10.15, p. 335]
17. 2.8168 **[Hint:** Write $y = \log_{10} x$; see Problem 10.2]
18. 8.93083 **[Hint:** Example 10 : 15, p. 335 and Example 10.6, p. 332]
19. 660.4 **[Hint:** Example 10.17, p. 334]
20. 0.7384 **[Hint:** Example 10.17, p. 334]

CALCUTTA UNIVERSITY

MODEL SHORT-ANSWER TYPE QUESTIONS— B.COM. (PASS)—STATISTICS

1. (a) Mention, giving reasons, the correct answer from the following:
Statistics consists of (i) a single figure: (ii) a group of homogeneous facts and figures; (iii) isolated fact.
- (b) What are the methods of collecting Primary data?
- (c) Which of the following is concerned with Primary data? (i) Annual Statement of the Foreign Trade of India, (ii) Annual Report of the Railway Board, (iii) International Labour Review, (iv) Reserve Bank of India Bulletin.
- (d) What methods would you employ in collection of data considering accuracy, time and cost involved when the field of enquiry is (i) small, (ii) fairly large, (iii) very large.

2. You are given the following distribution:

<i>Height (cm)</i> :	150.0–154.9	155.0–159.9	160.0–164.9	165.0–169.9	170.0–174.9
<i>No. of Students</i> :	5	10	16	12	7

- (a) Write down the class limits and class boundaries.
- (b) What are the frequency densities of the distributions?
- (c) Write down the percentage frequency in each class interval.
- (d) What are the cumulative frequencies from below?
3. State briefly, giving reasons, the kind of diagram you consider most appropriate for use with each of the following data:
 - (a) Monthly production of Hind Motor Cars in India for the year 1980.
 - (b) Budget allotments for the various departments.
 - (c) Different heads of income of a Railway Company during 1979 and 1980.
 - (d) Production and sale of electrical energy in West Bengal during the last 5 years.
4. (a) Find the arithmetic mean of 1, 2, 3, ..., n .
- (b) Find the geometric mean of 3, 12 and 48.
- (c) Find the H.M. of 1, $1/2$, $1/3$, $1/4$.
- (d) The A.M. of two observations is 36 and their G.M. is 24; find the H.M.



5. What type of average should be employed in each of the following cases? Give reasons for your answer:
 - (a) The marks obtained by 100 students in an examination.
 - (b) The size of shoes sold in largest number from a shop.
 - (c) A frequency distribution having open-end classes.
 - (d) Different rates of depreciation charged on diminishing balances.
6. (a) If the Mean and Median of a frequency distribution be 30 and 27 respectively, find the Mode.
 - (b) If the algebraic sum of the deviations of 10 observations measured from 20 is 30, find the Arithmetic Mean.
 - (c) If the A.M. of 3, 5, x , 12, 17 be 9, find the value of x .
 - (d) If for a symmetrical distribution $Q_1 = 24$, $Q_3 = 32$, find Median.
7. (a) Write down the value of Range of the daily wages of 6 persons given below: (Rs) 18, 25, 15, 20, 28, 22.
 - (b) If for a distribution, $Q_1 = 10$, $Q_2 = 22$, $Q_3 = 33$, find Quartile Deviation.
 - (c) If the absolute deviations of 7 observations from mean be 31, 35, 29, 63, 55, 72, 37, find Mean Deviation about mean.
 - (d) If for a frequency distribution Mean = 48, S.D. = 18.72, write down the value of Coefficient of Variation.
8. (a) If the means of two samples of sizes 40 and 60 are respectively 25 and 30, write down the value of Mean of the Combined Sample when the two samples are pooled together.
 - (b) The mean of 10 observations is 20. On checking it was found that two observations were wrongly copied as 15 and 17, while the correct observations were 18 and 19. What is the Correct Mean?
 - (c) Write down the value of S.D. of a distribution when Coefficient of Variation is 40 and Mean is 45.
 - (d) If for a distribution, $Q_1 = 5$, $Q_2 = 10$, $Q_3 = 15$, write down the value of Coefficient of Quartile Deviation.
9. Comment on the following statements:
 - (a) "The median of a distribution is $N/2$, the lower quartile is $N/4$ and the upper quartile is $3N/4$."
 - (b) "If Q_1 , Q_2 , Q_3 be respectively the lower quartile, the median and the upper quartile of a distribution, then $Q_3 - Q_2 = Q_2 - Q_1$."
 - (c) "The standard deviation of a distribution is invariant under changes of origin and scale."
 - (d) "Coefficient of variation is expressed in the same unit as the original data."
10. State with reasons whether the following statements are correct:
 - (a) "The standard deviation (s.d.) of 20 items is 8.5. If every item is increased by 4, the s.d. is increased by 4."
 - (b) "If the mean and S.D. of n observations $x_1, x_2, \dots, x_n$ be $\bar{x}$ and σ respectively, then the mean and the s.d. of $-x_1, -x_2, \dots, -x_n$ will be $-\bar{x}$ and $-\sigma$ respectively."
 - (c) "For a symmetrical distribution, all central moments of odd order are zero."
 - (d) "The 1st moment about origin is the mean and the 2nd moment about origin is the variance of the distribution."



11. (a) Write down the value of Mean, when the first moment of a distribution about the value 5 is 2
 (b) Write down the value of skewness of a distribution, when its Mean = 25, Mode = 22, S.D. = 6?
 (c) What is the value of Skewness of a distribution, when its Mean = 25, Median = 49, S.D. = 16?
 (d) What is the value of Kurtosis of a distribution, when its 4th central moment is 27 and $\sigma = \sqrt{3}$?
12. (a) If for a distribution $Q_1 = 5$, $Q_2 = 10$, $Q_3 = 13$, write down the value of Coefficient of Quartile Deviation.
 (b) What is the value of Coefficient of Mean Deviation when Mean Deviation about mean = 24 and Mean = 80 ?
 (c) Using Bowley's measure, write down with reasons the value of Skewness of a symmetrical distribution.
 (d) What is the value of Skewness of a distribution when the 2nd and the 3rd central moments are respectively 4 and 6?
13. (a) Write down the value of Correlation Coefficient r between x and y when $\text{cov}(x, y) = 9$, $\sigma_y = 3$, $\sigma_x = 4$.
 (b) If $r = 0.6$, $\text{cov}(x, y) = 12$, $\sigma_x = 5$, find the value of σ_y .
 (c) Write down the value of r when $b_{yx} = -0.4$, $b_{0.6} = -0.9$.
 (d) Write down the value of b_{yx} when $\sigma_x = 3$, $\sigma_y = 4$ and $r_{xy} = 0.75$.
14. (a) State in each case whether you would expect to find a positive correlation, a negative correlation: (i) Amount of rainfall and yield of crop, (ii) Shoe size and intelligence, (iii) Insurance companies profits and the number of claims they have to pay, (iv) The ages of husbands and wives.
 (b) State with reasons with measure of dispersion would you use to compare the variability of two or more distributions given in different units.
 (c) Mention the type of Kurtosis of a frequency distribution when (i) $\beta_2 = 4.56$, and (ii) $\beta_2 = 2.84$, where $\beta_2 = m_4/\sigma^4$.
 (d) Write down the value of Rank Correlation Coefficient when $V = 10$, $\Sigma D^2 = 30$, where D = difference between the ranks of two given series.
15. (a) You are given the following values of x and $f(x)$

x :	2	5	8	11	14
$f(x)$:	3	20	56	120	210

State with reasons which formula and would consider most appropriate to compute the value of $f(3.5)$ and $Q(9.3)$.

- (b) Write down the value of Coefficient of Skewness, when $Q_1 = 25$, $Q_2 = 36$, $Q_3 = 45$.
- (c) Draw the Scatter Diagram when $r = -1$ and $r = 1$.
- (d) The coefficient of correlation r of the following data is 0.8

X :	1	2	3	4	5
Y :	3	2	5	4	6

If the values of X are increased by 4 and those of Y decreased by 3, what will be the value of r after the changes? Give reasons for your answer.

ANSWERS

1. (a) (ii), since Statistics must be aggregates of facts and related to some specific field of enquiry (p. 2). (b) See p. 5. (c) (ii) and (iv). (d) (i) Direct personal observation, (ii) Schedules sent through investigators, (iii) Questionnaires sent by mail.
2. (a) *Class limits*: 150.0–154.9, 155.0–159.9, 160.0–164.9, 165.0–169.9, 170.0–174.9; *Class boundaries*: 149.95–154.95, 154.95–159.95, 159.95–164.95, 164.95–169.95, 169.95–174.95. (b) 1.0, 2.0, 3.2, 2.4, 1.4. (c) 10, 20, 32, 24, 14 (since $N = 50$). (d) 0, 5, 15, 31, 43, 50 corresponding to heights (cm) 149.95, 154.95, 159.95, 164.95, 169.95, 174.95 respectively (p. 83).
3. (a) Vertical Bar diagram (or Line diagram); since these are useful for the series data. (b) Pie diagram; since this is used for comparison among different components and also in relations to the total (p. 41). (c) Component Bar diagrams separately for each year; since comparison among different heads of income and also between the two years can be made (p. 39). (d) Grouped Bar diagram, one pair for each year showing Production and Sale; since comparison between them or year-to-year comparison is possible (p. 38).
4. (a) $(n + 1)/2$, (b) 12, (c) 0.4, (d) 16 (Example 5.44).
5. (a) A.M.; since it is simple to understand and based on all observations
(b) Mode; since it is representative because of its maximum frequency.
(c) Median; since midvalues and widths of open-end classes are not known.
(d) G.M.; since it is appropriate for averaging rates of change.
6. (a) 21 (Example 5.62), (b) 23, (c) 8, (d) 28 (Example 7.15).
7. (a) Rs 13, (b) 11.5, (c) 46, (d) 39.
8. (a) 28 (Example 5.25), (b) 20.5, (c) 18, (d) 50 (p. 216).
9. (a) and (b) See Nos. 50 (a) (b), p. 171,
(c) Invariant under change of origin, but depends on scale,
(d) Not correct; it is a pure number.
10. (a) Not correct; S.D. is unaffected by change of origin. (b) See No. 21, p. 220.
(c) Correct; (see p. 240). (d) 1st part correct, 2nd part not correct; since Variance = 2nd moment about *mean*.
11. (a) 7 (Example 7.1), (b) 0.5, (c) -0.75 , (d) $y_2 = 0$ (p. 241)
12. (a) 40 (p. 209), (b) 30, (c) 0, since for a symmetrical distribution $Q_1 - Q_2 = Q_2 - Q_1$. (d) $\gamma_1 = 0.75$ (p. 235).
13. (a) 0.75, (b) 4, (c) -0.6 , (d) 1
14. (a) See Example 9.6, (b) Coefficient of Variation; since absolute measures of dispersion will be in different units and hence not comparable.
(c) (i) Leptokurtic, (ii) Platykurtic (p. 242). (d) $+0.82$
15. (a) Since the given values of x are equidistant, Newton's Forward formula is appropriate for finding $f(3.5)$, because $x = 3.5$ lies near the beginning of the table, and Newton's Backward formula for $f(9.3)$, because $x = 9.3$ lies near the end of the table. (b) -0.1 , (c) See p. 275, (d) $r = 0.8$. since is unaffected by changes of origin (and also scale).

IMPORTANT FORMULAE AND RESULTS

Ref. Page

Measures of Central Tendency

Arithmetic Mean ($\bar{x}$) :

1. Simple A.M. = $\frac{1}{n} \sum x_i$
- Weighted A.M. = $\frac{1}{N} \sum f_i x_i$ (5.2.3) 106
2. $\sum x_i = n \bar{x}$; $\sum f_i x_i = N \bar{x}$ (5.3.1) 110
3. $\sum (x_i - \bar{x}) = 0$; $\sum f_i (x_i - \bar{x}) = 0$ (5.3.2) 110
4. If $y = (x - c)/d$, then $\bar{x} = c + d \bar{y}$ (5.4.2) 116
5. $N \bar{x} = n_1 \bar{x}_1 + n_2 \bar{x}_2$ (5.5.1) 124

Geometric Mean (G.M.):

6. Simple G.M. = $(x_1 \times x_2 \times \dots \times x_n)^{1/n}$
- Weighted G.M. = $(x_1^{f_1} \times x_2^{f_2} \times \dots \times x_n^{f_n})^{1/N}$ (5.6.3) 127
7. $\log (\text{simple G.M.}) = \frac{1}{n} \sum (\log x_i)$
- $\log (\text{weighted G.M.}) = \frac{1}{N} \sum f_i (\log x_i)$ (5.6.4) 127

Harmonic Mean (H.M.):

8. Simple H.M. = $\frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$ (5.8.1) 133
- Weighted H.M. = $\frac{N}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_n}{x_n}}$ (5.8.2) 133

Relation between A.M., G.M. and H.M.:

9. A.M. $\geq$ G.M. $\geq$ H.M. (5.10.1) 140



$$10. \frac{A.M.}{G.M.} = \frac{G.M.}{H.M.} \text{ (for 2 observations only)} \quad (5.10.2) \quad 137$$

$$11. \text{Median} = l_1 + \frac{N/2 - F}{f_m} \times c \quad (5.12.1) \quad 141$$

$$12. \frac{\text{Median} - l_1}{l_2 - l_1} = \frac{(N/2) - F_1}{F_2 - F_1} \quad (5.12.2) \quad 141$$

Mode:

$$13. \text{Mode} = l_1 + \frac{d_1}{d_1 + d_2} \times c$$

$$= l_1 + \frac{f_0 - f_{-1}}{2f_0 - f_{-1} - f_1} \times c \quad (5.15.1) \quad 152$$

$$14. \text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median}) \quad (5.17.1) \quad 157$$

Quartiles, Deciles and Percentiles:

$$15. \text{Median} = Q_2 = D_5 = P_{50};$$

$$Q_1 = P_{25}; Q_3 = P_{75} \quad (5.18.3) \quad 158$$

$$16. \text{(For simple series) Median, } Q_1, Q_3, \text{ have ranks } (n+1)/2, (n+1)/4 \text{ and } 3(n+1)/4 \quad (5.19.1) \quad 159$$

$$17. \text{(For Grouped Frequency distribution) Median, } Q_1, Q_3 \text{ correspond to cumulative frequencies } N/2, N/4, 3N/4 \quad (5.19.4) \quad 160$$

Measures of Dispersion

$$18. \text{Range} = \text{Maximum value} - \text{Minimum value} \quad (6.2.1) \quad 179$$

$$\text{Quartile Deviation} = (Q_3 - Q_1)/2 \quad (6.3.1) \quad 180$$

$$\text{Mean Deviation} = \frac{1}{n} \sum |x_i - \bar{x}|,$$

$$\text{or, } \frac{1}{N} \sum f_i |x_i - \bar{x}| \quad (6.4.2, 3) \quad 182$$

Standard Deviation (σ):

$$19. \sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2, \text{ or } \frac{1}{N} \sum f_i (x_i - \bar{x})^2 \quad (6.5.2, 3) \quad 185$$

$$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2, \text{ or } \frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N} \right)^2 \quad (6.7.1, 2) \quad 194$$

$$20. \text{If } y = x - c, \text{ then } \sigma_x = \sigma_y \quad (6.7.3) \quad 194$$

$$\text{If } y = (x - c)/d, \text{ then } \sigma_x = d \cdot \sigma_y \quad (6.7.4) \quad 194$$

$$21. N\sigma^2 = (n_1\sigma_1^2 + n_2\sigma_2^2) + (n_1d_1^2 + n_2d_2^2) \quad (6.8.1) \quad 203$$

Relative Measures:

$$22. \text{Coeff. of Variation} = 100 \times \frac{\text{S.D.}}{\text{Mean}}$$

$$\text{Coeff. of Quartile Deviation} = 100 \times \frac{\text{Q.D.}}{\text{Mean}} \quad 177$$

$$\text{Coeff. of Mean Deviation} = 100 \times \frac{\text{M.D.}}{\text{Mean (or Median)}}$$



Moments, Skewness and Kurtosis

$$23. \quad r\text{-th moment } (m_r') = \frac{1}{n} \sum (x_i - A)^r \quad (7.1.7) \quad 225$$

$$r\text{-th central moment } (m_r) = \frac{1}{n} \sum (x_i - \bar{x})^r \quad (7.1.7) \quad 225$$

$$24. \quad m_1' = \bar{x} - A, \quad (m_1 = 0, m_2 = \sigma^2) \quad (7.1.2, 6) \quad 224$$

$$\bar{x} = m_1' + A, \quad \sigma^2 = m_2' - m_1'^2 \quad (7.1.10) \quad 225$$

$$25. \quad m_2 = m_2' - m_1'^2$$

$$m_3 = m_3' - 3m_2'm_1' + 2m_1'^3$$

$$m_4 = m_4' - 4m_3'm_1' + 6m_2'm_1'^2 - 3m_1'^4 \quad (7.1.9) \quad 225$$

Beta-coefficients and Gamma-coefficients:

$$26. \quad \beta_1 = \frac{m_3'^2}{m_2'^3}; \quad \beta_2 = \frac{m_4'}{m_2'^2} \quad (7.3.1) \quad 229$$

$$\gamma_1 = \sqrt{\beta_1}; \quad \gamma_2 = \beta_2 - 3 \quad (7.3.2) \quad 229$$

Sheppard's Corrections for Grouping:

27. No corrections necessary for Mean and m_3 .

$$m_2 \text{ (corrected)} = m_2 - c^2/12$$

$$m_4 \text{ (corrected)} = m_4 - \frac{1}{2} c^2 m_2 + 7 c^4/240 \quad 233$$

Skewness:

$$28. \quad \text{Pearson's measure} = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$$

$$\text{or } \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}} \quad (7.7.1, 2) \quad 234$$

Bowley's measure

$$= \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)} = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1} \quad (7.7.3) \quad 234$$

$$\text{Moment measure } (\gamma_1) = \frac{m_3}{\sigma^3} = \frac{m_3}{(\sqrt{m_2})^3} \quad (7.7.4) \quad 234$$

Kurtosis:

$$29. \quad \text{Measure of kurtosis } (\gamma_2) = \frac{m_4}{\sigma^4} - 3 \quad (7.8.1) \quad 241$$

Curve Fitting and Method of Least Squares

Straight Line:

$$30. \quad y = a + bx; \text{ Normal equations } \sum y = an + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2 \quad (8.5.2) \quad 251$$

Parabola:

$$31. \quad y = a + bx + cx^2;$$

$$\text{Normal equations } \sum y = an + b \sum x + c \sum x^2$$



$$\begin{aligned}\Sigma xy &= a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \\ \Sigma xy &= a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4\end{aligned}\quad (8.7.2) \quad 259$$

32. When values of x have a common difference,

$$\text{write, } u = \frac{x - \text{central value of } x}{\text{common difference}}; (n \text{ odd}) \quad (8.6.3) \quad 252$$

$$u = \frac{x - (\text{mean of the two central values})}{\frac{1}{2} (\text{common difference})}; (n \text{ even}) \quad (8.6.5) \quad 252$$

and rewrite Straight line as $y = a + bu$, Parabola

as $y = a + bu + cu^2$, with u replacing x in normal equations.

Exponential and Geometric Curves:

33. $y = ab^x$; $y = ax^b$. Take logarithm of both sides, and transform equation into Straight line form with $Y = \log y$, $X = \log x$. 263

Correlation and Regression

Correlation:

$$34. \text{ Correlation coefficient } (r) = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \quad (9.7.1) \quad 276$$

$$\text{where } \text{cov}(x, y) = \frac{\Sigma xy}{n} - \left(\frac{\Sigma x}{n}\right) \left(\frac{\Sigma y}{n}\right) \quad (9.6.2) \quad 275$$

$$\sigma_x^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2; \sigma_y^2 = \frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n}\right)^2$$

$$35. \text{ If } u = (x - c)/d, v = (y - c')/d', \text{ then } r_{xy} = r_{uv} \quad (9.8.1) \quad 277$$

$$36. -1 \leq r \leq +1 \quad (9.8.3) \quad 277$$

$$\begin{aligned}37. \sigma_x^2 + y^2 &= \sigma_x^2 + \sigma_y^2 + 2r\sigma_x\sigma_y \\ \sigma_{x-y}^2 &= \sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y\end{aligned} \quad (9.11.3-4) \quad 286$$

Regression:

$$38. y - \bar{y} = b_{yx}(x - \bar{x}) \text{ Regression eqn. of } y \text{ on } x \quad (9.13.4) \quad 292$$

$$x - \bar{x} = b_{xy}(y - \bar{y}) \text{ Regression eqn. of } x \text{ on } y \quad (9.13.5) \quad 292$$

$$\text{where } b_{yx} = \frac{\text{cov}(x, y)}{\sigma_y^2} = r \frac{\sigma_x}{\sigma_y} \quad (9.13.6) \quad 292$$

$$b_{xy} = \frac{\text{cov}(x, y)}{\sigma_x^2} = r \frac{\sigma_y}{\sigma_x} \quad (9.113.7) \quad 292$$

$$39. b_{xy} \cdot b_{yx} = r^2 \quad (9.13.8) \quad 292$$

$$40. \Sigma(y_i - \bar{y})^2 = \Sigma(y_i - y_i')^2 + \Sigma(y_i' - \bar{y})^2 \quad 301$$

r^2 = proportion of variation explained by regression

Rank Correlation:

$$41. \text{ Rank correlation coefficient } (R) = 1 - \frac{6\Sigma d^2}{n^3 - n} \quad (9.16.1) \quad 304$$

Multiple and Partial Correlation:

$$42. R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}} \quad (9.17.3) \quad 309$$



$$43. \quad r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}} \quad (9.17.5) \quad 310$$

$$44. \quad 0 \leq R_{1.23} \leq 1; \quad -1 \leq r_{12.3} \leq 1 \quad (9.17.4, 6) \quad 310$$

$$45. \quad 1 - R_{1.23}^2 = (1 - r_{12}^2)(1 - r_{13.2}^2) \quad (9.17.7) \quad 310$$

$$= (1 - r_{13}^2)(1 - r_{12.3}^2) \quad (9.17.8) \quad 310$$

$$46. \quad R_{1.23} \geq |r_{12}|; \quad r_{1.23} \geq |r_{13}| \quad (9.17.8) \quad 310$$

Interpolation

$$47. \quad \Delta y_r = y_{r+1} - y_r; \quad (10.2.1) \quad 320$$

$$E y_r = y_{r+1}; \quad E = 1 + \Delta \quad (10.2.2) \quad 320$$

Newton's Forward formula: Writing $u = (x - x_0)/h$,

$$48. \quad y = y_0 + u \Delta y_0 + \frac{u(u-1)}{1 \times 2} \Delta^2 y_0 + \dots \dots$$

$$+ \frac{u(u-1) \dots (u-n+1)}{1 \times 2 \times \dots \times n} \Delta^n y_0 \quad (10.4.1) \quad 326$$

Newton's Backward formula: Writing $v = (x - x_n)/h$,

$$49. \quad y = y_n + v \Delta y_{n-1} + \frac{v(v+1)}{1 \times 2} \Delta^2 y_{n-2} + \dots \dots$$

$$+ \frac{v(v+1) \dots (v+n-1)}{1 \times 2 \times \dots \times n} \Delta^n y_0 \quad (10.5.1) \quad 329$$

Stirling's formula:

$$50. \quad y = y_0 + u \cdot \frac{(\Delta y_0 + \Delta y_{-1})}{2} + \frac{u^2}{2!} \Delta^2 y_{-1}$$

$$+ \frac{u(u^2 - 1^2)}{3!} \cdot \frac{(\Delta^3 y_{-1} + \Delta^3 y_{-2})}{2} + \dots \dots$$

where $u = (x - x_0)/h$. (10.6.1) 332

Bessel's formula:

$$51. \quad y = \frac{(y_0 + y_1)}{2} + v \cdot \Delta y_0 + \frac{(v^2 - \frac{1}{4})}{2!} \cdot \frac{(\Delta^2 y_0 + \Delta^2 y_{-1})}{2}$$

$$+ \frac{v(v^2 - \frac{1}{4})}{3!} \cdot \Delta^3 y_{-1} + \dots \dots$$

where $v = u - \frac{1}{2}$ and $u = (x - x_0)/h$. (10.6.2) 332

Lagrange's formula:

$$52. \quad y = \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0$$

$$+ \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1$$

$$+ \dots \dots \dots$$

$$+ \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} y_n \quad (10.7.1) \quad 333$$

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